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Online Instructor's Manual
to accompany

Introductory Circuit Analysis

Twelfth Edition

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Chapter 1

1. –

2. –

3. –

$$4. \quad v = \frac{d}{t} = \frac{20,000 \cancel{\text{ft}}}{10 \cancel{\text{s}}} \left[\frac{1 \text{ mi}}{5,280 \cancel{\text{ft}}} \right] \left[\frac{60 \cancel{\text{s}}}{1 \cancel{\text{min}}} \right] \left[\frac{60 \cancel{\text{min}}}{1 \text{ h}} \right] = \mathbf{1363.64 \text{ mph}}$$

$$5. \quad 4 \cancel{\text{min}} \left[\frac{1 \text{ h}}{60 \cancel{\text{min}}} \right] = \mathbf{0.067 \text{ h}}$$

$$v = \frac{d}{t} = \frac{31 \text{ mi}}{1.067 \text{ h}} = \mathbf{29.05 \text{ mph}}$$

$$6. \quad 100 \cancel{\text{yd}} \left[\frac{3 \cancel{\text{ft}}}{1 \cancel{\text{yd}}} \right] \left[\frac{1 \text{ mi}}{5,280 \cancel{\text{ft}}} \right] = 0.0568 \text{ mi}$$

$$\frac{60 \text{ mi}}{\cancel{\text{h}}} \left[\frac{1 \cancel{\text{h}}}{60 \cancel{\text{min}}} \right] \left[\frac{1 \cancel{\text{min}}}{60 \text{ s}} \right] = 0.0167 \text{ mi/s}$$

$$t = \frac{d}{v} = \frac{0.0568 \text{ mi}}{0.0167 \text{ mi/s}} = \mathbf{3.40 \text{ s}}$$

$$7. \quad \text{a.} \quad \frac{95 \cancel{\text{mi}}}{\cancel{\text{h}}} \left[\frac{5,280 \text{ ft}}{\cancel{\text{mi}}} \right] \left[\frac{1 \cancel{\text{h}}}{60 \cancel{\text{min}}} \right] \left[\frac{1 \cancel{\text{min}}}{60 \text{ s}} \right] = \mathbf{139.33 \text{ ft/s}}$$

$$\text{b.} \quad t = \frac{d}{v} = \frac{60 \text{ ft}}{139.33 \text{ ft/s}} = \mathbf{0.431 \text{ s}}$$

$$\text{c.} \quad v = \frac{d}{t} = \frac{60 \cancel{\text{ft}}}{1 \cancel{\text{s}}} \left[\frac{60 \cancel{\text{s}}}{1 \cancel{\text{min}}} \right] \left[\frac{60 \cancel{\text{min}}}{1 \text{ h}} \right] \left[\frac{1 \text{ mi}}{5,280 \cancel{\text{ft}}} \right] = \mathbf{40.91 \text{ mph}}$$

8. –

9. –

10. –

$$11. \quad \text{MKS, CGS, } ^\circ\text{C} = \frac{5}{9} (^{\circ}\text{F} - 32) = \frac{5}{9} (68 - 32) = \frac{5}{9} (36) = \mathbf{20^\circ}$$

$$\text{SI: } \text{K} = 273.15 + ^\circ\text{C} = 273.15 + 20 = \mathbf{293.15}$$

$$12. \quad 1000 \cancel{\text{J}} \left[\frac{0.7378 \text{ ft} \cdot \text{lb}}{1 \cancel{\text{J}}} \right] = \mathbf{737.8 \text{ ft} \cdot \text{lbs}}$$

13. $0.5 \cancel{\text{yd}} \left[\frac{3 \cancel{\text{ft}}}{1 \cancel{\text{yd}}} \right] \left[\frac{12 \cancel{\text{in.}}}{1 \cancel{\text{ft}}} \right] \left[\frac{2.54 \text{ cm}}{1 \cancel{\text{in.}}} \right] = 45.72 \text{ cm}$
14. a. $^{\circ}\text{F} = 2(^{\circ}\text{C}) + 30^{\circ} = 40^{\circ} + 30^{\circ} = 70^{\circ}$
- b. $^{\circ}\text{F} = \frac{9}{5}(^{\circ}\text{C}) + 32^{\circ} = \frac{9}{5}(20^{\circ}) + 32^{\circ} = 68^{\circ}$
- c. very close
- d. $30^{\circ}\text{C} \rightarrow 90^{\circ}\text{F}$ vs. 86°
 $5^{\circ}\text{C} \rightarrow 40^{\circ}\text{F}$ vs. 41°
15. a. **14.6** b. **56.0**
- c. **1,046.1** d. $\frac{1}{16} = 0.0625 = 0.1$
- e. $3.14159 = 3.1$
16. a. **14.60** b. **56.04**
- c. **1,046.06** d. $\frac{1}{16} = 0.0625 = 0.06$
- e. $3.14159 = 3.14$
17. a. **14.603** b. **56.042**
- c. **1,046.060** d. $\frac{1}{16} = 0.0625 = 0.063$
- e. $3.14159 = 3.142$
18. a. 10^4 b. 10^6 c. 10^3 d. 10^{-3}
- e. 10^0 f. 10^{-1}
19. a. 15×10^3 b. 5×10^{-3} c. 2.4×10^6 d. 60×10^3
- e. 4.02×10^{-4} f. 2×10^{-10}
20. a. $4.2 \times 10^3 + 48.0 \times 10^3 = 52.2 \times 10^3 = 5.22 \times 10^4$
- b. $90 \times 10^3 + 360 \times 10^3 = 450 \times 10^3 = 4.50 \times 10^5$
- c. $50 \times 10^{-5} - 6 \times 10^{-5} = 44 \times 10^{-5} = 4.4 \times 10^{-4}$
- d. $1.2 \times 10^3 + 0.05 \times 10^3 - 0.4 \times 10^3 = 0.85 \times 10^3 = 850$
21. a. $(10^2)(10^3) = 10^5 = 100 \times 10^3$
- b. $(10^{-2})(10^3) = 10^1 = 10$
- c. $(10^3)(10^6) = 1 \times 10^9$
- d. $(10^2)(10^{-5}) = 1 \times 10^{-3}$
- e. $(10^{-6})(10 \times 10^6) = 10$
- f. $(10^4)(10^{-8})(10^{28}) = 1 \times 10^{24}$

22. a. $(50 \times 10^3)(2 \times 10^{-3}) = 100 \times 10^0 = \mathbf{100}$
 b. $(2.2 \times 10^3)(2 \times 10^{-3}) = 4.4 \times 10^0 = \mathbf{4.4}$
 c. $(82 \times 10^{-6})(1.2 \times 10^{-6}) = \mathbf{98.4}$
 d. $(30 \times 10^{-4})(4 \times 10^{-3})(7 \times 10^8) = 840 \times 10^1 = \mathbf{8.40 \times 10^3}$
23. a. $10^2/10^4 = 10^{-2} = \mathbf{10 \times 10^{-3}}$
 b. $10^{-2}/10^3 = 10^{-5} = \mathbf{10 \times 10^{-6}}$
 c. $10^4/10^{-3} = 10^7 = \mathbf{10 \times 10^6}$
 d. $10^{-7}/10^2 = \mathbf{1.0 \times 10^{-9}}$
 e. $10^{38}/10^{-4} = \mathbf{1.0 \times 10^{42}}$
 f. $\sqrt{100}/10^{-2} = 10^1/10^{-2} = \mathbf{1 \times 10^3}$
24. a. $(2 \times 10^3)/(8 \times 10^{-5}) = 0.25 \times 10^8 = \mathbf{2.50 \times 10^7}$
 b. $(4 \times 10^{-3})/(4 \times 10^6) = 4/4 \times 10^{-9} = \mathbf{1 \times 10^{-9}}$
 c. $(22 \times 10^{-5})/(5 \times 10^{-5}) = 22/5 \times 10^0 = \mathbf{4.4}$
 d. $(78 \times 10^{18})/(4 \times 10^{-6}) = \mathbf{1.95 \times 10^{25}}$
25. a. $(10^2)^3 = \mathbf{1.0 \times 10^6}$ b. $(10^{-4})^{1/2} = \mathbf{10.0 \times 10^{-3}}$
 c. $(10^4)^8 = \mathbf{100.0 \times 10^{30}}$ d. $(10^{-7})^9 = \mathbf{1.0 \times 10^{-63}}$
26. a. $(2 \times 10^2)^2 = \mathbf{4 \times 10^4}$
 b. $(5 \times 10^{-3})^3 = \mathbf{125 \times 10^{-9}}$
 c. $(4 \times 10^{-3})(3 \times 10^{-3})^2 = (4 \times 10^{-3})(9 \times 10^4) = 36 \times 10^1 = \mathbf{360}$
 d. $((2 \times 10^{-3})(0.8 \times 10^4)(0.003 \times 10^5))^3 = (4.8 \times 10^3)^3 = (4.8)^3 \times (10^3)^3$
 $= 110.6 \times 10^9 = \mathbf{1.11 \times 10^{11}}$
27. a. $(-10^{-3})^2 = \mathbf{1.0 \times 10^{-6}}$
 b. $\frac{(10^2)(10^{-4})}{10^3} = 10^{-2}/10^3 = \mathbf{1.0 \times 10^{-5}}$
 c. $\frac{(10^{-3})^2(10^2)}{10^4} = \frac{(10^{-6})(10^2)}{10^4} = \frac{10^{-4}}{10^4} = \mathbf{1.0 \times 10^{-8}}$
 d. $\frac{(10^3)(10^4)}{10^{-4}} = 10^7/10^{-4} = \mathbf{1.0 \times 10^{11}}$
 e. $(1 \times 10^{-4})^3(10^2)/10^6 = (10^{-12})(10^2)/10^6 = 10^{-10}/10^6 = \mathbf{1.0 \times 10^{-16}}$
 f. $\frac{[(10^2)(10^{-2})]^3}{[(10^2)^2][10^{-3}]} = \frac{1}{(10^4)(10^{-3})} = \frac{1}{10} = \mathbf{1.0 \times 10^{-1}}$
28. a. $\frac{(3 \times 10^2)^2(10^2)}{3 \times 10^4} = (9 \times 10^4)(10^2)/(3 \times 10^4) = (9 \times 10^6)/(3 \times 10^4) = 3 \times 10^2 = \mathbf{300}$
 b. $\frac{(4 \times 10^4)^2}{(20)^3} = \frac{16 \times 10^8}{8 \times 10^3} = \mathbf{2 \times 10^5}$
 c. $\frac{(6 \times 10^4)^2}{(2 \times 10^{-2})^2} = \frac{36 \times 10^8}{4 \times 10^{-4}} = \mathbf{9.0 \times 10^{12}}$

$$d. \quad \frac{(27 \times 10^{-6})^{1/3}}{2 \times 10^5} = \frac{3 \times 10^{-2}}{2 \times 10^5} = 1.5 \times 10^{-7} = \mathbf{150.0 \times 10^{-9}}$$

$$e. \quad \frac{(4 \times 10^3)^2 (3 \times 10^2)}{2 \times 10^{-4}} = \frac{(16 \times 10^6)(3 \times 10^2)}{2 \times 10^{-4}} = \frac{48 \times 10^8}{2 \times 10^{-4}} = \mathbf{24.0 \times 10^{12}}$$

$$f. \quad (16 \times 10^{-6})^{1/2} (10^5)^5 (2 \times 10^{-2}) = (4 \times 10^{-3})(10^{25})(2 \times 10^{-2}) = 8 \times 10^{20} = \mathbf{800.0 \times 10^{18}}$$

$$g. \quad \frac{\left[(3 \times 10^{-3})^3 \right] \left[1.60 \times 10^2 \right]^2 \left[(2 \times 10^2)(8 \times 10^{-4}) \right]^{1/2}}{(7 \times 10^{-5})^2}$$

$$= \frac{(27 \times 10^{-9})(2.56 \times 10^4)(16 \times 10^{-2})^{1/2}}{49 \times 10^{-10}}$$

$$= \frac{(69.12 \times 10^{-5})(4 \times 10^{-1})}{49 \times 10^{-10}} = \frac{276.48 \times 10^{-6}}{49 \times 10^{-10}}$$

$$= 5.64 \times 10^4 = \mathbf{56.4 \times 10^3}$$

29. Scientific:
- a. 2.05×10^1
 - b. 5.04×10^4
 - c. 6.74×10^{-4}
 - d. 4.60×10^{-2}

- Engineering:
- a. 20.46×10^0
 - b. 50.42×10^3
 - c. 674.00×10^{-6}
 - d. 46.00×10^{-3}

30. Scientific
- a. 5.0×10^{-2}
 - b. 4.5×10^1
 - c. $1/32 = 0.03125 = 3.125 \times 10^{-2}$
 - d. $3.14159 = 3.142 \times 10^0$

- Engineering:
- a. 50.0×10^{-3}
 - b. 0.045×10^3
 - c. 31.25×10^{-3}
 - d. 3.142×10^0

31. a. $6 \times 10^4 = \underline{0.06} \times 10^{+6} = \mathbf{0.06 \times 10^{+6}}$
 (Curved arrows: $6 \times 10^4 \xrightarrow{+2} \underline{0.06} \times 10^{+6}$ and $6 \times 10^4 \xrightarrow{-3} \underline{0.06} \times 10^{+6}$)
- b. $0.4 \times 10^{-3} = \underline{400} \times 10^{-6} = \mathbf{400 \times 10^{-6}}$
 (Curved arrows: $0.4 \times 10^{-3} \xrightarrow{-3} \underline{400} \times 10^{-6}$ and $0.4 \times 10^{-3} \xrightarrow{+3} \underline{400} \times 10^{-6}$)
- c. $50 \times 10^5 = \underline{5000} \times 10^3 = \underline{5} \times 10^6 = \underline{0.005} \times 10^9 = \mathbf{0.005 \times 10^9}$
 (Curved arrows: $50 \times 10^5 \xrightarrow{-2} \underline{5000} \times 10^3$, $50 \times 10^5 \xrightarrow{+2} \underline{5000} \times 10^3$, $\underline{5000} \times 10^3 \xrightarrow{-3} \underline{5} \times 10^6$, $\underline{5000} \times 10^3 \xrightarrow{+3} \underline{5} \times 10^6$, $\underline{5} \times 10^6 \xrightarrow{-3} \underline{0.005} \times 10^9$, $\underline{5} \times 10^6 \xrightarrow{+3} \underline{0.005} \times 10^9$)
- d. $12 \times 10^{-7} = \underline{0.0012} \times 10^{-3} = \underline{1.2} \times 10^{-6} = \underline{1200} \times 10^{-9} = \mathbf{1200 \times 10^9}$
 (Curved arrows: $12 \times 10^{-7} \xrightarrow{+4} \underline{0.0012} \times 10^{-3}$, $12 \times 10^{-7} \xrightarrow{-4} \underline{0.0012} \times 10^{-3}$, $\underline{0.0012} \times 10^{-3} \xrightarrow{-3} \underline{1.2} \times 10^{-6}$, $\underline{0.0012} \times 10^{-3} \xrightarrow{+3} \underline{1.2} \times 10^{-6}$, $\underline{1.2} \times 10^{-6} \xrightarrow{-3} \underline{1200} \times 10^{-9}$, $\underline{1.2} \times 10^{-6} \xrightarrow{+3} \underline{1200} \times 10^{-9}$)
32. a. $0.05 \times 10^0 \text{ s} = \underline{50} \times 10^{-3} \text{ s} = \mathbf{50 \text{ ms}}$
 (Curved arrows: $0.05 \times 10^0 \text{ s} \xrightarrow{-3} \underline{50} \times 10^{-3} \text{ s}$ and $0.05 \times 10^0 \text{ s} \xrightarrow{+3} \underline{50} \times 10^{-3} \text{ s}$)
- b. $2000 \times 10^{-6} \text{ s} = \underline{2} \times 10^{-3} \text{ s} = \mathbf{2 \text{ ms}}$
 (Curved arrows: $2000 \times 10^{-6} \text{ s} \xrightarrow{+3} \underline{2} \times 10^{-3} \text{ s}$ and $2000 \times 10^{-6} \text{ s} \xrightarrow{-3} \underline{2} \times 10^{-3} \text{ s}$)
- c. $0.04 \times 10^{-3} \text{ s} = \underline{40} \times 10^{-6} \text{ s} = \mathbf{40 \mu\text{s}}$
 (Curved arrows: $0.04 \times 10^{-3} \text{ s} \xrightarrow{-3} \underline{40} \times 10^{-6} \text{ s}$ and $0.04 \times 10^{-3} \text{ s} \xrightarrow{+3} \underline{40} \times 10^{-6} \text{ s}$)
- d. $8400 \times 10^{-12} \text{ s} \Rightarrow \underline{0.0084} \times 10^{-6} \text{ s} = \mathbf{0.0084 \mu\text{s}}$
 (Curved arrows: $8400 \times 10^{-12} \text{ s} \xrightarrow{+6} \underline{0.0084} \times 10^{-6} \text{ s}$ and $8400 \times 10^{-12} \text{ s} \xrightarrow{-6} \underline{0.0084} \times 10^{-6} \text{ s}$)

$$\begin{array}{c}
 +3 \\
 \text{increase by 3} \\
 \downarrow \\
 10^0 \\
 \uparrow \\
 100 \times 10^3 \times 10^{-3} \text{ m} = \underline{0.1} \times 10^3 \text{ m} = \mathbf{0.1 \text{ km}} \\
 \downarrow \\
 -3
 \end{array}$$

33. a. $1.5 \cancel{\text{min}} \left[\frac{60 \text{ s}}{1 \cancel{\text{min}}} \right] = \mathbf{90 \text{ s}}$
- b. $2 \times 10^{-2} \cancel{\text{h}} \left[\frac{60 \cancel{\text{min}}}{1 \cancel{\text{h}}} \right] \left[\frac{60 \text{ s}}{1 \cancel{\text{min}}} \right] = \mathbf{72 \text{ s}}$
- c. $0.05 \cancel{\text{s}} \left[\frac{1 \mu\text{s}}{10^{-6} \cancel{\text{s}}} \right] = \mathbf{0.05 \times 10^6 \mu\text{s} = 50 \times 10^3 \mu\text{s}}$
- d. $0.16 \cancel{\text{m}} \left[\frac{1 \text{ mm}}{10^{-3} \cancel{\text{m}}} \right] = 0.16 \times 10^3 \text{ mm} = \mathbf{160 \text{ mm}}$
- e. $1.2 \times 10^{-7} \cancel{\text{s}} \left[\frac{1 \text{ ns}}{10^{-9} \cancel{\text{s}}} \right] = 1.2 \times 10^2 \text{ ns} = \mathbf{120 \text{ ns}}$
- f. $4 \times 10^8 \cancel{\text{s}} \left[\frac{1 \cancel{\text{min}}}{60 \cancel{\text{s}}} \right] \left[\frac{1 \cancel{\text{h}}}{60 \cancel{\text{min}}} \right] \left[\frac{1 \text{ day}}{24 \cancel{\text{h}}} \right] = \mathbf{4629.63 \text{ days}}$

34. a. $80 \times 10^{-3} \cancel{\text{m}} \left[\frac{100 \text{ cm}}{1 \cancel{\text{m}}} \right] = 8000 \times 10^{-3} \text{ cm} = \mathbf{8 \text{ cm}}$
- b. $60 \cancel{\text{cm}} \left[\frac{1 \cancel{\text{m}}}{100 \cancel{\text{cm}}} \right] \left[\frac{1 \text{ km}}{1000 \cancel{\text{m}}} \right] = \mathbf{60 \times 10^{-5} \text{ km}}$
- c. $12 \times 10^{-3} \cancel{\text{m}} \left[\frac{1 \mu\text{m}}{10^{-6} \cancel{\text{m}}} \right] = 12 \times 10^{-3} \times 10^{+6} \mu\text{m} = \mathbf{12 \times 10^3 \mu\text{m}}$
- d. $60 \cancel{\text{cm}^2} \left[\frac{1 \text{ m}}{100 \cancel{\text{cm}}} \right] \left[\frac{1 \text{ m}}{100 \cancel{\text{cm}}} \right] = \mathbf{60 \times 10^{-4} \text{ m}^2}$

35. a. $100 \cancel{\text{in.}} \left[\frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \right] = \mathbf{2.54 \text{ m}}$
- b. $4 \cancel{\text{ft}} \left[\frac{12 \cancel{\text{in.}}}{1 \cancel{\text{ft}}} \right] \left[\frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \right] = \mathbf{1.22 \text{ m}}$

$$c. \quad 6 \cancel{\text{N}} \left[\frac{4.45 \text{ N}}{1 \cancel{\text{N}}} \right] = \mathbf{26.7 \text{ N}}$$

$$d. \quad 60 \times 10^3 \cancel{\text{dynes}} \left[\frac{1 \cancel{\text{N}}}{10^5 \cancel{\text{dynes}}} \right] \left[\frac{1 \text{ lb}}{4.45 \cancel{\text{N}}} \right] = \mathbf{0.13 \text{ lb}}$$

$$e. \quad 150,000 \cancel{\text{cm}} \left[\frac{1 \cancel{\text{in.}}}{2.54 \cancel{\text{cm}}} \right] \left[\frac{1 \text{ ft}}{12 \cancel{\text{in.}}} \right] = \mathbf{4921.26 \text{ ft}}$$

$$f. \quad 0.002 \cancel{\text{mi}} \left[\frac{5280 \cancel{\text{ft}}}{1 \cancel{\text{mi}}} \right] \left[\frac{12 \cancel{\text{in.}}}{1 \cancel{\text{ft}}} \right] \left[\frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \right] = \mathbf{3.22 \text{ m}}$$

$$36. \quad \mathbf{5280 \text{ ft}}, \quad 5280 \cancel{\text{ft}} \left[\frac{1 \text{ yd}}{3 \cancel{\text{ft}}} \right] = \mathbf{1760 \text{ yds}}$$

$$5280 \cancel{\text{ft}} \left[\frac{12 \cancel{\text{in.}}}{1 \cancel{\text{ft}}} \right] \left[\frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \right] = \mathbf{1609.35 \text{ m}, 1.61 \text{ km}}$$

$$37. \quad \frac{60 \cancel{\text{mi}}}{1 \cancel{\text{h}}} \frac{5280 \cancel{\text{ft}}}{1 \cancel{\text{mi}}} \frac{12 \cancel{\text{in.}}}{1 \cancel{\text{ft}}} \frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \frac{1 \cancel{\text{h}}}{60 \cancel{\text{min}}} \frac{1 \cancel{\text{min}}}{60 \text{ s}} = \mathbf{26.82 \text{ m/s}}$$

$$38. \quad 10 \cancel{\text{km}} \left[\frac{1000 \cancel{\text{m}}}{1 \cancel{\text{km}}} \right] \left[\frac{39.37 \cancel{\text{in.}}}{1 \cancel{\text{m}}} \right] \left[\frac{1 \cancel{\text{ft}}}{12 \cancel{\text{in.}}} \right] \left[\frac{1 \text{ mi}}{5280 \cancel{\text{ft}}} \right] = 6.214 \text{ mi}$$

$$v = \frac{1 \text{ mi}}{6.5 \text{ min}}, \quad t = \frac{d}{v} = \frac{6.214 \cancel{\text{mi}}}{\frac{1 \cancel{\text{mi}}}{6.5 \text{ min}}} = \mathbf{40.39 \text{ min}}$$

$$39. \quad 100 \cancel{\text{yds}} \left[\frac{3 \cancel{\text{ft}}}{1 \cancel{\text{yd}}} \right] \left[\frac{12 \text{ in.}}{1 \cancel{\text{ft}}} \right] = 3600 \text{ in} \Rightarrow \mathbf{3600 \text{ quarters}}$$

$$40. \quad 60 \text{ mph:} \quad t = \frac{d}{v} = \frac{500 \text{ mi}}{60 \text{ mi/h}} = 8.33 \text{ h} = \mathbf{8 \text{ h}:19.8 \text{ min}}$$

$$75 \text{ mph:} \quad t = \frac{d}{v} = \frac{500 \text{ mi}}{75 \text{ mi/h}} = 6.67 \text{ h} = \mathbf{6 \text{ h}:40.2 \text{ min}}$$

$$\text{difference} = \mathbf{1 \text{ h}:28.6 \text{ minutes}}$$

$$41. \quad d = vt = \left[600 \frac{\cancel{\text{cm}}}{\cancel{\text{s}}} \right] [0.016 \cancel{\text{h}}] \left[\frac{60 \cancel{\text{min}}}{1 \cancel{\text{h}}} \right] \left[\frac{60 \cancel{\text{s}}}{1 \cancel{\text{min}}} \right] \left[\frac{1 \cancel{\text{m}}}{100 \cancel{\text{cm}}} \right] = \mathbf{345.6 \text{ m}}$$

42. $d = 86 \cancel{\text{ stories}} \left[\frac{14 \cancel{\text{ ft}}}{\cancel{\text{ story}}} \right] \left[\frac{1 \cancel{\text{ step}}}{\frac{9}{12} \cancel{\text{ ft}}} \right] = 1605 \text{ steps}$
 $v = \frac{d}{t} \Rightarrow t = \frac{d}{v} = \frac{1605 \text{ steps}}{\frac{2 \text{ steps}}{\text{second}}} = 802.5 \cancel{\text{ seconds}} \left[\frac{1 \cancel{\text{ minute}}}{60 \cancel{\text{ seconds}}} \right] = \mathbf{13.38 \text{ minutes}}$
43. $d = (86 \cancel{\text{ stories}}) \left[\frac{14 \cancel{\text{ ft}}}{\cancel{\text{ story}}} \right] = 1204 \cancel{\text{ ft}} \left[\frac{1 \cancel{\text{ mile}}}{5,280 \cancel{\text{ ft}}} \right] = 0.228 \text{ miles}$
 $\frac{\text{min}}{\text{mile}} = \frac{10.22 \text{ min}}{0.228 \text{ miles}} = \mathbf{44.82 \text{ min/mile}}$
44. $\frac{5 \text{ min}}{\text{mile}} \Rightarrow \frac{1 \cancel{\text{ mile}}}{5 \text{ min}} \left[\frac{5,280 \cancel{\text{ ft}}}{1 \cancel{\text{ mile}}} \right] = \frac{1056 \text{ ft}}{\text{minute}}, \quad \text{distance} = 86 \cancel{\text{ stories}} \left[\frac{14 \cancel{\text{ ft}}}{\cancel{\text{ story}}} \right] = 1204 \text{ ft}$
 $v = \frac{d}{t} \Rightarrow t = \frac{d}{v} = \frac{1204 \text{ ft}}{1056 \frac{\text{ft}}{\text{min}}} = \mathbf{1.14 \text{ minutes}}$
45. a. $5 \cancel{\text{ J}} \left[\frac{1 \cancel{\text{ Btu}}}{1054.35 \cancel{\text{ J}}} \right] = \mathbf{4.74 \times 10^{-3} \text{ Btu}}$
b. $24 \cancel{\text{ ounces}} \left[\frac{1 \cancel{\text{ gallon}}}{128 \cancel{\text{ ounces}}} \right] \left[\frac{1 \text{ m}^3}{264.172 \cancel{\text{ gallons}}} \right] = \mathbf{7.1 \times 10^{-4} \text{ m}^3}$
c. $1.4 \cancel{\text{ days}} \left[\frac{86,400 \cancel{\text{ s}}}{1 \cancel{\text{ day}}} \right] = \mathbf{1.21 \times 10^5 \text{ s}}$
d. $1 \cancel{\text{ m}^3} \left[\frac{264.172 \cancel{\text{ gallons}}}{1 \cancel{\text{ m}^3}} \right] \left[\frac{8 \cancel{\text{ pints}}}{1 \cancel{\text{ gallon}}} \right] = \mathbf{2113.38 \text{ pints}}$
46. $6(4 \times 2 + 8) = \mathbf{96}$
47. $(42 + 6/5)/3 = \mathbf{14.4}$
48. $\sqrt{5^2 + \frac{2}{3}^2} = \mathbf{5.044}$
49. MODE = DEGREES: $\cos 21.87^\circ = \mathbf{0.928}$
50. MODE = DEGREES: $\tan^{-1}(3/4) = \mathbf{36.87^\circ}$
51. $\sqrt{400/(6^2 + 10/5)} = \mathbf{7.071}$

52. 205×10^{-6}

53. 1.20×10^{12}

54. $6.667 \times 10^6 + 0.5 \times 10^6 = 7.17 \times 10^6$

Chapter 2

1. —

2. a. $F = k \frac{Q_1 Q_2}{r^2} = \frac{(9 \times 10^9)(1 \text{ C})(2 \text{ C})}{(1 \text{ m})^2} = \mathbf{18 \times 10^9 \text{ N}}$

b. $F = k \frac{Q_1 Q_2}{r^2} = \frac{(9 \times 10^9)(1 \text{ C})(2 \text{ C})}{(3 \text{ m})^2} = \mathbf{2 \times 10^9 \text{ N}}$

c. $F = k \frac{Q_1 Q_2}{r^2} = \frac{(9 \times 10^9)(1 \text{ C})(2 \text{ C})}{(10 \text{ m})^2} = \mathbf{0.18 \times 10^9 \text{ N}}$

d. Exponentially, $\frac{r_3}{r_1} = \frac{10 \text{ m}}{1 \text{ m}} = 10$ while $\frac{F_1}{F_2} = \frac{18 \times 10^9 \text{ N}}{0.18 \times 10^9 \text{ N}} = \mathbf{100}$

3. a. $r = 1 \text{ mi}$:

$$1 \cancel{\text{mi}} \left[\frac{5280 \text{ ft}}{1 \cancel{\text{mi}}} \right] \left[\frac{12 \cancel{\text{in.}}}{1 \text{ ft}} \right] \left[\frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \right] = 1609.35 \text{ m}$$

$$F = \frac{k Q_1 Q_2}{r^2} = \frac{(9 \times 10^9)(8 \times 10^{-6} \text{ C})(40 \times 10^{-6} \text{ C})}{(1609.35 \text{ m})^2} = \frac{2880 \times 10^{-3}}{2.59 \times 10^6}$$

$$= \mathbf{1.11 \mu\text{N}}$$

b. $r = 10 \text{ ft}$:

$$10 \cancel{\text{ft}} \left[\frac{12 \cancel{\text{in.}}}{1 \cancel{\text{ft}}} \right] \left[\frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \right] = 3.05 \text{ m}$$

$$F = \frac{k Q_1 Q_2}{r^2} = \frac{2880 \times 10^{-3}}{(3.05 \text{ m})^2} = \frac{2880 \times 10^{-3}}{9.30} = \mathbf{0.31 \text{ N}}$$

c. $\frac{1 \cancel{\text{in.}}}{16} \left[\frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \right] = 1.59 \text{ mm}$

$$F = \frac{k Q_1 Q_2}{r^2} = \frac{2880 \times 10^{-3}}{(1.59 \times 10^{-3} \text{ m})^2} = \frac{2880 \times 10^{-3}}{2.53 \times 10^{-6}} = 1138.34 \times 10^3 \text{ N}$$

$$= \mathbf{1138.34 \text{ kN}}$$

4. —

5. $Q_1 = Q_2 = Q$; $F_1 = \frac{k Q^2}{r_1^2} \Rightarrow Q^2 = \frac{F_1 r_1^2}{k}$; $F_2 = \frac{k Q^2}{r_2^2} = \frac{k}{r_2^2} \left[\frac{F_1 r_1^2}{k} \right]$ and $F_2 = \frac{r_1^2}{r_2^2} F_1$

6. $F = \frac{k Q_1 Q_2}{r^2} \Rightarrow r = \sqrt{\frac{k Q_1 Q_2}{F}} = \sqrt{\frac{(9 \times 10^9)(20 \times 10^{-6})^2}{3.6 \times 10^4}} = \mathbf{10 \text{ mm}}$

$$7. \quad F = \frac{kQ_1Q_2}{r^2} \Rightarrow 1.8 = \frac{kQ_1Q_2}{(2 \text{ m})^2} \Rightarrow kQ_1Q_2 = 4(1.8) = 7.2$$

$$a. \quad F = \frac{kQ_1Q_2}{r^2} = \frac{7.2}{(10)^2} = \mathbf{72 \text{ mN}}$$

$$b. \quad Q_1/Q_2 = 1/2 \Rightarrow Q_2 = 2Q_1$$

$$7.2 = kQ_1Q_2 = (9 \times 10^9)(Q_1)(2Q_1) = 9 \times 10^9(2Q_1^2)$$

$$\frac{7.2}{18 \times 10^9} = Q_1^2 \Rightarrow Q_1 = \sqrt{\frac{7.2}{18 \times 10^9}} = \mathbf{20 \mu C}$$

$$Q_2 = 2Q_1 = 2(2 \times 10^{-5} \text{ C}) = \mathbf{40 \mu C}$$

$$8. \quad V = \frac{W}{Q} = \frac{1.2 \text{ J}}{20 \mu C} = \mathbf{60 \text{ kV}}$$

$$9. \quad W = VQ = (60 \text{ V})(8 \text{ mC}) = \mathbf{0.48 \text{ J}}$$

$$10. \quad Q = \frac{W}{V} = \frac{120 \mu J}{20 \text{ mV}} = 6 \text{ mC} \left[\frac{6.242 \times 10^{18} \text{ electrons}}{1 \text{ C}} \right] = \mathbf{37.45 \times 10^{15} \text{ electrons}}$$

$$11. \quad Q = \frac{W}{V} = \frac{72 \text{ J}}{9 \text{ V}} = \mathbf{8 \text{ C}}$$

$$12. \quad a. \quad W = QV = (1 \times 10^{12} \text{ electrons})(40 \text{ V}) = \mathbf{40 \times 10^{12} \text{ eV}}$$

$$b. \quad 40 \times 10^{12} \text{ eV} \left[\frac{1 \text{ C}}{6.242 \times 10^{18} \text{ electrons}} \right] = \mathbf{6.41 \mu J}$$

$$13. \quad I = \frac{Q}{t} = \frac{12 \text{ mC}}{2.8 \text{ s}} = \mathbf{4.29 \text{ mA}}$$

$$14. \quad I = \frac{Q}{t} = \frac{312 \text{ C}}{(2)(60 \text{ s})} = \mathbf{2.60 \text{ A}}$$

$$15. \quad Q = It = (40 \text{ mA})(0.8)(60 \text{ s}) = \mathbf{1.92 \text{ C}}$$

$$16. \quad Q = It = (250 \text{ mA})(1.2)(60 \text{ s}) = \mathbf{18.0 \text{ C}}$$

$$17. \quad t = \frac{Q}{I} = \frac{6 \text{ mC}}{2 \text{ mA}} = \mathbf{3 \text{ s}}$$

$$18. \quad 21.847 \times 10^{18} \text{ electrons} \left[\frac{1 \text{ C}}{6.242 \times 10^{18} \text{ electrons}} \right] = 3.5 \text{ C}$$

$$I = \frac{Q}{t} = \frac{3.5 \text{ C}}{12 \text{ s}} = 0.29 \text{ A}$$

19. $Q = It = (4 \text{ mA})(90 \text{ s}) = 360 \text{ mC}$
 $360 \text{ mC} \left[\frac{6.242 \times 10^{18} \text{ electrons}}{1 \cancel{\text{C}}} \right] = 2.25 \times 10^{18} \text{ electrons}$
20. $I = \frac{Q}{t} = \frac{86 \text{ C}}{(1.2)(60 \text{ s})} = 1.194 \text{ A} > 1 \text{ A (yes)}$
21. $0.84 \times 10^{16} \cancel{\text{electrons}} \left[\frac{1 \text{ C}}{6.242 \times 10^{18} \cancel{\text{electrons}}} \right] = 1.346 \text{ mC}$
 $I = \frac{Q}{t} = \frac{1.346 \text{ mC}}{60 \text{ ms}} = 22.43 \text{ mA}$
22. a. $Q = It = (2 \text{ mA})(0.01 \mu\text{s}) = 2 \times 10^{-11} \text{ C}$
 $2 \times 10^{-11} \cancel{\text{C}} \left[\frac{6.242 \times 10^{18} \cancel{\text{electrons}}}{1 \cancel{\text{C}}} \right] \left[\frac{1 \text{ ¢}}{\cancel{\text{electron}}} \right]$
 $= 1.25 \times 10^8 \text{ ¢} = \$1.25 \times 10^6 = 1.25 \text{ million}$
- b. $Q = It = (100 \mu\text{A})(1.5 \text{ ns}) = 1.5 \times 10^{-13} \text{ C}$
 $1.5 \times 10^{-13} \cancel{\text{C}} \left[\frac{6.242 \times 10^{18} \cancel{\text{electrons}}}{1 \cancel{\text{C}}} \right] \left[\frac{\$1}{\cancel{\text{electron}}} \right] = 0.94 \text{ million}$
(a) > (b)
23. $Q = It = (200 \times 10^{-3} \text{ A})(30 \text{ s}) = 6 \text{ C}$
 $V = \frac{W}{Q} = \frac{40 \text{ J}}{6 \text{ C}} = 6.67 \text{ V}$
24. $Q = It = \left[\frac{420 \text{ C}}{\cancel{\text{min}}} \right] (0.5 \cancel{\text{min}}) = 210 \text{ C}$
 $V = \frac{W}{Q} = \frac{742 \text{ J}}{210 \text{ C}} = 3.53 \text{ V}$
25. $Q = \frac{W}{V} = \frac{0.4 \text{ J}}{24 \text{ V}} = 0.0167 \text{ C}$
 $I = \frac{Q}{t} = \frac{0.0167 \text{ C}}{5 \times 10^{-3} \text{ s}} = 3.34 \text{ A}$
26. $I = \frac{\text{Ah rating}}{t(\text{hours})} = \frac{200 \text{ Ah}}{40 \text{ h}} = 5 \text{ A}$
27. $\text{Ah} = (0.8 \text{ A})(75 \text{ h}) = 60.0 \text{ Ah}$
28. $t(\text{hours}) = \frac{\text{Ah rating}}{I} = \frac{32 \text{ Ah}}{1.28 \text{ A}} = 25 \text{ h}$

29. 40 Ah(for 1 h): $W_1 = VQ = V \cdot I \cdot t = (12 \text{ V})(40 \text{ A})(1 \text{ h}) \left[\frac{60 \cancel{\text{min}}}{1 \cancel{\text{h}}} \right] \left[\frac{60 \text{ s}}{1 \cancel{\text{min}}} \right] = 1.728 \times 10^6 \text{ J}$

60 Ah(for 1 h): $W_2 = (12 \text{ V})(60 \text{ A})(1 \text{ h}) \left[\frac{60 \cancel{\text{min}}}{1 \cancel{\text{h}}} \right] \left[\frac{60 \text{ s}}{1 \cancel{\text{min}}} \right] = 2.592 \times 10^6 \text{ J}$

Ratio $W_2/W_1 = 1.5$ or 50% more energy available with 60 Ah rating.

For 60 s discharge: $40 \text{ Ah} = It = I[60 \cancel{\text{s}}] \left[\frac{1 \cancel{\text{min}}}{60 \cancel{\text{s}}} \right] \left[\frac{1 \text{ h}}{60 \cancel{\text{min}}} \right] = I(16.67 \times 10^{-3} \text{ h})$

$$\text{and } I = \frac{40 \text{ Ah}}{16.67 \times 10^{-3} \text{ h}} = \mathbf{2400 \text{ A}}$$

$$60 \text{ Ah} = It = I[60 \cancel{\text{s}}] \left[\frac{1 \cancel{\text{min}}}{60 \cancel{\text{s}}} \right] \left[\frac{1 \text{ h}}{60 \cancel{\text{min}}} \right] = I(16.67 \times 10^{-3} \text{ h})$$

$$\text{and } I = \frac{60 \text{ Ah}}{16.67 \times 10^{-3} \text{ h}} = \mathbf{3600 \text{ A}}$$

$I_2/I_1 = 1.5$ or 50 % more starting current available at 60 Ah

30. $0.75(18 \text{ Ah}) = 13.5 \text{ Ah} \Rightarrow \cong \mathbf{250 \text{ mA}}$

31. $(18 \text{ Ah} - 15.5 \text{ Ah})/18 \text{ Ah} \times 100\% = \mathbf{13.89\%}$

32. At 100 mA, discharge time $\cong 120 \text{ h}$; At 25 mA, discharge time $\cong 425 \text{ h}$;
 $\cong \mathbf{300 \text{ h more at } 25 \text{ mA}}$

33. $I = \frac{3 \text{ Ah}}{6.0 \text{ h}} = 500 \text{ mA}$

$$Q = It = (500 \text{ mA})(6 \text{ h}) \left[\frac{60 \cancel{\text{min}}}{1 \cancel{\text{h}}} \right] \left[\frac{60 \text{ s}}{1 \cancel{\text{min}}} \right] = 10.80 \text{ kC}$$

$$W = QV = (10.8 \text{ kC})(12 \text{ V}) \cong \mathbf{129.6 \text{ kJ}}$$

34. —

35. —

36. —

37. a. $0.5 \cancel{\text{in}} \left[\frac{2.54 \text{ cm}}{1 \cancel{\text{in}}} \right] = 1.27 \text{ cm}$

$$1.27 \cancel{\text{cm}} \left[\frac{30 \text{ kV}}{\cancel{\text{cm}}} \right] = \mathbf{38.1 \text{ kV}}$$

b. $1.27 \text{ cm} \left[\frac{270 \text{ kV}}{\text{cm}} \right] = \mathbf{342.9 \text{ kV}}$

c. $342.9 \text{ kV}:38.1 \text{ kV} = \mathbf{9:1}$

- 38. —
- 39. —
- 40. —
- 41. —

Chapter 3

1. a. $0.5 \text{ in.} = \mathbf{500 \text{ mils}}$

b. $0.02 \cancel{\text{in.}} \left[\frac{1000 \text{ mils}}{1 \cancel{\text{in.}}} \right] = \mathbf{20 \text{ mils}}$

c. $\frac{1}{4} \text{ in.} = 0.25 \cancel{\text{in.}} \left[\frac{1000 \text{ mils}}{1 \cancel{\text{in.}}} \right] = \mathbf{250 \text{ mils}}$

d. $10 \text{ mm} = 10 \times 10^{-3} \cancel{\text{m}} \left[\frac{39.37 \text{ in}}{1 \cancel{\text{m}}} \right] \left[\frac{1000 \text{ mils}}{1 \text{ in}} \right] = \mathbf{393.7 \text{ mils}}$

e. $0.01 \cancel{\text{ft}} \left[\frac{12 \cancel{\text{in.}}}{1 \cancel{\text{ft}}} \right] \left[\frac{10^3 \text{ mils}}{1 \cancel{\text{in.}}} \right] = \mathbf{120 \text{ mils}}$

f. $0.1 \cancel{\text{cm}} \left[\frac{1 \cancel{\text{in.}}}{2.54 \cancel{\text{cm}}} \right] \left[\frac{1000 \text{ mils}}{1 \cancel{\text{in.}}} \right] = \mathbf{39.37 \text{ mils}}$

2. a. $A_{\text{CM}} = (30 \text{ mils})^2 = \mathbf{900 \text{ CM}}$

b. $0.016 \text{ in.} = 16 \text{ mils}, A_{\text{CM}} = (16 \text{ mils})^2 = \mathbf{256 \text{ CM}}$

c. $\frac{1}{8}'' = 0.125'' = 125 \text{ mils}, A_{\text{CM}} = (125 \text{ mils})^2 = \mathbf{15.63 \times 10^3 \text{ CM}}$

d. $1 \cancel{\text{cm}} \left[\frac{1 \cancel{\text{in.}}}{2.54 \cancel{\text{cm}}} \right] \left[\frac{1000 \text{ mils}}{1 \cancel{\text{in.}}} \right] = 393.7 \text{ mils}, A_{\text{CM}} = (393.7 \text{ mils})^2 = \mathbf{155 \times 10^3 \text{ CM}}$

e. $0.02 \cancel{\text{ft}} \left[\frac{12 \cancel{\text{in.}}}{1 \cancel{\text{ft}}} \right] \left[\frac{1000 \text{ mils}}{1 \cancel{\text{in.}}} \right] = 240 \text{ mils}, A_{\text{CM}} = (240 \text{ mils})^2 = \mathbf{57.60 \times 10^3 \text{ CM}}$

f. $4 \times 10^{-3} \cancel{\text{m}} \left[\frac{39.37 \cancel{\text{in.}}}{1 \cancel{\text{m}}} \right] \left[\frac{1000 \text{ mils}}{1 \cancel{\text{in.}}} \right] = \mathbf{157.48 \text{ mils}}, A_{\text{CM}} = (157.48 \text{ mils})^2 = \mathbf{24.8 \times 10^3 \text{ CM}}$

3. $A_{\text{CM}} = (d_{\text{mils}})^2 \rightarrow d_{\text{mils}} = \sqrt{A_{\text{CM}}}$

a. $d = \sqrt{1600 \text{ CM}} = 40 \text{ mils} = \mathbf{0.04 \text{ in.}}$

b. $d = \sqrt{820 \text{ CM}} = 28.64 \text{ mils} = \mathbf{0.029 \text{ in.}}$

c. $d = \sqrt{40,000 \text{ CM}} = 200 \text{ mils} = \mathbf{0.2 \text{ in.}}$

- d. $d = \sqrt{625 \text{ CM}} = 25 \text{ mils} = \mathbf{0.025 \text{ in.}}$
- e. $d = \sqrt{6.25 \text{ CM}} = 2.5 \text{ mils} = \mathbf{0.0025 \text{ in.}}$
- f. $d = \sqrt{3 \times 10^3 \text{ CM}} = 54.77 \text{ mils} = \mathbf{0.055 \text{ in.}}$
4. $0.02 \text{ in.} = 20 \text{ mils}, A_{\text{CM}} = (20 \text{ mils})^2 = 400 \text{ CM}$
 $R = \rho \frac{l}{A} = (10.37) \frac{(200')}{400 \text{ CM}} = \mathbf{5.19 \Omega}$
5. a. $A = \rho \frac{l}{R} = 17 \left(\frac{80'}{2.5 \Omega} \right) = \mathbf{544 \text{ CM}}$
b. $d = \sqrt{A_{\text{CM}}} = \sqrt{544 \text{ CM}} = 23.32 \text{ mils} = \mathbf{23.3 \times 10^{-3} \text{ in.}}$
6. $\frac{1}{32}'' = 0.03125'' = 31.25 \text{ mils}, A_{\text{CM}} = (31.25 \text{ mils})^2 = 976.56 \text{ CM}$
 $R = \rho \frac{l}{A} \Rightarrow l = \frac{RA}{\rho} = \frac{(2.2 \Omega)(976.56 \text{ CM})}{600} = \mathbf{3.58 \text{ ft}}$
7. a. $A_{\text{CM}} = \rho \frac{l}{A} = \frac{(10.37)(300')}{3.3 \Omega} = \mathbf{942.73 \text{ CM}}$
 $d = \sqrt{942.73 \text{ CM}} = 30.70 \text{ mils} = \mathbf{30.7 \times 10^{-3} \text{ in.}}$
b. larger
c. smaller
8. $\rho = \frac{RA}{l} = \frac{(500 \Omega)(94 \text{ CM})}{1000'} = 47 \Rightarrow \mathbf{\text{nickel}}$
9. a. $1/32'' = 0.03125'' = 31.25 \text{ mils}, A_{\text{CM}} = (31.25 \text{ mils})^2 = 976.56 \text{ CM}$
 $R = \frac{\rho l}{A} \Rightarrow l = \frac{RA}{\rho} = \frac{(3.12 \Omega)(976.56 \text{ CM})}{10.37} = \mathbf{293.82 \text{ ft}}$
b. $\frac{293.82'}{x} = \frac{1000'}{5 \text{ lb}} \Rightarrow x = \frac{(5)(293.82)}{1000} = \mathbf{1.47 \text{ lbs}}$
c. $-40^\circ \text{ C: } F = \frac{9}{5}C + 32^\circ = \frac{9}{5}(-40) + 32 = -40^\circ$
 $105^\circ \text{ C: } F = \frac{9}{5}C + 32^\circ = \frac{9}{5}(105) + 32 = 221^\circ$
 $\mathbf{F^\circ = -40^\circ \rightarrow 221^\circ}$

10. a. $\frac{3}{8}'' = 0.375'' = 375 \text{ mils}$
 $4.8'' = 4800 \text{ mils}$
 $A = (375 \text{ mils})(4800 \text{ mils}) = 1.8 \times 10^6 \text{ sq. mils} \left[\frac{4/\pi \text{ CM}}{1 \text{ sq. mil}} \right] = 2.29 \times 10^6 \text{ CM}$

b. $\frac{1}{12}'' = 0.083 \text{ in.} = 83 \text{ mils}$
 $A_{\text{CM}} = (83 \text{ mils})^2 = 6.89 \times 10^3 \text{ CM}$
 $(\#12) \frac{2.29 \times 10^6 \text{ CM}}{6.89 \times 10^3 \text{ CM}} = \mathbf{332.37 \text{ wires}}$

11. a. $3'' = 3000 \text{ mils}$, $1/2'' = 0.5 \text{ in.} = 500 \text{ mils}$
 $\text{Area} = (3 \times 10^3 \text{ mils})(5 \times 10^2 \text{ mils}) = 15 \times 10^5 \text{ sq. mils}$
 $15 \times 10^5 \text{ sq. mils} \left[\frac{4/\pi \text{ CM}}{1 \text{ sq. mil}} \right] = 19.108 \times 10^5 \text{ CM}$
 $R = \rho \frac{l}{A} = \frac{(10.37)(4')}{19.108 \times 10^5 \text{ CM}} = \mathbf{21.71 \mu\Omega}$

b. $R = \rho \frac{l}{A} = \frac{(17)(4')}{19.108 \times 10^5 \text{ CM}} = \mathbf{35.59 \mu\Omega}$
 Aluminum bus-bar has almost 64% higher resistance.

12. $l_2 = 2l_1$, $A_2 = A_1/4$, $\rho_2 = \rho_1$
 $\frac{R_2}{R_1} = \frac{\frac{\rho_2 l_2}{A_2}}{\frac{\rho_1 l_1}{A_1}} = \frac{\cancel{\rho_2} l_2 A_1}{\cancel{\rho_1} l_1 A_2} = \frac{2l_1 A_1}{l_1 A_1/4} = 8$
 and $R_2 = 8R_1 = 8(0.2 \Omega) = 1.6 \Omega$
 $\Delta R = 1.6 \Omega - 0.2 \Omega = \mathbf{1.4 \Omega}$

13. $A = \frac{\pi d^2}{4} \Rightarrow d = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(0.04 \text{ in.}^2)}{\pi}} = 0.2257 \text{ in.}$

$d_{\text{mils}} = 225.7 \text{ mils}$

$A_{\text{CM}} = (225.7 \text{ mils})^2 = 50,940.49 \text{ CM}$

$\frac{R_1}{R_2} = \frac{\frac{\rho_1 l_1}{A_1}}{\frac{\rho_2 l_2}{A_2}} = \frac{\cancel{\rho_1} l_1 A_2}{\cancel{\rho_2} l_2 A_1} = \frac{l_1 A_2}{l_2 A_1} \quad (\rho_1 = \rho_2)$

and $R_2 = \frac{R_1 l_2 A_1}{l_1 A_2} = \frac{(800 \text{ m}\Omega)(300 \text{ ft})(40,000 \text{ CM})}{(200 \text{ ft})(50,940.49 \text{ CM})} = \mathbf{942.28 \text{ m}\Omega}$

14. a. #12 = 6,529.9 CM, #14 = 4,106.8 CM

$$\frac{6,529.9 \text{ CM} - 4,106.8 \text{ CM}}{4,106.8 \text{ CM}} \times 100\% = \mathbf{59\% \text{ larger}}$$
- b. $\frac{\#12}{\#14} = \frac{20 \text{ A}}{15 \text{ A}} = 1.33$, $\frac{\#12}{\#14} = \frac{6,529.9 \text{ CM}}{4,106.8 \text{ CM}} = 1.59$
 $I_{\max}$ ratio = 1.33 vs Area ratio = 1.59

$$\frac{1.59 - 1.33}{1.33} \times 100\% = \mathbf{19.55\% \text{ higher ratio for area}}$$
15. a. $\frac{\#9}{\#12} = \frac{13,094 \text{ CM}}{6,529.9 \text{ CM}} = \mathbf{2 \text{ yes}}$
- b. $\frac{\#0}{\#12} = \frac{105,530 \text{ CM}}{6,529.9 \text{ CM}} = \mathbf{16.16 \text{ yes}}$

$$\frac{\#0}{\#12} = \frac{150 \text{ A}}{20 \text{ A}} = \mathbf{7.5}$$
16. a. $\frac{\#10}{\#20} = \frac{10,381 \text{ CM}}{1,021.5 \text{ CM}} = 10.16 \cong \mathbf{10 \text{ yes}}$
- b. $\frac{\#20}{\#40} = \frac{1,021.5 \text{ CM}}{9.89 \text{ CM}} = 103.28$
 $\mathbf{\text{yes} \cong 100}$
17. a. $A = \rho \frac{l}{R} = \frac{(10.37)(30')}{6 \text{ m}\Omega} = \frac{311.1 \text{ CM}}{6 \times 10^{-3}} = 51,850 \text{ CM} \Rightarrow \#3$
but 110 A $\Rightarrow \mathbf{\#2}$
- b. $A = \rho \frac{l}{R} = \frac{(10.37)(30')}{3 \text{ m}\Omega} = \frac{311.1 \text{ CM}}{3 \times 10^{-3}} = 103,700 \text{ CM} \Rightarrow \mathbf{\#0}$
18. a. $A/\text{CM} = 230 \text{ A}/211,600 \text{ CM} = \mathbf{1.09 \text{ mA/CM}}$
- b.
$$\frac{1.09 \text{ mA}}{\cancel{\text{CM}}} \left[\frac{1 \cancel{\text{CM}}}{\frac{\pi}{4} \text{ sq mils}} \right] \left[\frac{1000 \cancel{\text{mils}}}{1 \text{ in.}} \right] \left[\frac{1000 \cancel{\text{mils}}}{1 \text{ in.}} \right] = \mathbf{1.39 \text{ kA/in.}^2}$$
- c.
$$5 \text{ kA} \left[\frac{1 \text{ in.}^2}{1.39 \cancel{\text{kA}}} \right] = \mathbf{3.6 \text{ in.}^2}$$

$$19. \quad \frac{234.5 + 10}{2 \, \Omega} = \frac{234.5 + 80}{R_2}, \quad R_2 = \frac{(314.5)(2 \, \Omega)}{244.5} = \mathbf{2.57 \, \Omega}$$

$$20. \quad \frac{236 + 0}{0.02 \, \Omega} = \frac{236 + 100}{R_2}$$

$$R_2 = \frac{(0.02 \, \Omega)(336)}{236} = \mathbf{0.028 \, \Omega}$$

$$21. \quad C = \frac{5}{9}(\text{°F} - 32) = \frac{5}{9}(32 - 32) = 0^\circ (=32^\circ\text{F})$$

$$C = \frac{5}{9}(68 - 32) = 20^\circ (=68^\circ\text{F})$$

$$\frac{234.5^\circ + 20^\circ}{4 \, \Omega} = \frac{234.5^\circ + 0^\circ}{R_2}$$

$$R_2 = \frac{(234.5)(4 \, \Omega)}{254.5} = \mathbf{3.69 \, \Omega}$$

$$22. \quad \text{a.} \quad \text{°C} = \frac{5}{9}(\text{°F} - 32^\circ) = \frac{5}{9}(70^\circ - 32^\circ) = 21.11^\circ$$

$$\text{°C} = \frac{5}{9}(60^\circ - 32^\circ) = 15.56^\circ$$

$$\frac{234.5 + 21.11}{0.025 \, \Omega} = \frac{234.5 + 15.56}{R_2}$$

$$R_2 = \frac{(250.06)(0.025 \, \Omega)}{255.61} = \mathbf{24.46 \, m\Omega}$$

$$\text{b.} \quad \text{°C} = \frac{5}{9}(50^\circ - 32^\circ) = 10^\circ$$

$$\frac{234.5 + 21.11}{0.025 \, \Omega} = \frac{234.5 + 10}{R_2}$$

$$R_2 = \frac{(244.5)(0.025 \, \Omega)}{255.61} = \mathbf{23.91 \, m\Omega}$$

$$\text{c.} \quad \text{Part a: } 25 \, \text{m}\Omega - 24.46 \, \text{m}\Omega = 0.54 \, \text{m}\Omega$$

$$\text{Part b: } 24.45 \, \text{m}\Omega - 23.91 \, \text{m}\Omega = 0.55 \, \text{m}\Omega$$

$$\mathbf{\text{Linear } 40^\circ\text{F} \Rightarrow 23.91 \, \text{m}\Omega - 0.55 \, \text{m}\Omega = 23.36 \, \text{m}\Omega}$$

$$\text{d.} \quad \text{°C} = \frac{5}{9}(-30^\circ - 32^\circ) = -34.44^\circ$$

$$\frac{234.5 + 21.11}{25 \, \text{m}\Omega} = \frac{234.5 - 34.44}{R_2}$$

$$R_2 = \frac{(25 \, \text{m}\Omega)(200.06)}{255.61} = \mathbf{19.57 \, m\Omega}$$

$$\mathbf{\text{Yes, } 25 \, \text{m}\Omega - 19.57 \, \text{m}\Omega = 5.43 \, \text{m}\Omega}$$

$$e. \quad ^\circ\text{C} = \frac{5}{9}(120^\circ - 32^\circ) = 48.89^\circ$$

$$\frac{234.5 + 21.11}{25 \text{ m}\Omega} = \frac{234.5 + 48.89}{R_2}$$

$$R_2 = \frac{(25 \text{ m}\Omega)(283.39)}{255.61} = \mathbf{27.72 \text{ m}\Omega}$$

Yes, 2.72 mΩ

$$23. \quad a. \quad \frac{234.5 + 4}{1 \text{ }\Omega} = \frac{234.5 + t_2}{1.1 \text{ }\Omega}, \quad t_2 = \mathbf{27.85^\circ\text{C}}$$

$$b. \quad \frac{234.5 + 4}{1 \text{ }\Omega} = \frac{234.5 + t_2}{0.1 \text{ }\Omega}, \quad t_2 = \mathbf{-210.65^\circ\text{C}}$$

$$24. \quad a. \quad 68^\circ\text{F} = 20^\circ\text{C}$$

$$\frac{234.5 + 20}{1 \text{ }\Omega} = \frac{234.5 + T_2}{2 \text{ }\Omega}$$

$$\frac{2(254.5)}{1} - 234.5 = T_2$$

$$T_2 = \mathbf{274.5^\circ\text{C}}$$

$$b. \quad \#10 = \mathbf{0.9989 \text{ }\Omega/1000'}$$

$$c. \quad d_{\text{mils}} = \sqrt{A_{\text{CM}}} = \sqrt{10,381 \text{ CM}} = 101.89 \text{ mils}$$

$$d_{\text{in}} = 0.102 \text{ in} \cong \frac{\mathbf{1}}{\mathbf{10}}''$$

$$25. \quad a. \quad \alpha_{20} = \frac{1}{|T_i| + 20^\circ\text{C}} = \frac{1}{234.5 + 20} = \frac{1}{254.5} = 0.003929 \cong \mathbf{0.00393}$$

$$b. \quad R = R_{20}[1 + \alpha_{20}(t - 20^\circ\text{C})]$$

$$1 \text{ }\Omega = 0.8 \text{ }\Omega[1 + 0.00393(t - 20^\circ)]$$

$$1.25 = 1 + 0.00393t - 0.0786$$

$$1.25 - 0.9214 = 0.00393t$$

$$0.3286 = 0.00393t$$

$$t = \frac{0.3286}{0.00393} = \mathbf{83.61^\circ\text{C}}$$

$$26. \quad R = R_{20}[1 + \alpha_{20}(t - 20^\circ\text{C})]$$

$$= 0.4 \text{ }\Omega[1 + 0.00393(16 - 20)] = 0.4 \text{ }\Omega[1 - 0.01572] = \mathbf{0.39 \text{ }\Omega}$$

39. a. $621 = 62 \times 10^1 \Omega = 620 \Omega = \mathbf{0.62 \text{ k}\Omega}$
 b. $333 = 33 \times 10^3 \Omega = \mathbf{33 \text{ k}\Omega}$
 c. $Q2 = 3.9 \times 10^2 \Omega = \mathbf{390 \Omega}$
 d. $C6 = 1.2 \times 10^6 \Omega = \mathbf{1.2 \text{ M}\Omega}$
40. a. $G = \frac{1}{R} = \frac{1}{120 \Omega} = \mathbf{8.33 \text{ mS}}$
 b. $G = \frac{1}{4 \text{ k}\Omega} = \mathbf{0.25 \text{ mS}}$
 c. $G = \frac{1}{2.2 \text{ M}\Omega} = \mathbf{0.46 \mu\text{S}}$
 $G_a > G_b > G_c$ vs. $R_c > R_b > R_a$
41. a. Table 3.2, $\Omega/1000' = 1.588 \Omega$
 $G = \frac{1}{R} = \frac{1}{1.588 \Omega} = \mathbf{629.72 \text{ mS}}$
 or $G = \frac{A}{\rho l} = \frac{6529.9 \text{ CM (Table 3.2)}}{(10.37)(1000')} = \mathbf{629.69 \text{ mS (Cu)}}$
 b. $G = \frac{6529.9 \text{ CM}}{(17)(1000')} = \mathbf{384.11 \text{ mS (Al)}}$
42. a. $G_1 = \frac{1}{10 \Omega} = 100 \text{ mS}$, $G_2 = \frac{1}{20 \Omega} = 50 \text{ mS}$, $G_3 = \frac{1}{100 \Omega} = 10 \text{ mS}$
 b. $G_2:G_1 = 50 \text{ mS}: 100 \text{ mS} = 1:2$ whereas $R_2:R_1 = 20 \Omega:10 \Omega = 2:1$. The rate of change is the same although one is increasing and the other decreasing.
 c. inverse – linear
43. $A_2 = 1\frac{2}{3} A_1 = \frac{5}{3} A_1$, $l_2 = \left(1 - \frac{2}{3}\right)l_1 = \frac{l_1}{3}$, $\rho_2 = \rho_1$

$$\frac{G_1}{G_2} = \frac{\rho_1 \frac{A_1}{l_1}}{\rho_2 \frac{A_2}{l_2}} = \frac{\cancel{\rho_2} l_2 A_1}{\cancel{\rho_1} l_1 A_2} = \frac{\left(\frac{l_1}{3}\right) A_1}{l_1 \left(\frac{5}{3} A_1\right)} = \frac{1}{5}$$

$$G_2 = 5G_1 = 5(100 \text{ S}) = \mathbf{500 \text{ S}}$$
44. –
45. –
46. –

47. —

$$48. \quad \frac{1}{12} \text{ in.} = 0.083 \cancel{\text{in.}} \left(\frac{2.54 \text{ cm}}{1 \cancel{\text{in.}}} \right) = 0.21 \text{ cm}$$

$$A = \frac{\pi d^2}{4} = \frac{(3.14)(0.21 \text{ cm})^2}{4} = 0.035 \text{ cm}^2$$

$$l = \frac{RA}{\rho} = \frac{(2 \Omega)(0.035 \text{ cm}^2)}{1.724 \times 10^{-6}} = 40,603 \text{ cm} = \mathbf{406.03 \text{ m}}$$

$$49. \quad \text{a.} \quad \frac{1''}{2} \left[\frac{2.54 \text{ cm}}{1''} \right] = 1.27 \text{ cm}, \quad 3 \cancel{\text{in.}} \left[\frac{2.54 \text{ cm}}{1 \cancel{\text{in.}}} \right] = 7.62 \text{ cm}$$

$$4 \cancel{\text{ft}} \left[\frac{12 \cancel{\text{in.}}}{1 \cancel{\text{ft}}} \right] \left[\frac{2.54 \text{ cm}}{1 \cancel{\text{in.}}} \right] = 121.92 \text{ cm}$$

$$R = \rho \frac{l}{A} = \frac{(1.724 \times 10^{-6})(121.92 \text{ cm})}{(1.27 \text{ cm})(7.62 \text{ cm})} = \mathbf{21.71 \mu\Omega}$$

$$\text{b.} \quad R = \rho \frac{\ell}{A} = \frac{(2.825 \times 10^{-6})(121.92 \text{ cm})}{(1.27 \text{ cm})(7.62 \text{ cm})} = \mathbf{35.59 \mu\Omega}$$

c. increases

d. decreases

$$50. \quad R_s = \frac{\rho}{d} = 100 \Rightarrow d = \frac{\rho}{100} = \frac{250 \times 10^{-6}}{100} = \mathbf{2.5 \mu\text{cm}}$$

$$51. \quad R = R_s \frac{l}{w} \Rightarrow w = \frac{R_s l}{R} = \frac{(150 \Omega)(1/2 \text{ in.})}{500 \Omega} = \mathbf{0.15 \text{ in.}}$$

$$52. \quad \text{a.} \quad d = 1 \text{ in.} = 1000 \text{ mils}$$

$$A_{\text{CM}} = (10^3 \text{ mils})^2 = 10^6 \text{ CM}$$

$$\rho_1 = \frac{RA}{l} = \frac{(1 \text{ m}\Omega)(10^6 \text{ CM})}{10^3 \text{ ft}} = \mathbf{1 \text{ CM-}\Omega/\text{ft}}$$

$$\text{b.} \quad 1 \text{ in.} = 2.54 \text{ cm}$$

$$A = \frac{\pi d^2}{4} = \frac{\pi(2.54 \text{ cm})^2}{4} = 5.067 \text{ cm}^2$$

$$l = 1000 \text{ ft} \left[\frac{12 \text{ in.}}{1 \text{ ft}} \right] \left[\frac{2.54 \text{ cm}}{1 \text{ in.}} \right] = 30,480 \text{ cm}$$

$$\rho_2 = \frac{RA}{l} = \frac{(1 \text{ m}\Omega)(5.067 \text{ cm}^2)}{30,480 \text{ cm}} = \mathbf{1.66 \times 10^{-7} \Omega\text{-cm}}$$

c. $k = \frac{\rho_2}{\rho_1} = \frac{1.66 \times 10^{-7} \Omega\text{-cm}}{1 \text{ CM-}\Omega / \text{ft}} = 1.66 \times 10^{-7}$

53. —

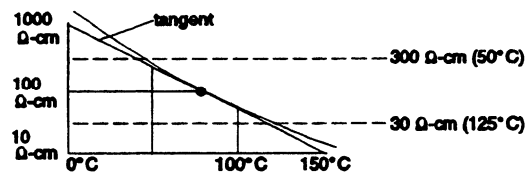
54. —

55. —

56. —

57. —

58. a. -50°C specific resistance $\cong 10^5 \Omega\text{-cm}$
 50°C specific resistance $\cong 500 \Omega\text{-cm}$
 200°C specific resistance $\cong 7 \Omega\text{-cm}$



b. negative

c. No

d. $\rho = \frac{\Delta\Omega\text{-cm}}{\Delta T} = \frac{300 - 30}{125 - 50} = \frac{270 \Omega\text{-cm}}{75^\circ\text{C}} \cong 3.6 \Omega\text{-cm}/^\circ\text{C}$

59. a. Log scale: $10 \text{ fc} \Rightarrow 3 \text{ k}\Omega$
 $100 \text{ fc} \Rightarrow 0.4 \text{ k}\Omega$

b. negative

c. no—log scales imply linearity

d. $1 \text{ k}\Omega \Rightarrow \cong 30 \text{ fc}$
 $10 \text{ k}\Omega \Rightarrow \cong 2 \text{ fc}$
 $\left| \frac{\Delta R}{\Delta \text{fc}} \right| = \frac{10 \text{ k}\Omega - 1 \text{ k}\Omega}{30 \text{ fc} - 2 \text{ fc}} = 321.43 \Omega/\text{fc}$
and $\frac{\Delta R}{\Delta \text{fc}} = -321.43 \Omega/\text{fc}$

60. a. @ 0.5 mA , $V \cong 195 \text{ V}$
@ 1 mA , $V \cong 200 \text{ V}$
@ 5 mA , $V \cong 215 \text{ V}$

b. $\Delta V_{\text{total}} = 215 \text{ V} - 195 \text{ V} = 20 \text{ V}$

- c. $5 \text{ mA} : 0.5 \text{ mA} = 10:1$
compared to
 $215 \text{ V} : 200 \text{ V} = 1.08:1$

Chapter 4

1. $V = IR = (5.6 \text{ mA})(220 \Omega) = \mathbf{1.23 \text{ V}}$

2. $I = \frac{V}{R} = \frac{24 \text{ V}}{6.8 \Omega} = \mathbf{3.53 \text{ A}}$

3. $R = \frac{V}{I} = \frac{24 \text{ V}}{1.5 \text{ mA}} = \mathbf{16 \text{ k}\Omega}$

4. $I = \frac{V}{R} = \frac{12 \text{ V}}{40 \times 10^{-3} \Omega} = \mathbf{300 \text{ A}}$

5. $V = IR = (3.6 \mu\text{A})(0.02 \text{ M}\Omega) = 0.072 \text{ V} = \mathbf{72 \text{ mV}}$

6. $I = \frac{V}{R} = \frac{120 \text{ V}}{50 \text{ k}\Omega} = \mathbf{2.4 \text{ mA}}$

7. $R = \frac{V}{I} = \frac{120 \text{ V}}{2.2 \text{ A}} = \mathbf{54.55 \Omega}$

8. $I = \frac{V}{R} = \frac{120 \text{ V}}{8 \text{ k}\Omega} = \mathbf{15 \text{ mA}}$

9. $R = \frac{V}{I} = \frac{120 \text{ V}}{4.2 \text{ A}} = \mathbf{28.57 \Omega}$

10. $R = \frac{V}{I} = \frac{4.5 \text{ V}}{125 \text{ mA}} = \mathbf{36 \Omega}$

11. $R = \frac{V}{I} = \frac{24 \text{ mV}}{20 \mu\text{A}} = \mathbf{1.2 \text{ k}\Omega}$

12. $V = IR = (12 \text{ A})(0.5 \Omega) = \mathbf{6 \text{ V}}$

13. a. $R = \frac{V}{I} = \frac{120 \text{ V}}{9.5 \text{ A}} = \mathbf{12.63 \Omega}$

b. $t = 2 \cancel{\text{h}} \left[\frac{60 \cancel{\text{min}}}{1 \cancel{\text{h}}} \right] \left[\frac{60 \text{ s}}{1 \cancel{\text{min}}} \right] = 7200 \text{ s}$

$$\begin{aligned} W &= Pt = VIt \\ &= (120 \text{ V})(9.5 \text{ A})(7200 \text{ s}) \\ &= \mathbf{8.21 \times 10^6 \text{ J}} \end{aligned}$$

14. $V = IR = (2.4 \mu\text{A})(3.3 \text{ M}\Omega) = \mathbf{7.92 \text{ V}}$

15. —

16. b. $(0.13 \text{ mA})(500 \text{ h}) = \mathbf{65 \text{ mAh}}$
17. –
18. –
19. –
20. $P = \frac{W}{t} = \frac{540 \text{ J}}{4 \cancel{\text{ min}} \frac{60 \text{ s}}{1 \cancel{\text{ min}}}} = \frac{540 \text{ J}}{240 \text{ s}} = \mathbf{2.25 \text{ W}}$
21. $t = \frac{W}{P} = \frac{640 \text{ J}}{40 \text{ J/s}} = \mathbf{16 \text{ s}}$
22. a. $8 \cancel{\text{ h}} \left[\frac{60 \cancel{\text{ min}}}{1 \cancel{\text{ h}}} \right] \left[\frac{60 \text{ s}}{1 \cancel{\text{ min}}} \right] = 28,800 \text{ s}$
 $W = Pt = (2 \text{ W})(28,800 \text{ s}) = \mathbf{57.6 \text{ kJ}}$
- b. $\text{kWh} = \frac{(2 \text{ W})(8 \text{ h})}{1000} = \mathbf{16 \times 10^{-3} \text{ kWh}}$
23. $P = VI = (3 \text{ V})(1.4 \text{ A}) = 4.20 \text{ W}$
 $t = \frac{W}{P} = \frac{12 \text{ J}}{4.2 \text{ W}} = \mathbf{2.86 \text{ s}}$
24. $P = EI = (6 \text{ V})(750 \text{ mA}) = \mathbf{4.5 \text{ W}}$
25. $P = I^2 R = (7.2 \text{ mA})^2 4 \text{ k}\Omega = \mathbf{207.36 \text{ mW}}$
26. $P = I^2 R \Rightarrow I = \sqrt{\frac{P}{R}} = \sqrt{\frac{240 \text{ mW}}{2.2 \text{ k}\Omega}} = \mathbf{10.44 \text{ mA}}$
27. $I = \sqrt{\frac{P}{R}} = \sqrt{\frac{2 \text{ W}}{120 \Omega}} = \mathbf{129.10 \text{ mA}}$
 $V = IR = (129.10 \text{ mA})(120 \Omega) = \mathbf{15.49 \text{ V}}$
28. $I = \frac{E}{R} = \frac{22 \text{ V}}{16.8 \text{ k}\Omega} = \mathbf{1.31 \text{ mA}}$
 $P = I^2 R = (1.31 \text{ mA})^2 16.8 \text{ k}\Omega = \mathbf{28.83 \text{ mW}}$
 $W = Pt = (28.83 \text{ mW}) \left(1 \cancel{\text{ h}} \left[\frac{60 \cancel{\text{ min}}}{1 \cancel{\text{ h}}} \right] \left[\frac{60 \text{ s}}{1 \cancel{\text{ min}}} \right] \right) = \mathbf{103.79 \text{ J}}$
29. $E = \frac{P}{I} = \frac{324 \text{ W}}{2.7 \text{ A}} = \mathbf{120 \text{ V}}$

$$30. \quad I = \sqrt{\frac{P}{R}} = \sqrt{\frac{1 \text{ W}}{4.7 \text{ M}\Omega}} = \mathbf{461.27 \mu A}$$

no

$$31. \quad V = \sqrt{PR} = \sqrt{(42 \text{ mW})(2.2 \text{ k}\Omega)} = \sqrt{92.40} = \mathbf{9.61 \text{ V}}$$

$$32. \quad P = VI, I = \frac{P}{V} = \frac{100 \text{ W}}{120 \text{ V}} = 0.833 \text{ A}$$

$$R = \frac{V}{I} = \frac{120 \text{ V}}{0.833 \text{ A}} = \mathbf{144.06 \Omega}$$

$$33. \quad V = \frac{P}{I} = \frac{450 \text{ W}}{3.75 \text{ A}} = \mathbf{120 \text{ V}}$$

$$R = \frac{V}{I} = \frac{120 \text{ V}}{3.75 \text{ A}} = \mathbf{32 \Omega}$$

$$34. \quad \text{a.} \quad P = EI \text{ and } I = \frac{P}{E} = \frac{0.4 \times 10^{-3} \text{ W}}{3 \text{ V}} = \mathbf{0.13 \text{ mA}}$$

$$\text{b.} \quad \text{Ah rating} = (0.13 \text{ mA})(500 \text{ h}) = \mathbf{66.5 \text{ mAh}}$$

$$35. \quad I = \sqrt{\frac{P}{R}} = \sqrt{\frac{100 \text{ W}}{20 \text{ k}\Omega}} = \sqrt{5 \times 10^{-3}} = \mathbf{70.71 \text{ mA}}$$

$$V = \sqrt{PR} = \sqrt{(100 \text{ W})(20 \text{ k}\Omega)} = \mathbf{1.42 \text{ kV}}$$

$$36. \quad P = EI = (220 \text{ V})(30 \text{ A}) = 6.6 \text{ kW}$$

$$P_{\text{HP}} = \frac{6.6 \text{ kW}}{746 \text{ W/HP}} = \mathbf{8.85 \text{ HP}}$$

$$37. \quad \text{a.} \quad W = Pt = \left(\frac{V^2}{R} \right) t = \left(\frac{12 \text{ V}}{10 \Omega} \right)^2 60 \text{ s} = \mathbf{86.4 \text{ J}}$$

$$\text{b.} \quad \text{Energy doubles, power the same}$$

$$39. \quad \frac{12 \text{ h}}{\cancel{\text{week}}} \left[\frac{4 \frac{1}{3} \cancel{\text{weeks}}}{1 \cancel{\text{month}}} \right] [5 \text{ months}] = 260 \text{ h}$$

$$\text{kWh} = \frac{(230 \text{ W})(260 \text{ h})}{1000} = \mathbf{59.80 \text{ kWh}}$$

$$40. \quad \text{a.} \quad W = Pt = (60 \text{ W})(10 \cancel{\text{h}}) \left(\frac{60 \cancel{\text{min}}}{1 \cancel{\text{h}}} \right) \left(\frac{60 \text{ s}}{1 \cancel{\text{min}}} \right) = \mathbf{2.16 \times 10^6 \text{ Ws}}$$

- b. $1 \text{ Ws} = 1 \text{ J} \therefore \mathbf{2.16 \times 10^6 \text{ J}}$
- c. $W = Pt = (60 \text{ W})(10 \text{ h}) = \mathbf{600 \text{ Wh}}$
- d. $\frac{600 \text{ Wh}}{1000 \text{ W/1 kWh}} = \mathbf{0.6 \text{ kWh}}$
- e. $\text{Cost} = (0.6 \text{ kWh})(11 \text{ ¢/kWh}) = \mathbf{6.6 \text{ ¢}}$
41. a. $\text{kWh} = \frac{Pt}{1000} \Rightarrow P = \frac{(1000)(\text{kWh})}{P} = \frac{(1000)(1200 \text{ kWh})}{10 \text{ h}} = \mathbf{120 \text{ kW}}$
- b. $I = \frac{P}{E} = \frac{120 \times 10^3 \text{ W}}{208 \text{ V}} = \mathbf{576.92 \text{ A}}$
- c. $P_{\text{lost}} = P_i - P_o = P_i - \eta P_i = P_i(1 - \eta) = 120 \text{ kW}(1 - 0.82) = 21.6 \text{ kW}$
 $\text{kWh}_{\text{lost}} = \frac{Pt}{1000} = \frac{(21.6 \text{ kW})(10 \text{ h})}{1000} = \mathbf{216 \text{ kWh}}$
42. $\# \text{kWh} = \frac{\$1.00}{11 \text{ ¢}} = 9.09$
 $\text{kWh} = \frac{Pt}{1000} \quad t = \frac{(\text{kWh})(1000)}{P} = \frac{(9.09)(1000)}{250 \text{ W}} = \mathbf{36.36 \text{ h}}$
 $t = \frac{(\text{kWh})(1000)}{P} = \frac{(11.11)(1000)}{4800} = \mathbf{2.32 \text{ h}}$
43. a. $\frac{\$74}{31 \text{ days}} = \mathbf{\$2.39/\text{day}}$
- b. $\frac{\$2.39/\text{day}}{15 \text{ h/day}} = \mathbf{16 \text{ ¢/h}}$
- c. $\frac{16 \text{ ¢}}{11 \text{ ¢/kWh}} = \mathbf{1.45 \text{ kWh}}$
- d. $\frac{1.45 \text{ kWh}}{60 \text{ W}} = 24.16 \cong \mathbf{24 \text{ bulbs}}$
- e. no
44. $\frac{\$1.00}{11 \text{ ¢/kWh}} = 9.09 \text{ kWh}$
 $\frac{9.09 \text{ kWh}}{187 \text{ W}} = \mathbf{48.61 \text{ h}}$

$$\begin{aligned}
 45. \quad t &= 5 \text{ h/day}(365 \text{ days}) = 1825 \text{ h} \\
 \text{kWh} &= \frac{P \cdot t}{1000} = \frac{(339 \text{ W})(1825 \text{ h})}{1000} = 618.68 \text{ kWh} \\
 \text{Cost} &= (618.68 \text{ kWh})(11\text{¢/kWh}) = \$68.05 \\
 \text{kWh} &= \frac{P \cdot t}{1000} = \frac{(213 \text{ W})(1825 \text{ h})}{1000} = 388.73 \text{ kWh} \\
 \text{Cost} &= (388.73 \text{ kWh})(11\text{¢/kWh}) = \$42.76 \\
 \text{Cost Savings} &= \$68.05 - \$42.76 = \mathbf{\$25.29}
 \end{aligned}$$

$$\begin{aligned}
 46. \quad \text{kWh} &= \frac{P \cdot t}{1000} = \frac{(78 \text{ W})(4 \text{ h/day})(31 \text{ days})}{1000} = 9.67 \text{ kWh} \\
 \text{Cost} &= (11\text{¢/kWh})(9.67 \text{ kWh}) = \mathbf{\$1.06}
 \end{aligned}$$

$$47. \quad a. \quad P = EI = (120 \text{ V})(100 \text{ A}) = \mathbf{12 \text{ kW}}$$

$$\begin{aligned}
 b. \quad P_T &= 5 \cancel{\text{hp}} \left[\frac{746 \text{ W}}{\cancel{\text{hp}}} \right] + 3000 \text{ W} + 2400 \text{ W} + 1000 \text{ W} \\
 &= 10,130 < 12,000 \text{ W} \quad \mathbf{(\text{Yes})}
 \end{aligned}$$

$$c. \quad W = Pt = (10.13 \text{ kW})(2 \text{ h}) = \mathbf{20.26 \text{ kWh}}$$

$$\begin{aligned}
 48. \quad \text{kWh} &= \frac{(1600 \text{ W})(6 \text{ h}) + (1200 \text{ W})(1/4 \text{ h}) + (4800 \text{ W}) \frac{1}{2} \text{ h} + (900 \text{ W}) \cancel{10 \text{ min}} \frac{1 \text{ h}}{\cancel{60 \text{ min}}} + (200 \text{ W})(2 \text{ h}) + (50 \text{ W})(3.5 \text{ h})}{1000} \\
 &= \frac{9600 \text{ Wh} + 300 \text{ Wh} + 2400 \text{ Wh} + 150 \text{ Wh} + 400 \text{ Wh} + 175 \text{ Wh}}{1000} = 13.025 \text{ kWh} \\
 (13.025 \text{ kWh})(11\text{¢/kWh}) &= \mathbf{1.43\text{¢}}
 \end{aligned}$$

$$\begin{aligned}
 49. \quad \text{kWh} &= \frac{(200 \text{ W})(4 \text{ h}) + (6)(60 \text{ W})(6 \text{ h}) + (1200 \text{ W}) \cancel{20 \text{ min}} \frac{1 \text{ h}}{\cancel{60 \text{ min}}} + (175 \text{ W})(3.5 \text{ h}) + (250 \text{ W}) \frac{2\frac{1}{6} \text{ h}}{} + (30 \text{ W})(8 \text{ h})}{1000} \\
 &= \frac{800 \text{ Wh} + 2160 \text{ Wh} + 400 \text{ Wh} + 612.5 \text{ Wh} + 541.67 \text{ Wh} + 240 \text{ Wh}}{1000} = \mathbf{4.754 \text{ kWh}} \\
 (4.754 \text{ kWh})(11\text{¢/kWh}) &= \mathbf{52.29\text{¢}}
 \end{aligned}$$

$$50. \quad \eta = \frac{P_o}{P_i} \times 100\% = \frac{(0.5 \cancel{\text{hp}}) \frac{746 \text{ W}}{\cancel{\text{hp}}}}{410 \text{ W}} \times 100\% = \frac{373}{410} \times 100\% = 90.98\%$$

$$\begin{aligned}
 51. \quad \eta &= \frac{P_o}{P_i}, P_i = \frac{P_o}{\eta} = \frac{(1.8 \cancel{\text{hp}})(746 \text{ W}/\cancel{\text{hp}})}{0.685} = 1960.29 \text{ W} \\
 P_i &= EI, I = \frac{P_i}{E} = \frac{1960.29 \text{ W}}{120 \text{ V}} = \mathbf{16.34 \text{ A}}
 \end{aligned}$$

$$52. \quad \eta = \frac{P_o}{P_i} \times 100\% = \frac{(0.8 \text{ hp})(746 \text{ W/hp})}{(4 \text{ A})(220 \text{ V})} \times 100\% = \frac{596.8}{880} \times 100\% = \mathbf{67.82\%}$$

$$53. \quad \text{a.} \quad P_i = EI = (120 \text{ V})(1.8 \text{ A}) = 216 \text{ W}$$

$$P_i = P_o + P_{\text{lost}}, \quad P_{\text{lost}} = P_i - P_o = 216 \text{ W} - 50 \text{ W} = \mathbf{166 \text{ W}}$$

$$\text{b.} \quad \eta\% = \frac{P_o}{P_i} \times 100\% = \frac{50 \text{ W}}{216 \text{ W}} \times 100\% = \mathbf{23.15\%}$$

$$54. \quad P_i = EI = \frac{P_o}{\eta} \Rightarrow I = \frac{P_o}{\eta E} = \frac{(3.6 \text{ hp})(746 \text{ W/hp})}{(0.76)(220 \text{ V})} = \mathbf{16.06 \text{ A}}$$

$$55. \quad \text{a.} \quad P_i = \frac{P_o}{\eta} = \frac{(2 \text{ hp})(746 \text{ W/hp})}{0.9} = \mathbf{1657.78 \text{ W}}$$

$$\text{b.} \quad P_i = EI = 1657.78 \text{ W}$$

$$(110 \text{ V})I = 1657.78 \text{ W}$$

$$I = \frac{1657.78 \text{ W}}{110 \text{ V}} = \mathbf{15.07 \text{ A}}$$

$$\text{c.} \quad P_i = \frac{P_o}{\eta} = \frac{(2 \text{ hp})(746 \text{ W/hp})}{0.7} = 2131.43 \text{ W}$$

$$P_i = EI = 2131.43 \text{ W}$$

$$(110 \text{ V})I = 2131.43 \text{ W}$$

$$I = \frac{2131.43 \text{ W}}{110 \text{ V}} = \mathbf{19.38 \text{ A}}$$

$$56. \quad P_i = \frac{P_o}{\eta} = \frac{(15 \text{ hp})(746 \text{ W/hp})}{0.9} = 12,433.33 \text{ W}$$

$$I = \frac{P_i}{E} = \frac{12,433.33 \text{ W}}{220 \text{ V}} = \mathbf{56.52 \text{ A}}$$

$$57. \quad \eta_T = \eta_1 \cdot \eta_2$$

$$0.75 = 0.85 \times \eta_2$$

$$\eta_2 = \mathbf{0.88}$$

$$58. \quad \eta_T = \eta_1 \cdot \eta_2 = (0.87)(0.75) = 0.6525 \Rightarrow \mathbf{65.25\%}$$

$$59. \quad \eta_T = \eta_1 \cdot \eta_2 = 0.78 = 0.9\eta_2$$

$$\eta_2 = \frac{0.78}{0.9} = 0.867 \Rightarrow \mathbf{86.7\%}$$

60. a. $\eta_T = \eta_1 \cdot \eta_2 \cdot \eta_3 = (0.93)(0.87)(0.21) = 0.170 \Rightarrow \mathbf{17\%}$

b. $\eta_T = \eta_1 \cdot \eta_2 \cdot \eta_3 = (0.93)(0.87)(0.80) = 0.647 \Rightarrow \mathbf{64.7\%}$
 $\frac{64.7\% - 17\%}{17\%} \times 100\% = \mathbf{280.59\%}$

61. $\eta_T = \frac{P_o}{P_i} = \eta_1 \cdot \eta_2 = \eta_1 \cdot 2\eta_1 = 2\eta_1^2$

$$\eta_1^2 = \frac{P_o}{2P_i} \Rightarrow \eta_1 = \sqrt{\frac{P_o}{2P_i}} = \sqrt{\frac{128 \text{ W}}{2(400 \text{ W})}} = 0.4$$

$$\eta_2 = 2\eta_1 = 2(0.4) = 0.8$$

$$\therefore \eta_1 = \mathbf{40\%}, \eta_2 = \mathbf{80\%}$$

Chapter 5

1.
 - a. E and R_1
 - b. R_1 and R_2
 - c. E_1 , E_2 , and R_1
 - d. E_1 and R_1 ; E_2 , R_3 and R_4
 - e. R_3 , R_4 and R_5 ; E and R_1
 - f. R_2 and R_3
2.
 - a. $R_T = 0.1 \text{ k}\Omega + 0.39 \text{ k}\Omega + 1.2 \text{ k}\Omega + 6.8 \text{ k}\Omega = \mathbf{8.49 \text{ k}\Omega}$
 - b. $R_T = 1.2 \text{ }\Omega + 2.7 \text{ }\Omega + 8.2 \text{ }\Omega = \mathbf{12.1 \text{ }\Omega}$
 - c. $R_T = 8.2 \text{ k}\Omega + 10 \text{ k}\Omega + 9.1 \text{ k}\Omega + 1.8 \text{ k}\Omega + 2.7 \text{ k}\Omega = \mathbf{31.8 \text{ k}\Omega}$
 - d. $R_T = 47 \text{ }\Omega + 820 \text{ }\Omega + 91 \text{ }\Omega + 1.2 \text{ k}\Omega = \mathbf{2158.0 \text{ }\Omega}$
3.
 - a. $R_T = 1.2 \text{ k}\Omega + 1 \text{ k}\Omega + 2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega = \mathbf{7.7 \text{ k}\Omega}$
 - b. $R_T = 1 \text{ k}\Omega + 2 \text{ k}\Omega + 3 \text{ k}\Omega + 4.7 \text{ k}\Omega + 6.8 \text{ k}\Omega = \mathbf{17.5 \text{ k}\Omega}$
4.
 - a. $\mathbf{1 \text{ M}\Omega}$
 - b. $\mathbf{100 \text{ }\Omega, 1 \text{ k}\Omega}$
 - c. $R_T = 100 \text{ }\Omega + 1 \text{ k}\Omega + 1 \text{ M}\Omega + 200 \text{ k}\Omega = \mathbf{1.2011 \text{ M}\Omega}$ vs. $\mathbf{1.2 \text{ M}\Omega}$ for part b.
5.
 - a. $R_T = 10 \text{ }\Omega + 33 \text{ }\Omega + 56 \text{ }\Omega$, Reading = $\mathbf{99 \text{ }\Omega}$
 - b. $R_T = 2.2 \text{ k}\Omega + 0.82 \text{ k}\Omega + 1.2 \text{ k}\Omega + 3.3 \text{ k}\Omega$, Reading = $\mathbf{7.52 \text{ k}\Omega}$
6.
 - a. $R_T = 129 \text{ k}\Omega = R + 56 \text{ k}\Omega + 22 \text{ k}\Omega + 33 \text{ k}\Omega$, Reading = $\mathbf{18 \text{ k}\Omega}$
 - b. $R_T = 103 \text{ k}\Omega = 24 \text{ k}\Omega + R_1 + 43 \text{ k}\Omega + 2R_1 = 67 \text{ k}\Omega + 3R_1$, $R_1 = \mathbf{12 \text{ k}\Omega}$
 $R_2 = \mathbf{24 \text{ k}\Omega}$
7.
 - a. $\mathbf{1.2 \text{ k}\Omega}$
 - b. $\mathbf{0 \text{ }\Omega}$
 - c. $\mathbf{\infty \text{ }\Omega}$
8.
 - a. $R_T = 10 \text{ }\Omega + 12 \text{ }\Omega + 18 \text{ }\Omega = \mathbf{40 \text{ }\Omega}$
 - b. $I_s = \frac{E}{R_T} = \frac{72 \text{ V}}{40 \text{ }\Omega} = \mathbf{1.8 \text{ A}}$
 - c. $V_1 = I_1 R_1 = (1.8 \text{ A})(10 \text{ }\Omega) = \mathbf{18 \text{ V}}$, $V_2 = I_2 R_2 = (1.8 \text{ A})(12 \text{ }\Omega) = \mathbf{21.6 \text{ V}}$,
 $V_3 = I_3 R_3 = (1.8 \text{ A})(18 \text{ }\Omega) = \mathbf{32.4 \text{ V}}$
 - d. $P_s = EI_s = (72 \text{ V})(1.8 \text{ A}) = \mathbf{129.6 \text{ W}}$
 - e. $P_{18 \text{ }\Omega} = V_3 I_3 = (32.4 \text{ V})(1.8 \text{ A}) = \mathbf{58.32 \text{ W}}$
9.
 - a. the most: R_3 , the least: R_1
 - b. R_3 , $R_T = 1.2 \text{ k}\Omega + 6.8 \text{ k}\Omega + 82 \text{ k}\Omega = \mathbf{90 \text{ k}\Omega}$
 $I_s = \frac{E}{R_T} = \frac{45 \text{ V}}{90 \text{ k}\Omega} = \mathbf{0.5 \text{ mA}}$
 - c. $V_1 = I_1 R_1 = (0.5 \text{ mA})(1.2 \text{ k}\Omega) = \mathbf{0.6 \text{ V}}$, $V_2 = I_2 R_2 = (0.5 \text{ mA})(6.8 \text{ k}\Omega) = \mathbf{3.4 \text{ V}}$,
 $V_3 = I_3 R_3 = (0.5 \text{ mA})(82 \text{ k}\Omega) = \mathbf{41 \text{ V}}$, results agree with part (a)

- d. $P_s = EI_s = (72 \text{ V})(1.8 \text{ A}) = \mathbf{129.6 \text{ W}}$
e. $P_{18\Omega} = V_3 I_3 = (32.4 \text{ V})(1.8 \text{ A}) = \mathbf{58.32 \text{ W}}$
10. a. $R_T = 12 \text{ k}\Omega + 4 \text{ k}\Omega + 6 \text{ k}\Omega = 22 \text{ k}\Omega$
 $E = IR_T = (4 \text{ mA})(22 \text{ k}\Omega) = \mathbf{88 \text{ V}}$
b. $R_T = 12 \Omega + 22 \Omega + 82 \Omega + 10 \Omega = 126 \Omega$
 $E = IR_T = (500 \text{ mA})(126 \Omega) = \mathbf{63 \text{ V}}$
11. a. a. $I = \frac{V}{R} = \frac{5.2 \text{ V}}{1.3 \Omega} = \mathbf{4 \text{ A}}$
b. $E = IR_T = (4 \text{ A})(9 \Omega) = \mathbf{36 \text{ V}}$
c. $R_T = 9 \Omega + 4.7 \Omega + 1.3 \Omega + R, \quad R = \mathbf{3 \Omega}$
d. $V_{4.7 \Omega} = (4 \text{ A})(4.7 \Omega) = \mathbf{18.8 \text{ V}}$
 $V_{1.3 \Omega} = (4 \text{ A})(1.3 \Omega) = \mathbf{5.2 \text{ V}}$
 $V_{3 \Omega} = (4 \text{ A})(3 \Omega) = \mathbf{12 \text{ V}}$
- b. a. $I = \frac{V}{R} = \frac{6.6 \text{ V}}{2.2 \text{ k}\Omega} = \mathbf{3 \text{ mA}}$
b. $V_{3.3 \text{ k}\Omega} = (3 \text{ mA})(3.3 \text{ k}\Omega) = 9.9 \text{ V}$
 $E = 6.6 \text{ V} + 9 \text{ V} + 9.9 \text{ V} = \mathbf{25.5 \text{ V}}$
c. $R = \frac{V}{I} = \frac{9 \text{ V}}{3 \text{ mA}} = \mathbf{3 \text{ k}\Omega}$
d. $V_{2.2 \text{ k}\Omega} = \mathbf{6.6 \text{ V}}, V_{3 \text{ k}\Omega} = \mathbf{9 \text{ V}}, V_{3.3 \text{ k}\Omega} = \mathbf{9.9 \text{ V}}$
12. a. $I_m = \frac{E}{R_T} = \frac{36 \text{ V}}{4.4 \text{ k}\Omega} = \mathbf{8.18 \text{ mA}}, V_m = \frac{1}{2}E = \frac{1}{2}(36 \text{ V}) = \mathbf{18 \text{ V}}$
- b. $R_T = 1 \text{ k}\Omega + 2.4 \text{ k}\Omega + 5.6 \text{ k}\Omega = 9 \text{ k}\Omega$
 $I_m = \frac{E}{R_T} = \frac{22.5 \text{ V}}{9 \text{ k}\Omega} = \mathbf{2.5 \text{ mA}}, V_m = 2.5 \text{ mA}(2.4 \text{ k}\Omega + 5.6 \text{ k}\Omega) = \mathbf{20 \text{ V}}$
- c. $V_{3.3 \text{ k}\Omega} = \frac{3.3 \text{ k}\Omega(12 \text{ V})}{4.5 \text{ k}\Omega} = 8.8 \text{ V}$
 $V_m = 12 \text{ V} - 8.8 \text{ V} = \mathbf{3.2 \text{ V}}$
 $I_m = \frac{12 \text{ V}}{4.5 \text{ k}\Omega} = \mathbf{2.67 \text{ mA}}$
13. a. $R_T = 3 \text{ k}\Omega + 1 \text{ k}\Omega + 2 \text{ k}\Omega = \mathbf{6 \text{ k}\Omega}$
 $I_s = \frac{E}{R_T} = \frac{120 \text{ V}}{6 \text{ k}\Omega} = \mathbf{20 \text{ mA}}$
 $V_{R_1} = (20 \text{ mA})(3 \text{ k}\Omega) = \mathbf{60 \text{ V}}$
 $V_{R_2} = (20 \text{ mA})(1 \text{ k}\Omega) = \mathbf{20 \text{ V}}$
 $V_{R_3} = (20 \text{ mA})(2 \text{ k}\Omega) = \mathbf{40 \text{ V}}$

- b. $P_{R_1} = I_1^2 R_1 = (20 \text{ mA})^2 \cdot 3 \text{ k}\Omega = \mathbf{1.2 \text{ W}}$
 $P_{R_2} = I_2^2 R_2 = (20 \text{ mA})^2 \cdot 1 \text{ k}\Omega = \mathbf{0.4 \text{ W}}$
 $P_{R_3} = I_3^2 R_3 = (20 \text{ mA})^2 \cdot 2 \text{ k}\Omega = \mathbf{0.8 \text{ W}}$
- c. $P_T = P_{R_1} + P_{R_2} + P_{R_3} = 1.2 \text{ W} + 0.4 \text{ W} + 0.8 \text{ W} = \mathbf{2.4 \text{ W}}$
- d. $P_T = EI_s = (120 \text{ V})(20 \text{ mA}) = \mathbf{2.4 \text{ W}}$
- e. the same
- f. R_1 – the largest
- g. dissipated
- h. R_1 : 2 W, R_2 : 1/2 W, R_3 : 1 W
14. $P = 21 \text{ W} = (1 \text{ A})^2 \cdot R$, $R = \mathbf{21 \Omega}$
 $V_1 = I_1 R_1 = (1 \text{ A})(2 \Omega) = \mathbf{2 \text{ V}}$, $V_2 = I_2 R_2 = (1 \text{ A})(1 \Omega) = \mathbf{1 \text{ V}}$
 $V_3 = I_3 R_3 = (1 \text{ A})(21 \Omega) = \mathbf{21 \text{ V}}$
 $E = V_1 + V_2 + V_3 = 2 \text{ V} + 1 \text{ V} + 21 \text{ V} = \mathbf{24 \text{ V}}$
15. $P = 8 \text{ W} = I^2 \cdot 1 \Omega$, $I = \sqrt{8} = \mathbf{2.828 \text{ A}}$
 $P = 16 \text{ W} = I^2 R_1 = (2.828 \text{ A})^2 R_1$, $R_1 = \mathbf{2 \Omega}$
 $R_T = 32 \Omega = 2 \Omega + R_2 + 1 \Omega = 3 \Omega + R_2$, $R_2 = \mathbf{29 \Omega}$
 $E = IR_T = (2.828 \text{ A})(32 \Omega) = \mathbf{90.5 \text{ V}}$
16. a. $R_T = NR_1 = 8 \left(28 \frac{1}{8} \Omega \right) = 225 \Omega$
 $I = \frac{E}{R_T} = \frac{120 \text{ V}}{225 \Omega} = \mathbf{0.53 \text{ A}}$
- b. $P = I^2 R = \left(\frac{8}{15} \text{ A} \right)^2 \left(28 \frac{1}{8} \Omega \right) = \left(\frac{64}{225} \right) \left(\frac{225}{8} \right) = \mathbf{8 \text{ W}}$
- c. $V = IR = \left(\frac{8}{15} \text{ A} \right) \left(\frac{225}{8} \Omega \right) = \mathbf{15 \text{ V}}$
- d. All go out!

17. $P_s = P_{R_1} + P_{R_2} + P_{R_3}$
 $E \cdot I = I^2 R_1 + I^2 R_2 + 24$
 $(R_1 + R_2)I^2 - E \cdot I + 24 = 0$
 $6I^2 - 24I + 24 = 0$
 $I^2 - 4I + 4 = 0$
 $I = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(4)}}{2(1)} = \frac{4 \pm \sqrt{16 - 16}}{2} = \frac{4}{2} = 2 \text{ A}$
 $P = 24 \text{ W} = (2 \text{ A})^2 R, \quad R = \frac{24 \Omega}{4} = \mathbf{6 \Omega}$
18. a. $V_{ab} + 4 \text{ V} + 24 \text{ V} - 12 \text{ V} = 0, V_{ab} = -28 \text{ V} + 12 \text{ V} = \mathbf{-16 \text{ V}}$
b. $V_{ab} + 4 \text{ V} + 8 \text{ V} - 16 \text{ V} = 0, V_{ab} = 16 \text{ V} - 12 \text{ V} = \mathbf{4 \text{ V}}$
c. $V_{ab} + 12 \text{ V} - 18 \text{ V} + 6 \text{ V} - 12 \text{ V} = 0, V_{ab} = 30 \text{ V} - 18 \text{ V} = \mathbf{12 \text{ V}}$
19. a. $E_T = 8 \text{ V} - 16 \text{ V} + 20 \text{ V} = \mathbf{12 \text{ V}}, I = \frac{12 \text{ V}}{10.3 \Omega} = \mathbf{1.17 \text{ A}}$
b. $E_T = -4 \text{ V} + 18 \text{ V} - 12 \text{ V} = \mathbf{2 \text{ V}}, I = \frac{2 \text{ V}}{11.5 \Omega} = \mathbf{173.91 \text{ mA}}$
20. a. $P = 8 \text{ mW} = I^2 R, R = \frac{8 \text{ mW}}{I^2} = \frac{8 \text{ mW}}{(2 \text{ mA})^2} = \mathbf{2 \text{ k}\Omega}$
 $I = \frac{E}{R_T} = \frac{20 \text{ V} - E}{3 \text{ k}\Omega + 2 \text{ k}\Omega} = 2 \text{ mA (CW)}, \quad E = \mathbf{10 \text{ V}}$
b. $I = \frac{16 \text{ V}}{2 \text{ k}\Omega} = 8 \text{ mA}, R = \frac{V}{I} = \frac{12 \text{ V}}{8 \text{ mA}} = \mathbf{1.5 \text{ k}\Omega}$
 $I = \frac{E}{R_T} = \frac{E - 4 \text{ V} - 10 \text{ V}}{2 \text{ k}\Omega + 1.5 \text{ k}\Omega} = \frac{E - 14 \text{ V}}{3.5 \text{ k}\Omega} = 8 \text{ mA (CCW)}$
 $E = \mathbf{42 \text{ V}}$
21. a. $+10 \text{ V} + 4 \text{ V} - 12 \text{ V} - V = 0$
 $V = 14 \text{ V} - 12 \text{ V} = \mathbf{2 \text{ V}}$
b. $+30 \text{ V} + 20 \text{ V} - 8 \text{ V} - V = 0$
 $V = 50 \text{ V} - 8 \text{ V} = \mathbf{42 \text{ V}}$
c. $-6 \text{ V} - 22 \text{ V} - V_1 + 36 \text{ V} = 0$
 $V = 36 \text{ V} - 28 \text{ V} = \mathbf{8 \text{ V}}$
22. a. $I = \frac{12 \text{ V}}{8 \Omega} = \mathbf{1.5 \text{ A}}$
b. $V_2 = IR = (1.5 \text{ A})(2 \Omega) = \mathbf{3 \text{ V}}$
c. $60 \text{ V} - 12 \text{ V} - V_1 - 3 \text{ V} = 0$
 $V_1 = 60 \text{ V} - 15 \text{ V} = \mathbf{45 \text{ V}}$

23. a. $+10\text{ V} - V_2 = 0$
 $V_2 = \mathbf{10\text{ V}}$
 $+10\text{ V} - 6\text{ V} - V_1 = 0$
 $V_1 = \mathbf{4\text{ V}}$
- b. $+24\text{ V} - 10\text{ V} - V_1 = 0$
 $V_1 = \mathbf{14\text{ V}}$
 $+10\text{ V} - V_2 + 8\text{ V} = 0$
 $V_2 = \mathbf{18\text{ V}}$
24. a. $V_{1.8\Omega} = IR = (3\text{ A})(1.8\ \Omega) = 5.4\text{ V}$
 $24\text{ V} - V_1 - 10\text{ V} - 5.4\text{ V} = 0, V_1 = 24\text{ V} - 15.4\text{ V} = \mathbf{8.6\text{ V}}$
 $V_{2.7\Omega} = IR = (3\text{ A})(2.7\ \Omega) = 8.1\text{ V}$
 $10\text{ V} - 8.1\text{ V} - V_2 = 0$
 $V_2 = 10\text{ V} - 8.1\text{ V} = \mathbf{1.9\text{ V}}$
- b. $+10\text{ V} - V_1 + 6\text{ V} - 2\text{ V} - 3\text{ V} = 0, V_1 = \mathbf{11\text{ V}}$
 $+10\text{ V} - V_2 - 3\text{ V} = 0, V_2 = \mathbf{7\text{ V}}$
25. $\frac{1\text{ V}}{2\ \Omega} = \frac{50\text{ V}}{R_2}, R_2 = \frac{(50\text{ V})(2\ \Omega)}{1\text{ V}} = \mathbf{100\ \Omega}$
 $\frac{1\text{ V}}{2\ \Omega} = \frac{100\text{ V}}{R_3}, R_3 = \frac{(100\text{ V})(2\ \Omega)}{1\text{ V}} = \mathbf{200\ \Omega}$
26. a. $\mathbf{10\text{ k}\Omega}$
- b. $V_3: V_2 = 10\text{ k}\Omega:1\text{ k}\Omega = \mathbf{10:1}$
 $V_3: V_1 = 10\text{ k}\Omega:100\ \Omega = \mathbf{100:1}$
- c. $V_3 = \frac{R_3 E}{R_T} = \frac{(10\text{ k}\Omega)(60\text{ V})}{0.1\text{ k}\Omega + 1\text{ k}\Omega + 10\text{ k}\Omega} = \mathbf{54.05\text{ V}}$
- d. $V' = \frac{(R_2 + R_3)E}{R_T} = \frac{(1\text{ k}\Omega + 10\text{ k}\Omega)(60\text{ V})}{11.1\text{ k}\Omega} = \mathbf{59.46\text{ V}}$
27. a. $V = \frac{40\ \Omega(30\text{ V})}{40\ \Omega + 20\ \Omega} = \mathbf{20\text{ V}}$
- b. $V = \frac{(2\text{ k}\Omega + 3\text{ k}\Omega)(40\text{ V})}{4\text{ k}\Omega + 1\text{ k}\Omega + 2\text{ k}\Omega + 3\text{ k}\Omega} = \frac{(5\text{ k}\Omega)(40\text{ V})}{10\text{ k}\Omega} = \mathbf{20\text{ V}}$
- c. $\frac{(1.5\ \Omega + 0.6\ \Omega + 0.9\ \Omega)(0.72\text{ V})}{(2.5\ \Omega + 1.5\ \Omega + 0.6\ \Omega + 0.9\ \Omega + 0.5\ \Omega)} = \frac{(3\ \Omega)(0.72\text{ V})}{6\text{ k}\Omega} = \mathbf{0.36\text{ V}}$

$$28. \quad a. \quad \frac{V_1}{1.2 \, \Omega} = \frac{20 \, \text{V}}{2 \, \Omega}, V_1 = \frac{(1.2 \, \Omega)(20 \, \text{V})}{2 \, \Omega} = \mathbf{12 \, \text{V}}$$

$$\frac{V_2}{6.8 \, \Omega} = \frac{20 \, \text{V}}{2 \, \Omega}, V_2 = \frac{(6.8 \, \Omega)(20 \, \text{V})}{2 \, \Omega} = \mathbf{68 \, \text{V}}$$

$$E = V_1 + 20 \, \text{V} + V_2 = 12 \, \text{V} + 20 \, \text{V} + 68 \, \text{V} = \mathbf{100 \, \text{V}}$$

$$b. \quad 120 \, \text{V} - V_1 - 80 \, \text{V} = 0, V_1 = \mathbf{40 \, \text{V}}$$

$$80 \, \text{V} - 10 \, \text{V} - V_3 = 0, V_3 = \mathbf{70 \, \text{V}}$$

$$c. \quad \frac{1000 \, \text{V}}{100 \, \Omega} = \frac{V_2}{68 \, \Omega}, V_2 = \frac{68 \, \Omega(1000 \, \text{V})}{100 \, \Omega} = \mathbf{680 \, \text{V}}$$

$$\frac{1000 \, \text{V}}{100 \, \Omega} = \frac{V_1}{2 \, \Omega}, V_1 = \frac{2 \, \Omega(1000 \, \text{V})}{100 \, \Omega} = \mathbf{20 \, \text{V}}$$

$$E = V_1 + V_2 + 1000 \, \text{V}$$

$$= 20 \, \text{V} + 680 \, \text{V} + 1000 \, \text{V}$$

$$= \mathbf{1700 \, \text{V}}$$

$$d. \quad V_1 = \mathbf{0 \, \text{V}}$$

$$V_2 = \frac{10 \, \text{k}\Omega(50 \, \text{V} - 30 \, \text{V})}{10 \, \text{k}\Omega + 3.3 \, \text{k}\Omega + 4.7 \, \text{k}\Omega}$$

$$= \frac{10 \, \text{k}\Omega(20 \, \text{V})}{18 \, \text{k}\Omega} = \mathbf{11.11 \, \text{V}}$$

$$V_x = E_1 - V_{3.3\text{k}\Omega}$$

$$V_{3.3\text{k}\Omega} = \frac{3.3 \, \text{k}\Omega(20 \, \text{V})}{18 \, \text{k}\Omega}$$

$$= 3.67 \, \text{V}$$

$$V_x = 50 \, \text{V} - 3.67 \, \text{V} = \mathbf{46.33 \, \text{V}}$$

$$29. \quad \frac{2 \, \text{V}}{1 \, \text{k}\Omega} = \frac{V_2}{2 \, \text{k}\Omega}, V_2 = \frac{2 \, \text{k}\Omega(2 \, \text{V})}{1 \, \text{k}\Omega} = \mathbf{4 \, \text{V}}$$

$$\frac{2 \, \text{V}}{1 \, \text{k}\Omega} = \frac{V_4}{3 \, \text{k}\Omega}, V_4 = \frac{3 \, \text{k}\Omega(2 \, \text{V})}{1 \, \text{k}\Omega} = \mathbf{6 \, \text{V}}$$

$$I = \frac{2 \, \text{V}}{1 \, \text{k}\Omega} = \mathbf{2 \, \text{mA}}$$

$$E = 2 \, \text{V} + 4 \, \text{V} + 12 \, \text{V} + 6 \, \text{V} = \mathbf{24 \, \text{V}}$$

30. a. $4 \text{ V} = \frac{R(20 \text{ V})}{2.2 \text{ k}\Omega + 1.8 \text{ k}\Omega + R}$
 $4(4 \text{ k}\Omega + R) = 20R$
 $16 \text{ k}\Omega + 4R = 20R$
 $16R = 16 \text{ k}\Omega$
 $R = \frac{16}{16} \text{ k}\Omega = \mathbf{1 \text{ k}\Omega}$
- b. $100 \text{ V} = \frac{(6 \text{ M}\Omega + R)(140 \text{ V})}{6 \text{ M}\Omega + R + 3 \text{ M}\Omega}$
 $100(9 \text{ M}\Omega + R) = 840 \text{ M}\Omega + 140R$
 $900 \text{ M}\Omega + 100R = 840 \text{ M}\Omega + 140R$
 $40R = 60 \text{ M}\Omega$
 $R = \frac{60}{40} \text{ M}\Omega = \mathbf{1.5 \text{ M}\Omega}$
31. a. $R_{\text{bulb}} = \frac{8 \text{ V}}{50 \text{ mA}} = 160 \Omega$
 $V_{\text{bulb}} = 8 \text{ V} = \frac{R_{\text{bulb}}(12 \text{ V})}{R_{\text{bulb}} + R_x} = \frac{160 \Omega(12 \text{ V})}{160 \Omega + R_x}, R_x = \mathbf{80 \Omega}$ in series with the bulb
- b. $V_R = 12 \text{ V} - 8 \text{ V} = 4 \text{ V}, P = \frac{V^2}{R} = \frac{(4 \text{ V})^2}{80 \Omega} = 0.2 \text{ W}, \therefore \mathbf{1/4 \text{ W okay}}$
32. $V_{R_1} + V_{R_2} = 72 \text{ V}$
 $\frac{1}{5}V_{R_2} + V_{R_2} = 72 \text{ V}$
 $V_{R_2} \left[1 + \frac{1}{5} \right] = 72 \text{ V}, V_{R_2} = \frac{72 \text{ V}}{1.2} = 60 \text{ V}$
 $R_2 = \frac{V_{R_2}}{I_{R_2}} = \frac{60 \text{ V}}{4 \text{ mA}} = \mathbf{15 \text{ k}\Omega}, R_1 = \frac{V_{R_1}}{I_{R_1}} = \frac{72 \text{ V} - 60 \text{ V}}{4 \text{ mA}} = \frac{12 \text{ V}}{4 \text{ mA}} = \mathbf{3 \text{ k}\Omega}$
33. $R_T = R_1 + R_2 + R_3 = 2R_3 + 7R_3 + R_3 = 10R_3$
 $V_{R_3} = \frac{R_3(60 \text{ V})}{10R_3} = \mathbf{6 \text{ V}}, V_{R_1} = 2V_{R_3} = 2(6 \text{ V}) = \mathbf{12 \text{ V}}, V_{R_2} = 7V_{R_3} = 7(6 \text{ V}) = \mathbf{42 \text{ V}}$
34. a. $V_{R_3} = 4V_{R_2} = 4(3V_{R_1}) = 12V_{R_1}$
 $E = V_{R_1} + 3V_{R_1} + 12V_{R_1} \therefore R_T = R_1 + 3R_1 + 12R_1 = 16R_1 = \frac{64 \text{ V}}{10 \text{ mA}} = 6.4 \text{ k}\Omega$
 $R_1 = \frac{6.4 \text{ k}\Omega}{16} = \mathbf{400 \Omega}, R_2 = 3R_1 = \mathbf{1.2 \text{ k}\Omega}, R_3 = 12R_1 = \mathbf{4.8 \text{ k}\Omega}$

- b. $R_T = \frac{64 \text{ V}}{10 \mu\text{A}} = 6.4 \text{ M}\Omega$, $R_1 = \frac{6.4 \text{ M}\Omega}{16} = \mathbf{400 \text{ k}\Omega}$, $R_2 = \mathbf{1.2 \text{ M}\Omega}$, $R_3 = \mathbf{4.8 \text{ M}\Omega}$
- $$\frac{I_1}{I'} = \frac{10 \text{ mA}}{10 \mu\text{A}} = \mathbf{10^3} \text{ and } \frac{R'_1}{R_1} = \frac{400 \text{ k}\Omega}{400 \Omega} = \mathbf{10^3} \text{ also}$$
35. a. $V_a = 12 \text{ V} + 5 \text{ V} = \mathbf{17 \text{ V}}$
 $V_b = 5 \text{ V} + 16 \text{ V} = \mathbf{21 \text{ V}}$
 $V_{ab} = 17 \text{ V} - 21 \text{ V} = \mathbf{-4 \text{ V}}$
- b. $V_a = \mathbf{14 \text{ V}}$
 $V_b = 14 \text{ V} + 6 \text{ V} + 10 \text{ V} = \mathbf{30 \text{ V}}$
 $V_{ab} = 14 \text{ V} - 30 \text{ V} = \mathbf{-16 \text{ V}}$
- c. $V_a = 10 \text{ V} + 3 \text{ V} = \mathbf{13 \text{ V}}$
 $V_b = \mathbf{-8 \text{ V}}$
 $V_{ab} = 13 \text{ V} - (-8 \text{ V}) = \mathbf{21 \text{ V}}$
36. a. $I \downarrow = \frac{60 \text{ V} + 20 \text{ V}}{18 \Omega + 82 \Omega} = \frac{80 \text{ V}}{100 \Omega} = \mathbf{0.8 \text{ A}}$
 $V_a = 60 \text{ V} - I(18 \Omega) = 60 \text{ V} - (0.8 \text{ A})(18 \Omega) = 60 \text{ V} - 14.4 \text{ V} = \mathbf{45.6 \text{ V}}$
- $$\overleftarrow{I} = \frac{100 \text{ V} + 60 \text{ V}}{2 \text{ k}\Omega + 2 \text{ k}\Omega + 2 \text{ k}\Omega + 2 \text{ k}\Omega} = \frac{160 \text{ V}}{8 \text{ k}\Omega} = \mathbf{20 \text{ mA}}$$
- $$V_a = 10 \text{ V} - I(2 \text{ k}\Omega) = 100 \text{ V} - (20 \text{ mA})(2 \text{ k}\Omega) = 100 \text{ V} - 40 \text{ V} = \mathbf{60 \text{ V}}$$
37. $I = \frac{47 \text{ V} - 20 \text{ V}}{2 \text{ k}\Omega + 3 \text{ k}\Omega + 4 \text{ k}\Omega} = \frac{27 \text{ V}}{9 \text{ k}\Omega} = 3 \text{ mA (CCW)}$
 $V_{2\text{k}\Omega} = 6 \text{ V}$, $V_{3\text{k}\Omega} = 9 \text{ V}$, $V_{4\text{k}\Omega} = 12 \text{ V}$
- a. $V_a = \mathbf{20 \text{ V}}$, $V_b = 20 \text{ V} + 6 \text{ V} = \mathbf{26 \text{ V}}$, $V_c = 20 \text{ V} + 6 \text{ V} + 9 \text{ V} = \mathbf{35 \text{ V}}$
 $V_d = \mathbf{-12 \text{ V}}$, $V_e = \mathbf{0 \text{ V}}$
- b. $V_{ab} = \mathbf{-6 \text{ V}}$, $V_{dc} = \mathbf{-47 \text{ V}}$, $V_{cb} = \mathbf{9 \text{ V}}$
- c. $V_{ac} = \mathbf{-15 \text{ V}}$, $V_{db} = -47 \text{ V} + 9 \text{ V} = \mathbf{-38 \text{ V}}$
38. $I_{R_2} = \frac{4 \text{ V} + 4 \text{ V}}{8 \Omega} = \frac{8 \text{ V}}{8 \Omega} = 1 \text{ A}$, $R_1 = \frac{V_{R_1}}{I} = \frac{12 \text{ V} - 4 \text{ V}}{1 \text{ A}} = \frac{8 \text{ V}}{1 \text{ A}} = \mathbf{8 \Omega}$,
 $R_3 = \frac{V_{R_3}}{I} = \frac{8 \text{ V} - 4 \text{ V}}{1 \text{ A}} = \frac{4 \text{ V}}{1 \text{ A}} = \mathbf{4 \Omega}$

39. $V_{R_2} = 48 \text{ V} - 12 \text{ V} = 36 \text{ V}$
 $R_2 = \frac{V_{R_2}}{I} = \frac{36 \text{ V}}{16 \text{ mA}} = \mathbf{2.25 \text{ k}\Omega}$
 $V_{R_3} = 12 \text{ V} - 0 \text{ V} = 12 \text{ V}$
 $R_3 = \frac{V_{R_3}}{I} = \frac{12 \text{ V}}{16 \text{ mA}} = \mathbf{0.75 \text{ k}\Omega}$
 $V_{R_4} = 20 \text{ V}$
 $R_4 = \frac{V_{R_4}}{I} = \frac{20 \text{ V}}{16 \text{ mA}} = \mathbf{1.25 \text{ k}\Omega}$
 $V_{R_1} = E - V_{R_2} - V_{R_3} - V_{R_4}$
 $= 100 \text{ V} - 36 \text{ V} - 12 \text{ V} - 20 \text{ V} = 32 \text{ V}$
 $R_1 = \frac{V_{R_1}}{I} = \frac{32 \text{ V}}{16 \text{ mA}} = \mathbf{2 \text{ k}\Omega}$
40. a. $V_a = -8 \text{ V} + 14 \text{ V} = \mathbf{+6 \text{ V}}$, $V_b = \mathbf{14 \text{ V}}$
 $V_c = +I(10 \Omega) - 6 \text{ V}$ with
 $I = \frac{14 \text{ V} + 6 \text{ V}}{10 \Omega + 10 \Omega} = \frac{20 \text{ V}}{20 \Omega} = 1 \text{ A}$
Therefore, $V_c = (1 \text{ A})(10 \Omega) - 6 \text{ V} = 10 \text{ V} - 6 \text{ V} = \mathbf{4 \text{ V}}$
 $V_d = \mathbf{0 \text{ V}}$
- b. $V_{ab} = V_a - V_b = 6 \text{ V} - 14 \text{ V} = \mathbf{-8 \text{ V}}$
 $V_{cb} = V_c - V_b = 4 \text{ V} - 14 \text{ V} = \mathbf{-10 \text{ V}}$
 $V_{cd} = V_c - V_d = 4 \text{ V} - 0 \text{ V} = \mathbf{4 \text{ V}}$
- c. $V_{ad} = V_a - V_d = 6 \text{ V} - 0 \text{ V} = \mathbf{6 \text{ V}}$
 $V_{ca} = V_c - V_a = 4 \text{ V} - 6 \text{ V} = \mathbf{-2 \text{ V}}$
41. $V_0 = \mathbf{0 \text{ V}}$, $V_4 = (2 \text{ k}\Omega)(6 \text{ mA}) + 3 \text{ V} = 12 \text{ V} + 3 \text{ V} = \mathbf{15 \text{ V}}$, $V_7 = \mathbf{4 \text{ V}}$
 $V_{10} = V_1 - V_0 = 12 \text{ V} - 0 \text{ V} = \mathbf{12 \text{ V}}$, $V_{23} = V_2 - V_3 = 4 \text{ V} - (-8 \text{ V}) = 4 \text{ V} + 8 \text{ V} = \mathbf{12 \text{ V}}$
 $V_{30} = V_3 - V_0 = -8 \text{ V} - 0 \text{ V} = \mathbf{-8 \text{ V}}$, $V_{67} = V_6 - V_7 = 4 \text{ V} - 4 \text{ V} = \mathbf{0 \text{ V}}$
 $V_{56} = V_5 - V_6 = 3 \text{ V} - 4 \text{ V} = \mathbf{-1 \text{ V}}$, $I \uparrow = \frac{4 \text{ V} + 8 \text{ V}}{4 \Omega} = \frac{12 \text{ V}}{4 \Omega} = \mathbf{3 \text{ A}}$
42. $V_0 = \mathbf{0 \text{ V}}$, $V_{03} = V_0 - V_3 = 0 \text{ V} - 0 \text{ V} = \mathbf{0 \text{ V}}$, $V_2 = (3 \text{ mA})(3.3 \text{ k}\Omega) = \mathbf{9.9 \text{ V}}$
 $V_{23} = V_2 - V_3 = 9.9 \text{ V} - 0 \text{ V} = \mathbf{9.9 \text{ V}}$, $V_{12} = V_1 - V_2 = 20 \text{ V} - 9.9 \text{ V} = \mathbf{10.1 \text{ V}}$
 $\sum I_i = \sum I_o$
 $I_i = 4 \text{ mA} + 3 \text{ mA} + 10 \text{ mA} = \mathbf{17 \text{ mA}}$

43. a. $V_L = I_L R_L = (2 \text{ A})(28 \Omega) = 56 \text{ V}$
 $V_{\text{int}} = 60 \text{ V} - 56 \text{ V} = 4 \text{ V}$
 $R_{\text{int}} = \frac{V_{\text{int}}}{I} = \frac{4 \text{ V}}{2 \text{ A}} = \mathbf{2 \Omega}$
- b. $VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\% = \frac{60 \text{ V} - 56 \text{ V}}{56 \text{ V}} \times 100\% = \mathbf{7.14\%}$
44. a. $V_L = \frac{3.3 \Omega (12 \text{ V})}{3.3 \Omega + 43 \text{ m}\Omega} = \frac{39.6 \text{ V}}{3.343 \Omega} = \mathbf{11.85 \text{ V}}$
- b. $VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\% = \frac{12 \text{ V} - 11.85 \text{ V}}{11.85 \text{ V}} \times 100\% = \mathbf{1.27\%}$
- c. $I_s = I_L = \frac{11.85 \text{ V}}{3.3 \Omega} = 3.59 \text{ A}$
 $P_s = EI_s = (12 \text{ V})(3.59 \text{ A}) = \mathbf{43.08 \text{ W}}$
 $P_{\text{int}} = I^2 R_{\text{int}} = (3.59 \text{ A})^2 43 \Omega = \mathbf{0.554 \text{ W}}$
45. a. $I = \frac{E}{R_T} = \frac{12 \text{ V}}{2 \text{ k}\Omega + 6.8 \text{ k}\Omega} = \frac{12 \text{ V}}{8.8 \text{ k}\Omega} = \mathbf{1.36 \text{ mA}}$
- b. $I = \frac{E}{R_T} = \frac{12 \text{ V}}{8.8 \text{ k}\Omega + 0.25 \text{ k}\Omega} = \frac{12 \text{ V}}{9.05 \text{ k}\Omega} = \mathbf{1.33 \text{ mA}}$
- c. not for most applications.

Chapter 6

1.
 - a. R_2 and R_3
 - b. E and R_3
 - c. R_2 and R_3
 - d. R_2 and R_3
 - e. E , R_1 , R_2 , R_3 , and R_4
 - f. E , R_1 , R_2 , and R_3
 - g. E_2 , R_2 and R_3

2.
 - a. R_3 and R_4 , R_5 and R_6
 - b. E and R_1 , R_6 and R_7

3.
 - a. $R_T = \frac{(36\ \Omega)(18\ \Omega)}{36\ \Omega + 18\ \Omega} = \mathbf{12\ \Omega}$

- b. $R_T = \frac{1}{\frac{1}{1\ \text{k}\Omega} + \frac{1}{2\ \text{k}\Omega} + \frac{1}{30\ \text{k}\Omega}} = \frac{1}{1 \times 10^{-3}\text{S} + 0.5 \times 10^{-3}\text{S} + 33.33 \times 10^{-6}\text{S}}$
 $= \frac{1}{1.533 \times 10^{-3}\text{S}} = \mathbf{0.652\ \text{k}\Omega}$

- c. $R_T = \frac{1}{\frac{1}{1.2\ \Omega} + \frac{1}{120\ \text{k}\Omega} + \frac{1}{12\ \text{k}\Omega}} = \frac{1}{833.33 \times 10^{-6}\text{S} + 8.33 \times 10^{-3}\text{S} + 83.33 \times 10^{-3}\text{S}} = \frac{1}{92.49 \times 10^{-3}\text{S}}$
 $= \mathbf{10.81\ \Omega}$

- d. $R'_T = \frac{18\ \text{k}\Omega}{3} = 6\ \text{k}\Omega$
 $R_T = \frac{(6\ \text{k}\Omega)(6\ \text{k}\Omega)}{6\ \text{k}\Omega + 6\ \text{k}\Omega} = \mathbf{3\ \text{k}\Omega}$

- e. $R'_T = \frac{22\ \Omega}{4} = 5.5\ \Omega$, $R_{T''} = \frac{10\ \Omega}{2} = 5\ \Omega$
 $R_T = \frac{(5.5\ \Omega)(5\ \Omega)}{5.5\ \Omega + 5\ \Omega} = \mathbf{2.62\ \Omega}$

- f. $R_T = \frac{1}{\frac{1}{1\ \Omega} + \frac{1}{1\ \text{k}\Omega} + \frac{1}{1\ \text{M}\Omega}} = \frac{1}{1000 \times 10^{-3}\text{S} + 1 \times 10^{-3}\text{S} + 0.001 \times 10^{-3}\text{S}}$
 $= \frac{1}{1001.001 \times 10^{-3}\text{S}} = \mathbf{0.99\ \Omega}$

4.
 - a. $R_T = \frac{1}{\frac{1}{1\ \text{k}\Omega} + \frac{1}{1.2\ \text{k}\Omega} + \frac{1}{0.3\ \text{k}\Omega}} = \frac{1}{1 \times 10^{-3}\text{S} + 0.833 \times 10^{-3}\text{S} + 3.333 \times 10^{-3}\text{S}}$
 $= \frac{1}{5.166 \times 10^{-3}\text{S}} = \mathbf{193.57\ \Omega}$

- b.
$$R_T = \frac{1}{\frac{1}{1\text{ k}\Omega} + \frac{1}{1.2\text{ k}\Omega} + \frac{1}{2.2\text{ k}\Omega} + \frac{1}{1\text{ k}\Omega}} = \frac{1}{1 \times 10^{-3}\text{ S} + 0.833 \times 10^{-3}\text{ S} + 0.455 \times 10^{-3}\text{ S} + 1 \times 10^{-3}\text{ S}}$$
$$= \frac{1}{3.288 \times 10^{-3}\text{ S}} = \mathbf{304.14\ \Omega}$$
5. a. $R'_T = 3\ \Omega \parallel 6\ \Omega = 2\ \Omega$
 $R_T = 1.6\ \Omega = \frac{(2\ \Omega)(R)}{2\ \Omega + R}, \quad R = \mathbf{8\ \Omega}$
- b. $R'_T = \frac{6\text{ k}\Omega}{3} = 2\text{ k}\Omega$
 $R_T = 1.8\text{ k}\Omega = \frac{(2\text{ k}\Omega)(R)}{2\text{ k}\Omega + R}, \quad R = \mathbf{18\text{ k}\Omega}$
- c. $R_T = 5.08\text{ k}\Omega = \frac{(20\text{ k}\Omega)(R)}{20\text{ k}\Omega + R}, \quad R = \mathbf{6.8\text{ k}\Omega}$
- d. $R_T = 1.02\ \Omega = \frac{1}{\frac{1}{2.4\text{ k}\Omega} + \frac{1}{R} + \frac{1}{6.8\text{ k}\Omega}} = \frac{1}{416.67 \times 10^{-6}\text{ S} + \frac{1}{R} + 147.06 \times 10^{-6}\text{ S}}$

$$1.02\text{ k}\Omega = \frac{1}{563.73 \times 10^{-6} + \frac{1}{R}}$$

$$575 \times 10^{-3} + \frac{1.020\text{ k}\Omega}{R} = 1$$

$$R = \frac{1.020\text{ k}\Omega}{425 \times 10^{-3}} = \mathbf{2.4\text{ k}\Omega}$$
- e. $R_T = 6\text{ k}\Omega = \frac{R_1}{4}$
 $R_1 = \mathbf{24\text{ k}\Omega}$
6. a. $1.2\text{ k}\Omega$
b. about $1\text{ k}\Omega$
- c.
$$R_T = \frac{1}{\frac{1}{1.2\text{ k}\Omega} + \frac{1}{22\text{ k}\Omega} + \frac{1}{220\text{ k}\Omega} + \frac{1}{2.2\text{ M}\Omega}}$$
$$= \frac{1}{833.333 \times 10^{-6}\text{ S} + 45.455 \times 10^{-6}\text{ S} + 4.545 \times 10^{-6}\text{ S} + 0.455 \times 10^{-6}\text{ S}}$$
$$= \frac{1}{883.788 \times 10^{-6}\text{ S}} = \mathbf{1.131\text{ k}\Omega}$$
- d. $220\text{ k}\Omega, 2.2\text{ M}\Omega: R_T = \frac{(1.2\text{ k}\Omega)(22\text{ k}\Omega)}{1.2\text{ k}\Omega + 22\text{ k}\Omega} = \mathbf{1.138\text{ k}\Omega}$
- e. R_T reduced.

7. a. $R_T = \frac{1}{\frac{1}{4\ \Omega} + \frac{1}{2\ \Omega} + \frac{1}{10\ \Omega}} = \frac{1}{0.25\ \text{S} + 0.50\ \text{S} + 0.10\ \text{S}} = \frac{1}{0.85\ \text{S}} = \mathbf{1.18\ \Omega}$
- b. $\infty\ \Omega$
- c. $R_T = 3\ \Omega \parallel 6\ \Omega = \mathbf{2\ \Omega}$
8. $24\ \Omega \parallel 24\ \Omega = 12\ \Omega$
 $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{12\ \Omega} + \frac{1}{120\ \Omega}$
 $0.1\ \text{S} = \frac{1}{R_1} + 0.08333\ \text{S} + 0.00833\ \text{S}$
 $0.1\ \text{S} = \frac{1}{R_1} + 0.09167\ \text{S}$
 $\frac{1}{R_1} = 0.1\ \text{S} - 0.09167\ \text{S} = 0.00833\ \text{S}$
 $R_1 = \frac{1}{0.00833\ \text{S}} = \mathbf{120\ \Omega}$
9. a. $R_T = \frac{(8\ \Omega)(24\ \Omega)}{8\ \Omega + 24\ \Omega} = \mathbf{6\ \Omega}$
- b. $V_{R_1} = V_{R_2} = \mathbf{36\ \text{V}}$
- c. $I_s = \frac{E}{R_T} = \frac{36\ \text{V}}{6\ \Omega} = \mathbf{6\ \text{A}}$
 $I_1 = \frac{V_{R_1}}{R_1} = \frac{36\ \text{V}}{8\ \Omega} = \mathbf{4.5\ \text{A}}$
 $I_2 = \frac{V_{R_2}}{R_2} = \frac{36\ \text{V}}{24\ \Omega} = \mathbf{1.5\ \text{A}}$
- d. $I_s = I_1 + I_2$
 $6\ \text{A} = 4.5\ \text{A} + 1.5\ \text{A} = 6\ \text{A}\ (\text{checks})$
10. a. $I_1 = \frac{V_{R_1}}{R_1} = \frac{18\ \text{V}}{3\ \Omega} = \mathbf{6\ \text{A}}, I_2 = \frac{V_{R_2}}{R_2} = \frac{18\ \text{V}}{9\ \Omega} = \mathbf{2\ \text{A}}, I_3 = \frac{V_{R_3}}{R_3} = \frac{18\ \text{V}}{36\ \Omega} = \mathbf{0.5\ \text{A}}$
- b. $R_T = \frac{1}{\frac{1}{3\ \Omega} + \frac{1}{9\ \Omega} + \frac{1}{36\ \Omega}} = \frac{1}{0.333\ \text{S} + 0.111\ \text{S} + 0.028\ \text{S}}$
 $= \frac{1}{472 \times 10^{-3}\ \text{S}} = \mathbf{2.12\ \Omega}$
- c. $I_s = \frac{E}{R_T} = \frac{18\ \text{V}}{2.12\ \Omega} = \mathbf{8.5\ \text{A}}$

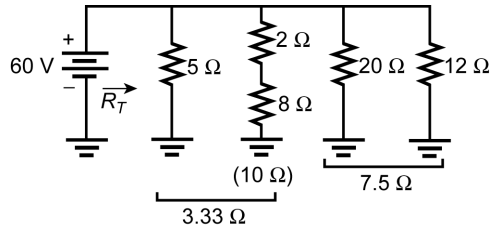
- d. $I_s = I_1 + I_2 + I_3 = 6 \text{ A} + 2 \text{ A} + 0.5 \text{ A} = \mathbf{8.5 \text{ A}}$
- e. they match
11. a. $I_{R_1} = \frac{V_{R_1}}{R_1} = \frac{24 \text{ V}}{10 \text{ k}\Omega} = \mathbf{2.4 \text{ mA}}$, $I_{R_2} = \frac{V_{R_2}}{R_2} = \frac{24 \text{ V}}{1.2 \text{ k}\Omega} = \mathbf{20 \text{ mA}}$,
 $I_{R_3} = \frac{V_{R_3}}{R_3} = \frac{24 \text{ V}}{6.8 \text{ k}\Omega} = \mathbf{3.53 \text{ mA}}$
- b. $R_T = \frac{1}{\frac{1}{10 \text{ k}\Omega} + \frac{1}{1.2 \text{ k}\Omega} + \frac{1}{6.8 \text{ k}\Omega}} = \frac{1}{100 \times 10^{-6} \text{ S} + 833.333 \times 10^{-6} \text{ S} + 147.06 \times 10^{-6} \text{ S}}$
 $= \frac{1}{1.08 \times 10^{-3} \text{ S}} = \mathbf{925.93 \Omega}$
- c. $I_s = \frac{E}{R_T} = \frac{24 \text{ V}}{925.93 \Omega} = \mathbf{25.92 \text{ mA}}$
- d. $I_s = I_1 + I_2 + I_3 = 2.4 \text{ mA} + 20 \text{ mA} + 3.53 \text{ mA} = \mathbf{25.93 \text{ mA}}$
- e. they match
12. a. $R_T \cong 900 \Omega$
- b. $R_T = \frac{1}{\frac{1}{20 \text{ k}\Omega} + \frac{1}{10 \text{ k}\Omega} + \frac{1}{1 \text{ k}\Omega} + \frac{1}{91 \text{ k}\Omega}}$
 $= \frac{1}{50 \times 10^{-6} \text{ S} + 100 \times 10^{-6} \text{ S} + 1 \times 10^{-3} \text{ S} + 10.99 \times 10^{-6} \text{ S}}$
 $= \frac{1}{1.16 \times 10^{-3} \text{ S}} = \mathbf{862.07 \Omega}$, very close
- c. I_3 the most, I_4 the least
- d. $I_{R_1} = \frac{V_{R_1}}{R_1} = \frac{60 \text{ V}}{20 \text{ k}\Omega} = \mathbf{3.0 \text{ mA}}$, $I_{R_2} = \frac{V_{R_2}}{R_2} = \frac{60 \text{ V}}{10 \text{ k}\Omega} = \mathbf{6 \text{ mA}}$
 $I_{R_3} = \frac{V_{R_3}}{R_3} = \frac{60 \text{ V}}{1 \text{ k}\Omega} = \mathbf{60.0 \text{ mA}}$, $I_{R_4} = \frac{V_{R_4}}{R_4} = \frac{60 \text{ V}}{91 \text{ k}\Omega} = \mathbf{0.659 \text{ mA}}$
- e. $I_s = \frac{E}{R_T} = \frac{60 \text{ V}}{862.07 \text{ k}\Omega} = \mathbf{69.6 \text{ mA}}$
 $I_s = 3 \text{ mA} + 6 \text{ mA} + 60 \text{ mA} + 0.659 \text{ mA} = \mathbf{69.66 \text{ mA}}$ (checks)
- f. always greater

13. a. $R_T = 6\ \Omega = \frac{(18\ \Omega)(R_2)}{18\ \Omega + R_2}$
 $108\ \Omega + 6R_2 = 18R_2$
 $12R_2 = 108\ \Omega$
 $R_2 = \frac{108\ \Omega}{12} = \mathbf{9\ \Omega}$
- b. $P = 81\ \text{W} = \frac{V^2}{R} = \frac{E^2}{R} = \frac{E^2}{9\ \Omega}$
and $E^2 = (9)(81)$
or $E = \sqrt{729} = \mathbf{27\ \text{V}}$
14. a. $P = \frac{V^2}{R} = \frac{E^2}{R}$ and $E = \sqrt{PR} = \sqrt{(100\ \text{W})(4\ \Omega)} = \sqrt{400} = \mathbf{20\ \text{V}}$
- b. $R_2 = \frac{E}{I_2} = \frac{20\ \text{V}}{2\ \text{A}} = \mathbf{10\ \Omega}$
- c. $I_1 = \frac{V_1}{R_1} = \frac{E}{R_1} = \frac{20\ \text{V}}{10\ \Omega} = \mathbf{2\ \text{A}}$
- d. $I_s = I_1 + I_2 + I_3 = 2\ \text{A} + 2\ \text{A} + \frac{20\ \text{V}}{4\ \Omega} = 4\ \text{A} + 5\ \text{A} = \mathbf{9\ \text{A}}$
- e. $P_s = EI_s = (20\ \text{V})(9\ \text{A}) = \mathbf{180\ \text{W}}$
- f. $P_{R_1} = \frac{E^2}{R_1} = \frac{(20\ \text{V})^2}{10\ \Omega} = \mathbf{40\ \text{W}}$, $P_{R_2} = \frac{E^2}{R_2} = \frac{(20\ \text{V})^2}{\left(\frac{20\ \text{V}}{2\ \text{A}}\right)} = \frac{400\ \text{W}}{10} = \mathbf{40\ \text{W}}$,
- g. $P_s = P_1 + P_2 + P_3$
 $180\ \text{W} = 40\ \text{W} + 40\ \text{W} + 100\ \text{W} = 180\ \text{W}$ (checks)
15. $I_3 = \frac{(20\ \Omega)(10.8\ \text{A})}{20\ \Omega + 4\ \Omega} = \mathbf{9\ \text{A}}$
 $E = V_{R_3} = I_3 R_3 = (9\ \text{A})(4\ \Omega) = \mathbf{36\ \text{V}}$
 $I_{R_1} = 12.3\ \text{A} - 10.8\ \text{A} = 1.5\ \text{A}$
 $R_1 = \frac{V_{R_1}}{I_{R_1}} = \frac{36\ \text{V}}{1.5\ \text{A}} = \mathbf{24\ \Omega}$
16. a. $V = \mathbf{48\ \text{V}}$
- b. $I_2 = \frac{48\ \text{V}}{18\ \text{k}\Omega} = \mathbf{2.67\ \text{mA}}$

- c. $I_s = \frac{48 \text{ V}}{3 \text{ k}\Omega} + \frac{48 \text{ V}}{12 \text{ k}\Omega} + I_2 = 16 \text{ mA} + 4 \text{ mA} + 2.67 \text{ mA} = \mathbf{22.67 \text{ mA}}$
- d. $P = \frac{V^2}{R} = \frac{E^2}{R} = \frac{(48 \text{ V})^2}{12 \text{ k}\Omega} = \mathbf{192 \text{ mW}}$
17. a. $I_{R_2} \uparrow = 4 \text{ A} - 1 \text{ A} = 3 \text{ A}, R_2 = \frac{V_{R_2}}{I_2} = \frac{E}{I_2} = \frac{12 \text{ V}}{3 \text{ A}} = \mathbf{4 \Omega}$
- b. $R_3 = \frac{V_{R_3}}{I_3} = \frac{E}{I_3} = \frac{12 \text{ V}}{1 \text{ A}} = \mathbf{12 \Omega}$
- c. $I_1 \uparrow = \frac{12 \text{ V}}{2 \Omega} = 6 \text{ A}, I_s = I_1 + 4 \text{ A} = 6 \text{ A} + 4 \text{ A} = \mathbf{10 \text{ A}}$
18. –
19. a. $R_T = \frac{1}{\frac{1}{1 \text{ k}\Omega} + \frac{1}{4.7 \text{ k}\Omega} + \frac{1}{10 \text{ k}\Omega}} = \frac{1}{1000 \times 10^{-6} \text{ S} + 212.77 \times 10^{-6} \text{ S} + 100 \times 10^{-6} \text{ S}}$
 $= \frac{1}{1.313 \times 10^{-3} \text{ S}} = \mathbf{761.61 \Omega}$
 $I_{R_1} = \frac{V_{R_1}}{R_1} = \frac{60 \text{ V}}{1 \text{ k}\Omega} = \mathbf{60 \text{ mA}}, I_{R_2} = \frac{V_{R_2}}{R_2} = \frac{60 \text{ V}}{4.7 \text{ k}\Omega} = \mathbf{12.77 \text{ mA}}$
 $I_{R_3} = \frac{V_{R_3}}{R_3} = \frac{60 \text{ V}}{10 \text{ k}\Omega} = \mathbf{6 \text{ mA}}$
- b. $P_{R_1} = V_{R_1} \cdot I_{R_1} = (60 \text{ V})(60 \text{ mA}) = \mathbf{3.6 \text{ W}}$
 $P_{R_2} = V_{R_2} \cdot I_{R_2} = (60 \text{ V})(12.77 \text{ mA}) = \mathbf{766.2 \text{ mW}}$
 $P_{R_3} = V_{R_3} \cdot I_{R_3} = (60 \text{ V})(6 \text{ mA}) = \mathbf{360 \text{ W}}$
- c. $I_s = \frac{E}{R_T} = \frac{60 \text{ V}}{761.61 \Omega} = \mathbf{78.78 \text{ mA}}$
 $P_s = E_s I_s = (60 \text{ V})(78.78 \text{ mA}) = \mathbf{4.73 \text{ W}}$
- d. $P_s = \mathbf{4.73 \text{ W}} = 3.6 \text{ W} + 766.2 \text{ mW} + 360 \text{ mW} = \mathbf{4.73 \text{ W}}$ (checks)
- e. R_1 = the smallest parallel resistor
20. a. $I_{\text{bulb}} = \frac{E}{R_{\text{bulb}}} = \frac{120 \text{ V}}{1.8 \text{ k}\Omega} = \mathbf{66.667 \text{ mA}}$
- b. $R_T = \frac{R}{N} = \frac{1.8 \text{ k}\Omega}{8} = \mathbf{225 \Omega}$
- c. $I_s = \frac{E}{R_T} = \frac{120 \text{ V}}{225 \Omega} = \mathbf{0.533 \text{ A}}$

- d. $P = \frac{V^2}{R} = \frac{(120 \text{ V})^2}{1.8 \text{ k}\Omega} = \mathbf{8 \text{ W}}$
- e. $P_s = 8(8 \text{ W}) = \mathbf{64 \text{ W}}$
- f. none, I_s drops by 66.667 mA

21. Network redrawn:



$$R_T = 3.33 \Omega \parallel 7.5 \Omega = 2.31 \Omega$$

$$P_s = \frac{E^2}{R_T} = \frac{(60 \text{ V})^2}{2.31 \Omega} = \mathbf{1.56 \text{ kW}}$$

22. a. $5 \times 60 \text{ W} = 300 \text{ W}$

$$I_{\text{bulbs}} = \frac{300 \text{ W}}{120 \text{ V}} = \mathbf{2.5 \text{ A}}$$

$$I_{\text{micro}} = \frac{1200 \text{ W}}{120 \text{ V}} = \mathbf{10 \text{ A}}$$

$$I_{\text{TV}} = \frac{320 \text{ W}}{120 \text{ V}} = \mathbf{2.67 \text{ A}}$$

$$I_{\text{DVD}} = \frac{25 \text{ W}}{120 \text{ V}} = \mathbf{208.33 \text{ mA}}$$

b. $I_s = \sum I = 2.5 \text{ A} + 10 \text{ A} + 2.67 \text{ A} + 208.33 \text{ mA} = \mathbf{15.38 \text{ A}}$
No

c. $R_T = \frac{E}{I_s} = \frac{120 \text{ V}}{15.38 \text{ A}} = \mathbf{7.8 \Omega}$

d. $P_s = E I_s = (120 \text{ V})(15.38 \text{ A}) = \mathbf{1,845.60 \text{ W}}$

e. $P_s = 1845.60 \text{ W} = 300 \text{ W} + 1200 \text{ W} + 320 \text{ W} + 25 \text{ W} = \mathbf{1845 \text{ W}}$ (checks)

23. a. $8\ \Omega \parallel 12\ \Omega = 4.8\ \Omega$, $4.8\ \Omega \parallel 4\ \Omega = 2.182\ \Omega$

$$I_1 = \frac{24\ \text{V} + 8\ \text{V}}{2.182\ \Omega} = \mathbf{14.67\ \text{A}}$$

b. $P_4 = \frac{V^2}{R} = \frac{(24\ \text{V} + 8\ \text{V})^2}{4\ \Omega} = \mathbf{256\ \text{W}}$

c. $I_2 = I_1 = \mathbf{14.67\ \text{A}}$

24. $I_s = 8\ \text{mA} + 6\ \text{mA} = \mathbf{14\ \text{mA}}$

$$I_2 = 6\ \text{mA} - 2\ \text{mA} = \mathbf{4\ \text{mA}}$$

25. a. $\sum I_i = \sum I_o$

$$2\ \text{A} + 3\ \text{A} + 9\ \text{A} = 6\ \text{A} + I$$

$$14\ \text{A} = 6\ \text{A} + I$$

$$I = 14\ \text{A} - 6\ \text{A} = \mathbf{8\ \text{A}}$$

b. $\sum I_i = \sum I_o$

$$8\ \text{mA} = 2\ \text{mA} + I_1$$

$$I_1 = 8\ \text{mA} - 2\ \text{mA} = \mathbf{6\ \text{mA}}$$

$$\sum I_i = \sum I_o$$

$$I_1 + 9\ \text{mA} = I_2$$

$$I_2 = 6\ \text{mA} + 9\ \text{mA} = \mathbf{15\ \text{mA}}$$

$$\sum I_i = \sum I_o$$

$$I_2 = 10\ \text{mA} + I_3$$

$$I_3 = 15\ \text{mA} - 10\ \text{mA} = \mathbf{5\ \text{mA}}$$

26. a. $\sum I_i = \sum I_o$

$$8\ \text{A} = 3\ \text{A} + I_2$$

$$I_2 = 8\ \text{A} - 3\ \text{A} = \mathbf{5\ \text{A}}, I_3 = \mathbf{3\ \text{A}}$$

$$\sum I_i = \sum I_o$$

$$I_2 + I_3 = I_4$$

$$I_4 = 5\ \text{A} + 3\ \text{A} = \mathbf{8\ \text{A}}$$

b. $\sum I_i = \sum I_o$

$$I_s = 36\ \text{mA} + 4\ \text{mA} = \mathbf{40\ \text{mA}}$$

$$\sum I_i = \sum I_o$$

$$36\ \text{mA} = I_3 + 20\ \text{mA}$$

$$I_3 = 36\ \text{mA} - 20\ \text{mA} = \mathbf{16\ \text{mA}}$$

$$\sum I_i = \sum I_o$$

$$4\ \text{mA} + 20\ \text{mA} = I_4$$

$$I_4 = \mathbf{24\ \text{mA}}$$

$$I_5 = I_s = \mathbf{40\ \text{mA}}$$

27. $I_{R_2} = 5 \text{ mA} - 2 \text{ mA} = 3 \text{ mA}$
 $E = V_{R_2} = (3 \text{ mA})(4 \text{ k}\Omega) = \mathbf{12 \text{ V}}$
 $R_1 = \frac{V_{R_1}}{I_{R_1}} = \frac{12 \text{ V}}{(9 \text{ mA} - 5 \text{ mA})} = \frac{12 \text{ V}}{4 \text{ mA}} = \mathbf{3 \text{ k}\Omega}$
 $R_3 = \frac{V_{R_3}}{I_{R_3}} = \frac{12 \text{ V}}{2 \text{ mA}} = \mathbf{6 \text{ k}\Omega}$
 $R_T = \frac{E}{I_T} = \frac{12 \text{ V}}{9 \text{ mA}} = \mathbf{1.33 \text{ k}\Omega}$
28. a. $R_1 = \frac{E}{I_1} = \frac{10 \text{ V}}{2 \text{ A}} = \mathbf{5 \Omega}$
 $I_2 = I - I_1 = 3 \text{ A} - 2 \text{ A} = \mathbf{1 \text{ A}}$
 $R = \frac{E}{I_2} = \frac{10 \text{ V}}{1 \text{ A}} = \mathbf{10 \Omega}$
- b. $E = I_1 R_1 = (2 \text{ A})(6 \Omega) = \mathbf{12 \text{ V}}$
 $I_2 = \frac{E}{R_2} = \frac{12 \text{ V}}{9 \Omega} = \mathbf{1.33 \text{ A}}$
 $I_3 = \frac{P}{V} = \frac{12 \text{ W}}{12 \text{ V}} = \mathbf{1 \text{ A}}$
 $R_3 = \frac{E}{I_3} = \frac{12 \text{ V}}{1 \text{ A}} = \mathbf{12 \Omega}$
 $I = I_1 + I_2 + I_3 = 2 \text{ A} + 1.33 \text{ A} + 1 \text{ A} = \mathbf{4.33 \text{ A}}$
29. a. $I_1 = \frac{64 \text{ V}}{1 \text{ k}\Omega} = \mathbf{64 \text{ mA}}$
 $I_3 = \frac{64 \text{ V}}{4 \text{ k}\Omega} = \mathbf{16 \text{ mA}}$
 $I_s = I_1 + I_2 + I_3$
 $I_2 = I_s - I_1 - I_3 = 100 \text{ mA} - 64 \text{ mA} - 16 \text{ mA} = \mathbf{20 \text{ mA}}$
 $R = \frac{E}{I_2} = \frac{64 \text{ V}}{20 \text{ mA}} = \mathbf{3.2 \text{ k}\Omega}$
 $I = I_2 + I_3 = 20 \text{ mA} + 16 \text{ mA} = \mathbf{36 \text{ mA}}$

$$\text{b. } P = \frac{V_1^2}{R_1} \Rightarrow V_1 = \sqrt{PR_1} = \sqrt{(30 \text{ W})(30 \Omega)} = \mathbf{30 \text{ V}}$$

$$E = V_1 = \mathbf{30 \text{ V}}$$

$$I_1 = \frac{E}{R_1} = \frac{30 \text{ V}}{30 \Omega} = \mathbf{1 \text{ A}}$$

$$\text{Because } R_3 = R_2, I_3 = I_2, \text{ and } I_s = I_1 + I_2 + I_3 = I_1 + 2I_2$$

$$2 \text{ A} = 1 \text{ A} + 2I_2$$

$$I_2 = \frac{1}{2}(1 \text{ A}) = 0.5 \text{ A}$$

$$I_3 = \mathbf{0.5 \text{ A}}$$

$$R_2 = R_3 = \frac{E}{I_2} = \frac{30 \text{ V}}{0.5 \text{ A}} = \mathbf{60 \Omega}$$

$$P_{R_2} = I_2^2 R_2 = (0.5 \text{ A})^2 \cdot 60 \Omega = \mathbf{15 \text{ W}}$$

$$30. \quad I_2 = \frac{6 \Omega}{12 \Omega} I_1 = \frac{1}{2} I_1 = \frac{1}{2}(9 \text{ A}) = \mathbf{4.5 \text{ A}}$$

$$I_3 = \frac{6 \Omega}{2 \Omega} I_1 = 3I_1 = 3(9 \text{ A}) = \mathbf{27 \text{ A}}$$

$$I_4 = \frac{6 \Omega}{18 \Omega} I_1 = \frac{1}{3} I_1 = \frac{1}{3}(9 \text{ A}) = 3 \text{ A}$$

$$I_T = I_1 + I_2 + I_3 + I_4 = 9 \text{ A} + 4.5 \text{ A} + 27 \text{ A} + 3 \text{ A} = \mathbf{43.5 \text{ A}}$$

$$31. \quad \text{a. } I_1 = \frac{8 \text{ k}\Omega(20 \text{ mA})}{2 \text{ k}\Omega + 8 \text{ k}\Omega} = \mathbf{16 \text{ mA}}$$

$$I_2 = 20 \text{ mA} - 16 \text{ mA} = \mathbf{4 \text{ mA}}$$

$$\text{b. } I_{2.4 \text{ k}\Omega} = 2.5 \text{ A} = \frac{1 \text{ k}\Omega(I_T)}{1 \text{ k}\Omega + 2.4 \text{ k}\Omega} = \frac{1 \text{ k}\Omega(I_T)}{3.4 \text{ k}\Omega}$$

$$\text{and } I_T = \frac{3.4 \text{ k}\Omega(2.5 \text{ A})}{1 \text{ k}\Omega} = \mathbf{8.5 \text{ A}}$$

$$I_1 = I_T - 2.5 \text{ A} = 8.5 \text{ A} - 2.5 \text{ A} = \mathbf{6 \text{ A}}$$

$$32. \quad \text{a. } R_T = \frac{1}{\frac{1}{4 \Omega} + \frac{1}{8 \Omega} + \frac{1}{12 \Omega}} = \frac{1}{250 \times 10^{-3} \text{ S} + 125 \times 10^{-3} \text{ S} + 83.333 \times 10^{-3} \text{ S}}$$

$$= \frac{1}{458.333 \times 10^{-3}} = 2.18 \Omega$$

$$I_x = \frac{R_T}{R_x} I, \quad I_1 = \frac{2.18 \Omega}{4 \Omega} (6 \text{ A}) = \mathbf{3.27 \text{ A}}$$

$$I_2 = \frac{2.18 \Omega}{8 \Omega} (6 \text{ A}) = \mathbf{1.64 \text{ A}}$$

$$I_3 = \frac{2.18 \Omega}{12 \Omega} (6 \text{ A}) = \mathbf{1.09 \text{ A}}$$

$$I_4 = \mathbf{6 \text{ A}}$$

- b. $4\ \Omega \parallel 4\ \Omega = 2\ \Omega$
 $I_2 = \frac{20\ \Omega(8\ \text{A})}{20\ \Omega + 2\ \Omega + 8\ \Omega} = \frac{20\ \Omega(8\ \text{A})}{30\ \Omega} = \mathbf{5.33\ \text{A}}$
 $I_1 = \frac{I_2}{2} = \frac{5.33\ \text{A}}{2} = \mathbf{2.67\ \text{A}}$
 $I_3 = 8\ \text{A} - I_2 = 8\ \text{A} - 5.33\ \text{A} = \mathbf{2.67\ \text{A}}$
 $I_4 = \mathbf{8\ \text{A}}$
33. a. $I_1 \cong \frac{9}{10}(10\ \text{A}) = \mathbf{9\ \text{A}}$
- b. $I_1/I_2 = 10\ \Omega/1\ \Omega = 10, \quad I_2 = \frac{I_1}{10} = \frac{9\ \text{A}}{10} \cong \mathbf{0.9\ \text{A}}$
- c. $I_1/I_3 = 1\ \text{k}\Omega/1\ \Omega = 1000, \quad I_3 = I_1/1000 = 9\ \text{A}/1000 \cong \mathbf{9\ \text{mA}}$
- d. $I_1/I_4 = 100\ \text{k}\Omega/1\ \Omega = 100,000, \quad I_4 = I_1/100,000 = 9\ \text{A}/100,000 \cong \mathbf{90\ \mu\text{A}}$
- e. very little effect, $1/100,000$
- f. $R_T = \frac{1}{\frac{1}{1\ \Omega} + \frac{1}{10\ \Omega} + \frac{1}{1\ \text{k}\Omega} + \frac{1}{100\ \text{k}\Omega}}$
 $= \frac{1}{1\ \text{S} + 0.1\ \text{S} + 1 \times 10^{-3}\ \text{S} + 10 \times 10^{-6}\ \text{S}}$
 $= \frac{1}{1.10\ \text{S}} = 0.91\ \Omega$
 $I_x = \frac{R_T}{R_x} I, \quad I_1 = \frac{0.91\ \Omega}{1\ \Omega}(10\ \text{A}) = \mathbf{9.1\ \text{A}}$ excellent (9 A)
- g. $I_2 = \frac{0.91\ \Omega}{10\ \Omega}(10\ \text{A}) = \mathbf{0.91\ \text{A}}$ excellent (0.9 A)
- h. $I_3 = \frac{0.91\ \Omega}{1\ \text{k}\Omega}(10\ \text{A}) = \mathbf{9.1\ \text{mA}}$ excellent (9 mA)
- i. $I_4 = \frac{0.91\ \Omega}{100\ \text{k}\Omega}(10\ \text{A}) = \mathbf{91\ \mu\text{A}}$ excellent (90 μA)
34. a. CDR: $I_{36\Omega} = \frac{3\ \Omega I}{3\ \Omega + 36\ \Omega} = 1\ \text{A}, \quad I = \frac{(39\ \Omega)(1\ \text{A})}{3\ \Omega} = \mathbf{13\ \text{A}} = I_2$
 $I_1 = I - 1\ \text{A} = 13\ \text{A} - 1\ \text{A} = \mathbf{12\ \text{A}}$
- b. $I_3 = I = 24\ \text{mA}, \quad V_{12\text{k}\Omega} = IR = (4\ \text{mA})(12\ \text{k}\Omega) = 48\ \text{V}$
 $I_2 = \frac{V}{R} = \frac{48\ \text{V}}{4\ \text{k}\Omega} = \mathbf{12\ \text{mA}}$
 $I_1 = I - 4\ \text{mA} - I_2$
 $= 24\ \text{mA} - 4\ \text{mA} - 12\ \text{mA}$
 $= \mathbf{8\ \text{mA}}$

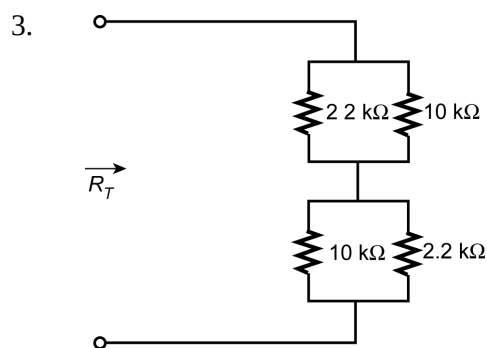
35. a. $R = 3(2 \text{ k}\Omega) = \mathbf{6 \text{ k}\Omega}$
 b. $I_1 = \frac{6 \text{ k}\Omega(32 \text{ mA})}{6 \text{ k}\Omega + 2 \text{ k}\Omega} = \mathbf{24 \text{ mA}}$
 $I_2 = \frac{I_1}{3} = \frac{24 \text{ mA}}{3} = \mathbf{8 \text{ mA}}$
36. $84 \text{ mA} = I_1 + I_2 + I_3 = I_1 + 2I_1 + 2I_2 = I_1 + 2I_1 + 2(2I_1)$
 $84 \text{ mA} = I_1 + 2I_1 + 4I_1 = 7I_1$
 and $I_1 = \frac{84 \text{ mA}}{7} = 12 \text{ mA}$
 $I_2 = 2I_1 = 2(12 \text{ mA}) = 24 \text{ mA}$
 $I_3 = 2I_2 = 2(24 \text{ mA}) = 48 \text{ mA}$
 $R_1 = \frac{V_{R_1}}{I_1} = \frac{24 \text{ V}}{12 \text{ mA}} = \mathbf{2 \text{ k}\Omega}$
 $R_2 = \frac{V_{R_2}}{I_2} = \frac{24 \text{ V}}{24 \text{ mA}} = \mathbf{1 \text{ k}\Omega}$
 $R_3 = \frac{V_{R_3}}{I_3} = \frac{24 \text{ V}}{48 \text{ mA}} = \mathbf{0.5 \text{ k}\Omega}$
37. a. $P_L = V_L I_L$
 $72 \text{ W} = 12 \text{ V} \cdot I_L$
 $I_L = \frac{72 \text{ W}}{12 \text{ V}} = 6 \text{ A}$
 $I_1 = I_2 = \frac{I_L}{2} = \frac{6 \text{ A}}{2} = \mathbf{3 \text{ A}}$
 b. $P_{\text{source}} = EI = (12 \text{ V})(3 \text{ A}) = \mathbf{36 \text{ W}}$
 c. $P_{s_1} + P_{s_2} = 36 \text{ W} + 36 \text{ W} = \mathbf{72 \text{ W}}$ (the same)
 d. $I_{\text{drain}} = \mathbf{6 \text{ A}}$ (twice as much)
38. $R_T = 8 \Omega \parallel 56 \Omega = 7 \Omega$
 $I_2 = I_3 = \frac{E}{R_T} = \frac{12 \text{ V}}{7 \Omega} = \mathbf{1.71 \text{ A}}$
 $I_1 = \frac{1}{2} I_2 = \frac{1}{2} (1.71 \text{ A}) = \mathbf{0.86 \text{ A}}$
39. $I_{8 \Omega} = \frac{16 \text{ V}}{8 \Omega} = 2 \text{ A}, \quad I = 5 \text{ A} - 2 \text{ A} = \mathbf{3 \text{ A}}$
 $I_R = 5 \text{ A} + 3 \text{ A} = 8 \text{ A}, \quad R = \frac{V_R}{I_R} = \frac{16 \text{ V}}{8 \text{ A}} = \mathbf{2 \Omega}$

40. a. $I_s = \frac{E}{R_T} = \frac{12 \text{ V}}{0.1 \text{ k}\Omega + 10 \text{ k}\Omega} = \frac{12 \text{ V}}{10.1 \text{ k}\Omega} = \mathbf{1.188 \text{ mA}}$
 $V_L = I_s R_L = (1.19 \text{ mA})(10 \text{ k}\Omega) = \mathbf{11.90 \text{ V}}$
 b. $I_s = \frac{12 \text{ V}}{100 \Omega} = \mathbf{120 \text{ mA}}$
 c. $V_L = E = \mathbf{12 \text{ V}}$
41. a. $V_L = \frac{4.7 \text{ k}\Omega(9 \text{ V})}{4.7 \text{ k}\Omega + 2.2 \text{ k}\Omega} = \frac{42.3 \text{ V}}{6.9} = \mathbf{6.13 \text{ V}}$
 b. $V_L = E = \mathbf{9 \text{ V}}$
 c. $V_L = E = \mathbf{9 \text{ V}}$
42. a. $I_1 = \frac{20 \text{ V}}{4 \Omega} = \mathbf{5 \text{ A}}, I_2 = \mathbf{0 \text{ A}}$
 b. $V_1 = \mathbf{0 \text{ V}}, V_2 = \mathbf{20 \text{ V}}$
 c. $I_s = I_1 = \mathbf{5 \text{ A}}$
43. a. $V_2 = \frac{22 \text{ k}\Omega(20 \text{ V})}{22 \text{ k}\Omega + 4.7 \text{ k}\Omega} = \mathbf{16.48 \text{ V}}$
 b. $R'_T = 11 \text{ M}\Omega \parallel 22 \text{ k}\Omega = 21.956 \text{ k}\Omega$
 $V_2 = \frac{21.956 \text{ k}\Omega(20 \text{ V})}{21.956 \text{ k}\Omega + 4.7 \text{ k}\Omega} = \mathbf{16.47 \text{ V}}$ (very close to ideal)
 c. $R_m = 20 \text{ V}[20,000 \Omega/\text{V}] = 400 \text{ k}\Omega$
 $R'_T = 400 \text{ k}\Omega \parallel 22 \text{ k}\Omega = 20.853 \text{ k}\Omega$
 $V_2 = \frac{20.853 \text{ k}\Omega(20 \text{ V})}{20.853 \text{ k}\Omega + 4.7 \text{ k}\Omega} = \mathbf{16.32 \text{ V}}$ (still very close to ideal)
 d: a. $V_2 = \frac{200 \text{ k}\Omega(20 \text{ V})}{200 \text{ k}\Omega + 100 \text{ k}\Omega} = \mathbf{13.33 \text{ V}}$
 b. $R'_T = 200 \text{ k}\Omega \parallel 11 \text{ M}\Omega = 196.429 \text{ k}\Omega$
 $V_2 = \frac{(196.429 \text{ k}\Omega)(20 \text{ V})}{196.429 \text{ k}\Omega + 100 \text{ k}\Omega} = \mathbf{13.25 \text{ V}}$ (very close to ideal)
 c. $R_m = 400 \text{ k}\Omega$
 $R'_T = 400 \text{ k}\Omega \parallel 200 \text{ k}\Omega = 133.333 \text{ k}\Omega$
 $V_2 = \frac{(133.333 \text{ k}\Omega)(20 \text{ V})}{133.333 \text{ k}\Omega + 100 \text{ k}\Omega} = \mathbf{11.43 \text{ V}}$ (a 1.824 V drop from $R_{\text{int}} = 11 \text{ M}\Omega$ level)
 e. DMM level of 11 M Ω not a problem for most situations
 VOM level of 400 k Ω can be a problem for some situations.

44. a. $V_{ab} = 20 \text{ V}$
- b. $V_{ab} = \frac{11 \text{ M}\Omega(20 \text{ V})}{11 \text{ M}\Omega + 1 \text{ M}\Omega} = 18.33 \text{ V}$
- c. $R_m = 200 \text{ V}[20,000 \text{ }\Omega/\text{V}] = 4 \text{ M}\Omega$
 $V_{ab} = \frac{4 \text{ M}\Omega(20 \text{ V})}{4 \text{ M}\Omega + 1 \text{ M}\Omega} = 16.0 \text{ V}$ (significant drop from ideal)
 $R_m = 20 \text{ V}[20,000 \text{ }\Omega/\text{V}] = 400 \text{ k}\Omega$
 $V_{ab} = \frac{400 \text{ k}\Omega(20 \text{ V})}{400 \text{ k}\Omega + 1 \text{ M}\Omega} = 5.71 \text{ V}$ (significant error)
45. not operating properly, $6 \text{ k}\Omega$ not connected at both ends
 $R_T = \frac{6 \text{ V}}{3.5 \text{ mA}} = 1.71 \text{ k}\Omega$
 $R_T = 3 \text{ k}\Omega \parallel 4 \text{ k}\Omega = 1.71 \text{ k}\Omega$
46. $V_{ab} = E + I_{4 \text{ k}\Omega} \cdot R_{4 \text{ k}\Omega}$
 $I_{4 \text{ k}\Omega} = \frac{12 \text{ V} - 4 \text{ V}}{1 \text{ k}\Omega + 4 \text{ k}\Omega} = \frac{8 \text{ V}}{5 \text{ k}\Omega} = 1.6 \text{ mA}$
 $V_{ab} = 4 \text{ V} + (1.6 \text{ mA})(4 \text{ k}\Omega) = 4 \text{ V} + 6.4 \text{ V} = 10.4 \text{ V}$
 4 V supply connected in reverse so that
 $I = \frac{12 \text{ V} + 4 \text{ V}}{1 \text{ k}\Omega + 4 \text{ k}\Omega} = \frac{16 \text{ V}}{5 \text{ k}\Omega} = 3.2 \text{ mA}$
and $V_{ab} = 12 \text{ V} - (3.2 \text{ mA})(1 \text{ k}\Omega) = 12 \text{ V} - 3.2 \text{ V} = 8.8 \text{ V}$ obtained

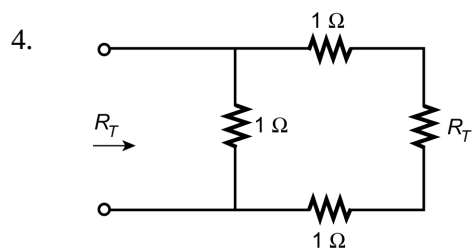
Chapter 7

1.
 - a. $R_1, R_2,$ and E are in series; R_3, R_4 and R_5 are in parallel
 - b. E and R_1 are in series; R_2, R_3 and R_4 are in parallel.
 - c. E and R_1 are in series; R_2, R_3 and R_4 are in parallel.
 - d. E_1 and R_1 are in series; E_2 and R_4 in parallel.
 - e. E and R_1 are in series; R_2 and R_3 are in parallel.
 - f. E, R_1, R_4 and R_6 are in parallel; R_2 and R_5 are in parallel.
2.
 - a. $R_T = 4\ \Omega + 10\ \Omega \parallel (4\ \Omega + 4\ \Omega) + 4\ \Omega = 4\ \Omega + 10\ \Omega \parallel 8\ \Omega + 4\ \Omega$
 $= 4\ \Omega + 4.44\ \Omega + 4\ \Omega = \mathbf{12.44\ \Omega}$
 - b. $R_T = 10\ \Omega + \frac{10\ \Omega}{2} = 10\ \Omega + 5\ \Omega = \mathbf{15\ \Omega}$
 - c. $R_T = \frac{4\ \Omega}{2} + 10\ \Omega = 2\ \Omega + 10\ \Omega = \mathbf{12\ \Omega}$
 - d. $R_T = \mathbf{10\ \Omega}$



$$2.2\ \text{k}\Omega \parallel 10\ \text{k}\Omega = 1.8\ \text{k}\Omega$$

$$R_T = 2 \times 1.8\ \text{k}\Omega = \mathbf{3.6\ \text{k}\Omega}$$



$$R_T = 1\ \Omega \parallel (1\ \Omega + 1\ \Omega + R_T) = 1\ \Omega \parallel (2\ \Omega + R_T)$$

$$= \frac{2\ \Omega + R_T}{1\ \Omega + 2\ \Omega + R_T} = \frac{2\ \Omega + R_T}{3\ \Omega + R_T}$$

$$R_T(3\ \Omega + R_T) = 2\ \Omega + R_T$$

$$3R_T + R_T^2 = 2\ \Omega + R_T$$

$$R_T^2 + 2R_T - 2\ \Omega = 0$$

$$R_T = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-2)}}{2}$$

$$= \frac{-2 \pm \sqrt{4+8}}{2} = \frac{2 \pm \sqrt{12}}{2} = \frac{-2 \pm 3.464}{2}$$

$$R_T = -1 \pm 1.732 = 0.732\ \Omega \text{ or } -2.732\ \Omega$$

Since $R_T < 1\ \Omega$ and positive choose $R_T = \mathbf{0.732\ \Omega}$

$$5. \quad R_T = 7.2 \text{ k}\Omega = R_1 \parallel \left(R_1 + \frac{R_1}{2} \right) = R_1 \parallel 1.5R_1$$

$$\text{so that } 7.2 \text{ k}\Omega = \frac{(R_1)(1.5R_1)}{R_1 + 1.5R_1} = \frac{1.5R_1^2}{2.5R_1} = \frac{1.5R_1}{2.5}$$

$$\text{and } R_1 = \frac{2.5(7.2 \text{ k}\Omega)}{1.5} \quad \mathbf{1.2 \text{ k}\Omega}$$

$$6. \quad \text{a. } \mathbf{\text{yes}}$$

$$\text{b. } I_2 = I_s - I_1 = 10 \text{ A} - 4 \text{ A} = \mathbf{6 \text{ A}}$$

$$\text{c. } \mathbf{\text{yes}}$$

$$\text{d. } V_3 = E - V_2 = 14 \text{ V} - 8 \text{ V} = \mathbf{6 \text{ V}}$$

$$\text{e. } R'_T = 4 \Omega \parallel 2 \Omega = 1.33 \Omega, \quad R''_T = 4 \Omega \parallel 6 \Omega = 2.4 \Omega$$

$$R_T = R'_T + R''_T = 1.33 \Omega + 2.4 \Omega = \mathbf{3.73 \Omega}$$

$$\text{f. } R'_T = R''_T = \frac{20 \Omega}{2} = 10 \Omega, \quad R_T = R'_T + R''_T = 10 \Omega + 10 \Omega = 20 \Omega$$

$$I_s = \frac{E}{R_T} = \frac{20 \text{ V}}{20 \Omega} = \mathbf{1 \text{ A}}$$

$$\text{g. } P_s = EI_s = P_{\text{absorbed}} = (20 \text{ V})(1 \text{ A}) = \mathbf{20 \text{ W}}$$

$$7. \quad \text{a. } R'_T = R_1 \parallel R_2 = 10 \Omega \parallel 15 \Omega = 6 \Omega$$

$$R_T = R'_T \parallel (R_3 + R_4) = 6 \Omega \parallel (10 \Omega + 2 \Omega) = 6 \Omega \parallel 12 \Omega = \mathbf{4 \Omega}$$

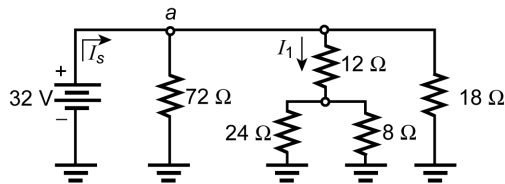
$$\text{b. } I_s = \frac{E}{R_T} = \frac{36 \text{ V}}{4 \Omega} = \mathbf{9 \text{ A}}, \quad I_1 = \frac{E}{R'_T} = \frac{36 \text{ V}}{6 \Omega} = \mathbf{6 \text{ A}}$$

$$I_2 = \frac{E}{R_3 + R_4} = \frac{36 \text{ V}}{10 \Omega + 2 \Omega} = \frac{36 \text{ V}}{12 \Omega} = \mathbf{3 \text{ A}}$$

$$I_1 = I_s - I_2 = 6 \text{ A} - 3 \text{ A} = \mathbf{3 \text{ A}}$$

$$\text{c. } V_a = I_2 R_4 = (3 \text{ A})(2 \Omega) = \mathbf{6 \text{ V}}$$

8. Redrawn:



$$\text{a. } V_a = \mathbf{32 \text{ V}}$$

$$8 \Omega \parallel 24 \Omega = 6 \Omega$$

$$V_b = \frac{6 \Omega (32 \text{ V})}{6 \Omega + 12 \Omega} = \mathbf{10.67 \text{ V}}$$

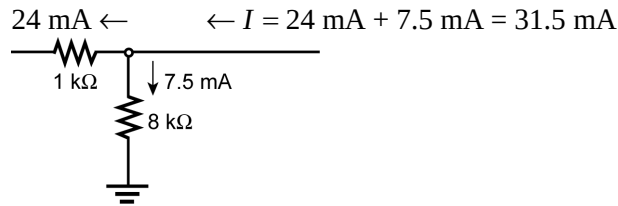
$$\text{b. } I_1 = \frac{32 \text{ V}}{12 \Omega + 6 \Omega} = \frac{32 \text{ V}}{18 \Omega} = \mathbf{1.78 \text{ A}}$$

$$R_T = 72 \Omega \parallel \underbrace{18 \Omega \parallel 18 \Omega}_{9 \Omega} = 8.12 \Omega$$

$$I_s = \frac{E}{R_T} = \frac{32 \text{ V}}{8.12 \Omega} = \mathbf{3.94 \text{ A}}$$

9. a. $V_a = \mathbf{36 \text{ V}}$, $V_b = \mathbf{60 \text{ V}}$ $V_c = \frac{5 \text{ k}\Omega(60 \text{ V})}{5 \text{ k}\Omega + 10 \text{ k}\Omega} = \mathbf{20 \text{ V}}$

b. $I_1 = \frac{60 \text{ V} - 36 \text{ V}}{1 \text{ k}\Omega} = \mathbf{24 \text{ mA}}$,
 $I_{8\text{k}\Omega} = \frac{60 \text{ V}}{8 \text{ k}\Omega} = 7.5 \text{ mA}$, $I_{10\text{k}\Omega} = \frac{60 \text{ V}}{15 \text{ k}\Omega} = 4 \text{ mA}$



$$\leftarrow I_2 = 31.5 \text{ mA} + 4 \text{ mA} = \mathbf{35.5 \text{ mA}}$$

10. a. $R'_T = 1.2 \text{ k}\Omega + 6.8 \text{ k}\Omega = 8 \text{ k}\Omega$, $R''_T = 2 \text{ k}\Omega \parallel R'_T = 2 \text{ k}\Omega \parallel 8 \text{ k}\Omega = 1.6 \text{ k}\Omega$
 $R'''_T = R''_T + 2.4 \text{ k}\Omega = 1.6 \text{ k}\Omega + 2.4 \text{ k}\Omega = 4 \text{ k}\Omega$
 $R_T = 1 \text{ k}\Omega \parallel R'''_T = 1 \text{ k}\Omega \parallel 4 \text{ k}\Omega = \mathbf{0.8 \text{ k}\Omega}$

b. $I_s = \frac{E}{R_T} = \frac{48 \text{ V}}{0.8 \text{ k}\Omega} = \mathbf{60 \text{ mA}}$

c. $V = \frac{R''_T E}{R''_T + 2.4 \text{ k}\Omega} = \frac{(1.6 \text{ k}\Omega)(48 \text{ V})}{1.6 \text{ k}\Omega + 2.4 \text{ k}\Omega} = \mathbf{19.2 \text{ V}}$

11. $R_T = 2 R \parallel 2 R \parallel (R + R) = 2 R \parallel 2 R \parallel 2 R = \frac{2 R}{3}$

$$R_T = \frac{E}{I} = \frac{120 \text{ V}}{8 \text{ A}} = 15 \Omega$$

$$15 \Omega = \frac{2 R}{3} \text{ and } R = \frac{3}{2}(15 \Omega) = \mathbf{22.5 \Omega}$$

$$2 R = \mathbf{45 \Omega}$$

12. a. $R_T = (R_1 \parallel R_2 \parallel R_3) \parallel (R_6 + R_4 \parallel R_5)$
 $= (12 \text{ k}\Omega \parallel 12 \text{ k}\Omega \parallel 3 \text{ k}\Omega) \parallel (10.4 \text{ k}\Omega + 9 \text{ k}\Omega \parallel 6 \text{ k}\Omega)$
 $= (6 \text{ k}\Omega \parallel 3 \text{ k}\Omega) \parallel (10.4 \text{ k}\Omega + 3.6 \text{ k}\Omega)$
 $= 2 \text{ k}\Omega \parallel 14 \text{ k}\Omega = 1.75 \text{ k}\Omega$
 $I_s = \frac{E}{R_T} = \frac{28 \text{ V}}{1.75 \text{ k}\Omega} = \mathbf{16 \text{ mA}}$, $I_2 = \frac{E}{R_2} = \frac{28 \text{ V}}{12 \text{ k}\Omega} = \mathbf{2.33 \text{ mA}}$
 $R' = R_1 \parallel R_2 \parallel R_3 = 2 \text{ k}\Omega$
 $R'' = R_6 + R_4 \parallel R_5 = 14 \text{ k}\Omega$

$$I_6 = \frac{R'(I_s)}{R' + R''} = \frac{2 \text{ k}\Omega(16 \text{ mA})}{2 \text{ k}\Omega + 14 \text{ k}\Omega} = \mathbf{2 \text{ mA}}$$

- b. $V_1 = E = \mathbf{28\text{ V}}$
 $R' = R_4 \parallel R_5 = 6\text{ k}\Omega \parallel 9\text{ k}\Omega = 3.6\text{ k}\Omega$
 $V_5 = I_6 R' = (2\text{ mA})(3.6\text{ k}\Omega) = \mathbf{7.2\text{ V}}$
- c. $P = \frac{V_{R_3}^2}{R_3} = \frac{(28\text{ V})^2}{3\text{ k}\Omega} = \mathbf{261.33\text{ mW}}$
13. a. $I_1 \downarrow = \frac{24\text{ V}}{4\text{ }\Omega} = \mathbf{6\text{ A}}$; $V_{R_2} = 24\text{ V} - 8\text{ V} = 16\text{ V}$, $I_2 \downarrow = V_{R_2} / R_2 = 16\text{ V} / 2\text{ }\Omega = \mathbf{8\text{ A}}$
 $I_3 \downarrow = \frac{8\text{ V}}{10\text{ }\Omega} = \mathbf{0.8\text{ A}}$, $I = I_1 + I_2 = 6\text{ A} + 8\text{ A} = \mathbf{14\text{ A}}$
14. $I_1 = \frac{20\text{ V}}{47\text{ }\Omega} = \mathbf{425.5\text{ mA}}$
 $I_2 = \frac{14\text{ V}}{160\text{ }\Omega \parallel 270\text{ }\Omega} = \frac{14\text{ V}}{100.47\text{ }\Omega} = \mathbf{139.35\text{ mA}}$
15. a. $R' = R_4 + R_5 = 14\text{ }\Omega + 6\text{ }\Omega = 20\text{ }\Omega$
 $R'' = R_2 \parallel R' = 20\text{ }\Omega \parallel 20\text{ }\Omega = 10\text{ }\Omega$
 $R''' = R'' + R_1 = 10\text{ }\Omega + 10\text{ }\Omega = 20\text{ }\Omega$
 $R_T = R_3 \parallel R''' = 5\text{ }\Omega \parallel 20\text{ }\Omega = \mathbf{4\text{ }\Omega}$
 $I_s = \frac{E}{R_T} = \frac{20\text{ V}}{4\text{ }\Omega} = \mathbf{5\text{ A}}$
 $I_1 = \frac{20\text{ V}}{R_1 + R''} = \frac{20\text{ V}}{10\text{ }\Omega + 10\text{ }\Omega} = \frac{20\text{ V}}{20\text{ }\Omega} = \mathbf{1\text{ A}}$
 $I_3 = \frac{20\text{ V}}{5\text{ }\Omega} = \mathbf{4\text{ A}}$
 $I_4 = \frac{I_1}{2} = (\text{since } R' = R_2) = \frac{1\text{ A}}{2} = \mathbf{0.5\text{ A}}$
- b. $V_a = I_3 R_3 - I_4 R_5 = (4\text{ A})(5\text{ }\Omega) - (0.5\text{ A})(6\text{ }\Omega) = 20\text{ V} - 3\text{ V} = \mathbf{17\text{ V}}$
 $V_{bc} = \left(\frac{I_1}{2}\right) R_2 = (0.5\text{ A})(20\text{ }\Omega) = \mathbf{10\text{ V}}$
16. a. $I_1 = \frac{E}{R_1 + R_4 \parallel (R_2 + R_3 \parallel R_5)} = \frac{20\text{ V}}{3\text{ }\Omega + 3\text{ }\Omega \parallel (3\text{ }\Omega + 6\text{ }\Omega \parallel 6\text{ }\Omega)}$
 $= \frac{20\text{ V}}{3\text{ }\Omega + 3\text{ }\Omega \parallel (3\text{ }\Omega + 3\text{ }\Omega)} = \frac{20\text{ V}}{3\text{ }\Omega + 3\text{ }\Omega \parallel 6\text{ }\Omega} = \frac{20\text{ V}}{3\text{ }\Omega + 2\text{ }\Omega}$
 $= \mathbf{4\text{ A}}$

$$\begin{aligned} \text{b. CDR: } I_2 &= \frac{R_4(I_1)}{R_4 + R_2 + R_3 \parallel R_5} = \frac{3 \, \Omega(4 \, \text{A})}{3 \, \Omega + 3 \, \Omega + 6 \, \Omega \parallel 6 \, \Omega} \\ &= \frac{12 \, \text{A}}{6 + 3} = \mathbf{1.33 \, \text{A}} \\ I_3 &= \frac{I_2}{2} = \mathbf{0.67 \, \text{A}} \end{aligned}$$

$$\begin{aligned} \text{c. } I_4 &= I_1 - I_2 = 4 \, \text{A} - 1.33 \, \text{A} = 2.67 \, \text{A} \\ V_a &= I_4 R_4 = (2.67 \, \text{A})(3 \, \Omega) = \mathbf{8 \, \text{V}} \\ V_b &= I_3 R_3 = (0.67 \, \text{A})(6 \, \Omega) = \mathbf{4 \, \text{V}} \end{aligned}$$

$$\begin{aligned} 17. \quad \text{a. } I_E &= \frac{V_E}{R_E} = \frac{2 \, \text{V}}{1 \, \text{k}\Omega} = \mathbf{2 \, \text{mA}} \\ I_C &= I_E = \mathbf{2 \, \text{mA}} \end{aligned}$$

$$\begin{aligned} \text{b. } I_B &= \frac{V_{R_B}}{R_B} = \frac{V_{CC} - (V_{BE} + V_E)}{R_B} = \frac{8 \, \text{V} - (0.7 \, \text{V} + 2 \, \text{V})}{220 \, \text{k}\Omega} \\ &= \frac{8 \, \text{V} - 2.7 \, \text{V}}{220 \, \text{k}\Omega} = \frac{5.3 \, \text{V}}{220 \, \text{k}\Omega} = \mathbf{24 \, \mu\text{A}} \end{aligned}$$

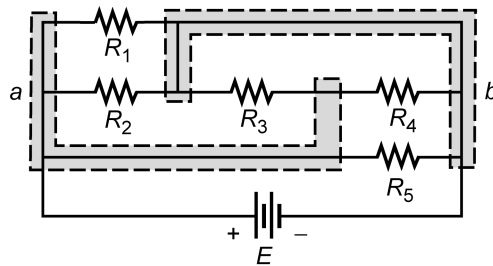
$$\begin{aligned} \text{c. } V_B &= V_{BE} + V_E = \mathbf{2.7 \, \text{V}} \\ V_C &= V_{CC} - I_C R_C = 8 \, \text{V} - (2 \, \text{mA})(2.2 \, \text{k}\Omega) = 8 \, \text{V} - 4.4 \, \text{V} = \mathbf{3.6 \, \text{V}} \end{aligned}$$

$$\begin{aligned} \text{d. } V_{CE} &= V_C - V_E = 3.6 \, \text{V} - 2 \, \text{V} = \mathbf{1.6 \, \text{V}} \\ V_{BC} &= V_B - V_C = 2.7 \, \text{V} - 3.6 \, \text{V} = \mathbf{-0.9 \, \text{V}} \end{aligned}$$

$$18. \quad \text{a. } I = \frac{22 \, \text{V}}{4 \, \Omega + 18 \, \Omega} = \frac{22 \, \text{V}}{22 \, \Omega} = \mathbf{1 \, \text{A}}$$

$$\text{b. } -22 \, \text{V} + V_i - 22 \, \text{V} = 0, V_i = \mathbf{44 \, \text{V}}$$

$$19. \quad \text{a. All resistors in parallel (between terminals a & b)}$$



$$\begin{aligned} R_T &= 16 \, \Omega \parallel 16 \, \Omega \parallel 8 \, \Omega \parallel 4 \, \Omega \parallel 32 \, \Omega \\ &\quad \underbrace{\hspace{1.5cm}}_{8 \, \Omega \parallel 8 \, \Omega \parallel 4 \, \Omega \parallel 32 \, \Omega} \\ &\quad \underbrace{\hspace{1.5cm}}_{4 \, \Omega \parallel 4 \, \Omega \parallel 32 \, \Omega} \\ &\quad \underbrace{\hspace{1.5cm}}_{2 \, \Omega \parallel 32 \, \Omega} = \mathbf{1.88 \, \Omega} \end{aligned}$$

b. All in parallel. Therefore, $V_1 = V_4 = E = \mathbf{32\text{ V}}$

c. $I_3 = V_3/R_3 = 32\text{ V}/4\ \Omega = \mathbf{8\text{ A}} \leftarrow$

d.
$$\begin{aligned} I_s &= I_1 + I_2 + I_3 + I_4 + I_5 \\ &= \frac{32\text{ V}}{16\ \Omega} + \frac{32\text{ V}}{8\ \Omega} + \frac{32\text{ V}}{4\ \Omega} + \frac{32\text{ V}}{32\ \Omega} + \frac{32\text{ V}}{16\ \Omega} \\ &= 2\text{ A} + 4\text{ A} + 8\text{ A} + 1\text{ A} + 2\text{ A} \\ &= 17\text{ A} \\ R_T &= \frac{E}{I_s} = \frac{32\text{ V}}{17\text{ A}} = \mathbf{1.88\ \Omega} \text{ as above} \end{aligned}$$

20. $I = \frac{12\text{ V}}{10\text{ k}\Omega} = \mathbf{1.2\text{ mA}}$

$$V_{ab} = V_a - V_b = 12\text{ V} - (-18\text{ V}) = \mathbf{30\text{ V}}$$

21. a. $V_a = -6\text{ V}$, $V_b = -20\text{ V}$
 $V_{ab} = V_a - V_b = (-6\text{ V}) - (-20\text{ V}) = -6\text{ V} + 20\text{ V} = \mathbf{+14\text{ V}}$

b. $I_{5\Omega} \downarrow = \frac{20\text{ V}}{5\ \Omega} = 4\text{ A}$
 $I_{2\Omega} \rightarrow = \frac{V_{ab}}{2\ \Omega} = \frac{14\text{ V}}{2\ \Omega} = 7\text{ A}$
 $I_{3\Omega} \uparrow = \frac{6\text{ V}}{3\ \Omega} = 2\text{ A}$
 $I_{3\Omega} = I_{2\Omega} + I_{6V} \uparrow, I_{6V} = I_{3\Omega} - I_{2\Omega} = 2\text{ A} - 7\text{ A} = -5\text{ A}$
 $I + I_{6V} = I_{5\Omega}, I = I_{5\Omega} - I_{6V} = 4\text{ A} - (-5\text{ A}) = \mathbf{9\text{ A}}$

22. a. Applying Kirchhoff's voltage law in the CCW direction in the upper "window":

$$\begin{aligned} +18\text{ V} + 20\text{ V} - V_{8\Omega} &= 0 \\ V_{8\Omega} &= 38\text{ V} \\ I_{8\Omega} &= \frac{38\text{ V}}{8\ \Omega} = 4.75\text{ A} \\ I_{3\Omega} &= \frac{18\text{ V}}{3\ \Omega + 6\ \Omega} = \frac{18\text{ V}}{9\ \Omega} = 2\text{ A} \end{aligned}$$

$$\text{KCL: } I_{18V} = 4.75\text{ A} + 2\text{ A} = \mathbf{6.75\text{ A}}$$

b. $V = (I_{3\Omega})(6\ \Omega) + 20\text{ V} = (2\text{ A})(6\ \Omega) + 20\text{ V} = 12\text{ V} + 20\text{ V} = \mathbf{32\text{ V}}$

$$23. \quad I_2 R_2 = I_3 R_3 \text{ and } I_2 = \frac{I_3 R_3}{R_2} = \frac{2R_3}{20} = \frac{R_3}{10} \text{ (since the voltage across parallel elements is the same)}$$

$$I_1 = I_2 + I_3 = \frac{R_3}{10} + 2$$

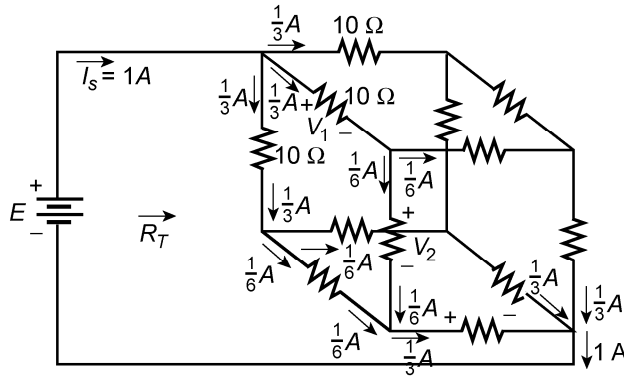
$$\text{KVL: } 120 = I_1 12 + I_3 R_3 = \left(\frac{R_3}{10} + 2 \right) 12 + 2R_3$$

$$\text{and } 120 = 1.2R_3 + 24 + 2R_3$$

$$3.2R_3 = 96 \, \Omega$$

$$R_3 = \frac{96 \, \Omega}{3.2} = 30 \, \Omega$$

24. Assuming $I_s = 1 \text{ A}$, the current I_s will divide as determined by the load appearing in each branch. Since balanced I_s will split equally between all three branches.



$$V_1 = \left(\frac{1}{3} \text{ A} \right) (10 \, \Omega) = \frac{10}{3} \text{ V}$$

$$V_2 = \left(\frac{1}{6} \text{ A} \right) (10 \, \Omega) = \frac{10}{6} \text{ V}$$

$$V_3 = \left(\frac{1}{3} \text{ A} \right) (10 \, \Omega) = \frac{10}{3} \text{ V}$$

$$E = V_1 + V_2 + V_3 = \frac{10}{3} \text{ V} + \frac{10}{6} \text{ V} + \frac{10}{3} \text{ V} = 8.33 \text{ V}$$

$$R_T = \frac{E}{I} = \frac{8.33 \text{ V}}{1 \text{ A}} = 8.33 \, \Omega$$

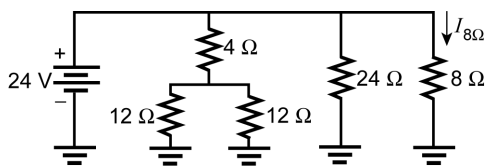
$$25. \quad 36 \, \text{k}\Omega \parallel 6 \, \text{k}\Omega \parallel 12 \, \text{k}\Omega = 3.6 \, \text{k}\Omega$$

$$V = \frac{3.6 \, \text{k}\Omega (45 \text{ V})}{3.6 \, \text{k}\Omega + 6 \, \text{k}\Omega} = 16.88 \text{ V} \neq 27 \text{ V. Therefore, **not** operating properly!}$$

6 kΩ resistor "open"

$$R' = 12 \, \text{k}\Omega \parallel 36 \, \text{k}\Omega = 9 \, \text{k}\Omega, \quad V = \frac{R' (45 \text{ V})}{R' + 6 \, \text{k}\Omega} = \frac{9 \, \text{k}\Omega (45 \text{ V})}{9 \, \text{k}\Omega + 6 \, \text{k}\Omega} = 27 \text{ V}$$

26. a. $R'_T = R_5 \parallel (R_6 + R_7) = 6 \, \Omega \parallel 3 \, \Omega = 2 \, \Omega$
 $R''_T = R_3 \parallel (R_4 + R'_T) = 4 \, \Omega \parallel (2 \, \Omega + 2 \, \Omega) = 2 \, \Omega$
 $R_T = R_1 + R_2 + R''_T = 3 \, \Omega + 5 \, \Omega + 2 \, \Omega = 10 \, \Omega$
 $I = \frac{240 \, \text{V}}{10 \, \Omega} = \mathbf{24 \, A}$
- b. $I_4 = \frac{4 \, \Omega(I)}{4 \, \Omega + 4 \, \Omega} = \frac{4 \, \Omega(24 \, \text{A})}{8 \, \Omega} = 12 \, \text{A}$
 $I_7 = \frac{6 \, \Omega(12 \, \text{A})}{6 \, \Omega + 3 \, \Omega} = \frac{72 \, \text{A}}{9} = \mathbf{8 \, A}$
- c. $V_3 = I_3 R_3 = (I - I_4) R_3 = (24 \, \text{A} - 12 \, \text{A}) 4 \, \Omega = \mathbf{48 \, V}$
 $V_5 = I_5 R_5 = (I_4 - I_7) R_5 = (4 \, \text{A}) 6 \, \Omega = \mathbf{24 \, V}$
 $V_7 = I_7 R_7 = (8 \, \text{A}) 2 \, \Omega = \mathbf{16 \, V}$
- d. $P = I_7^2 R_7 = (8 \, \text{A})^2 2 \, \Omega = \mathbf{128 \, W}$
 $P = EI = (240 \, \text{V})(24 \, \text{A}) = \mathbf{5760 \, W}$
27. a. $R'_T = R_4 \parallel (R_6 + R_7 + R_8) = 2 \, \Omega \parallel 7 \, \Omega = 1.56 \, \Omega$
 $R''_T = R_2 \parallel (R_3 + R_5 + R'_T) = 2 \, \Omega \parallel (4 \, \Omega + 1 \, \Omega + 1.56 \, \Omega) = 1.53 \, \Omega$
 $R_T = R_1 + R''_T = 4 \, \Omega + 1.53 \, \Omega = \mathbf{5.53 \, \Omega}$
- b. $I = 40 \, \text{V} / 5.53 \, \Omega = \mathbf{7.23 \, A}$
- c. $I_3 = \frac{2 \, \Omega(I)}{2 \, \Omega + 6.56 \, \Omega} = \frac{2 \, \Omega(7.23 \, \text{A})}{2 \, \Omega + 6.56 \, \Omega} = 1.69 \, \text{A}$
 $I_7 = \frac{2 \, \Omega(1.69 \, \text{A})}{2 \, \Omega + 7 \, \Omega} = 0.375 \, \text{mA}$
 $P_{R_7} = I^2 R = (0.375 \, \text{A})^2 2 \, \Omega = \mathbf{0.281 \, W}$
28. Network redrawn:



$$I_{8\Omega} = I_{6\Omega} = \frac{24 \, \text{V}}{8 \, \Omega} = 3 \, \text{A}$$

$$P_{6\Omega} = I^2 R = (3 \, \text{A})^2 \cdot 6 \, \Omega = \mathbf{54 \, W}$$

29. a. $R_{10} + R_{11} \parallel R_{12} = 1 \, \Omega + 2 \, \Omega \parallel 2 \, \Omega = 2 \, \Omega$
 $R_4 \parallel (R_5 + R_6) = 10 \, \Omega \parallel 10 \, \Omega = 5 \, \Omega$
 $R_1 + R_2 \parallel (R_3 + 5 \, \Omega) = 3 \, \Omega + 6 \, \Omega \parallel 6 \, \Omega = 6 \, \Omega$
 $R_T = 2 \, \Omega \parallel 3 \, \Omega \parallel 6 \, \Omega = 2 \, \Omega \parallel 2 \, \Omega = 1 \, \Omega$
 $I = 12 \, \text{V} / 1 \, \Omega = \mathbf{12 \, A}$
- b. $I_1 = 12 \, \text{V} / 6 \, \Omega = 2 \, \text{A}$
 $I_3 = \frac{6 \, \Omega (2 \, \text{A})}{6 \, \Omega + 6 \, \Omega} = 1 \, \text{A}$
 $I_4 = \frac{1 \, \text{A}}{2} = \mathbf{0.5 \, A}$
- c. $I_6 = I_4 = \mathbf{0.5 \, A}$
- d. $I_{10} = \frac{12 \, \text{A}}{2} = \mathbf{6 \, A}$
30. a. $E = (40 \, \text{mA})(1.6 \, \text{k}\Omega) = \mathbf{64 \, V}$
- b. $R_{L_2} = \frac{48 \, \text{V}}{12 \, \text{mA}} = \mathbf{4 \, \text{k}\Omega}$
 $R_{L_3} = \frac{24 \, \text{V}}{8 \, \text{mA}} = \mathbf{3 \, \text{k}\Omega}$
- c. $I_{R_1} = 72 \, \text{mA} - 40 \, \text{mA} = 32 \, \text{mA}$
 $I_{R_2} = 32 \, \text{mA} - 12 \, \text{mA} = 20 \, \text{mA}$
 $I_{R_3} = 20 \, \text{mA} - 8 \, \text{mA} = 12 \, \text{mA}$
 $R_1 = \frac{V_{R_1}}{I_{R_1}} = \frac{64 \, \text{V} - 48 \, \text{V}}{32 \, \text{mA}} = \frac{16 \, \text{V}}{32 \, \text{mA}} = \mathbf{0.5 \, \text{k}\Omega}$
 $R_2 = \frac{V_{R_2}}{I_{R_2}} = \frac{48 \, \text{V} - 24 \, \text{V}}{20 \, \text{mA}} = \frac{24 \, \text{V}}{20 \, \text{mA}} = \mathbf{1.2 \, \text{k}\Omega}$
 $R_3 = \frac{V_{R_3}}{I_{R_3}} = \frac{24 \, \text{V}}{12 \, \text{mA}} = \mathbf{2 \, \text{k}\Omega}$
31. $I_{R_1} = 40 \, \text{mA}$
 $I_{R_2} = 40 \, \text{mA} - 10 \, \text{mA} = 30 \, \text{mA}$
 $I_{R_3} = 30 \, \text{mA} - 20 \, \text{mA} = 10 \, \text{mA}$
 $I_{R_5} = 40 \, \text{mA}$
 $I_{R_4} = 40 \, \text{mA} - 4 \, \text{mA} = 36 \, \text{mA}$
 $R_1 = \frac{V_{R_1}}{I_{R_1}} = \frac{120 \, \text{V} - 100 \, \text{V}}{40 \, \text{mA}} = \frac{20 \, \text{V}}{40 \, \text{mA}} = \mathbf{0.5 \, \text{k}\Omega}$
 $R_2 = \frac{V_{R_2}}{I_{R_2}} = \frac{100 \, \text{V} - 40 \, \text{V}}{30 \, \text{mA}} = \frac{60 \, \text{V}}{30 \, \text{mA}} = \mathbf{2 \, \text{k}\Omega}$

$$R_3 = \frac{V_{R_3}}{I_{R_3}} = \frac{40 \text{ V}}{10 \text{ mA}} = \mathbf{4 \text{ k}\Omega}$$

$$R_4 = \frac{V_{R_4}}{I_{R_4}} = \frac{36 \text{ V}}{36 \text{ mA}} = \mathbf{1 \text{ k}\Omega}$$

$$R_5 = \frac{V_{R_5}}{I_{R_5}} = \frac{60 \text{ V} - 36 \text{ V}}{40 \text{ mA}} = \frac{24 \text{ V}}{40 \text{ mA}} = \mathbf{0.6 \text{ k}\Omega}$$

$$P_1 = I_1^2 R_1 = (40 \text{ mA})^2 0.5 \text{ k}\Omega = \mathbf{0.8 \text{ W}} \text{ (1 watt resistor)}$$

$$P_2 = I_2^2 R_2 = (30 \text{ mA})^2 2 \text{ k}\Omega = \mathbf{1.8 \text{ W}} \text{ (2 watt resistor)}$$

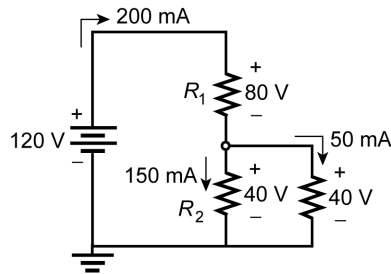
$$P_3 = I_3^2 R_3 = (10 \text{ mA})^2 4 \text{ k}\Omega = \mathbf{0.4 \text{ W}} \text{ (1/2 watt or 1 watt resistor)}$$

$$P_4 = I_4^2 R_4 = (36 \text{ mA})^2 1 \text{ k}\Omega = \mathbf{1.3 \text{ W}} \text{ (2 watt resistor)}$$

$$P_5 = I_5^2 R_5 = (40 \text{ mA})^2 0.6 \text{ k}\Omega = \mathbf{0.96 \text{ W}} \text{ (1 watt resistor)}$$

All power levels less than **2 W**. Four less than **1 W**.

32.



$$R_1 = \frac{80 \text{ V}}{200 \text{ mA}} = 400 \Omega \Rightarrow \mathbf{390 \Omega}$$

$$R_2 = \frac{40 \text{ V}}{150 \text{ mA}} = 266.67 \Omega \Rightarrow \mathbf{270 \Omega}$$

33.

a. **yes**, $R_L \gg R_{\max}$ (potentiometer)

$$\text{b. VDR: } V_{R_2} = 3 \text{ V} = \frac{R_2(12 \text{ V})}{R_1 + R_2} = \frac{R_2(12 \text{ V})}{1 \text{ k}\Omega}$$

$$R_2 = \frac{3 \text{ V}(1 \text{ k}\Omega)}{12 \text{ V}} = 0.25 \text{ k}\Omega = \mathbf{250 \Omega}$$

$$R_1 = 1 \text{ k}\Omega - 0.25 \text{ k}\Omega = 0.75 \text{ k}\Omega = \mathbf{750 \Omega}$$

c. $V_{R_1} = E - V_L = 12 \text{ V} - 3 \text{ V} = 9 \text{ V}$ (Chose V_{R_1} rather than $V_{R_2 \parallel R_L}$ since numerator of VDR

$$V_{R_1} = 9 \text{ V} = \frac{R_1(12 \text{ V})}{R_1 + (R_2 \parallel R_L)} \quad \text{equation "cleaner"}$$

$$9R_1 + 9(R_2 \parallel R_L) = 12R_1$$

$$\left. \begin{aligned} R_1 &= 3(R_2 \parallel R_L) \\ R_1 + R_2 &= 1 \text{ k}\Omega \end{aligned} \right\} \text{2 eq. 2 unk (} R_L = 10 \text{ k}\Omega \text{)}$$

$$R_1 = \frac{3R_2 R_L}{R_2 + R_L} \Rightarrow \frac{3R_2 \cdot 10 \text{ k}\Omega}{R_2 + 10 \text{ k}\Omega}$$

$$\text{and } R_1(R_2 + 10 \text{ k}\Omega) = 30 \text{ k}\Omega R_2$$

$$\begin{aligned}
 R_1 R_2 + 10 \text{ k}\Omega R_1 &= 30 \text{ k}\Omega R_2 \\
 R_1 + R_2 &= 1 \text{ k}\Omega: (1 \text{ k}\Omega - R_2)R_2 + 10 \text{ k}\Omega (1 \text{ k}\Omega - R_2) = 30 \text{ k}\Omega R_2 \\
 R_2^2 + 39 \text{ k}\Omega R_2 - 10 \text{ k}\Omega^2 &= 0 \\
 R_2 &= 0.255 \text{ k}\Omega, -39.255 \text{ k}\Omega \\
 R_2 &= \mathbf{255 \Omega} \\
 R_1 &= 1 \text{ k}\Omega - R_2 = \mathbf{745 \Omega}
 \end{aligned}$$

34. a. $V_{ab} = \frac{80 \Omega (40 \text{ V})}{100 \Omega} = \mathbf{32 \text{ V}}$
 $V_{bc} = 40 \text{ V} - 32 \text{ V} = \mathbf{8 \text{ V}}$
- b. $80 \Omega \parallel 1 \text{ k}\Omega = 74.07 \Omega$
 $20 \Omega \parallel 10 \text{ k}\Omega = 19.96 \Omega$
 $V_{ab} = \frac{74.07 \Omega (40 \text{ V})}{74.07 \Omega + 19.96 \Omega} = \mathbf{31.51 \text{ V}}$
 $V_{bc} = 40 \text{ V} - 31.51 \text{ V} = \mathbf{8.49 \text{ V}}$
- c. $P = \frac{(31.51 \text{ V})^2}{80 \Omega} + \frac{(8.49 \text{ V})^2}{20 \Omega} = 12.411 \text{ W} + 3.604 \text{ W} = \mathbf{16.02 \text{ W}}$
- d. $P = \frac{(32 \text{ V})^2}{80 \Omega} + \frac{(8 \text{ V})^2}{20 \Omega} = 12.8 \text{ W} + 3.2 \text{ W} = \mathbf{16 \text{ W}}$
The applied loads dissipate less than 20 mW of power.
35. a. $I_{CS} = \mathbf{1 \text{ mA}}$
- b. $R_{\text{shunt}} = \frac{R_m I_{CS}}{I_{\text{max}} - I_{CS}} = \frac{(100 \Omega)(1 \text{ mA})}{20 \text{ A} - 1 \text{ mA}} \cong \frac{0.1}{20} \Omega = \mathbf{5 \text{ m}\Omega}$
36. 25 mA: $R_{\text{shunt}} = \frac{(1 \text{ k}\Omega)(50 \mu\text{A})}{25 \text{ mA} - 0.05 \text{ mA}} \cong \mathbf{2 \Omega}$
50 mA: $R_{\text{shunt}} = \frac{(1 \text{ k}\Omega)(50 \mu\text{A})}{50 \text{ mA} - 0.05 \text{ mA}} = \mathbf{1 \Omega}$
100 mA: $R_{\text{shunt}} \cong \mathbf{0.5 \Omega}$
37. a. $R_s = \frac{V_{\text{max}} - V_{VS}}{I_{CS}} = \frac{15 \text{ V} - (50 \mu\text{A})(1 \text{ k}\Omega)}{50 \mu\text{A}} = \mathbf{300 \text{ k}\Omega}$
- b. $\Omega/\text{V} = 1/I_{CS} = 1/50 \mu\text{A} = \mathbf{20,000}$
38. 5 V: $R_s = \frac{5 \text{ V} - (1 \text{ mA})(1000 \Omega)}{1 \text{ mA}} = \mathbf{4 \text{ k}\Omega}$
50 V: $R_s = \frac{50 \text{ V} - 1 \text{ V}}{1 \text{ mA}} = \mathbf{49 \text{ k}\Omega}$
500 V: $R_s = \frac{500 \text{ V} - 1 \text{ V}}{1 \text{ mA}} = \mathbf{499 \text{ k}\Omega}$

39. $10 \text{ M}\Omega = (0.5 \text{ V})(\Omega/\text{V}) \Rightarrow \Omega/\text{V} = 20 \times 10^6$

$$I_{CS} = 1/(\Omega/\text{V}) = \frac{1}{20 \times 10^6} = \mathbf{0.05 \mu A}$$

40. a. $R_s = \frac{E}{I_m} - R_m - \frac{\text{zero adjust}}{2} = \frac{3 \text{ V}}{100 \mu A} - 1 \text{ k}\Omega - \frac{2 \text{ k}\Omega}{2} = \mathbf{28 \text{ k}\Omega}$

b. $xI_m = \frac{E}{R_{\text{series}}} + R_m + \frac{\text{zero adjust}}{2} + R_{\text{unk}}$

$$R_{\text{unk}} = \frac{E}{xI_m} - \left(R_{\text{series}} + R_m + \frac{\text{zero adjust}}{2} \right)$$

$$= \frac{3 \text{ V}}{x100 \mu A} - 30 \text{ k}\Omega \Rightarrow \frac{30 \times 10^3}{x} - 30 \times 10^3$$

$x = \frac{3}{4}, R_{\text{unk}} = \mathbf{10 \text{ k}\Omega}; x = \frac{1}{2}, R_{\text{unk}} = \mathbf{30 \text{ k}\Omega}; x = \frac{1}{4}, R_{\text{unk}} = \mathbf{90 \text{ k}\Omega}$

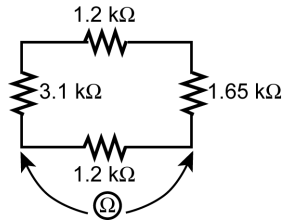
41. —

40. a. Carefully redrawing the network will reveal that all three resistors are in parallel

and $R_T = \frac{R}{N} = \frac{12 \Omega}{3} = \mathbf{4 \Omega}$

b. Again, all three resistors are in parallel and $R_T = \frac{R}{N} = \frac{18 \Omega}{3} = \mathbf{6 \Omega}$

42. a. Network redrawn:



$$R_{\text{ohmmeter}} = 1.2 \text{ k}\Omega \parallel (3.1 \text{ k}\Omega + 1.2 \text{ k}\Omega + 1.65 \text{ k}\Omega)$$

$$= 1.2 \text{ k}\Omega \parallel 5.95 \text{ k}\Omega$$

$$= \mathbf{1 \text{ k}\Omega}$$

b. All three resistors are in parallel

$$R_{\text{ohmmeter}} = \frac{R}{N} = \frac{18 \Omega}{3} = \mathbf{6 \Omega}$$

Chapter 8

1.
 - a. $I_1 = \frac{8\ \Omega(6\ \text{A})}{8\ \Omega + 2\ \Omega} = \mathbf{4.8\ \text{A}}$ $I_2 = 6\ \text{A} - I_1 = 6\ \text{A} - 4.8\ \text{A} = \mathbf{1.2\ \text{A}}$
 - b. $V_s = I_1 R_1 = (4.8\ \text{A})(2\ \Omega) = \mathbf{9.6\ \text{V}}$
2.
 - a. $I_1 = I_2 = \mathbf{20\ \text{mA}}$
 - b. $V_2 = I_2 R_2 = (20\ \text{mA})(3.3\ \text{k}\Omega) = \mathbf{66\ \text{V}}$
 $V_s = IR_T = (20\ \text{mA})((1.2\ \text{k}\Omega + 3.3\ \text{k}\Omega) = 20\ \text{mA}(4.5\ \text{k}\Omega) = \mathbf{90\ \text{V}}$
3. $E + V_{R_1} - V_s = 0$, $V_{R_1} = (8\ \text{mA})(2.7\ \text{k}\Omega) = 21.6\ \text{V}$
 $V_s = E + V_{R_1} = 10\ \text{V} + 21.6\ \text{V} = \mathbf{31.6\ \text{V}}^+$
4.
 - a. $V_s = E = \mathbf{24\ \text{V}}$
 - b. $I_2 = \frac{E}{R_1 + R_2} = \frac{24\ \text{V}}{1\ \Omega + 3\ \Omega} = \frac{24\ \text{V}}{4\ \Omega} = \mathbf{6\ \text{A}}$
 - c. $I + I_s = I_2$, $I_s = I_2 - I = 6\ \text{A} - 2\ \text{A} = \mathbf{4\ \text{A}}$
5. $V_1 = V_2 = V_s = IR_T = 0.6\ \text{A}[6\ \Omega \parallel 24\ \Omega \parallel 24\ \Omega] = 0.6\ \text{A}[6\ \Omega \parallel 12\ \Omega] = 2.4\ \text{V}$
 $I_2 = \frac{V_2}{R_2} = \frac{2.4\ \text{V}}{24\ \Omega} = \mathbf{0.1\ \text{A}}$
 $V_3 = \frac{R_3 V_s}{R_3 + R_4} = \frac{16\ \Omega(2.4\ \text{V})}{24\ \Omega} = \mathbf{1.6\ \text{V}}$
6.
 - a. $I_1 = \frac{E}{R_1} = \frac{24\ \text{V}}{2\ \Omega} = \mathbf{12\ \text{A}}$, $I_{R_2} = \frac{E}{R_2 + R_3} = \frac{24\ \text{V}}{6\ \Omega + 2\ \Omega} = \frac{24\ \Omega}{8\ \Omega} = 3\ \text{A}$
 KCL: $I + I_s - I_1 - I_{R_2} = 0$
 $I_s = I_1 + I_{R_2} - I = 12\ \text{A} + 3\ \text{A} - 4\ \text{A} = \mathbf{11\ \text{A}}$
 - b. $V_s = E = 24\ \text{V}$
 VDR: $V_3 = \frac{R_3 E}{R_2 + R_3} = \frac{2\ \Omega(24\ \text{V})}{6\ \Omega + 2\ \Omega} = \frac{48\ \text{V}}{8\ \Omega} = \mathbf{6\ \text{V}}$
7.
 - a. $I = \frac{E}{R_s} = \frac{22\ \text{V}}{4.7\ \Omega} = \mathbf{4.68\ \text{A}}$, $R_p = R_s = \mathbf{4.7\ \Omega}$
 - b. $I = \frac{E}{R_s} = \frac{9\ \text{V}}{2.2\ \text{k}\Omega} = \mathbf{4.09\ \text{mA}}$, $R_p = R_s = \mathbf{2.2\ \text{k}\Omega}$

8. a. $E = IR_s = (6 \text{ A})(12 \Omega) = \mathbf{72 \text{ V}}$, $R_s = \mathbf{12 \Omega}$
b. $E = IR_s = (18 \text{ mA})(5.6 \text{ k}\Omega) = \mathbf{100.8 \text{ V}}$, $R_s = \mathbf{5.6 \text{ k}\Omega}$
9. a. CDR: $I_L = \frac{R_s(I)}{R_s + R_L} = \frac{100 \Omega(20 \text{ A})}{100 \Omega + 10 \Omega} = \mathbf{18.18 \text{ A}}$, $I_L \cong I$
b. $E_s = IR = (20 \text{ A})(100 \Omega) = \mathbf{2 \text{ kV}}$
 $R_s = 100 \Omega$
 $I = \frac{E_s}{R_s + R_L} = \frac{2 \text{ kV}}{100 \Omega + 10 \Omega} = \mathbf{18.18 \text{ A}}$
10. a. $E = IR_2 = (2 \text{ A})(5.6 \Omega) = \mathbf{11.2 \text{ V}}$, $R = \mathbf{5.6 \Omega}$
b. $E_T = 12 \text{ V} + 11.2 \text{ V} = \mathbf{23.2 \text{ V}}$, $R_T = 10 \Omega + 5.6 \Omega = \mathbf{15.6 \Omega}$
c. $I_3 = \frac{E_T}{R_T + 91 \Omega} = \frac{23.2 \text{ V}}{15.6 \Omega + 91 \Omega} = \mathbf{217.64 \text{ mA}}$
11. a. $I_T = 6.2 \text{ A} - 1.2 \text{ A} - 0.8 \text{ A} = \mathbf{4.2 \text{ A}}$
b. $V_s = I_T R = (4.2 \text{ A})(4 \Omega) = \mathbf{16.8 \text{ V}}$
12. $I_T \uparrow = 7 \text{ A} - 3 \text{ A} = 4 \text{ A}$
CDR: $I_1 = \frac{R_2(I_T)}{R_1 + R_2} = \frac{6 \Omega(4 \text{ A})}{4 \Omega + 6 \Omega} = \mathbf{2.4 \text{ A}}$
 $V_2 = I_1 R_1 = (2.4 \text{ A})(4 \Omega) = \mathbf{9.6 \text{ V}}$
13. a. Conversions: $I_1 = E_1/R_1 = 9 \text{ V}/3 \Omega = \mathbf{3 \text{ A}}$, $R_1 = \mathbf{3 \Omega}$
 $I_2 = E_2/R_2 = 20 \text{ V}/2 \Omega = \mathbf{10 \text{ A}}$, $R_2 = \mathbf{2 \Omega}$
b. $I_T \downarrow = 10 \text{ A} - 3 \text{ A} = 7 \text{ A}$, $R_T = 3 \Omega \parallel 6 \Omega \parallel 2 \Omega$
 $= 2 \Omega \parallel 2 \Omega$
 $= 1 \Omega$
 $V_{ab} = -I_T R_T = -(7 \text{ A})(1 \Omega) = \mathbf{-7 \text{ V}}$
c. $I_3 \uparrow = \frac{7 \text{ V}}{6 \Omega} = \mathbf{1.17 \text{ A}}$
14. a. $I = \frac{E}{R_2} = \frac{12 \text{ V}}{2.2 \text{ k}\Omega} = \mathbf{5.45 \text{ mA}}$, $R_p = \mathbf{2.2 \text{ k}\Omega}$
b. $I_T \uparrow = 8 \text{ mA} + 5.45 \text{ mA} - 3 \text{ mA} = 10.45 \text{ mA}$
 $R' = 6.8 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega = 1.66 \text{ k}\Omega$
 $V_1 = I_T R' = (10.45 \text{ mA})(1.66 \text{ k}\Omega) = \mathbf{17.35 \text{ V}}$

c. $V_1 = V_2 + 12 \text{ V} \Rightarrow V_2 = V_1 - 12 \text{ V} = 17.35 \text{ V} - 12 \text{ V} = 5.35 \text{ V}$

d. $I_2 = \frac{V_2}{R_2} = \frac{5.35 \text{ V}}{2.2 \text{ k}\Omega} = 2.43 \text{ mA}$

15. a. $\begin{array}{l} \overrightarrow{I_1} \downarrow \overleftarrow{I_3} \nearrow I_2 \\ 4 - 4I_1 - 8I_3 = 0 \\ 6 - 2I_2 - 8I_3 = 0 \\ I_1 + I_2 = I_3 \end{array}$

$$I_1 = -\frac{1}{7} \text{ A}, I_2 = \frac{5}{7} \text{ A}, I_3 = \frac{4}{7} \text{ A}$$

$$I_{R_1} = I_1 = -\frac{1}{7} \text{ A}, I_{R_2} = I_2 = \frac{5}{7} \text{ A}, I_{R_3} = I_3 = \frac{4}{7} \text{ A}$$

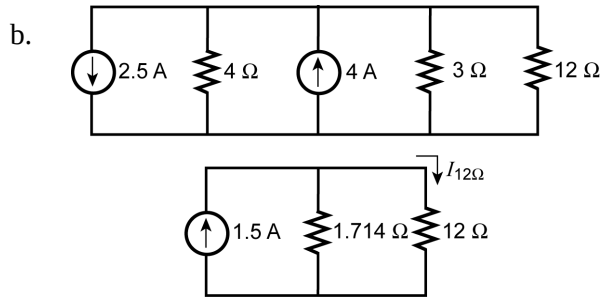
b. $V_a = I_3 R_3 = \left(\frac{4}{7} \text{ A}\right)(8 \Omega) = 4.57 \text{ V}$

16. a. $\begin{array}{l} \overrightarrow{I_1} \uparrow \overrightarrow{I_3} \searrow I_2 \\ 10 + 12 - 3I_3 - 4I_1 = 0 \\ 12 - 3I_3 - 12I_2 = 0 \\ I_1 + I_2 = I_3 \end{array}$

$$\begin{array}{l} I_1 = 3.06 \text{ A} \\ I_2 = 0.19 \text{ A} \\ I_3 = 3.25 \text{ A} \end{array}$$

$$I_{R_1} = I_1 = 3.06 \text{ A}, I_{R_3} = I_2 = 0.19 \text{ A} = I_{12\Omega}$$

$$I_{R_2} = I_3 = 3.25 \text{ A}$$

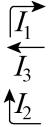


$$I_{12\Omega} = \frac{1.714 \Omega (1.5 \text{ A})}{1.714 \Omega + 12 \Omega} = 0.19 \text{ A}$$

c. the same

17. $\begin{array}{l} \overrightarrow{I_1} \downarrow \overleftarrow{I_3} \nearrow I_2 \\ 10 - I_1 5.6 \text{ k}\Omega - I_3 2.2 \text{ k}\Omega + 20 = 0 \\ -20 + I_3 2.2 \text{ k}\Omega + I_2 3.3 \text{ k}\Omega - 30 = 0 \\ I_1 + I_2 = I_3 \end{array}$

$$I_1 = I_{R_1} = 1.45 \text{ mA}, I_2 = I_{R_2} = 8.51 \text{ mA}, I_3 = I_{R_3} = 9.96 \text{ mA}$$

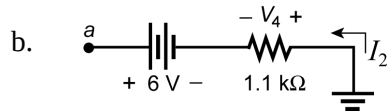
18. a. 
$$\begin{aligned} -1.2 \text{ k}\Omega I_1 + 9 - 8.2 \text{ k}\Omega I_3 &= 0 \\ -10.2 \text{ k}\Omega I_2 + 8.2 \text{ k}\Omega I_3 + 6 &= 0 \\ I_2 + I_3 &= I_1 \end{aligned}$$

$I_1 = 2.03 \text{ mA}, I_2 = 1.23 \text{ mA}, I_3 = 0.8 \text{ mA}$

$I_{R_1} = I_1 = 2.03 \text{ mA}$

$I_{R_2} = I_3 = 0.8 \text{ mA}$

$I_{R_3} = I_{R_4} = I_2 = 1.23 \text{ mA} = I_{9.1 \text{ k}\Omega}$



19. $V_4 = I_2 R_4 = (1.23 \text{ mA})(1.1 \text{ k}\Omega) = 1.35 \text{ V}$
 $V_a = 6 \text{ V} - V_4 = 6 \text{ V} - 1.35 \text{ V} = 4.65 \text{ V}$

$I_1 = I_{R_1} \text{ (CW)}, I_2 = I_{R_2} \text{ (down)}, I_3 = I_{R_3} \text{ (CW)}, I_4 = I_{R_4} \text{ (down)}$

$I_5 = I_{R_5} \text{ (CW)}$

a.
$$\begin{aligned} E_1 - I_1 R_1 - I_2 R_2 &= 0 \\ I_2 R_2 - I_3 R_3 - I_4 R_4 &= 0 \\ I_4 R_4 - I_5 R_5 - E_2 &= 0 \\ I_1 &= I_2 + I_3 \\ I_3 &= I_4 + I_5 \end{aligned}$$

b.
$$\begin{aligned} E_1 - I_2(R_1 + R_2) - I_3 R_1 &= 0 \\ I_2 R_2 - I_3(R_3 + R_4) + I_5 R_4 &= 0 \\ I_3 R_4 - I_5(R_4 + R_5) - E_2 &= 0 \end{aligned}$$

c.
$$\begin{aligned} I_2(R_1 + R_2) + I_3 R_1 + 0 &= E_1 \\ I_2(R_2) - I_3(R_3 + R_4) + I_5 R_4 &= 0 \\ 0 + I_3 R_4 - I_5(R_4 + R_5) &= E_2 \end{aligned}$$

$3I_2 + 2I_3 + 0 = 10$

$1I_2 - 9I_3 + 5I_5 = 0$

$0 + 5I_3 - 8I_5 = 6$

d. $I_3 = I_{R_3} = -63.69 \text{ mA (CW)}$

20. a. $\overrightarrow{I_1} \downarrow \overrightarrow{I_2} \downarrow$
$$\begin{aligned} 4 - 4I_1 - 8(I_1 - I_2) &= 0 \\ -8(I_2 - I_1) - 2I_2 - 6 &= 0 \end{aligned}$$

$$I_1 = -\frac{1}{7} \text{ A}, I_2 = -\frac{5}{7} \text{ A}$$
$$I_{R_1} = I_1 = -\frac{1}{7} \text{ A}$$
$$I_{R_2} = I_2 = -\frac{5}{7} \text{ A}$$
$$I_{R_3} = I_1 - I_2 = \left(-\frac{1}{7} \text{ A}\right) - \left(-\frac{5}{7} \text{ A}\right) = \frac{4}{7} \text{ A (dir. of } I_1)$$
- b. $V_a = I_{R_3} R_3 = \left(\frac{4}{7} \text{ A}\right)(8 \Omega) = \mathbf{4.57 \text{ V}}$
21. a. $\overrightarrow{I_1} \downarrow \overrightarrow{I_2} \downarrow$
$$\begin{aligned} -10 - 4I_1 - 3(I_1 - I_2) - 12 &= 0 \\ 12 - 3(I_2 - I_1) - 12I_2 &= 0 \end{aligned}$$

$$I_1 = -3.06 \text{ A}, I_2 = 0.19 \text{ A}$$
$$I_{E_1} = \mathbf{3.06 \text{ A (CCW)}}$$
$$I_{E_2} = 3.06 \text{ A} + 0.19 \text{ A} = \mathbf{3.25 \text{ A (up)}}$$
$$I_{R_2} = I_1 - I_2 = (-3.06 \text{ A}) - (0.19 \text{ A}) = \mathbf{-3.25 \text{ A}}$$
- b. $P_{E_2} = I_{E_2} E_2 = (3.25 \text{ A})(12 \text{ V}) = \mathbf{39 \text{ W}}$
$$P_{R_3} = I_{R_3}^2 R_3 = (0.19 \text{ A})^2 12 \Omega = \mathbf{433.2 \text{ mW}}$$
22. a. $\overrightarrow{I_1} \downarrow \overrightarrow{I_2} \downarrow$
$$\begin{aligned} 10 - I_1(5.6 \text{ k}\Omega) - 2.2 \text{ k}\Omega(I_1 - I_2) + 20 &= 0 \\ -20 - 2.2 \text{ k}\Omega(I_2 - I_1) - I_2 3.3 \text{ k}\Omega - 30 &= 0 \end{aligned}$$

$$I_1 = 1.45 \text{ mA}, I_2 = 8.51 \text{ mA}$$
$$I_{R_1} = I_1 = \mathbf{1.45 \text{ mA}}, I_{R_2} = I_2 = \mathbf{8.51 \text{ mA}}$$
$$I_{R_3} = I_2 - I_1 = \mathbf{7.06 \text{ mA (direction of } I_2)}$$
- b. $V_{3.3\text{k}\Omega} = I_2 R_2 = (8.51 \text{ mA})(3.3 \text{ k}\Omega) = \mathbf{28.1 \text{ V}}$
23. a. $\overrightarrow{I_1} \downarrow$
 $\overrightarrow{I_2} \downarrow$
$$\begin{aligned} -I_1(1.2 \text{ k}\Omega) + 9 - 8.2 \text{ k}\Omega(I_1 - I_2) &= 0 \\ -I_2(1.1 \text{ k}\Omega) + 6 - I_2(9.1 \text{ k}\Omega) - 8.2 \text{ k}\Omega(I_2 - I_1) &= 0 \end{aligned}$$

$$I_1 = 2.03 \text{ mA}, I_2 = 1.23 \text{ mA}$$
$$I_{R_1} = I_1 = \mathbf{2.03 \text{ mA}}, I_{R_3} = I_{R_4} = I_2 = \mathbf{1.23 \text{ mA}}$$
$$I_{R_2} = I_1 - I_2 = 2.03 \text{ mA} - 1.23 \text{ mA} = \mathbf{0.80 \text{ mA (direction of } I_1)}$$
- b. $V_a = 6 \text{ V} - I_2(1.1 \text{ k}\Omega) = 6 \text{ V} - (1.23 \text{ mA})(1.1 \text{ k}\Omega) = 6 \text{ V} - 1.35 \text{ V} = \mathbf{4.65 \text{ V}}$

24. a. $I_1 \downarrow I_2 \downarrow I_3 \downarrow$

$$\begin{aligned}
 10 - I_1 - 2 - 1(I_1 - I_2) &= 0 \\
 -1(I_2 - I_1) - I_2 - 4 - 5(I_2 - I_3) &= 0 \\
 -5(I_3 - I_2) - I_3 - 3 - 6 &= 0
 \end{aligned}$$

$$\begin{aligned}
 3I_1 - 1I_2 + 0 &= 10 \\
 -1I_1 + 10I_2 - 5I_3 &= 0 \\
 0 - 5I_2 + 8I_3 &= -6
 \end{aligned}$$

$I_1 = 3.31 \text{ A}, I_2 = -63.69 \text{ mA}, I_3 = -789.8 \text{ mA}$

b, c. Ignore

d. $I_{10V} \uparrow = I_1 = 3.31 \text{ A}$
 $I_{6V} \uparrow = -I_3 = -(-789.8 \text{ mA}) = 789.8 \text{ mA}$

25. a. $I_1 \downarrow I_2 \downarrow I_3 \downarrow$

$$\begin{aligned}
 -I_1 - 2.2 \text{ k}\Omega - (I_1 - I_2)9.1 \text{ k}\Omega + 18 \text{ V} &= 0 \\
 -18 \text{ V} - (I_2 - I_1)9.1 \text{ k}\Omega - 7.5 \text{ k}\Omega I_2 - (I_2 - I_3)6.8 \text{ k}\Omega &= 0 \\
 -6.8 \text{ k}\Omega(I_3 - I_2) - 3 \text{ V} - 3.3 \text{ k}\Omega I_3 &= 0
 \end{aligned}$$

with

$$\begin{aligned}
 11.3 \text{ k}\Omega I_1 - 9.1 \text{ k}\Omega I_2 &= 18 \text{ V} \\
 23.4 \text{ k}\Omega I_2 - 9.1 \text{ k}\Omega I_1 - 6.8 \text{ k}\Omega I_3 &= -18 \text{ V} \\
 10.1 \text{ k}\Omega I_3 - 6.8 \text{ k}\Omega I_2 &= -3 \text{ V}
 \end{aligned}$$

or

$$\begin{aligned}
 11.3 \text{ k}\Omega I_1 - 9.1 \text{ k}\Omega I_2 &= 18 \text{ V} \\
 -9.1 \text{ k}\Omega I_2 + 23.4 \text{ k}\Omega I_2 - 6.8 \text{ k}\Omega I_3 &= -18 \text{ V} \\
 -6.8 \text{ k}\Omega I_2 + 10.1 \text{ k}\Omega I_3 &= -3 \text{ V}
 \end{aligned}$$

b. $I_1 = 1.21 \text{ mA}, I_2 = -0.48 \text{ mA}, I_3 = -0.62 \text{ mA}$

c. $I_{E_1} \downarrow = I_1 - I_2 = 1.21 \text{ mA} - (-0.48 \text{ mA}) = 1.69 \text{ mA}$
 $I_{E_2} \uparrow = -I_3 = -(-0.62 \text{ mA}) = 0.62 \text{ mA}$

26. a. $I_1 \downarrow I_2 \downarrow I_3 \downarrow$

$$\begin{aligned}
 16 - 4I_1 - 3(I_1 - I_2) - 12 - 4(I_1 - I_3) &= 0 \\
 12 - 3(I_2 - I_1) - 10 I_2 - 15 - 4(I_2 - I_3) &= 0 \\
 -16 - 4(I_3 - I_1) - 4(I_3 - I_2) - 7I_3 &= 0
 \end{aligned}$$

b. $I_1 = -0.24 \text{ A}, I_2 = -0.52 \text{ A}, I_3 = -1.28 \text{ A}$

c. $I_{R_5} = I_1 = -0.24 \text{ A}$

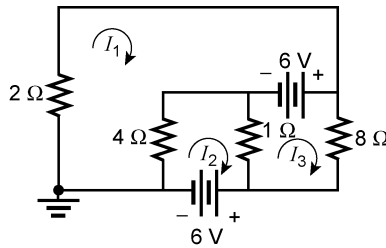
27. a. $\begin{matrix} I_1 \rightarrow \\ I_2 \rightarrow \\ I_3 \rightarrow \\ I_4 \rightarrow \end{matrix}$

$$\begin{aligned} -6.8 \text{ k}\Omega I_1 - 4.7 \text{ k}\Omega(I_1 - I_2) + 6 - 2.2 \text{ k}\Omega(I_1 - I_4) &= 0 \\ -6 - 4.7 \text{ k}\Omega(I_2 - I_1) - 2.7 \text{ k}\Omega I_2 - 8.2 \text{ k}\Omega(I_2 - I_3) &= 0 \\ -1.1 \text{ k}\Omega I_3 - 22 \text{ k}\Omega(I_3 - I_4) - 8.2 \text{ k}\Omega(I_3 - I_2) - 9 &= 0 \\ 5 - 1.2 \text{ k}\Omega I_4 - 2.2 \text{ k}\Omega(I_4 - I_1) - 22 \text{ k}\Omega(I_4 - I_3) &= 0 \end{aligned}$$

b. $I_1 = 0.03 \text{ mA}, I_2 = -0.88 \text{ mA}, I_3 = -0.97 \text{ mA}, I_4 = -0.64 \text{ mA}$

c. $I_{6V} = I_1 - I_2 = 0.03 \text{ mA} - (-0.88 \text{ mA}) = 0.91 \text{ mA},$
 $P_{6V} = E I_{6V} = (6 \text{ V})(0.91 \text{ mA}) = \mathbf{5.46 \text{ mW}}$

28. a. Network redrawn:



b. $-2I_1 - 6 - 4I_1 + 4I_2 = 0$
 $-4I_2 + 4I_1 - 1I_2 + 1I_3 - 6 = 0$
 $-1I_3 + 1I_2 + 6 - 8I_3 = 0$

c. $I_1 = -3.8 \text{ A}, I_2 = -4.20 \text{ A}, I_3 = 0.20 \text{ A}$

d. $P_{E_2} = E_2 I_3 = (6 \text{ V})(0.2 \text{ A}) = 1.2 \text{ W}$
 $P_{E_1} = E_1 I_2 = (6 \text{ V})(4.2 \text{ A}) = 25.2 \text{ W}$
 $P_T = P_{E_1} + P_{E_2} = 1.2 \text{ W} + 25.2 \text{ W} = \mathbf{26.4 \text{ W}}$

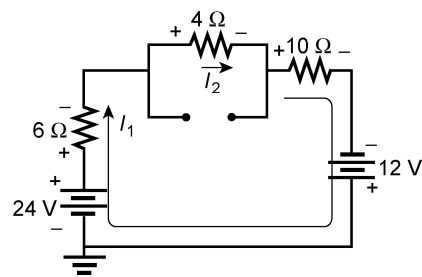
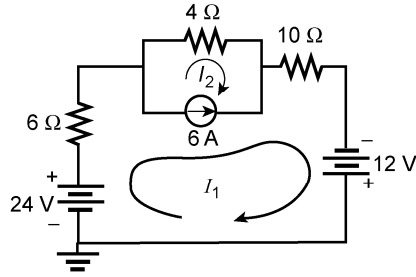
29. a. $20 \text{ V} - I_B(270 \text{ k}\Omega) - 0.7 \text{ V} - I_E(0.51 \text{ k}\Omega) = 0$
 $I_E(0.51 \text{ k}\Omega) + 8 \text{ V} + I_C(2.2 \text{ k}\Omega) - 20 \text{ V} = 0$
 $I_E = I_B + I_C$

$I_B = 63.02 \text{ }\mu\text{A}, I_C = 4.42 \text{ mA}, I_E = 4.48 \text{ mA}$

b. $V_B = 20 \text{ V} - I_B(270 \text{ k}\Omega) = 20 \text{ V} - (63.02 \text{ }\mu\text{A})(270 \text{ k}\Omega) = 20 \text{ V} - 17.02 \text{ V} = \mathbf{2.98 \text{ V}}$
 $V_E = I_E R_E = (4.48 \text{ mA})(510 \text{ }\Omega) = \mathbf{2.28 \text{ V}}$
 $V_C = 20 \text{ V} - I_C(2.2 \text{ k}\Omega) = 20 \text{ V} - (4.42 \text{ mA})(2.2 \text{ k}\Omega) = 20 \text{ V} - 9.72 \text{ V} = \mathbf{10.28 \text{ V}}$

c. $\beta \cong I_C/I_B = 4.42 \text{ mA}/63.02 \text{ }\mu\text{A} = \mathbf{70.14}$

30.



$$24 \text{ V} - 6I_1 - 4I_2 - 10I_1 + 12 \text{ V} = 0$$

$$\text{and } 16I_1 + 4I_2 = 36$$

$$I_1 - I_2 = 6 \text{ A}$$

$$I_1 = I_2 + 6 \text{ A}$$

$$16[I_2 + 6 \text{ A}] + 4I_2 = 36$$

$$16I_2 + 96 + 4I_2 = 36$$

$$20I_2 = -60$$

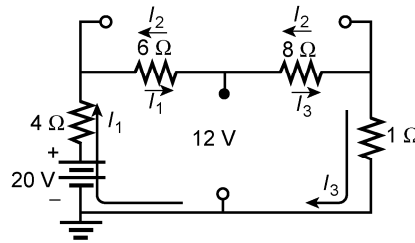
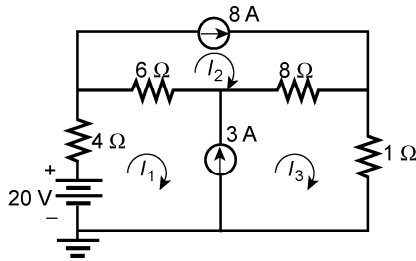
$$I_2 = -3 \text{ A}$$

$$I_1 = I_2 + 6 \text{ A} = -3 \text{ A} + 6 \text{ A} = 3 \text{ A}$$

$$I_{24\text{V}} = I_{6\Omega} = I_{10\Omega} = I_{12\text{V}} = 3 \text{ A (CW)}$$

$$I_{4\Omega} = 3 \text{ A (CCW)}$$

31.



$$20 \text{ V} - 4I_1 - 6(I_1 - I_2) - 8(I_3 - I_2) - 1I_3 = 0$$

$$10I_1 - 14I_2 + 9I_3 = 20$$

$$I_3 - I_1 = 3 \text{ A}$$

$$I_2 = 8 \text{ A}$$

$$10I_1 - 14(8 \text{ A}) + 9[I_1 + 3 \text{ A}] = 20$$

$$19I_1 = 105$$

$$I_1 = 5.526 \text{ A}$$

$$I_3 = I_1 + 3 \text{ A} = 5.526 \text{ A} + 3 \text{ A} = 8.526 \text{ A}$$

$$I_2 = 8 \text{ A}$$

$$I_{20\text{V}} = I_{4\Omega} = 5.53 \text{ A (dir. of } I_1)$$

$$I_{6\Omega} = I_2 - I_1 = 2.47 \text{ A (dir. of } I_2)$$

$$I_{8\Omega} = I_3 - I_2 = 0.53 \text{ A (dir. of } I_3)$$

$$I_{1\Omega} = 8.53 \text{ A (dir. of } I_3)$$

32. a. $\overrightarrow{I_1} \overrightarrow{I_2}$
$$\begin{aligned} (4 + 8)I_1 - 8I_2 &= 4 \\ (8 + 2)I_2 - 8I_1 &= -6 \end{aligned}$$

b. $I_{8\Omega} \downarrow = I_1 - I_2 = \left(-\frac{1}{7} \text{ A}\right) - \left(-\frac{5}{7} \text{ A}\right) = \frac{4}{7} \text{ A}$

33. a. $\overrightarrow{I_1} \overrightarrow{I_2}$
$$\begin{aligned} (4 + 3)I_1 - 3I_2 &= -10 - 12 \\ (3 + 12)I_2 - 3I_1 &= 12 \end{aligned}$$

b. $I_{3\Omega} \uparrow = I_2 - I_1 = 0.19 \text{ A} - (-3.06 \text{ A}) = \mathbf{3.25 \text{ A}}$

34. a. $\overrightarrow{I_1} \overrightarrow{I_2}$

a.
$$\begin{aligned} I_1(5.6 \text{ k}\Omega + 2.2 \text{ k}\Omega) - 2.2 \text{ k}\Omega (I_2) &= 10 + 20 \\ I_2(2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega) - 2.2 \text{ k}\Omega (I_1) &= -20 - 30 \end{aligned}$$

b. $I_{E_1} = I_1 = \mathbf{1.45 \text{ mA}}$, $I_{E_2} \uparrow = \mathbf{8.51 \text{ mA}}$,
 $I_{E_3} \downarrow = I_1 - I_2 = (1.45 \text{ mA}) - (-8.5 \text{ mA}) = \mathbf{9.96 \text{ mA}}$

35. a. $I_1 \downarrow I_2 \downarrow I_3 \downarrow$
$$\begin{aligned} I_1(2 + 1) - 1I_2 &= 10 \\ I_2(1 + 4 + 5) - 1I_1 - 5I_3 &= 0 \\ I_3(5 + 3) - 5I_2 &= -6 \end{aligned}$$

b. $I_1 = \mathbf{3.31 \text{ A}}$, $I_2 = \mathbf{-63.69 \text{ mA}}$, $I_3 = \mathbf{-789.8 \text{ mA}}$

c. $I_{R_2} = I_1 - I_2 = (3.31 \text{ A}) - (-63.69 \text{ mA}) = \mathbf{3.37 \text{ A}}$

36. a. $I_1 \downarrow I_2 \downarrow I_3 \downarrow$

$$\begin{aligned} (2.2 \text{ k}\Omega + 9.1 \text{ k}\Omega)I_1 - 9.1 \text{ k}\Omega I_2 &= 18 \\ (9.1 \text{ k}\Omega + 7.5 \text{ k}\Omega + 6.8 \text{ k}\Omega)I_2 - 9.1 \text{ k}\Omega I_1 - 6.8 \text{ k}\Omega I_3 &= -18 \\ (6.8 \text{ k}\Omega + 3.3 \text{ k}\Omega)I_3 - 6.8 \text{ k}\Omega I_2 &= -3 \end{aligned}$$

b. $I_1 = \mathbf{1.21 \text{ mA}}$, $I_2 = \mathbf{-0.48 \text{ mA}}$, $I_3 = \mathbf{-0.62 \text{ mA}}$

c. $I_{E_1} \downarrow = I_1 - I_2 = 1.21 \text{ mA} - (-0.48 \text{ mA}) = \mathbf{1.69 \text{ mA}}$
 $I_{E_2} \uparrow = -I_3 = -(-0.62 \text{ mA}) = \mathbf{0.62 \text{ mA}}$

37. a. $(3\ \Omega + 6\ \Omega)I_1 - 6\ \Omega I_2 = 9\ \text{V}$
 $(6\ \Omega + 2\ \Omega)I_2 - 6\ \Omega I_1 = 20\ \text{V}$

$$\begin{aligned} 9\ \Omega I_1 - 6\ \Omega I_2 &= 9\ \text{V} \\ -6\ \Omega I_1 + 8\ \Omega I_2 &= 20\ \text{V} \end{aligned}$$

$$I_1 = \mathbf{5.33\ \text{A}}, I_2 = \mathbf{6.5\ \text{A}}$$

b. $V_{ab} = 2\ \Omega (I_2) - 20\ \text{V} = 2\ \Omega (6.5\ \text{A}) - 20\ \text{V} = 13\ \text{V} - 20\ \text{V} = \mathbf{-7\ \text{V}}$

38. a. $\begin{array}{c} I_1 \searrow I_2 \searrow \\ I_4 \searrow I_3 \searrow \end{array}$ $\begin{aligned} I_1(6.8\ \text{k}\Omega + 4.7\ \text{k}\Omega + 2.2\ \text{k}\Omega) - 4.7\ \text{k}\Omega I_2 - 2.2\ \text{k}\Omega I_4 &= 6 \\ I_2(2.7\ \text{k}\Omega + 8.2\ \text{k}\Omega + 4.7\ \text{k}\Omega) - 4.7\ \text{k}\Omega I_1 - 8.2\ \text{k}\Omega I_3 &= -6 \\ I_3(8.2\ \text{k}\Omega + 1.1\ \text{k}\Omega + 22\ \text{k}\Omega) - 22\ \text{k}\Omega I_4 - 8.2\ \text{k}\Omega I_2 &= -9 \\ I_4(2.2\ \text{k}\Omega + 22\ \text{k}\Omega + 1.2\ \text{k}\Omega) - 2.2\ \text{k}\Omega I_1 - 22\ \text{k}\Omega I_3 &= 5 \end{aligned}$

b. $I_1 = \mathbf{0.03\ \text{mA}}, I_2 = \mathbf{-0.88\ \text{mA}}, I_3 = \mathbf{-0.97\ \text{mA}}, I_4 = \mathbf{-0.64\ \text{mA}}$

c. $I_{22\text{k}\Omega} \downarrow = I_4 - I_3 = (-0.64\ \text{mA}) - (-0.97\ \text{mA}) = 0.33\ \text{mA}$
 $V_{22\text{k}\Omega} = I_{22\text{k}\Omega} \cdot 22\ \text{k}\Omega = (0.33\ \text{mA})(22\ \text{k}\Omega) = \mathbf{7.26\ \text{V}}$

39. a. $\begin{array}{c} \curvearrowright I_1 \curvearrowright I_2 \\ \curvearrowright I_3 \end{array}$ $\begin{aligned} (1\ \Omega + 2\ \Omega + 4\ \Omega)I_1 - 2\ \Omega I_2 - 4\ \Omega I_3 &= 12\ \text{V} \\ (2\ \Omega + 2\ \Omega + 10\ \Omega)I_2 - 2\ \Omega I_1 - 10\ \Omega I_3 &= -20\ \text{V} \\ (4\ \Omega + 10\ \Omega + 8\ \Omega)I_3 - 10\ \Omega I_2 - 4\ \Omega I_1 &= 20\ \text{V} \end{aligned}$

$$\begin{aligned} 7I_1 - 2I_2 - 4I_3 &= 12 \\ 2I_1 - 14I_2 + 10I_3 &= 20 \\ 4I_1 + 10I_2 - 22I_3 &= -20 \end{aligned}$$

b. $I_1 = \mathbf{2.38\ \text{A}}, I_2 = \mathbf{-0.195\ \text{A}}, I_3 = \mathbf{1.25\ \text{A}}$

c. $V_a = (I_1 - I_3)4\ \Omega = (2.38\ \text{A} - 1.25\ \text{A})4\ \Omega = \mathbf{4.5\ \text{V}}$
 $V_b = I_3 8\ \Omega = (1.25\ \text{A})(8\ \Omega) = \mathbf{10\ \text{V}}$

d. $V_{ab} = V_a - V_b = 4.5\ \text{V} - 10\ \text{V} = \mathbf{-5.5\ \text{V}}$

40. a. $\begin{matrix} V_1 \\ \circ \end{matrix} \quad \begin{matrix} V_2 \\ \circ \end{matrix}$

At V_1 : $\sum I_i = \sum I_o$

$$0 = \frac{V_1}{2 \Omega} + 5 \text{ A} + \frac{V_1 - V_2}{8 \Omega}$$

At V_2 : $\sum I_i = \sum I_o$

$$\frac{V_1 - V_2}{8 \Omega} = 3 \text{ A} + \frac{V_2}{4 \Omega}$$

and $V_1 \left[\frac{1}{2} + \frac{1}{8} \right] - V_2 \left[\frac{1}{8} \right] = -5$
 $-V_1 \left[\frac{1}{8} \right] + V_2 \left[\frac{1}{8} + \frac{1}{4} \right] = -3$

b. $V_1 = -10.27 \text{ V}$, $V_2 = -11.36 \text{ V}$

c. $V_{8\Omega} = V_1 - V_2 = -10.27 \text{ V} - (-11.36 \text{ V}) = 1.09 \text{ V}$

d. $I_{2\Omega} \uparrow = \frac{V_1}{2 \Omega} = \frac{10.27 \text{ V}}{2 \Omega} = 5.14 \text{ A}$
 $I_{4\Omega} \uparrow = \frac{V_2}{4 \Omega} = \frac{11.36 \text{ V}}{4 \Omega} = 2.84 \text{ A}$

41. a. $\begin{matrix} V_1 \\ \circ \end{matrix} \quad \begin{matrix} V_2 \\ \circ \end{matrix}$

At V_1 : $\sum I_i = \sum I_o$

$$0 = \frac{V_1}{8 \Omega} + 12 \text{ A} + I_{6\Omega} \text{ and } V_1 - I_{6\Omega} - 54 \text{ V} - V_2 = 0$$

$$\text{or } I = \frac{V_1 - V_2 - 54 \text{ V}}{6 \Omega} = \frac{V_1}{6 \Omega} + \frac{V_2}{6 \Omega} - 9 \text{ A}$$

so that $0 = \frac{V_1}{8 \Omega} + 12 \text{ A} + \frac{V_1}{6 \Omega} - \frac{V_2}{6 \Omega} - 9 \text{ A}$

or $V_1 \left[\frac{1}{8 \Omega} + \frac{1}{6 \Omega} \right] - V_2 \left[\frac{1}{6 \Omega} \right] = -12 \text{ A} + 9 \text{ A} = -3 \text{ A}$

At V_2 : $\sum I_i = \sum I_o$

$$I = \frac{V_2}{20 \Omega} + \frac{V_2}{5 \Omega}$$

or $\frac{V_1}{6 \Omega} - \frac{V_2}{6 \Omega} - 9 \text{ A} = \frac{V_2}{20 \Omega} + \frac{V_2}{5 \Omega}$

and $V_2 \left[\frac{1}{6 \Omega} + \frac{1}{20 \Omega} + \frac{1}{5 \Omega} \right] - V_1 \left[\frac{1}{6 \Omega} \right] = -9 \text{ A}$

$$\begin{aligned}\text{resulting in } V_1 \left[\frac{1}{8 \Omega} + \frac{1}{6 \Omega} \right] - V_2 \left[\frac{1}{6 \Omega} \right] &= -3 \text{ A} \\ -V_1 \left[\frac{1}{6 \Omega} \right] + V_2 \left[\frac{1}{6 \Omega} + \frac{1}{20 \Omega} + \frac{1}{5 \Omega} \right] &= -9 \text{ A}\end{aligned}$$

b. $V_1 = -29.29 \text{ V}$, $V_2 = -33.34 \text{ V}$

c. $I_{20\Omega} \uparrow = \frac{V_2}{20 \Omega} = \frac{33.34 \text{ V}}{20 \Omega} = 1.67 \text{ A}$

42. a. $\begin{matrix} V_1 \\ \circ \end{matrix} \quad \begin{matrix} V_2 \\ \circ \end{matrix}$

At V_1 : $\sum I_i = \sum I_o$

$$4 \text{ A} = \frac{V_1}{2 \Omega} + \frac{V_1 - V_2}{4 \Omega} + 2 \text{ A}$$

Ar V_2 : $\sum I_i = \sum I_o$

$$2 \text{ A} + \frac{V_1 - V_2}{4 \Omega} = \frac{V_2}{20 \Omega} + \frac{V_2}{5 \Omega}$$

or $V_1 \left[\frac{1}{2} + \frac{1}{4} \right] - V_2 \left[\frac{1}{4} \right] = 2$

$$-V_1 \left[\frac{1}{4} \right] + V_2 \left[\frac{1}{4} + \frac{1}{20} + \frac{1}{5} \right] = 2$$

b. $V_1 = 4.8 \text{ V}$, $V_2 = 6.4 \text{ V}$

c. I_1 : $P = V_1 I_1 = (4.8 \text{ V})(4 \text{ A}) = 19.2 \text{ W}$

I_2 : $P = |(V_1 - V_2) I_2| = |(4.8 \text{ V} - 6.4 \text{ V})(2 \text{ A})| = 3.2 \text{ W}$

43. a. $\begin{matrix} V_1 \\ \circ \end{matrix} \quad \begin{matrix} V_2 \\ \circ \end{matrix}$

At V_1 : $\sum I_i = \sum I_o$

$$0 = 6 \text{ A} + \frac{V_1}{5 \Omega} + \frac{V_1 - V_2}{3 \Omega} + \frac{V_1 - V_2}{2 \Omega}$$

At V_2 : $\sum I_i = \sum I_o$

$$7 \text{ A} + \frac{V_1 - V_2}{3 \Omega} + \frac{V_1 - V_2}{2 \Omega} = \frac{V_2}{4 \Omega} + \frac{V_2}{8 \Omega}$$

so that $V_1 \left[\frac{1}{5 \Omega} + \frac{1}{3 \Omega} + \frac{1}{2 \Omega} \right] - V_2 \left[\frac{1}{3 \Omega} + \frac{1}{2 \Omega} \right] = -6 \text{ A}$

$$V_2 \left[\frac{1}{4 \Omega} + \frac{1}{8 \Omega} + \frac{1}{3 \Omega} + \frac{1}{2 \Omega} \right] - V_1 \left[\frac{1}{3 \Omega} + \frac{1}{2 \Omega} \right] = 7 \text{ A}$$

or $1.03V_1 - 0.833V_2 = -6$

$-0.833V_1 + 1.21V_2 = 7$

b. $V_1 = -2.59 \text{ V}$, $V_2 = 4 \text{ V}$

c. $V_{2\Omega} = V_{3\Omega} = V_2 - V_1 = 4 \text{ V} - (-2.59 \text{ V}) = 6.59 \text{ V}$

$V_{5\Omega} = V_1 = -2.59 \text{ V}$

$V_{4\Omega} = V_{8\Omega} = V_2 = 4 \text{ V}$

44. a. $\begin{array}{c} V_1 \\ \circ \end{array} \quad \begin{array}{c} V_2 \\ \circ \end{array}$

Source conversion: $I_3 = \frac{12 \text{ V}}{4 \Omega} = 3 \text{ A}$, $R_p = R_3 = 4 \Omega$

At V_1 : $\sum I_i = \sum I_o$

$$0 = \frac{V_1}{3 \Omega} + \frac{V_1}{6 \Omega} + 5 \text{ A} + \frac{V_1 - V_2}{4 \Omega} + 3 \text{ A}$$

At V_2 : $\sum I_i = \sum I_o$

$$3 \text{ A} + \frac{V_1 - V_2}{4 \Omega} = \frac{V_2}{8 \Omega} + 4 \text{ A}$$

Rewritten: $V_1 \left[\frac{1}{3 \Omega} + \frac{1}{6 \Omega} + \frac{1}{4 \Omega} \right] - \frac{V_2}{4 \Omega} = -5 \text{ A} - 3 \text{ A}$

$$-V_1 \left[\frac{1}{4 \Omega} \right] + V_2 \left[\frac{1}{4 \Omega} + \frac{1}{8 \Omega} \right] = -4 \text{ A} + 3 \text{ A}$$

b. $V_1 = -14.86 \text{ V}$, $V_2 = -12.57 \text{ V}$

c. $I_{6\Omega} \uparrow = \frac{14.86 \text{ V}}{6 \Omega} = 2.48 \text{ A}$

45. a. $\begin{array}{c} V_1 \\ \circ \end{array} \quad \begin{array}{c} V_2 \\ \circ \end{array} \quad \begin{array}{c} V_3 \\ \circ \end{array}$

Source Conversion: $I_2 = \frac{15 \text{ V}}{3 \Omega} = 5 \text{ A}$, $R_p = R_1 = 3 \Omega$

At V_1 : $\sum I_i = \sum I_o$

$$5 \text{ A} = \frac{V_1}{3 \Omega} + \frac{V_1 - V_2}{6 \Omega} + \frac{V_1 - V_3}{6 \Omega}$$

At V_2 : $\sum I_i = \sum I_o$

$$\frac{V_1 - V_2}{6 \Omega} = \frac{V_2}{4 \Omega} + 3 \text{ A} + \frac{V_2 - V_3}{5 \Omega}$$

At V_3 : $\sum I_i = \sum I_o$

$$\frac{V_1 - V_3}{6 \Omega} + \frac{V_2 - V_3}{5 \Omega} = \frac{V_3}{7 \Omega}$$

Rewritten:

$$\begin{aligned} V_1 \left[\frac{1}{3 \Omega} + \frac{1}{6 \Omega} + \frac{1}{6 \Omega} \right] - \frac{1}{6 \Omega} V_2 - \frac{1}{6 \Omega} V_3 &= 5 \text{ A} \\ V_2 \left[\frac{1}{6 \Omega} + \frac{1}{4 \Omega} + \frac{1}{5 \Omega} \right] - \frac{1}{6 \Omega} V_1 - \frac{1}{5 \Omega} V_3 &= -3 \text{ A} \\ V_3 \left[\frac{1}{6 \Omega} + \frac{1}{5 \Omega} + \frac{1}{7 \Omega} \right] - \frac{1}{5 \Omega} V_2 - \frac{1}{6 \Omega} V_1 &= 0 \end{aligned}$$

b. $V_1 = \mathbf{7.24 \text{ V}}$, $V_2 = \mathbf{-2.45 \text{ V}}$, $V_3 = \mathbf{1.41 \text{ V}}$

c. $V_{5\Omega} = V_3 - V_2 = 1.41 \text{ V} - (-2.45 \text{ V}) = \mathbf{3.86 \text{ V}}$
 $\quad \quad \quad - \quad +$

46. a. $\circ V_1 \quad \circ V_3$
 $\quad \quad \quad \circ V_2$

Source Conversion: $I_2 = \frac{16 \text{ V}}{4 \Omega} = 4 \text{ A}$, $R_p = R_2 = 4 \Omega$

At V_1 : $\sum I_i = \sum I_o$

$$0 = 2 \text{ A} + \frac{V_1}{9 \Omega} + \frac{V_1 - V_3}{20 \Omega} + \frac{V_1 - V_2}{20 \Omega}$$

At V_2 : $\sum I_i = \sum I_o$

$$\frac{V_1 - V_2}{20 \Omega} = \frac{V_2 - V_3}{20 \Omega} + \frac{V_2}{18 \Omega}$$

At V_3 : $\sum I_i = \sum I_o$

$$\frac{V_1 - V_3}{20 \Omega} + \frac{V_2 - V_3}{20 \Omega} + 4 \text{ A} = \frac{V_3}{4 \Omega}$$

Rewritten:

$$\begin{aligned} V_1 \left[\frac{1}{9 \Omega} + \frac{1}{20 \Omega} + \frac{1}{20 \Omega} \right] - \frac{1}{20 \Omega} V_2 - \frac{1}{20 \Omega} V_3 &= -2 \text{ A} \\ V_2 \left[\frac{1}{20 \Omega} + \frac{1}{20 \Omega} + \frac{1}{18 \Omega} \right] - \frac{1}{20 \Omega} V_1 - \frac{1}{20 \Omega} V_3 &= 0 \\ V_3 \left[\frac{1}{20 \Omega} + \frac{1}{20 \Omega} + \frac{1}{4 \Omega} \right] - \frac{1}{20 \Omega} V_2 - \frac{1}{20 \Omega} V_1 &= 4 \text{ A} \end{aligned}$$

b. $V_1 = \mathbf{-6.64 \text{ V}}$, $V_2 = \mathbf{1.29 \text{ V}}$, $V_3 = \mathbf{10.66 \text{ V}}$

c. $V_{R_6} = V_3 - V_1 = 10.66 \text{ V} - (-6.64 \text{ V}) = \mathbf{17.30 \text{ V}}$
 $\quad \quad \quad - \quad +$

47. a. $\circ V_1 \quad \circ V_2 \quad \circ V_3$

At V_1 : $\sum I_i = \sum I_o$

$$0 = 5 \text{ A} + \frac{V_1}{2 \Omega} + \frac{V_1 - V_2}{2 \Omega}$$

At V_2 : $\sum I_i = \sum I_o$

$$\frac{V_1 - V_2}{2 \Omega} = \frac{V_2}{9 \Omega} + \frac{V_2}{7 \Omega} + \frac{V_2 - V_3}{2 \Omega}$$

At V_3 : $\sum I_i = \sum I_o$

$$\frac{V_2 - V_3}{2 \Omega} + 5 \text{ A} = \frac{V_3}{2 \Omega} + \frac{V_3}{4 \Omega}$$

Rewritten:

$$\left[\frac{1}{2 \Omega} + \frac{1}{2 \Omega} \right] V_1 - \frac{1}{2 \Omega} V_2 + 0 = -5 \text{ A}$$

$$\left[\frac{1}{2 \Omega} + \frac{1}{9 \Omega} + \frac{1}{7 \Omega} + \frac{1}{2 \Omega} \right] V_2 - \frac{1}{2 \Omega} V_1 - \frac{1}{2 \Omega} V_3 = 0$$

$$V_3 \left[\frac{1}{2 \Omega} + \frac{1}{2 \Omega} + \frac{1}{4 \Omega} \right] V_3 - \frac{1}{2 \Omega} V_2 = 5 \text{ A}$$

b. $V_1 = -5.31 \text{ V}, V_2 = -0.62 \text{ V}, V_3 = 3.75 \text{ V}$

c. $I_{9\Omega} = \frac{V_2}{9 \Omega} = \frac{0.62 \text{ V}}{9 \Omega} = \mathbf{68.9 \text{ mA}}$

48. a. $\circ V_1 \quad \circ V_2$

$\circ V_3$

At V_1 : $\sum I_i = \sum I_o$

$$0 = 5 \text{ A} + \frac{V_1}{2 \Omega} + \frac{V_1 - V_3}{6 \Omega}$$

At V_2 : $\sum I_i = \sum I_o$

$$5 \text{ A} = 2 \text{ A} + \frac{V_2}{4 \Omega}$$

At V_3 : $\sum I_i = \sum I_o$

$$\frac{V_1 - V_3}{6 \Omega} + 2 \text{ A} = \frac{V_3}{5 \Omega}$$

Rewritten:

$$V_1 \left[\frac{1}{2 \Omega} + \frac{1}{6 \Omega} \right] - \frac{1}{6 \Omega} V_3 = -5 \text{ A}$$

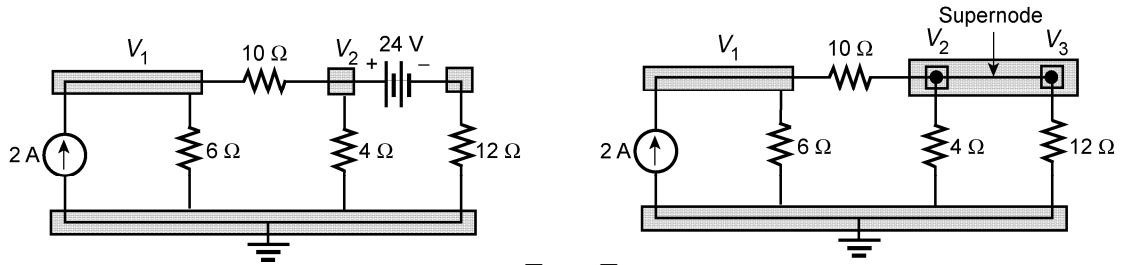
$$V_2 \left[\frac{1}{4 \Omega} \right] = 5 \text{ A} - 2 \text{ A}$$

$$V_3 \left[\frac{1}{6 \Omega} + \frac{1}{5 \Omega} \right] - \frac{1}{6 \Omega} V_1 = 2 \text{ A}$$

b. $V_1 = -6.92 \text{ V}$, $V_2 = 12 \text{ V}$, $V_3 = 2.3 \text{ V}$

c. $I_{4\Omega} = \frac{V_2}{4 \Omega} = \frac{12 \text{ V}}{4 \Omega} = 3 \text{ A}$

49. a.



$$\sum I_i = \sum I_o$$

Node V_1 :

$$2 \text{ A} = \frac{V_1}{6 \Omega} + \frac{V_1 - V_2}{10 \Omega}$$

Supernode V_2, V_3 :

$$0 = \frac{V_2 - V_1}{10 \Omega} + \frac{V_2}{4 \Omega} + \frac{V_3}{12 \Omega}$$

Independent source:

$$V_2 - V_3 = 24 \text{ V or } V_3 = V_2 - 24 \text{ V}$$

2 eq. 2 unknowns:

$$\frac{V_1}{6 \Omega} + \frac{V_1 - V_2}{10 \Omega} = 2 \text{ A}$$

$$\frac{V_2 - V_1}{10 \Omega} + \frac{V_2}{4 \Omega} + \frac{V_2 - 24 \text{ V}}{12 \Omega} = 0$$

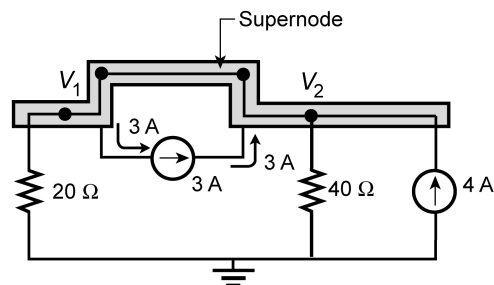
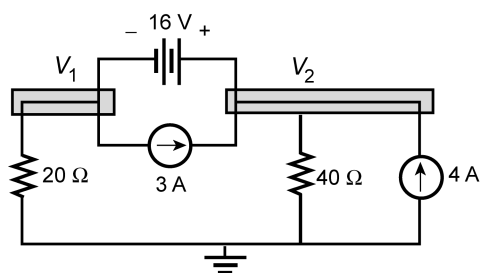
$$0.267V_1 - 0.1V_2 = 2$$

$$+0.1V_1 - 0.433V_2 = -2$$

$$V_1 = 10.08 \text{ V}, V_2 = 6.94 \text{ V}$$

$$V_3 = V_2 - 24 \text{ V} = -17.06 \text{ V}$$

50.



$$\sum I_i = \sum I_o$$

Supernode:

$$3 \text{ A} + 4 \text{ A} = 3 \text{ A} + \frac{V_1}{20 \Omega} + \frac{V_2}{40 \Omega}$$

$$2 \text{ eq. } 2 \text{ unk.} \begin{cases} 4 \text{ A} = \frac{V_1}{20 \Omega} + \frac{V_2}{40 \Omega} \\ V_2 - V_1 = 16 \text{ V} \end{cases}$$

$$\text{Subt. } V_2 = 16 \text{ V} + V_1$$

$$4 \text{ A} = \frac{V_1}{20 \Omega} + \frac{(16 \text{ V} + V_1)}{40 \Omega}$$

$$\text{and } V_1 = \mathbf{48 \text{ V}}$$

$$V_2 = 16 \text{ V} + V_1 = \mathbf{64 \text{ V}}$$

51.

$$\begin{array}{cc} V_1 & V_2 \\ \circ & \circ \end{array}$$

$$V_1 \left[\frac{1}{2} + \frac{1}{8} \right] - V_2 \left[\frac{1}{8} \right] = -5$$

$$-V_1 \left[\frac{1}{8} \right] + V_2 \left[\frac{1}{8} + \frac{1}{4} \right] = -3$$

$$V_1 = \mathbf{-10.27 \text{ V}}, V_2 = \mathbf{-11.36 \text{ V}}$$

b. $V_{I_1} = V_1 = \mathbf{-10.27 \text{ V}}, V_{I_2} = V_2 = \mathbf{-11.36 \text{ V}}$

52.

$$\begin{array}{cc} V_1 & V_2 \\ \circ & \circ \end{array}$$

$$V_1 \left[\frac{1}{8} + \frac{1}{6} \right] - V_2 \left[\frac{1}{6} \right] = -12 \text{ A} + 9 \text{ A} = -3 \text{ A}$$

$$V_2 \left[\frac{1}{20} + \frac{1}{5} + \frac{1}{6} \right] - V_1 \left[\frac{1}{6} \right] = -9 \text{ A}$$

$$V_1 = \mathbf{-29.29 \text{ V}}, V_2 = \mathbf{-33.34 \text{ V}}$$

$$\begin{aligned} \text{b. } V_1 - V_{6\Omega} - 54 \text{ V} - V_2 &= 0 \\ V_{6\Omega} &= V_1 - V_2 - 54 \text{ V} = -29.29 \text{ V} - (-33.34 \text{ V}) - 54 \text{ V} = \mathbf{-49.95 \text{ V}} \end{aligned}$$

$$53. \quad \text{a. } \begin{matrix} \circ V_1 & \circ V_3 \\ & \circ V_2 \end{matrix}$$

$$V_1 \left[\frac{1}{9 \Omega} + \frac{1}{20 \Omega} + \frac{1}{20 \Omega} \right] - \frac{1}{20 \Omega} V_2 - \frac{1}{20 \Omega} V_3 = -2 \text{ A}$$

$$V_2 \left[\frac{1}{20 \Omega} + \frac{1}{20 \Omega} + \frac{1}{18 \Omega} \right] - \frac{1}{20 \Omega} V_1 - \frac{1}{20 \Omega} V_3 = 0$$

$$V_3 \left[\frac{1}{20 \Omega} + \frac{1}{20 \Omega} + \frac{1}{4 \Omega} \right] - \frac{1}{20 \Omega} V_2 - \frac{1}{20 \Omega} V_1 = 4 \text{ A}$$

$$V_1 = \mathbf{-6.64 \text{ V}}, \quad V_2 = \mathbf{1.29 \text{ V}}, \quad V_3 = \mathbf{10.66 \text{ V}}$$

$$\begin{aligned} \text{b. } \text{Original Network:} \\ V_{4\Omega} &= 16 \text{ V} - V_3 = 16 \text{ V} - 10.66 \text{ V} = 5.34 \text{ V} \\ I_{4\Omega} &= \frac{5.34 \text{ V}}{4 \Omega} = \mathbf{1.34 \text{ A}} \end{aligned}$$

$$54. \quad \text{a. } \begin{matrix} \circ V_1 & \circ V_2 & \circ V_3 \end{matrix}$$

$$\left[\frac{1}{2 \Omega} + \frac{1}{2 \Omega} \right] V_1 - \frac{1}{2 \Omega} V_2 + 0 = -5 \text{ A}$$

$$\left[\frac{1}{2 \Omega} + \frac{1}{9 \Omega} + \frac{1}{7 \Omega} + \frac{1}{2 \Omega} \right] V_2 - \frac{1}{2 \Omega} V_1 - \frac{1}{2 \Omega} V_3 = 0$$

$$\left[\frac{1}{2 \Omega} + \frac{1}{2 \Omega} + \frac{1}{4 \Omega} \right] V_3 - \frac{1}{2 \Omega} V_2 = 5 \text{ A}$$

$$\text{b. } V_1 = \mathbf{-5.31 \text{ V}}, \quad V_2 = \mathbf{-0.62 \text{ V}}, \quad V_3 = \mathbf{3.75 \text{ V}}$$

$$\text{c. } I_{9\Omega} = \frac{V_2}{9 \Omega} = \frac{0.62 \text{ V}}{9 \Omega} = \mathbf{68.89 \text{ mA}}$$

$$\begin{aligned}
 55. \quad a. \quad & V_1 \left[\frac{1}{2 \, \Omega} + \frac{1}{6 \, \Omega} \right] - \frac{1}{6 \, \Omega} V_3 = -5 \, \text{A} \\
 & V_2 \left[\frac{1}{4 \, \Omega} \right] = 5 \, \text{A} - 2 \, \text{A} \\
 & V_3 \left[\frac{1}{6 \, \Omega} + \frac{1}{5 \, \Omega} \right] - \frac{1}{6 \, \Omega} V_1 = 2 \, \text{A}
 \end{aligned}$$

$$b. \quad V_1 = -6.92 \, \text{V}, \, V_2 = 12 \, \text{V}, \, V_3 = 2.3 \, \text{V}$$

$$c. \quad I_{2\Omega} = \frac{V_2}{2 \, \Omega} = \frac{6.92 \, \text{V}}{2 \, \Omega} = 3.46 \, \text{A}$$

$$\begin{aligned}
 56. \quad a. \quad & \textcircled{V}_1 \\
 & \textcircled{V}_2 \quad \textcircled{V}_3 \\
 & V_1 \left[\frac{1}{1 \, \Omega} + \frac{1}{2 \, \Omega} + \frac{1}{2 \, \Omega} \right] - \frac{1}{2 \, \Omega} V_2 - \frac{1}{2 \, \Omega} V_3 = 12 \, \text{A} \\
 & V_2 \left[\frac{1}{2 \, \Omega} + \frac{1}{4 \, \Omega} + \frac{1}{10 \, \Omega} \right] - \frac{1}{2 \, \Omega} V_1 - \frac{1}{10 \, \Omega} V_3 = -2 \, \text{A} \\
 & V_3 \left[\frac{1}{2 \, \Omega} + \frac{1}{10 \, \Omega} + \frac{1}{8 \, \Omega} \right] - \frac{1}{2 \, \Omega} V_1 - \frac{1}{10 \, \Omega} V_2 = 2 \, \text{A}
 \end{aligned}$$

$$\begin{aligned}
 \text{and} \quad & 2V_1 - 0.5V_2 - 0.5V_3 = 12 \\
 & -0.5V_1 + 0.85V_2 - 0.1V_3 = -2 \\
 & -0.5V_1 - 0.1V_2 + 0.725V_3 = 2
 \end{aligned}$$

$$V_1 = 9.63 \, \text{V}, \, V_2 = 4.49 \, \text{V}, \, V_3 = 10.02 \, \text{V}$$

$$b. \quad V_{ab} = V_2 - V_3 = 4.49 \, \text{V} - 10.02 \, \text{V} = -5.53 \, \text{V}$$

$$\begin{aligned}
 57. \quad a. \quad & \begin{array}{c} \textcircled{I_1} \quad \textcircled{I_2} \\ \textcircled{I_3} \end{array} \\
 & I_1(6 \, \Omega + 2 \, \Omega + 10 \, \Omega) - 2 \, \Omega I_2 - 10 \, \Omega I_3 = 12 \, \text{V} \\
 & I_2(2 \, \Omega + 5 \, \Omega + 5 \, \Omega) - 2 \, \Omega I_1 - 5 \, \Omega I_3 = 0 \\
 & I_3(5 \, \Omega + 20 \, \Omega + 10 \, \Omega) - 10 \, \Omega I_1 - 5 \, \Omega I_2 = 0
 \end{aligned}$$

$$\begin{aligned}
 \text{or} \quad & 18I_1 - 2I_2 - 10I_3 = 12 \\
 & -2I_1 + 12I_2 - 5I_3 = 0 \\
 & -10I_1 - 5I_2 + 35I_3 = 0
 \end{aligned}$$

$$I_1 = 850.99 \, \text{mA}, \, I_2 = 258.53 \, \text{mA}, \, I_3 = 280.07 \, \text{mA}$$

b. $I_{R_5} = I_3 - I_2 = 280.07 \text{ mA} - 258.53 \text{ mA} = \mathbf{21.54 \text{ mA}}$

c. **no**

d. **no** $2 \Omega / 10 \Omega = \frac{1}{5} \neq 5 \Omega / 20 \Omega = \frac{1}{4}$

58. a. $\circ V_1$

$\circ V_2$ $\circ V_3$

$$V_1 \left[\frac{1}{6 \Omega} + \frac{1}{2 \Omega} + \frac{1}{5 \Omega} \right] - \frac{1}{2 \Omega} V_2 - \frac{1}{5 \Omega} V_3 = 2 \text{ A}$$

$$V_2 \left[\frac{1}{2 \Omega} + \frac{1}{5 \Omega} + \frac{1}{10 \Omega} \right] - \frac{1}{2 \Omega} V_1 - \frac{1}{5 \Omega} V_3 = 0$$

$$V_3 \left[\frac{1}{5 \Omega} + \frac{1}{5 \Omega} + \frac{1}{20 \Omega} \right] - \frac{1}{5 \Omega} V_2 - \frac{1}{5 \Omega} V_1 = 0$$

or $0.867V_1 - 0.5V_2 - 0.2V_3 = 12$

$$-0.5V_1 + 0.8V_2 - 0.2V_3 = 0$$

$$-0.2V_1 - 0.2V_2 + 0.45V_3 = 0$$

$$V_2 = \mathbf{5.7 \text{ V}}, V_3 = \mathbf{5.6 \text{ V}}$$

b. $V_{5\Omega} = V_2 - V_3 = 5.7 \text{ V} - 5.6 \text{ V} = \mathbf{0.1 \text{ V}}$

c. **no**

d. **no** $2 \Omega / 10 \Omega = \frac{1}{5} \neq 5 \Omega / 20 \Omega = \frac{1}{4}$

59. a. 

Source conversion: $E = IR = (12 \text{ m})(2 \text{ k}\Omega) = 24 \text{ V}$
 $R_s = 2 \text{ k}\Omega$

$$I_1(2 \text{ k}\Omega + 33 \text{ k}\Omega + 3.3 \text{ k}\Omega) - 33 \text{ k}\Omega I_2 - 3.3 \text{ k}\Omega I_3 = 24 \text{ V}$$

$$I_2(33 \text{ k}\Omega + 56 \text{ k}\Omega + 36 \text{ k}\Omega) - 33 \text{ k}\Omega I_1 - 36 \text{ k}\Omega I_3 = 0$$

$$I_3(36 \text{ k}\Omega + 3.3 \text{ k}\Omega + 5.6 \text{ k}\Omega) - 3.3 \text{ k}\Omega I_1 - 36 \text{ k}\Omega I_2 = 0$$

$$I_1 = \mathbf{0.97 \text{ mA}}, I_2 = I_3 = \mathbf{0.36 \text{ mA}}$$

b. $I_5 = I_2 - I_3 = 0.36 \text{ mA} - 0.36 \text{ mA} = \mathbf{0 \text{ mA}}$

c, d. **yes**

60. a. $\begin{matrix} & \circ V_1 \\ \circ V_2 & & \circ V_3 \end{matrix}$

$$\begin{aligned} V_1 \left[\frac{1}{2 \text{ k}\Omega} + \frac{1}{33 \text{ k}\Omega} + \frac{1}{56 \text{ k}\Omega} \right] - \frac{1}{33 \text{ k}\Omega} V_2 - \frac{1}{56 \text{ k}\Omega} V_3 &= 12 \text{ mA} \\ V_2 \left[\frac{1}{33 \text{ k}\Omega} + \frac{1}{3.3 \text{ k}\Omega} + \frac{1}{36 \text{ k}\Omega} \right] - \frac{1}{33 \text{ k}\Omega} V_1 - \frac{1}{36 \text{ k}\Omega} V_3 &= 0 \\ V_3 \left[\frac{1}{56 \text{ k}\Omega} + \frac{1}{36 \text{ k}\Omega} + \frac{1}{5.6 \text{ k}\Omega} \right] - \frac{1}{36 \text{ k}\Omega} V_2 - \frac{1}{56 \text{ k}\Omega} V_1 &= 0 \end{aligned}$$

$$\begin{aligned} \text{Rewritten: } 548.16V_1 - 30.3V_2 - 17.86V_3 &= 12 \times 10^3 \\ -30.3V_1 + 361.11V_2 - 27.78V_3 &= 0 \\ -17.86V_1 - 27.78V_2 + 224.21V_3 &= 0 \end{aligned}$$

$$V_2 = \mathbf{2.01 \text{ V}}, V_3 = \mathbf{2.01 \text{ V}}$$

b. $V_{R_5} = V_2 - V_3 = 2.01 \text{ V} - 2.01 \text{ V} = \mathbf{0 \text{ V}}$

c, d. **yes**

61. Mesh Analysis

$$\begin{matrix} I_2 \downarrow \\ I_1 \downarrow \\ I_3 \downarrow \end{matrix} \quad \begin{aligned} (1 \text{ k}\Omega + 2 \text{ k}\Omega + 2 \text{ k}\Omega)I_1 - 2 \text{ k}\Omega I_2 - 2 \text{ k}\Omega I_3 &= 10 \\ (2 \text{ k}\Omega + 2 \text{ k}\Omega + 2 \text{ k}\Omega)I_2 - 2 \text{ k}\Omega I_1 - 2 \text{ k}\Omega I_3 &= 0 \\ (2 \text{ k}\Omega + 2 \text{ k}\Omega + 2 \text{ k}\Omega)I_3 - 2 \text{ k}\Omega I_1 - 2 \text{ k}\Omega I_2 &= 0 \end{aligned}$$

$$I_1 = I_{10V} = \mathbf{3.33 \text{ mA}}$$

Nodal Analysis:

Source conversion: $I = 10 \text{ V}/1 \text{ k}\Omega = 10 \text{ mA}$, $R = 1 \text{ k}\Omega$

$$\begin{matrix} \circ V_1 \\ \circ V_3 & \circ V_2 \\ \text{---} \end{matrix} \quad \begin{aligned} V_1 \left[\frac{1}{1 \text{ k}\Omega} + \frac{1}{2 \text{ k}\Omega} + \frac{1}{2 \text{ k}\Omega} \right] - \frac{1}{2 \text{ k}\Omega} V_2 - \frac{1}{2 \text{ k}\Omega} V_3 &= 10 \text{ mA} \\ V_2 \left[\frac{1}{2 \text{ k}\Omega} + \frac{1}{2 \text{ k}\Omega} + \frac{1}{2 \text{ k}\Omega} \right] - \frac{1}{2 \text{ k}\Omega} V_1 - \frac{1}{2 \text{ k}\Omega} V_3 &= 0 \\ V_3 \left[\frac{1}{2 \text{ k}\Omega} + \frac{1}{2 \text{ k}\Omega} + \frac{1}{2 \text{ k}\Omega} \right] - \frac{1}{2 \text{ k}\Omega} V_2 - \frac{1}{2 \text{ k}\Omega} V_1 &= 0 \end{aligned}$$

$$V_1 = 6.67 \text{ V} = E - IR_s = 10 \text{ V} - I(1 \text{ k}\Omega)$$

$$I = \frac{10 - 6.67 \text{ V}}{1 \text{ k}\Omega} = \mathbf{3.33 \text{ mA}}$$

62.

Mesh Analysis

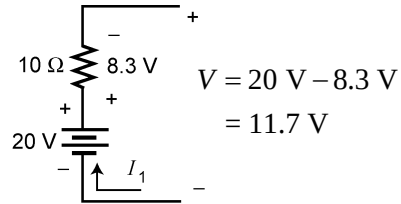
Source conversion: $E = 20 \text{ V}$, $R = 10 \Omega$

$$(10 + 10 + 20)I_1 - 10I_2 - 20I_3 = 20$$

$$(10 + 20 + 20)I_2 - 10I_1 - 20I_3 = 0$$

$$(20 + 20 + 10)I_3 - 20I_1 - 20I_2 = 0$$

$$I_1 = I_{20V} = 0.83 \text{ A}$$



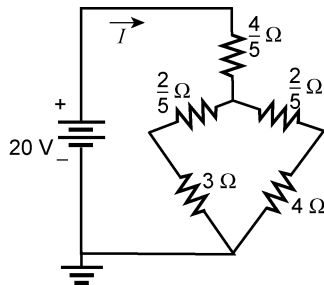
$$I_s = \frac{V}{R_s} = \frac{11.70 \text{ V}}{10 \Omega} = \mathbf{1.17 \text{ A}}$$

Nodal Analysis:

$$\begin{aligned} \text{Node } 1: & V_1 \left[\frac{1}{10} + \frac{1}{10} + \frac{1}{20} \right] - \left[\frac{1}{20} \right] V_2 - \left[\frac{1}{10} \right] V_3 = 2 \\ \text{Node } 2: & V_2 \left[\frac{1}{20} + \frac{1}{20} + \frac{1}{10} \right] - \left[\frac{1}{20} \right] V_1 - \left[\frac{1}{20} \right] V_3 = 0 \\ \text{Node } 3: & V_3 \left[\frac{1}{10} + \frac{1}{20} + \frac{1}{20} \right] - \left[\frac{1}{10} \right] V_1 - \left[\frac{1}{20} \right] V_2 = 0 \end{aligned}$$

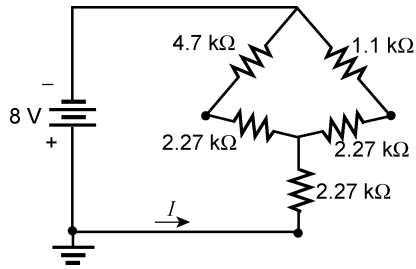
$$I_{R_s} = \frac{V_1}{R_s} = \mathbf{1.17 \text{ A}}$$

63.



$$\begin{aligned} I &= \frac{20 \text{ V}}{\frac{4}{5} \Omega + \left[\frac{2}{5} \Omega + 3 \Omega \right] \parallel \left[\frac{2}{5} \Omega + 4 \Omega \right]} \\ &= \frac{20 \text{ V}}{\frac{4}{5} \Omega + (3.14 \Omega) \parallel (4.4 \Omega)} \\ &= \mathbf{7.36 \text{ A}} \end{aligned}$$

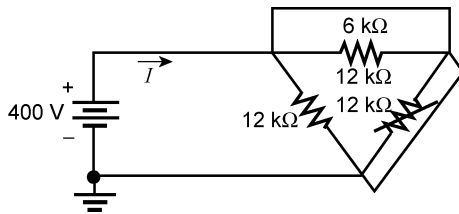
64.



$$\begin{aligned} R_T &= 2.27 \text{ k}\Omega + [4.7 \text{ k}\Omega + 2.27 \text{ k}\Omega] \parallel [1.1 \text{ k}\Omega + 2.27 \text{ k}\Omega] \\ &= 2.27 \text{ k}\Omega + [6.97 \text{ k}\Omega] \parallel [3.37 \text{ k}\Omega] \\ &= 2.27 \text{ k}\Omega + 2.27 \text{ k}\Omega \\ &= 4.54 \text{ k}\Omega \end{aligned}$$

$$I = \frac{8 \text{ V}}{4.54 \text{ k}\Omega} = \mathbf{1.76 \text{ mA}}$$

65.



(Y-Δ conversion)

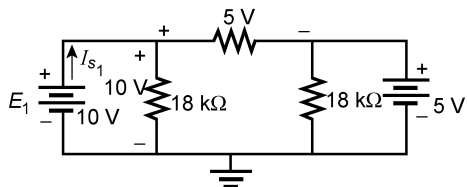
$$\begin{aligned} I &= \frac{400 \text{ V}}{12 \text{ k}\Omega \parallel 12 \text{ k}\Omega \parallel 6 \text{ k}\Omega} = \frac{400 \text{ V}}{3 \text{ k}\Omega} \\ &= \mathbf{133.33 \text{ mA}} \end{aligned}$$

66.

$$\begin{aligned} \text{a. } I &= \frac{42 \text{ V}}{(18 \Omega \parallel 18 \Omega) \parallel [(18 \Omega \parallel 18 \Omega) + (18 \Omega \parallel 18 \Omega)]} = \frac{42 \text{ V}}{9 \Omega \parallel [9 \Omega + 9 \Omega]} \\ &= \mathbf{7 \text{ A}} \text{ (Y-}\Delta \text{ conversion)} \end{aligned}$$

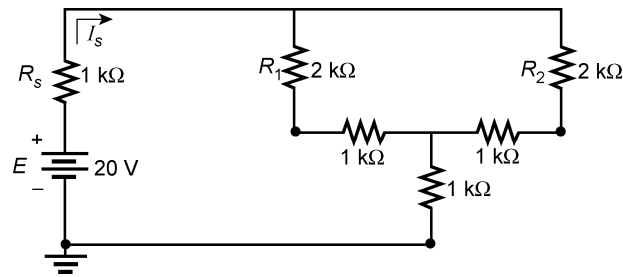
b. Δ – Y conversion

67.



$$I_{s_1} = \frac{10 \text{ V}}{18 \text{ k}\Omega} + \frac{5 \text{ V}}{18 \text{ k}\Omega} = \frac{15 \text{ V}}{18 \text{ k}\Omega} = \mathbf{0.83 \text{ mA}}$$

68. a.



b.

$$R' = R_1 + 1 \text{ k}\Omega = 3 \text{ k}\Omega$$

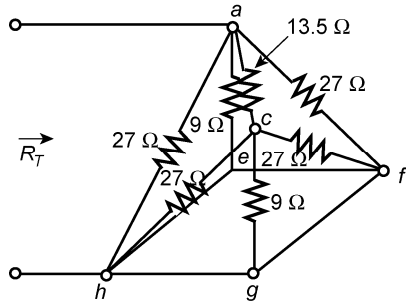
$$R'' = R_2 + 1 \text{ k}\Omega = 3 \text{ k}\Omega$$

$$R'_T = \frac{3 \text{ k}\Omega}{2} = 1.5 \text{ k}\Omega$$

$$R_T = 1 \text{ k}\Omega + 1.5 \text{ k}\Omega + 1 \text{ k}\Omega = 3.5 \text{ k}\Omega$$

$$I_s = \frac{E}{R_T} = \frac{20 \text{ V}}{3.5 \text{ k}\Omega} = \mathbf{5.71 \text{ mA}}$$

69. Using two $\Delta - Y$ conversions:



$$c - g: 27 \Omega \parallel 9 \Omega \parallel 27 \Omega = 5.4 \Omega$$

$$a - h: 27 \Omega \parallel 9 \Omega \parallel 27 \Omega = 5.4 \Omega$$

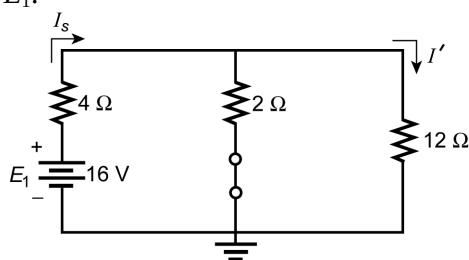
$$R_T = 5.4 \Omega \parallel (13.5 \Omega + 5.4 \Omega)$$

$$= 5.4 \Omega \parallel 18.9 \Omega$$

$$= \mathbf{4.2 \Omega}$$

Chapter 9

1. a. E_1 :



$$R_T \text{ (from source)} = 4\Omega + 2\Omega \parallel 12\Omega$$

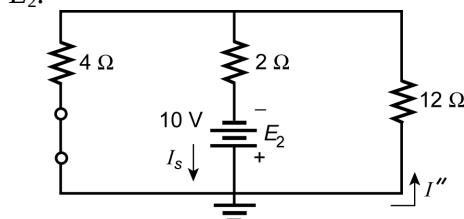
$$= 4\Omega + 1.71\Omega$$

$$= 5.71\Omega$$

$$I_s = \frac{E_1}{R_T} = \frac{16\text{ V}}{5.71\Omega} = 2.8\text{ A}$$

$$I'_{12\Omega} = \frac{2\Omega(2.8\text{ A})}{2\Omega + 12\Omega} = 0.4\text{ A}$$

E_2 :



$$R_T \text{ (from source)} = 2\Omega + 4\Omega \parallel 12\Omega$$

$$= 2\Omega + 3\Omega$$

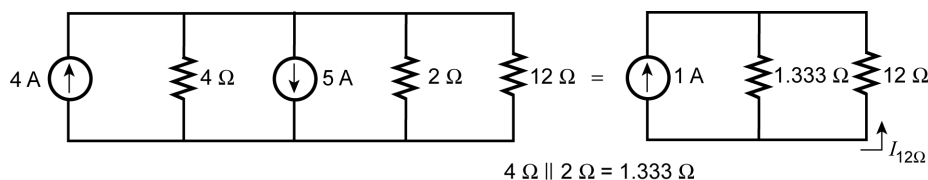
$$= 5\Omega$$

$$I_s = \frac{E_2}{R_T} = \frac{10\text{ V}}{5\Omega} = 2\text{ A}$$

$$I''_{12\Omega} = \frac{4\Omega(2\text{ A})}{4\Omega + 12\Omega} = 0.5\text{ A}$$

$$I_{12\Omega} \uparrow = 0.5\text{ A} - 0.4\text{ A} = \mathbf{0.1\text{ A}}$$

b.

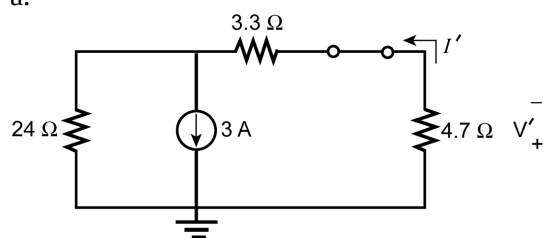


$$I_{12\Omega} = \frac{1.333\Omega(1\text{A})}{1.333\Omega + 12\Omega} = \mathbf{0.1\text{ A}}$$

c. the same

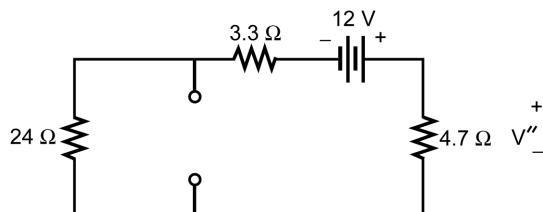
2.

a.



$$I' = \frac{24\Omega(3\text{ A})}{24\Omega + 8\Omega} = 2.25\text{ A}$$

$$V' = I'R = (2.25)(4.7\Omega) = 10.575\text{ V}$$



$$V'' = \frac{4.7\Omega(12\text{ V})}{4.7\Omega + 3.3\Omega + 24\Omega} = 1.763\text{ V}$$

$$V''_+ = 10.575\text{ V} - 1.763\text{ V} = \mathbf{8.81\text{ V}}$$

$$\text{b. } P = \frac{V'^2}{R} = \frac{(10.575 \text{ V})^2}{4.7 \Omega} = \mathbf{23.79 \text{ W}}$$

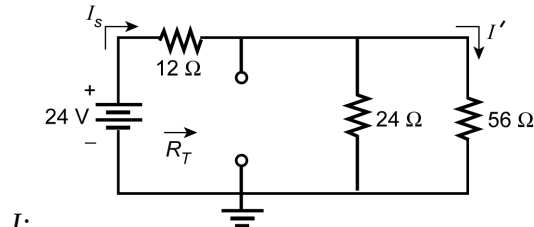
$$\text{c. } P = \frac{V''^2}{R} = \frac{(1.763 \text{ V})^2}{4.7 \Omega} = \mathbf{0.661 \text{ W}}$$

$$\text{d. } P = \frac{V^2}{R} = \frac{(8.81 \text{ V})^2}{4.7 \Omega} = \mathbf{16.51 \text{ W}}$$

$$\text{e. } 23.79 \text{ W} + 0.661 \text{ W} \neq 16.51 \text{ W}$$

$$24.45 \text{ W} \neq 16.51 \text{ W}$$

3. E :



$$R_T = 12 \Omega + 24 \Omega \parallel 56 \Omega = 28.8 \Omega$$

$$I_s = \frac{E}{R_T} = \frac{24 \text{ V}}{28.8 \Omega} = 0.833 \text{ A}$$

$$I'_{56\Omega} = \frac{24 \Omega (0.833 \text{ A})}{24 \Omega + 56 \Omega} = 0.25 \text{ A}$$

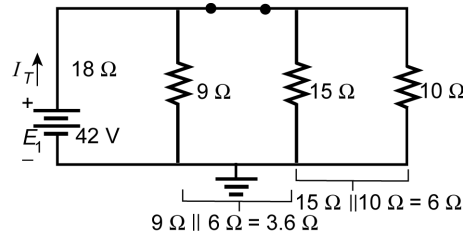
$$24 \Omega \parallel 56 \Omega = 16.8 \Omega$$

$$I''' = \frac{12 \Omega (8 \text{ A})}{12 \Omega + 16.8 \Omega} = 3.33 \text{ A}$$

$$I''_{56\Omega} = \frac{24 \Omega (3.33 \text{ A})}{24 \Omega + 56 \Omega} = 1 \text{ A}$$

$$I_{56\Omega} = I' + I'' = 0.25 \text{ A} + 1 \text{ A} = \mathbf{1.25 \text{ A} \downarrow}$$

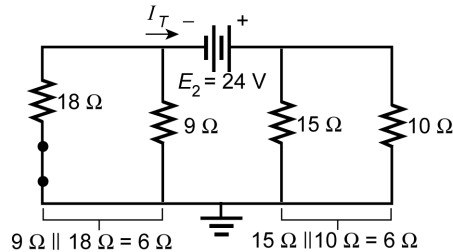
4. E_1 :



$$I_T = \frac{42 \text{ V}}{18 \Omega + 3.6 \Omega} = 1.944 \text{ A}$$

$$I_1 = \frac{9 \Omega (I_T)}{9 \Omega + 6 \Omega} = \frac{9 \Omega (1.944 \text{ A})}{15 \Omega} = 1.17 \text{ A}$$

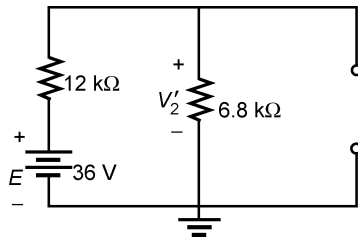
E_2 :



$$I_T = \frac{E_2}{R_T} = \frac{24 \text{ V}}{12 \Omega} = 2 \text{ A}$$

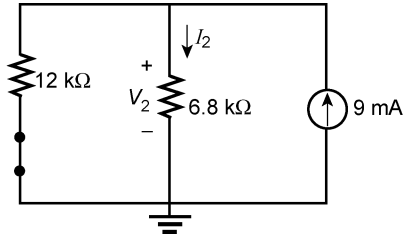
$$I_{24\text{V}} = I_T + I_1 = 2 \text{ A} + 1.17 \text{ A} = \mathbf{3.17 \text{ A (dir. of } I_1)}$$

5. E :



$$V'_2 = \frac{6.8 \text{ k}\Omega (36 \text{ V})}{6.8 \text{ k}\Omega + 12 \text{ k}\Omega} = 13.02 \text{ V}$$

I :

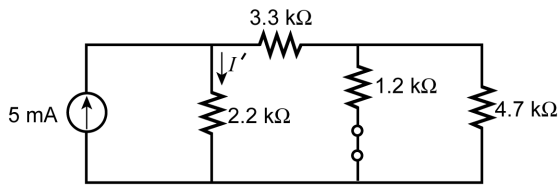


$$I_2 = \frac{12 \text{ k}\Omega (9 \text{ mA})}{12 \text{ k}\Omega + 6.8 \text{ k}\Omega} = 5.75 \text{ mA}$$

$$V_2'' = I_2 R_2 = (5.75 \text{ mA})(6.8 \text{ k}\Omega) = 39.10 \text{ V}$$

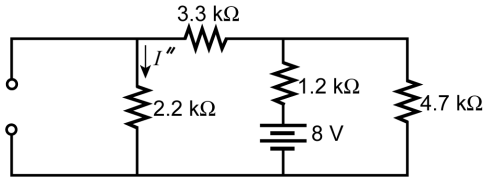
$$V_2 = V'_2 + V_2'' = 13.02 \text{ V} + 39.10 \text{ V} = \mathbf{52.12 \text{ V}}$$

6. I :



$$\begin{aligned} 1.2 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega &= 0.956 \text{ k}\Omega \\ 3.3 \text{ k}\Omega + 0.956 \text{ k}\Omega &= 4.256 \text{ k}\Omega \\ I' &= \frac{4.256 \text{ k}\Omega (5 \text{ mA})}{2.2 \text{ k}\Omega + 4.256 \text{ k}\Omega} \\ &= 3.3 \text{ mA} \end{aligned}$$

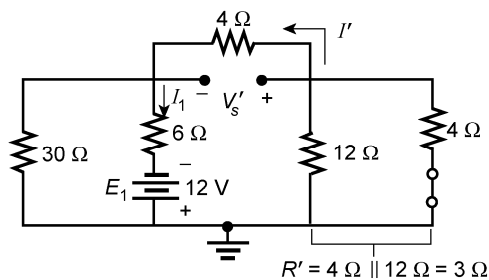
E :



$$\begin{aligned} 2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega &= 5.5 \text{ k}\Omega \\ 5.5 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega &= 2.53 \text{ k}\Omega \\ R_T &= 2.54 \text{ k}\Omega + 1.2 \text{ k}\Omega = 3.73 \text{ k}\Omega \\ I_s &= \frac{8 \text{ V}}{3.73 \text{ k}\Omega} = 2.14 \text{ mA} \\ I'' &= \frac{4.7 \text{ k}\Omega (2.14 \text{ mA})}{4.7 \text{ k}\Omega + 5.5 \text{ k}\Omega} = 0.986 \text{ mA} \end{aligned}$$

$$I = I' + I'' = 3.3 \text{ mA} + 0.986 \text{ mA} = \mathbf{4.286 \text{ mA}}$$

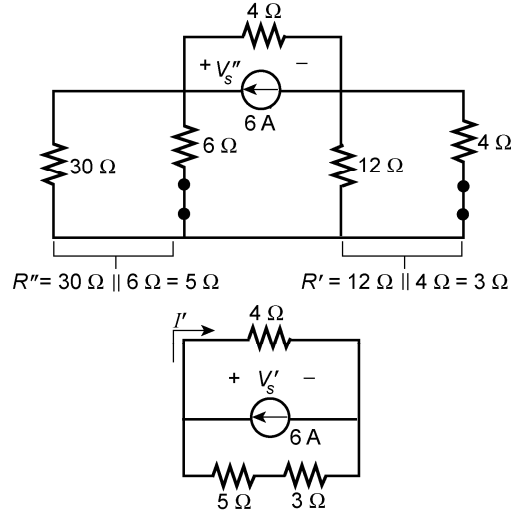
7. E_1 :



$$I_1 = \frac{E_1}{R_T} = \frac{12 \text{ V}}{6 \Omega + 5.88 \Omega} = 1.03 \text{ A}$$

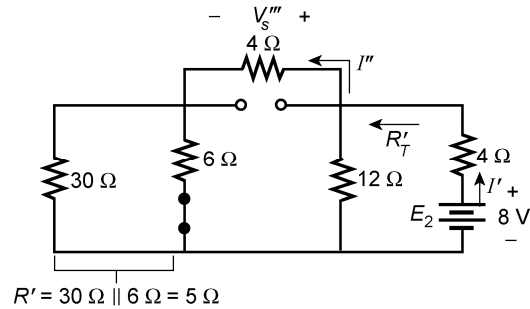
$$\begin{aligned}
 I' &= \frac{30 \, \Omega (I_1)}{30 \, \Omega + 7 \, \Omega} = \frac{30 \, \Omega (1.03 \, \text{A})}{37 \, \Omega} \\
 &= 835.14 \, \text{mA} \\
 V_s' &= I'(4 \, \Omega) = (835.14 \, \text{mA})(4 \, \Omega) \\
 &= 3.34 \, \text{V}
 \end{aligned}$$

I :



$$\begin{aligned}
 I' &= \frac{8 \, \Omega (6 \, \text{A})}{8 \, \Omega + 4 \, \Omega} = 4 \, \text{A} \\
 V_s'' &= I'(4 \, \Omega) = 4 \, \text{A}(4 \, \Omega) = 16 \, \text{V}
 \end{aligned}$$

E_2 :



$$\begin{aligned}
 R_T' &= 12 \, \Omega \parallel (4 \, \Omega + 5 \, \Omega) = 12 \, \Omega \parallel 9 \, \Omega = 5.14 \, \Omega \\
 I' &= \frac{E_2}{R_T' + 4 \, \Omega} = \frac{8 \, \text{V}}{4 \, \Omega + 5.14 \, \Omega} = 0.875 \, \text{A} \\
 I'' &= \frac{12 \, \Omega (I')}{12 \, \Omega + 9 \, \Omega} = \frac{12 \, \Omega (0.875 \, \text{A})}{21 \, \Omega} = 0.5 \, \text{A} \\
 V_s''' &= I''(4 \, \Omega) = 0.5 \, \text{A}(4 \, \Omega) = 2 \, \text{V} \\
 V_s &= V_s'' - V_s' - V_s''' = 16 \, \text{V} - 3.34 \, \text{V} - 2 \, \text{V} = \mathbf{10.66 \, \text{V}}
 \end{aligned}$$

8. a. $R_{Th} = R_3 + R_1 \parallel R_2 = 4 \, \Omega + 6 \, \Omega \parallel 3 \, \Omega = 4 \, \Omega + 2 \, \Omega = \mathbf{6 \, \Omega}$

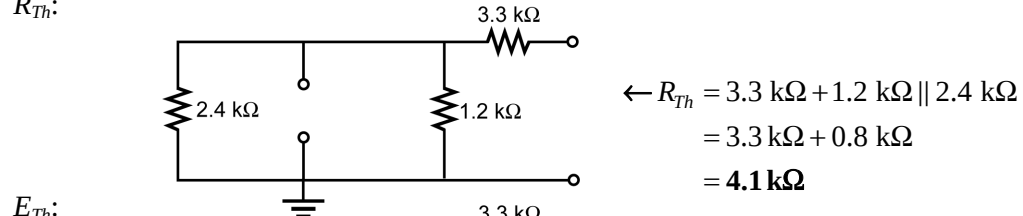
$$E_{Th} = \frac{R_2 E}{R_2 + R_1} = \frac{3 \, \Omega (18 \, \text{V})}{3 \, \Omega + 6 \, \Omega} = \mathbf{6 \, \text{V}}$$

$$b. \quad I_1 = \frac{E_{Th}}{R_{Th} + R} = \frac{6 \text{ V}}{6 \Omega + 2 \Omega} = \mathbf{0.75 \text{ A}}$$

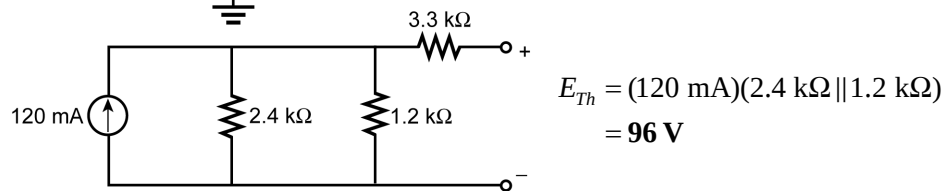
$$I_2 = \frac{6 \text{ V}}{6 \Omega + 30 \Omega} = \mathbf{166.67 \text{ mA}}$$

$$I_3 = \frac{6 \text{ V}}{6 \Omega + 100 \Omega} = \mathbf{56.60 \text{ mA}}$$

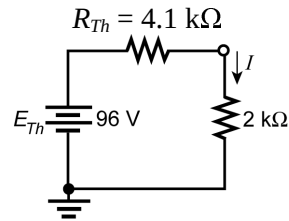
9. a. R_{Th} :



E_{Th} :



b.



$$I = \frac{96 \text{ V}}{6.1 \text{ k}\Omega} = 15.74 \text{ mA}$$

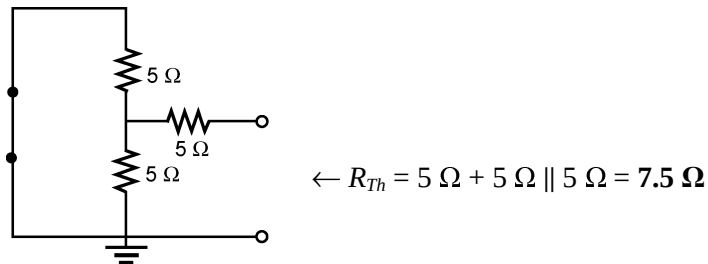
$$P = I^2 R = (15.74 \text{ mA})^2 2 \text{ k}\Omega = \mathbf{0.495 \text{ W}}$$

$R = 100 \text{ k}\Omega$:

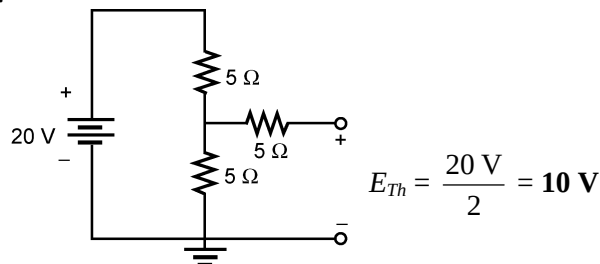
$$I = \frac{96 \text{ V}}{104.1 \text{ k}\Omega} = 0.922 \text{ mA}$$

$$P = I^2 R = (0.922 \text{ mA})^2 100 \text{ k}\Omega = \mathbf{85 \text{ mW}}$$

10. a. R_{Th} :



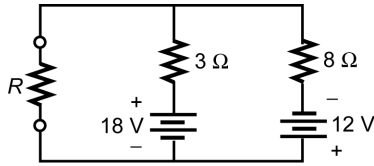
E_{Th} :



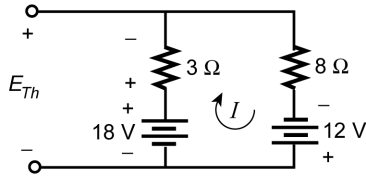
$$\text{b. } R = 2 \, \Omega: P = \left(\frac{E_{Th}}{R_{Th} + R} \right)^2 R = \left(\frac{10 \, \text{V}}{7.5 \, \Omega + 2 \, \Omega} \right)^2 2 \, \Omega = \mathbf{2.22 \, \text{W}}$$

$$R = 100 \, \Omega: P = \left(\frac{10 \, \text{V}}{7.5 \, \Omega + 100 \, \Omega} \right)^2 100 \, \Omega = \mathbf{0.87 \, \text{W}}$$

11.



$$R_{Th} = 3 \, \Omega \parallel 8 \, \Omega = \mathbf{2.18 \, \Omega}$$

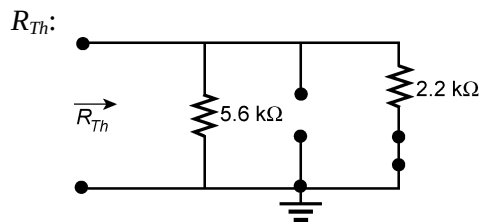


$$I = \frac{18 \, \text{V} + 12 \, \text{V}}{3 \, \Omega + 8 \, \Omega} = 2.73 \, \text{A}$$

$$V_{3\Omega} = IR = (2.73 \, \text{A})(3 \, \Omega) = 8.19 \, \text{V}$$

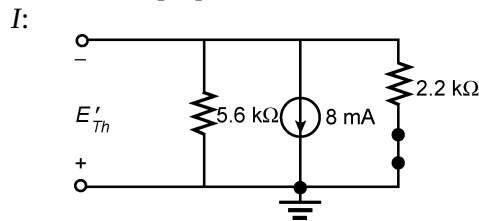
$$E_{Th} = 18 \, \text{V} - 8.19 \, \text{V} = \mathbf{9.81 \, \text{V}}$$

12.



$$R_{Th} = 5.6 \, \text{k}\Omega \parallel 2.2 \, \text{k}\Omega = \mathbf{1.58 \, \text{k}\Omega}$$

E_{Th} : Superposition:

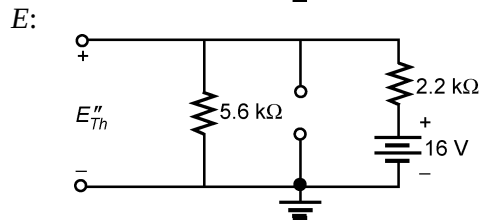


$$E'_{Th} = IR_T$$

$$= 8 \, \text{mA}(5.6 \, \text{k}\Omega \parallel 2.2 \, \text{k}\Omega)$$

$$= 8 \, \text{mA}(1.579 \, \text{k}\Omega)$$

$$= 12.64 \, \text{V}$$

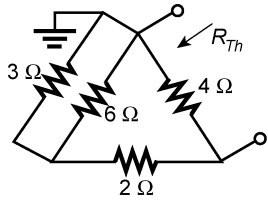


$$E''_{Th} = \frac{5.6 \, \text{k}\Omega(16 \, \text{V})}{5.6 \, \text{k}\Omega + 2.2 \, \text{k}\Omega}$$

$$= 11.49 \, \text{V}$$

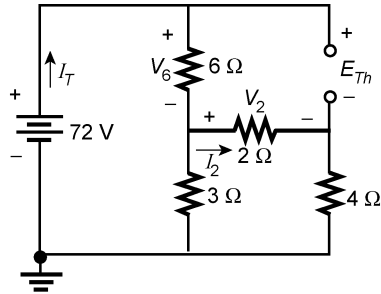
$$E_{Th} = 11.49 \, \text{V} - 12.64 \, \text{V} = \mathbf{-1.15 \, \text{V}}$$

13. R_{Th} :



$$R_{Th} = 4\ \Omega \parallel (2\ \Omega + 6\ \Omega \parallel 3\ \Omega) = 2\ \Omega$$

E_{Th} :

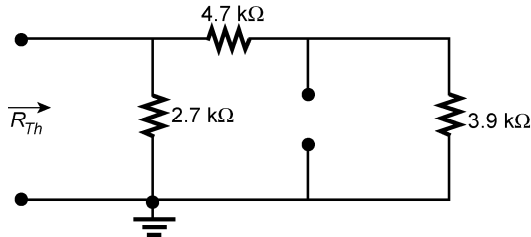


$$I_T = \frac{72\text{ V}}{6\ \Omega + 3\ \Omega \parallel (2\ \Omega + 4\ \Omega)} = 9\text{ A}$$

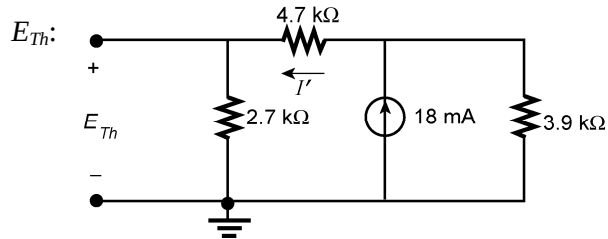
$$I_2 = \frac{3\ \Omega(I_T)}{3\ \Omega + 6\ \Omega} = \frac{3\ \Omega(9\text{ A})}{9\ \Omega} = 3\text{ A}$$

$$E_{Th} = V_6 + V_2 = (I_T)(6\ \Omega) + I_2(2\ \Omega) = (9\text{ A})(6\ \Omega) + (3\text{ A})(2\ \Omega) = 60\text{ V}$$

14. a. R_{Th} :



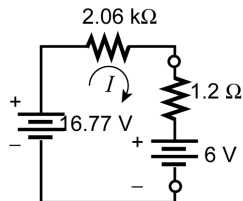
$$R_{Th} = 2.7\text{ k}\Omega \parallel (4.7\text{ k}\Omega + 3.9\text{ k}\Omega) = 2.7\text{ k}\Omega \parallel 8.6\text{ k}\Omega = 2.06\text{ k}\Omega$$



$$I' = \frac{3.9\text{ k}\Omega(18\text{ mA})}{3.9\text{ k}\Omega + 7.4\text{ k}\Omega} = 6.21\text{ mA}$$

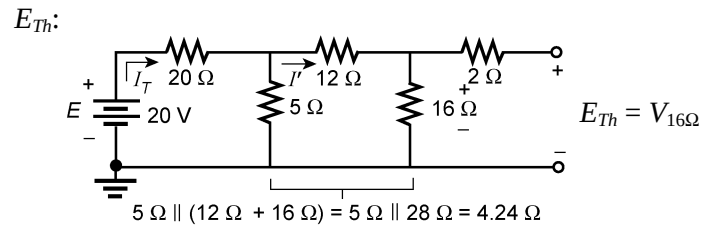
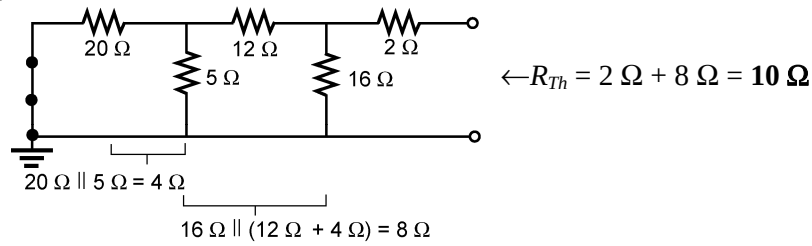
$$E_{Th} = I'(2.7\text{ k}\Omega) = (6.21\text{ mA})(2.7\text{ k}\Omega) = 16.77\text{ V}$$

b.



$$I = \frac{16.77\text{ V}}{2.06\text{ k}\Omega + 1.2\text{ k}\Omega} = \frac{16.77\text{ V}}{3.26\text{ k}\Omega} = 5.14\text{ mA}$$

15. a. R_{Th} :



$$I_T = \frac{20 \text{ V}}{20 \Omega + 4.24 \Omega} = 825.08 \text{ mA}$$

$$I' = \frac{5 \Omega (I_T)}{5 \Omega + 28 \Omega} = \frac{5 \Omega (825.08 \text{ mA})}{33 \Omega} = 125.01 \text{ mA}$$

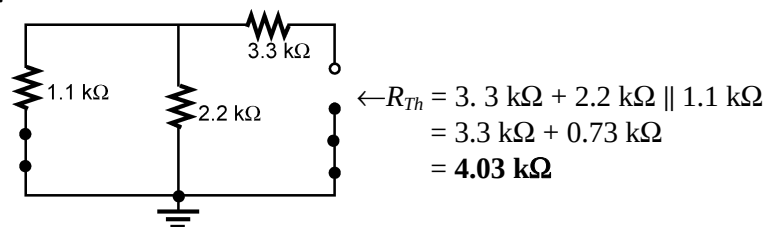
$$E_{Th} = V_{16\Omega} = (I')(16\ \Omega) = (125.01\ \text{mA})(16\ \Omega) = \mathbf{2\ V}$$

$$\text{b. } 20 \, \Omega: \quad I = \frac{E_{Th}}{R_{Th} + R} = \frac{2 \, \text{V}}{10 \, \Omega + 20 \, \Omega} = \frac{2 \, \text{V}}{30 \, \Omega} = \mathbf{66.67 \, \text{mA}}$$

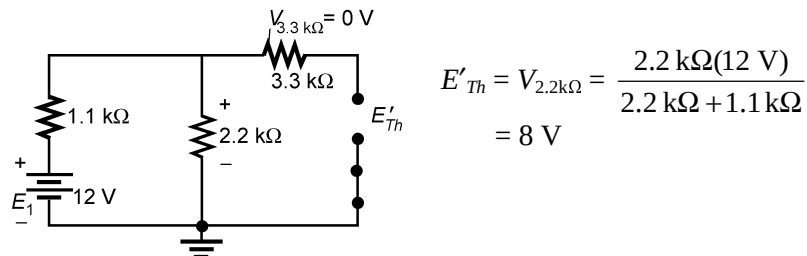
$$50\ \Omega: \quad I = \frac{2\ \text{V}}{10\ \Omega + 50\ \Omega} = \frac{2\ \text{V}}{60\ \Omega} = 33.33\ \text{mA}$$

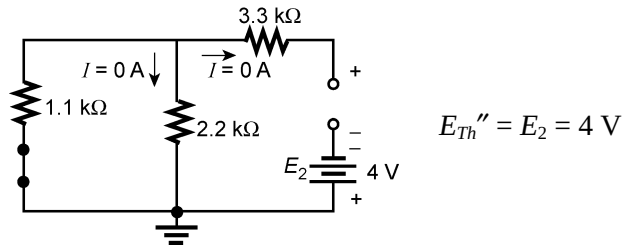
$$100\ \Omega: \quad I = \frac{2\text{ V}}{10\ \Omega + 100\ \Omega} = \frac{2\text{ V}}{110\ \Omega} = \mathbf{18.18\text{ mA}}$$

16. a. R_{Th} :



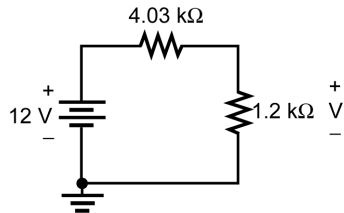
E_{Th} : Superposition:

 $E_1:$ 



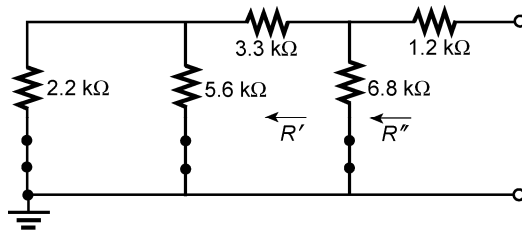
$$E_{Th} = E'_{Th} + E''_{Th} = 8 \text{ V} + 4 \text{ V} = 12 \text{ V}$$

b.



$$V = \frac{1.2 \text{ k}\Omega (12 \text{ V})}{1.2 \text{ k}\Omega + 4.03 \text{ k}\Omega} = 2.75 \text{ V}$$

17. R_{Th} :



$$\begin{aligned} \leftarrow R_{Th} &= 2.2 \text{ k}\Omega \parallel 5.6 \text{ k}\Omega = 1.58 \text{ k}\Omega \\ R' &= 1.58 \text{ k}\Omega + 3.3 \text{ k}\Omega \\ &= 4.88 \text{ k}\Omega \end{aligned}$$

$$R'' = 4.88 \text{ k}\Omega \parallel 6.8 \text{ k}\Omega = 2.84 \text{ k}\Omega$$

$$R_{Th} = 1.2 \text{ k}\Omega + R'' = 1.2 \text{ k}\Omega + 2.84 \text{ k}\Omega = 4.04 \text{ k}\Omega$$

E_{Th} : Source conversions:

$$I_1 = \frac{22 \text{ V}}{2.2 \text{ k}\Omega} = 10 \text{ mA}, R_s = 2.2 \text{ k}\Omega$$

$$I_2 = \frac{12 \text{ V}}{5.6 \text{ k}\Omega} = 2.14 \text{ mA}, R_s = 5.6 \text{ k}\Omega$$

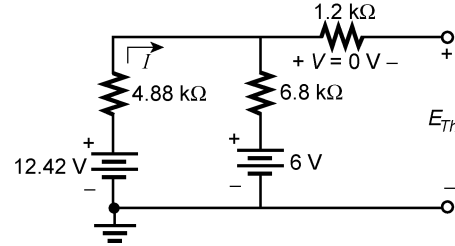
Combining parallel current sources: $I'_T = I_1 - I_2 = 10 \text{ mA} - 2.14 \text{ mA} = 7.86 \text{ mA}$

$$2.2 \text{ k}\Omega \parallel 5.6 \text{ k}\Omega = 1.58 \text{ k}\Omega$$

Source conversion:

$$E = (7.86 \text{ mA})(1.58 \text{ k}\Omega) = 12.42 \text{ V}$$

$$R' = R_s + 3.3 \text{ k}\Omega = 1.58 \text{ k}\Omega + 3.3 \text{ k}\Omega = 4.88 \text{ k}\Omega$$

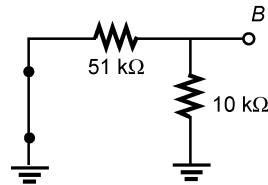


$$I = \frac{12.42 \text{ V} - 6 \text{ V}}{4.88 \text{ k}\Omega + 6.8 \text{ k}\Omega} = \frac{6.42 \text{ V}}{11.68 \text{ k}\Omega} = 549.66 \mu\text{A}$$

$$V_{6.8\text{k}\Omega} = I(6.8 \text{ k}\Omega) = (549.66 \mu\text{A})(6.8 \text{ k}\Omega) = 3.74 \text{ V}$$

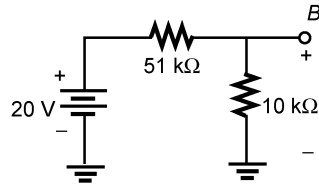
$$E_{Th} = 6 \text{ V} + V_{6.8\text{k}\Omega} = 6 \text{ V} + 3.74 \text{ V} = \mathbf{9.74 \text{ V}}$$

18. a. R_{Th} :



$$\leftarrow R_{Th} = 51 \text{ k}\Omega \parallel 10 \text{ k}\Omega = \mathbf{8.36 \text{ k}\Omega}$$

E_{Th} :



$$E_{Th} = \frac{10 \text{ k}\Omega(20 \text{ V})}{10 \text{ k}\Omega + 51 \text{ k}\Omega} = 3.28 \text{ V}$$

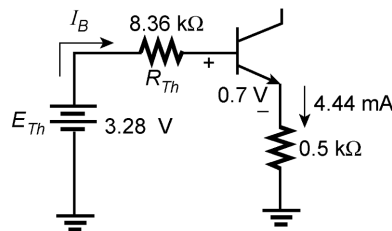
b. $I_E R_E + V_{CE} + I_C R_C = 20 \text{ V}$

but $I_C = I_E$

and $I_E(R_C + R_E) + V_{CE} = 20 \text{ V}$

$$\text{or } I_E = \frac{20 \text{ V} - V_{CE}}{R_C + R_E} = \frac{20 \text{ V} - 8 \text{ V}}{2.2 \text{ k}\Omega + 0.5 \text{ k}\Omega} = \frac{12 \text{ V}}{2.7 \text{ k}\Omega} = \mathbf{4.44 \text{ mA}}$$

c.



$$E_{Th} - I_B R_{Th} - V_{BE} - V_E = 0$$

$$\begin{aligned} \text{and } I_B &= \frac{E_{Th} - V_{BE} - V_E}{R_{Th}} = \frac{3.28 \text{ V} - 0.7 \text{ V} - (4.44 \text{ mA})(0.5 \text{ k}\Omega)}{8.36 \text{ k}\Omega} \\ &= \frac{2.58 \text{ V} - 2.22 \text{ V}}{8.36 \text{ k}\Omega} = \frac{0.36 \text{ V}}{8.36 \text{ k}\Omega} = \mathbf{43.06 \mu\text{A}} \end{aligned}$$

$$\begin{aligned} \text{d. } V_C &= 20 \text{ V} - I_C R_C = 20 \text{ V} - (4.44 \text{ mA})(2.2 \text{ k}\Omega) \\ &= 20 \text{ V} - 9.77 \text{ V} \\ &= \mathbf{10.23 \text{ V}} \end{aligned}$$

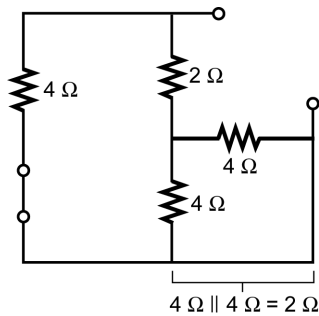
19. a. $E_{Th} = \mathbf{20 \text{ V}}$

$$I = 1.6 \text{ mA} = \frac{E_{Th}}{R_{Th}} = \frac{20 \text{ V}}{R_{Th}}, R_{Th} = \frac{20 \text{ V}}{1.6 \text{ mA}} = \mathbf{12.5 \text{ k}\Omega}$$

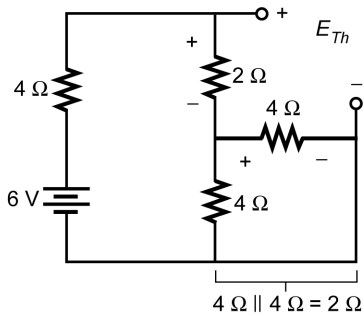
b. $E_{Th} = \mathbf{60 \text{ mV}}, R_{Th} = \mathbf{2.72 \text{ k}\Omega}$

c. $E_{Th} = \mathbf{16 \text{ V}}, R_{Th} = \mathbf{2.2 \text{ k}\Omega}$

20.



$$R_{Th} = 4 \Omega \parallel (2 \Omega + 2 \Omega) = \frac{4 \Omega}{2} = \mathbf{2 \Omega}$$

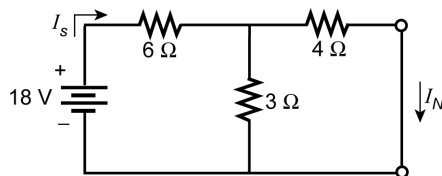


$$V_{4\Omega} = \frac{2 \Omega (6 \text{ V})}{2 \Omega + 4 \Omega + 2 \Omega} = \frac{12 \text{ V}}{8 \Omega} = 1.5 \text{ V}$$

$$V_{2\Omega} = V_{4\Omega} = 1.5 \text{ V}$$

$$E_{Th} = V_{4\Omega} + V_{2\Omega} = 1.5 \text{ V} + 1.5 \text{ V} = \mathbf{3 \text{ V}}$$

21. a. From Problem 8, $R_N = R_{Th} = \mathbf{6 \Omega}$



$$R_T = 6 \Omega + 3 \Omega \parallel 4 \Omega$$

$$= 6 \Omega + 1.714 \Omega = 7.714 \Omega$$

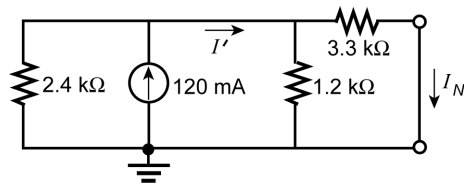
$$I_s = \frac{E}{R_T} = \frac{18 \text{ V}}{7.714 \Omega} = 2.333 \text{ A}$$

$$I_N = \frac{3 \Omega (2.333 \text{ A})}{3 \Omega + 4 \Omega} = \mathbf{1 \text{ A}}$$

b. $R_{Th} = \mathbf{6 \Omega}, E_{Th} = I_N R_N = (1 \text{ A})(6 \Omega) = \mathbf{6 \text{ V}}$

c. same results

22. a. From Problem 9, $R_N = R_{Th} = 4.1 \text{ k}\Omega$

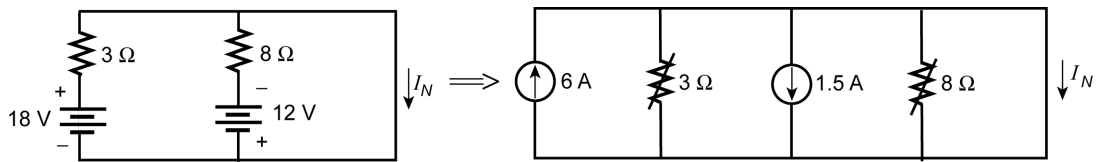


$$I' = \frac{2.4 \text{ k}\Omega(120 \text{ mA})}{2.4 \text{ k}\Omega + (1.2 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega)} = 87.80 \text{ mA}$$

$$I_N = \frac{1.2 \text{ k}\Omega(87.80 \text{ mA})}{1.2 \text{ k}\Omega + 3.3 \text{ k}\Omega} = 23.41 \text{ mA}$$

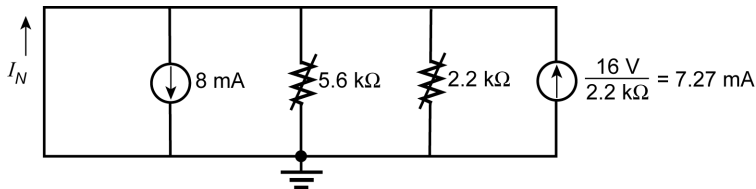
- b. $R_{Th} = 4.1 \text{ k}\Omega$, $E_{Th} = I_N R_N = (23.41 \text{ mA})(4.1 \text{ k}\Omega) = 96 \text{ V}$
 c. same results.

23. From Problem 11, $R_N = R_{Th} = 2.18 \Omega$



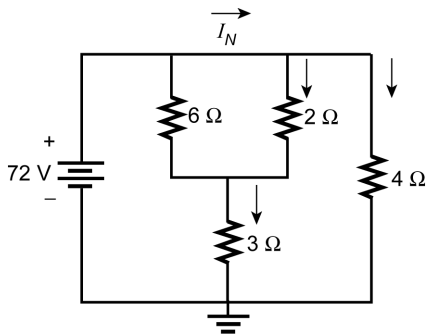
$$I_N = 6 \text{ A} - 1.5 \text{ A} = 4.5 \text{ A}$$

24. From Problem 12, $R_N = R_{Th} = 1.58 \text{ k}\Omega$



$$I_N = 8 \text{ mA} - 7.27 \text{ mA} = 0.73 \text{ mA}$$

25. From Problem 13, $R_N = R_{Th} = 2 \Omega$



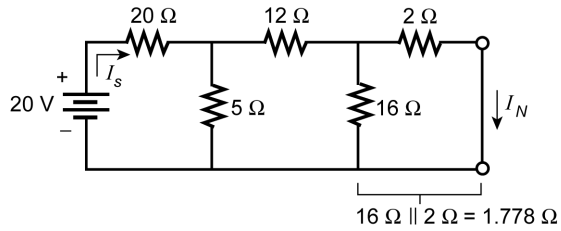
$$I_{4\Omega} = \frac{72 \text{ V}}{4 \Omega} = 18 \text{ A}$$

$$I_{3\Omega} = \frac{72 \text{ V}}{3 \Omega + 6 \Omega \parallel 2 \Omega} = 16 \text{ A}$$

$$I_{2\Omega} = \frac{6 \Omega(16 \text{ A})}{6 \Omega + 2 \Omega} = 12 \text{ A}$$

$$I_N = I_{4\Omega} + I_{2\Omega} = 18 \text{ A} + 12 \text{ A} = 30 \text{ A}$$

26. From Problem 15, $R_N = R_{Th} = 10\ \Omega$



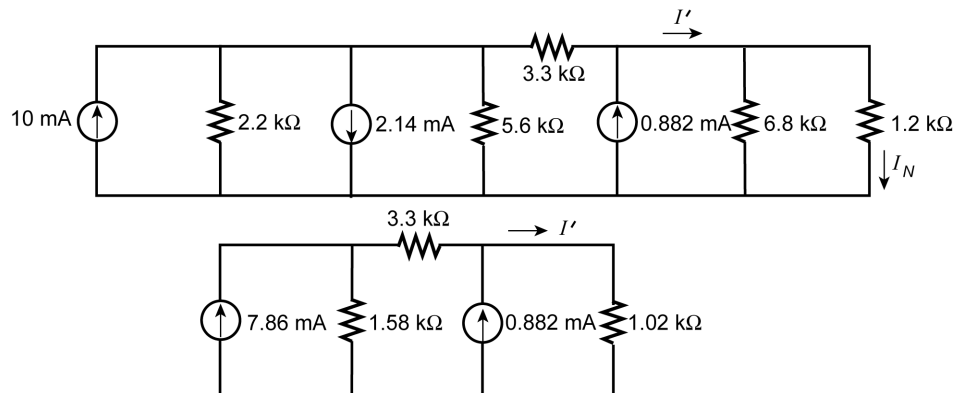
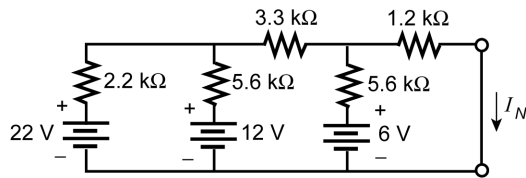
$$R_T = 20\ \Omega + 5\ \Omega \parallel (12\ \Omega + 1.778\ \Omega) = 23.67\ \Omega$$

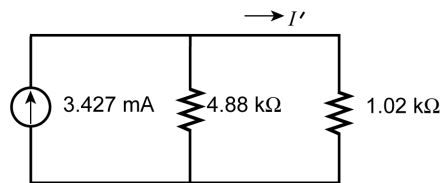
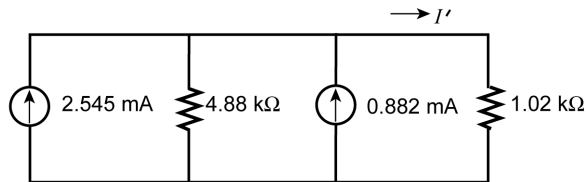
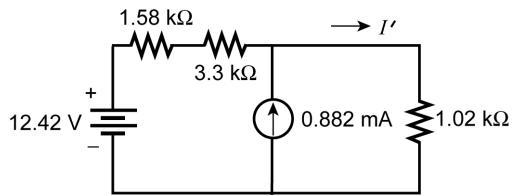
$$I_s = \frac{E}{R_T} = \frac{20\ \text{V}}{23.67\ \Omega} = 844.95\ \text{mA}$$

$$I_{12\Omega} = \frac{5\ \Omega(844.95\ \text{mA})}{5\ \Omega + (12\ \Omega + 1.778\ \Omega)} = 224.98\ \text{mA}$$

$$I_N = \frac{16\ \Omega(224.98\ \text{mA})}{16\ \Omega + 2\ \Omega} = 200\ \text{mA}$$

27. From Problem 17, $R_N = R_{Th} = 4.04\ \text{k}\Omega$

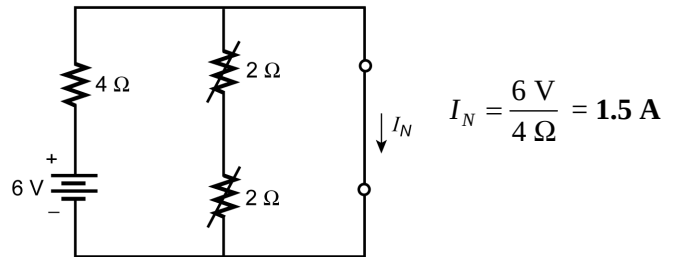
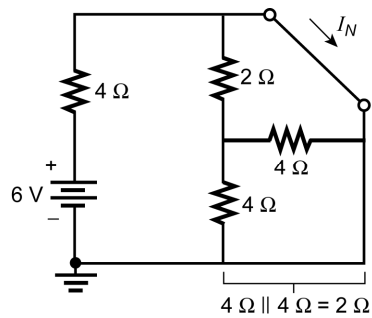




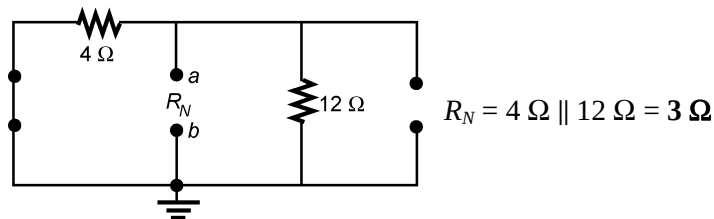
$$I' = \frac{4.88 \text{ k}\Omega(3.427 \text{ mA})}{4.88 \text{ k}\Omega + 1.02 \text{ k}\Omega} = 2.83 \text{ mA}$$

$$I_N = \frac{6.8 \text{ k}\Omega(2.83 \text{ mA})}{6.8 \text{ k}\Omega + 1.2 \text{ k}\Omega} = 2.41 \text{ mA}$$

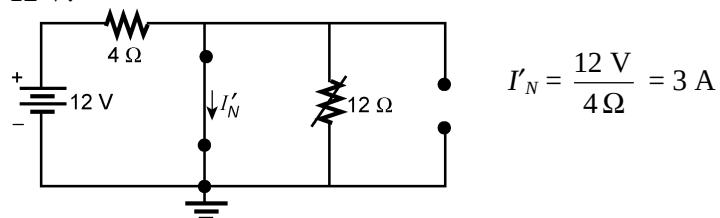
28. From Problem 20, $R_N = R_{Th} = 2 \Omega$



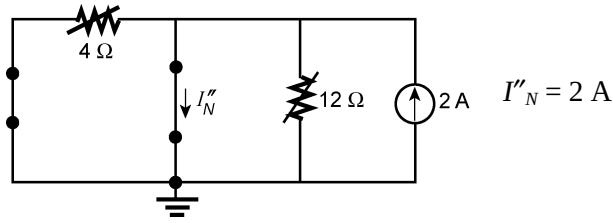
29. a. R_N :



$E = 12 \text{ V}$:

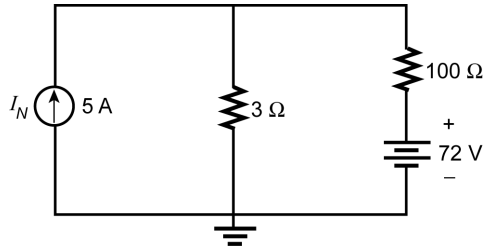


$I = 2 \text{ A}$:

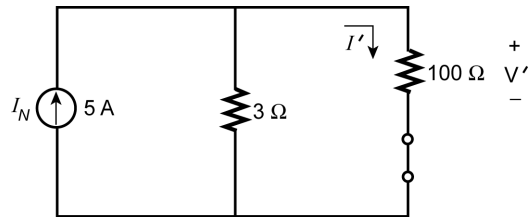


$$I_N = I'_N + I''_N = 3 \text{ A} + 2 \text{ A} = 5 \text{ A}$$

b.



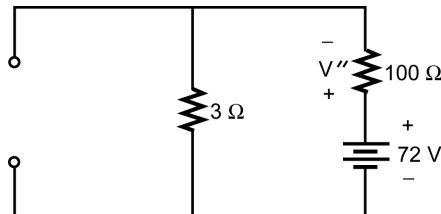
I :



$$I' = \frac{3 \Omega (5 \text{ A})}{3 \Omega + 100 \Omega} = 145.63 \text{ mA}$$

$$V' = IR = (145.63 \text{ mA})(100 \Omega) = 14.56 \text{ V}$$

E :



$$V'' = \frac{100 \Omega (72 \text{ V})}{100 \Omega + 3 \Omega} = 69.9 \text{ V}$$

$$V_{100\Omega+} = V'' - V' = 69.9 \text{ V} - 14.56 \text{ V} = 55.34 \text{ V}$$

30. a. $R = R_{Th} = 6 \Omega$ from Problem 8

b. $E_{Th} = 6 \text{ V}$ from Problem 8

$$P_{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{(6 \text{ V})^2}{4(6 \Omega)} = 1.5 \text{ W}$$

31. a. $R = R_{Th} = 2.18 \Omega$ from Problem 11

b. $E_{Th} = 9.81 \text{ V}$ from Problem 11

$$P_{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{(9.81 \text{ V})^2}{4(2.18 \Omega)} = 11.06 \text{ W}$$

32. a. $R = R_{Th} = 2 \Omega$ from Problem 13

b. $E_{Th} = 60 \text{ V}$ from Problem 13

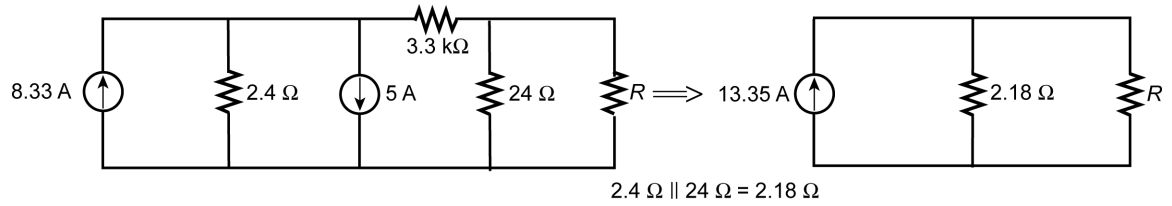
$$P_{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{(60 \text{ V})^2}{4(2 \Omega)} = 450 \text{ W}$$

33. a. $R = R_{Th} = 4.04 \text{ k}\Omega$ from Problem 17

b. $E_{Th} = 9.74 \text{ V}$ from Problem 17

$$P_{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{(9.74 \text{ V})^2}{4(4.04 \text{ k}\Omega)} = 5.87 \text{ mW}$$

34. a.



$$R = R_N = R_{Th} = 2.18 \Omega$$

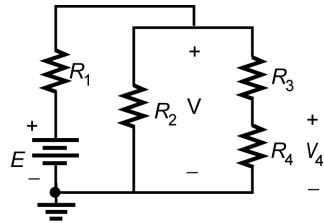
b. $P_{\max} = \frac{I_N^2 R_N}{4} = \frac{(13.33 \text{ A})^2 2.18 \Omega}{4} = 96.84 \text{ W}$

35. $P_{\max} = \left[\frac{E_{Th}}{R_{Th} + R_4} \right]^2 R_4$

with $R_1 = 0 \Omega$, E_{Th} is a maximum and R_{Th} is a minimum.

$$\therefore R_1 = 0 \Omega$$

36. a.



V , and therefore V_4 will be its largest value when R_2 is as large as possible. Therefore, choose $R_2 = \text{open-circuit } (\infty \Omega)$ and

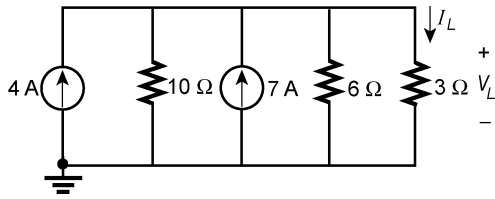
$$P_4 = \frac{V_4^2}{R_4} \text{ will be a maximum.}$$

b. No, examine each individually.

37. The voltage V_L will be a maximum when $R = 500 \Omega$ because the full voltage, E , will appear across R_L .

$$P_{\max} = \frac{V_L^2}{R_L} = \frac{E^2}{R_L} = \frac{(12 \text{ V})^2}{500 \Omega} = 1.44 \text{ W}$$

38.



$$I_T \uparrow = 4 \text{ A} + 7 \text{ A} = 11 \text{ A}$$

$$R_T = 10 \Omega \parallel 6 \Omega \parallel 3 \Omega = 1.67 \Omega$$

$$V_L = I_T R_T = (11 \text{ A})(1.67 \Omega) = \mathbf{18.37 \text{ V}}$$

$$I_L = \frac{V_L}{R_L} = \frac{18.37 \text{ V}}{3 \Omega} = \mathbf{6.12 \text{ A}}$$

$$39. \quad E_{\text{eq}} = \frac{-5 \text{ V} / 2.2 \text{ k}\Omega + 20 \text{ V} / 8.2 \text{ k}\Omega}{1 / 2.2 \text{ k}\Omega + 1 / 8.2 \text{ k}\Omega} = 0.2879 \text{ V}$$

$$R_{\text{eq}} = \frac{1}{1 / 2.2 \text{ k}\Omega + 1 / 8.2 \text{ k}\Omega} = 1.7346 \text{ k}\Omega$$

$$I_L = \frac{E_{\text{eq}}}{R_{\text{eq}} + R_L} = \frac{0.2879 \text{ V}}{1.7346 \text{ k}\Omega + 5.6 \text{ k}\Omega} = \mathbf{39.3 \mu\text{A}}$$

$$V_L = I_L R_L = (39.3 \mu\text{A})(5.6 \text{ k}\Omega) = \mathbf{220 \text{ mV}}$$

$$40. \quad I_T \downarrow = 5 \text{ A} - 0.4 \text{ A} - 0.2 \text{ A} = 4.40 \text{ A}$$

$$R_T = 200 \Omega \parallel 80 \Omega \parallel 50 \Omega \parallel 50 \Omega = 17.39 \Omega$$

$$V_L = I_T R_T = (4.40 \text{ A})(17.39 \Omega) = \mathbf{75.52 \text{ V}}$$

$$I_L = \frac{V_L}{R_L} = \frac{76.52 \text{ V}}{200 \Omega} = \mathbf{0.38 \text{ A}}$$

$$41. \quad I_{\text{eq}} = \frac{(4 \text{ A})(4.7 \Omega) + (1.6 \text{ A})(3.3 \Omega)}{4.7 \Omega + 3.3 \Omega} = \frac{18.8 \text{ V} + 5.28 \text{ V}}{8 \Omega} = 3.01 \text{ A}$$

$$R_{\text{eq}} = 4.7 \Omega + 3.3 \Omega = 8 \Omega$$

$$I_L = \frac{R_{\text{eq}}(I_{\text{eq}})}{R_{\text{eq}} + R_L} = \frac{8 \Omega(3.01 \text{ A})}{8 \Omega + 2.7 \Omega} = \mathbf{2.25 \text{ A}}$$

$$V_L = I_L R_L = (2.25 \text{ A})(2.7 \Omega) = \mathbf{6.08 \text{ V}}$$

$$42. \quad I_{\text{eq}} = \frac{\leftarrow (4 \text{ mA})(8.2 \text{ k}\Omega) + (8 \text{ mA})(4.7 \text{ k}\Omega) - (10 \text{ mA})(2 \text{ k}\Omega)}{8.2 \text{ k}\Omega + 4.7 \text{ k}\Omega + 2 \text{ k}\Omega}$$

$$= \frac{32.8 \text{ V} + 37.6 \text{ V} - 20 \text{ V}}{14.9 \text{ k}\Omega} = 3.38 \text{ mA}$$

$$R_{\text{eq}} = 8.2 \text{ k}\Omega + 4.7 \text{ k}\Omega + 2 \text{ k}\Omega = 14.9 \text{ k}\Omega$$

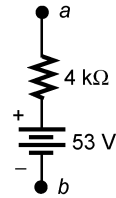
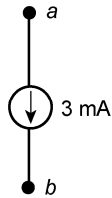
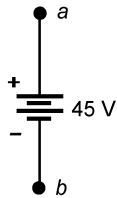
$$I_L = \frac{R_{\text{eq}} I_{\text{eq}}}{R_{\text{eq}} + R_L} = \frac{(14.9 \text{ k}\Omega)(3.38 \text{ mA})}{14.9 \text{ k}\Omega + 6.8 \text{ k}\Omega} = \mathbf{2.32 \text{ mA}}$$

$$V_L = I_L R_L = (2.32 \text{ mA})(6.8 \text{ k}\Omega) = \mathbf{15.78 \text{ V}}$$

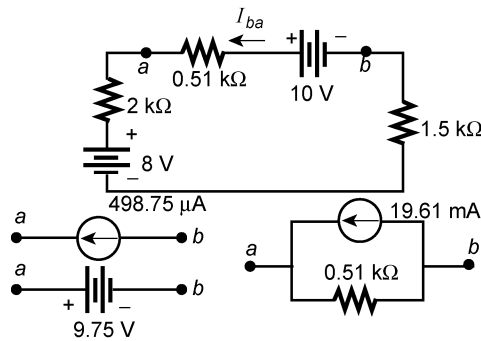
43. $15 \text{ k}\Omega \parallel (8 \text{ k}\Omega + 7 \text{ k}\Omega) = 15 \text{ k}\Omega \parallel 15 \text{ k}\Omega = 7.5 \text{ k}\Omega$

$$V_{ab} = \frac{7.5 \text{ k}\Omega (60 \text{ V})}{7.5 \text{ k}\Omega + 2.5 \text{ k}\Omega} = 45 \text{ V}$$

$$I_{ab} = \frac{45 \text{ V}}{15 \text{ k}\Omega} = 3 \text{ mA}$$



44.



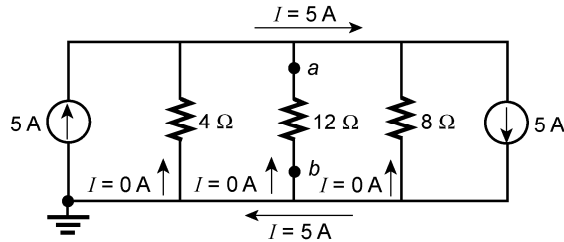
$$I_{ba} = \frac{10 \text{ V} - 8 \text{ V}}{2 \text{ k}\Omega + 0.51 \text{ k}\Omega + 1.5 \text{ k}\Omega}$$

$$= 498.75 \text{ } \mu\text{A}$$

$$V_{0.51 \text{ k}\Omega} = (498.75 \text{ } \mu\text{A})(0.51 \text{ k}\Omega) = 0.25 \text{ V}$$

$$V_{ab} = 10 \text{ V} - 0.25 \text{ V} = 9.75 \text{ V}$$

45.



$$V_{ab} = 0 \text{ V (short)}$$

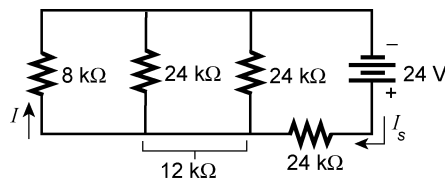
$$I_{ab} = 0 \text{ A (open)}$$

R_2 any resistive value

$\therefore R_2 = \text{short-circuit, open-circuit, any value}$

46. a. $I_s = \frac{24 \text{ V}}{8 \text{ k}\Omega + \frac{24 \text{ k}\Omega}{3}} = 1.5 \text{ mA}, I = \frac{I_s}{3} = 0.5 \text{ mA}$

b.

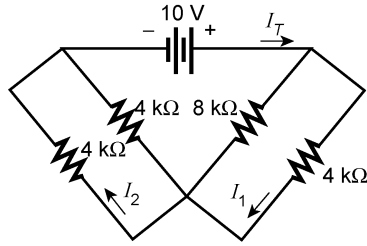


$$I_s = \frac{24 \text{ V}}{24 \text{ k}\Omega + 8 \text{ k}\Omega \parallel 12 \text{ k}\Omega} = 0.83 \text{ mA}$$

$$I = \frac{12 \text{ k}\Omega (I_s)}{12 \text{ k}\Omega + 8 \text{ k}\Omega} = 0.5 \text{ mA}$$

c. yes

47. (a)



$$I_T = \frac{10 \text{ V}}{4 \text{ k}\Omega \parallel 8 \text{ k}\Omega + 4 \text{ k}\Omega \parallel 4 \text{ k}\Omega}$$

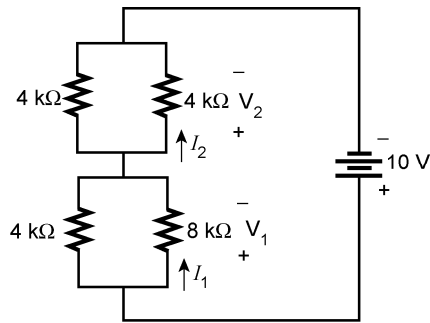
$$= \frac{10 \text{ V}}{2.67 \text{ k}\Omega + 2 \text{ k}\Omega}$$

$$= \frac{10 \text{ V}}{4.67 \text{ k}\Omega} = 2.14 \text{ mA}$$

$$I_1 = \frac{8 \Omega(I_T)}{8 \Omega + 4 \Omega} = 1.43 \text{ mA}, I_2 = I_T/2 = 1.07 \text{ mA}$$

$$I = I_1 - I_2 = 1.43 \text{ mA} - 1.07 \text{ mA} = \mathbf{0.36 \text{ mA}}$$

(b)



$$V_1 = \frac{(8 \text{ k}\Omega \parallel 4 \text{ k}\Omega)(10 \text{ V})}{8 \text{ k}\Omega \parallel 4 \text{ k}\Omega + 4 \text{ k}\Omega \parallel 4 \text{ k}\Omega}$$

$$= 5.72 \text{ V}$$

$$I_1 = \frac{V_1}{8 \text{ k}\Omega} = 0.71 \text{ mA}$$

$$V_2 = E - V_1 = 10 \text{ V} - 5.72 \text{ V}$$

$$= 4.28 \text{ V}$$

$$I_2 = \frac{V_2}{4 \text{ k}\Omega} = 1.07 \text{ mA}$$

$$I = I_2 - I_1 = 1.07 \text{ mA} - 0.71 \text{ mA}$$

$$= \mathbf{0.36 \text{ mA}}$$

48. a. $I_{R_2} = \frac{R_1(I)}{R_1 + R_2 + R_3} = \frac{3 \Omega(6 \text{ A})}{3 \Omega + 2 \Omega + 4 \Omega} = 2 \text{ A}$

$$V = I_{R_2} R_2 = (2 \text{ A})(2 \Omega) = \mathbf{4 \text{ V}}$$

b. $I_{R_1} = \frac{R_2(I)}{R_1 + R_2 + R_3} = \frac{2 \Omega(6 \text{ A})}{3 \Omega + 2 \Omega + 4 \Omega} = 1.33 \text{ A}$

$$V = I_{R_1} R_1 = (1.33 \text{ A})(3 \Omega) = \mathbf{4 \text{ V}}$$

Chapter 10

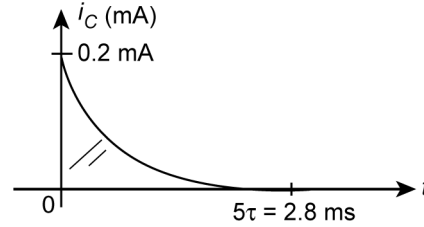
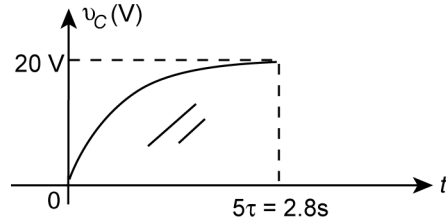
1. (a) $\mathcal{E} = k \frac{Q_1}{r^2} = \frac{(9 \times 10^9)(4 \mu\text{C})}{(1 \text{ m})^2} = \mathbf{36 \times 10^3 \text{ N/C}}$
 (b) $\mathcal{E} = k \frac{Q_1}{r^2} = \frac{(9 \times 10^9)(4 \mu\text{C})}{(1 \text{ mm})^2} = \mathbf{36 \times 10^9 \text{ N/C}}$
 $\mathcal{E}(1 \text{ mm}): \mathcal{E}(2 \text{ m}) = 36 \times 10^9: 36 \times 10^3 = \mathbf{1 \times 10^6}$
2. $\mathcal{E} = \frac{kQ}{r^2} \Rightarrow r = \sqrt{\frac{kQ}{\mathcal{E}}} = \sqrt{\frac{(9 \times 10^9)(2 \mu\text{C})}{72 \text{ N/C}}} = \mathbf{15.81 \text{ m}}$
3. $C = \frac{Q}{V} = \frac{1200 \mu\text{C}}{24 \text{ V}} = \mathbf{50 \mu\text{F}}$
4. $Q = CV = (0.15 \mu\text{F})(45 \text{ V}) = \mathbf{6.75 \mu\text{C}}$
5. a. $1'' \left[\frac{1 \text{ m}}{39.37''} \right] = 25.4 \text{ mm}$
 $\mathcal{E} = \frac{V}{d} = \frac{500 \text{ mV}}{25.4 \text{ mm}} = \mathbf{19.69 \text{ V/m}}$
 b. $\frac{25.4 \text{ mm}}{100} = 0.254 \text{ mm}$
 $\mathcal{E} = \frac{V}{d} = \frac{500 \text{ mV}}{0.254 \text{ mm}} = \mathbf{1.97 \text{ kV/m}}$
6. $V = \frac{Q}{C} = \frac{160 \mu\text{C}}{6.8 \mu\text{F}} = 23.53 \text{ V}$
 $\mathcal{E} = \frac{V}{d} = \frac{23.53 \text{ V}}{5 \text{ mm}} = \mathbf{4.71 \text{ kV/m}}$
7. $0.1'' \left[\frac{1 \text{ m}}{39.37''} \right] = 2.54 \text{ mm}$
 $C = 8.85 \times 10^{-12} \epsilon_r \frac{A}{d} = 8.85 \times 10^{-12} (1) \frac{(0.1 \text{ m}^2)}{2.54 \text{ mm}} = \mathbf{348.43 \text{ pF}}$
8. $C = 8.85 \times 10^{-12} \epsilon_r \frac{A}{d} = 8.85 \times 10^{-12} (2.5) \frac{(0.1 \text{ m}^2)}{2.54 \text{ mm}} = \mathbf{871.06 \text{ pF}}$
9. $C = 8.85 \times 10^{-12} \epsilon_r \frac{A}{d} \Rightarrow d = \frac{8.85 \times 10^{-12} (4) (0.15 \text{ m}^2)}{2 \mu\text{F}} = \mathbf{2.66 \mu\text{m}}$
10. $C = \epsilon_r C_o \Rightarrow \epsilon_r = \frac{C}{C_o} = \frac{6.8 \text{ nF}}{1360 \text{ pF}} = \mathbf{5 \text{ (mica)}}$

11. a. $C = 8.85 \times 10^{-12}(7) \frac{(0.08 \text{ m}^2)}{0.2 \text{ mm}} = \mathbf{24.78 \text{ nF}}$
- b. $\mathcal{E} = \frac{V}{d} = \frac{200 \text{ V}}{0.2 \text{ mm}} = \mathbf{10^6 \text{ V/m}}$
- c. $Q = CV = (24.78 \text{ nF})(200 \text{ V}) = \mathbf{4.96 \mu\text{C}}$
12. a. $C = \frac{1}{2}(4.7 \mu\text{F}) = \mathbf{2.35 \mu\text{F}}$
- b. $C = 2(4.7 \mu\text{F}) = \mathbf{9.4 \mu\text{F}}$
- c. $C = 20(4.7 \mu\text{F}) = \mathbf{94 \mu\text{F}}$
- d. $C = \frac{(4)\left(\frac{1}{3}\right)}{\left(\frac{1}{4}\right)}(4.7 \mu\text{F}) = \mathbf{25.1 \mu\text{F}}$
13. $d = \frac{8.85 \times 10^{-12} \epsilon_r A}{C} = \frac{(8.85 \times 10^{-12})(5)(0.02 \text{ m}^2)}{6800 \text{ pF}} = 130.15 \mu\text{m}$
- $d = 130.15 \mu\text{m} \left[\frac{10^{-6} \cancel{\text{m}}}{1 \cancel{\mu\text{m}}} \right] \left[\frac{39.37 \cancel{\text{in.}}}{1 \cancel{\text{in.}}} \right] \left[\frac{1000 \text{ mils}}{1 \cancel{\text{in.}}} \right] = 5.12 \text{ mils}$
- $5.12 \cancel{\text{mils}} \left[\frac{5000 \text{ V}}{\cancel{\text{mil}}} \right] = \mathbf{25.6 \text{ kV}}$
14. mica: $\frac{1200 \text{ V}}{5000 \text{ V}} = 1200 \text{ V} \left[\frac{\cancel{\text{mil}}}{5000 \text{ V}} \right] = 0.24 \text{ mils}$
- $0.24 \cancel{\text{mils}} \left[\frac{1 \cancel{\text{in.}}}{1000 \cancel{\text{mils}}} \right] \left[\frac{1 \text{ m}}{39.37 \cancel{\text{in.}}} \right] = \mathbf{6.10 \mu\text{m}}$
15. $\frac{200}{1 \times 10^6} (22 \mu\text{F}) / ^\circ\text{C} = 4400 \text{ pF}/^\circ\text{C}$
- $\frac{4400 \text{ pF}}{^\circ\text{C}} [\Delta T] = \frac{4400 \text{ pF}}{^\circ\text{C}} [80^\circ\text{C}] = \mathbf{0.35 \mu\text{F}}$
16. $J = \pm 5\%$, Size $\Rightarrow 40 \text{ pF} \pm 2 \text{ pF}$, $\mathbf{38 \text{ pF} \rightarrow 42 \text{ pF}}$
17. $F = \pm 1\%$, Size $\Rightarrow 47 \times 10^1 = 470 \mu\text{F} \pm 4.7 \mu\text{F}$, $\mathbf{465.3 \mu\text{F} \rightarrow 474.7 \mu\text{F}}$
18. $K = \pm 10\%$, Size $\Rightarrow 18 \times 10^2 \text{ pF} = \mathbf{1800 \text{ pF} \pm 180 \text{ pF}}$, $\mathbf{1620 \text{ pF} \rightarrow 1980 \text{ pF}}$
19. a. $\tau = RC = (10^5 \Omega)(5.6 \mu\text{F}) = \mathbf{0.56 \text{ s}}$
- b. $v_C = E(1 - e^{-t/\tau}) = \mathbf{20 \text{ V}(1 - e^{-t/0.56 \text{ s}})}$

c. $1\tau = 0.632(20 \text{ V}) = \mathbf{12.64 \text{ V}}$, $3\tau = 0.95(20 \text{ V}) = \mathbf{19 \text{ V}}$
 $5\tau = 0.993(20 \text{ V}) = \mathbf{19.87 \text{ V}}$

d. $i_C = \frac{20 \text{ V}}{100 \text{ k}\Omega} e^{-t/\tau} = \mathbf{0.2 \text{ mA} e^{-t/0.56 \text{ s}}}$
 $v_R = E e^{-t/\tau} = \mathbf{20 \text{ V} e^{-t/0.56 \text{ s}}}$

e.



20. a. $\tau = RC = (10^6 \Omega)(5.6 \mu\text{F}) = \mathbf{5.6 \text{ s}}$

b. $v_C = E(1 - e^{-t/\tau}) = \mathbf{20 \text{ V}(1 - e^{-t/5.6 \text{ s}})}$

c. $1\tau = \mathbf{12.64 \text{ V}}$, $3\tau = \mathbf{19 \text{ V}}$, $5\tau = \mathbf{19.87 \text{ V}}$

d. $i_C = \frac{20 \text{ V}}{1 \text{ M}\Omega} e^{-t/\tau} = \mathbf{20 \mu\text{A} e^{-t/5.6 \text{ s}}}$
 $v_R = E e^{-t/\tau} = \mathbf{20 \text{ V} e^{-t/5.6 \text{ s}}}$

e. Same as problem 21 with $5\tau = 28 \text{ s}$ and $I_m = 20 \mu\text{A}$

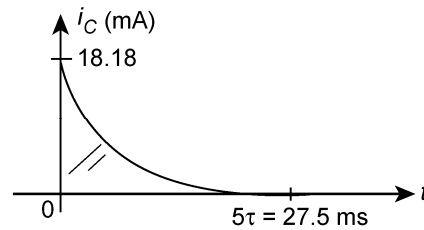
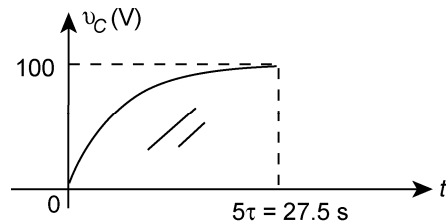
21. a. $\tau = RC = (2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega)1 \mu\text{F} = (5.5 \text{ k}\Omega)(1 \mu\text{F}) = \mathbf{5.5 \text{ ms}}$

b. $v_C = E(1 - e^{-t/\tau}) = \mathbf{100 \text{ V}(1 - e^{-t/5.5 \text{ ms}})}$

c. $1\tau = \mathbf{63.21 \text{ V}}$, $3\tau = \mathbf{95.02 \text{ V}}$, $5\tau = \mathbf{99.33 \text{ V}}$

d. $i_C = \frac{E}{R_T} e^{-t/\tau} = \frac{100 \text{ V}}{5.5 \text{ k}\Omega} e^{-t/\tau} = \mathbf{18.18 \text{ mA} e^{-t/5.5 \text{ ms}}}$
 $V_{R_2} = \frac{3.3 \text{ k}\Omega(100 \text{ V})}{3.3 \text{ k}\Omega + 2.2 \text{ k}\Omega} = 60 \text{ V}$
 $v_{R_2} = \mathbf{60 \text{ V} e^{-t/5.5 \text{ ms}}}$

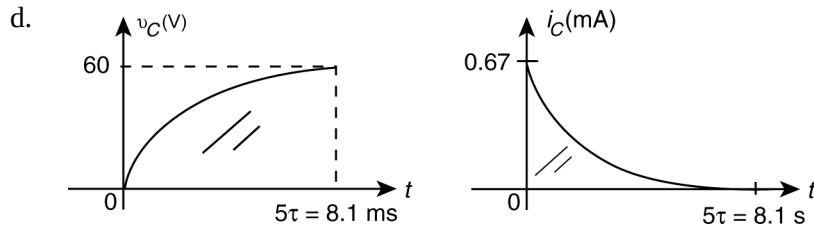
e.



22. a. $R = 68 \text{ k}\Omega + 22 \text{ k}\Omega = 90 \text{ k}\Omega$
 $\tau = RC = (90 \text{ k}\Omega)(18 \text{ }\mu\text{F}) = \mathbf{1.62 \text{ s}}$

b. $v_C = E(1 - e^{-t/\tau}) = (20 \text{ V} + 40 \text{ V})(1 - e^{-t/\tau})$
 $v_C = \mathbf{60 \text{ V}(1 - e^{-t/1.62\text{s}})}$

c. $i_C = \frac{E}{R} e^{-t/\tau} = \frac{60 \text{ V}}{90 \text{ k}\Omega} e^{-t/\tau} = \mathbf{0.67 \text{ mA}e^{-t/1.62\text{s}}}$



23. a. $\mathbf{100 \text{ }\mu\text{s}}$

b. $v_C = 12 \text{ V}(1 - e^{-50\mu\text{s}/100\mu\text{s}}) = 12 \text{ V}(1 - e^{-0.5}) = 12 \text{ V}(1 - 0.607)$
 $= 12 \text{ V}(0.393) = \mathbf{4.72 \text{ V}}$

c. $v_C = 12 \text{ V}(1 - e^{-1\text{ms}/100\mu\text{s}}) = 12 \text{ V}(1 - e^{-10}) = 12 \text{ V}(1 - 45.4 \times 10^{-6})$
 $\cong 12 \text{ V}(999.95 \times 10^{-3}) \cong \mathbf{12 \text{ V}}$

24. a. $\tau = 20 \text{ ms}, 5\tau = 5(20 \text{ ms}) = \mathbf{100 \text{ ms}}$

b. $\tau = RC, R = \frac{\tau}{C} = \frac{20 \text{ ms}}{10 \text{ }\mu\text{F}} = \mathbf{2 \text{ k}\Omega}$

c. $v_C(20 \text{ ms}) = 40 \text{ mV}(1 - e^{-20\text{ms}/20\text{ms}}) = 40 \text{ mV}(1 - e^{-1})$
 $= 40 \text{ mV}(1 - 0.368) = 40 \text{ mV}(0.632) = \mathbf{25.28 \text{ mV}}$

d. $v_C = 40 \text{ mV}(1 - e^{-10}) = 40 \text{ mV}(1 - 45 \times 10^{-6}) \cong \mathbf{40 \text{ mV}}$

e. $Q = CV = (10 \text{ }\mu\text{F})(40 \text{ mV}) = \mathbf{0.4 \text{ }\mu\text{C}}$

f. $\tau = RC = (1000 \times 10^6 \Omega)(10 \text{ }\mu\text{F}) = 10 \times 10^3 \text{ s}$

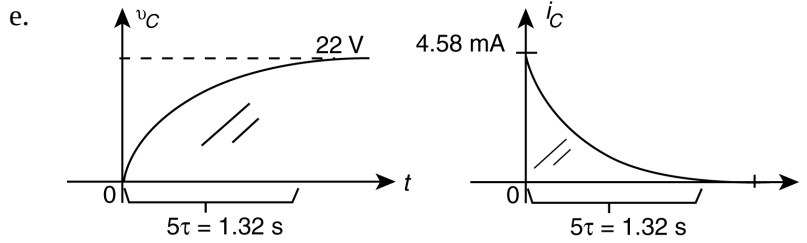
$5\tau = 50 \times 10^3 \cancel{\text{s}} \left[\frac{1 \cancel{\text{min}}}{60 \cancel{\text{s}}} \right] \left[\frac{1 \text{ h}}{60 \cancel{\text{min}}} \right] = \mathbf{13.89 \text{ h}}$

25. a. $\tau = RC = (4.7 \text{ k}\Omega)(56 \text{ }\mu\text{F}) = \mathbf{263.2 \text{ ms}}$

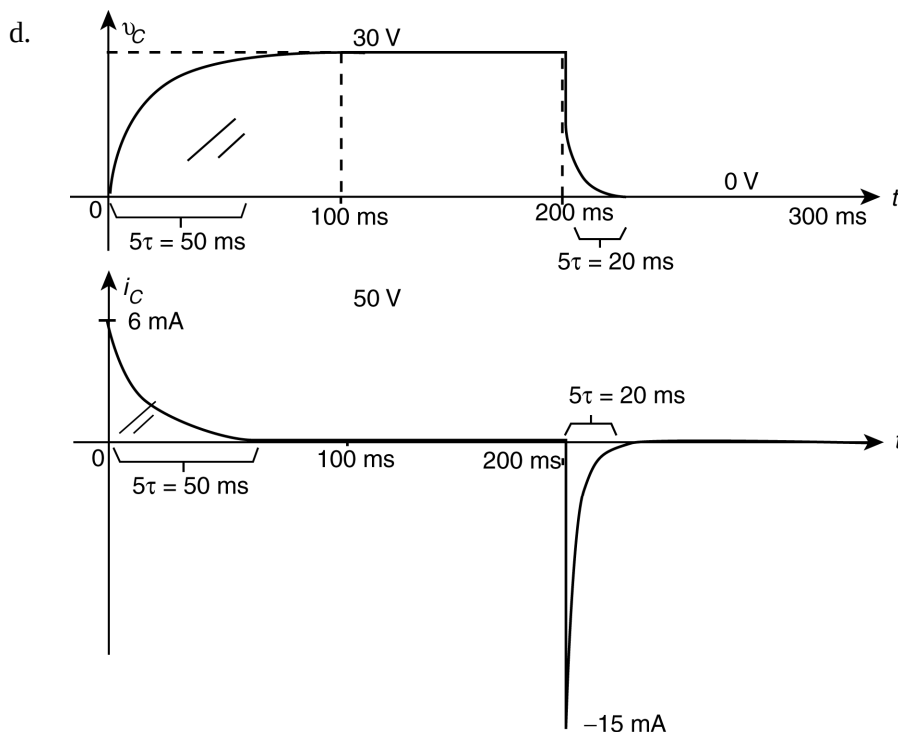
b. $v_C = E(1 - e^{-t/\tau}) = \mathbf{22 \text{ V}(1 - e^{-t/263.2\text{ms}})}$
 $i_C = \frac{E}{R} e^{-t/\tau} = \frac{22 \text{ V}}{4.7 \text{ k}\Omega} e^{-t/263.2\text{ms}} = \mathbf{4.68 \text{ mA}e^{-t/263.2\text{ms}}}$

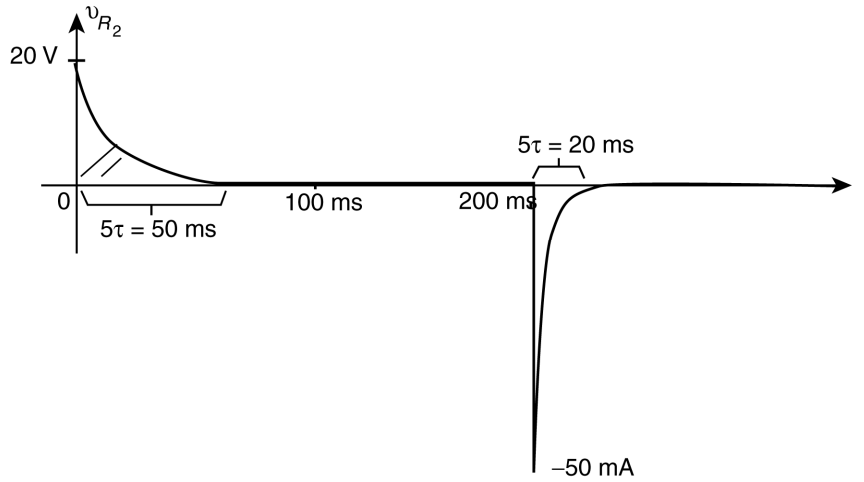
c. $v_C(1 \text{ s}) = 22 \text{ V}(1 - e^{-1\text{s}/263.2\text{ms}}) = 22 \text{ V}(1 - e^{-3.8})$
 $= 22 \text{ V}(1 - 22.37 \times 10^{-3}) = \mathbf{21.51 \text{ V}}$
 $i_C(1 \text{ s}) = 4.68 \text{ mA}e^{-1\text{s}/263.2\text{ms}} = 4.68 \text{ mA}(22.37 \times 10^{-3}) = \mathbf{0.105 \text{ mA}}$

d. $v_C = \mathbf{21.51 \text{ V}e^{-t/263.2\text{ms}}}$
 $i_C = \frac{21.51 \text{ V}}{4.7 \text{ k}\Omega} e^{-t/263.2\text{ms}} = \mathbf{4.58 \text{ mA}e^{-t/263.2\text{ms}}}$



26. a. $\tau = RC = (3 \text{ k}\Omega + 2 \text{ k}\Omega)(2 \text{ }\mu\text{F}) = 10 \text{ ms}$
 $v_C = 30 \text{ V}(1 - e^{-t/10\text{ms}})$
 $i_C = \frac{30 \text{ V}}{5 \text{ k}\Omega} e^{-t/10\text{ms}} = 6 \text{ mA} e^{-t/10\text{ms}}$
 $v_{R_1} = i_C R_1 = (6 \text{ mA})(3 \text{ k}\Omega) e^{-t/10\text{ms}} = 18 \text{ V} e^{-t/10\text{ms}}$
- b. 100ms: $e^{-10} = 45.4 \times 10^{-6}$
 $v_C = 30 \text{ V}(1 - 45.4 \times 10^{-6}) = 30 \text{ V}$
 $i_C = 6 \text{ mA}(45.4 \times 10^{-6}) = 0.27 \text{ }\mu\text{A}$
 $v_{R_1} = 18 \text{ V}(45.4 \times 10^{-6}) = 0.82 \text{ mV}$
- c. 200 ms: $\tau' = R_2 C = (2 \text{ k}\Omega)(2 \text{ }\mu\text{F}) = 4 \text{ ms}$
 $v_C = 30 \text{ V} e^{-t/4\text{ms}}$
 $i_C = -\frac{30 \text{ V}}{2 \text{ k}\Omega} e^{-t/4\text{ms}} = -15 \text{ mA} e^{-t/4\text{ms}}$
 At $t = 0$: $v_{R_2} = i_C R_2 = (6 \text{ mA})(2 \text{ k}\Omega) e^{-t/10 \text{ ms}}$
 $= 12 \text{ V} e^{-t/10 \text{ ms}}$
 At $t = 200 \text{ ms}$: $v_{R_2} = -(15 \text{ mA})(2 \text{ k}\Omega) e^{-t/4 \text{ ms}}$
 $= -30 \text{ V} e^{-t/4 \text{ ms}}$





27. a. $\tau = RC = (220 \text{ k}\Omega)(22 \text{ pF}) = 4.84 \text{ }\mu\text{s}$

$$v_C = 60 \text{ V} (1 - e^{-t/4.84 \mu\text{s}})$$

$$i_C = \frac{60 \text{ V}}{220 \text{ k}\Omega} e^{-t/\tau} = 272.73 \text{ }\mu\text{A} e^{-t/4.84 \mu\text{s}}$$

b. $\tau' = R'C = (220 \text{ k}\Omega + 470 \text{ k}\Omega)(22 \text{ pF}) = (690 \text{ k}\Omega)(22 \text{ pF}) = 15.18 \text{ }\mu\text{s}$

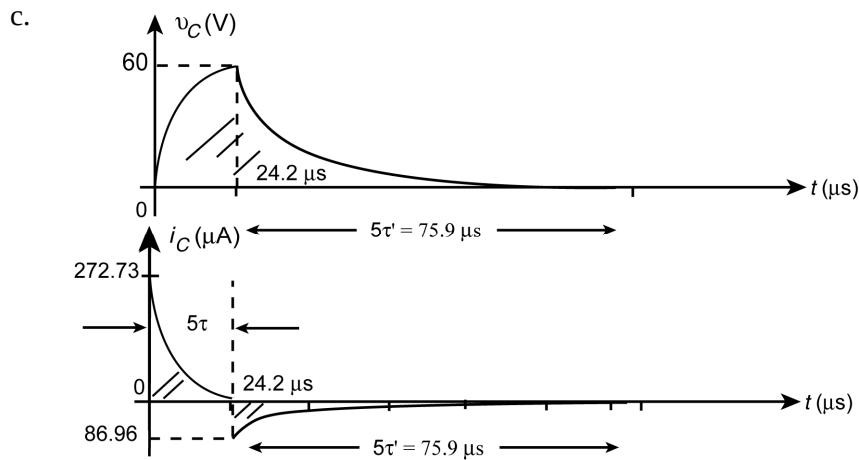
$$v_C = 60 \text{ V} (1 - e^{-5}) = 60 \text{ V} (1 - 6.73 \times 10^{-3}) = 59.6 \text{ V}$$

$$v_C = 59.6 \text{ V} e^{-t/15.18 \mu\text{s}}$$

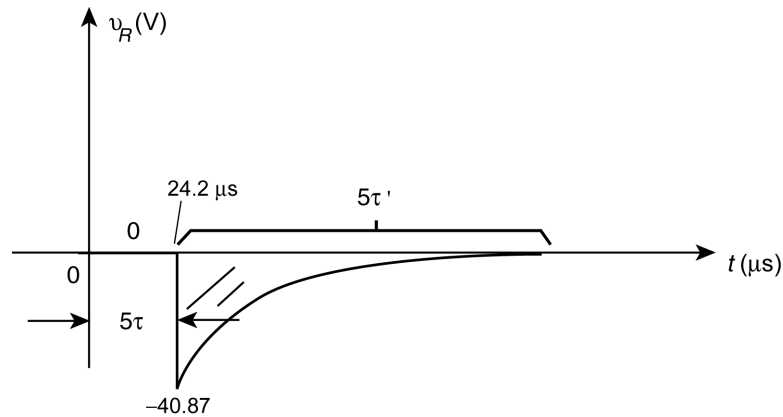
$$i_C = 272.73 \text{ }\mu\text{A} e^{-5} = 272.73 \text{ }\mu\text{A} (6.73 \times 10^{-3}) = 1.84 \text{ }\mu\text{A}$$

$$i_{C(\text{max})} = \frac{-60 \text{ V}}{690 \text{ k}\Omega} = -86.96 \text{ }\mu\text{A}$$

$$i_C = -86.96 \text{ }\mu\text{A} e^{-t/15.18 \mu\text{s}}$$



d.
$$v_R = -(470 \text{ k}\Omega)(86.96 \text{ }\mu\text{A})e^{-t/15.18 \text{ ms}}$$
$$= -40.87 \text{ V}e^{-t/15.18 \text{ }\mu\text{s}}$$



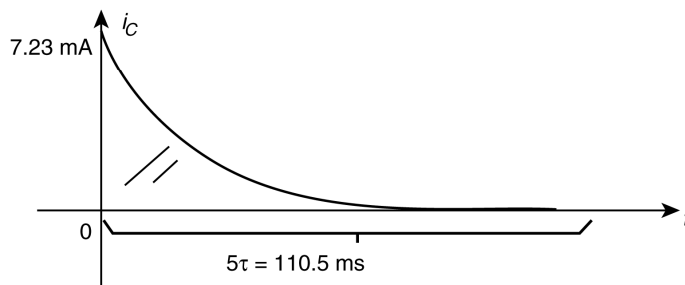
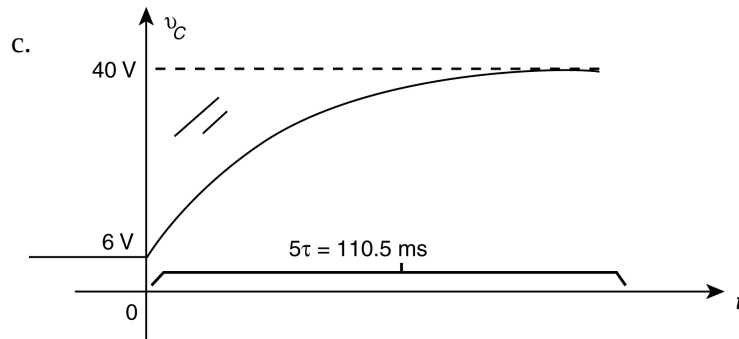
28. a.
$$\tau = RC = (2 \text{ m}\Omega)(1000 \text{ }\mu\text{F}) = 2 \text{ }\mu\text{s}$$
$$5\tau = 10 \text{ }\mu\text{s}$$

b.
$$I_m = \frac{V}{R} = \frac{12 \text{ V}}{2 \text{ m}\Omega} = 6 \text{ kA}$$

c. yes

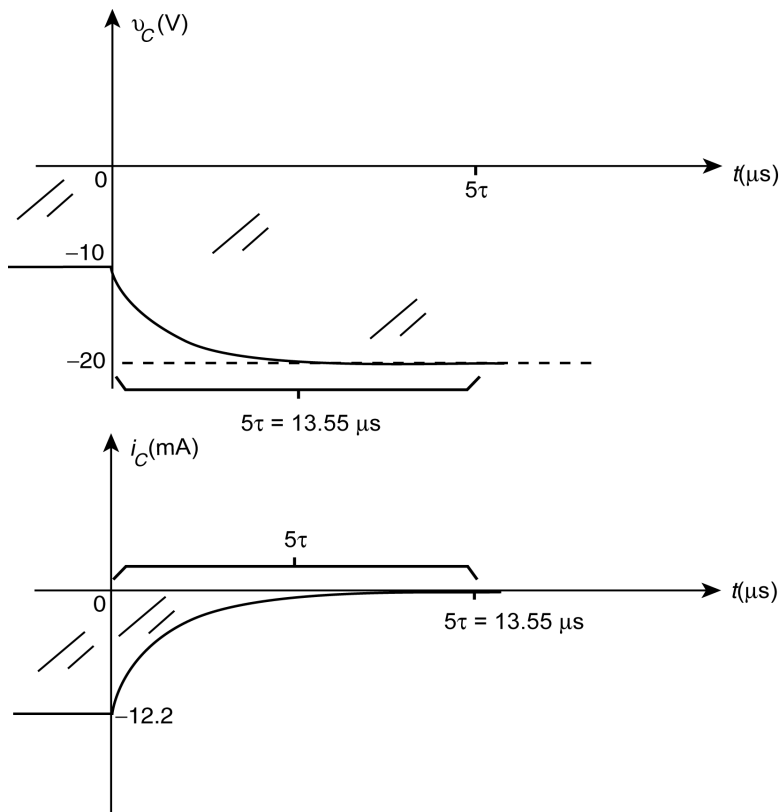
29. a.
$$v_C = V_f + (V_i - V_f)e^{-t/\tau}$$
$$\tau = RC = (4.7 \text{ k}\Omega)(4.7 \text{ }\mu\text{F}) = 22.1 \text{ ms}, V_f = 40 \text{ V}, V_i = 6 \text{ V}$$
$$v_C = 40 \text{ V} + (6 \text{ V} - 40 \text{ V})e^{-t/22.1 \text{ ms}}$$
$$v_C = 40 \text{ V} - 34 \text{ V}e^{-t/22.1 \text{ ms}}$$

b. Initially $V_R = E + v_C = 40 \text{ V} - 6 \text{ V} = 34 \text{ V}$
$$i_C = \frac{V_R}{R}e^{-t/\tau} = \frac{34 \text{ V}}{4.7 \text{ k}\Omega}e^{-t/22.1 \text{ ms}} = 7.23 \text{ mA}e^{-t/22.1 \text{ ms}}$$



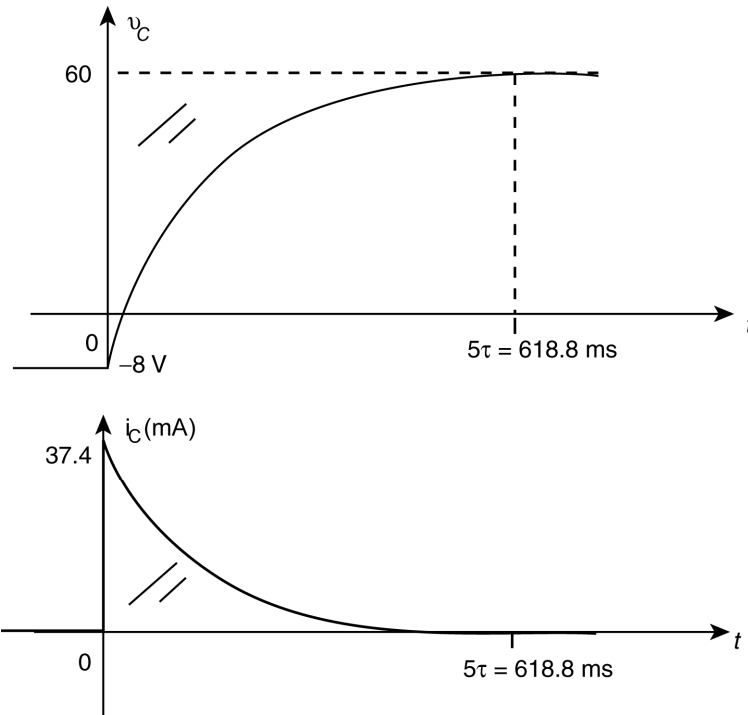
30. $\tau = RC = (2.2 \text{ k}\Omega)(2000 \text{ }\mu\text{F}) = 4.4 \text{ s}$
 $v_C = V_C e^{-t/\tau} = 40 \text{ V} e^{-t/4.4 \text{ s}}$
 $I_C = \frac{V_C}{R} e^{-t/\tau} = \frac{40 \text{ V}}{2.2 \text{ k}\Omega} e^{-t/4.4 \text{ s}} = 18.18 \text{ mA} e^{-t/4.4 \text{ s}}$
 $v_R = v_C = 40 \text{ V} e^{-t/4.4 \text{ s}}$

31. $v_C = V_f + (V_i - V_f) e^{-t/\tau}$
 $\tau = RC = (820 \text{ }\Omega)(3300 \text{ pF}) = 2.71 \text{ }\mu\text{s}, V_f = -20 \text{ V}, V_i = -10 \text{ V}$
 $v_C = -20 \text{ V} + (-10 \text{ V} - (-20 \text{ V})) e^{-t/2.71 \text{ }\mu\text{s}}$
 $v_C = -20 \text{ V} + 10 \text{ V} e^{-t/2.71 \text{ }\mu\text{s}}$
 $I_m = \frac{-(20 \text{ V} - 10 \text{ V})}{820 \text{ }\Omega} = \frac{-10 \text{ V}}{820 \text{ }\Omega} = -12.2 \text{ mA}$
 $i_C = i_R = -12.2 \text{ mA} e^{-t/2.71 \text{ }\mu\text{s}}$



32. a. $R = 10 \text{ k}\Omega + 8.2 \text{ k}\Omega = 18.2 \text{ k}\Omega$
 $\tau = RC = (18.2 \text{ k}\Omega)(6.8 \text{ }\mu\text{F}) = 123.76 \text{ ms}$
 $v_C = V_f + (V_i - V_f) e^{-t/\tau}$
 $V_f = 20 \text{ V} + 40 \text{ V} = 60 \text{ V}$
 $V_i = -8 \text{ V}$
 $v_C = 60 \text{ V} + (-8 \text{ V} - 60 \text{ V}) e^{-t/123.76 \text{ ms}}$
 $v_C = 60 \text{ V} - 68 \text{ V} e^{-t/123.76 \text{ ms}}$
 $I_m = \frac{8 \text{ V} + 20 \text{ V} + 40 \text{ V}}{18.2 \text{ k}\Omega} = 3.74 \text{ mA}$
 $i_C = 3.74 \text{ mA} e^{-t/123.76 \text{ ms}}$

b.



33. a. $v_C = 140 \text{ mV}(1 - e^{-1\text{ms}/2 \text{ ms}}) = 140 \text{ mV}(1 - e^{-0.5}) = 140 \text{ mV}(1 - 0.6065)$
 $= 140 \text{ mV}(0.3935) = \mathbf{55.59 \text{ mV}}$
- b. $v_C = 140 \text{ mV}(1 - e^{-10}) = 140 \text{ mV}(1 - 45.4 \times 10^{-6})$
 $\cong \mathbf{139.99 \text{ mV}}$
- c. $100 \text{ mV} = 140 \text{ mV}(1 - e^{-t/2 \text{ ms}})$
 $0.714 = 1 - e^{-t/2 \text{ ms}}$
 $0.286 = e^{-t/2 \text{ ms}}$
 $\log_e 0.286 = \log_e e^{-t/2 \text{ ms}}$
 $1.252 = -t/2 \text{ ms}$
 $t = 1.252 (2 \text{ ms}) = \mathbf{2.5 \text{ ms}}$
- d. $v_C = 138 \text{ mV} = 140 \text{ mV}(1 - e^{t/2 \text{ ms}})$
 $0.986 = 1 - e^{-t/2 \text{ ms}}$
 $-14 \times 10^{-3} = -e^{-t/2 \text{ ms}}$
 $\log_e 14 \times 10^{-3} = -t/2 \text{ ms}$
 $-4.268 = -t/2 \text{ ms}$
 $t = (4.268)(2 \text{ ms}) = \mathbf{8.54 \mu\text{s}}$

$$\begin{aligned}
34. \quad \tau &= RC = (33 \text{ k}\Omega)(20 \text{ }\mu\text{F}) = 0.66 \text{ s} \\
v_C &= 12 \text{ V}(1 - e^{-t/0.66 \text{ s}}) \\
8 \text{ V} &= 12 \text{ V}(1 - e^{-t/0.66 \text{ s}}) \\
8 \text{ V} &= 12 \text{ V} - 12 \text{ V}e^{-t/0.66 \text{ s}} \\
-4 \text{ V} &= -12 \text{ V}e^{-t/0.66 \text{ s}} \\
0.333 &= e^{-t/0.66 \text{ s}} \\
\log_e 0.333 &= -t/0.66 \text{ s} \\
-1.0996 &= -t/0.66 \text{ s} \\
t &= 1.0996(0.66 \text{ s}) = \mathbf{0.73 \text{ s}}
\end{aligned}$$

$$\begin{aligned}
35. \quad t &= -\tau \log_e \left(1 - \frac{v_C}{E} \right) \\
10 \text{ s} &= -\tau \log_e \left(1 - \frac{12 \text{ V}}{20 \text{ V}} \right) \\
&\quad \underbrace{\hspace{1.5cm}}_{.4} \\
&\quad \underbrace{\hspace{1.5cm}}_{-916.29 \times 10^{-3}} \\
\tau &= \frac{10 \text{ s}}{0.916} = 10.92 \text{ s} \\
\tau = RC \Rightarrow R &= \frac{\tau}{C} = \frac{10.92 \text{ s}}{200 \text{ }\mu\text{F}} = \mathbf{54.60 \text{ k}\Omega}
\end{aligned}$$

$$\begin{aligned}
36. \quad \text{a.} \quad \tau &= RC = (12 \text{ k}\Omega + 8.2 \text{ k}\Omega)(6.8 \text{ }\mu\text{F}) = 137.36 \text{ ms} \\
v_C &= 60 \text{ V}(1 - e^{-t/\tau}) \\
48 \text{ V} &= 60 \text{ V}(1 - e^{-t/\tau}) \\
0.8 &= 1 - e^{-t/\tau} \\
0.2 &= e^{-t/\tau} \\
\log_e 0.2 &= \log_e e^{-t/\tau} \\
-1.61 &= -t/\tau \\
t &= (1.61)\tau = (1.61)(137.36 \text{ ms}) = \mathbf{221.15 \text{ ms}} \\
\text{b.} \quad i_C &= \frac{E}{R} e^{-t/\tau} = \frac{60 \text{ V}}{20.2 \text{ k}\Omega} e^{-t/\tau} \\
&= 2.97 \text{ mA} e^{-t/137.36 \text{ ms}} \\
i_C(221.15 \text{ ms}) &= 2.97 \text{ mA} e^{-221.15 \text{ ms}/137.36 \text{ ms}} \\
&= 2.97 \text{ mA} e^{-1.61} \\
&= 2.97 \text{ mA} (199.89 \times 10^{-3}) \\
&= \mathbf{0.594 \text{ mA}} \\
\text{c.} \quad t &= 2\tau \\
i_C &= 2.97 \text{ mA} e^{-2t/\tau} = 2.97 \text{ mA} e^{-2} \\
&= 0.4 \text{ mA} \quad \underbrace{\hspace{1.5cm}}_{0.135}
\end{aligned}$$

$$P = EI = (60 \text{ V})(0.4 \text{ mA}) = \mathbf{24 \text{ mW}}$$

37. a. $v_m = v_R = Ee^{-t/\tau} = 60 \text{ V}e^{-t/2} = 60 \text{ V}e^{-1}$
 $= 60 \text{ V}(0.3679)$
 $= \mathbf{22.07 \text{ V}}$

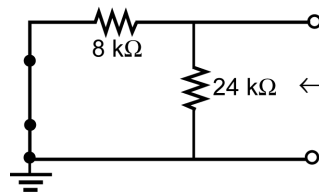
b. $i_C = \frac{E}{R}e^{-t/\tau} = \frac{60 \text{ V}}{10 \text{ M}\Omega}e^{-2\tau/\tau} = 6 \mu\text{A}e^{-2}$
 $= 6 \mu\text{A}(0.1353)$
 $= \mathbf{0.81 \mu\text{A}}$

c. $v_C = E(1 - e^{-t/\tau})$
 $50 \text{ V} = 60 \text{ V}(1 - e^{-t/2 \text{ s}})$
 $0.8333 = 1 - e^{-t/2 \text{ s}}$
 $\log_e 0.1667 = -t/2 \text{ s}$
 $t = -(2 \text{ s})(-1.792)$
 $= \mathbf{3.58 \text{ s}}$

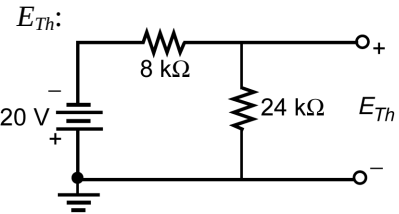
$\tau = RC = (10 \text{ M}\Omega)(0.2 \mu\text{F}) = 2 \text{ s}$

38. a. Thevenin's theorem:

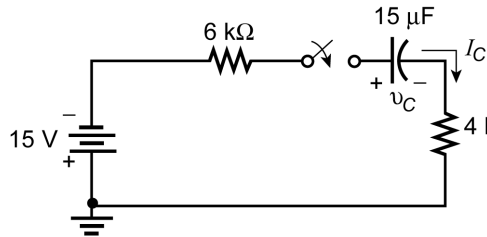
R_{Th} :



$\leftarrow R_{Th} = 8 \text{ k}\Omega \parallel 24 \text{ k}\Omega$
 $= 6 \text{ k}\Omega$



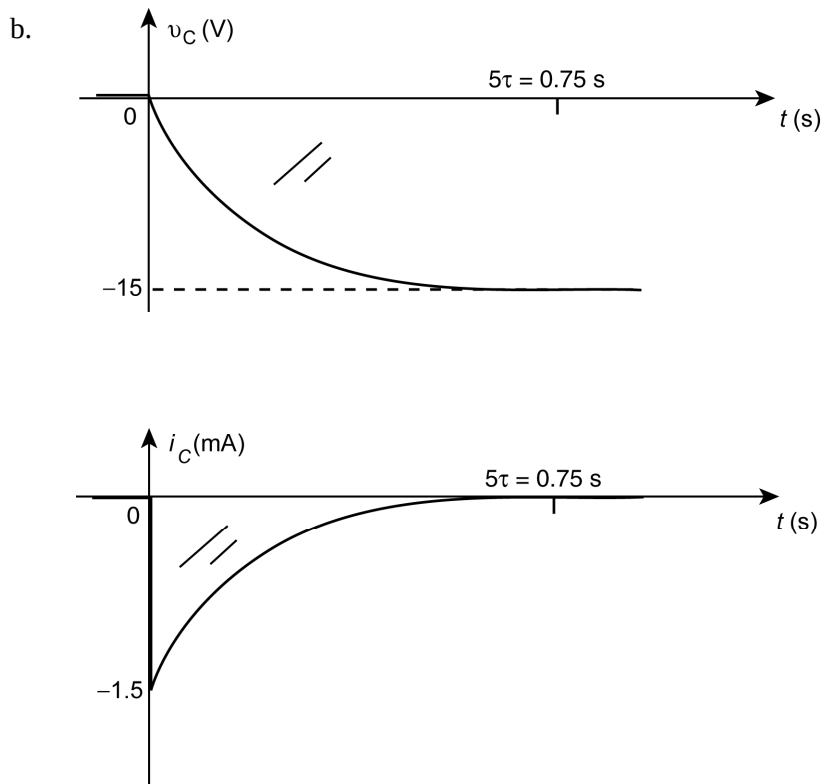
$E_{Th} = -\frac{24 \text{ k}\Omega(20 \text{ V})}{24 \text{ k}\Omega + 8 \text{ k}\Omega} = -15 \text{ V}$



$\tau = RC = (10 \text{ k}\Omega)(15 \mu\text{F}) = 0.15 \text{ s}$

$v_C = E(1 - e^{-t/\tau})$
 $= -15 \text{ V}(1 - e^{-t/0.15 \text{ s}})$

$i_C = \frac{E}{R}e^{-t/\tau} = -\frac{15 \text{ V}}{10 \text{ k}\Omega}e^{-t/0.15} = \mathbf{-1.5 \text{ mA}e^{-t/0.15 \text{ s}}}$



39. a. Source conversion and combining series resistors:

$$E = -(4 \text{ mA})(6.8 \text{ k}\Omega) = -27.2 \text{ V}$$

$$R_T = 6.8 \text{ k}\Omega + 1.5 \text{ k}\Omega = 8.3 \text{ k}\Omega$$

$$V_f = -27.2 \text{ V}, V_i = 10 \text{ V}$$

$$\tau = RC = (8.3 \text{ k}\Omega)(2.2 \text{ }\mu\text{F}) = 18.26 \text{ ms}$$

$$v_C = V_f + (V_i - V_f)e^{-t/\tau}$$

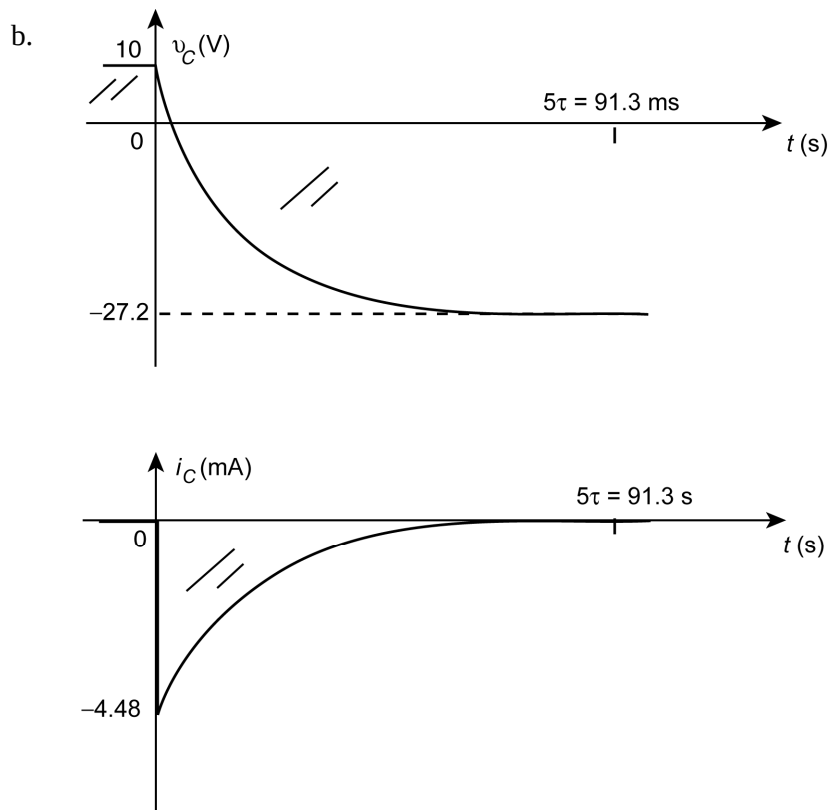
$$= -27.2 \text{ V} + (10 \text{ V} - (-27.2 \text{ V}))e^{-t/18.26 \text{ ms}}$$

$$v_C = -27.2 \text{ V} + 37.2 \text{ V}e^{-t/18.26 \text{ ms}}$$

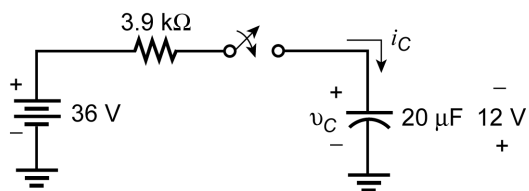
$$v_R(0+) = -27.2 \text{ V} - (-27.2 \text{ V})e^{-t/18.26 \text{ ms}} = -37.2 \text{ V}$$

$$i_C = -\frac{37.2 \text{ V}}{8.3 \text{ k}\Omega}e^{-t/18.26 \text{ ms}}$$

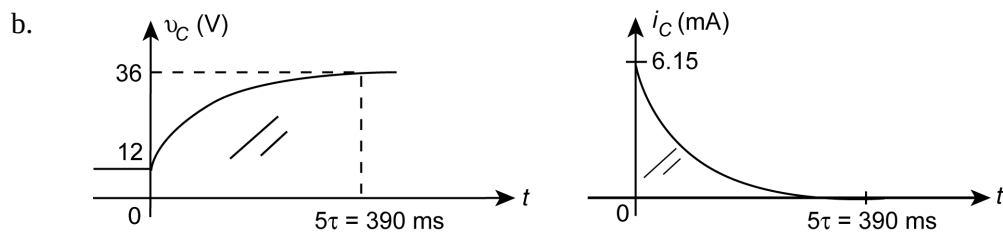
$$i_C = -4.48 \text{ mA}e^{-t/18.26 \text{ ms}}$$



40. a. $R_{Th} = 3.9 \text{ k}\Omega + 0 \text{ }\Omega \parallel 1.8 \text{ k}\Omega = 3.9 \text{ k}\Omega$
 $E_{Th} = 36 \text{ V}$



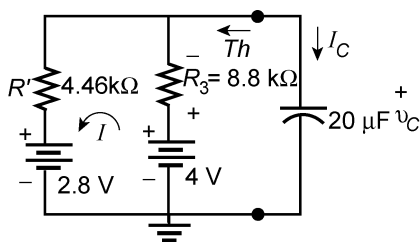
$$\begin{aligned}\tau &= RC = (3.9 \text{ k}\Omega)(20 \text{ }\mu\text{F}) = 78 \text{ ms} \\ v_C &= V_f + (V_i - V_f)e^{-t/\tau} \\ &= 36 \text{ V} + (-12 \text{ V} - 36 \text{ V})e^{-t/78 \text{ ms}} \\ v_C &= \mathbf{36 \text{ V} - 24 \text{ V}e^{-t/78 \text{ ms}}} \\ v_R(0+) &= 24 \text{ V} - 12 \text{ V} = 24 \text{ V} \\ i_C &= \frac{24 \text{ V}}{3.9 \text{ k}\Omega}e^{-t/78 \text{ ms}} \\ i_C &= \mathbf{6.15 \text{ mA}e^{-t/78 \text{ ms}}}\end{aligned}$$



41. Source conversion:

$$E = IR_1 = (5 \text{ mA})(0.56 \text{ k}\Omega) = 2.8 \text{ V}$$

$$R' = R_1 + R_2 = 0.56 \text{ k}\Omega + 3.9 \text{ k}\Omega = 4.46 \text{ k}\Omega$$



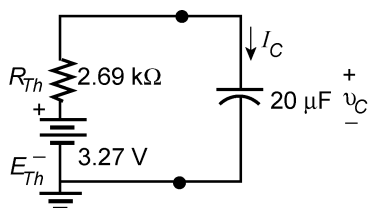
$$R_{Th} = 4.46 \text{ k}\Omega \parallel 8.8 \text{ k}\Omega = 2.69 \text{ k}\Omega$$

$$I = \frac{4 \text{ V} - 2.8 \text{ V}}{8.8 \text{ k}\Omega + 4.46 \text{ k}\Omega} = \frac{1.2 \text{ V}}{13.26 \text{ k}\Omega} = 0.0905 \text{ mA}$$

$$E_{Th} = 4 \text{ V} - (0.0905 \text{ mA})(8.8 \text{ k}\Omega)$$

$$= 4 \text{ V} - 0.796 \text{ V}$$

$$= 3.204 \text{ V}$$

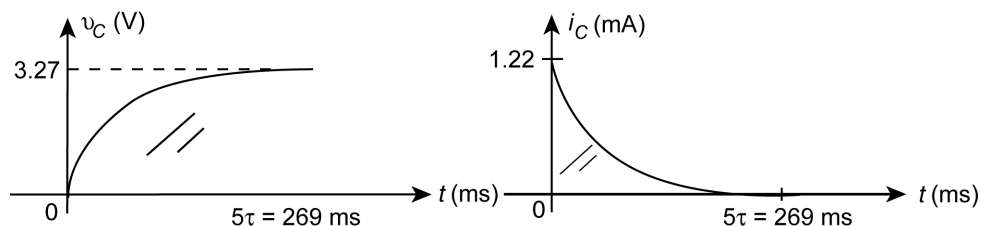


$$v_C = 3.204 \text{ V}(1 - e^{-t/\tau})$$

$$\tau = RC = (2.69 \text{ k}\Omega)(20 \mu\text{F}) = 53.8 \text{ ms}$$

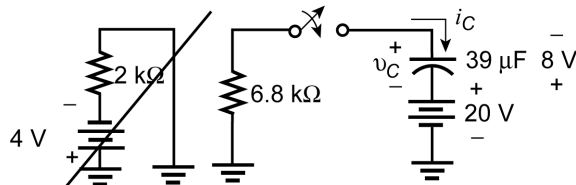
$$v_C = 3.204 \text{ V}(1 - e^{-t/53.8 \text{ ms}})$$

$$i_C = \frac{3.204 \text{ V}}{2.69 \text{ k}\Omega} e^{-t/\tau} = 1.191 \text{ mA} e^{-t/53.8 \text{ ms}}$$



42.

a.



$$\tau = RC = (6.8 \text{ k}\Omega)(39 \mu\text{F}) = 265.2 \text{ ms}$$

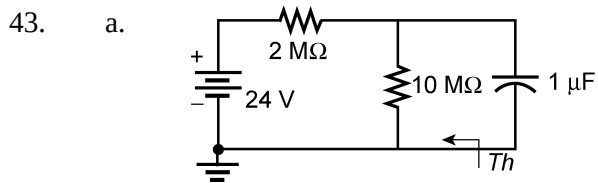
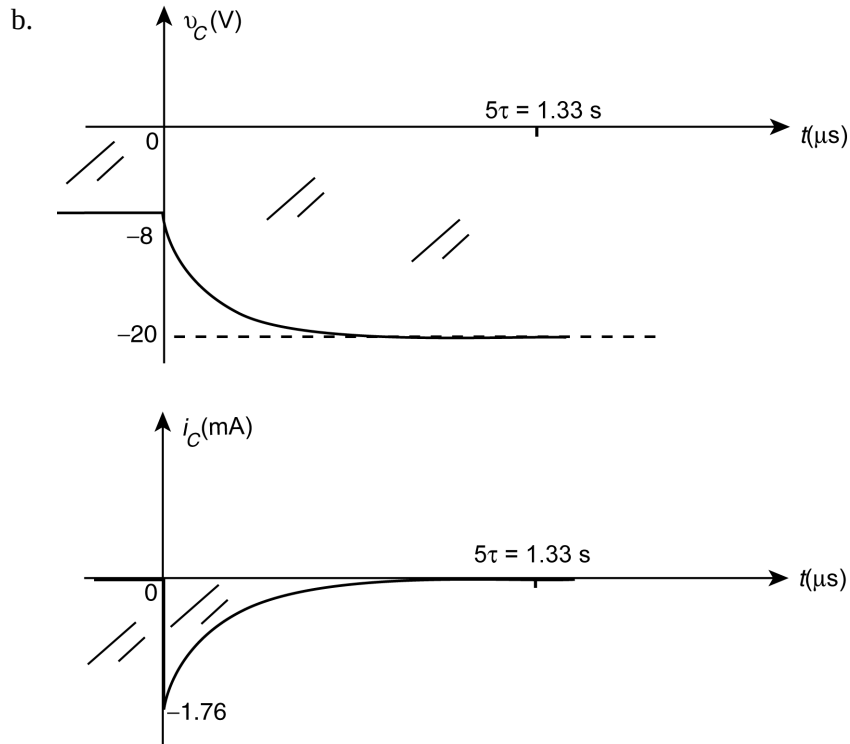
$$v_C = V_f + (V_i - V_f)e^{-t/\tau} = -20 \text{ V} + (-8 \text{ V} - (-20 \text{ V}))e^{-t/265.2 \text{ ms}}$$

$$v_C = -20 \text{ V} + 12 \text{ V}e^{-t/265.2 \text{ ms}}$$

$$v_C(0^+) = -20 \text{ V} + 12 \text{ V} = -8 \text{ V}$$

$$i_C = -\frac{12 \text{ V}}{6.8 \text{ k}\Omega} e^{-t/265.2 \text{ ms}}$$

$$i_C = -1.76 \text{ mA} e^{-t/265.2 \text{ ms}}$$



$$R_{Th} = 2 \text{ M}\Omega \parallel 10 \text{ M}\Omega = 1.67 \text{ M}\Omega$$

$$E_{Th} = \frac{10 \text{ M}\Omega(24 \text{ V})}{10 \text{ M}\Omega + 2 \text{ M}\Omega} = 20 \text{ V}$$

$$\begin{aligned} v_C &= E_{Th}(1 - e^{-t/\tau}) \\ &= 20 \text{ V}(1 - e^{-4\pi\tau}) \\ &= 20 \text{ V}(1 - e^{-4}) \\ &= 20 \text{ V}(1 - 0.0183) \\ &= \mathbf{19.63 \text{ V}} \end{aligned}$$

$$\tau = R_{Th}C = (1.67 \text{ M}\Omega)(1 \mu\text{F}) = 1.67 \text{ s}$$

$$i_C = \frac{E}{R} e^{-t/\tau}$$

$$3 \mu\text{A} = \frac{20 \text{ V}}{1.67 \text{ M}\Omega} e^{-t/1.67\text{s}}$$

$$0.25 = e^{-t/1.67\text{s}}$$

$$\log_e 0.25 = -t/1.67 \text{ s}$$

$$\begin{aligned} t &= -(1.67 \text{ s})(-1.39) \\ &= \mathbf{2.32 \text{ s}} \end{aligned}$$

c. $v_{\text{meter}} = v_C$

$$v_C = E_{Th}(1 - e^{-t/\tau})$$

$$10 \text{ V} = 20 \text{ V}(1 - e^{-t/1.67\text{s}})$$

$$0.5 = 1 - e^{-t/1.67\text{s}}$$

$$-0.5 = -e^{-t/1.67\text{s}}$$

$$\log_e 0.5 = -t/1.67 \text{ s}$$

$$\begin{aligned} t &= -(1.67 \text{ s})(-0.69) \\ &= \mathbf{1.15 \text{ s}} \end{aligned}$$

44. $i_{C_{ao}} = C \frac{\Delta v_C}{\Delta t}$

$$0 \rightarrow 1 \text{ ms: } i_C = 2 \times 10^{-6} \frac{(40 \text{ V})}{1 \text{ ms}} = \mathbf{80 \text{ mA}}$$

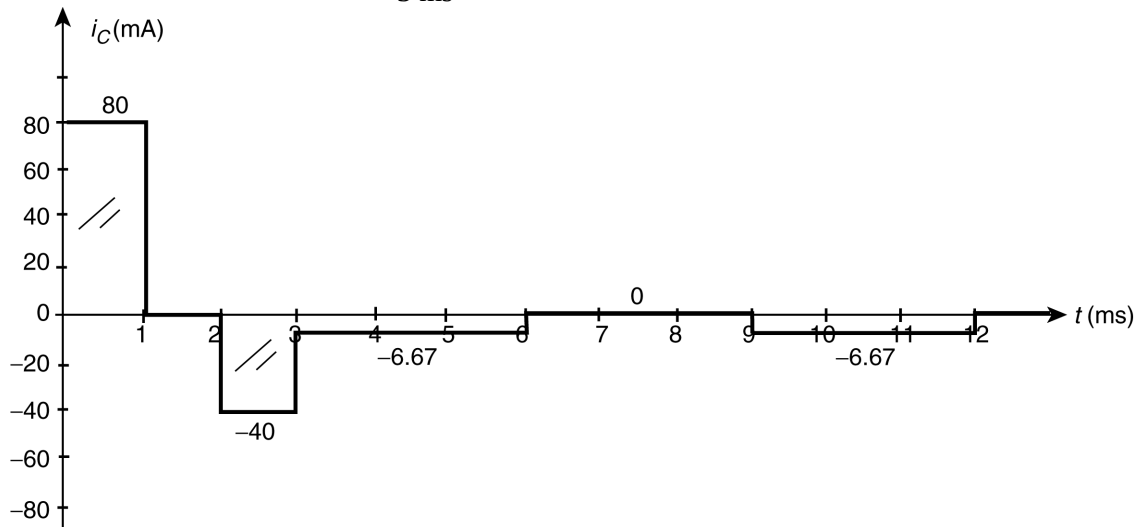
$$1 \rightarrow 2 \text{ ms: } i_C = 2 \times 10^{-6} \frac{(0 \text{ V})}{1 \text{ ms}} = \mathbf{0 \text{ mA}}$$

$$2 \rightarrow 3 \text{ ms: } i_C = -2 \times 10^{-6} \frac{(20 \text{ V})}{1 \text{ ms}} = \mathbf{-40 \text{ mA}}$$

$$3 \rightarrow 6 \text{ ms: } i_C = -2 \times 10^{-6} \frac{(10 \text{ V})}{3 \text{ ms}} = \mathbf{-6.67 \text{ mA}}$$

$$6 \rightarrow 9 \text{ ms: } i_C = 2 \times 10^{-6} \frac{(0 \text{ V})}{3 \text{ ms}} = \mathbf{0 \text{ mA}}$$

$$9 \rightarrow 12 \text{ ms: } i_C = -2 \times 10^{-6} \frac{(10 \text{ V})}{3 \text{ ms}} = \mathbf{-6.67 \text{ mA}}$$



45. $i_{C_{ao}} = C \frac{\Delta v_C}{\Delta t}$

$$0 \rightarrow 20 \mu\text{s: } i_C = -4.7 \mu\text{F} \frac{(5 \text{ V})}{20 \mu\text{s}} = \mathbf{-1.18 \text{ A}}$$

$$20 \rightarrow 30 \mu\text{s: } i_C = 4.7 \mu\text{F} \frac{(15 \text{ V})}{10 \mu\text{s}} = \mathbf{7.05 \text{ A}}$$

$$30 \rightarrow 40 \mu\text{s: } i_C = -4.7 \mu\text{F} \frac{(15 \text{ V})}{10 \mu\text{s}} = \mathbf{-7.05 \text{ A}}$$

$$40 \rightarrow 50 \mu\text{s: } i_C = 4.7 \mu\text{F} \frac{(0 \text{ V})}{10 \mu\text{s}} = \mathbf{0 \text{ A}}$$

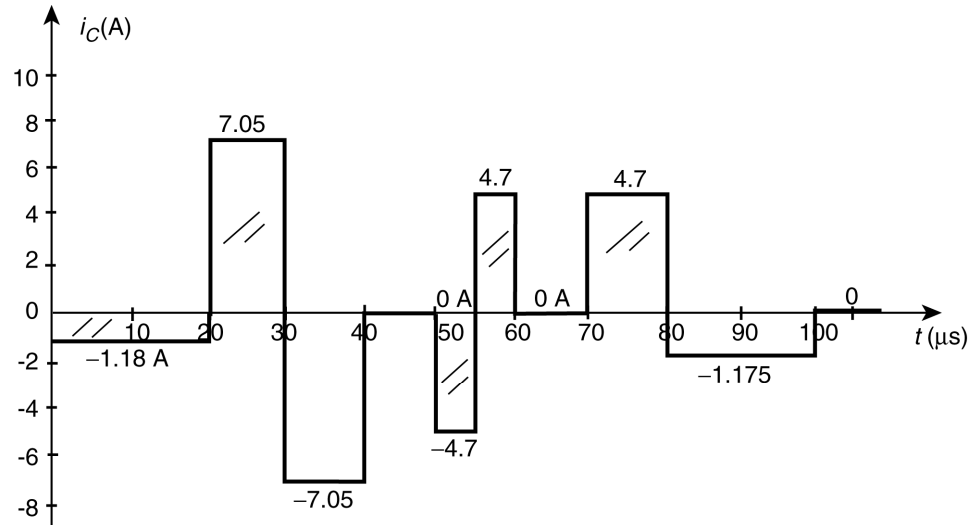
$$50 \rightarrow 55 \mu\text{s: } i_C = -4.7 \mu\text{F} \frac{(5 \text{ V})}{5 \mu\text{s}} = \mathbf{-4.7 \text{ A}}$$

$$55 \mu\text{s} \rightarrow 60 \mu\text{s: } i_C = 4.7 \mu\text{F} \frac{(5 \text{ V})}{5 \mu\text{s}} = \mathbf{4.7 \text{ A}}$$

$$60 \mu\text{s} \rightarrow 70 \mu\text{s}: i_C = 4.7 \mu\text{F} \frac{(0 \text{ V})}{10 \mu\text{s}} = 0 \text{ A}$$

$$70 \mu\text{s} \rightarrow 80 \mu\text{s}: i_C = 4.7 \mu\text{F} \frac{(10 \text{ V})}{10 \mu\text{s}} = 4.7 \text{ A}$$

$$80 \mu\text{s} \rightarrow 100 \mu\text{s}: i_C = -4.7 \mu\text{F} \frac{(5 \text{ V})}{20 \mu\text{s}} = -1.175 \text{ A}$$



$$46. \quad i_C = C \frac{\Delta v_C}{\Delta t} \Rightarrow \Delta v_C = \frac{\Delta t}{C} (i_C)$$

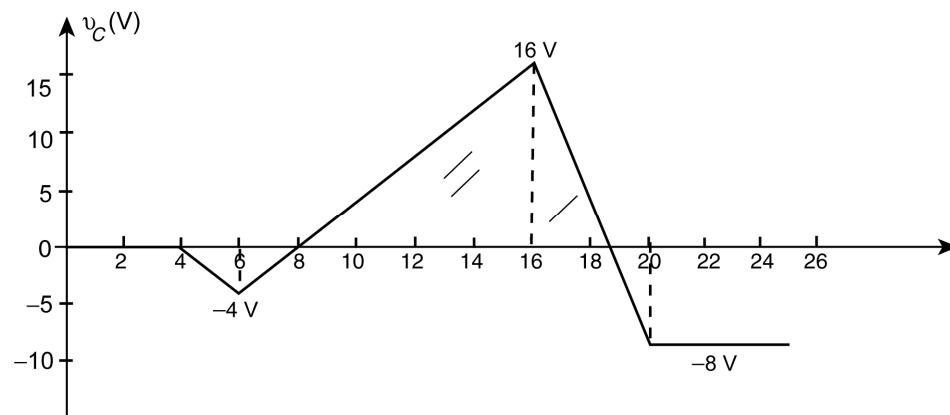
$$0 \rightarrow 4 \text{ ms}: i_C = 0 \text{ mA} \quad \Delta v_C = 0 \text{ V}$$

$$4 \rightarrow 6 \text{ ms}: i_C = -40 \text{ mA} \quad \Delta v_C = \frac{(2 \text{ ms})}{20 \mu\text{F}} (-40 \text{ mA}) = -4 \text{ V}$$

$$6 \rightarrow 16 \text{ ms}: i_C = +40 \text{ mA} \quad \Delta v_C = \frac{(10 \text{ ms})}{20 \mu\text{F}} (40 \text{ mA}) = +20 \text{ V}$$

$$16 \rightarrow 20 \text{ ms}: i_C = -120 \text{ mA} \quad \Delta v_C = \frac{(4 \text{ ms})}{20 \mu\text{F}} (-120 \text{ mA}) = -24 \text{ V}$$

$$20 \rightarrow 25 \text{ ms}: i_C = 0 \text{ mA} \quad \Delta v_C = 0 \text{ V}$$



47. $6 \mu\text{F} + 4 \mu\text{F} = 10 \mu\text{F}$, $8 \mu\text{F} + 12 \mu\text{F} = 20 \mu\text{F}$
 $10 \mu\text{F} \parallel 20 \mu\text{F} = \mathbf{6.67 \mu\text{F}}$
48. $C'_T = 6 \mu\text{F} \parallel 12 \mu\text{F} = 4 \mu\text{F}$
 $C''_T = C'_T + 12 \mu\text{F} = 4 \mu\text{F} + 12 \mu\text{F} = 16 \mu\text{F}$
 $C_T = 6 \mu\text{F} \parallel C''_T = \frac{6 \mu\text{F} \cdot C''_T}{6 \mu\text{F} + C''_T} = \frac{(6 \mu\text{F})(16 \mu\text{F})}{6 \mu\text{F} + 16 \mu\text{F}} = \mathbf{4.36 \mu\text{F}}$
49. $V_1 = \mathbf{10 \text{ V}}$, $Q_1 = V_1 C_1 = (10 \text{ V})(6 \mu\text{F}) = \mathbf{60 \mu\text{C}}$
 $C_T = 6 \mu\text{F} \parallel 12 \mu\text{F} = 4 \mu\text{F}$, $Q_T = C_T E = (4 \mu\text{F})(10 \text{ V}) = 40 \mu\text{C}$
 $Q_2 = Q_3 = \mathbf{40 \mu\text{C}}$
 $V_2 = \frac{Q_2}{C_2} = \frac{40 \mu\text{C}}{6 \mu\text{F}} = \mathbf{6.67 \text{ V}}$
 $V_3 = \frac{Q_3}{C_3} = \frac{40 \mu\text{C}}{12 \mu\text{F}} = \mathbf{3.33 \text{ V}}$
50. $360 \mu\text{F} + 200 \mu\text{F} = 560 \mu\text{F}$
 $470 \mu\text{F} \parallel 560 \mu\text{F} = 255.53 \mu\text{F}$
 $Q_T = Q_3 = C_T E = (255.53 \mu\text{F})(56 \text{ V}) = \mathbf{14.5 \text{ mC}}$
 $V_3 = \frac{Q_3}{C_3} = \frac{14.5 \text{ mC}}{470 \mu\text{F}} = \mathbf{30.4 \text{ V}}$
 $V_1 = V_2 = E - V_3 = 56 \text{ V} - 30.4 \text{ V} = \mathbf{25.6 \text{ V}}$
 $Q_1 = V_1 C_1 = (25.6 \text{ V})(360 \mu\text{F}) = \mathbf{9.2 \text{ mC}}$
 $Q_2 = V_2 C_2 = (25.6 \text{ V})(200 \mu\text{F}) = \mathbf{5.1 \text{ mC}}$
51. steady – state – ignore $10 \text{ k}\Omega$ resistor
 $330 \mu\text{F} + 120 \mu\text{F} = 450 \mu\text{F}$
 $C_T = 220 \mu\text{F} \parallel 450 \mu\text{F} = 147.76 \mu\text{F}$
 $Q_T = Q_1 = C_T E = (147.76 \mu\text{F})(20 \text{ V}) = \mathbf{2.96 \text{ mC}}$
 $V_1 = \frac{Q_1}{C_1} = \frac{2.96 \text{ mC}}{220 \mu\text{F}} = \mathbf{13.45 \text{ V}}$
 $V_3 = V_2 = E - V_1 = 20 \text{ V} - 13.45 \text{ V} = \mathbf{6.55 \text{ V}}$
 $Q_2 = C_2 V_2 = (330 \mu\text{F})(6.55 \text{ V}) = \mathbf{2.16 \text{ mC}}$
 $Q_3 = C_3 V_3 = (120 \mu\text{F})(6.55 \text{ V}) = \mathbf{0.786 \text{ mC}}$
52. $V_{4\text{k}\Omega} = \frac{4 \text{ k}\Omega(48 \text{ V})}{4 \text{ k}\Omega + 2 \text{ k}\Omega} = \mathbf{32 \text{ V}} = V_{0.08\mu\text{F}}$
 $Q_{0.08\mu\text{F}} = (0.08 \mu\text{F})(32 \text{ V}) = \mathbf{2.56 \mu\text{C}}$
 $V_{0.04\mu\text{F}} = \mathbf{48 \text{ V}}$
 $Q_{0.04\mu\text{F}} = (0.04 \mu\text{F})(48 \text{ V}) = \mathbf{1.92 \mu\text{C}}$
53. $W_C = \frac{1}{2} C V^2 = \frac{1}{2} (120 \text{ pF})(12 \text{ V})^2 = \mathbf{8,640 \text{ pJ}}$
54. $W = \frac{Q^2}{2C} \Rightarrow Q = \sqrt{2CW} = \sqrt{2(6 \mu\text{F})(1200 \text{ J})} = \mathbf{0.12 \text{ C}}$

55. a. $V_{200\mu\text{F}} = \frac{(220 \text{ k}\Omega + 3.3 \text{ k}\Omega)(12 \text{ V})}{2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega + 1.2 \text{ k}\Omega} = 9.85 \text{ V}$
 $V_{100\mu\text{F}} = \frac{(3.3 \text{ k}\Omega)(12 \text{ V})}{2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega + 1.2 \text{ k}\Omega} = 5.91 \text{ V}$
 $W_{200\mu\text{F}} = \frac{1}{2}(200 \mu\text{F})(9.85 \text{ V})^2 = \mathbf{970 \text{ mJ}}$
 $W_{100\mu\text{F}} = \frac{1}{2}(100 \mu\text{F})(5.91 \text{ V})^2 = \mathbf{1.75 \text{ mJ}}$
56. a. $W_C = \frac{1}{2}CV^2 = \frac{1}{2}(1000 \mu\text{F})(100 \text{ V})^2 = \mathbf{5 \text{ pJ}}$
- b. $Q = CV = (1000 \mu\text{F})(100 \text{ V}) = \mathbf{0.1 \text{ C}}$
- c. $I = Q/t = 0.1 \text{ C}/(1/2000) = \mathbf{200 \text{ A}}$
- d. $P = V_{av}I_{av} = W/t = 5 \text{ J}/(1/2000 \text{ s}) = \mathbf{10,000 \text{ W}}$
- e. $t = Q/I = 0.1 \text{ C}/10 \text{ mA} = \mathbf{10 \text{ s}}$

Chapter 11

1.
 - a. $B = \frac{\Phi}{A} = \frac{4 \times 10^{-4} \text{ Wb}}{0.01 \text{ m}^2} = 4 \times 10^{-2} \text{ Wb/m}^2 = \mathbf{0.04 \text{ Wb/m}^2}$
 - b. $\mathbf{0.04 \text{ T}}$
 - c. $F = NI = (40 \text{ t})(2.2 \text{ A}) = \mathbf{88 \text{ At}}$
 - d. $0.04 \cancel{\text{A}} \left[\frac{10^4 \text{ gauss}}{1 \cancel{\text{A}}} \right] = \mathbf{0.4 \times 10^3 \text{ gauss}}$

2.

$$0.2'' \left[\frac{2.54 \text{ cm}}{1''} \right] \left[\frac{1 \text{ m}}{100 \text{ cm}} \right] = 5.08 \text{ mm}$$

$$1'' \left[\frac{2.54 \text{ cm}}{1''} \right] \left[\frac{1 \text{ m}}{100 \text{ cm}} \right] = 25.4 \text{ mm}$$

$$A = \frac{\pi d^2}{4} = \frac{\pi (5.08 \text{ mm})^2}{4} = 20.27 \times 10^{-6} \text{ m}^2$$

$$L = \frac{N^2 \mu A}{\ell} = \frac{(200 \text{ t})^2 (4\pi \times 10^{-7}) (20.27 \times 10^{-6} \text{ m}^2)}{25.4 \text{ mm}} = \mathbf{40.1 \mu\text{H}}$$

3.
 - a. $L = \frac{N^2 \mu_r \mu_o A}{\ell} = \frac{(200 \text{ t})^2 (500) (4\pi \times 10^{-7}) (20.27 \times 10^{-6} \text{ m}^2)}{25.4 \text{ mm}} = \mathbf{20.06 \text{ mH}}$
 - b. increase = change in μ_r
 $L_{\text{new}} = \mu_r L_o$

4. $L = N^2 \frac{\mu_r \mu_o}{\ell} = \frac{(200 \text{ t})^2 (1000) (4\pi \times 10^{-7}) (1.5 \times 10^{-4} \text{ m}^2)}{0.15 \text{ m}} = \mathbf{50.27 \text{ mH}}$

5. $L = \frac{N^2 \mu_r \mu_o A}{\ell}$
 - a. $L' = (3)^2 L_o = 9 L_o = 9(4.7 \text{ mH}) = \mathbf{42.3 \text{ mH}}$
 - b. $L' = \frac{1}{3} L_o = \frac{1}{3} (4.7 \text{ mH}) = \mathbf{1.57 \text{ mH}}$
 - c. $L' = \frac{(2)(2)^2}{\frac{1}{2}} L_o = 16 (4.7 \text{ mH}) = \mathbf{75.2 \text{ mH}}$
 - d. $L' = \frac{\left(\frac{1}{2}\right)^2 \frac{1}{2} (1500) L_o}{\frac{1}{2}} = 375 (4.7 \text{ mH}) = \mathbf{1.76 \text{ mH}}$

6.
 - a. $39 \times 10^2 \mu\text{H} \pm 10\% \Rightarrow 3900 \mu\text{H} \pm 10\% \Rightarrow \mathbf{3.9 \text{ mH} \pm 10\%}$
 - b. $68 \times 10^0 \mu\text{H} \pm 5\% = \mathbf{68 \mu\text{F} \pm 5\%}$

c. **$47 \mu\text{H} \pm 10\%$**

d. $15 \times 10^2 \mu\text{H} \pm 10\% = 1500 \mu\text{H} \pm 10\% = \mathbf{15 \text{ mH} \pm 10\%}$

7. $e = N \frac{d\phi}{dt} = (50 \text{ t})(120 \text{ mWb/s}) = \mathbf{6.0 \text{ V}}$

8. $e = N \frac{d\phi}{dt} \Rightarrow \frac{d\phi}{dt} = \frac{e}{N} = \frac{20 \text{ V}}{200 \text{ t}} = \mathbf{100 \text{ mWb/s}}$

9. $e = N \frac{d\phi}{dt} \Rightarrow N = e \left(\frac{1}{\frac{d\phi}{dt}} \right) = 42 \text{ mV} \left(\frac{1}{3 \text{ mWb/s}} \right) = \mathbf{14 \text{ turns}}$

10. a. $e = L \frac{di_L}{dt} = (22 \text{ mH})(1 \text{ A/s}) = \mathbf{22 \text{ mV}}$

b. $e = L \frac{di_L}{dt} = (22 \text{ mH})(1 \text{ mA/ms}) = \mathbf{22 \text{ mV}}$

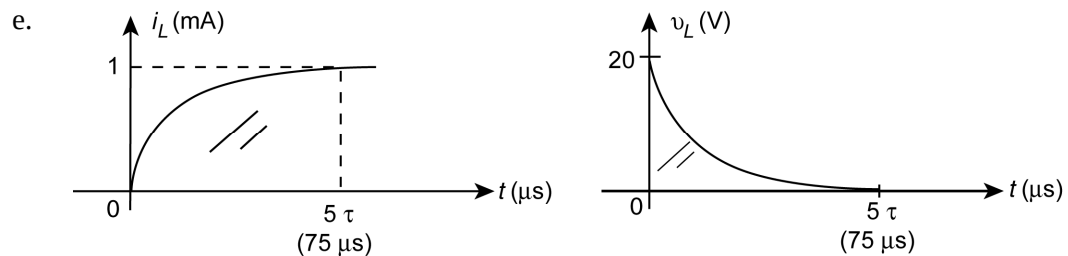
$e = L \frac{di_L}{dt} = (22 \text{ mH}) \left(\frac{2 \text{ mA}}{10 \mu\text{s}} \right) = \mathbf{4.4 \text{ V}}$

11. a. $\tau = \frac{L}{R} = \frac{300 \text{ mH}}{20 \text{ k}\Omega} = \mathbf{15 \mu\text{s}}$

b. $i_L = \frac{E}{R}(1 - e^{-t/\tau}) = \frac{20 \text{ mV}}{20 \text{ k}\Omega}(1 - e^{-t/\tau})$
 $= \mathbf{1 \text{ mA}(1 - e^{-t/15 \mu\text{s}})}$

c. $v_L = Ee^{-t/\tau} = \mathbf{20 \text{ V}e^{-t/15 \mu\text{s}}}$
 $v_R = i_R R = i_L R = E(1 - e^{-t/\tau}) = \mathbf{20 \text{ V}(1 - e^{-t/15 \mu\text{s}})}$

d. i_L : $1\tau = \mathbf{0.632 \text{ mA}}$, $3\tau = \mathbf{0.951 \text{ mA}}$, $5\tau = \mathbf{0.993 \text{ mA}}$
 v_L : $1\tau = \mathbf{7.36 \text{ V}}$, $3\tau = \mathbf{0.98 \text{ V}}$, $5\tau = \mathbf{140 \text{ mV}}$

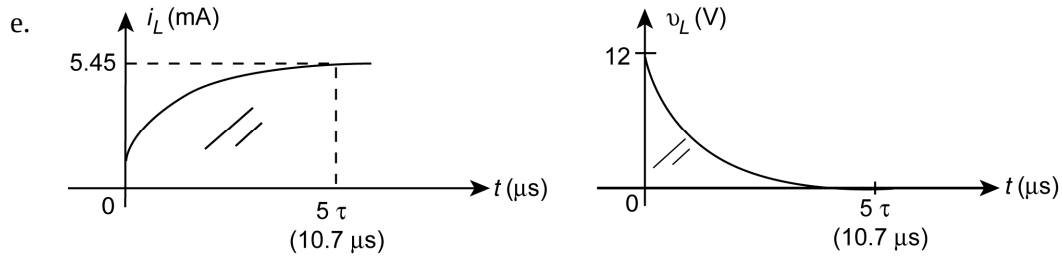


12. a. $\tau = \frac{L}{R} = \frac{4.7 \text{ mH}}{2.2 \text{ k}\Omega} = 2.14 \mu\text{s}$

b. $i_L = \frac{E}{R}(1 - e^{-t/\tau}) = \frac{12 \text{ V}}{2.2 \text{ k}\Omega}(1 - e^{-t/2.14 \mu\text{s}}) = 5.45 \text{ mA}(1 - e^{-t/2.14 \mu\text{s}})$

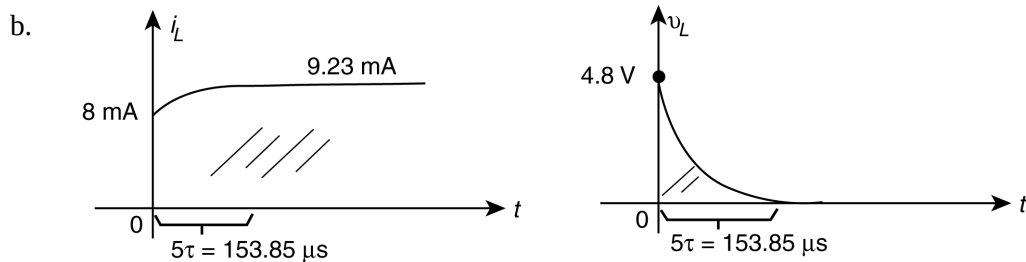
c. $v_L = Ee^{-t/\tau} = 12 \text{ V}e^{-t/2.14 \mu\text{s}}$
 $v_R = i_R R = i_L R = E(1 - e^{-t/\tau}) = 12 \text{ V}(1 - e^{-t/2.14 \mu\text{s}})$

d. i_L : $1\tau = 3.45 \text{ mA}$, $3\tau = 5.18 \text{ mA}$, $5\tau = 5.41 \text{ mA}$
 v_L : $1\tau = 4.42 \text{ V}$, $3\tau = 0.60 \text{ V}$, $5\tau = 0.08 \text{ V}$



13. $i_L = \frac{18 \text{ V}}{R_T}(1 - e^{-t/\tau})$
 $R_T = \frac{18 \text{ V}}{15 \text{ mA}} = 1.2 \text{ k}\Omega$
 $5\tau = 15 \left(\frac{L}{R} \right) = 15 \mu\text{s}$: $L = \frac{R(15 \mu\text{s})}{5} = \frac{1.2 \text{ k}\Omega(15 \mu\text{s})}{5} = 3.6 \text{ mH}$

14. a. $i_L = I_f + (I_i - I_f)e^{-t/\tau}$
 $I_i = 8 \text{ mA}$, $I_f = \frac{E}{R} = \frac{36 \text{ V}}{3.9 \text{ k}\Omega} = 9.23 \text{ mA}$, $\tau = \frac{L}{R} = \frac{120 \text{ mH}}{3.9 \text{ k}\Omega} = 30.77 \mu\text{s}$
 $i_L = 9.23 \text{ mA} + (8 \text{ mA} - 9.23 \text{ mA})e^{-t/30.77 \mu\text{s}}$
 $i_L = 9.23 \text{ mA} - 1.23 \text{ mA}e^{-t/30.77 \mu\text{s}}$
 $+E - v_L - v_R = 0$ and $v_L = E - v_R$
 $v_R = i_R R = i_L R = (8 \text{ mA})(3.9 \text{ k}\Omega) = 31.2 \text{ V}$
 $v_L = E - v_R = 36 \text{ V} - 31.2 \text{ V} = 4.8 \text{ V}$
 $v_L = 4.8 \text{ V}e^{-t/30.77 \mu\text{s}}$



15. a. $I_i = -8 \text{ mA}$, $I_f = 9.23 \text{ mA}$, $\tau = \frac{L}{R} = \frac{120 \text{ mH}}{3.9 \text{ k}\Omega} = 30.77 \mu\text{s}$

$$i_L = I_f + (I_i - I_f)e^{-t/\tau}$$

$$= 9.23 \text{ mA} + (-8 \text{ mA} - 9.23 \text{ mA})e^{-t/30.77 \mu\text{s}}$$

$$i_L = \mathbf{9.23 \text{ mA} - 17.23 \text{ mA} e^{-t/30.77 \mu\text{s}}}$$

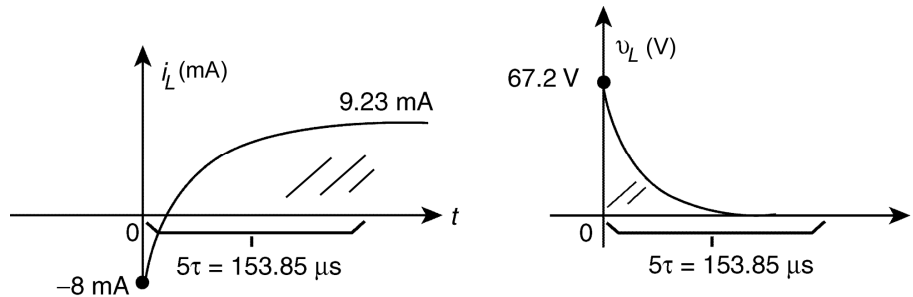
$$+E - v_L - v_R = 0 \text{ (at } t = 0^-)$$

$$\text{but, } v_R = i_R R = -i_L R = (-8 \text{ mA})(3.9 \text{ k}\Omega) = -31.2 \text{ V}$$

$$v_L = E - v_R = 36 \text{ V} - (-31.2 \text{ V}) = 67.2 \text{ V}$$

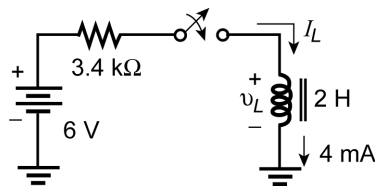
$$v_L = \mathbf{67.2 \text{ V} e^{-t/30.77 \mu\text{s}}}$$

b.



c. Final levels are the same. Transition period defined by 5τ is also the same.

16. a. Source conversion:



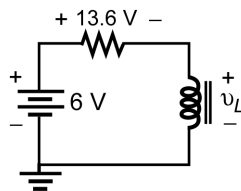
$$\tau = \frac{L}{R} = \frac{2 \text{ H}}{3.4 \text{ k}\Omega} = 588.2 \mu\text{s}$$

$$i_L = I_f + (I_i - I_f)e^{-t/\tau}$$

$$I_f = \frac{6 \text{ V}}{3.4 \text{ k}\Omega} = 1.76 \text{ mA}$$

$$i_L = 1.76 \text{ mA} + (4 \text{ mA} - 1.76 \text{ mA})e^{-t/588.2 \mu\text{s}}$$

$$i_L = \mathbf{1.76 \text{ mA} + 2.24 \text{ mA} e^{-t/588.2 \mu\text{s}}}$$



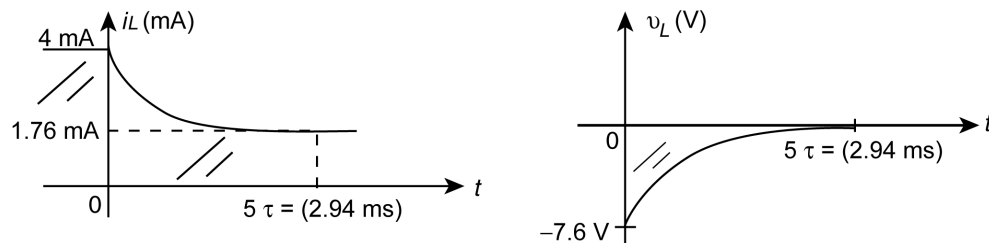
$$v_R(0+) = 4 \text{ mA}(3.4 \text{ k}\Omega) = 13.6 \text{ V}$$

$$\text{KVL: } +6 \text{ V} - 13.6 \text{ V} - v_L(0+) = 0$$

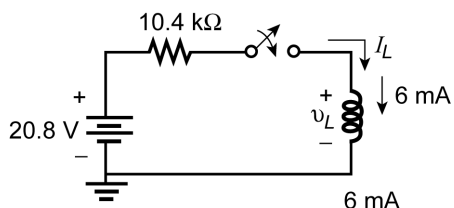
$$v_L(0+) = -7.6 \text{ V}$$

$$v_L = \mathbf{-7.6 \text{ V} e^{-t/588.2 \mu\text{s}}}$$

b.



17. a.



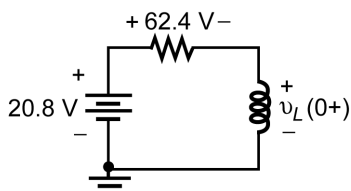
$$I_f = \frac{20.8 \text{ V}}{10.4 \text{ k}\Omega} = 2 \text{ mA}$$

$$\tau = \frac{L}{R} = \frac{200 \text{ mH}}{10.4 \text{ k}\Omega} = 19.23 \mu\text{s}$$

$$i_L = I_f + (I_i - I_f)e^{-t/\tau}$$

$$= 2 \text{ mA} + (6 \text{ mA} - 2 \text{ mA})e^{-t/19.23 \mu\text{s}}$$

$$i_L = 2 \text{ mA} + 4 \text{ mA}e^{-t/19.23 \mu\text{s}}$$

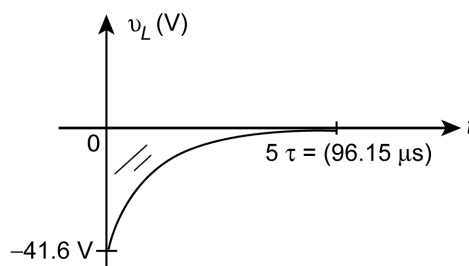
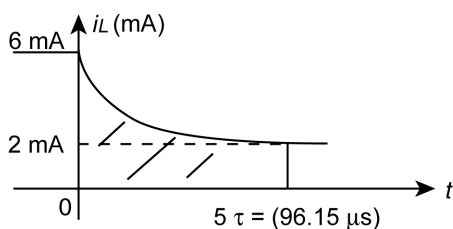


$$\text{KVL: } 20.8 \text{ V} - 62.4 \text{ V} - v_L(0+) = 0$$

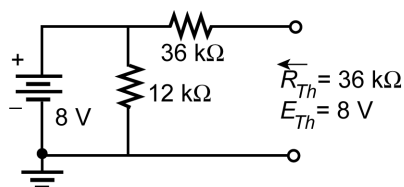
$$v_L(0+) = -41.60 \text{ V}$$

$$v_L = -41.6 \text{ V}e^{-t/19.23 \mu\text{s}}$$

b.



18. a.



$$\tau = \frac{L}{R} = \frac{10 \text{ mH}}{36 \text{ k}\Omega} = 0.278 \mu\text{s}$$

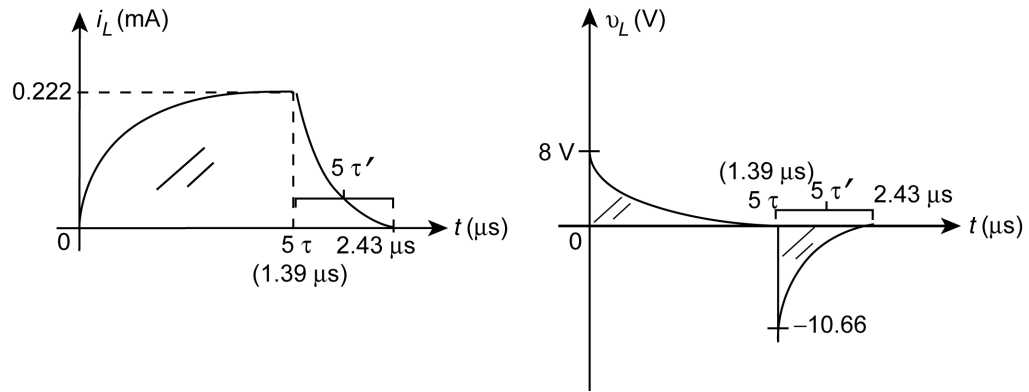
$$v_L = 8 \text{ V}e^{-t/0.278 \mu\text{s}}, i_L = \frac{E}{R}(1 - e^{-t/\tau}) = 0.222 \text{ mA}(1 - e^{-t/0.278 \mu\text{s}})$$

b. $5\tau \Rightarrow$ steady state

$$\tau' = \frac{L}{R} = \frac{10 \text{ mH}}{12 \text{ k}\Omega + 36 \text{ k}\Omega} = 0.208 \mu\text{s}$$

$$i_L = I_m e^{-t/\tau'} = 0.222 \text{ mA}e^{-t/0.208 \mu\text{s}}$$

$$v_L = -(0.222 \text{ mA})(48 \text{ k}\Omega)e^{-t/\tau} = -10.66 \text{ V}e^{-t/0.308 \mu\text{s}}$$



19. a. $\tau = \frac{L}{R} = \frac{1 \text{ mH}}{2 \text{ k}\Omega} = 0.5 \mu\text{s}$

$$i_L = \frac{E}{R}(1 - e^{-t/\tau}) = \frac{12 \text{ V}}{2 \text{ k}\Omega}(1 - e^{-t/0.5 \mu\text{s}}) = 6 \text{ mA}(1 - e^{-t/0.5 \mu\text{s}})$$

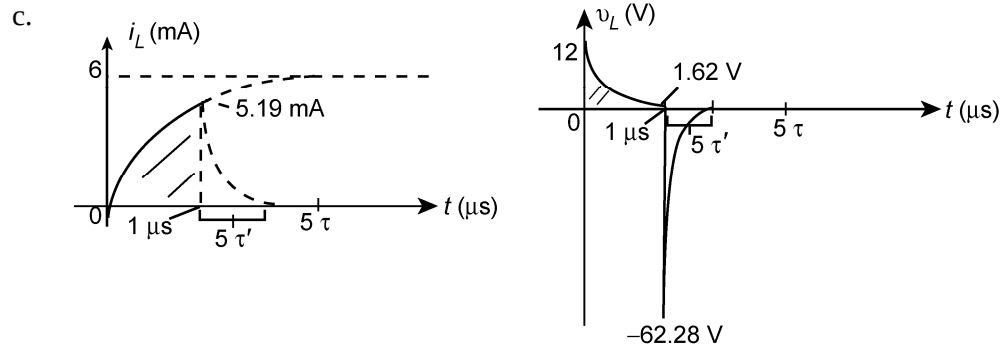
$$v_L = Ee^{-t/\tau} = 12 \text{ V}e^{-t/0.5 \mu\text{s}}$$

b. $i_L = 6 \text{ mA}(1 - e^{-t/0.5 \mu\text{s}}) = 6 \text{ mA}(1 - e^{-1 \mu\text{s}/0.5 \mu\text{s}})$
 $= 6 \text{ mA}(1 - e^{-2}) = 5.19 \text{ mA}$

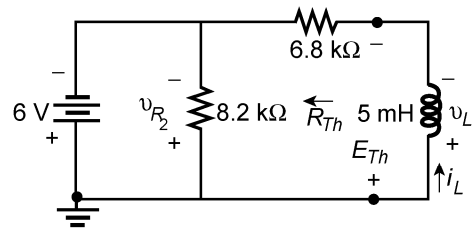
$$i_L = I_m e^{-t'/\tau'} \quad t' = \frac{L}{R} = \frac{1 \text{ mH}}{12 \text{ k}\Omega} = 83.3 \text{ ns}$$

$$i_L = 5.19 \text{ mA}e^{-t'/83.3 \text{ ns}}$$

$t = 1 \mu\text{s}$: $v_L = 12 \text{ V}e^{-t/0.5 \mu\text{s}} = 12 \text{ V}e^{-2} = 12 \text{ V}(0.1353) = 1.62 \text{ V}$
 $V_L' = (5.19 \text{ mA})(12 \text{ k}\Omega) = 62.28 \text{ V}$
 $v_L = -62.28 \text{ V}e^{-t'/83.3 \text{ ns}}$

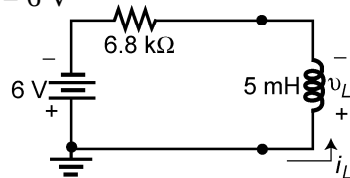


20. a.



$$R_{Th} = 6.8 \text{ k}\Omega$$

$$E_{Th} = 6 \text{ V}$$

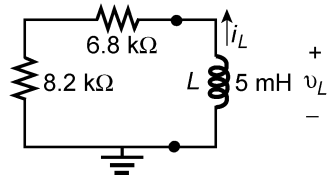


$$\tau = \frac{L}{R} = \frac{5 \text{ mH}}{6.8 \text{ k}\Omega} = 0.74 \mu\text{s}$$

$$i_L = \frac{E}{R} (1 - e^{-t/\tau}) = \frac{6 \text{ V}}{6.8 \text{ k}\Omega} (1 - e^{-t/0.74 \mu\text{s}}) = 0.88 \text{ mA} (1 - e^{-t/0.74 \mu\text{s}})$$

$$v_L = E e^{-t/\tau} = 6 \text{ V} e^{-t/0.74 \mu\text{s}}$$

b. Assume steady state and $I_L = 0.88 \text{ mA}$



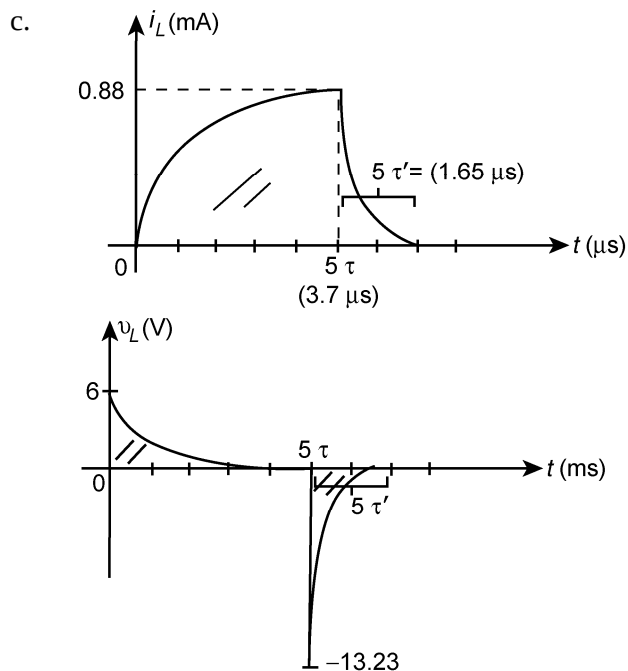
$$\tau' = \frac{L}{R} = \frac{5 \text{ mH}}{15 \text{ k}\Omega} = 0.33 \mu\text{s}$$

$$i_L = I_m e^{-t/\tau'} = 0.88 \text{ mA} e^{-t/0.33 \mu\text{s}}$$

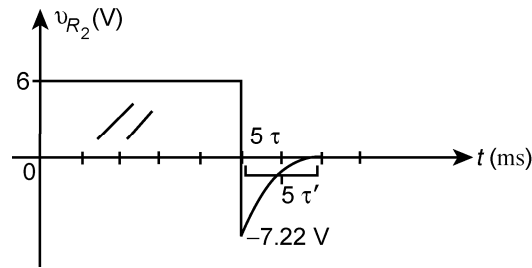
$$v_L = -V_m e^{-t/\tau'}$$

$$V_m = I_m R = (0.88 \text{ mA})(15 \text{ k}\Omega) = 13.23 \text{ V}$$

$$v_L = -13.23 \text{ V} e^{-t/0.33 \mu\text{s}}$$



d. $V_{R_2 \max} = I_m R_2 = (0.88 \text{ mA})(8.2 \text{ k}\Omega) = 7.22 \text{ V}$



21. a. $R_{Th} = 2 \text{ k}\Omega + 2.2 \text{ k}\Omega + 6.2 \text{ k}\Omega \parallel 3 \text{ k}\Omega = 6.22 \text{ k}\Omega$

$$E_{Th} = \frac{6.2 \text{ k}\Omega(12 \text{ V})}{6.2 \text{ k}\Omega + 3 \text{ k}\Omega} = 8.09 \text{ V}$$

$$I_f = \frac{8.09 \text{ V}}{6.22 \text{ k}\Omega} = 1.3 \text{ mA}, \quad \tau = \frac{L}{R} = \frac{47 \text{ mH}}{6.22 \text{ k}\Omega} = 7.56 \mu\text{s}$$

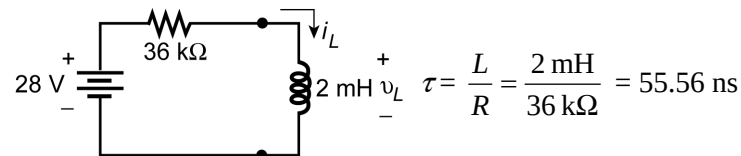
$$i_L = 1.3 \text{ mA}(1 - e^{-t/7.56 \mu\text{s}})$$

$$v_L = 8.09 \text{ V}e^{-t/7.56 \mu\text{s}}$$

b. $0.632(1.3 \text{ mA}) = \mathbf{0.822 \text{ mA}}$

$$0.368(8.09 \text{ V}) = \mathbf{2.98 \text{ V}}$$

22. a. Source conversion: $E = IR = (4 \text{ mA})(12 \text{ k}\Omega) = 48 \text{ V}$, $E_{\text{Net}} = 48 \text{ V} - 20 \text{ V} = 28 \text{ V}$



$$i_L = \frac{E}{R}(1 - e^{-t/\tau}) = \frac{28 \text{ V}}{36 \text{ k}\Omega}(1 - e^{-t/\tau}) = \mathbf{0.778 \text{ mA}(1 - e^{-t/55.56 \text{ ns}})}$$

$$v_L = Ee^{-t/\tau} = \mathbf{28 \text{ V}e^{-t/55.56 \text{ ns}}}$$

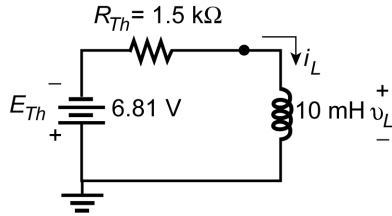
b. $t = 100 \text{ ns}$:

$$i_L = 0.778 \text{ mA}(1 - e^{-100 \text{ ns}/55.56 \text{ ns}}) = 0.778 \text{ mA}(\underbrace{1 - e^{-1.8}}_{0.165}) = \mathbf{0.65 \text{ mA}}$$

$$v_L = 28 \text{ V}e^{-1.8} = \mathbf{4.62 \text{ V}}$$

23.

a.



$$R_{Th} = 2.2 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega = 1.50 \text{ k}\Omega$$

$$E_{Th} = -\frac{4.7 \text{ k}\Omega(10 \text{ V})}{4.7 \text{ k}\Omega + 2.2 \text{ k}\Omega} = -6.81 \text{ V}$$

$$\tau = \frac{L}{R} = \frac{10 \text{ mH}}{1.50 \text{ k}\Omega} = 6.67 \mu\text{s}$$

$$i_L = -\frac{E}{R}(1 - e^{-t/\tau}) = -\frac{6.81 \text{ V}}{1.5 \text{ k}\Omega}(1 - e^{-t/6.67 \mu\text{s}}) = -4.54 \text{ mA}(1 - e^{-t/6.67 \mu\text{s}})$$

$$v_L = Ee^{-t/\tau} = -E_{Th}e^{-t/\tau} = -6.81 \text{ V}e^{-t/6.67 \mu\text{s}}$$

b. $t = 10 \mu\text{s}$:

$$i_L = -4.54 \text{ mA}(1 - e^{-10 \mu\text{s}/6.67 \mu\text{s}}) = -4.54 \text{ mA}(1 - \underbrace{e^{-1.5}}_{0.223})$$

$$= -3.53 \text{ mA}$$

$$v_L = -6.81 \text{ V}(0.223) = 1.52 \text{ V}$$

c. $\tau' = \frac{L}{R} = \frac{10 \text{ mH}}{4.7 \text{ k}\Omega} = 2.13 \mu\text{s}$

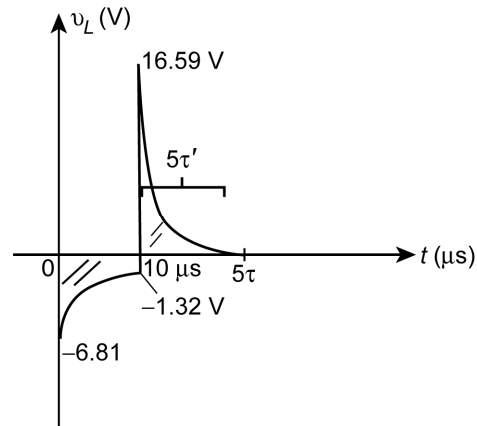
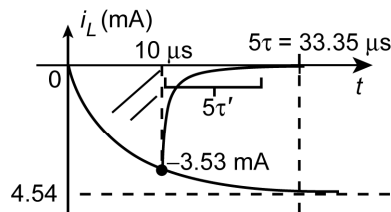
$$i_L = -3.53 \text{ mA}e^{-t/2.13 \mu\text{s}}$$

At $t = 10 \mu\text{s}$

$$V_L = (3.53 \text{ mA})(4.7 \text{ k}\Omega) = 16.59 \text{ V}$$

$$v_L = -16.59 \text{ V}e^{-t/2.13 \mu\text{s}}$$

d.



24.

a.

$$v_L = Ee^{-t/\tau}$$

$$\tau = \frac{L}{R_1 + R_3} = \frac{0.6 \text{ H}}{100 \Omega + 20 \Omega} = \frac{0.6 \text{ H}}{120 \Omega} = 5 \text{ ms}$$

$$v_L = 36 \text{ V}e^{-t/5 \text{ ms}}$$

$$v_L = 36 \text{ V}e^{-25 \text{ ms}/5 \text{ ms}} = 36 \text{ V}e^{-5} = 36 \text{ V}(0.00674) = 0.24 \text{ V}$$

b.

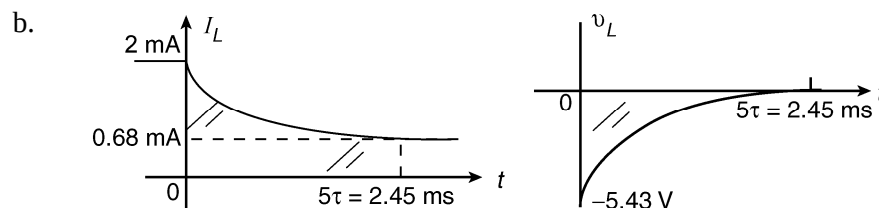
$$v_L = 36 \text{ V}e^{-1 \text{ ms}/5 \text{ ms}} = 36 \text{ V}e^{-0.2} = 36 \text{ V}(0.819) = 29.47 \text{ V}$$

$$\begin{aligned}
 \text{c. } v_{R_1} &= i_{R_1} R_1 = i_L R_1 = \left(\frac{E}{R_1 + R_3} (1 - e^{-t/\tau}) \right) R_1 \\
 &= \left(\frac{36 \text{ V}}{120 \Omega} (1 - e^{-t/5 \text{ ms}}) \right) 100 \Omega \\
 &= (300 \text{ mA} (1 - e^{-t/5 \text{ ms}})) 100 \Omega \\
 &= 30 \text{ V} (1 - e^{-5 \text{ ms}/5 \text{ ms}}) = 30 \text{ V} (1 - e^{-1}) \\
 &= 30 \text{ V} (1 - 0.368) = \mathbf{18.96 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 \text{d. } i_L &= 300 \text{ mA} (1 - e^{-t/5 \text{ ms}}) \\
 100 \text{ mA} &= 300 \text{ mA} (1 - e^{-t/5 \text{ ms}}) \\
 0.333 &= 1 - e^{-t/5 \text{ ms}} \\
 0.667 &= e^{-t/5 \text{ ms}} \\
 \log_e 0.667 &= -t/5 \text{ ms} \\
 0.405 &= t/5 \text{ ms} \\
 t &= 0.405(5 \text{ ms}) = \mathbf{2.03 \text{ ms}}
 \end{aligned}$$

e. None-In parallel with supply.

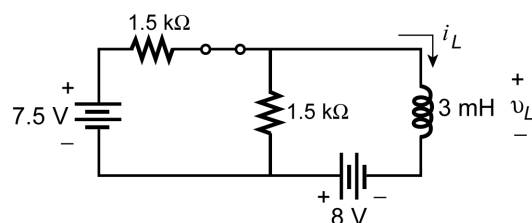
$$\begin{aligned}
 \text{25. a. } I_i &= \frac{16 \text{ V}}{4.7 \text{ k}\Omega + 3.3 \text{ k}\Omega} = 2 \text{ mA} \\
 t = 0 \text{ s: Thevenin:} \\
 R_{Th} &= 3.3 \text{ k}\Omega + 1 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega = 3.3 \text{ k}\Omega + 0.82 \text{ k}\Omega = 4.12 \text{ k}\Omega \\
 E_{Th} &= \frac{1 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 4.7 \text{ k}\Omega} = 2.81 \text{ V} \\
 i_L &= I_f + (I_i - I_f) e^{-t/\tau} \\
 I_f &= \frac{2.81 \text{ V}}{4.12 \text{ k}\Omega} = 0.68 \text{ mA}, \tau = \frac{L}{R} = \frac{2 \text{ H}}{4.12 \text{ k}\Omega} = 0.49 \text{ ms} \\
 i_L &= 0.68 \text{ mA} + (2 \text{ mA} - 0.68 \text{ mA}) e^{-t/0.49 \text{ ms}} \\
 i_L &= \mathbf{0.68 \text{ mA} + 1.32 \text{ mA} e^{-t/0.49 \text{ ms}}} \\
 v_R(0+) &= 2 \text{ mA} (4.12 \text{ k}\Omega) = 8.24 \text{ V} \\
 \text{KVL}(0+): \quad 2.81 \text{ V} - 8.24 \text{ V} - v_L &= 0 \\
 v_L &= -5.43 \text{ V} \\
 v_L &= \mathbf{-5.43 \text{ V} e^{-t/0.49 \text{ ms}}}
 \end{aligned}$$



26. a. Steady-state: $I_L = \frac{8 \text{ V}}{1.5 \text{ k}\Omega} = 5.33 \text{ mA}$, $V_L = 0 \text{ V}$

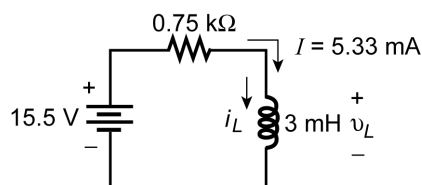
$$R_{Th} = (3 \text{ k}\Omega \parallel 12 \text{ k}\Omega) \parallel 4 \text{ k}\Omega = 1.5 \text{ k}\Omega$$

$$E_{Th} = \frac{2.4 \text{ k}\Omega(20 \text{ V})}{2.4 \text{ k}\Omega + 4 \text{ k}\Omega} = 7.5 \text{ V}$$



$$R'_{Th} = 1.5 \text{ k}\Omega \parallel 1.5 \text{ k}\Omega = 0.75 \text{ k}\Omega$$

$$E'_{Th} = 8 \text{ V} + 7.5 \text{ V} = 15.5 \text{ V}$$



$$\tau = \frac{L}{R} = \frac{3 \text{ mH}}{0.75 \text{ k}\Omega} = 4 \mu\text{s}$$

$$I_f = \frac{15.5 \text{ V}}{0.75 \text{ k}\Omega} = 20.67 \text{ mA} \quad I_i = 5.33 \text{ mA}$$

$$i_L = I_f + (I_i - I_f)e^{-t/\tau} = 20.67 \text{ mA} + (5.33 \text{ mA} - 20.67 \text{ mA})e^{-t/4 \mu\text{s}}$$

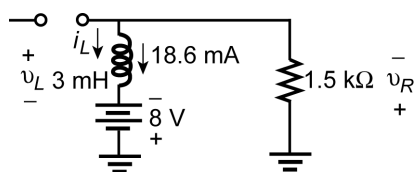
$$i_L = 20.67 \text{ mA} - 15.34 \text{ mA}e^{-t/4 \mu\text{s}}$$

$$v_L = 15.5 \text{ V}e^{-t/4 \mu\text{s}}$$

b. $i_L(2\tau) = 20.67 \text{ mA} - 15.34 \text{ mA} \underbrace{e^{-2}}_{0.135}$
 $= 18.6 \text{ mA}$

$$v_L(2\tau) = 15.5 \text{ V}e^{-2} = 15.5 \text{ V}(0.135) = 2.09 \text{ V}$$

c. $I_i = 18.6 \text{ mA}$



$$v_L + v_R - 8 \text{ V} = 0$$

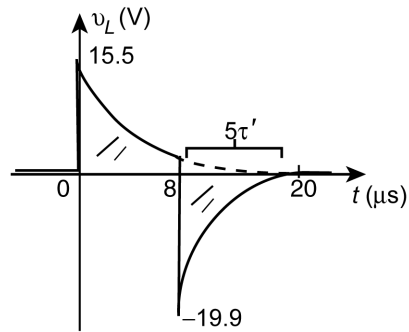
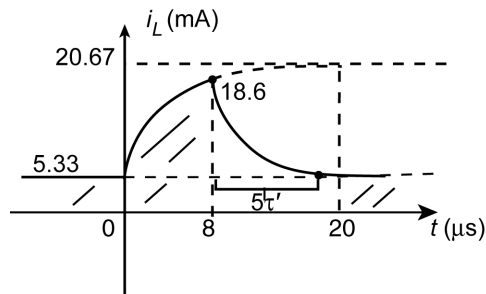
$$v_L = 8 \text{ V} - v_R = 8 \text{ V} - (18.6 \text{ mA})(1.5 \text{ k}\Omega) = -19.9 \text{ V}$$

$$\tau' = \frac{L}{R} = \frac{3 \text{ mH}}{1.5 \text{ k}\Omega} = 2 \mu\text{s}$$

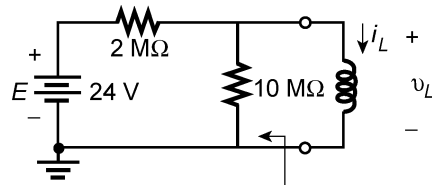
$$I_i = 18.6 \text{ mA} \quad I_f = \frac{8 \text{ V}}{1.5 \text{ k}\Omega} = 5.33 \text{ mA}$$

$$i_L = I_f + (I_i - I_f)e^{-t/\tau} = 5.33 \text{ mA} + (18.6 \text{ mA} - 5.33 \text{ mA})e^{-t/2 \mu\text{s}} = 5.33 \text{ mA} + 13.27 \text{ mA}e^{-t/2 \mu\text{s}}$$

$$v_L = -19.9 \text{ V}e^{-t/2 \mu\text{s}}$$



27. a.



$$R_{Th} = 2 \text{ M}\Omega \parallel 10 \text{ M}\Omega = 1.67 \text{ M}\Omega$$

$$E_{Th} = \frac{10 \text{ M}\Omega(24 \text{ V})}{10 \text{ M}\Omega + 2 \text{ M}\Omega} = 20 \text{ V}$$

$$I_L(0^-) = \frac{E_{Th}}{R_{Th}} = \frac{20 \text{ V}}{1.67 \text{ M}\Omega} = 12 \text{ }\mu\text{A}$$

$$\tau' = \frac{L}{R_{\text{meter}}} = \frac{5 \text{ H}}{10 \text{ M}\Omega} = 5 \mu\text{s}$$

$$i_L = 12 \text{ }\mu\text{A}e^{-t/5 \mu\text{s}}$$

$$10 \text{ }\mu\text{A} = 12 \text{ }\mu\text{A}e^{-t/5 \mu\text{s}}$$

$$0.833 = e^{-t/5 \mu\text{s}}$$

$$\log_e 0.833 = -t/5 \mu\text{s}$$

$$0.183 = t/5 \mu\text{s}$$

$$t = 0.183(5 \mu\text{s}) = 0.92 \mu\text{s}$$

b. $v_L(0^+) = i_L(0^+)R_m = (12 \text{ }\mu\text{A})(10 \text{ M}\Omega) = 120 \text{ V}$

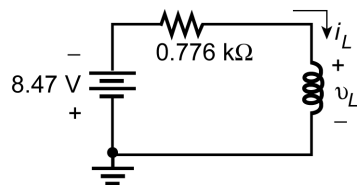
$$v_L = 120 \text{ V}e^{-t/5 \mu\text{s}} = 120 \text{ V}e^{-10 \mu\text{s}/5 \mu\text{s}} = 120 \text{ V}e^{-2} = 120 \text{ V}(0.135) = 16.2 \text{ V}$$

c. $v_L = 120 \text{ V}e^{-5 \tau' \tau} = 120 \text{ V}e^{-5} = 120 \text{ V}(6.74 \times 10^{-3}) = 0.81 \text{ V}$

28. a. Closed Switch:

$$R_{Th} = 1.2 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega = 0.776 \text{ k}\Omega$$

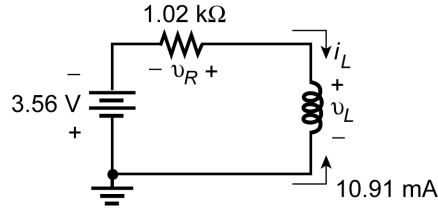
$$E_{Th} = \frac{1.2 \text{ k}\Omega(24 \text{ V})}{1.2 \text{ k}\Omega + 2.2 \text{ k}\Omega} = 8.47 \text{ V}$$



Open Switch:

$$R'_{Th} = 6.9 \text{ k}\Omega \parallel 1.2 \text{ k}\Omega = 1.02 \text{ k}\Omega$$

$$E'_{Th} = -\frac{1.2 \text{ k}\Omega(24 \text{ V})}{8.1 \text{ k}\Omega} = -3.56 \text{ V}$$



$$-3.56 \text{ V} + v_R - v_L = 0$$

$$v_L = -3.56 \text{ V} + (1.09 \text{ mA})(1.02 \text{ k}\Omega)$$

$$= 7.57 \text{ V}$$

$$v_L = 7.57 \text{ V} e^{-t/1.18 \text{ ms}}$$

$$\tau = \frac{L}{R} = \frac{1.24}{1.02 \text{ k}\Omega} = 1.18 \text{ ms}$$

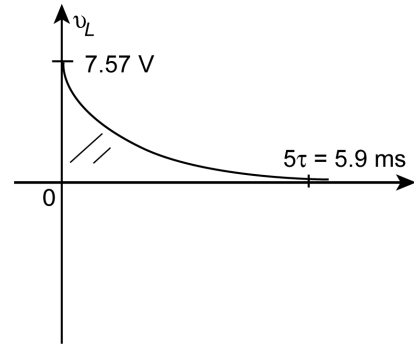
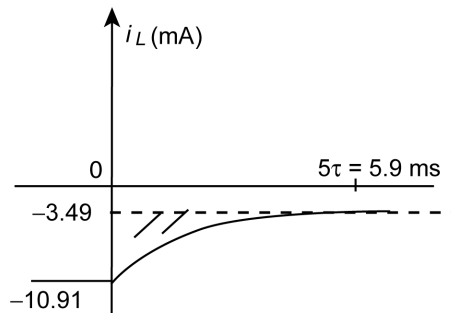
$$I_{ss} = \frac{3.56 \text{ V}}{1.02 \text{ k}\Omega} = 3.49 \text{ mA} = I_f$$

$$i_L = I_f + (I_i - I_f)e^{-t/\tau}$$

$$= -3.49 \text{ mA} + ((-10.91 \text{ mA}) - (-3.49 \text{ mA}))e^{-t/1.18 \text{ ms}}$$

$$i_L = -3.49 \text{ mA} - 7.42 \text{ mA} e^{-t/1.18 \text{ ms}}$$

b.



29. a. $i_L = 100 \text{ mA}(1 - e^{-1\text{ms}/20\text{ms}}) = 100 \text{ mA}(1 - e^{-1/20})$
 $= 100 \text{ mA}(1 - e^{-0.05}) = 100 \text{ mA}(1 - 951.23 \times 10^{-3}) = 100 \text{ mA}(48.77 \times 10^{-3})$
 $= \mathbf{4.88 \text{ mA}}$

b. $i_L = 100 \text{ mA}(1 - e^{-100\text{ms}/20\text{ms}}) = 100 \text{ mA}(1 - e^{-5})$
 $= \mathbf{99.33 \text{ mA}}$

c. $50 \text{ mA} = 100 \text{ mA}(1 - e^{-t/\tau})$
 $0.5 = 1 - e^{-t/\tau}$
 $-0.5 = -e^{-t/\tau}$
 $0.5 = e^{-t/\tau}$
 $\log_e 0.5 = -t/\tau$
 $t = -(\tau)(\log_e 0.5) = -(20 \text{ ms})(\log_e 0.5) = -(20 \text{ ms})(-693.15 \times 10^{-3})$
 $= \mathbf{13.86 \text{ ms}}$

$$\begin{aligned}
 \text{d. } 99 \text{ mA} &= 100 \text{ mA}(1 - e^{-t/20 \text{ ms}}) \\
 0.99 &= 1 - e^{-t/20 \text{ ms}} \\
 -0.01 &= -e^{-t/20 \text{ ms}} \\
 0.01 &= e^{-t/20 \text{ ms}} \\
 \log_e 0.01 &= -t/20 \text{ ms} \\
 t &= -(20 \text{ ms})(\log_e 0.01) = -(20 \text{ ms})(-4.605) = \mathbf{92.1 \text{ ms}}
 \end{aligned}$$

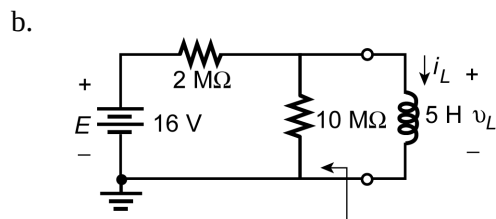
$$\begin{aligned}
 30. \quad \text{a. } I_L(1\tau) &= 0.632 I_{\max} = 126.4 \mu\text{A} \\
 I_{\max} &= \frac{126.4}{0.632} = \mathbf{200 \mu\text{A}}
 \end{aligned}$$

$$\begin{aligned}
 \text{b. } i_L &= I_m(1 - e^{-t/\tau}) \\
 160 \mu\text{A} &= 200 \mu\text{A} \left(1 - e^{\frac{-64.4 \mu\text{s}}{\tau}} \right) \\
 0.8 &= 1 - e^{\frac{-64.4 \mu\text{s}}{\tau}} \\
 0.2 &= e^{\frac{-64.4 \mu\text{s}}{\tau}} \\
 \log_e 0.2 &= -1.61 = \frac{-64.4 \mu\text{s}}{\tau} \\
 \tau &= \frac{64.4 \mu\text{s}}{1.61} = \mathbf{40 \mu\text{s}}
 \end{aligned}$$

$$\begin{aligned}
 \text{c. } \tau &= \frac{L}{R} = 40 \mu\text{s} = \frac{L}{500 \Omega} \\
 L &= \mathbf{20 \text{ mH}}
 \end{aligned}$$

$$\text{d. } I_m = \frac{E}{R} \Rightarrow E = (200 \mu\text{A})(500 \Omega) = \mathbf{100 \text{ mV}}$$

$$\begin{aligned}
 31. \quad \text{a. } L &\Rightarrow \text{open circuit equivalent} \\
 V_L &= \frac{10 \text{ M}\Omega(16 \text{ V})}{10 \text{ M}\Omega + 2 \text{ M}\Omega} = \mathbf{13.33 \text{ V}}
 \end{aligned}$$



$$\begin{aligned}
 R_{Th} &= 2 \text{ M}\Omega \parallel 10 \text{ M}\Omega = 1.67 \text{ M}\Omega \\
 E_{Th} &= \frac{10 \text{ M}\Omega(16 \text{ V})}{10 \text{ M}\Omega + 2 \text{ M}\Omega} = 13.33 \text{ V} \\
 I_{L_{\text{final}}} &= \frac{E_{Th}}{R_{Th}} = \frac{13.33 \text{ V}}{1.67 \text{ M}\Omega} = \mathbf{7.98 \mu\text{A}}
 \end{aligned}$$

$$c. \quad i_L = 7.98 \mu\text{A}(1 - e^{-t/3 \mu\text{s}})$$

$$\tau = \frac{L}{R} = \frac{5 \text{ H}}{1.67 \text{ M}\Omega} = 3 \mu\text{s}$$

$$10 \mu\text{A} = 7.98 \mu\text{A}(1 - e^{-t/3 \mu\text{s}})$$

$$1.253 = 1 - e^{-t/3 \mu\text{s}}$$

$$0.253 = e^{-t/3 \mu\text{s}}$$

$$\log_e(0.253) = -t/3 \mu\text{s}$$

$$1.374 = t/3 \mu\text{s}$$

$$t = 1.374(3 \mu\text{s}) = \mathbf{4.12 \mu\text{s}}$$

$$d. \quad v_L = 13.33 \text{ V } e^{-t/3 \mu\text{s}} = 13.33 \text{ V } e^{-12 \mu\text{s}/3 \mu\text{s}} = 13.33 \text{ V } e^{-4}$$

$$= 13.33 \text{ V}(0.0183) = \mathbf{0.244 \text{ V}}$$

$$32. \quad e_L = L \frac{\Delta i}{\Delta t} : \quad 0 - 2 \text{ ms}, e_L = -(200 \text{ mH}) \left[\frac{15 \text{ mA}}{2 \text{ ms}} \right] = \mathbf{-1.5 \text{ V}}$$

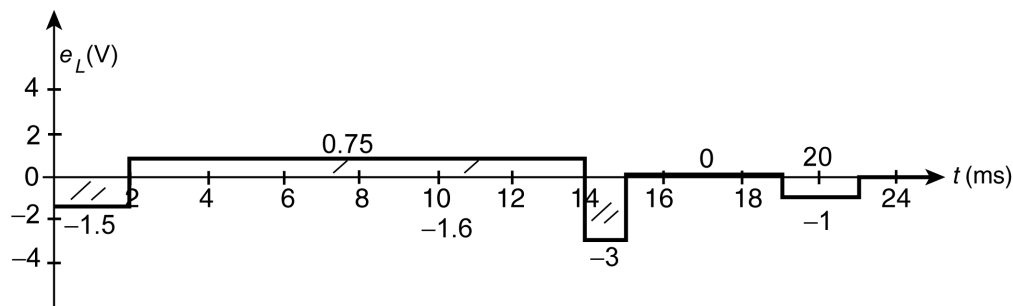
$$2 - 14 \text{ ms}, e_L = (200 \text{ mH}) \left(\frac{45 \text{ mA}}{12 \text{ ms}} \right) = \mathbf{0.75 \text{ V}}$$

$$14 - 15 \text{ ms}, e_L = -(200 \text{ mH}) \left(\frac{15 \text{ mA}}{1 \text{ ms}} \right) = \mathbf{-3 \text{ V}}$$

$$15 - 19 \text{ ms}, e_L = \mathbf{0 \text{ V}}$$

$$19 - 22 \text{ ms}, e_L = -(200 \text{ mH}) \left(\frac{15 \text{ mA}}{3 \text{ ms}} \right) = \mathbf{-1 \text{ V}}$$

$$22 - 24 \text{ ms}, e_L = \mathbf{0 \text{ V}}$$



$$33. \quad v_L = L \frac{\Delta i_L}{\Delta t}$$

$$0 \rightarrow 2 \text{ ms}: v_L = (5 \text{ mH}) \left(\frac{15 \text{ mA}}{2 \text{ ms}} \right) = \mathbf{37.5 \text{ mV}}$$

$$2 \rightarrow 6 \text{ ms}: v_L = -(5 \text{ mH}) \left(\frac{30 \text{ mA}}{4 \text{ ms}} \right) = \mathbf{-37.5 \text{ mV}}$$

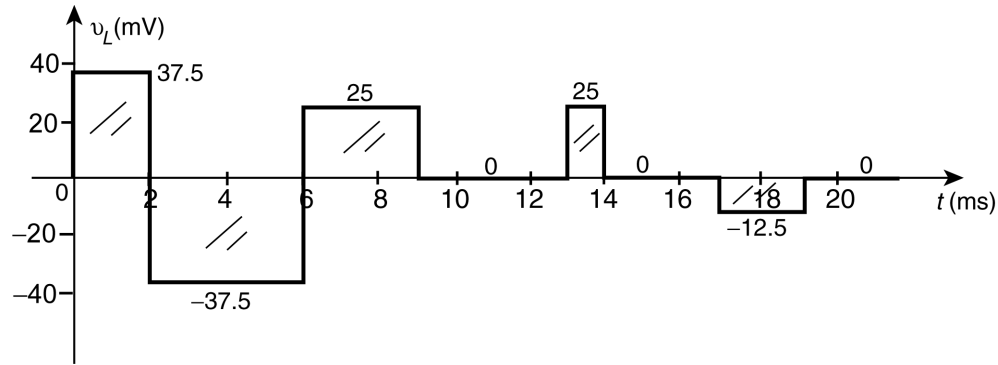
$$6 \rightarrow 9 \text{ ms}: v_L = (5 \text{ mH}) \left(\frac{15 \text{ mA}}{3 \text{ ms}} \right) = \mathbf{25 \text{ mV}}$$

$$9 \rightarrow 13 \text{ ms}: v_L = \mathbf{0 \text{ V}}$$

$$13 \rightarrow 14 \text{ ms}: v_L = (5 \text{ mH}) \left(\frac{5 \text{ mA}}{1 \text{ ms}} \right) = \mathbf{25 \text{ mV}}$$

$$14 \rightarrow 17 \text{ ms}: v_L = \mathbf{0 \text{ V}}$$

$$17 \rightarrow 19 \text{ ms: } v_L = -(5 \text{ mH}) \left(\frac{5 \text{ mA}}{2 \text{ ms}} \right) = -12.5 \text{ mV}$$



34. $L = 10 \text{ mH}$, 4 mA at $t = 0 \text{ s}$

$$v_L = L \frac{\Delta i}{\Delta t} \Rightarrow \Delta i = \frac{\Delta t}{L} v_L$$

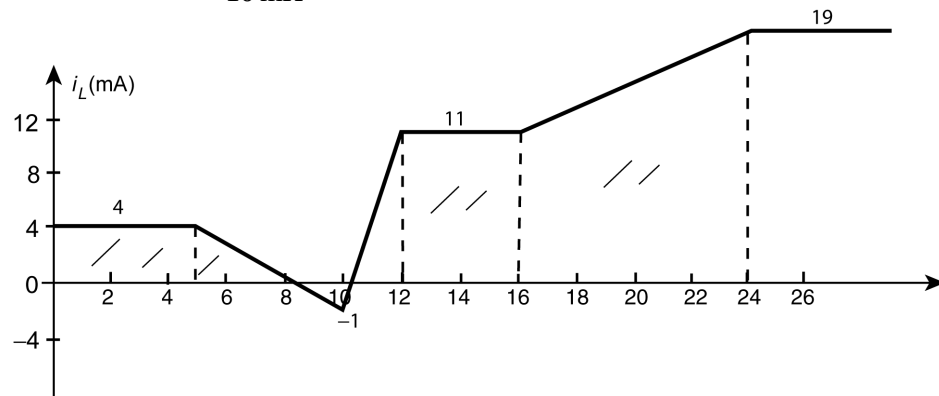
$$0 - 5 \mu\text{s: } v_L = 0 \text{ V, } \Delta i_L = 0 \text{ mA and } i_L = 4 \text{ mA}$$

$$5 - 10 \mu\text{s: } \Delta i_L = \frac{5 \mu\text{s}}{10 \text{ mH}} (-10 \text{ V}) = -5 \text{ mA}$$

$$10 - 12 \mu\text{s: } \Delta i_L = \frac{2 \mu\text{s}}{10 \text{ mH}} (+60 \text{ V}) = +12 \text{ mA}$$

$$12 - 16 \mu\text{s: } v_L = 0 \text{ V, } \Delta i_L = 0 \text{ mA and } i_L = 11 \text{ mA}$$

$$16 - 24 \mu\text{s: } \Delta i_L = \frac{8 \mu\text{s}}{10 \text{ mH}} 10 \text{ V} = 8 \text{ mA}$$



35. a. $L_3 \parallel L_4 = 3.5 \text{ mH} \parallel 5.6 \text{ mH} = 1.953 \text{ mH}$
 $L_2 + L_3 \parallel L_4 = 3.3 \text{ mH} + 1.953 \text{ mH} = 5.253 \text{ mH}$

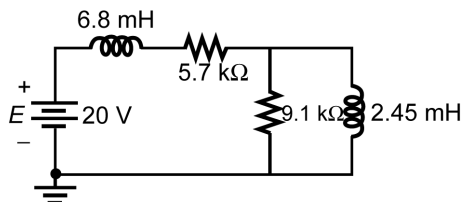
$$L' = L_1 \parallel (L_2 + L_3 \parallel L_4) = 2.4 \text{ mH} \parallel 5.253 \text{ mH} = 1.647 \text{ mH}$$

$$L_T = L_4 + L' = 9.1 \text{ mH} + 1.647 \text{ mH}$$

$$L_T = \mathbf{10.75 \text{ mH}}$$

36. $L_2 \parallel L_3 = 10 \text{ mH} \parallel 30 \text{ mH} = 7.5 \text{ mH}$
 $L' = L_1 + L_2 \parallel L_3 = 47 \text{ mH} + 7.5 \text{ mH} = 54.5 \text{ mH}$
 $L' \parallel L_4 = 54.5 \text{ mH} \parallel 22 \text{ mH} = \mathbf{15.67 \text{ mH}}$

37. $33 \text{ mH} + 1.8 \text{ mH} = 5.1 \text{ mH}$
 $4.7 \text{ mH} \parallel 5.1 \text{ mH} = \mathbf{2.45 \text{ mH}}$



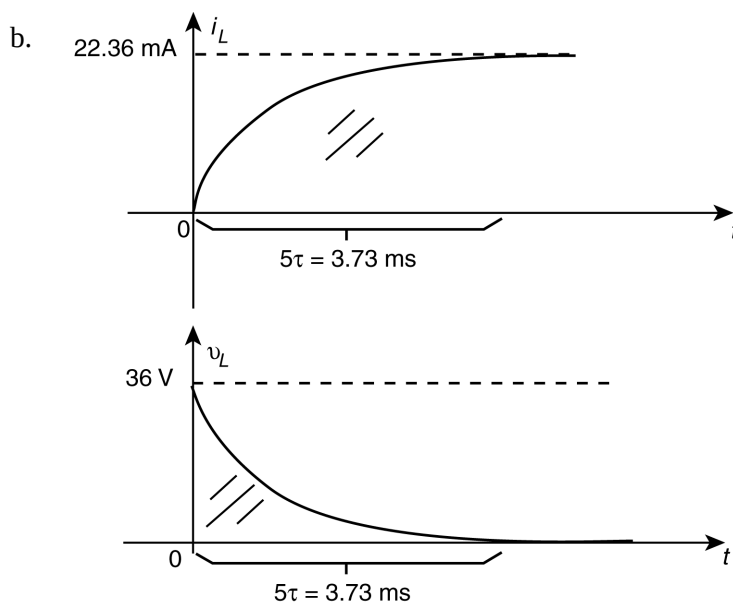
38. $L'_T = 6.2 \text{ mH} + 12 \text{ mH} \parallel 36 \text{ mH} + 24 \text{ mH} = \mathbf{39.2 \text{ mH}}$
 $C'_T = 9.1 \mu\text{F} + 10 \mu\text{F} \parallel 91 \mu\text{F} = 9.1 \mu\text{F} + 9.01 \mu\text{F} = \mathbf{18.11 \mu\text{F}}$
 39.2 mH in series with $18.11 \mu\text{F}$
39. $7 \mu\text{F} \parallel 42 \mu\text{F} = 6 \mu\text{F}$
 $12 \mu\text{F} + 6 \mu\text{F} = \mathbf{18 \mu\text{F}}$
 $5 \text{ mH} + 20 \text{ mH} = \mathbf{25 \text{ mH}}$
 Series combination of $2.2 \text{ k}\Omega$ resistor, 25 mH coil, $18 \mu\text{F}$ capacitor
40. a. $R'_T = 2 \text{ k}\Omega \parallel 8.2 \text{ k}\Omega = 1.61 \text{ k}\Omega$, $L'_T = 3 \text{ mH} \parallel 2 \text{ mH} = 1.2 \text{ mH}$

$$\tau = \frac{L'_T}{R'_T} = \frac{1.2 \text{ mH}}{1.61 \text{ k}\Omega} = 745.3 \mu\text{s}$$

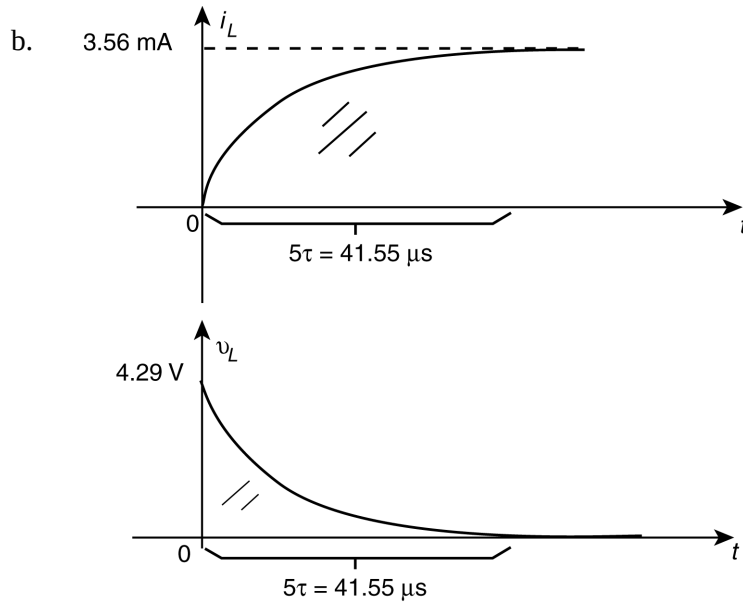
$$i_L = \frac{E}{R'_T} (1 - e^{-t/\tau})$$

$$= \frac{36 \text{ V}}{1.61 \text{ k}\Omega} (1 - e^{-t/745.3 \mu\text{s}}) = \mathbf{22.36 \text{ mA} (1 - e^{-t/745.3 \mu\text{s}})}$$

$$v_L = E e^{-t/\tau} = \mathbf{36 \text{ V} e^{-t/745.3 \mu\text{s}}}$$

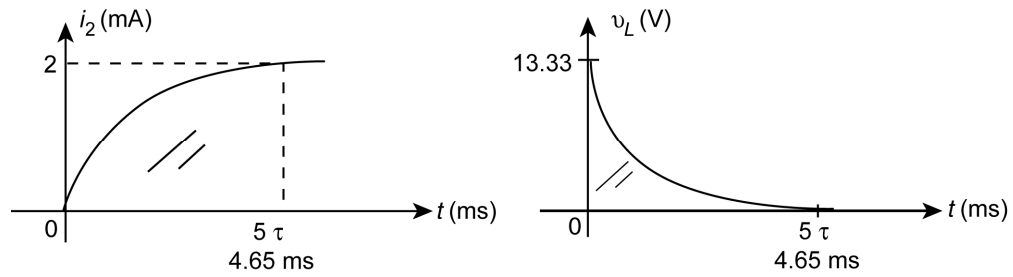


41. a. Source conversion: $E = 16 \text{ V}$, $R_s = 2 \text{ k}\Omega$
 $R_{Th} = 2 \text{ k}\Omega + 2 \text{ k}\Omega \parallel 8.2 \text{ k}\Omega = 2 \text{ k}\Omega + 1.61 \text{ k}\Omega = 3.61 \text{ k}\Omega$
 $E_{Th} = \frac{8.2 \text{ k}\Omega(16 \text{ V})}{8.2 \text{ k}\Omega + 2 \text{ k}\Omega} = 12.86 \text{ V}$
 $I_m = \frac{E_{Th}}{R_{Th}} = \frac{12.86 \text{ V}}{3.61 \text{ k}\Omega} = 3.56 \text{ mA}$, $\tau = \frac{L}{R} = \frac{30 \text{ mH}}{3.61 \text{ k}\Omega} = 8.31 \mu\text{s}$
 $i_L = 3.56 \text{ mA}(1 - e^{-t/8.31\mu\text{s}})$
 $v_{L_1} + v_{L_2} = 12.86 \text{ V}$ initially ($t = 0+$)
 $v_L = \frac{10 \text{ mH}}{10 \text{ mH} + 20 \text{ mH}}$ of total $= \frac{1}{3}(12.86 \text{ V}) = 4.29 \text{ V}$
 $v_L = 4.29 \text{ V}e^{-t/8.31\mu\text{s}}$

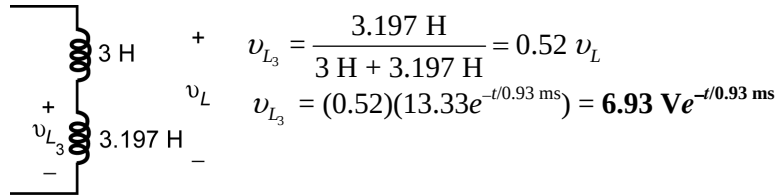


42. a.
-
- $\leftarrow R_{Th} = 10 \text{ k}\Omega \parallel 20 \text{ k}\Omega = 6.67 \text{ k}\Omega$
 $E_{Th} = \frac{20 \text{ k}\Omega(20 \text{ V})}{20 \text{ k}\Omega + 10 \text{ k}\Omega} = 13.33 \text{ V}$
- $L_T = 3 \text{ H} + 4.7 \text{ H} \parallel 10 \text{ H} = 3 \text{ H} + 3.197 \text{ H} = \mathbf{6.197 \text{ H}}$
 $\tau = \frac{L_T}{R} = \frac{6.197 \text{ H}}{6.67 \text{ k}\Omega} = 0.93 \text{ ms}$
 $v_L = 13.33 \text{ V}e^{-t/0.93 \text{ ms}}$
 $i_L = \frac{13.33 \text{ V}}{6.67 \text{ k}\Omega} (1 - e^{-t/\tau}) = \mathbf{2 \text{ mA}(1 - e^{-t/0.93 \text{ ms}})}$

b.



c.



$$v_{L_3} = \frac{3.197\text{ H}}{3\text{ H} + 3.197\text{ H}} = 0.52 v_L$$

$$v_{L_3} = (0.52)(13.33e^{-t/0.93\text{ ms}}) = \mathbf{6.93\text{ V}e^{-t/0.93\text{ ms}}}$$

43. $I_{R_1} = \frac{E}{R_1} = \frac{20\text{ V}}{4\ \Omega} = 5\text{ A}$

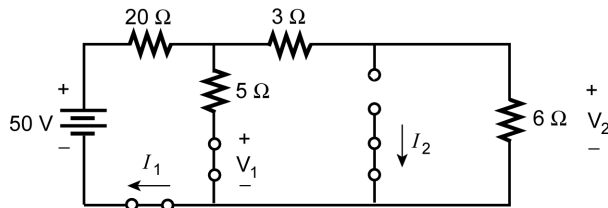
$$I_2 = I_{R_2} = \frac{E}{R_2 + R_3} = \frac{20\text{ V}}{6\ \Omega + 4\ \Omega} = \frac{20\text{ V}}{10\ \Omega} = \mathbf{2\text{ A}}$$

$$I_1 = I_{R_1} + I_2 = 5\text{ A} + 2\text{ A} = \mathbf{7\text{ A}}$$

44. $I_1 = I_2 = \mathbf{0\text{ A}}$
 $V_1 = V_2 = E = \mathbf{60\text{ V}}$

45. $I_1 = \frac{12\text{ V}}{4\ \Omega} = \mathbf{3\text{ A}}, I_2 = \mathbf{0\text{ A}}$
 $V_1 = \mathbf{12\text{ V}}, V_2 = \mathbf{0\text{ V}}$

46.



$$V_2 = \frac{6\ \Omega(50\text{ V})}{6\ \Omega + 20\ \Omega + 3\ \Omega} = \mathbf{10.34\text{ V}}$$

$$V_1 = \frac{(3\ \Omega + 6\ \Omega)(50\text{ V})}{29\ \Omega} = \mathbf{15.52\text{ V}}$$

$$I_1 = \frac{50\text{ V}}{20\ \Omega + 3\ \Omega + 6\ \Omega} = \mathbf{1.72\text{ A}}$$

$$I_2 = \mathbf{0\text{ A}}$$

Chapter 12

1. Φ : CGS: 5×10^4 **Maxwells**, English: 5×10^4 **lines**
 B : CGS: **8 Gauss**, English: **51.62 lines/in.²**
2. Φ : SI 6×10^{-4} **Wb**, English **60,000 lines**
 B : SI **0.465 T**, CGS 4.65×10^3 **Gauss**, English **30,000 lines/in.²**

3. a. $B = \frac{\Phi}{A} = \frac{4 \times 10^{-4} \text{ Wb}}{0.01 \text{ m}^2} = \mathbf{0.04 \text{ T}}$

4. a. $\mathcal{R} = \frac{l}{\mu A} = \frac{0.06 \text{ m}}{\mu 2 \times 10^{-4} \text{ m}^2} = \frac{300}{\mu \text{ m}}$

- b. $\mathcal{R} = \frac{l}{\mu A} = \frac{0.0762 \text{ m}}{\mu 5 \times 10^{-4} \text{ m}^2} = \frac{152.4}{\mu \text{ m}}$

- c. $\mathcal{R} = \frac{l}{\mu A} = \frac{0.1 \text{ m}}{\mu 1 \times 10^{-4} \text{ m}^2} = \frac{1000}{\mu \text{ m}}$

from the above $\mathcal{R}_{(c)} > \mathcal{R}_{(a)} > \mathcal{R}_{(b)}$

5. $\mathcal{R} = \frac{F}{\Phi} = \frac{400 \text{ At}}{4.2 \times 10^{-4} \text{ Wb}} = \mathbf{952.4 \times 10^3 \text{ At/Wb}}$

6. $\mathcal{R} = \frac{F}{\Phi} = \frac{120 \text{ gilberts}}{72,000 \text{ maxwells}} = \mathbf{1.67 \times 10^{-3} \text{ rels (CGS)}}$

7. $\cancel{6 \text{ in.}} \left[\frac{1 \text{ m}}{39.37 \cancel{\text{ in.}}} \right] = 0.1524 \text{ m}$
 $H = \frac{F}{l} = \frac{400 \text{ At}}{0.1524 \text{ m}} = \mathbf{2624.67 \text{ At/m}}$

8. $\mu = \frac{2B}{H} = \frac{2(1200 \times 10^{-4} \text{ T})}{600 \text{ At/m}} = \mathbf{4 \times 10^{-4} \text{ Wb/Am}}$

9. $B = \frac{\Phi}{A} = \frac{10 \times 10^{-4} \text{ Wb}}{3 \times 10^{-3} \text{ m}^2} = 0.33 \text{ T}$

Fig. 12.7: $H \cong 800 \text{ At/m}$

$$NI = Hl \Rightarrow I = Hl/N = (800 \text{ At/m})(0.2 \text{ m})/75 = \mathbf{2.13 \text{ A}}$$

$$10. \quad B = \frac{\Phi}{A} = \frac{3 \times 10^{-4} \text{ Wb}}{5 \times 10^{-4} \text{ m}^2} = 0.6 \text{ T}$$

Fig. 12.7, $H_{\text{iron}} = 2500 \text{ At/m}$

Fig. 12.8, $H_{\text{steel}} = 70 \text{ At/m}$

$$NI = Hl_{(\text{iron})} + Hl_{(\text{steel})}$$

$$(100 \text{ t})I = (H_{\text{iron}} + H_{\text{steel}})l$$

$$(100 \text{ t})I = (2500 \text{ At/m} + 70 \text{ At/m})0.3 \text{ m}$$

$$I = \frac{771 \text{ A}}{100} = \mathbf{7.71 \text{ A}}$$

$$11. \quad a. \quad N_1 I_1 + N_2 I_2 = Hl$$

$$B = \frac{\Phi}{A} = \frac{12 \times 10^{-4} \text{ Wb}}{12 \times 10^{-4} \text{ m}^2} = 1 \text{ T}$$

Fig. 12.7: $H \cong 750 \text{ At/m}$

$$N_1(2 \text{ A}) + 30 \text{ At} = (750 \text{ At/m})(0.2 \text{ m})$$

$$N_1 = \mathbf{60 \text{ t}}$$

$$b. \quad \mu = \frac{B}{H} = \frac{1 \text{ T}}{750 \text{ At/m}} = \mathbf{13.34 \times 10^{-4} \text{ Wb/Am}}$$

$$12. \quad a. \quad 80,000 \cancel{\text{ lines}} \left[\frac{1 \text{ Wb}}{10^8 \cancel{\text{ lines}}} \right] = 8 \times 10^4 \times 10^{-8} \text{ Wb} = 8 \times 10^{-4} \text{ Wb}$$

$$l_{(\text{cast steel})} = 5.5 \cancel{\text{ in.}} \left[\frac{1 \text{ m}}{39.37 \cancel{\text{ in.}}} \right] = 0.14 \text{ m}$$

$$l_{(\text{sheet steel})} = 0.5 \cancel{\text{ in.}} \left[\frac{1 \text{ m}}{39.37 \cancel{\text{ in.}}} \right] = 0.013 \text{ m}$$

$$\text{Area} = 1 \cancel{\text{ in.}^2} \left[\frac{1 \text{ m}}{39.37 \cancel{\text{ in.}}} \right] \left[\frac{1 \text{ m}}{39.37 \cancel{\text{ in.}}} \right] = 6.45 \times 10^{-4} \text{ m}^2$$

$$B = \frac{\Phi}{A} = \frac{8 \times 10^{-4} \text{ Wb}}{6.45 \times 10^{-4} \text{ m}^2} = 1.24 \text{ T}$$

Fig 12.8: $H_{\text{sheet steel}} \cong 460 \text{ At/m}$, Fig. 12.7: $H_{\text{cast steel}} \cong 1275 \text{ At/m}$

$$\begin{aligned} NI &= Hl_{(\text{sheet steel})} + Hl_{(\text{cast iron})} \\ &= (460 \text{ At/m})(0.013 \text{ m}) + (1275 \text{ At/m})(0.14 \text{ m}) \\ &= 5.98 \text{ At} + 178.50 \text{ At} \\ NI &= \mathbf{184.48 \text{ At}} \end{aligned}$$

$$b. \quad \text{Cast steel: } \mu = \frac{B}{H} = \frac{1.24 \text{ T}}{1275 \text{ At/m}} = \mathbf{9.73 \times 10^{-4} \text{ Wb/Am}}$$

$$\text{Sheet steel: } \mu = \frac{B}{H} = \frac{1.24 \text{ T}}{460 \text{ At/m}} = \mathbf{26.96 \times 10^{-4} \text{ Wb/Am}}$$

$$13. \quad N_1 I + N_2 = \underbrace{Hl}_{\text{cast steel}} + \underbrace{Hl}_{\text{cast iron}}$$

$$(20 \text{ t})I + (30 \text{ t})I = "$$

$$(50 \text{ t})I = "$$

$$B = \frac{\Phi}{A} \text{ with } 0.25 \text{ in.}^2 \left[\frac{1 \text{ m}}{39.37 \text{ in.}} \right] \left[\frac{1 \text{ m}}{39.37 \text{ in.}} \right] = 1.6 \times 10^{-4} \text{ m}^2$$

$$B = \frac{0.8 \times 10^{-4} \text{ Wb}}{1.6 \times 10^{-4} \text{ m}^2} = 0.5 \text{ T}$$

$$\text{Fig. 12.8: } H_{\text{cast steel}} \cong 280 \text{ At/m}$$

$$\text{Fig. 12.7: } H_{\text{cast iron}} \cong 1500 \text{ At/m}$$

$$l_{\text{cast steel}} = 5.5 \text{ in.} \left[\frac{1 \text{ m}}{39.37 \text{ in.}} \right] = 0.14 \text{ m}$$

$$l_{\text{cast iron}} = 2.5 \text{ in.} \left[\frac{1 \text{ m}}{39.37 \text{ in.}} \right] = 0.064 \text{ m}$$

$$(50 \text{ t})I = (280 \text{ At/m})(0.14 \text{ m}) + (1500 \text{ At/m})(0.064 \text{ m})$$

$$50I = 39.20 + 96.00 = 135.20$$

$$I = \mathbf{2.70 \text{ A}}$$

$$14. \quad \text{a. } l_{ab} = l_{ef} = 0.05 \text{ m}, l_{af} = 0.02 \text{ m}, l_{bc} = l_{de} = 0.0085 \text{ m}$$

$$NI = 2H_{ab}l_{ab} + 2H_{bc}l_{bc} + H_{fa}l_{fa} + H_g l_g$$

$$B = \frac{\Phi}{A} = \frac{2.4 \times 10^{-4} \text{ Wb}}{2 \times 10^{-4} \text{ m}^2} = 1.2 \text{ T} \Rightarrow H \cong 360 \text{ At/m (Fig. 12.8)}$$

$$100I = 2(360 \text{ At/m})(0.05 \text{ m}) + 2(360 \text{ At/m})(0.0085 \text{ m})$$

$$+ (360 \text{ At/m})(0.02 \text{ m}) + 7.97 \times 10^5 (1.2 \text{ T})(0.003 \text{ m})$$

$$= 36 \text{ At} + 6.12 \text{ At} + 7.2 \text{ At} + 2869 \text{ At}$$

$$100I = 2918.32 \text{ At}$$

$$I \cong \mathbf{29.18 \text{ A}}$$

$$\text{b. air gap: metal} = 2869 \text{ At}:49.72 \text{ At} = \mathbf{58.17:1}$$

$$\mu_{\text{sheet steel}} = \frac{B}{H} = \frac{1.2 \text{ T}}{360 \text{ At/m}} = \mathbf{3.33 \times 10^{-3} \text{ Wb/Am}}$$

$$\mu_{\text{air}} = \mathbf{4\pi \times 10^{-7} \text{ Wb/Am}}$$

$$\mu_{\text{sheet steel}} : \mu_{\text{air}} = 3.33 \times 10^{-3} \text{ Wb/Am} : 4\pi \times 10^{-7} \cong \mathbf{2627:1}$$

$$15. \quad 4 \text{ cm} \left[\frac{1 \text{ m}}{100 \text{ cm}} \right] = 0.04 \text{ m}$$

$$f = \frac{1}{2} NI \frac{d\phi}{dx} = \frac{1}{2} (80 \text{ t})(0.9 \text{ A}) \frac{(8 \times 10^{-4} \text{ Wb} - 0.5 \times 10^{-4} \text{ Wb})}{\frac{1}{2}(0.04 \text{ m})} = \frac{36(7.5 \times 10^{-4})}{0.02}$$

$$= \mathbf{1.35 \text{ N}}$$

16. $C = 2\pi r = (6.28)(0.3 \text{ m}) = 1.88 \text{ m}$

$$B = \frac{\Phi}{A} = \frac{2 \times 10^{-4} \text{ Wb}}{1.3 \times 10^{-4} \text{ m}^2} = 1.54 \text{ T}$$

Fig. 12.7: $H_{\text{sheet steel}} \cong 2100 \text{ At/m}$

$$H_g = 7.97 \times 10^5 B_g = (7.97 \times 10^5)(1.54 \text{ T}) = 1.23 \times 10^6 \text{ At/m}$$

$$N_1 I_1 + N_2 I_2 = H_g l_g + H l_{(\text{sheet steel})}$$

$$(200 \text{ t}) I_1 + (40 \text{ t})(0.3 \text{ A}) = (1.23 \times 10^6 \text{ At/m})(2 \text{ mm}) + (2100 \text{ At/m})(1.88 \text{ m})$$

$$I_1 = \mathbf{31.98 \text{ A}}$$

17. a. $0.2 \text{ cm} \left[\frac{1 \text{ m}}{100 \text{ cm}} \right] = 2 \times 10^{-3} \text{ m}$

$$A = \frac{\pi d^2}{4} = \frac{(3.14)(0.01 \text{ m})^2}{4} = 0.79 \times 10^{-4} \text{ m}^2$$

$$NI = H_g l_g, H_g = 7.96 \times 10^5 B_g$$

$$(200 \text{ t}) I = \left[(7.96 \times 10^5) \left(\frac{0.2 \times 10^{-4} \text{ Wb}}{0.79 \times 10^{-4} \text{ m}^2} \right) \right] 2 \times 10^{-3} \text{ m}$$

$$I = \mathbf{2.02 \text{ A}}$$

b. $B_g = \frac{\Phi}{A} = \frac{2 \times 10^{-4} \text{ Wb}}{0.79 \times 10^{-4} \text{ m}^2} = 0.25 \text{ T}$

$$F \cong \frac{1}{2} \frac{B_g^2 A}{\mu_o} = \frac{1}{2} \frac{(0.25 \text{ T})^2 (0.79 \times 10^{-4} \text{ m}^2)}{4\pi \times 10^{-7}}$$

$$\cong \mathbf{2 \text{ N}}$$

18. **Table:**

Section	$\Phi(\text{Wb})$	$A(\text{m}^2)$	$B(\text{T})$	H	$l(\text{m})$	Hl
a-b, g-h		5×10^{-4}			0.2	
b-c, f-g	2×10^{-4}	5×10^{-4}			0.1	
c-d, e-f	2×10^{-4}	5×10^{-4}			0.099	
a-h		5×10^{-4}			0.2	
b-g		2×10^{-4}			0.2	
d-e	2×10^{-4}	5×10^{-4}			0.002	

$$B_{bc} = B_{cd} = B_g = B_{ef} = B_{fg} = \frac{\Phi}{A} = \frac{2 \times 10^{-4} \text{ Wb}}{5 \times 10^{-4} \text{ m}^2} = 0.4 \text{ T}$$

$$\text{Air gap: } H_g = 7.97 \times 10^5 (0.4 \text{ T}) = 3.19 \times 10^5 \text{ At/m}$$

$$H_g l_g = (3.19 \times 10^5 \text{ At/m})(2 \text{ mm}) = 638 \text{ At}$$

$$\text{Fig 12.8: } H_{bc} = H_{cd} = H_{ef} = H_{fg} = 55 \text{ At/m}$$

$$H_{bc} l_{bc} = H_{fg} l_{fg} = (55 \text{ At/m})(0.1 \text{ m}) = 5.5 \text{ At}$$

$$H_{cd} l_{cd} = H_{ef} l_{ef} = (55 \text{ At/m})(0.099 \text{ m}) = 5.45 \text{ At}$$

For loop 2: $\sum \mathcal{F} = 0$

$$H_{bc} l_{bc} + H_{cd} l_{cd} + H_g l_g + H_{ef} l_{ef} + H_{fg} l_{fg} - H_{gb} l_{gb} = 0$$

$$5.5 \text{ At} + 5.45 \text{ At} + 638 \text{ At} + 5.45 \text{ At} + 5.50 \text{ At} - H_{gb} l_{gb} = 0$$

$$H_{gb} l_{gb} = 659.90 \text{ At}$$

$$\text{and } H_{gb} = \frac{659.90 \text{ At}}{0.2 \text{ m}} = 3300 \text{ At/m}$$

$$\text{Fig 12.7: } B_{gb} \cong 1.55 \text{ T}$$

$$\text{with } \Phi_2 = B_{gb} A = (1.55 \text{ T})(2 \times 10^{-4} \text{ m}^2) = 3.1 \times 10^{-4} \text{ Wb}$$

$$\Phi_T = \Phi_1 + \Phi_2$$

$$= 2 \times 10^{-4} \text{ Wb} + 3.1 \times 10^{-4} \text{ Wb}$$

$$= 5.1 \times 10^{-4} \text{ Wb} = \Phi_{ab} = \Phi_{ha} = \Phi_{gh}$$

$$B_{ab} = B_{ha} = B_{gh} = \frac{\Phi_T}{A} = \frac{5.1 \times 10^{-4} \text{ Wb}}{5 \times 10^{-4} \text{ m}^2} = 1.02 \text{ T}$$

B-H curve: (Fig 12.8):

$$H_{ab} = H_{ha} = H_{gh} \cong 180 \text{ At/m}$$

$$H_{ab} l_{ab} = (180 \text{ At/m})(0.2 \text{ m}) = 36 \text{ At}$$

$$H_{ha} l_{ha} = (180 \text{ At/m})(0.2 \text{ m}) = 36 \text{ At}$$

$$H_{gh} l_{gh} = (180 \text{ At/m})(0.2 \text{ m}) = 36 \text{ At}$$

which **completes** the table!

Loop #1: $\sum \mathcal{F} = 0$

$$NI = H_{ab} l_{ab} + H_{bg} l_{bg} + H_{gh} l_{gh} + H_{ah} l_{ah}$$

$$(200 \text{ t})I = 36 \text{ At} + 659.49 \text{ At} + 36 \text{ At} + 36 \text{ At}$$

$$(200 \text{ t})I = 767.49 \text{ At}$$

$$I \cong \mathbf{3.84 \text{ A}}$$

$$19. \quad NI = HI$$

$$l = 2\pi r = (6.28)(0.08 \text{ m}) = 0.50 \text{ m}$$

$$(100 \text{ t})(2 \text{ A}) = H(0.50 \text{ m})$$

$$H = 400 \text{ At/m}$$

$$\text{Fig. 12.8: } B \cong 0.68 \text{ T}$$

$$\Phi = BA = (0.68 \text{ T})(0.009 \text{ m}^2)$$

$$\Phi = \mathbf{6.12 \text{ mWb}}$$

20. $NI = H_{ab}(l_{ab} + l_{bc} + l_{de} + l_{ef} + l_{fa}) + H_g l_g$
 $300 \text{ At} = H_{ab}(0.8 \text{ m}) + 7.97 \times 10^5 B_g(0.8 \text{ mm})$
 $300 \text{ At} = H_{ab}(0.8 \text{ m}) + 637.6 B_g$
Assuming $637.6 B_g \gg H_{ab}(0.8 \text{ m})$
then $300 \text{ At} = 637.6 B_g$
and $B_g = 0.47 \text{ T}$
 $\Phi = BA = (0.47 \text{ T})(2 \times 10^{-4} \text{ m}^2) = 0.94 \times 10^{-4} \text{ Wb}$
 $B_{ab} = B_g = 0.47 \text{ T} \Rightarrow H \cong 270 \text{ At/m}$ (Fig. 12.8)
 $300 \text{ At} = (270 \text{ At/m})(0.8 \text{ m}) + 637.6(0.47 \text{ T})$
 $300 \text{ At} \neq 515.67 \text{ At}$
 $\therefore$ Poor approximation!
 $\frac{300 \text{ At}}{515.67 \text{ At}} \times 100\% \cong 58\%$
Reduce Φ to 58%
 $0.58(0.94 \times 10^{-4} \text{ Wb}) = 0.55 \times 10^{-4} \text{ Wb}$
 $B = \frac{\Phi}{A} = \frac{0.55 \times 10^{-4} \text{ Wb}}{2 \times 10^{-4} \text{ m}^2} = 0.28 \text{ T} \Rightarrow H \cong 190 \text{ At/m}$ (Fig. 12.8)
 $300 \text{ At} = (190 \text{ At/m})(0.8 \text{ m}) + 637.6(0.28 \text{ T})$
 $300 \text{ At} \neq 330.53 \text{ At}$
Reduce Φ another 10% $= 0.55 \times 10^{-4} \text{ Wb} - 0.1(0.55 \times 10^{-4} \text{ Wb})$
 $= 0.495 \times 10^{-4} \text{ Wb}$
 $B = \frac{\Phi}{A} = \frac{0.495 \times 10^{-4} \text{ Wb}}{2 \times 10^{-4} \text{ m}^2} = 0.25 \text{ T} \Rightarrow H \cong 175 \text{ At/m}$ (Fig. 12.7)
 $300 \text{ At} = (175 \text{ At/m})(0.8) + 637.6(0.25 \text{ T})$
 $300 \text{ At} \neq 318.53 \text{ At}$ but within 5% $\therefore$ OK
 $\Phi \cong \mathbf{0.55 \times 10^{-4} \text{ Wb}}$

21. a. $1\tau = 0.632 T_{\max}$
 $T_{\max} \cong 1.5 \text{ T}$ for cast steel
 $0.632(1.5 \text{ T}) = 0.945 \text{ T}$
At 0.945 T, $H \cong 700 \text{ At/m}$ (Fig. 12.7)
 $\therefore \mathbf{B = 1.5 \text{ T}(1 - e^{-H/700 \text{ At/m}})}$

b. $H = 900 \text{ At/m}$:
 $B = 1.5 \text{ T} \left(1 - e^{-\frac{900 \text{ At/m}}{700 \text{ At/m}}}\right) = \mathbf{1.09 \text{ T}}$
Graph: $\cong \mathbf{1.1 \text{ T}}$
 $H = 1800 \text{ At/m}$:
 $B = 1.5 \text{ T} \left(1 - e^{-\frac{1800 \text{ At/m}}{700 \text{ At/m}}}\right) = \mathbf{1.39 \text{ T}}$
Graph: $\cong \mathbf{1.38 \text{ T}}$
 $H = 2700 \text{ At/m}$:
 $B = 1.5 \text{ T} \left(1 - e^{-\frac{2700 \text{ At/m}}{700 \text{ At/m}}}\right) = \mathbf{1.47 \text{ T}}$
Graph: $\cong \mathbf{1.47 \text{ T}}$
Excellent comparison!

c. $B = 1.5 \text{ T}(1 - e^{-H/700 \text{ At/m}}) = 1.5 \text{ T} - 1.5 \text{ T}e^{-H/700 \text{ At/m}}$
 $B - 1.5 \text{ T} = -1.5 \text{ T}e^{-H/700 \text{ At/m}}$
 $1.5 - B = 1.5 \text{ T}e^{-H/700 \text{ At/m}}$
 $\frac{1.5 \text{ T} - B}{1.5 \text{ T}} = e^{-H/700 \text{ At/m}}$
 $\log_e \left(1 - \frac{B}{1.5 \text{ T}} \right) = \frac{-H}{700 \text{ At/m}}$
and $H = -700 \log_e \left(1 - \frac{B}{1.5 \text{ T}} \right)$

d. $B = 1 \text{ T}$:
 $H = -700 \log_e \left(1 - \frac{1 \text{ T}}{1.5 \text{ T}} \right) = \mathbf{769.03 \text{ At/m}}$

Graph: $\cong \mathbf{750 \text{ At/m}}$
 $B = 1.4 \text{ T}$:

$$H = -700 \log_e \left(1 - \frac{1.4 \text{ T}}{1.5 \text{ T}} \right) = \mathbf{1895.64 \text{ At/m}}$$

Graph: $\cong \mathbf{1920 \text{ At/m}}$

e. $H = -700 \log_e \left(1 - \frac{B}{1.5 \text{ T}} \right)$
 $= -700 \log_e \left(1 - \frac{0.2 \text{ T}}{1.5 \text{ T}} \right)$
 $= 100.2 \text{ At/m}$
 $I = \frac{Hl}{N} = \frac{(100.2 \text{ At/m})(0.16 \text{ m})}{400 \text{ t}} = \mathbf{40.1 \text{ mA}}$

vs 44 mA for Ex. 12.1

Chapter 13

1.
 - a. **10 V**
 - b. 15 ms: **-10 V**, 20 ms: **0 V**
 - c. **20 V**
 - d. **20 ms**
 - e. **2 cycles**

2.
 - a. **200 μA**
 - b. 1 μs : **200 μA** , 7 μs : **-200 μA**
 - c. **400 μA**
 - d. **4 μs**
 - e. **2.5 cycles**

3.
 - a. **40 mV**
 - b. 1.5 ms: **-40 mV**, 5.1 ms: **-40 mV**
 - c. **80 mV**
 - d. **2 ms**
 - e. **3.5 cycles**

4.
 - a. $T = \frac{1}{f} = \frac{1}{200 \text{ Hz}} = \mathbf{5 \text{ ms}}$
 - b. $T = \frac{1}{f} = \frac{1}{40 \text{ MHz}} = \mathbf{25 \text{ ns}}$
 - c. $T = \frac{1}{f} = \frac{1}{20 \text{ kHz}} = \mathbf{50 \mu\text{s}}$
 - d. $T = \frac{1}{f} = \frac{1}{1 \text{ Hz}} = \mathbf{1 \text{ s}}$

5.
 - a. $f = \frac{1}{T} = \frac{1}{1 \text{ s}} = \mathbf{1 \text{ Hz}}$
 - b. $f = \frac{1}{T} = \frac{1}{\frac{1}{16} \text{ s}} = \mathbf{16 \text{ Hz}}$
 - c. $f = \frac{1}{T} = \frac{1}{40 \text{ ms}} = \mathbf{25 \text{ Hz}}$
 - d. $f = \frac{1}{T} = \frac{1}{25 \mu\text{s}} = \mathbf{40 \text{ kHz}}$

6. $T = \frac{1}{1 \text{ kHz}} = 1 \text{ ms}$, $5(1 \text{ ms}) = \mathbf{5 \text{ ms}}$

7. $T = \frac{24 \text{ ms}}{80 \text{ cycles}} = \mathbf{0.3 \text{ ms}}$

8. $f = \frac{42 \text{ cycles}}{6 \text{ s}} = \mathbf{7 \text{ Hz}}$
9.
 - a. $V_{\text{peak}} = (2.5 \text{ div.})(50 \text{ mV/div}) = \mathbf{125 \text{ mV}}$
 - b. $T = (3.2 \text{ div.})(10 \text{ }\mu\text{s/div.}) = \mathbf{32 \text{ }\mu\text{s}}$
 - c. $f = \frac{1}{T} = \frac{1}{32 \text{ }\mu\text{s}} = \mathbf{31.25 \text{ kHz}}$
10.
 - a. $\text{Radians} = \left(\frac{\pi}{180^\circ}\right)40^\circ = \mathbf{0.22 \pi \text{ rad}}$
 - b. $\text{Radians} = \left(\frac{\pi}{180^\circ}\right)60^\circ = \mathbf{\frac{\pi}{3} \text{ rad}}$
 - c. $\text{Radians} = \left(\frac{\pi}{180^\circ}\right)135^\circ = \mathbf{0.75\pi \text{ rad}}$
 - d. $\text{Radians} = \left(\frac{\pi}{180^\circ}\right)170^\circ = \mathbf{0.94\pi \text{ rad}}$
11.
 - a. $\text{Degrees} = \left(\frac{180^\circ}{\pi}\right)\frac{\pi}{3} = \mathbf{60^\circ}$
 - b. $\text{Degrees} = \left(\frac{180^\circ}{\pi}\right)1.2\pi = \mathbf{216^\circ}$
 - c. $\text{Degrees} = \left(\frac{180^\circ}{\pi}\right)\frac{1}{10}\pi = \mathbf{18^\circ}$
 - d. $\text{Degrees} = \left(\frac{180^\circ}{\pi}\right)0.6 \pi = \mathbf{108^\circ}$
12.
 - a. $\omega = \frac{2\pi}{T} = \frac{2\pi}{1.8 \text{ s}} = \mathbf{3.49 \text{ rad/s}}$
 - b. $\omega = \frac{2\pi}{0.3 \times 10^{-3} \text{ s}} = \mathbf{20.94 \times 10^3 \text{ rad/s}}$
 - c. $\omega = \frac{2\pi}{8 \times 10^{-6} \text{ s}} = \mathbf{785.4 \times 10^3 \text{ rad/s}}$
 - d. $\omega = \frac{2\pi}{4 \times 10^{-6} \text{ s}} = \mathbf{1.57 \times 10^6 \text{ rad/s}}$
13.
 - a. $\omega = 2\pi f = 2\pi(100 \text{ Hz}) = \mathbf{628.32 \text{ rad/s}}$
 - b. $\omega = 2\pi f = 2\pi(0.25 \text{ kHz}) = \mathbf{1.57 \times 10^3 \text{ rad/s}}$
 - c. $\omega = 2\pi f = 2\pi(2 \text{ kHz}) = \mathbf{12.56 \times 10^3 \text{ rad/s}}$
 - d. $\omega = 2\pi f = 2\pi(0.004 \text{ MHz}) = \mathbf{25.13 \times 10^3 \text{ rad/s}}$

14. a. $\omega = 2\pi f = \frac{2\pi}{T} \Rightarrow f = \frac{\omega}{2\pi}$
 $T = \frac{2\pi}{\omega} = \frac{1}{f}$
 $f = \frac{\omega}{2\pi} = \frac{754 \text{ rad/s}}{2\pi} = \mathbf{120 \text{ Hz}}, T = \mathbf{8.33 \text{ ms}}$
- b. $f = \frac{\omega}{2\pi} = \frac{12 \text{ rad/s}}{2\pi} = \mathbf{1.91 \text{ Hz}}, T = \mathbf{523.6 \text{ ms}}$
- c. $f = \frac{\omega}{2\pi} = \frac{6000 \text{ rad/s}}{2\pi} = \mathbf{954.93 \text{ Hz}}, T = \mathbf{1.05 \text{ ms}}$
- d. $f = \frac{\omega}{2\pi} = \frac{0.16 \text{ rad/s}}{2\pi} = \mathbf{25.46 \times 10^{-3} \text{ Hz}}, T = \mathbf{39.28 \text{ ms}}$
15. $(60^\circ) \left(\frac{\pi}{180^\circ} \right) = \frac{\pi}{3} \text{ radians}$
 $t = \frac{\theta}{\omega} = \frac{\pi/3 \text{ rad}}{2\pi f} = \frac{\pi/3 \text{ rad}}{2\pi(60 \text{ Hz})} = \frac{1}{(6)(60)} = \frac{1}{360} = \mathbf{2.78 \text{ ms}}$
16. $(30^\circ) \left(\frac{\pi}{180^\circ} \right) = \frac{\pi}{6}, \alpha = \omega t \Rightarrow \omega = \frac{\alpha}{t} = \frac{\pi/6}{5 \times 10^{-3} \text{ s}} = \mathbf{104.7 \text{ rad/s}}$
17. a. Amplitude = **20**, $f = \frac{\omega}{2\pi} = \frac{377 \text{ rad/s}}{2\pi} = \mathbf{60 \text{ Hz}}$
b. Amplitude = **12**, $f = \mathbf{120 \text{ Hz}}$
c. Amplitude = **10^6** , $f = \frac{\omega}{2\pi} = \frac{10,000 \text{ rad/s}}{2\pi} = \mathbf{1591.55 \text{ Hz}}$
d. Amplitude = **8**, $f = \frac{\omega}{2\pi} = \frac{10,058 \text{ rad/s}}{2\pi} = \mathbf{1.6 \text{ kHz}}$
18. –
19. –
20. $T = \frac{2\pi}{\omega} = \frac{2\pi}{157} = 40 \text{ ms}, \frac{1}{2} \text{ cycle} = \mathbf{20 \text{ ms}}$
21. $i = 0.5 \sin 72^\circ = 0.5(0.9511) = \mathbf{0.48 \text{ A}}$
22. $1.2\pi \left(\frac{180^\circ}{\pi} \right) = 216^\circ$
 $v = 20 \sin 216^\circ = 20(-0.588) = \mathbf{-11.76 \text{ V}}$

23. $6 \times 10^{-3} = 30 \times 10^{-3} \sin \alpha$
 $0.2 = \sin \alpha$
 $\alpha = \sin^{-1} 0.2 = \mathbf{11.54^\circ}$ and $180^\circ - 11.54^\circ = \mathbf{168.46^\circ}$
24. $v = V_m \sin \alpha$
 $40 = V_m \sin 30^\circ = V_m (0.5)$
 $\therefore V_m = \frac{40}{0.5} = \mathbf{80 \text{ V}}$
- $\frac{30^\circ}{360^\circ} = \frac{1 \text{ ms}}{T}$
 $T = 1 \text{ ms} \left(\frac{360}{30} \right) = \mathbf{12 \text{ ms}}$
 $f = \frac{1}{T} = \frac{1}{12 \times 10^{-3} \text{ s}} = \mathbf{83.33 \text{ Hz}}$
 $\omega = 2\pi f = (2\pi)(83.33 \text{ Hz}) = \mathbf{523.58 \text{ rad/s}}$
and $v = \mathbf{80 \sin 523.58t}$
25. —
26. —
27. a. $v = \mathbf{6 \times 10^{-3} \sin (2\pi 2000t + 30^\circ)}$
b. $i = \mathbf{20 \times 10^{-3} \sin (2\pi 60t - 60^\circ)}$
28. a. $v = \mathbf{120 \times 10^{-6} \sin (2\pi 1000t - 80^\circ)}$
29. $v = \mathbf{12 \times 10^{-3} \sin (2\pi 2000t + 135^\circ)}$
30. $v = \mathbf{8 \times 10^{-3} \sin (2\pi 500t + \pi/6)}$
31. v leads i by $\mathbf{90^\circ}$
32. i leads v by $\mathbf{40^\circ}$
33. $v = 2 \sin (\omega t - \underbrace{30^\circ + 90^\circ}_{+60^\circ})$
 $i = 5 \sin (\omega t + 60^\circ)$ } **in phase**
34. $v = 4 \sin (\omega t + 90^\circ + 90^\circ + 180^\circ) = 4 \sin \omega t$
 $i = \sin (\omega t + 10^\circ + 180^\circ) = \sin (\omega t + 190^\circ)$ } **i leads v by 190°**
35. $T = \frac{1}{f} = \frac{1}{1000 \text{ Hz}} = 1 \text{ ms}$
 $t_1 = \frac{120^\circ}{180^\circ} \left(\frac{T}{2} \right) = \frac{2}{3} \left(\frac{1 \text{ ms}}{2} \right) = \mathbf{\frac{1}{3} \text{ ms}}$

36. $\omega = 2\pi f = \frac{2\pi}{T}$
 $T = \frac{2\pi}{\omega} = \frac{2\pi}{50,000 \text{ rad/s}} = 125.66 \mu\text{s}$
 $t_1 = \frac{40^\circ}{360^\circ}(T) = \frac{40^\circ}{360^\circ}(125.66 \mu\text{s})$
 $= \mathbf{13.96 \mu\text{s}}$
37. $T = 1 \text{ ms}$
 $t_{\text{peak}} @ 30^\circ$
 $t_{\text{peak}} = \frac{30^\circ}{360^\circ}(T) = \frac{1}{12} \text{ ms}$
38. a. $T = (8 \text{ div.})(1 \text{ ms/div.}) = \mathbf{8 \text{ ms}}$ (both waveforms)
b. $f = \frac{1}{T} = \frac{1}{8 \text{ ms}} = \mathbf{125 \text{ Hz}}$ (both)
c. Peak = (2.5 div)(0.5 V/div.) = 1.25 V
 $V_{\text{rms}} = 0.707(1.25 \text{ V}) = \mathbf{0.884 \text{ V}}$
d. Phase shift = 4.6 div., $T = 8 \text{ div.}$
 $\theta = \frac{4.6 \text{ div.}}{8 \text{ div.}} \times 360^\circ = \mathbf{207^\circ i \text{ leads } e}$
or $\mathbf{e \text{ leads } i \text{ by } 153^\circ}$
39. $G = \frac{0 + (6 \text{ V})(5 \text{ ms}) + (3 \text{ V})(10 \text{ ms}) - (3 \text{ V})(10 \text{ ms})}{30 \text{ ms}}$
 $= \frac{30 \text{ V} + 30 \text{ V} - 30 \text{ V}}{30} = \mathbf{1 \text{ V}}$
40. $G = \frac{\left[\frac{1}{2}(4 \text{ ms})(20 \text{ mA}) \right] - (2 \text{ ms})(8 \text{ mA}) - \frac{1}{2}(2 \text{ ms})(8 \text{ mA})}{8 \text{ ms}}$
 $= \frac{40 \text{ mA} - 16 \text{ mA} - 8 \text{ mA}}{8} = \frac{16 \text{ mA}}{8}$
 $= \mathbf{2 \text{ mA}}$
41. $G = \frac{(35 \text{ V})(5 \text{ ms}) - \frac{1}{2}(20 \text{ V})(20 \text{ ms}) - (20 \text{ V})(15 \text{ ms}) - \frac{1}{2}(20 \text{ V})(7.5 \text{ ms}) + \frac{1}{2}(20 \text{ V})(7.5 \text{ ms}) + \frac{1}{2}(20 \text{ V})(15 \text{ ms}) + 0}{75 \text{ ms}}$
 $= \frac{175 \text{ V} - 200 \text{ V} - 300 \text{ V} - 75 \text{ V} + 75 \text{ V} + 150 \text{ V}}{75}$
 $= \frac{400 \text{ V} - 575 \text{ V}}{75} = \mathbf{2.33 \text{ V}}$

$$42. \quad G = \frac{0 + \frac{1}{2}(30 \text{ mA})(3 \text{ ms}) - \frac{1}{2}(20 \text{ mA})(2 \text{ ms})}{7 \text{ ms}}$$

$$= \frac{45 \text{ mA} - 20 \text{ mA}}{7} = \mathbf{3.57 \text{ mA}}$$

$$43. \quad \text{a. } 0 \text{ V}$$

$$\text{b. } G = \frac{(4 \text{ V})(5 \text{ ms}) + \frac{1}{2}(8 \text{ V})(5 \text{ ms}) - (8 \text{ V})(5 \text{ ms}) + (4 \text{ V})(5 \text{ ms}) + \frac{1}{2}(8 \text{ V})(5 \text{ ms}) - (8 \text{ V})(5 \text{ ms})}{25 \text{ ms}}$$

$$= \frac{20 \text{ V} + 20 \text{ V} - 40 \text{ V} + 20 \text{ V} + 20 \text{ V} - 40 \text{ V}}{25}$$

$$= \mathbf{0 \text{ V}}$$

$$\text{c. } \text{The same}$$

$$44. \quad \text{Area} = \frac{1}{2}(\pi r^2) = \frac{1}{2}(\pi 20^2) = 628.32$$

$$\text{Area} = \frac{628.32}{d} = \frac{628.32}{40} = 15.71 \Rightarrow 15.71 \text{ mA}$$

$$G = \frac{(15.71 \text{ mA})(\pi) - (5 \text{ mA})(\pi)}{2\pi}$$

$$= \mathbf{5.36 \text{ mA}}$$

$$45. \quad \text{a. } T = (2 \text{ div.})(0.2 \text{ ms/div}) = \mathbf{0.4 \text{ ms}}$$

$$\text{b. } f = \frac{1}{T} = \frac{1}{0.4 \text{ ms}} = \mathbf{2.5 \text{ kHz}}$$

$$\text{c. } \text{Average} = (-2.5 \text{ div.})(10 \text{ mV/div.}) = \mathbf{-25 \text{ mV}}$$

$$46. \quad \text{a. } T = (4 \text{ div.})(10 \mu\text{s/div.}) = \mathbf{40 \mu\text{s}}$$

$$\text{b. } f = \frac{1}{T} = \frac{1}{40 \mu\text{s}} = \mathbf{25 \text{ kHz}}$$

$$\text{c. } G = \frac{(2.5 \text{ div.})(1.5 \text{ div.}) + (1 \text{ div.})(0.5 \text{ div.}) + (1 \text{ div.})(0.6 \text{ div.}) + (2.5 \text{ div.})(0.4 \text{ div.}) + (1 \text{ div.})(1 \text{ div.})}{4 \text{ div.}}$$

$$= \frac{3.75 \text{ div.} + 0.5 \text{ div.} + 0.6 \text{ div.} + 1 \text{ div.} + 1 \text{ div.}}{4}$$

$$= \frac{6.85 \text{ div.}}{4} = 1.713 \text{ div.}$$

$$1.713 \text{ div.}(10 \text{ mV/div.}) = \mathbf{17.13 \text{ mV}}$$

$$47. \quad \text{a. } V_{\text{rms}} = 0.7071(120 \text{ V}) = \mathbf{84.85 \text{ V}}$$

$$\text{b. } I_{\text{rms}} = 0.7071(6 \text{ mA}) = \mathbf{4.24 \text{ mA}}$$

$$\text{c. } V_{\text{rms}} = 0.7071(8 \mu\text{V}) = \mathbf{5.66 \mu\text{V}}$$

48. a. $v = 6.79 \sin 377t$
 b. $i = 70.7 \times 10^{-3} \sin 377t$
 c. $v = 2.83 \times 10^3 \sin 377t$

49.
$$V_{\text{rms}} = \frac{\sqrt{(2 \text{ V})^2(4 \text{ s}) + (-2 \text{ V})^2(1 \text{ s}) + (3 \text{ V})^2\left(\frac{1}{2} \text{ s}\right)}}{12 \text{ s}} = 1.43 \text{ V}$$

50.
$$V_{\text{rms}} = \frac{\sqrt{(3 \text{ V})^2(2 \text{ s}) + (2 \text{ V})^2(2 \text{ s}) + 0 + (-1 \text{ V})^2(2 \text{ s}) + (-3 \text{ V})^2(2 \text{ s}) + (-2 \text{ V})^2(2 \text{ s})}}{12 \text{ s}}$$

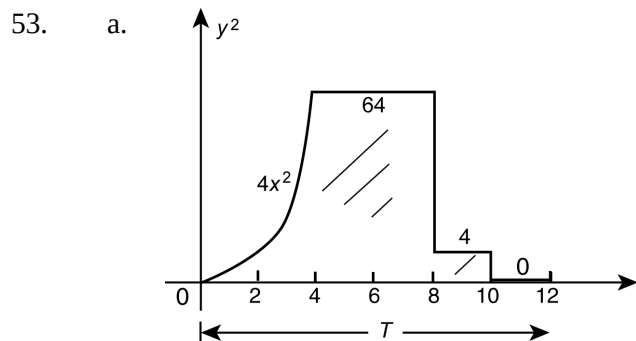
$$= \sqrt{\frac{54}{12}} \text{ V}^2 = \sqrt{4.5 \text{ V}^2} = 2.12 \text{ V}$$

51.
$$G = \frac{(8 \text{ V})(4 \text{ ms}) - (8 \text{ V})(4 \text{ ms})}{8 \text{ ms}} = \frac{0}{8 \text{ ms}} = 0 \text{ V}$$

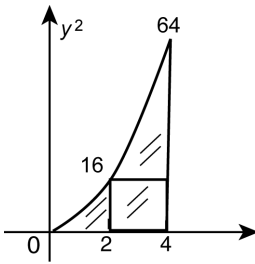
$$V_{\text{rms}} = \sqrt{\frac{(8 \text{ V})^2(4 \text{ ms}) + (-8 \text{ V})^2(4 \text{ ms})}{8 \text{ ms}}} = 8 \text{ V}$$

52. a. $T = (4 \text{ div.})(10 \mu\text{s/div.}) = 40 \mu\text{s}$
 $f = \frac{1}{T} = \frac{1}{40 \mu\text{s}} = 25 \text{ kHz}$
 $\text{Av.} = (1 \text{ div.})(20 \text{ mV/div.}) = 20 \text{ mV}$
 $\text{Peak} = (2 \text{ div.})(20 \text{ mV/div.}) = 40 \text{ mV}$
 $\text{rms} = \sqrt{V_0^2 + \frac{V_{\text{max}}^2}{2}} = \sqrt{(20 \text{ mV})^2 + \frac{(40 \text{ mV})^2}{2}} = 34.64 \text{ mV}$

b. $T = (2 \text{ div.})(50 \mu\text{s}) = 100 \mu\text{s}$
 $f = \frac{1}{T} = \frac{1}{100 \mu\text{s}} = 10 \text{ kHz}$
 $\text{Av.} = (-1.5 \text{ div.})(0.2 \text{ V/div.}) = -0.3 \text{ V}$
 $\text{Peak} = (1.5 \text{ div.})(0.2 \text{ V/div.}) = 0.3 \text{ mV}$
 $\text{rms} = \sqrt{V_0^2 + \frac{V_{\text{max}}^2}{2}} = \sqrt{(.3 \text{ V})^2 + \frac{(.3 \text{ V})^2}{2}} = 367.42 \text{ mV}$



b.



$$A_1 = \frac{1}{2}(2)(16) + (2)(16) + \frac{1}{2}(2)(48) = 96$$

$$\text{Area} = 96 + (4)(64) + (2)(4) = 96 + 256 + 8 = \mathbf{360}$$

c. $\text{rms} = \sqrt{\frac{360}{12}} = \sqrt{30} = \mathbf{5.48}$

d. $G = \frac{\frac{1}{2}(4)(8) + 4(8) - 2(2)}{12} = \frac{16 + 32 - 4}{12} = \mathbf{3.67}$

e. $\text{rms} \cong \mathbf{1.5 \text{ (average value)}}$

54. a. $V_{dc} = IR = (4 \text{ mA})(2 \text{ k}\Omega) = 8 \text{ V}$
Meter indication = $2.22(8 \text{ V}) = \mathbf{17.76 \text{ V}}$

b. $V_{\text{rms}} = 0.707(16 \text{ V}) = \mathbf{11.31 \text{ V}}$

Chapter 14

1. —
2. —
3.
 - a. $(377)(10) \cos 377t = \mathbf{3770 \cos 377t}$
 - b. $(200)(0.6) \cos(754t + 20^\circ) = \mathbf{120 \cos(754t + 20^\circ)}$
 - c. $(\sqrt{2} 20)(157) \cos(157t - 20^\circ) = \mathbf{4440.63 \cos(157t - 20^\circ)}$
 - d. $(-200)(1) \cos(t + 180^\circ) = -200 \cos(t + 180^\circ) = \mathbf{200 \cos t}$
4.
 - a. $I_m = V_m/R = 150 \text{ V}/3 \Omega = 50 \text{ A}, i = \mathbf{50 \sin 200t}$
 - b. $I_m = V_m/R = 30 \text{ V}/3 \Omega = 10 \text{ A}, i = \mathbf{10 \sin(377t + 20^\circ)}$
 - c. $I_m = V_m/R = 6 \text{ V}/3 \Omega = 2 \text{ A}, i = \mathbf{2 \sin(\omega t + 100^\circ)}$
 - d. $I_m = V_m/R = 12 \text{ V}/3 \Omega = 4 \text{ A}, i = \mathbf{4 \sin(\omega t + 220^\circ)}$
5.
 - a. $V_m = I_m R = (0.1 \text{ A})(7 \times 10^3 \Omega) = 700 \text{ V}$
 $v = \mathbf{700 \sin 1000t}$
 - b. $V_m = I_m R = (2 \times 10^{-3} \text{ A})(7 \times 10^3 \Omega) = 14.8 \text{ V}$
 $v = \mathbf{14.8 \sin(400t - 120^\circ)}$
6.
 - a. $\mathbf{0 \Omega}$
 - b. $X_L = 12.56f = 12.56(60 \text{ Hz}) = \mathbf{753.6 \Omega}$
 - c. $X_L = 12.56f = 12.56(4 \text{ kHz}) = \mathbf{50.24 \text{ k}\Omega}$
 - d. $X_L = 12.56f = 12.56(1.2 \text{ MHz}) = \mathbf{15.07 \text{ M}\Omega}$
7.
 - a. $L = \frac{X_L}{2\pi f} = \frac{2 \text{ k}\Omega}{2\pi(14.47 \text{ kHz})} = \mathbf{22 \text{ mH}}$
 - b. $L = \frac{X_L}{2\pi f} = \frac{40 \text{ k}\Omega}{2\pi(5.3 \text{ kHz})} = \mathbf{1.2 \text{ H}}$
8.
 - a. $X_L = 2\pi fL \Rightarrow f = \frac{X_L}{2\pi L} = \frac{X_L}{(6.28)(1 \text{ mH})} = \frac{X_L}{6.28 \times 10^{-3} \text{ H}}$
 $f = \frac{10 \Omega}{6.28 \times 10^{-3} \text{ H}} = \mathbf{1.59 \text{ kHz}}$

- b. $f = \frac{X_L}{6.28 \times 10^{-3} \text{ H}} = \frac{4 \text{ k}\Omega}{6.28 \times 10^{-3} \text{ H}} = \mathbf{636.94 \text{ kHz}}$
- c. $f = \frac{X_L}{6.28 \times 10^{-3} \text{ H}} = \frac{12 \text{ k}\Omega}{6.28 \times 10^{-3} \text{ H}} = \mathbf{1.91 \text{ MHz}}$
9. a. $V_m = I_m X_L = (5 \text{ A})(20 \Omega) = 100 \text{ V}$
 $v = \mathbf{100 \sin(\omega t + 90^\circ)}$
- b. $V_m = I_m X_L = (40 \times 10^{-3} \text{ A})(20 \Omega) = 0.8 \text{ V}$
 $v = \mathbf{0.8 \sin(\omega t + 150^\circ)}$
- c. $i = 6 \sin(\omega t + 150^\circ)$, $V_m = I_m X_L = (6 \text{ A})(20 \Omega) = 120 \text{ V}$
 $v = 120 \sin(\omega t + 240^\circ) = \mathbf{120 \sin(\omega t - 120^\circ)}$
10. a. $X_L = \omega L = (100 \text{ rad/s})(0.1 \text{ H}) = 10 \Omega$
 $V_m = I_m X_L = (10 \text{ A})(10 \Omega) = 100 \text{ V}$
 $v = \mathbf{100 \sin(100t + 90^\circ)}$
- b. $X_L = \omega L = (400 \text{ rad/s})(0.1 \text{ H}) = 40 \Omega$
 $V_m = I_m X_L = (5 \times 10^{-6} \text{ A})(40 \Omega) = 200 \mu\text{V}$
 $v = \mathbf{200 \times 10^{-6} \sin(400t + 110^\circ)}$
11. a. $I_m = \frac{V_m}{X_L} = \frac{120 \text{ V}}{50 \Omega} = 2.4 \text{ A}$, $i = \mathbf{2.4 \sin(\omega t - 90^\circ)}$
- b. $I_m = \frac{V_m}{X_L} = \frac{30 \text{ V}}{50 \Omega} = 0.6 \text{ A}$, $i = \mathbf{0.6 \sin(\omega t - 70^\circ)}$
12. a. $X_L = \omega L = (60 \text{ rad/s})(0.2 \text{ H}) = 12 \Omega$
 $I_m = V_m / X_L = 1.5 \text{ V} / 12 \Omega = 0.125 \text{ A}$
 $i = \mathbf{0.125 \sin(60t - 90^\circ)}$
- b. $X_L = \omega L = (10 \text{ rad/s})(0.2 \text{ H}) = 2 \Omega$
 $I_m = V_m / X_L = 16 \text{ mV} / 2 \Omega = 8 \text{ mA}$
 $i = 8 \times 10^{-3} \sin(t + 2^\circ - 90^\circ) = \mathbf{8 \times 10^{-3} \sin(t - 88^\circ)}$
13. a. $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(0 \text{ Hz})(5 \times 10^{-6} \text{ F})} = \mathbf{\infty \Omega}$
- b. $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(60 \text{ Hz})(5 \times 10^{-6} \text{ F})} = \mathbf{530.79 \Omega}$
- c. $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(2 \text{ kHz})(5 \times 10^{-6} \text{ F})} = \mathbf{15.92 \Omega}$

- d. $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(2 \times 10^6 \text{ Hz})(5 \times 10^{-6} \text{ F})} = \mathbf{62.83 \, \Omega}$
14. $X_C = \frac{1}{2\pi fC} \Rightarrow C = \frac{1}{2\pi fX_C}$
- a. $C = \frac{1}{2\pi(265 \text{ Hz})(60 \, \Omega)} = \mathbf{10 \, \mu F}$
- b. $C = \frac{1}{2\pi(34 \text{ kHz})(1.2 \text{ k}\Omega)} = \mathbf{3900 \text{ pF}}$
15. a. $f = \frac{1}{2\pi CX_C} = \frac{1}{2\pi(3.9 \times 10^{-6} \text{ F})(10 \, \Omega)} = \mathbf{4.08 \text{ kHz}}$
- b. $f = \frac{1}{2\pi CX_C} = \frac{1}{2\pi(3.9 \times 10^{-6} \text{ F})(1.2 \text{ k}\Omega)} = \mathbf{34 \text{ Hz}}$
- c. $f = \frac{1}{2\pi CX_C} = \frac{1}{2\pi(3.9 \times 10^{-6} \text{ F})(0.1 \, \Omega)} = \mathbf{408.1 \text{ kHz}}$
- d. $f = \frac{1}{2\pi CX_C} = \frac{1}{2\pi(3.9 \times 10^{-6} \text{ F})(2000 \, \Omega)} = \mathbf{20.40 \text{ Hz}}$
16. a. $I_m = V_m/X_C = 120 \text{ V}/2.5 \, \Omega = 48 \text{ A}$
 $i = \mathbf{48 \sin(\omega t + 90^\circ)}$
- b. $I_m = V_m/X_C = 4 \times 10^{-3} \text{ V}/2.5 \, \Omega = 0.16 \text{ A}$
 $i = \mathbf{1.6 \times 10^{-3} \sin(\omega t + 130^\circ)}$
17. a. $v = 30 \sin 200t$, $X_C = \frac{1}{\omega C} = \frac{1}{(200)(1 \times 10^{-6} \text{ F})} = 5 \text{ k}\Omega$
 $I_m = \frac{V_m}{X_C} = \frac{30 \text{ V}}{5 \text{ k}\Omega} = 6 \text{ mA}$, $i = \mathbf{6 \times 10^{-3} \sin(200t + 90^\circ)}$
- b. $v = 60 \times 10^{-3} \sin 377t$, $X_C = \frac{1}{\omega C} = \frac{1}{(377)(1 \times 10^{-6})} = 2.65 \text{ k}\Omega$
 $I_m = \frac{V_m}{X_C} = \frac{60 \times 10^{-3} \text{ V}}{2.65 \text{ k}\Omega} = 22.64 \, \mu\text{A}$, $i = \mathbf{22.64 \times 10^{-6} \sin(377t + 90^\circ)}$

18. a. $V_m = I_m X_C = (50 \times 10^{-3} \text{ A})(10 \Omega) = 0.5 \text{ V}$
 $v = \mathbf{0.5 \sin(\omega t - 90^\circ)}$
- b. $V_m = I_m X_C = (2 \times 10^{-6})(10 \Omega) = 20 \mu\text{V}$
 $v = \mathbf{20 \times 10^{-6} \sin(\omega t - 30^\circ)}$
19. a. $i = 0.2 \sin 300t$, $X_C = \frac{1}{\omega C} = \frac{1}{(300)(0.56 \times 10^{-6} \text{ F})} = 5.952 \text{ k}\Omega$
 $V_m = I_m X_C = (0.2 \text{ A})(5.952 \text{ k}\Omega) = 1190.48 \text{ V}$, $v = \mathbf{1190.48 \sin(300t - 90^\circ)}$
- b. $i = 8 \times 10^{-3} \sin(377t - 30^\circ)$, $X_C = \frac{1}{\omega C} = \frac{1}{(377)(0.56 \times 10^{-6} \text{ F})} = 4.737 \text{ k}\Omega$
 $V_m = I_m X_C = (8 \times 10^{-3} \text{ A})(4.737 \text{ k}\Omega) = 37.81 \text{ V}$
 $v = \mathbf{37.81 \sin(377t - 120^\circ)}$
20. a. v leads i by $90^\circ \Rightarrow L$, $X_L = V_m/I_m = 550 \text{ V}/11 \text{ A} = 50 \Omega$
 $L = \frac{X_L}{\omega} = \frac{50 \Omega}{377 \text{ rad/s}} = \mathbf{132.63 \text{ mH}}$
- b. v leads i by $90^\circ \Rightarrow L$, $X_L = V_m/I_m = 36 \text{ V}/4 \text{ A} = 9 \Omega$
 $L = \frac{1}{\omega X_L} = \frac{1}{(754 \text{ rad/s})(9 \Omega)} = \mathbf{147.36 \mu\text{H}}$
- c. v and i are in phase $\Rightarrow R$
 $R = \frac{V_m}{I_m} = \frac{10.5 \text{ V}}{1.5 \text{ A}} = \mathbf{7 \Omega}$
21. a. $\left. \begin{array}{l} i = 5 \sin(\omega t + 90^\circ) \\ v = 2000 \sin \omega t \end{array} \right\} i \text{ leads } v \text{ by } 90^\circ \Rightarrow \mathbf{C}$
 $X_C = \frac{V_m}{I_m} = \frac{2000 \text{ V}}{5 \text{ A}} = 400 \Omega$
- b. $\left. \begin{array}{l} i = 2 \sin(157t + 60^\circ) \\ v = 80 \sin(157t + 150^\circ) \end{array} \right\} v \text{ leads } i \text{ by } 90^\circ \Rightarrow \mathbf{L}$
 $X_L = \frac{V_m}{I_m} = \frac{80 \text{ V}}{2 \text{ A}} = 40 \Omega$, $L = \frac{X_L}{\omega} = \frac{40 \Omega}{157 \text{ rad/s}} = \mathbf{254.78 \text{ mH}}$
- c. $\left. \begin{array}{l} v = 35 \sin(\omega t - 20^\circ) \\ i = 7 \sin(\omega t - 20^\circ) \end{array} \right\} \text{in phase} \Rightarrow \mathbf{R}$
 $R = \frac{V_m}{I_m} = \frac{35 \text{ V}}{7 \text{ A}} = \mathbf{5 \Omega}$
22. —

23. –

$$24. \quad X_C = \frac{1}{2\pi fC} = R \Rightarrow f = \frac{1}{2\pi RC} = \frac{1}{2\pi(2 \times 10^3 \Omega)(1 \times 10^{-6} \text{ F})} = \frac{1}{12.56 \times 10^{-3}} \\ \cong \mathbf{79.62 \text{ Hz}}$$

$$25. \quad X_L = 2\pi fL = R \\ L = \frac{R}{2\pi f} = \frac{10,000 \Omega}{2\pi(5 \times 10^3 \text{ Hz})} = \mathbf{318.47 \text{ mH}}$$

$$26. \quad X_C = X_L \\ \frac{1}{2\pi fC} = 2\pi fL \\ f^2 = \frac{1}{4\pi^2 LC} \\ \text{and } f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(10 \times 10^{-3} \text{ H})(1 \times 10^{-6} \text{ F})}} = \mathbf{1.59 \text{ kHz}}$$

$$27. \quad X_C = X_L \\ \frac{1}{2\pi fC} = 2\pi fL \Rightarrow C = \frac{1}{4\pi^2 f^2 L} = \frac{1}{4(9.86)(2500 \times 10^6)(2 \times 10^{-3})} = \mathbf{5.07 \text{ nF}}$$

$$28. \quad \text{a. } P = \frac{(60 \text{ V})(15 \text{ A})}{2} \cos 30^\circ = \mathbf{389.7 \text{ W}}, F_p = \mathbf{0.866}$$

$$\text{b. } P = \frac{(50 \text{ V})(2 \text{ A})}{2} \cos 0^\circ = \mathbf{50 \text{ W}}, F_p = \mathbf{1.0}$$

$$\text{c. } P = \frac{(50 \text{ V})(3 \text{ A})}{2} \cos 10^\circ = \mathbf{73.86 \text{ W}}, F_p = \mathbf{0.985}$$

$$\text{d. } P = \frac{(75 \text{ V})(0.08 \text{ A})}{2} \cos 40^\circ = \mathbf{2.30 \text{ W}}, F_p = \mathbf{0.766}$$

$$29. \quad R = \frac{V_m}{I_m} = \frac{48 \text{ V}}{8 \text{ A}} = 6 \Omega, P = I^2 R = \left(\frac{8 \text{ A}}{\sqrt{2}}\right)^2 6 \Omega = \mathbf{192 \text{ W}} \\ P = \frac{V_m I_m}{2} \cos \theta = \frac{(48 \text{ V})(8 \text{ A})}{2} \cos 0^\circ = \mathbf{192 \text{ W}} \\ P = VI \cos \theta = \left(\frac{48 \text{ V}}{\sqrt{2}}\right)\left(\frac{8 \text{ A}}{\sqrt{2}}\right) \cos 0^\circ = \mathbf{192 \text{ W}}$$

All the same!

30. $P = 100 \text{ W}$: $F_p = \cos \theta = P/VI = 100 \text{ W}/(150 \text{ V})(2 \text{ A}) = \mathbf{0.333}$
 $P = 0 \text{ W}$: $F_p = \cos \theta = \mathbf{0}$
 $P = 300 \text{ W}$: $F_p = \frac{300}{300} = \mathbf{1}$
31. $P = \frac{V_m I_m}{2} \cos \theta$
 $500 \text{ W} = \frac{(50 \text{ V}) I_m}{2} (0.5) \Rightarrow I_m = 40 \text{ A}$
 $i = \mathbf{40 \sin(\omega t - 50^\circ)}$
32. a. $I_m = E_m/R = 34 \text{ V}/6.8 \Omega = 3.53 \text{ A}$, $i = \mathbf{3.53 \sin(2\pi 60t + 20^\circ)}$
b. $P = I^2 R = \frac{3.53 \text{ A}}{\sqrt{2}}^2 6.8 \Omega = \mathbf{42.38 \text{ W}}$
c. $T = \frac{2\pi}{\omega} = \frac{6.28}{2\pi 60 \text{ rad/s}} = \mathbf{16.67 \text{ ms}}$
 $6(16.67 \text{ ms}) = 100.02 \text{ ms} \cong \mathbf{0.1 \text{ s}}$
33. a. $I_m = \frac{V_m}{X_L} = \frac{128 \text{ V}}{30 \Omega} = 4.27 \text{ A}$, $i = \mathbf{4.27 \sin(1000t - 30^\circ)}$
b. $L = \frac{X_L}{\omega} = \frac{30 \Omega}{1000 \text{ rad/s}} = \mathbf{30 \text{ mH}}$, standard = $\mathbf{30 \text{ mH}}$
c. $L \Rightarrow \mathbf{0 \text{ W}}$
34. a. $E_m = I_m X_C = (30 \times 10^{-3} \text{ A})(2.4 \text{ k}\Omega) = 72 \text{ V}$
 $e = 72 \sin(2\pi 500t - 20^\circ - 90^\circ) = \mathbf{72 \sin(2\pi 500t - 110^\circ)}$
b. $C = \frac{1}{\omega X_C} = \frac{1}{(3.14 \times 10^3 \text{ rad/s})(2.4 \text{ k}\Omega)} = \mathbf{0.133 \mu\text{F}}$ standard = $\mathbf{0.13 \mu\text{F}}$
c. $P = 0 \text{ W}$
35. a. $X_{C_1} = \frac{1}{2\pi f C_1} = \frac{1}{\omega C_1} = \frac{1}{(10^4 \text{ rad/s})(2 \mu\text{F})} = 50 \Omega$
 $X_{C_2} = \frac{1}{\omega C_2} = \frac{1}{(10^4)(10 \mu\text{F})} = 10 \Omega$
 $E = 84.85 \text{ V} \angle 60^\circ$ $I_1 = \frac{E}{Z_{C_1}} = \frac{84.85 \text{ V} \angle 60^\circ}{50 \Omega \angle -90^\circ} = 1.697 \text{ A} \angle 150^\circ$
 $I_2 = \frac{E}{Z_{C_2}} = \frac{84.85 \text{ V} \angle 60^\circ}{10 \Omega \angle -90^\circ} = 8.485 \text{ A} \angle 150^\circ$
 $i_1 = \mathbf{2.4 \sin(10^4 t + 150^\circ)}$
 $i_2 = \mathbf{12 \sin(10^4 t + 150^\circ)}$

- b. $C_T = 2 \mu\text{F} \parallel 10 \mu\text{F} = 12 \mu\text{F}$

$$X_C = \frac{1}{\omega C} = \frac{1}{(10^4 \text{ rad/s})(12 \mu\text{F})}$$

$$= 8.33 \Omega$$

$$\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{X}_{C_T}} = \frac{84.85 \angle 60^\circ}{8.33 \Omega \angle -90^\circ}$$

$$= 10.19 \text{ A} \angle 150^\circ$$

$$i_s = \mathbf{14.4 \sin(10^4 t + 150^\circ)}$$
36. a. $L_1 \parallel L_2 = 60 \text{ mH} \parallel 120 \text{ mH} = 40 \text{ mH}$
 $X_{L_T} = 2\pi f L_T = 2\pi(10^3 \text{ Hz})(40 \text{ mH}) = 251.33 \Omega$
 $V_m = I_m X_{L_T} = (24 \text{ A})(251.33 \Omega) = 6.03 \text{ kV}$
and $v_s = 6.03 \text{ kV} \sin(10^3 t + 30^\circ + 90^\circ)$
or $v_s = \mathbf{6.03 \times 10^3 \sin(10^3 t + 120^\circ)}$
- b. $I_{m_1} = \frac{V_m}{X_{L_1}}, X_{L_1} = 2\pi f L_1 = 2\pi(10^3 \text{ Hz})(60 \text{ mH}) = 376.99 \Omega$

$$I_{m_1} = \frac{6.03 \times 10^3 \text{ V}}{376.99 \Omega} = 16 \text{ A}$$
and $i_1 = \mathbf{16 \sin(10^3 t + 30^\circ)}$
 $X_{L_2} = 2\pi f L_2 = 2\pi(10^3 \text{ Hz})(120 \text{ mH}) = 753.98 \Omega$

$$I_{m_2} = \frac{6.03 \times 10^3 \text{ V}}{753.98 \Omega} = 8 \text{ A}$$
and $i_2 = \mathbf{8 \text{ A} \sin(10^3 t + 30^\circ)}$
37. a. $\mathbf{5.0 \angle 36.87^\circ}$ b. $\mathbf{2.83 \angle 45^\circ}$
c. $\mathbf{12.65 \angle 7.57^\circ}$ d. $\mathbf{1001.25 \angle 2.86^\circ}$
e. $\mathbf{4123.11 \angle 104.04^\circ}$ f. $\mathbf{0.894 \angle 116.57^\circ}$
38. a. $\mathbf{17.89 \angle -116.57^\circ}$ b. $\mathbf{8.94 \angle -26.57^\circ}$
c. $\mathbf{20.22 \times 10^{-3} \angle -8.53^\circ}$ d. $\mathbf{8.49 \times 10^{-3} \angle -135^\circ}$
e. $\mathbf{\cong 200 \angle 0^\circ}$ f. $\mathbf{1000 \angle -178.85^\circ}$
39. a. $\mathbf{4.6 + j3.86}$ b. $\mathbf{-6.0 + j10.39}$
c. $\mathbf{-j2000}$ d. $\mathbf{-6 \times 10^{-3} - j2.2 \times 10^{-3}}$
e. $\mathbf{47.97 + j1.68}$ f. $\mathbf{4.7 \times 10^{-4} - j1.71 \times 10^{-4}}$
40. a. $\mathbf{42 + j0.11}$ b. $\mathbf{1 \times 10^3 - j1.73 \times 10^3}$
c. $\mathbf{-3 \times 10^{-3} - j5.20 \times 10^{-3}}$ d. $\mathbf{-6.13 \times 10^{-3} + j5.14 \times 10^{-3}}$
e. $\mathbf{-15}$ f. $\mathbf{2.09 \times 10^{-3} - j1.20}$

41. a. $11.8 + j7.0$
 b. $151.90 + j49.90$
 c. $4.72 \times 10^{-6} + j71$
42. a. $5.20 + j1.60$
 b. $209.30 + j311.0$
 c. $-21.20 + j12.0$
43. a. $12.17 \angle 54.70^\circ$
 b. $98.37 \angle 13.38^\circ$
 c. $28.07 \angle -115.91^\circ$
44. a. $-12.0 + j34.0$
 b. $86.80 + j312.40$
 c. $-283.90 - j637.65$
45. a. $8.00 \angle 20^\circ$
 b. $49.68 \angle -64.0^\circ$
 c. $40 \times 10^{-3} \angle 40^\circ$
46. a. $6.0 \angle -50^\circ$
 b. $200 \times 10^{-6} \angle 60^\circ$
 c. $109 \angle -170^\circ$
47. a. 4
 b. $-4.15 - j4.23$
 c. $6.69 - j6.46$
48. a. $\frac{10 - j5}{1 + j0} = 10.0 - j5.0$
 b. $\frac{8 \angle 60^\circ}{102 + j400} = \frac{8 \angle 60^\circ}{412.80 \angle 75.69^\circ} = 19.38 \times 10^{-3} \angle -15.69^\circ$
 c. $\frac{(6 \angle 20^\circ)(120 \angle -40^\circ)(8.54 \angle 69.44^\circ)}{2 \angle -30^\circ} = \frac{6.15 \times 10^3 \angle 49.44^\circ}{2 \angle -30^\circ} = 3.07 \times 10^3 \angle 79.44^\circ$

$$49 \quad a. \quad \frac{(0.16 \angle 120^\circ)(300 \angle 40^\circ)}{9.487 \angle 71.565^\circ} = \frac{48 \angle 160^\circ}{9.487 \angle 71.565^\circ} = \mathbf{5.06 \angle 88.44^\circ}$$

$$b. \quad \left(\frac{1}{4 \times 10^{-4} \angle 20^\circ} \right) \left(\frac{8}{j(j^2)} \right) \left(\frac{1}{36 - j30} \right)$$

$$(2500 \angle -20^\circ) \left(\frac{8}{-j} \right) \left(\frac{1}{46.861 \angle -39.81^\circ} \right)$$

$$(2500 \angle -20^\circ)(8j)(0.0213 \angle 39.81^\circ) = \mathbf{426 \angle 109.81^\circ}$$

$$50. \quad a. \quad \begin{array}{l} x + j4 + 3x + jy - j7 = 16 \\ (x + 3x) + j(4 + y - 7) = 16 + j0 \\ x + 3x = 16 \qquad \qquad 4 + y - 7 = 0 \\ 4x = 16 \qquad \qquad y = +7 - 4 \\ x = \mathbf{4} \qquad \qquad y = \mathbf{3} \end{array}$$

$$b. \quad \begin{array}{l} (10 \angle 20^\circ)(x \angle -60^\circ) = 30.64 - j25.72 \\ 10x \angle -40^\circ = 40 \angle -40^\circ \\ 10x = 40 \\ x = \mathbf{4} \end{array}$$

$$51. \quad a. \quad \frac{5x + j10}{2 - jy}$$

$$\begin{array}{l} 10x + j20 - j5xy - j^2 10y = 90 - j70 \\ (10x + 10y) + j(20 - 5xy) = 90 - j70 \\ 10x + 10y = 90 \\ x + y = 9 \\ x = 9 - y \Rightarrow \end{array}$$

$$\begin{array}{l} 20 - 5xy = -70 \\ 20 - 5(9 - y)y = -70 \\ 5y(9 - y) = 90 \\ y^2 - 9y + 18 = 0 \\ y = \frac{-(-9) \pm \sqrt{(-9)^2 - 4(1)(18)}}{2} \\ y = \frac{9 \pm 3}{2} = 6, 3 \end{array}$$

$$\begin{array}{l} \text{For } y = 6, x = 3 \\ y = 3, x = 6 \\ \mathbf{(x = 3, y = 6) \text{ or } (x = 6, y = 3)} \end{array}$$

$$b. \quad \frac{80 \angle 0^\circ}{40 \angle \theta} = 4 \angle -\theta = 3.464 - j2 = 4 \angle -30^\circ$$

$$\theta = \mathbf{30^\circ}$$

$$52. \quad \begin{array}{l} a. \quad \mathbf{160.0 \angle 30^\circ} \\ b. \quad \mathbf{25 \times 10^{-3} \angle -40^\circ} \\ c. \quad \mathbf{70.71 \angle -90^\circ} \end{array}$$

53. a. $14.14 \angle -180^\circ$
 b. $4.24 \times 10^{-6} \angle 90^\circ$
 c. $2.55 \times 10^{-6} \angle 70^\circ$
54. a. $56.57 \sin(377t + 20^\circ)$ b. $169.68 \sin(377t + 10^\circ)$
 c. $11.31 \times 10^{-3} \sin(377t - 110^\circ)$ d. $6000 \sin(377t - 180^\circ)$
55. (Using peak values)

$$e_{in} = v_a + v_b \Rightarrow v_a = e_{in} - v_b$$

$$= 60 \text{ V } \angle 45^\circ - 20 \text{ V } \angle -45^\circ$$

$$= 63.25 \text{ V } \angle 63.43^\circ$$
and $e_{in} = 63.25 \sin(377t + 63.43^\circ)$
56. $i_s = i_1 + i_2 \Rightarrow i_1 = i_s - i_2$
 (Using peak values) $= (20 \times 10^{-6} \text{ A } \angle 60^\circ) - (6 \times 10^{-6} \text{ A } \angle -30^\circ) = 20.88 \times 10^{-6} \text{ A } \angle 76.70^\circ$
 $i_1 = 20.88 \times 10^{-6} \sin(\omega t + 76.70^\circ)$
57. $e_{in} = v_a + v_b + v_c$
 $v_a = e_{in} - v_b - v_c$
 $= 120 \text{ V } \angle 30^\circ - 30 \text{ V } \angle 60^\circ - 40 \text{ V } \angle 120^\circ$
 $= 108.92 \text{ V } \angle -0.33^\circ$
 $e_{in} = 108.92 \sin(377t - 0.33^\circ)$
58. $\mathbf{I}_s = \mathbf{I}_1 + \mathbf{I}_2 + \mathbf{I}_3$
 $\mathbf{I}_1 = \mathbf{I}_s - \mathbf{I}_2 - \mathbf{I}_3$
 $= 12.73 \text{ A } \angle 180^\circ - 5.66 \text{ A } \angle -180^\circ - 2[5.66 \text{ A } \angle -180^\circ]$
 $= 12.73 \text{ A } \angle 180^\circ - 5.66 \text{ A } \angle -180^\circ - 11.32 \text{ A } \angle 180^\circ$
 $= 4.25 \text{ A } \angle 0^\circ$
 $i_1 = 6.01 \sin 377t$

Chapter 15

1.
 - a. $R \angle 0^\circ = 6.8 \Omega \angle 0^\circ = 6.8 \Omega$
 - b. $X_L = \omega L = (377 \text{ rad/s})(1.2 \text{ H}) = 452.4 \Omega$
 $X_L \angle 90^\circ = 452.4 \Omega \angle 90^\circ = +j452.4 \Omega$
 - c. $X_L = 2\pi fL = (6.28)(50 \text{ Hz})(47 \text{ mH}) = 1.48 \Omega$
 $X_L \angle 90^\circ = 1.48 \Omega \angle 90^\circ = +j1.48 \Omega$
 - d. $X_C = \frac{1}{\omega C} = \frac{1}{(100 \text{ rad/s})(10 \times 10^{-6} \text{ F})} = 1 \text{ k}\Omega$
 $X_C \angle -90^\circ = 1 \text{ k}\Omega \angle -90^\circ = -j1 \text{ k}\Omega$
 - e. $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(10 \times 10^3 \text{ Hz})(0.47 \mu\text{F})} = 33.86 \Omega$
 $X_C \angle -90^\circ = 33.86 \Omega \angle -90^\circ = -j33.86 \Omega$
 - f. $R \angle 0^\circ = 220 \Omega \angle 0^\circ = 220 \Omega$
2.
 - a. $V = 10.61 \text{ V} \angle 10^\circ, I = \frac{V \angle \theta}{R \angle 0^\circ} = \frac{10.61 \text{ V} \angle 10^\circ}{3 \Omega \angle 0^\circ} = 3.54 \text{ A} \angle 10^\circ$
 $i = 5 \sin(\omega t + 10^\circ)$
 - b. $V = 4.24 \text{ V} \angle 10^\circ, I = \frac{V \angle \theta}{X_L \angle 90^\circ} = \frac{4.24 \text{ V} \angle 10^\circ}{1 \Omega \angle 90^\circ} = 4.24 \text{ A} \angle -80^\circ$
 $i = 6 \sin(\omega t - 80^\circ)$
 - c. $V = 84.84 \text{ V} \angle 0^\circ, X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(5 \text{ kHz})(2 \mu\text{F})} = 15.924 \Omega$
 $I = \frac{V \angle \theta}{X_C \angle -90^\circ} = \frac{84.84 \text{ V} \angle 0^\circ}{15.924 \Omega \angle -90^\circ} = 5.328 \text{ A} \angle 90^\circ$
 $i = 7.534 \sin(\omega t + 90^\circ)$
3.
 - a. $I = (0.707)(4 \text{ mA} \angle 0^\circ) = 2.828 \text{ mA} \angle 0^\circ$
 $V = (I \angle 0^\circ)(R \angle 0^\circ) = 2.828 \text{ mA} \angle 0^\circ(22 \Omega \angle 0^\circ) = 62.216 \text{ mV} \angle 0^\circ$
 $v = 88 \times 10^{-3} \sin 1000t$
 - b. $I = (0.707)(1.5 \text{ A} \angle 60^\circ) = 1.061 \text{ A} \angle 60^\circ$
 $X_L = 2\pi fL = 2\pi(200 \text{ Hz})(12 \text{ mH}) = 15.08 \Omega$
 $V = (I \angle \theta)(X_L \angle 90^\circ) = (1.061 \text{ A} \angle 60^\circ)(15.08 \Omega \angle 90^\circ) = 16 \text{ V} \angle 150^\circ$
 $v = 22.62 \sin(2\pi 200t + 150^\circ)$

- c. $\mathbf{I} = (0.707)(2 \text{ mA} \angle 40^\circ) = 1.414 \text{ mA} \angle 40^\circ$

$$X_C = \frac{1}{\omega C} = \frac{1}{(157 \text{ rad/s})(0.047 \mu\text{F})} = 135.52 \text{ k}\Omega$$

$$\mathbf{V} = (I \angle \theta)(X_C \angle -90^\circ) = (1.414 \text{ mA} \angle 40^\circ)(135.52 \text{ k}\Omega \angle -90^\circ) = 191.63 \text{ V} \angle -50^\circ$$

$$V_p = \sqrt{2}(191.63 \text{ V}) = 270.96 \text{ V}$$

and $v = 270.96 \sin(157t - 50^\circ)$
4. a. $\mathbf{Z}_T = 6.8 \Omega + j8.2 \Omega = 10.65 \Omega \angle 50.33^\circ$
b. $\mathbf{Z}_T = 2 \Omega - j6 \Omega + 10 \Omega = 12 \Omega - j6 \Omega = 13.42 \Omega \angle -26.57^\circ$
c. $\mathbf{Z}_T = 1 \text{ k}\Omega + j3.2 \text{ k}\Omega + 5.6 \text{ k}\Omega + j6.8 \text{ k}\Omega = 6.6 \text{ k}\Omega + j10 \text{ k}\Omega = 11.98 \text{ k}\Omega \angle 56.58^\circ$
5. a. $\mathbf{Z}_T = 3 \Omega + j4 \Omega - j5 \Omega = 3 \Omega - j1 \Omega = 3.16 \Omega \angle -18.43^\circ$
b. $\mathbf{Z}_T = 1 \text{ k}\Omega + j8 \text{ k}\Omega - j4 \text{ k}\Omega = 1 \text{ k}\Omega + j4 \text{ k}\Omega = 4.12 \text{ k}\Omega \angle 75.96^\circ$
c. $L_T = 247 \text{ mH}$
 $X_L = \omega L = 2\pi f L = 2\pi(10^3 \text{ Hz})(247 \times 10^{-3} \text{ H}) = 1.55 \text{ k}\Omega$

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(10^3 \text{ Hz})(0.1 \times 10^{-6} \text{ F})} = 1.59 \text{ k}\Omega$$

$$= 470 \Omega + j1.55 \text{ k}\Omega - j1.59 \text{ k}\Omega$$

$$= 470 \Omega - j40 \Omega = 471.70 \Omega \angle -4.86^\circ$$
6. a. $\mathbf{Z}_T = \frac{\mathbf{E}}{\mathbf{I}} = \frac{120 \text{ V} \angle 0^\circ}{60 \text{ A} \angle 70^\circ} = 2 \Omega \angle -70^\circ = 0.684 \Omega - j1.879 \Omega = R - jX_C$
b. $\mathbf{Z}_T = \frac{\mathbf{E}}{\mathbf{I}} = \frac{80 \text{ V} \angle 320^\circ}{20 \text{ mA} \angle 40^\circ} = 4 \text{ k}\Omega \angle 280^\circ = 4 \text{ k}\Omega \angle -80^\circ = 0.695 \text{ k}\Omega - j3.939 \Omega$

$$= R - jX_C$$

c. $\mathbf{Z}_T = \frac{\mathbf{E}}{\mathbf{I}} = \frac{8 \text{ kV} \angle 0^\circ}{0.2 \text{ A} \angle -60^\circ} = 40 \text{ k}\Omega \angle 60^\circ = 20 \text{ k}\Omega + j34.64 \text{ k}\Omega = R + jX_L$
7. a. $\mathbf{Z}_T = 8 \Omega + j6 \Omega = 10 \Omega \angle 36.87^\circ$
c. $\mathbf{I} = \mathbf{E}/\mathbf{Z}_T = 100 \text{ V} \angle 0^\circ / 10 \Omega \angle 36.87^\circ = 10 \text{ A} \angle -36.87^\circ$
 $\mathbf{V}_R = (I \angle \theta)(R \angle 0^\circ) = (10 \text{ A} \angle -36.87^\circ)(8 \Omega \angle 0^\circ) = 80 \text{ V} \angle -36.87^\circ$
 $\mathbf{V}_L = (I \angle \theta)(X_L \angle 90^\circ) = (10 \text{ A} \angle -36.87^\circ)(6 \Omega \angle 90^\circ) = 60 \text{ V} \angle 53.13^\circ$
f. $P = I^2 R = (10 \text{ A})^2 8 \Omega = 800 \text{ W}$
g. $F_p = \cos \theta_T = R/Z_T = 8 \Omega / 10 \Omega = 0.8 \text{ lagging}$
h. $v_R = 113.12 \sin(\omega t - 36.87^\circ)$
 $v_L = 84.84 \sin(\omega t + 53.13^\circ)$
 $i = 14.14 \sin(\omega t - 36.87^\circ)$

8. a. $\mathbf{Z}_T = 18 \Omega - j29.15 \Omega = \mathbf{34.26 \Omega \angle -58.30^\circ}$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(60 \text{ Hz})(91 \mu\text{F})} = 29.15 \Omega$$
- c. $\mathbf{I} = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{120 \text{ V} \angle 20^\circ}{34.26 \Omega \angle -58.30^\circ} = \mathbf{3.50 \text{ A} \angle 78.30^\circ}$

$$\mathbf{V}_R = (I \angle \theta)(R \angle 0^\circ) = (3.50 \text{ A} \angle 78.30^\circ)(18 \Omega \angle 0^\circ) = \mathbf{63.0 \text{ V} \angle 78.30^\circ}$$

$$\mathbf{V}_C = (I \angle \theta)(X_C \angle -90^\circ) = (3.50 \text{ A} \angle 78.30^\circ)(29.15 \Omega \angle -90^\circ) = \mathbf{102.03 \text{ V} \angle -11.70^\circ}$$
- f. $P = I^2 R = (3.50 \text{ A})^2 18 \Omega = \mathbf{220.5 \text{ W}}$
- g. $F_p = R/Z_T = 18 \Omega / 34.26 \Omega = \mathbf{0.525 \text{ leading}}$
- h. $i = \mathbf{4.95 \sin(377t + 78.30^\circ)}$
 $v_R = \mathbf{89.1 \sin(377t + 78.30^\circ)}$
 $v_C = \mathbf{144.27 \sin(377t - 11.70^\circ)}$
9. a. $\mathbf{Z}_T = 4 \Omega + j6 \Omega - j10 \Omega = 4 \Omega - j4 \Omega = \mathbf{5.66 \Omega \angle -45^\circ}$
- c. $X_L = \omega L \Rightarrow L = \frac{X_L}{\omega} = \frac{6 \Omega}{377 \text{ rad/s}} = \mathbf{16 \text{ mH}}$

$$X_C = \frac{1}{\omega C} \Rightarrow C = \frac{1}{\omega X_C} = \frac{1}{(377 \text{ rad/s})(10 \Omega)} = \mathbf{265 \mu\text{F}}$$
- d. $\mathbf{I} = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{50 \text{ V} \angle 0^\circ}{5.66 \Omega \angle -45^\circ} = \mathbf{8.83 \text{ A} \angle 45^\circ}$

$$\mathbf{V}_R = (I \angle \theta)(R \angle 0^\circ) = (8.83 \text{ A} \angle 45^\circ)(4 \Omega \angle 0^\circ) = \mathbf{35.32 \text{ V} \angle 45^\circ}$$

$$\mathbf{V}_L = (I \angle \theta)(X_L \angle 90^\circ) = (8.83 \text{ A} \angle 45^\circ)(6 \Omega \angle 90^\circ) = \mathbf{52.98 \text{ V} \angle 135^\circ}$$

$$\mathbf{V}_C = (I \angle \theta)(X_C \angle -90^\circ) = (8.83 \text{ A} \angle 45^\circ)(10 \Omega \angle -90^\circ) = \mathbf{88.30 \text{ V} \angle -45^\circ}$$
- f. $\mathbf{E} = \mathbf{V}_R + \mathbf{V}_L + \mathbf{V}_C$

$$50 \text{ V} \angle 0^\circ = 35.32 \text{ V} \angle 45^\circ + 52.98 \text{ V} \angle 135^\circ + 88.30 \text{ V} \angle -45^\circ$$

$$50 \text{ V} \angle 0^\circ = 49.95 \text{ V} \angle 0^\circ \cong 50 \text{ V} \angle 0^\circ$$
- g. $P = I^2 R = (8.83 \text{ A})^2 4 \Omega = \mathbf{311.88 \text{ W}}$
- h. $F_p = \cos \theta_T = \frac{R}{Z_T} = 4 \Omega / 5.66 \Omega = \mathbf{0.707 \text{ leading}}$
- i. $i = \mathbf{12.49 \sin(377t + 45^\circ)}$
 $e = \mathbf{70.7 \sin 377t}$
 $v_R = \mathbf{49.94 \sin(377t + 45^\circ)}$
 $v_L = \mathbf{74.91 \sin(377t + 135^\circ)}$
 $v_C = \mathbf{124.86 \sin(377t - 45^\circ)}$

10. a. $X_L = \omega L = (20 \times 10^3 \text{ rad/s})(0.1 \text{ H}) = 2 \text{ k}\Omega$
 $X_C = \frac{1}{\omega C} = \frac{1}{(20 \times 10^3 \text{ rad/s})(8200 \text{ pF})} = 6.1 \text{ k}\Omega$
 $Z_T = 1.2 \text{ k}\Omega + j2 \text{ k}\Omega - j6.1 \text{ k}\Omega$
 $= \mathbf{1.2 \text{ k}\Omega - j4.1 \text{ k}\Omega = 4.27 \text{ k}\Omega \angle -73.69^\circ}$
- b. —
- c. —
- d. $\mathbf{I = \frac{E}{Z_T} = \frac{4.24 \text{ V} \angle 60^\circ}{4.27 \text{ k}\Omega \angle -73.69^\circ} = 0.993 \text{ mA} \angle 133.69^\circ}$
 $\mathbf{V_R = IR = (0.993 \text{ mA} \angle 133.69^\circ)(1.2 \text{ k}\Omega \angle 0^\circ) = 1.19 \text{ V} \angle 133.69^\circ}$
 $\mathbf{V_L = IX_L = (0.993 \text{ mA} \angle 133.69^\circ)(2 \text{ k}\Omega \angle 90^\circ) = 1.99 \text{ V} \angle 223.69^\circ}$
 $\mathbf{V_C = IX_C = (0.993 \text{ mA} \angle 133.69^\circ)(6.1 \text{ k}\Omega \angle -90^\circ) = 6.06 \text{ V} \angle 43.69^\circ}$
- e. —
- f. $\mathbf{E = V_R + V_L + V_C}$
 $4.24 \text{ V} \angle 60^\circ = 1.19 \text{ V} \angle 133.69^\circ + 1.99 \text{ V} \angle 223.69^\circ + 6.06 \text{ V} \angle 43.69^\circ$
 $= (-0.822 \text{ V} + j0.80 \text{ V}) + (-1.44 \text{ V} - j1.37 \text{ V}) + (4.38 \text{ V} + j 4.19 \text{ V})$
 $= 2.12 \text{ V} + j3.62 \text{ V}$
 $\mathbf{4.24 \text{ V} \angle 60^\circ \cong 4.20 \text{ V} \angle 59.65^\circ}$
- g. $\mathbf{P = I^2 R = (0.993 \text{ mA})^2(1.2 \text{ k}\Omega) = 1.18 \text{ mW}}$
- h. $\mathbf{F_p = \frac{R}{Z_T} = \frac{1.2 \text{ k}\Omega}{4.27 \text{ k}\Omega} = 0.281 \text{ leading}}$
- i. $\mathbf{i = 1.4 \times 10^{-3} \sin(20,000t + 133.69^\circ)}$
 $\mathbf{v_R = 1.68 \sin(20,000t + 133.69^\circ)}$
 $\mathbf{v_L = 2.81 \sin(20,000t + 223.69^\circ)}$
 $\mathbf{v_C = 8.57 \sin(20,000t + 43.69^\circ)}$
11. $20 \text{ V (rms)} \Rightarrow 28.28 \text{ V (peak)}$
 $43.20 \text{ V}(p-p) \Rightarrow 21.60 \text{ V (peak)}$
- $\mathbf{V_{\text{scope}} = 21.60 \text{ V} = \frac{22 \text{ }\Omega(28.28 \text{ V})}{22 \text{ }\Omega + R}}$
 $475.20 + 21.60R = 622.16$
 $\mathbf{R = \frac{146.96 \text{ }\Omega}{21.60} = 6.8 \text{ }\Omega}$

12. a. $V_L(\text{rms}) = 0.7071 \left(\frac{22.8 \text{ V}}{2} \right) = 8.06 \text{ V}$
 $X_L = \frac{V_L(\text{rms})}{I(\text{rms})} = \frac{8.06 \text{ V}}{1.3 \text{ mA}} = 6.2 \text{ k}\Omega$
 $X_L = \omega L = (1000 \text{ rad/s})L = 6.2 \text{ k}\Omega \Rightarrow L = \frac{6.2 \text{ k}\Omega}{1000 \text{ rad/s}} = \mathbf{6.2 \text{ H}}$
- b. $E^2 = V_R^2 + V_L^2$
 $(22 \text{ V})^2 = V_R^2 + (8.06 \text{ V})^2$
 $484 = V_R^2 + 64.96$
 $V_R^2 = 419.04$
 $V_R = \sqrt{419.04} = 20.47 \text{ V}$
 $R = \frac{V_R(\text{rms})}{I(\text{rms})} = \frac{20.47 \text{ V}}{1.3 \text{ mA}} = \mathbf{15.75 \text{ k}\Omega}$
- c. $\mathbf{6.2 \text{ H}}$
13. a. $V_R(\text{rms}) = 0.7071 \left(\frac{8.27 \text{ V}}{2} \right) = 2.924 \text{ V}$
 $I(\text{rms}) = \frac{V_R(\text{rms})}{R_2} = \frac{2.924 \text{ V}}{10 \text{ k}\Omega} = \mathbf{292.4 \mu\text{A}}$
- b. $E^2 = V_R^2 + V_C^2$
 $(12 \text{ V})^2 = (2.924 \text{ V})^2 + V_C^2$
 $144 = 8.55 + V_C^2$
 $V_C^2 = 135.35$
 $V_C = \sqrt{135.45} = 11.64 \text{ V}$
 $X_C = \frac{V_C(\text{rms})}{I(\text{rms})} = \frac{11.64 \text{ V}}{292.4 \mu\text{A}} = 39.81 \text{ k}\Omega$
 $X_C = \frac{1}{2\pi fC} \Rightarrow C = \frac{1}{2\pi fX_C} = \frac{1}{2\pi(40 \text{ kHz})(39.81 \text{ k}\Omega)} = \mathbf{100 \text{ pF}}$
14. a. $V_1 = \frac{(2 \text{ k}\Omega \angle 0^\circ)(120 \text{ V} \angle 60^\circ)}{2 \text{ k}\Omega + j8 \text{ k}\Omega} = \frac{240 \text{ V} \angle 60^\circ}{8.25 \angle 75.96^\circ} = \mathbf{29.09 \text{ V} \angle -15.96^\circ}$
 $V_2 = \frac{(8 \text{ k}\Omega \angle 90^\circ)(120 \text{ V} \angle 60^\circ)}{8.25 \text{ k}\Omega \angle 75.96^\circ} = \mathbf{116.36 \text{ V} \angle 74.04^\circ}$
- b. $V_1 = \frac{(40 \Omega \angle 90^\circ)(60 \text{ V} \angle 5^\circ)}{6.8 \Omega + j40 \Omega + 22 \Omega} = \frac{2400 \text{ V} \angle 95^\circ}{28.8 + j40} = \mathbf{48.69 \text{ V} \angle 40.75^\circ}$
 $V_2 = \frac{(22 \Omega \angle 0^\circ)(60 \text{ V} \angle 5^\circ)}{49.29 \Omega \angle 54.25^\circ} = \frac{1.32 \text{ kV} \angle 5^\circ}{49.29 \Omega \angle 54.25^\circ} = \mathbf{26.78 \text{ V} \angle -49.25^\circ}$

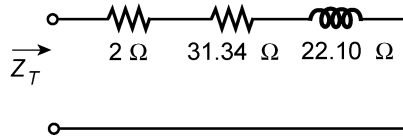
15. a.
$$\mathbf{V}_1 = \frac{(20 \Omega \angle 90^\circ)(20 \text{ V} \angle 70^\circ)}{20 \Omega + j20 \Omega - j40 \Omega} = \mathbf{14.14 \text{ V} \angle -155^\circ}$$
$$\mathbf{V}_2 = \frac{(40 \Omega \angle -90^\circ)(20 \text{ V} \angle 70^\circ)}{28.28 \Omega \angle -45^\circ} = \mathbf{28.29 \text{ V} \angle 25^\circ}$$
- b.
$$\mathbf{Z}_T = 4.7 \text{ k}\Omega + j30 \text{ k}\Omega + 3.3 \text{ k}\Omega - j10 \text{ k}\Omega = 8 \text{ k}\Omega + j20 \text{ k}\Omega = 21.541 \text{ k}\Omega \angle 68.199^\circ$$
$$\mathbf{Z}'_T = 3.3 \text{ k}\Omega + j30 \text{ k}\Omega - j10 \text{ k}\Omega = 3.3 \text{ k}\Omega + j20 \text{ k}\Omega = 20.27 \text{ k}\Omega \angle 80.631^\circ$$
$$\mathbf{V}_1 = \frac{\mathbf{Z}'_T \mathbf{E}}{\mathbf{Z}_T} = \frac{(20.27 \text{ k}\Omega \angle 80.631^\circ)(120 \text{ V} \angle 0^\circ)}{21.541 \text{ k}\Omega \angle 68.199^\circ} = \mathbf{112.92 \text{ V} \angle 12.432^\circ}$$
$$\mathbf{V}_2 = \frac{\mathbf{Z}''_T \mathbf{E}}{\mathbf{Z}_T} \quad \mathbf{Z}''_T = 3.3 \text{ k}\Omega - j10 \text{ k}\Omega = 10.53 \text{ k}\Omega \angle -71.737^\circ$$
$$= \frac{(10.53 \text{ k}\Omega \angle -71.737^\circ)(120 \text{ V} \angle 0^\circ)}{21.541 \text{ k}\Omega \angle 68.199^\circ} = \mathbf{58.66 \text{ V} \angle -139.94^\circ}$$
16. a.
$$X_L = \omega L = (1000 \text{ rad/s})(20 \text{ mH}) = 20 \Omega$$
$$X_C = \frac{1}{\omega C} = \frac{1}{(1000 \text{ rad/s})(39 \mu\text{F})} = 25.64 \Omega$$
$$\mathbf{Z}_T = 30 \Omega + j20 \Omega - j25.64 \Omega = 30 \Omega - j5.64 \Omega = \mathbf{30.53 \Omega \angle -10.65^\circ}$$
$$\mathbf{I} = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{20 \text{ V} \angle 40^\circ}{30.53 \Omega \angle -10.65^\circ} = \mathbf{655.1 \text{ mA} \angle 50.65^\circ}$$
$$\mathbf{V}_R = (\mathbf{I} \angle \theta)(R \angle 0^\circ) = (655.1 \text{ mA} \angle 50.65^\circ)(30 \Omega \angle 0^\circ) = \mathbf{19.65 \text{ V} \angle 50.65^\circ}$$
$$\mathbf{V}_C = (655.1 \text{ mA} \angle 50.65^\circ)(25.64 \Omega \angle -90^\circ) = \mathbf{16.80 \text{ V} \angle -39.35^\circ}$$
- b.
$$\cos \theta_T = \frac{R}{Z_T} = \frac{30 \Omega}{30.53 \Omega} = \mathbf{0.983 \text{ leading}}$$
- c.
$$P = I^2 R = (655.1 \text{ mA})^2 30 \Omega = \mathbf{12.87 \text{ W}}$$
- f.
$$\mathbf{V}_R = \frac{(30 \Omega \angle 0^\circ)(20 \text{ V} \angle 40^\circ)}{\mathbf{Z}_T} = \frac{600 \text{ V} \angle 40^\circ}{30.53 \Omega \angle -10.65^\circ} = \mathbf{19.66 \text{ V} \angle 50.65^\circ}$$
$$\mathbf{V}_C = \frac{(25.64 \Omega \angle -90^\circ)(20 \text{ V} \angle 40^\circ)}{30.53 \Omega \angle -10.65^\circ} = \mathbf{16.80 \text{ V} \angle -39.35^\circ}$$
- g.
$$\mathbf{Z}_T = \mathbf{30 \Omega - j5.64 \Omega} = R - jX_C$$
17.
$$P = VI \cos \theta \Rightarrow 8000 \text{ W} = (200 \text{ V})(I)(0.8)$$
$$I = \frac{8000 \text{ A}}{160} = 50 \text{ A}$$
$$0.8 = \cos \theta$$
$$\theta = 36.87^\circ$$
$$\mathbf{V} = 200 \text{ V} \angle 0^\circ, \mathbf{I} = 50 \text{ A} \angle -36.87^\circ$$
$$\mathbf{Z}_T = \frac{\mathbf{V}}{\mathbf{I}} = \frac{200 \text{ V} \angle 0^\circ}{50 \text{ A} \angle -36.87^\circ} = 4 \Omega \angle 36.87^\circ = \mathbf{3.2 \Omega + j2.4 \Omega}$$

18. $P = VI \cos \theta \Rightarrow 300 \text{ W} = (120 \text{ V})(3 \text{ A}) \cos \theta$
 $\cos \theta = 0.833 \Rightarrow \theta = 33.59^\circ$

$\mathbf{V} = 120 \text{ V} \angle 0^\circ, \mathbf{I} = 3 \text{ A} \angle -33.59^\circ$

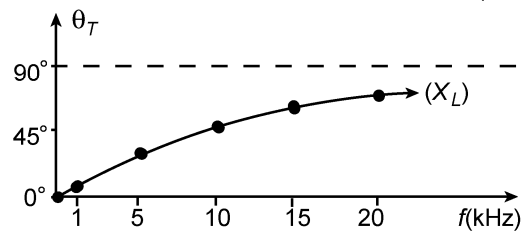
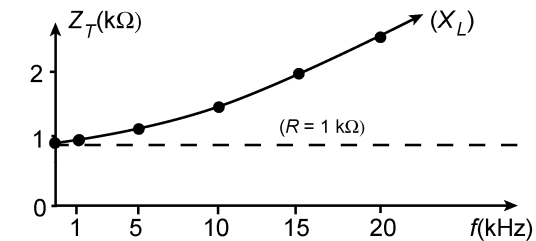
$\mathbf{Z}_T = \frac{\mathbf{V}}{\mathbf{I}} = \frac{120 \text{ V} \angle 0^\circ}{3 \text{ A} \angle -33.59^\circ} = 40 \Omega \angle 33.59^\circ = 33.34 \Omega + j22.10 \Omega$

$R_T = 33.34 \Omega = 2 \Omega + R \Rightarrow R = 31.34 \Omega$



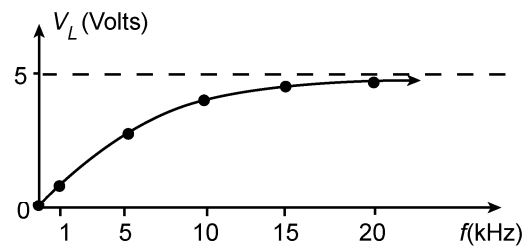
19. a. $\mathbf{Z}_T = \sqrt{R^2 + X_L^2} \angle \tan^{-1} X_L/R$

f	Z_T	θ_T
0 Hz	1.0 k Ω	0.0°
1 kHz	1.008 k Ω	7.16°
5 kHz	1.181 k Ω	32.14°
10 kHz	1.606 k Ω	51.49°
15 kHz	2.134 k Ω	62.05°
20 kHz	2.705 k Ω	68.3°



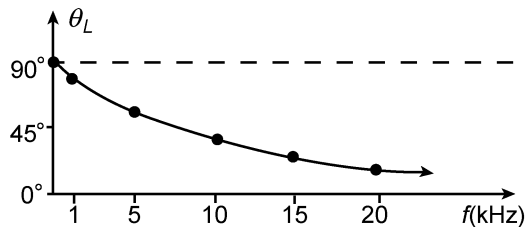
b. $V_L = \frac{X_L E}{Z_T}$

f	V_L
0 Hz	0.0 V
1 kHz	0.623 V
5 kHz	2.66 V
10 kHz	3.888 V
15 kHz	4.416 V
20 kHz	4.646 V



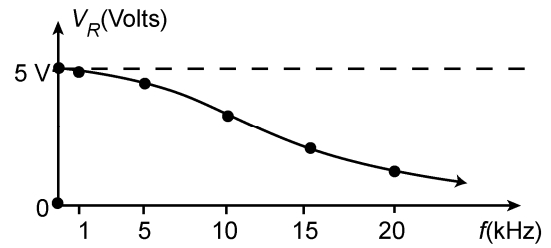
c.

f	$\theta_L = 90^\circ - \tan^{-1} X_L/R$
0 Hz	90.0°
1 kHz	82.84°
5 kHz	57.85°
10 kHz	38.5°
15 kHz	27.96°
20 kHz	21.7°



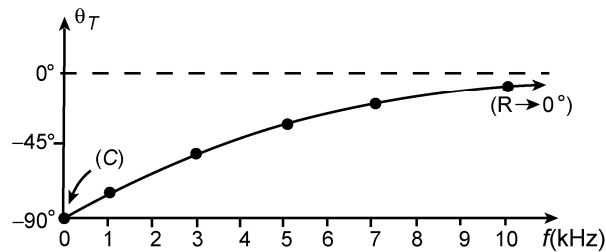
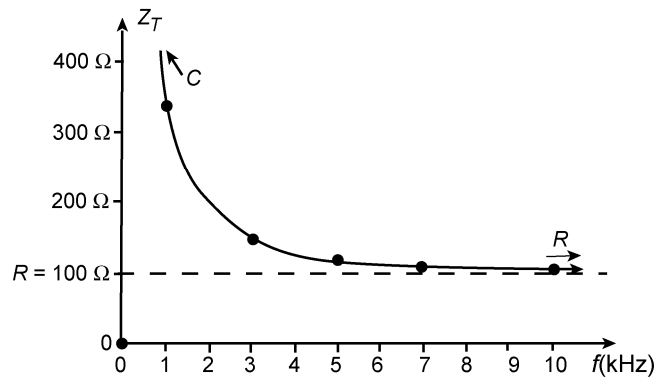
d.

f	$V_R = RE/Z_T$
0 Hz	5.0 V
1 kHz	4.96 V
5 kHz	4.23 V
10 kHz	3.11 V
15 kHz	2.34 V
20 kHz	1.848 V



20. a. $Z_T = \sqrt{R^2 + X_C^2} \angle -\tan^{-1} X_C/R$
 $|Z_T| = \sqrt{R^2 + X_C^2}$, $\theta_T = -\tan^{-1} X_C/R$

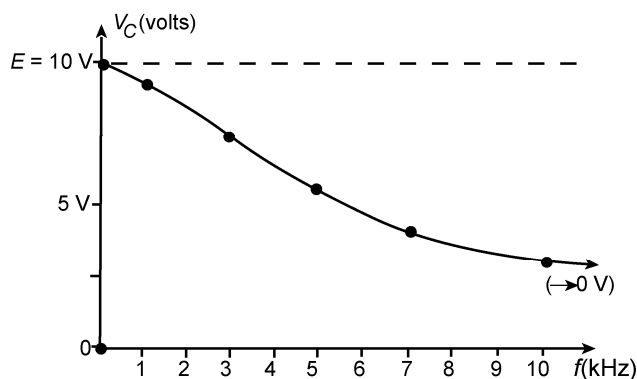
f	$ Z_T $	θ_T
0 kHz	$\infty \Omega$	-90.0°
1 kHz	353.1Ω	-73.55°
3 kHz	150.80Ω	-48.46°
5 kHz	120.78Ω	-34.11°
7 kHz	111.09Ω	-25.82°
10 kHz	105.58Ω	-18.71°



$$\mathbf{V}_C = \frac{(X_C \angle -90^\circ)(E \angle 0^\circ)}{R - jX_C} = \frac{X_C E}{\sqrt{R^2 + X_C^2}} \angle -90^\circ + \tan^{-1} X_C/R$$

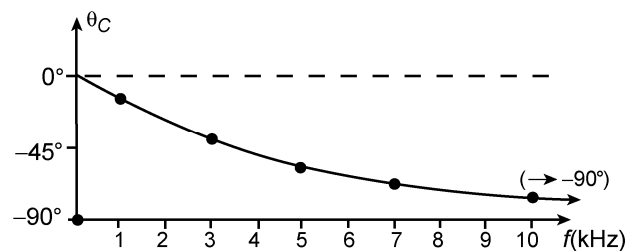
$$|V_C| = \frac{X_C E}{\sqrt{R^2 + X_C^2}}$$

f	$ V_C $
0 Hz	10.0 V
1 kHz	9.59 V
3 kHz	7.49 V
5 kHz	5.61 V
7 kHz	4.36 V
10 kHz	3.21 V



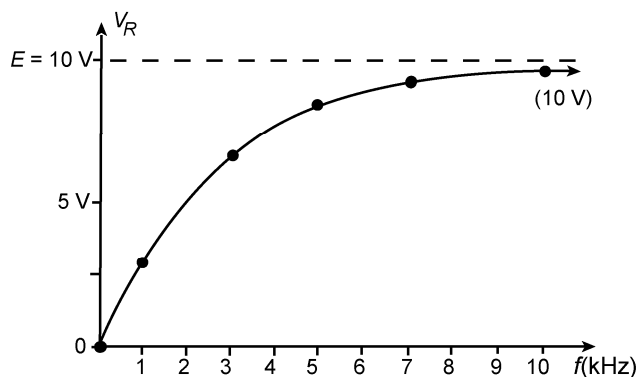
$$\theta_C = -90^\circ + \tan^{-1} X_C/R$$

f	θ_C
0 Hz	0.0°
1 kHz	-16.45°
3 kHz	-41.54°
5 kHz	-55.89°
7 kHz	-64.18°
10 kHz	-71.29°



$$|V_R| = \frac{RE}{\sqrt{R^2 + X_C^2}}$$

f	$ V_R $
0 Hz	0.0 V
1 kHz	2.83 V
3 kHz	6.63 V
5 kHz	8.28 V
7 kHz	9.00 V
10 kHz	9.47 V



21. a. $Z_T = \sqrt{R^2 + (X_L - X_C)^2} \angle \tan^{-1}(X_L - X_C)/R$

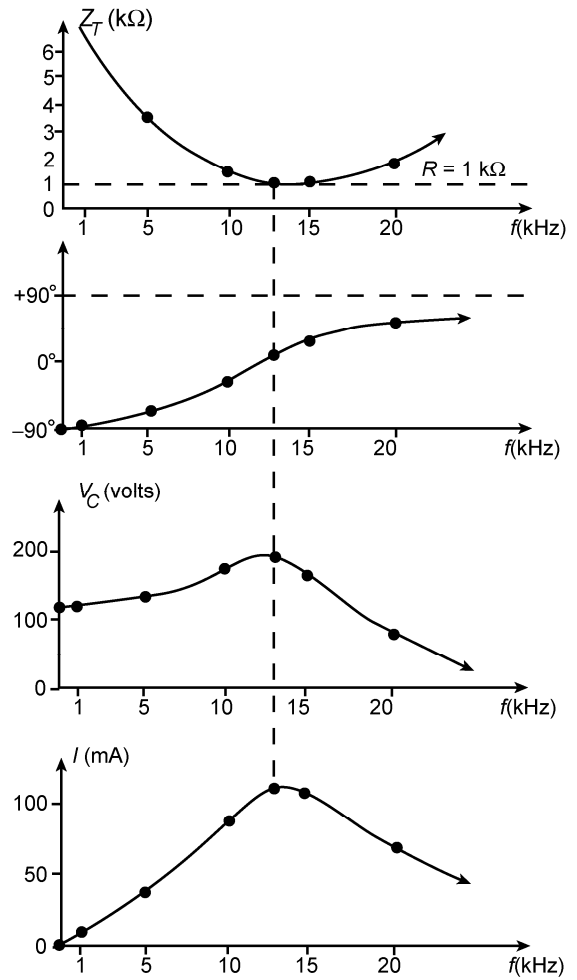
f	Z_T	θ_T
0 Hz	$\infty \Omega$	-90.0°
1 kHz	$19.31 \times 10^3 \Omega$	-87.03°
5 kHz	$3.40 \times 10^3 \Omega$	-72.91°
10 kHz	$1.21 \times 10^3 \Omega$	-34.33°
15 kHz	$1.16 \times 10^3 \Omega$	$+30.75^\circ$
20 kHz	$1.84 \times 10^3 \Omega$	$+56.99^\circ$

b. $|V_C| = \frac{X_C E}{Z_T}$

f	$ V_C $
0 Hz	120 V
1 kHz	120.62 V
5 kHz	136.94 V
10 kHz	192.4 V
15 kHz	133.45 V
20 kHz	63.29 V

c. $|I| = \frac{E}{Z_T}$

f	I
0 Hz	0.0 mA
1 kHz	6.21 mA
5 kHz	35.29 mA
10 kHz	99.17 mA
15 kHz	103.45 mA
20 kHz	65.22 mA



22. a. $X_C = \frac{1}{2\pi fC} = R \Rightarrow f = \frac{1}{2\pi RC} = \frac{1}{2\pi(220\ \Omega)(0.47\ \mu\text{F})} = \mathbf{1.54\ \text{kHz}}$
- b. Low frequency: X_C very large resulting in large Z_T
 High frequency: X_C approaches zero ohms and Z_T approaches R
- c. $f = 100\ \text{Hz}$: $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(100\ \text{Hz})(0.47\ \mu\text{F})} = 3.39\ \text{k}\Omega$
 $Z_T \cong X_C$
- $f = 10\ \text{kHz}$: $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(10\ \text{kHz})(0.47\ \mu\text{F})} = 33.86\ \Omega$
 $Z_T \cong R$
- d. –
- e. $f = 40\ \text{kHz}$: $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(40\ \text{kHz})(0.47\ \mu\text{F})} = 8.47\ \text{k}\Omega$
 $\theta = -\tan^{-1} \frac{X_C}{R} = -\tan^{-1} \frac{8.47\ \Omega}{220\ \Omega} = \mathbf{-2.2^\circ}$
23. a. $\mathbf{Y_T = \frac{1}{R} \angle 0^\circ = \frac{1}{6.8\ \Omega} \angle 0^\circ = 0.147\ \text{S} \angle 0^\circ}$
- b. $\mathbf{Y_T = \frac{1}{X_L \angle 90^\circ} = \frac{1}{200\ \Omega} \angle -90^\circ = 5\ \text{mS} \angle -90^\circ}$
- c. $\mathbf{Y_T = \frac{1}{Z_T} = \frac{1}{X_C \angle -90^\circ} = \frac{1}{2\ \text{k}\Omega} \angle 90^\circ = 0.5\ \text{mS} \angle 90^\circ}$
24. a. $\mathbf{Z_T = \frac{(10\ \Omega \angle 0^\circ)(60\ \Omega \angle 90^\circ)}{10\ \Omega + j60\ \Omega} = 9.86\ \Omega \angle 9.46^\circ}$
 $\mathbf{Y_T = \frac{1}{Z_T} = \frac{1}{9.86\ \Omega \angle 9.46^\circ} = 0.10\ \text{S} \angle -9.46^\circ = 0.1\ \text{S} - 0.02\ \text{S} = G - jB_L}$
- b. $22\ \Omega \parallel 2.2\ \Omega = 2\ \Omega$
 $\mathbf{Z_T = \frac{(2\ \Omega \angle 0^\circ)(6\ \Omega \angle -90^\circ)}{2\ \Omega - j6\ \Omega} = \frac{12\ \Omega \angle -90^\circ}{6.32\ \Omega \angle -71.57^\circ} = 1.90\ \Omega \angle -18.43^\circ}$
 $\mathbf{Y_T = \frac{1}{Z_T} = \frac{1}{1.90\ \Omega \angle -18.43^\circ} = 0.53\ \text{S} \angle 18.43^\circ = 0.5\ \text{S} + j0.17\ \text{S} = G + jB_C}$
- c. $\mathbf{Y_T = \frac{1}{3\ \text{k}\Omega \angle 0^\circ} + \frac{1}{6\ \text{k}\Omega \angle 90^\circ} + \frac{1}{9\ \text{k}\Omega \angle -90^\circ}}$
 $= 0.333 \times 10^{-3} \angle 0^\circ + 0.167 \times 10^{-3} \angle -90^\circ + 0.111 \times 10^{-3} \angle 90^\circ$
 $= \mathbf{0.333 \times 10^{-3}\ \text{S} - j0.056 \times 10^{-3}\ \text{S} = 0.34\ \text{mS} \angle -9.55^\circ}$
 $= G - jB_L$

25. a. $\mathbf{Z_T = 4.7\ \Omega + j8\ \Omega = 9.28\ \Omega \angle 59.57^\circ}$, $\mathbf{Y_T = 0.108\ S \angle -59.57^\circ}$
 $\mathbf{Y_T = 54.7\ mS - j93.12\ mS = G - jB_L}$
- b. $\mathbf{Z_T = 33\ \Omega + 20\ \Omega - j70\ \Omega = 53\ \Omega - j70\ \Omega = 87.80\ \Omega \angle -52.87^\circ}$
 $\mathbf{Y_T = 11.39\ mS \angle 52.87^\circ = 6.88\ mS + j9.08\ mS = G + jB_C}$
- c. $\mathbf{Z_T = 200\ \Omega + j500\ \Omega - j600\ \Omega = 200\ \Omega - j100\ \Omega = 223.61\ \Omega \angle -26.57^\circ}$
 $\mathbf{Y_T = 4.47\ mS \angle 26.57^\circ = 4\ mS + j2\ mS = G + jB_C}$
26. a. $\mathbf{Y_T = \frac{I}{E} = \frac{60\ A \angle 70^\circ}{120\ V \angle 0^\circ} = 0.5\ S \angle 70^\circ = 0.171\ S + j0.470\ S = G + jB_C}$
 $\mathbf{R = \frac{1}{G} = 5.85\ \Omega}$, $\mathbf{X_C = \frac{1}{B_C} = 2.13\ \Omega}$
- b. $\mathbf{Y_T = \frac{I}{E} = \frac{20\ mA \angle 40^\circ}{80\ V \angle 320^\circ} = 0.25\ mS \angle -280^\circ = 0.25\ mS \angle 80^\circ}$
 $\mathbf{= 0.043\ mS + j0.246\ mS = G + jB_C}$
 $\mathbf{R = \frac{1}{G} = 23.26\ k\Omega}$, $\mathbf{X_C = \frac{1}{B_C} = 4.07\ k\Omega}$
- c. $\mathbf{Y_T = \frac{I}{E} = \frac{0.2\ A \angle -60^\circ}{8\ kV \angle 0^\circ} = 0.25\ mS \angle -60^\circ = 0.0125\ mS - j0.02165\ mS = G - jB_L}$
 $\mathbf{R = \frac{1}{G} = 80\ k\Omega}$, $\mathbf{X_L = \frac{1}{B_L} = 46.19\ k\Omega}$
27. a. $\mathbf{Y_T = \frac{1}{10\ \Omega \angle 0^\circ} + \frac{1}{20\ \Omega \angle 90^\circ} = 0.1\ S - j0.05\ S = 111.8\ mS \angle -26.57^\circ}$
- c. $\mathbf{E = I_s Y_T = 2\ A \angle 0^\circ / 111.8\ mS \angle -26.57^\circ = 17.89\ V \angle 26.57^\circ}$
 $\mathbf{I_R = \frac{E \angle \theta}{R \angle 0^\circ} = 17.89\ V \angle 26.57^\circ / 10\ \Omega \angle 0^\circ = 1.79\ A \angle 26.57^\circ}$
 $\mathbf{I_L = \frac{E \angle \theta}{X_L \angle 90^\circ} = 17.89\ V \angle 26.57^\circ / 20\ \Omega \angle 90^\circ = 0.89\ A \angle -63.43^\circ}$
- f. $\mathbf{P = I^2 R = (1.79\ A)^2 10\ \Omega = 32.04\ W}$
- g. $\mathbf{F_p = \frac{G}{Y_T} = \frac{0.1\ S}{111.8\ mS} = 0.894\ \text{lagging}}$
- h. $\mathbf{e = 25.30\ sin(377t + 26.57^\circ)}$
 $\mathbf{i_R = 2.53\ sin(377t + 26.57^\circ)}$
 $\mathbf{i_L = 1.26\ sin(377t - 63.43^\circ)}$
 $\mathbf{i_s = 2.83\ sin\ 377t}$

28. a. $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(60 \text{ Hz})(0.13 \mu\text{F})} = 20.4 \text{ k}\Omega$

b. $Y_T = \frac{1}{10 \text{ k}\Omega \angle 0^\circ} + \frac{1}{20.4 \text{ k}\Omega \angle -90^\circ} = 0.1 \text{ mS} \angle 0^\circ + 0.049 \text{ mS} \angle -90^\circ$
 $= \mathbf{0.111 \text{ mS} \angle 26.10^\circ}$

c. $E = \frac{I_s}{Y_T} = \frac{2 \text{ mA} \angle 20^\circ}{0.111 \text{ mS} \angle 26.10^\circ} = \mathbf{18.02 \text{ V} \angle -6.1^\circ}$
 $I_R = \frac{E}{Z_R} = \frac{18.02 \text{ V} \angle -6.1^\circ}{10 \text{ k}\Omega \angle 0^\circ} = \mathbf{1.80 \text{ mA} \angle -6.1^\circ}$
 $I_C = \frac{E}{Z_C} = \frac{18.02 \text{ V} \angle -6.1^\circ}{20.4 \text{ k}\Omega \angle -90^\circ} = \mathbf{0.883 \text{ mA} \angle 83.90^\circ}$

e. $I_s = I_R + I_C$
 $2 \text{ mA} \angle 20^\circ = 1.80 \text{ mA} \angle -6.1^\circ + 0.883 \text{ mA} \angle 83.90^\circ$
 $= (1.79 \text{ mA} - j0.191 \text{ mA}) + (0.094 \text{ mA} + j0.878 \text{ mA})$
 $= 1.88 \text{ mA} + j0.687 \text{ mA}$
 $2 \text{ mA} \angle 20^\circ \cong 2 \text{ mA} \angle 20.07^\circ$

f. $P = I^2 R = (1.80 \text{ mA})^2 10 \text{ k}\Omega = \mathbf{32.4 \text{ mW}}$

g. $F_p = \frac{G}{Y_T} = \frac{0.1 \text{ mS}}{0.111 \text{ mS}} = \mathbf{0.9 \text{ leading}}$

h. $\omega = 2\pi f = 377 \text{ rad/s}$
 $i_s = 2.83 \times 10^{-3} \sin(\omega t + 20^\circ)$
 $i_R = 2.55 \times 10^{-3} \sin(\omega t - 6.57^\circ)$
 $i_C = 1.25 \times 10^{-3} \sin(\omega t + 83.44^\circ)$
 $e = 25.48 \sin(\omega t - 6.57^\circ)$

29. a. $Y_T = \frac{1}{1.2 \Omega \angle 0^\circ} + \frac{1}{2 \Omega \angle 90^\circ} + \frac{1}{5 \Omega \angle -90^\circ}$
 $= 0.833 \text{ S} \angle 0^\circ + 0.5 \text{ S} \angle -90^\circ + 0.2 \text{ S} \angle 90^\circ$
 $= 0.833 \text{ S} - j0.3 \text{ S} = \mathbf{0.89 \text{ S} \angle -19.81^\circ}$
 $Z_T = \mathbf{1.12 \Omega \angle 19.81^\circ}$

c. $X_C = \frac{1}{\omega C} \Rightarrow C = \frac{1}{\omega X_C} = \frac{1}{(377 \text{ rad/s})(5 \Omega)} = \mathbf{531 \mu\text{F}}$
 $X_L = \omega L \Rightarrow L = \frac{X_L}{\omega} = \frac{2 \Omega}{377 \text{ rad/s}} = \mathbf{5.31 \text{ mH}}$

d.
$$\mathbf{E} = \frac{\mathbf{I}_s}{\mathbf{Y}_T} = \frac{(0.707)(3 \text{ A}) \angle 60^\circ}{0.885 \text{ S} \angle -19.81^\circ} = \frac{2.121 \text{ A} \angle 60^\circ}{0.885 \text{ S} \angle -19.81^\circ} = \mathbf{2.40 \text{ V} \angle 79.81^\circ}$$

$$\mathbf{I}_R = \frac{\mathbf{E} \angle \theta}{R \angle 0^\circ} = \frac{2.397 \text{ V} \angle 79.81^\circ}{1.2 \Omega \angle 0^\circ} = \mathbf{2.00 \text{ A} \angle 79.81^\circ}$$

$$\mathbf{I}_L = \frac{\mathbf{E} \angle \theta}{X_L \angle 90^\circ} = \frac{2.397 \text{ V} \angle 79.81^\circ}{2 \Omega \angle 90^\circ} = \mathbf{1.20 \text{ A} \angle -10.19^\circ}$$

$$\mathbf{I}_C = \frac{\mathbf{E} \angle \theta}{X_C \angle -90^\circ} = \frac{2.397 \text{ V} \angle 79.81^\circ}{5 \Omega \angle -90^\circ} = \mathbf{0.48 \text{ A} \angle 169.81^\circ}$$

f.
$$\mathbf{I}_s = \mathbf{I}_R + \mathbf{I}_L + \mathbf{I}_C$$

$$2.121 \text{ A} \angle 60^\circ = 2.00 \text{ A} \angle 79.81^\circ + 1.20 \text{ A} \angle -10.19^\circ + 0.48 \text{ A} \angle 169.81^\circ$$

$$2.121 \text{ A} \angle 60^\circ = 2.13 \text{ A} \angle 60.01^\circ$$

g.
$$P = I^2 R = (2.00 \text{ A})^2 1.2 \Omega = \mathbf{4.8 \text{ W}}$$

h.
$$F_p = \frac{G}{Y_T} = \frac{0.833 \text{ S}}{0.885 \text{ S}} = \mathbf{0.941 \text{ lagging}}$$

i.
$$e = \mathbf{3.39 \sin(377t + 79.81^\circ)}$$

$$i_R = \mathbf{2.83 \sin(377t + 79.81^\circ)}$$

$$i_L = \mathbf{1.70 \sin(377t - 10.19^\circ)}$$

$$i_C = \mathbf{0.68 \sin(377t + 169.81^\circ)}$$

30. a.
$$X_L = \omega L = (1000 \text{ rad/s})(3.9 \text{ H}) = 3.9 \text{ k}\Omega,$$

$$X_C = \frac{1}{\omega C} = \frac{1}{(1000 \text{ rad/s})(0.12 \mu\text{F})} = 8.33 \text{ k}\Omega$$

$$\mathbf{Y}_T = \frac{1}{3 \text{ k}\Omega \angle 0^\circ} + \frac{1}{3.9 \text{ k}\Omega \angle 90^\circ} + \frac{1}{8.33 \text{ k}\Omega \angle -90^\circ}$$

$$= 0.333 \text{ mS} \angle 0^\circ + 0.256 \text{ mS} \angle -90^\circ + 0.120 \text{ mS} \angle 90^\circ$$

$$= \mathbf{0.333 \text{ mS} - j0.136 \text{ mS} = 0.36 \text{ mS} \angle -22.22^\circ}$$

d.
$$\mathbf{E} = \mathbf{I/Y}_T = 3.54 \text{ mA} \angle -20^\circ / 0.36 \text{ mS} \angle -22.22^\circ = \mathbf{9.83 \text{ V} \angle 2.22^\circ}$$

$$\mathbf{I}_R = \frac{\mathbf{E} \angle \theta}{R \angle 0^\circ} = 9.83 \text{ V} \angle 2.22^\circ / 3 \text{ k}\Omega \angle 0^\circ = \mathbf{3.28 \text{ mA} \angle 2.22^\circ}$$

$$\mathbf{I}_L = \frac{\mathbf{E} \angle \theta}{X_L \angle 90^\circ} = 9.83 \text{ V} \angle 2.22^\circ / 3.9 \text{ k}\Omega \angle 90^\circ = \mathbf{2.52 \text{ mA} \angle -87.78^\circ}$$

$$\mathbf{I}_C = \frac{\mathbf{E} \angle \theta}{X_C \angle -90^\circ} = 9.83 \text{ V} \angle 2.22^\circ / 8.33 \text{ k}\Omega \angle -90^\circ = \mathbf{1.18 \text{ mA} \angle 92.22^\circ}$$

g.
$$P = I^2 R = (3.28 \text{ mA})^2 3 \text{ k}\Omega = \mathbf{32.28 \text{ mW}}$$

h.
$$F_p = G/Y_T = 0.333 \text{ mS} / 0.36 \text{ mS} = \mathbf{0.925 \text{ leading}}$$

i.
$$e = \mathbf{13.9 \sin(1000t + 2.22^\circ)}$$

$$i_R \cong \mathbf{4.64 \times 10^{-3} \sin(1000t + 2.22^\circ)}$$

$$i_L \cong \mathbf{3.56 \times 10^{-3} \sin(1000t - 87.78^\circ)}$$

$$i_C = \mathbf{1.67 \times 10^{-3} \sin(1000t + 92.22^\circ)}$$

$$31. \quad a. \quad \mathbf{I}_1 = \frac{(60 \, \Omega \angle 90^\circ)(20 \, \text{A} \angle 40^\circ)}{22 \, \Omega + j60 \, \Omega} = \frac{1200 \, \text{A} \angle 130^\circ}{63.91 \angle 69.86^\circ} = \mathbf{18.78 \angle 60.14^\circ}$$

$$\mathbf{I}_2 = \frac{(22 \, \Omega \angle 0^\circ)(20 \, \text{A} \angle 40^\circ)}{63.91 \angle 69.86^\circ} = \frac{440 \, \text{A} \angle 40^\circ}{63.91 \angle 69.86^\circ} = \mathbf{6.88 \angle -29.86^\circ}$$

$$b. \quad \mathbf{I}_1 = \frac{(12 \, \Omega - j6 \, \Omega)(6 \, \text{A} \angle 30^\circ)}{12 \, \Omega - j6 \, \Omega + j4 \, \Omega} = \frac{(13.42 \angle -26.57^\circ)(6 \, \text{A} \angle 30^\circ)}{12 - j2}$$

$$= \frac{80.52 \, \text{A} \angle 3.43^\circ}{12.17 \angle -9.46^\circ} = \mathbf{6.62 \, \text{A} \angle 12.89^\circ}$$

$$\mathbf{I}_2 = \frac{(4 \, \Omega \angle 90^\circ)(6 \, \text{A} \angle 30^\circ)}{12.17 \angle -9.46^\circ} = \frac{24 \, \text{A} \angle 120^\circ}{12.17 \angle -9.46^\circ} = \mathbf{1.97 \, \text{A} \angle 129.46^\circ}$$

$$c. \quad \mathbf{I}_1 = \frac{(-j10 \, \Omega + j40 \, \Omega)(4 \, \text{A} \angle 0^\circ)}{+j20 \, \Omega - j10 \, \Omega + j40 \, \Omega} = \frac{(30 \, \Omega \angle 90^\circ)(4 \, \text{A} \angle 0^\circ)}{50 \, \Omega \angle 90^\circ}$$

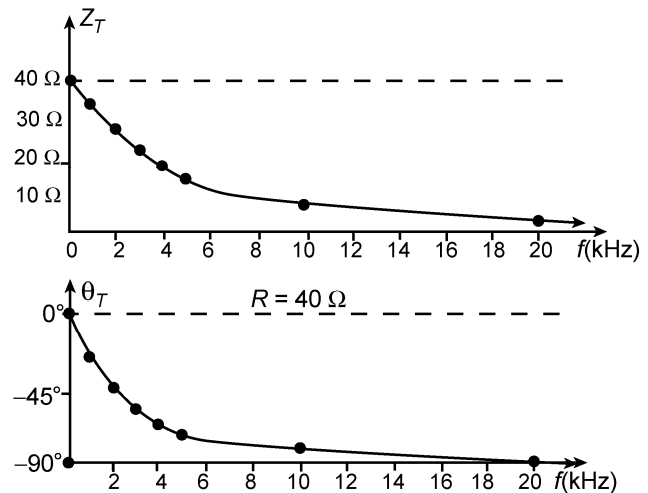
$$= \mathbf{2.4 \, \text{A} \angle 0^\circ}$$

$$\mathbf{I}_2 = \frac{(20 \, \Omega \angle 90^\circ)(4 \, \text{A} \angle 0^\circ)}{50 \, \Omega \angle 90^\circ} = \mathbf{1.6 \, \text{A} \angle 0^\circ}$$

$$32. \quad a. \quad \mathbf{Z}_T = \frac{(R \angle 0^\circ)(X_C \angle -90^\circ)}{R - jX_C} = \frac{RX_C}{\sqrt{R^2 + X_C^2}} \angle -90^\circ + \tan^{-1} X_C/R$$

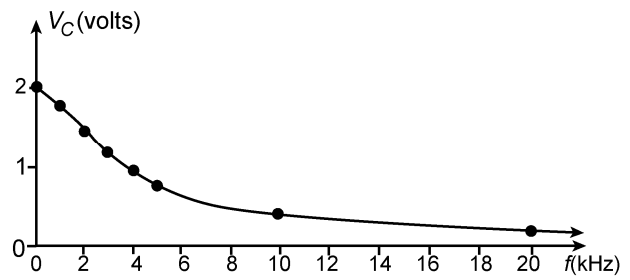
$$|Z_T| = \frac{RX_C}{\sqrt{R^2 + X_C^2}} \quad \theta_T = -90^\circ + \tan^{-1} X_C/R$$

f	$ Z_T $	θ_T
0 Hz	40.0 Ω	0.0°
1 kHz	35.74 Ω	-26.67°
2 kHz	28.22 Ω	-45.14°
3 kHz	22.11 Ω	-56.44°
4 kHz	17.82 Ω	-63.55°
5 kHz	14.79 Ω	-68.30°
10 kHz	7.81 Ω	-78.75°
20 kHz	3.959 Ω	-89.86°



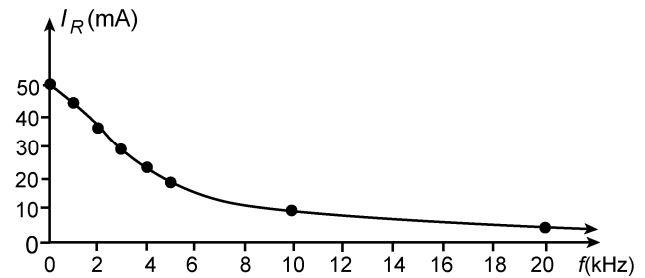
b. $|V_C| = \frac{IRX_C}{\sqrt{R^2 + X_C^2}} = I[Z_T(f)]$

f	$ V_C $
0 kHz	2.0 V
1 kHz	1.787 V
2 kHz	1.411 V
3 kHz	1.105 V
4 kHz	0.891 V
5 kHz	0.740 V
10 kHz	0.391 V
20 kHz	0.198 V



c. $|I_R| = \left| \frac{V_C}{R} \right|$

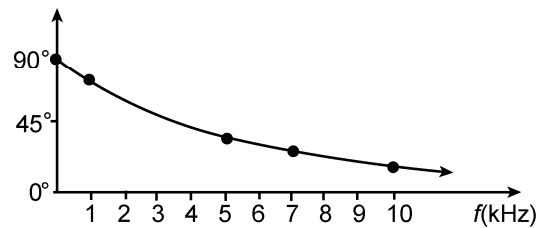
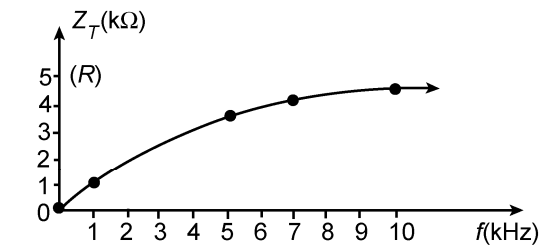
f	$ I_R $
0 kHz	50.0 mA
1 kHz	44.7 mA
2 kHz	35.3 mA
3 kHz	27.64 mA
4 kHz	22.28 mA
5 kHz	18.50 mA
10 kHz	9.78 mA
20 kHz	4.95 mA



33. a. $Z_T = \frac{Z_R Z_L}{Z_R + Z_L} = \frac{(R \angle 0^\circ)(X_L \angle 90^\circ)}{R + jX_L} = \frac{RX_L}{\sqrt{R^2 + X_L^2}} \angle 90^\circ - \tan^{-1} X_L/R$

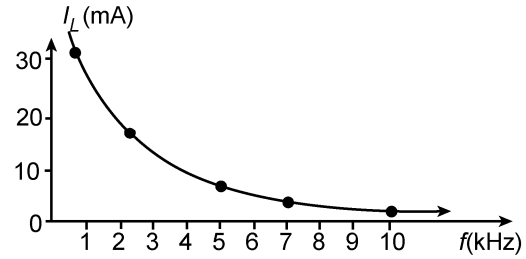
$|Z_T| = \frac{RX_L}{\sqrt{R^2 + X_L^2}} \theta_T = 90^\circ - \tan^{-1} X_L/R$

f	$ Z_T $	θ_T
0 Hz	0.0 kΩ	90.0°
1 kHz	1.22 kΩ	75.86°
5 kHz	3.91 kΩ	38.53°
7 kHz	4.35 kΩ	29.6°
10 kHz	4.65 kΩ	21.69°

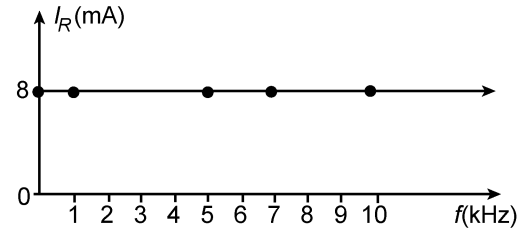


b. $|I_L| = \frac{E}{X_L}$

f	$ I_L $
0 Hz	∞
1 kHz	31.75 mA
5 kHz	6.37 mA
7 kHz	4.55 mA
10 kHz	3.18 mA

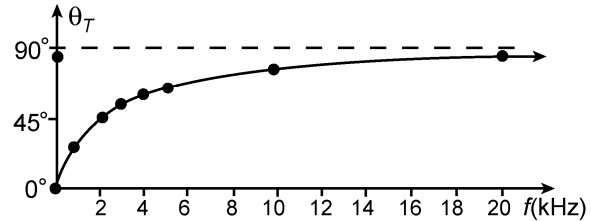
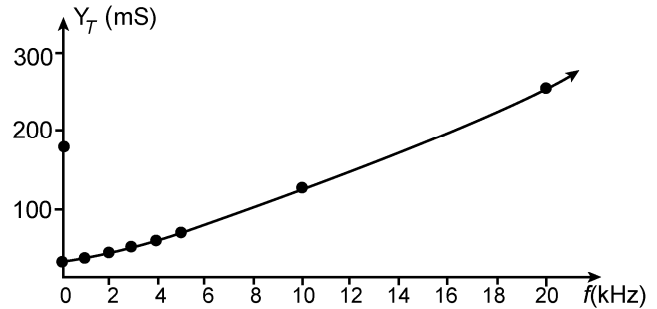


c. $I_R = \frac{E}{R} = \frac{40 \text{ V}}{5 \text{ k}\Omega} = 8 \text{ mA (constant)}$



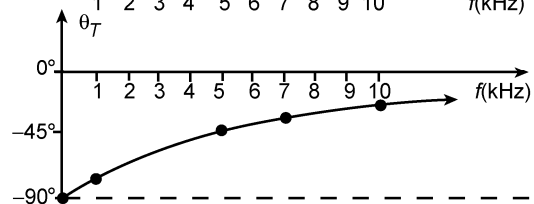
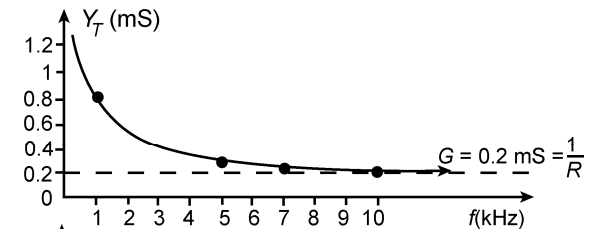
34. $\mathbf{Y}_T = \frac{\sqrt{R^2 + X_C^2}}{RX_C} \angle 90^\circ - \tan^{-1} X_C/R$

f	$ Y_T $	θ_T
0 Hz	25.0 mS	0.0°
1 kHz	27.98 mS	26.67°
2 kHz	35.44 mS	45.14°
3 kHz	45.23 mS	56.44°
4 kHz	56.12 mS	63.55°
5 kHz	67.61 mS	68.30°
10 kHz	128.04 mS	78.75°
20 kHz	252.59 mS	89.86°



35. $Y_T = \frac{1}{Z_T}$ (use data of Prob. 36), $\theta_{T_y} = -\theta_{T_z}$

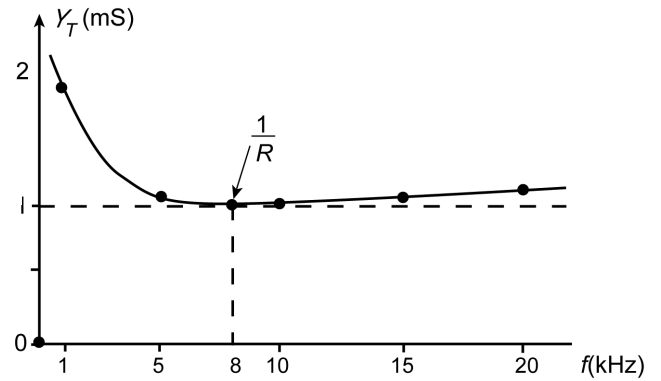
f	Y_T	θ_T
0 Hz	∞	-90.0°
1 kHz	0.82 mS	-75.86°
5 kHz	0.256 mS	-38.53°
7 kHz	0.23 mS	-29.6°
10 kHz	0.215 mS	-21.69°



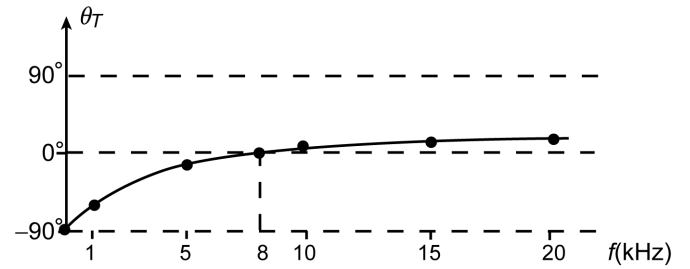
36. a.
$$\mathbf{Y}_T = G \angle 0^\circ + B_L \angle -90^\circ + B_C \angle 90^\circ$$

$$= \sqrt{G^2 + (B_C - B_L)^2} \angle \tan^{-1} \frac{B_C - B_L}{G}$$

f	$ \mathbf{Y}_T $
0 Hz	$X_L \Rightarrow 0 \Omega, Z_T = 0 \Omega,$ $\mathbf{Y}_T = \infty \Omega$
1 kHz	1.86 mS
5 kHz	1.02 mS
10 kHz	1.00 mS
15 kHz	1.02 mS
20 kHz	1.04 mS

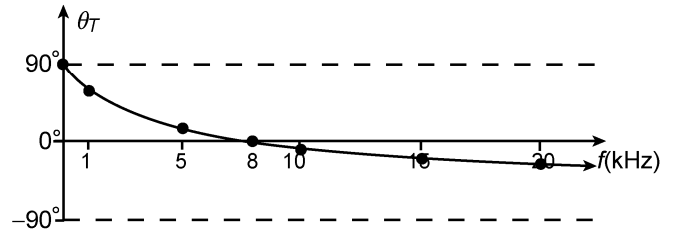
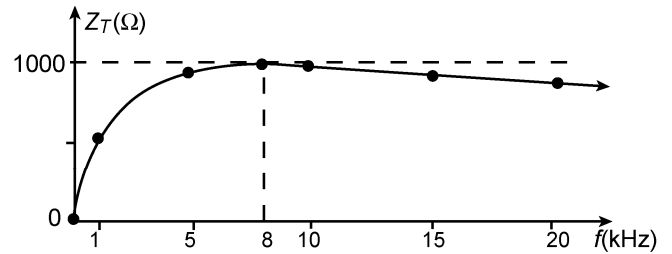


f	$ \theta_T $
0 Hz	-90.0°
1 kHz	-57.51°
5 kHz	-12.63°
10 kHz	$+1.66^\circ$
15 kHz	$+9.98^\circ$
20 kHz	$+16.54^\circ$



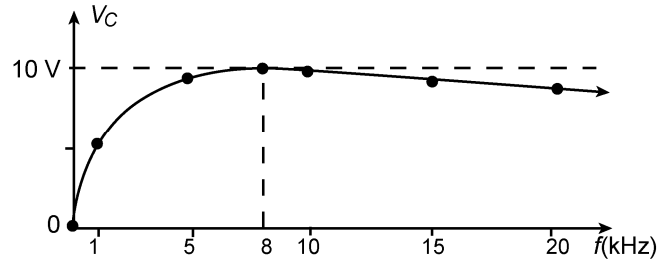
b.
$$\mathbf{Z}_T = \frac{1}{\mathbf{Y}_T}, \theta_{T_z} = -\theta_{T_y}$$

f	Z_T	θ_T
0 kHz	0.0Ω	90.0°
1 kHz	537.63Ω	57.52°
5 kHz	980.39Ω	12.63°
10 kHz	$1 \text{ k}\Omega$	-1.66°
15 kHz	980.39Ω	-9.98°
20 kHz	961.54Ω	-16.54°



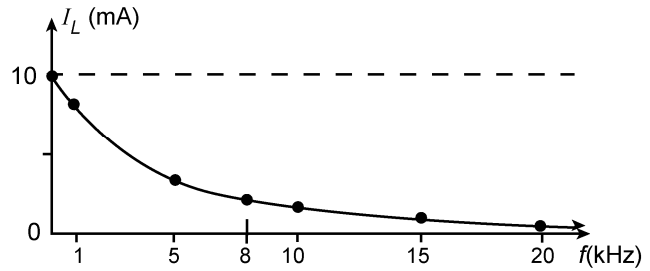
c. $V_C(f) = I[Z_T(f)]$

f	$ V_C $
0 kHz	0.0 V
1 kHz	5.38 V
5 kHz	9.80 V
10 kHz	10 V
15 kHz	9.80 V
20 kHz	9.62 V



d. $I_L = \frac{V_L(f)}{X_L} = \frac{V_C(f)}{X_L}$

f	I_L
0 kHz	10.0 mA
1 kHz	8.56 mA
5 kHz	3.12 mA
10 kHz	1.59 mA
15 kHz	1.04 mA
20 kHz	0.765 mA



37. a. $R_p = \frac{R_s^2 + X_s^2}{R_s} = \frac{(20 \Omega)^2 + (40 \Omega)^2}{20 \Omega} = 100 \Omega (R)$
 $X_p = \frac{R_s^2 + X_s^2}{X_s} = \frac{2000 \Omega}{40} = 50 \Omega (C)$

b. $R_p = \frac{R_s^2 + X_s^2}{R_s} = \frac{(2 \text{ k}\Omega)^2 + (8 \text{ k}\Omega)^2}{2 \text{ k}\Omega} = 34 \text{ k}\Omega (R)$
 $X_p = \frac{R_s^2 + X_s^2}{X_s} = \frac{(2 \text{ k}\Omega)^2 + (8 \text{ k}\Omega)^2}{8 \text{ k}\Omega} = 8.5 \text{ k}\Omega (L)$

38. a. $R_s = \frac{R_p X_p^2}{X_p^2 + R_p^2} = \frac{(8.2 \text{ k}\Omega)(20 \text{ k}\Omega)^2}{(20 \text{ k}\Omega)^2 + (8.2 \text{ k}\Omega)^2} = 7.02 \text{ k}\Omega$
 $X_s = \frac{R_p^2 X_p}{X_p^2 + R_p^2} = \frac{(8.2 \text{ k}\Omega)^2(20 \text{ k}\Omega)}{467.24 \text{ k}\Omega} = 2.88 \text{ k}\Omega$
 $Z_T = 7.02 \text{ k}\Omega - j2.88 \text{ k}\Omega$

b. $R_s = \frac{R_p X_p^2}{X_p^2 + R_p^2} = \frac{(68 \Omega)(40 \Omega)^2}{(40 \Omega)^2 + (68 \Omega)^2} = 17.48 \Omega$
 $X_s = \frac{R_p^2 X_p}{X_p^2 + R_p^2} = \frac{(68 \Omega)^2(40 \Omega)}{6224 \Omega^2} = 29.72 \Omega$
 $Z_T = 17.48 \Omega + j29.72 \Omega$

39. a. $C_T = 2 \mu\text{F}$
 $X_C = \frac{1}{\omega C} = \frac{1}{2\pi(10^3 \text{ Hz})(2 \mu\text{F})} = 79.62 \Omega$
 $X_L = \omega L = 2\pi(10^3 \text{ Hz})(10 \text{ mH}) = 62.80 \Omega$

$$\mathbf{Y}_T = \frac{1}{220 \Omega \angle 0^\circ} + \frac{1}{79.62 \Omega \angle -90^\circ} + \frac{1}{62.8 \Omega \angle 90^\circ}$$

$$= 4.55 \text{ mS} \angle 0^\circ + 12.56 \text{ mS} \angle 90^\circ + 15.92 \text{ mS} \angle -90^\circ$$

$$= \mathbf{4.55 \text{ mS} - j3.36 \text{ mS} = 5.66 \text{ mS} \angle -36.44^\circ}$$

$$\mathbf{E = I/Y}_T = 1 \text{ A} \angle 0^\circ / 5.66 \text{ mS} \angle -36.44^\circ = \mathbf{176.68 \text{ V} \angle 36.44^\circ}$$

$$\mathbf{I}_R = \frac{E \angle \theta}{R \angle 0^\circ} = 176.68 \text{ V} \angle 36.44^\circ / 220 \Omega \angle 0^\circ = \mathbf{0.803 \text{ A} \angle 36.44^\circ}$$

$$\mathbf{I}_L = \frac{E \angle \theta}{X_L \angle 90^\circ} = 176.68 \text{ V} \angle 36.44^\circ / 62.80 \Omega \angle 90^\circ = \mathbf{2.813 \text{ A} \angle -53.56^\circ}$$

b. $F_p = G/Y_T = 4.55 \text{ mS} / 5.66 \text{ mS} = \mathbf{0.804 \text{ lagging}}$

c. $P = I^2 R = (0.803 \text{ A})^2 220 \Omega = \mathbf{141.86 \text{ W}}$

f. $\mathbf{I}_s = \mathbf{I}_R + 2\mathbf{I}_C + \mathbf{I}_L$
and $\mathbf{I}_C = \frac{\mathbf{I}_s - \mathbf{I}_R - \mathbf{I}_L}{2}$

$$= \frac{1 \text{ A} \angle 0^\circ - 0.803 \text{ A} \angle 36.44^\circ - 2.813 \text{ A} \angle -53.56^\circ}{2}$$

$$= \frac{1 - (0.646 + j0.477) - (1.671 - j2.263)}{2} = \frac{-1.317 + j1.786}{2}$$

$$\mathbf{I}_C = -0.657 + j0.893 = \mathbf{1.11 \text{ A} \angle 126.43^\circ}$$

g. $\mathbf{Z}_T = \frac{1}{\mathbf{Y}_T} = \frac{1}{5.66 \text{ mS} \angle -36.44^\circ} = 176.7 \Omega \angle 36.44^\circ$

$$= \mathbf{142.15 \Omega + j104.96 \Omega = R + jX_L}$$

40. $P = VI \cos \theta = 3000 \text{ W}$

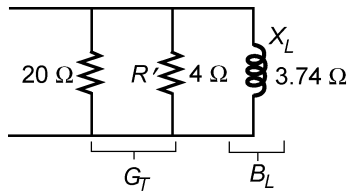
$$\cos \theta = \frac{3000 \text{ W}}{VI} = \frac{3000 \text{ W}}{(100 \text{ V})(40 \text{ A})} = \frac{3000}{4000} = 0.75 \text{ (lagging)}$$

$$\theta = \cos^{-1} 0.75 = 41.41^\circ$$

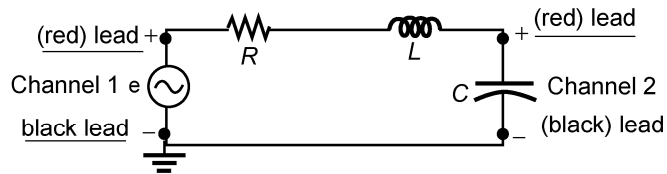
$$\mathbf{Y}_T = \frac{\mathbf{I}}{\mathbf{E}} = \frac{40 \text{ A} \angle -41.41^\circ}{100 \text{ V} \angle 0^\circ} = 0.4 \text{ S} \angle -41.41^\circ = 0.3 \text{ S} - j0.265 \text{ S} = G_T - jB_L$$

$$G_T = 0.3 \text{ S} = \frac{1}{20 \Omega} + \frac{1}{R'} = 0.05 \text{ S} + \frac{1}{R'}$$
and $R' = \frac{1}{0.25 \text{ S}} = 4 \Omega$

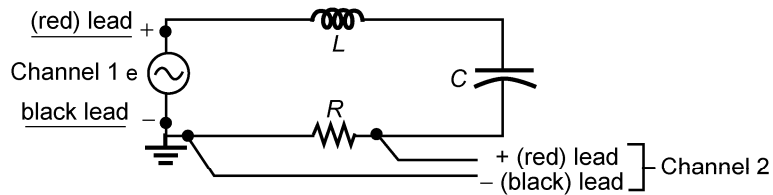
$$X_L = \frac{1}{B_L} = \frac{1}{0.265 \text{ S}} = 3.74 \Omega$$



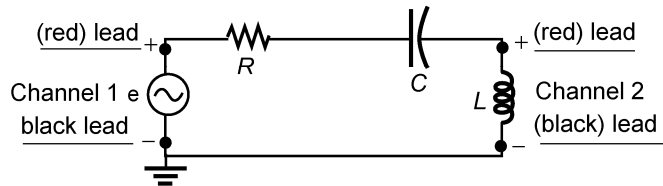
41. a.



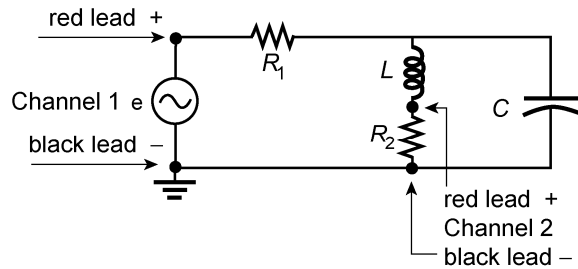
b.



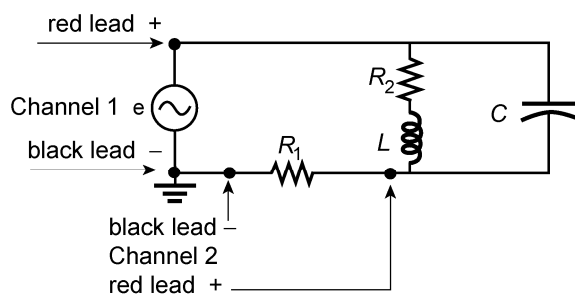
c.



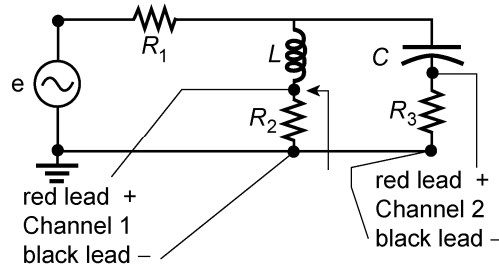
42. a. e and v_{R_2}



b. e and i_s



c. i_L and i_C



43. (I): (a) $\theta_{\text{div.}} = 0.8 \text{ div.}, \theta_T = 4 \text{ div.}$
 $\theta = \frac{0.8 \text{ div.}}{4 \text{ div.}} \times 360^\circ = 72^\circ$
 v_1 leads v_2 by 72°
- (b) v_1 : peak-to-peak = $(5 \text{ div.})(0.5 \text{ V/div.}) = 2.5 \text{ V}$
 $V_1(\text{rms}) = 0.7071 \left(\frac{2.5 \text{ V}}{2} \right) = 0.88 \text{ V}$
 v_2 : peak-to-peak = $(2.4 \text{ div.})(0.5 \text{ V/div.}) = 1.2 \text{ V}$
 $V_2(\text{rms}) = 0.7071 \left(\frac{1.2 \text{ V}}{2} \right) = 0.42 \text{ V}$
- (c) $T = (4 \text{ div.})(0.2 \text{ ms/div.}) = 0.8 \text{ ms}$
 $f = \frac{1}{T} = \frac{1}{0.8 \text{ ms}} = 1.25 \text{ kHz (both)}$
- (II): (a) $\theta_{\text{div.}} = 2.2 \text{ div.}, \theta_T = 6 \text{ div.}$
 $\theta = \frac{2.2 \text{ div.}}{6 \text{ div.}} \times 360^\circ = 132^\circ$
 v_1 leads v_2 by 132°
- (b) v_1 : peak-to-peak = $(2.8 \text{ div.})(2 \text{ V/div.}) = 5.6 \text{ V}$
 $V_1(\text{rms}) = 0.7071 \left(\frac{5.6 \text{ V}}{2} \right) = 1.98 \text{ V}$
 v_2 : peak-to-peak = $(4 \text{ div.})(2 \text{ V/div.}) = 8 \text{ V}$
 $V_2(\text{rms}) = 0.7071 \left(\frac{8 \text{ V}}{2} \right) = 2.83 \text{ V}$
- (c) $T = (6 \text{ div.})(10 \mu\text{s/div.}) = 60 \mu\text{s}$
 $f = \frac{1}{T} = \frac{1}{60 \mu\text{s}} = 16.67 \text{ kHz}$

Chapter 16

1. a.
$$\mathbf{Z}_T = j4 \, \Omega + \frac{(8 \, \Omega \angle -90^\circ)(12 \, \Omega \angle 0^\circ)}{-j8 \, \Omega + 12 \, \Omega} = 3.69 \, \Omega - j1.54 \, \Omega \cong 4 \, \Omega \angle -22.65^\circ$$
- b.
$$\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{14 \, \text{V} \angle 0^\circ}{4 \, \Omega \angle -22.65^\circ} = 3.5 \, \text{A} \angle 22.65^\circ$$
- c.
$$\mathbf{I}_1 = 3.5 \, \text{A} \angle 22.65^\circ$$
- d.
$$\mathbf{I}_2 = \frac{(8 \, \Omega \angle -90^\circ)(3.5 \, \text{A} \angle 22.65^\circ)}{12 \, \Omega - j8 \, \Omega} = 1.94 \, \text{A} \angle -33.66^\circ$$
- e.
$$\mathbf{V}_L = \mathbf{I}_s \mathbf{X}_L = (3.5 \, \text{A} \angle 22.65^\circ)(4 \, \Omega \angle 90^\circ) = 14 \, \text{V} \angle 112.65^\circ$$
2. a.
$$\begin{aligned} \mathbf{Z}_T &= 3 \, \Omega + j6 \, \Omega + 2 \, \Omega \angle 0^\circ \parallel 8 \, \Omega \angle -90^\circ \\ &= 3 \, \Omega + j6 \, \Omega + 1.94 \, \Omega \angle -14.04^\circ \\ &= 3 \, \Omega + j6 \, \Omega + 1.88 \, \Omega - j0.47 \, \Omega \\ &= 4.88 \, \Omega + j5.53 \, \Omega = 7.38 \, \Omega \angle 48.57^\circ \end{aligned}$$
- b.
$$\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{30 \, \text{V} \angle 0^\circ}{7.38 \, \Omega \angle 48.57^\circ} = 4.07 \, \text{A} \angle -48.57^\circ$$
- c.
$$\begin{aligned} \mathbf{I}_C &= \frac{\mathbf{Z}_{R_2} \mathbf{I}_s}{\mathbf{Z}_{R_2} + \mathbf{Z}_C} = \frac{(2 \, \Omega \angle 0^\circ)(4.07 \, \text{A} \angle -48.57^\circ)}{2 \, \Omega - j8 \, \Omega} \\ &= \frac{8.14 \, \text{A} \angle -48.57^\circ}{8.25 \angle -75.96^\circ} = 0.987 \, \text{A} \angle 27.39^\circ \end{aligned}$$
- d.
$$\begin{aligned} \mathbf{V}_L &= \frac{\mathbf{Z}_L \mathbf{E}}{\mathbf{Z}_T} = \frac{(6 \, \Omega \angle 90^\circ)(30 \, \text{V} \angle 0^\circ)}{7.38 \, \Omega \angle 48.57^\circ} = \frac{180 \, \text{V} \angle 90^\circ}{7.38 \, \Omega \angle 48.57^\circ} \\ &= 24.39 \, \text{V} \angle 41.43^\circ \end{aligned}$$
3. a.
$$\begin{aligned} \mathbf{Z}_T &= 12 \, \Omega \angle 90^\circ \parallel (9.1 \, \Omega - j12 \, \Omega) = 12 \, \Omega \angle 90^\circ \parallel 15.06 \, \Omega \angle -52.826^\circ \\ &= \frac{180.72 \, \Omega \angle 37.17^\circ}{9.10 \angle 0^\circ} \\ &= 19.86 \, \Omega \angle 37.17^\circ \end{aligned}$$
- b.
$$\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{60 \, \text{V} \angle 0^\circ}{19.86 \, \Omega \angle 37.17^\circ} = 3.02 \, \text{A} \angle -37.17^\circ$$
- c.
$$\begin{aligned} (\text{CDR}) \mathbf{I}_2 &= \frac{(12 \, \Omega \angle 90^\circ)(3.02 \, \text{A} \angle -37.17^\circ)}{j12 \, \Omega + 9.1 \, \Omega - j12 \, \Omega} = \frac{36.24 \, \text{A} \angle 52.83^\circ}{9.1 \angle 0^\circ} \\ &= 3.98 \, \text{A} \angle 52.83^\circ \end{aligned}$$

$$\begin{aligned} \text{d. } (\text{VDR}) \mathbf{V}_C &= \frac{(12 \Omega \angle -90^\circ)(60 \text{ V} \angle 0^\circ)}{9.1 \Omega - j12 \Omega} = \frac{720 \text{ V} \angle -90^\circ}{15.06 \angle -52.826^\circ} \\ &= \mathbf{47.81 \text{ V} \angle -37.17^\circ} \end{aligned}$$

$$\begin{aligned} \text{e. } P &= EI \cos \theta = (60 \text{ V})(3.02 \text{ A}) \cos(37.17^\circ) \\ &= 181.20(0.797) = \mathbf{144.42 \text{ W}} \end{aligned}$$

$$\begin{aligned} 4. \quad \text{a. } \mathbf{Z}_T &= 2 \text{ k}\Omega \angle 0^\circ + \frac{(4 \text{ k}\Omega \angle -90^\circ)(6 \text{ k}\Omega \angle 90^\circ)}{-j4 \text{ k}\Omega + j6 \text{ k}\Omega} + \frac{\left(\frac{6.8 \text{ k}\Omega}{2} \angle 0^\circ\right)(8 \text{ k}\Omega \angle 90^\circ)}{\frac{6.8 \text{ k}\Omega}{2} + j8 \text{ k}\Omega} \\ &= 2 \text{ k}\Omega + \frac{24 \text{ k}\Omega \angle 0^\circ}{2 \angle 90^\circ} + \frac{27.2 \text{ k}\Omega \angle 90^\circ}{\underbrace{3.4 \text{ k}\Omega + j8 \text{ k}\Omega}_{8.69 \text{ k}\Omega \angle 66.97^\circ}} \\ &= 2 \text{ k} + j12 \text{ k}\Omega \quad \underbrace{3.13 \text{ k}\Omega \angle 23.03^\circ}_{2.88 \text{ k}\Omega + j1.22 \text{ k}\Omega} \\ &= 2 \text{ k}\Omega + 7.88 \text{ k}\Omega - j12 \text{ k}\Omega + j1.22 \text{ k}\Omega \\ \mathbf{Z_T} &= \mathbf{4.88 \text{ k}\Omega - j10.78 \text{ k}\Omega = 11.83 \text{ k}\Omega \angle -65.64^\circ} \end{aligned}$$

$$\begin{aligned} \text{b. } \mathbf{V}_2 &= \mathbf{I Z_T} = (4 \text{ mA} \angle 0^\circ)(3.13 \text{ k}\Omega \angle 23.03^\circ) = \mathbf{12.52 \text{ V} \angle 23.03^\circ} \\ \mathbf{I_L} &= \frac{\mathbf{V}_2}{\mathbf{X_{L_2}}} = \frac{12.52 \text{ V} \angle 23.03^\circ}{8 \text{ k}\Omega \angle 90^\circ} = \mathbf{1.57 \text{ A} \angle -66.97^\circ} \end{aligned}$$

$$\text{c. } F_p = \frac{R}{Z_T} = \frac{4.88 \text{ k}\Omega}{11.83 \text{ k}\Omega} = \mathbf{0.413 \text{ (leading)}}$$

$$\begin{aligned} 5. \quad \text{a. } 400 \Omega \angle -90^\circ \parallel 400 \Omega \angle -90^\circ &= \frac{400 \Omega \angle -90^\circ}{2} = 200 \Omega \angle -90^\circ \\ \mathbf{Z'} &= 100 \Omega - j200 \Omega = 223.61 \Omega \angle -63.43^\circ \\ \mathbf{Z''} &= -j200 \Omega + j600 \Omega = j400 \Omega = 400 \Omega \angle 90^\circ \\ \mathbf{Z_T} &= \mathbf{Z' \parallel Z''} = \frac{(223.61 \Omega \angle -63.43^\circ)(400 \Omega \angle 90^\circ)}{(100 \Omega - j200 \Omega) + j400 \Omega} = \frac{89444.00 \Omega \angle 26.57^\circ}{223.61 \angle 63.43^\circ} \\ &= 400 \Omega \angle -36.86^\circ \\ \mathbf{I} &= \frac{\mathbf{E}}{\mathbf{Z_T}} = \frac{100 \text{ V} \angle 0^\circ}{400 \Omega \angle -36.86^\circ} = \mathbf{0.25 \text{ A} \angle 36.86^\circ} \end{aligned}$$

$$\text{b. } \mathbf{V_C} = \frac{(200 \Omega \angle -90^\circ)(100 \text{ V} \angle 0^\circ)}{100 \Omega - j200 \Omega} = \frac{20,000 \text{ V} \angle -90^\circ}{223.61 \angle -63.43^\circ} = \mathbf{89.44 \text{ V} \angle -26.57^\circ}$$

$$\begin{aligned} \text{c. } P &= EI \cos \theta = (100 \text{ V})(0.25 \text{ A}) \cos 36.86^\circ \\ &= (25)(0.8) = \mathbf{20 \text{ W}} \end{aligned}$$

6. a. $Z_1 = 3 \Omega + j4 \Omega = 5 \Omega \angle 53.13^\circ$

$$I_1 = \frac{E}{Z_1} = \frac{120 \text{ V} \angle 0^\circ}{5 \Omega \angle 53.13^\circ} = \mathbf{24 \text{ A} \angle 53.13^\circ}$$

b. $V_C = \frac{(13 \Omega \angle -90^\circ)(120 \text{ V} \angle 0^\circ)}{-j13 \Omega + j7 \Omega} = \frac{1560 \text{ V} \angle -90^\circ}{6 \angle -90^\circ} = \mathbf{260 \text{ V} \angle 0^\circ}$

c. $V_{R_1} = (I_1 \angle \theta)R \angle 0^\circ = (24 \text{ A} \angle -53.13^\circ)(3 \Omega \angle 0^\circ) = 72 \text{ V} \angle -53.13^\circ$

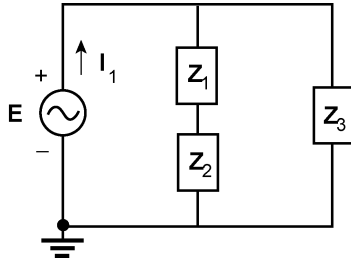
$$V_{ab} + V_{R_1} - V_C = 0$$

$$V_{ab} = V_C - V_{R_1} = 260 \text{ V} \angle 0^\circ - 72 \text{ V} \angle -53.13^\circ$$

$$= 260 \text{ V} - (43.20 \text{ V} + j57.60 \text{ V})$$

$$= \mathbf{216.80 \text{ V} + j57.60 \text{ V} = 224.32 \text{ V} \angle -14.88^\circ}$$

7. a.



$$Z_1 = 10 \Omega \angle 0^\circ$$

$$Z_2 = 80 \Omega \angle 90^\circ \parallel 20 \Omega \angle 0^\circ$$

$$= \frac{1600 \Omega \angle 90^\circ}{20 + j80} = \frac{1600 \Omega \angle 90^\circ}{82.462 \angle 75.964^\circ}$$

$$= 19.403 \Omega \angle 14.036^\circ$$

$$Z_3 = 60 \Omega \angle -90^\circ$$

$$Z_T = (Z_1 + Z_2) \parallel Z_3$$

$$= (10 \Omega + 18.824 \Omega + j4.706 \Omega) \parallel 60 \Omega \angle -90^\circ$$

$$= 29.206 \Omega \angle 9.273^\circ \parallel 6 \Omega \angle -90^\circ = \frac{1752.36 \Omega \angle -80.727^\circ}{28.824 + j4.706 - j60}$$

$$= \frac{1752.36 \Omega \angle -80.727^\circ}{62.356 \angle -62.468^\circ} = \mathbf{28.103 \Omega \angle -18.259^\circ}$$

$$I_1 = \frac{E}{Z_T} = \frac{40 \text{ V} \angle 0^\circ}{28.103 \Omega \angle -18.259^\circ} = \mathbf{1.42 \text{ A} \angle 18.26^\circ}$$

b. $V_1 = \frac{Z_2 E}{Z_2 + Z_1} = \frac{(19.403 \Omega \angle 14.036^\circ)(40 \text{ V} \angle 0^\circ)}{29.206 \Omega \angle 9.273^\circ} = \frac{776.12 \text{ V} \angle 14.036^\circ}{29.206 \angle 9.273^\circ}$

$$= \mathbf{26.57 \text{ V} \angle 4.76^\circ}$$

c. $P = EI \cos \theta = (40 \text{ V})(1.423 \text{ A}) \cos 18.259^\circ$

$$= \mathbf{54.07 \text{ W}}$$

8. a. $\mathbf{Z}_1 = 2 \Omega + j1 \Omega = 2.236 \Omega \angle 26.565^\circ$, $\mathbf{Z}_2 = 3 \Omega \angle 0^\circ$

$$\mathbf{Z}_3 = 16 \Omega + j15 \Omega - j7 \Omega = 16 \Omega + j8 \Omega = 17.889 \Omega \angle 26.565^\circ$$

$$\begin{aligned}\mathbf{Y}_T &= \frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2} + \frac{1}{\mathbf{Z}_3} = \frac{1}{2.236 \Omega \angle 26.565^\circ} + \frac{1}{3 \Omega \angle 0^\circ} + \frac{1}{17.889 \Omega \angle 26.565^\circ} \\ &= 0.447 \text{ S} \angle -26.565^\circ + 0.333 \text{ S} \angle 0^\circ + 0.056 \text{ S} \angle -26.565^\circ \\ &= (0.4 \text{ S} - j0.2 \text{ S}) + (0.333 \text{ S}) + (0.05 \text{ S} - j0.025 \text{ S}) \\ &= 0.783 \text{ S} - j0.225 \text{ S} = \mathbf{0.82 \text{ S} \angle -16.03^\circ}\end{aligned}$$

$$\mathbf{Z}_T = \frac{1}{\mathbf{Y}_T} = \frac{1}{0.82 \text{ S} \angle -16.03^\circ} = \mathbf{1.23 \Omega \angle 16.03^\circ}$$

b. $\mathbf{I}_1 = \frac{\mathbf{E}}{\mathbf{Z}_1} = \frac{60 \text{ V} \angle 0^\circ}{2.236 \Omega \angle 26.565^\circ} = \mathbf{26.83 \text{ A} \angle -26.57^\circ}$

$$\mathbf{I}_2 = \frac{\mathbf{E}}{\mathbf{Z}_2} = \frac{60 \text{ V} \angle 0^\circ}{3 \Omega \angle 0^\circ} = \mathbf{20 \text{ A} \angle 0^\circ}$$

$$\mathbf{I}_3 = \frac{\mathbf{E}}{\mathbf{Z}_3} = \frac{60 \text{ V} \angle 0^\circ}{17.889 \Omega \angle 26.565^\circ} = \mathbf{3.35 \text{ A} \angle -26.57^\circ}$$

c. $\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{60 \text{ V} \angle 0^\circ}{1.227 \Omega \angle 16.032^\circ} = \mathbf{48.9 \text{ A} \angle -16.03^\circ}$

$$\mathbf{I}_s \stackrel{?}{=} \mathbf{I}_1 + \mathbf{I}_2 + \mathbf{I}_3$$

$$\begin{aligned}48.9 \text{ A} \angle -16.03^\circ &\stackrel{?}{=} 26.83 \text{ A} \angle -26.57^\circ + 20 \text{ A} \angle 0^\circ + 3.35 \text{ A} \angle -26.57^\circ \\ &= (24 \text{ A} - j12 \text{ A}) + (20 \text{ A}) + (3 \text{ A} - j1.5 \text{ A}) \\ &\checkmark = 47 \text{ A} + j13.5 \text{ A} = \mathbf{48.9 \text{ A} \angle -16.03^\circ} \text{ (checks)}\end{aligned}$$

d. $F_p = \frac{G}{Y_T} = \frac{0.783 \text{ S}}{0.820 \text{ S}} = \mathbf{0.955 \text{ (lagging)}}$

9. a. $X_{L_1} = \omega L_1 = 2\pi(10^3 \text{ Hz})(0.1 \text{ H}) = 628 \Omega$

$$X_{L_2} = \omega L_2 = 2\pi(10^3 \text{ Hz})(0.2 \text{ H}) = 1.256 \text{ k}\Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi(10^3 \text{ Hz})(1 \mu\text{F})} = 0.159 \text{ k}\Omega$$

$$\begin{aligned}\mathbf{Z}_T &= R \angle 0^\circ + X_{L_1} \angle 90^\circ + X_C \angle -90^\circ \parallel X_{L_2} \angle 90^\circ \\ &= 300 \Omega + j628 \Omega + 0.159 \text{ k}\Omega \angle -90^\circ \parallel 1.256 \text{ k}\Omega \angle 90^\circ \\ &= 300 \Omega + j628 \Omega - j182 \Omega \\ &= \mathbf{300 \Omega + j446 \Omega = 537.51 \Omega \angle 56.07^\circ}\end{aligned}$$

b. $\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{50 \text{ V} \angle 0^\circ}{537.51 \Omega \angle 56.07^\circ} = \mathbf{93 \text{ mA} \angle -56.07^\circ}$

c. (CDR):
$$\mathbf{I}_1 = \frac{\mathbf{Z}_{L_2} \mathbf{I}_s}{\mathbf{Z}_{L_2} + \mathbf{Z}_C} = \frac{(1.256 \text{ k}\Omega \angle 90^\circ)(93 \text{ mA} \angle -56.07^\circ)}{+j1.256 \text{ k}\Omega - j0.159 \text{ k}\Omega}$$
$$= \frac{116.81 \text{ mA} \angle 33.93^\circ}{1.097 \angle 90^\circ} = \mathbf{106.48 \text{ mA} \angle -56.07^\circ}$$
$$\mathbf{I}_2 = \frac{\mathbf{Z}_C \mathbf{I}_s}{\mathbf{Z}_{L_2} + \mathbf{Z}_C} = \frac{(0.159 \text{ k}\Omega \angle -90^\circ)(93 \text{ mA} \angle -56.07^\circ)}{1.097 \text{ k}\Omega \angle 90^\circ}$$
$$= \frac{14.79 \text{ mA} \angle -146.07^\circ}{1.097 \angle 90^\circ} = 13.48 \text{ mA} \angle -236.07^\circ$$
$$= \mathbf{13.48 \text{ mA} \angle 123.93^\circ}$$

d.
$$\mathbf{V}_1 = (\mathbf{I}_2 \angle \theta)(X_{L_2} \angle 90^\circ) = (13.48 \text{ mA} \angle 123.92^\circ)(1.256 \text{ k}\Omega \angle 90^\circ)$$
$$= \mathbf{16.93 \text{ V} \angle 213.93^\circ}$$
$$\mathbf{V}_{ab} = \mathbf{E} - (\mathbf{I}_s \angle \theta)(R \angle 0^\circ) = 50 \text{ V} \angle 0^\circ - (93 \text{ mA} \angle -56.07^\circ)(300 \Omega \angle 0^\circ)$$
$$= 50 \text{ V} - 27.9 \text{ V} \angle -56.07^\circ$$
$$= 50 \text{ V} - (15.573 \text{ V} - j23.149 \text{ V})$$
$$= 34.43 \text{ V} + j23.149 \text{ V} = \mathbf{41.49 \text{ V} \angle 33.92^\circ}$$

e.
$$P = I_s^2 R = (93 \text{ mA})^2 300 \Omega = \mathbf{2.595 \text{ W}}$$

f.
$$F_p = \frac{R}{Z_T} = \frac{300 \Omega}{537.51 \Omega} = \mathbf{0.558 \text{ (lagging)}}$$

10. a.
$$\mathbf{Z}_T = 1.2 \text{ k}\Omega + \frac{(1.2 \text{ k}\Omega \angle 0^\circ)(1.8 \text{ k}\Omega \angle -90^\circ)}{1.2 \text{ k}\Omega - j1.8 \text{ k}\Omega} + \frac{2.4 \text{ k}\Omega \angle 90^\circ}{2}$$
$$= 1.2 \text{ k}\Omega + \frac{2.16 \text{ k}\Omega \angle -90^\circ}{2.16 \angle -56.31^\circ} + 1.2 \text{ k}\Omega \angle 90^\circ$$
$$= 1.2 \text{ k}\Omega + 1 \text{ k}\Omega \angle -33.69^\circ + j1.2 \text{ k}\Omega$$
$$= 1.2 \text{ k}\Omega + 832.05 \Omega - j554.70 \Omega + j1.2 \text{ k}\Omega$$
$$= \mathbf{2.03 \text{ k}\Omega + j645.30 \Omega = 2.13 \text{ k}\Omega \angle 17.63^\circ}$$

b.
$$\mathbf{V}_1 = \mathbf{I} \mathbf{R}_1 = (20 \text{ mA} \angle 0^\circ)(1.2 \text{ k}\Omega \angle 0^\circ) = \mathbf{24 \text{ V} \angle 0^\circ}$$

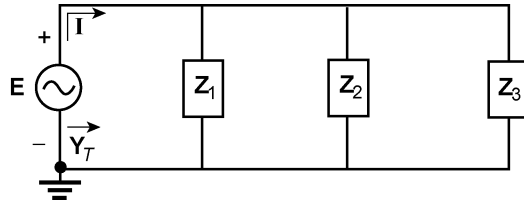
c.
$$\mathbf{I}_1 = \frac{(1.2 \text{ k}\Omega \angle 0^\circ)(20 \text{ mA} \angle 0^\circ)}{1.2 \text{ k}\Omega - j1.8 \text{ k}\Omega} = \frac{2.4 \text{ A} \angle 0^\circ}{2.16 \times 10^3 \angle -56.31^\circ} = \mathbf{11.11 \text{ mA} \angle 56.31^\circ}$$

d.
$$\mathbf{V}_2 = \mathbf{I}(\mathbf{X}_{L_1} \parallel \mathbf{X}_{L_2}) = (20 \text{ mA} \angle 0^\circ) \left(\frac{(2.4 \text{ k}\Omega)}{2} \angle 90^\circ \right) = \mathbf{24 \text{ V} \angle 90^\circ}$$

e.
$$\mathbf{V}_s = \mathbf{I} \mathbf{Z}_T = (20 \text{ mA} \angle 0^\circ)(2.13 \text{ k}\Omega \angle 17.63^\circ) = \mathbf{42.60 \text{ V} \angle 17.63^\circ}$$

11.

a.



$$\mathbf{Z}_1 = 2 \, \Omega - j2 \, \Omega = 2.828 \, \Omega \angle -45^\circ$$

$$\mathbf{Z}_2 = 3 \, \Omega - j9 \, \Omega + j6 \, \Omega$$

$$= 3 \, \Omega - j3 \, \Omega = 4.243 \, \Omega \angle -45^\circ$$

$$\mathbf{Z}_3 = 10 \, \Omega \angle 0^\circ$$

$$\begin{aligned} \mathbf{Y}_T &= \frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2} + \frac{1}{\mathbf{Z}_3} = \frac{1}{2.828 \, \Omega \angle -45^\circ} + \frac{1}{4.243 \, \Omega \angle -45^\circ} + \frac{1}{10 \, \Omega \angle 0^\circ} \\ &= 0.354 \, \text{S} \angle 45^\circ + 0.236 \, \text{S} \angle 45^\circ + 0.1 \, \text{S} \angle 0^\circ = 0.59 \, \text{S} \angle 45^\circ + 0.1 \, \text{S} \angle 0^\circ \\ &= 0.417 \, \text{S} + j0.417 \, \text{S} + 0.1 \, \text{S} \end{aligned}$$

$$\mathbf{Y}_T = 0.517 \, \text{S} + j0.417 \, \text{S} = \mathbf{0.66 \, S} \angle \mathbf{38.89^\circ}$$

$$\mathbf{Z}_T = \frac{1}{\mathbf{Y}_T} = \frac{1}{0.66 \, \text{S} \angle 38.89^\circ} = \mathbf{1.52 \, \Omega} \angle \mathbf{-38.89^\circ}$$

$$\mathbf{b.} \quad \mathbf{V}_1 = \frac{(2 \, \Omega \angle 0^\circ)(60 \, \text{V} \angle 0^\circ)}{2 \, \Omega - j2 \, \Omega} = \frac{120 \, \text{V} \angle 0^\circ}{2.828 \angle -45^\circ} = \mathbf{42.43 \, V} \angle \mathbf{45^\circ}$$

$$\begin{aligned} \mathbf{c.} \quad \mathbf{I}_1 &= \frac{\mathbf{E}}{\mathbf{Z}'} = \frac{60 \, \text{V} \angle 0^\circ}{3 \, \Omega - j9 \, \Omega + j6 \, \Omega} = \frac{60 \, \text{V} \angle 0^\circ}{3 \, \Omega - j3 \, \Omega} = \frac{60 \, \text{V} \angle 0^\circ}{4.243 \angle -45^\circ} \\ &= \mathbf{14.14 \, A} \angle \mathbf{45^\circ} \end{aligned}$$

$$\mathbf{d.} \quad \mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{60 \, \text{V} \angle 0^\circ}{1.52 \, \Omega \angle -38.89^\circ} = \mathbf{39.47 \, A} \angle \mathbf{38.89^\circ}$$

$$12. \quad \mathbf{Z}' = 12 \, \Omega - j20 \, \Omega = 23.32 \, \Omega \angle -59.04^\circ$$

$$R_4 \angle 0^\circ \parallel \mathbf{Z}' = 20 \, \Omega \angle 0^\circ \parallel 23.32 \, \Omega \angle -59.04^\circ = 12.36 \, \Omega \angle -27.03^\circ$$

$$\begin{aligned} \mathbf{Z}'' &= R_3 \angle 0^\circ + R_4 \angle 0^\circ \parallel \mathbf{Z}' = 12 \, \Omega + 12.36 \, \Omega \angle -27.03^\circ \\ &= 12 \, \Omega + (11.01 \, \Omega - j5.62 \, \Omega) \\ &= 23.01 \, \Omega - j5.62 \, \Omega = 23.69 \, \Omega \angle -13.73^\circ \end{aligned}$$

$$R_2 \angle 0^\circ \parallel \mathbf{Z}'' = 20 \, \Omega \angle 0^\circ \parallel 23.69 \, \Omega \angle -13.73^\circ = 10.92 \, \Omega \angle -6.29^\circ$$

$$\begin{aligned} \mathbf{Z}_T &= R_1 \angle 0^\circ + R_2 \angle 0^\circ \parallel \mathbf{Z}'' = 12 \, \Omega + 10.92 \, \Omega \angle -6.29^\circ \\ &= 12 \, \Omega + (10.85 \, \Omega - j1.25 \, \Omega) \\ &= 22.85 \, \Omega - j1.2 \, \Omega = \mathbf{22.88 \, \Omega} \angle \mathbf{-3.01^\circ} \end{aligned}$$

$$\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{100 \, \text{V} \angle 0^\circ}{22.88 \, \Omega \angle -3.01^\circ} = \mathbf{4.37 \, A} \angle \mathbf{+3.01^\circ}$$

$$\mathbf{I}_{R_1} = \mathbf{I}$$

$$\begin{aligned} \mathbf{I}_{R_3} &= \frac{R_2 \angle 0^\circ \mathbf{I}_s}{R_2 \angle 0^\circ + \mathbf{Z}''} = \frac{(20 \, \Omega \angle 0^\circ)(4.37 \, \text{A} \angle 3.01^\circ)}{20 \, \Omega + 23.01 \, \Omega - j5.62 \, \Omega} = \frac{87.40 \, \text{A} \angle 3.01^\circ}{43.38 \angle -7.44^\circ} \\ &= 2.01 \, \text{A} \angle 10.45^\circ \end{aligned}$$

$$\begin{aligned} \mathbf{I}_4 &= \frac{R_4 \angle 0^\circ \mathbf{I}_{R_3}}{R_4 \angle 0^\circ + \mathbf{Z}'} = \frac{(20 \, \Omega \angle 0^\circ)(2.01 \, \text{A} \angle 10.45^\circ)}{20 \, \Omega + 12 \, \Omega - j20 \, \Omega} = \frac{40.20 \, \text{A} \angle 10.45^\circ}{32 \, \Omega - j20 \, \Omega} = \frac{40.20 \, \text{A} \angle 10.45^\circ}{37.74 \angle -32.01^\circ} \\ &= \mathbf{1.07 \, A} \angle \mathbf{42.46^\circ} \end{aligned}$$

$$\begin{aligned}
13. \quad R_3 + R_4 &= 2.7 \text{ k}\Omega + 4.3 \text{ k}\Omega = 7 \text{ k}\Omega \\
R' &= 3 \text{ k}\Omega \parallel 7 \text{ k}\Omega = 2.1 \text{ k}\Omega \\
Z' &= 2.1 \text{ k}\Omega - j10 \Omega
\end{aligned}$$

$$\begin{aligned}
(\text{CDR}) \quad \mathbf{I}' \text{ (of } 10 \Omega \text{ cap.)} &= \frac{(40 \text{ k}\Omega \angle 0^\circ)(20 \text{ mA} \angle 0^\circ)}{40 \text{ k}\Omega + 2.1 \text{ k}\Omega - j10 \Omega} \\
&= 19 \text{ mA} \angle +0.014^\circ \text{ as expected since } R_1 \gg Z'
\end{aligned}$$

$$\begin{aligned}
(\text{CDR}) \quad \mathbf{I}_4 &= \frac{(3 \text{ k}\Omega \angle 0^\circ)(19 \text{ mA} \angle 0.014^\circ)}{3 \text{ k}\Omega + 7 \text{ k}\Omega} = \frac{57 \text{ mA} \angle 0.014^\circ}{10} \\
&= 5.7 \text{ mA} \angle 0.014^\circ \\
P &= I^2 R = (5.7 \text{ mA})^2 4.3 \text{ k}\Omega = \mathbf{139.71 \text{ mW}}
\end{aligned}$$

$$\begin{aligned}
14. \quad Z' &= X_{C_2} \angle -90^\circ \parallel R_1 \angle 0^\circ = 2 \Omega \angle -90^\circ \parallel 1 \Omega \angle 0^\circ \\
&= \frac{2 \Omega \angle -90^\circ}{1 - j2} = \frac{2 \Omega \angle -90^\circ}{2.236 \angle -63.435^\circ} \\
&= 0.894 \Omega \angle -26.565^\circ \\
Z'' &= X_{L_2} \angle 90^\circ + Z' = +j8 \Omega + 0.894 \Omega \angle -26.565^\circ \\
&= +j8 \Omega + (0.8 \Omega - j4 \Omega) \\
&= 0.8 \Omega + j4 = 4.079 \Omega \angle 78.69^\circ
\end{aligned}$$

$$\begin{aligned}
\mathbf{I}_{X_{L_2}} &= \frac{X_{C_1} \angle -90^\circ \mathbf{I}}{X_{C_1} \angle -90^\circ + Z''} = \frac{(2 \Omega \angle -90^\circ)(0.5 \text{ A} \angle 0^\circ)}{-j2 \Omega + (0.8 \Omega + j4 \Omega)} = \frac{1 \text{ A} \angle -90^\circ}{0.8 + j2} \\
&= \frac{1 \text{ A} \angle -90^\circ}{2.154 \angle 68.199^\circ} = 0.464 \text{ A} \angle -158.99^\circ
\end{aligned}$$

$$\begin{aligned}
\mathbf{I}_1 &= \frac{X_{C_2} \angle -90^\circ \mathbf{I}_{X_{C_2}}}{X_{C_2} \angle -90^\circ + R_1} = \frac{(2 \Omega \angle -90^\circ)(0.464 \text{ A} \angle -158.99^\circ)}{-j2 \Omega + 1 \Omega} = \frac{0.928 \text{ A} \angle -248.99^\circ}{2.236 \angle -63.435^\circ} \\
&= \mathbf{0.42 \text{ A} \angle 174.45^\circ}
\end{aligned}$$

Chapter 17

1. –

2. $\mathbf{Z} = -j5 \, \Omega + 2 \, \Omega \angle 0^\circ \parallel 5 \, \Omega \angle 90^\circ = -j5 \, \Omega + 1.72 \, \Omega + j0.69 \, \Omega = \mathbf{4.64 \, \Omega \angle -68.24^\circ}$

$$\mathbf{I} = \frac{\mathbf{E}}{\mathbf{Z}} = \frac{60 \, \text{V} \angle 30^\circ}{4.64 \, \Omega \angle -68.24^\circ} = \mathbf{12.93 \, \text{A} \angle 98.24^\circ}$$

3. $\mathbf{Z} = 10 \, \Omega \angle 0^\circ \parallel 6 \, \Omega \angle 90^\circ = \mathbf{5.15 \, \Omega \angle 59.04^\circ}$

$$\mathbf{E} = \mathbf{IZ} = (2 \, \text{A} \angle 120^\circ)(5.15 \, \Omega \angle 59.04^\circ) = \mathbf{10.30 \, \text{V} \angle 179.04^\circ}$$

4. a. $\mathbf{I} = \frac{\mu\text{V}}{R} = \frac{16 \, \text{V}}{4 \times 10^3} = \mathbf{4 \times 10^{-3} \, \text{V}}$
 $\mathbf{Z} = \mathbf{4 \, k\Omega \angle 0^\circ}$

b. $\mathbf{V} = (h\mathbf{I})(R) = (50 \, \text{I})(50 \, \text{k}\Omega) = \mathbf{2.5 \times 10^6 \, \text{I}}$
 $\mathbf{Z} = \mathbf{50 \, k\Omega \angle 0^\circ}$

5. Clockwise mesh currents:

$$\mathbf{E} - \mathbf{I}_1 \mathbf{Z}_1 - \mathbf{I}_1 \mathbf{Z}_2 + \mathbf{I}_2 \mathbf{Z}_2 = 0$$

$$-\mathbf{I}_2 \mathbf{Z}_2 + \mathbf{I}_1 \mathbf{Z}_2 - \mathbf{I}_2 \mathbf{Z}_3 - \mathbf{E}_2 = 0$$

$$[\mathbf{Z}_1 + \mathbf{Z}_2]\mathbf{I}_1 - \mathbf{Z}_2 \mathbf{I}_2 = \mathbf{E}_1$$

$$-\mathbf{Z}_2 \mathbf{I}_1 + [\mathbf{Z}_2 + \mathbf{Z}_3]\mathbf{I}_2 = -\mathbf{E}_2$$

$$\mathbf{Z}_1 = R_1 \angle 0^\circ = 4 \, \Omega \angle 0^\circ$$

$$\mathbf{Z}_2 = X_L \angle 90^\circ = 6 \, \Omega \angle 90^\circ$$

$$\mathbf{Z}_3 = X_C \angle -90^\circ = 8 \, \Omega \angle -90^\circ$$

$$\mathbf{E}_1 = 10 \, \text{V} \angle 0^\circ, \mathbf{E}_2 = 40 \, \text{V} \angle 60^\circ$$

$$\mathbf{I}_{R_1} = \mathbf{I}_1 = \frac{\begin{vmatrix} \mathbf{E}_1 & -\mathbf{Z}_2 \\ -\mathbf{E}_2 & [\mathbf{Z}_2 + \mathbf{Z}_3] \end{vmatrix}}{\begin{vmatrix} [\mathbf{Z}_1 + \mathbf{Z}_2] & -\mathbf{Z}_2 \\ -\mathbf{Z}_2 & [\mathbf{Z}_2 + \mathbf{Z}_3] \end{vmatrix}} = \frac{[\mathbf{Z}_2 + \mathbf{Z}_3]\mathbf{E}_1 - \mathbf{Z}_2 \mathbf{E}_2}{\mathbf{Z}_1 \mathbf{Z}_2 + \mathbf{Z}_1 \mathbf{Z}_3 + \mathbf{Z}_2 \mathbf{Z}_3} = \mathbf{5.15 \, \text{A} \angle -24.5^\circ}$$

6. By interchanging the right two branches, the general configuration of Problem 5 will result and

$$\mathbf{I}_{50\Omega} = \mathbf{I}_1 = \frac{[\mathbf{Z}_2 + \mathbf{Z}_3]\mathbf{E}_1 - \mathbf{Z}_2 \mathbf{E}_2}{\mathbf{Z}_1 \mathbf{Z}_2 + \mathbf{Z}_1 \mathbf{Z}_3 + \mathbf{Z}_2 \mathbf{Z}_3} = \mathbf{0.44 \, \text{A} \angle 143.48^\circ}$$

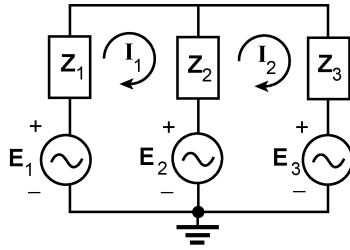
$$\mathbf{Z}_1 = R_1 = 50 \, \Omega \angle 0^\circ$$

$$\mathbf{Z}_2 = X_C \angle -90^\circ = 60 \, \Omega \angle -90^\circ$$

$$\mathbf{Z}_3 = X_L \angle 90^\circ = 20 \, \Omega \angle 90^\circ$$

$$\mathbf{E}_1 = 5 \, \text{V} \angle 30^\circ, \mathbf{E}_2 = 20 \, \text{V} \angle 0^\circ$$

7. a.



$$\begin{aligned} Z_1 &= 12 \Omega + j12 \Omega = 16.971 \Omega \angle 45^\circ \\ Z_2 &= 3 \Omega \angle 0^\circ \\ Z_3 &= -j1 \Omega \\ E_1 &= 20 \text{ V} \angle 50^\circ \\ E_2 &= 60 \text{ V} \angle 70^\circ \\ E_3 &= 40 \text{ V} \angle 0^\circ \end{aligned}$$

$$\begin{aligned} I_1[Z_1 + Z_2] - Z_2 I_2 &= E_1 - E_2 \\ I_2[Z_2 + Z_3] - Z_3 I_1 &= E_2 - E_3 \end{aligned}$$

$$\begin{aligned} (Z_1 + Z_2)I_1 - Z_2 I_2 &= E_1 - E_2 \\ -Z_3 I_1 + (Z_2 + Z_3)I_2 &= E_2 - E_3 \end{aligned}$$

Using determinants:

$$I_{R_1} = I_1 = \frac{(E_1 - E_2)(Z_2 + Z_3) + Z_2(E_2 - E_3)}{Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_3} = 2.55 \text{ A} \angle 132.72^\circ$$

8. Clockwise mesh currents:

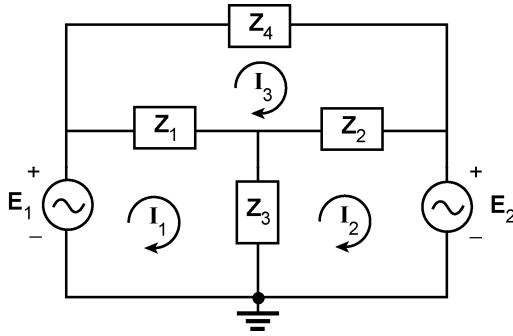
$$\begin{aligned} E_1 - I_1 Z_1 - I_1 Z_2 + I_2 Z_2 &= 0 \\ -I_2 Z_2 + I_1 Z_2 - I_2 Z_3 - I_2 Z_4 + I_3 Z_4 &= 0 \\ -I_3 Z_4 + I_2 Z_4 - I_3 Z_5 - E_2 &= 0 \end{aligned}$$

$$\begin{aligned} Z_1 &= 4 \Omega + j3 \Omega, Z_2 = -j1 \Omega \\ Z_3 &= +j6 \Omega, Z_4 = j2 \Omega \\ Z_5 &= 8 \Omega \\ E_1 &= 60 \text{ V} \angle 0^\circ, E_2 = 120 \text{ V} \angle 120^\circ \end{aligned}$$

$$\begin{aligned} [Z_1 + Z_2]I_1 - Z_2 I_2 + 0 &= E_1 \\ -Z_2 I_1 + [Z_2 + Z_3 + Z_4]I_2 - Z_4 I_3 &= 0 \\ 0 - Z_4 I_2 + [Z_4 + Z_5]I_3 &= -E_2 \end{aligned}$$

$$\begin{aligned} I_{R_1} = I_3 &= \frac{[Z_2 Z_4]E_1 + [Z_2^2 - [Z_1 + Z_2][Z_2 + Z_3 + Z_4]]E_2}{[Z_1 + Z_2][Z_2 + Z_3 + Z_4][Z_4 + Z_5] - [Z_1 + Z_2]Z_4^2 - [Z_4 + Z_5]Z_2^2} \\ &= 13.07 \text{ A} \angle -33.71^\circ \end{aligned}$$

9.



$$\begin{aligned} Z_1 &= 15 \Omega \angle 0^\circ, Z_2 = 15 \Omega \angle 0^\circ \\ Z_3 &= -j10 \Omega = 10 \Omega \angle -90^\circ \\ Z_4 &= 3 \Omega + j4 \Omega = 5 \Omega \angle 53.13^\circ \\ E_1 &= 220 \text{ V } \angle 0^\circ \\ E_2 &= 100 \text{ V } \angle 90^\circ \end{aligned}$$

$$\begin{aligned} I_1(Z_1 + Z_3) - I_2Z_3 - I_3Z_1 &= E_1 \\ I_2(Z_2 + Z_3) - I_1Z_3 - I_3Z_2 &= -E_2 \\ I_3(Z_1 + Z_2 + Z_4) - I_1Z_1 - I_2Z_2 &= 0 \end{aligned}$$

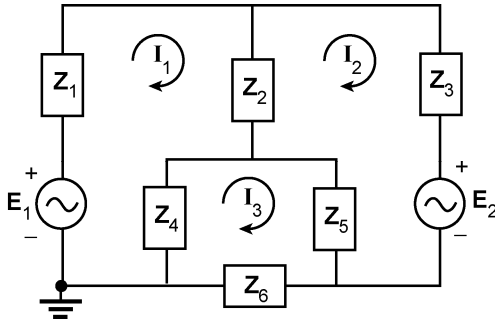
$$\begin{array}{rrr} I_1(Z_1 + Z_3) & - I_2Z_3 & - I_3Z_1 & = E_1 \\ -I_1Z_3 & + I_2(Z_2 + Z_3) & - I_3Z_2 & = -E_2 \\ -I_1Z_1 & - I_2Z_2 & + I_3(Z_1 + Z_2 + Z_4) & = 0 \end{array}$$

Applying determinants:

$$\begin{aligned} I_3 &= \frac{-(Z_1 + Z_3)(Z_2)E_2 - Z_1Z_3E_2 + E_1[Z_2Z_3 + Z_1(Z_2 + Z_3)]}{(Z_1 + Z_3)[(Z_2 + Z_3)(Z_1 + Z_2 + Z_4) - Z_2^2] + Z_3[Z_3(Z_1 + Z_2 + Z_4) - Z_1Z_2] - Z_1[-Z_2Z_3 - Z_1(Z_2 + Z_3)]} \\ &= 48.33 \text{ A } \angle -77.57^\circ \end{aligned}$$

$$\text{or } I_3 = \frac{E_1 - E_2}{Z_4} \text{ if one carefully examines the network!}$$

10.



$$\begin{aligned} Z_1 &= 5 \Omega \angle 0^\circ, Z_2 = 5 \Omega \angle 90^\circ \\ Z_3 &= 4 \Omega \angle 0^\circ, Z_4 = 6 \Omega \angle -90^\circ \\ Z_5 &= 4 \Omega \angle 0^\circ, Z_6 = 6 \Omega + j8 \Omega \\ E_1 &= 20 \text{ V } \angle 0^\circ, E_2 = 40 \text{ V } \angle 60^\circ \end{aligned}$$

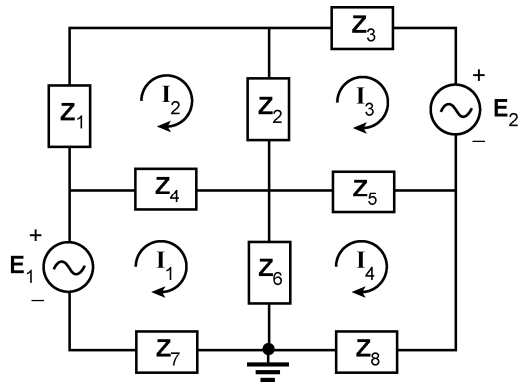
$$\begin{aligned} I_1(Z_1 + Z_2 + Z_4) - I_2Z_2 - I_3Z_4 &= E_1 \\ I_2(Z_2 + Z_3 + Z_5) - I_1Z_2 - I_3Z_5 &= -E_2 \\ I_3(Z_4 + Z_5 + Z_6) - I_1Z_4 - I_2Z_5 &= 0 \end{aligned}$$

$$\begin{array}{rrr} (Z_1 + Z_2 + Z_4)I_1 & - Z_2I_2 & - Z_4I_3 & = E_1 \\ -Z_2I_1 & + (Z_2 + Z_3 + Z_5)I_2 & - Z_5I_3 & = -E_2 \\ -Z_4I_1 & - Z_5I_2 & + (Z_4 + Z_5 + Z_6)I_3 & = 0 \end{array}$$

Using $Z' = Z_1 + Z_2 + Z_4$, $Z'' = Z_2 + Z_3 + Z_5$, $Z''' = Z_4 + Z_5 + Z_6$ and determinants:

$$\begin{aligned} I_{R_1} = I_1 &= \frac{E_1(Z''Z''' - Z_5^2) - E_2(Z_2Z''' + Z_4Z_5)}{Z'(Z''Z''' - Z_5^2) - Z_2(Z_2Z''' + Z_4Z_5) - Z_4(Z_2Z_5 + Z_4Z'')} \\ &= 3.04 \text{ A } \angle 169.12^\circ \end{aligned}$$

11.



$$\begin{aligned} Z_1 &= 10 \, \Omega + j20 \, \Omega \\ Z_3 &= 80 \, \Omega \angle 0^\circ \\ Z_5 &= 15 \, \Omega \angle 90^\circ \\ Z_7 &= 5 \, \Omega \angle 0^\circ \\ E_1 &= 25 \, \text{V} \angle 0^\circ \end{aligned}$$

$$\begin{aligned} Z_2 &= -j20 \, \Omega \\ Z_4 &= 6 \, \Omega \angle 0^\circ \\ Z_6 &= 10 \, \Omega \angle 0^\circ \\ Z_8 &= 5 \, \Omega - j20 \, \Omega \\ E_2 &= 75 \, \text{V} \angle 20^\circ \end{aligned}$$

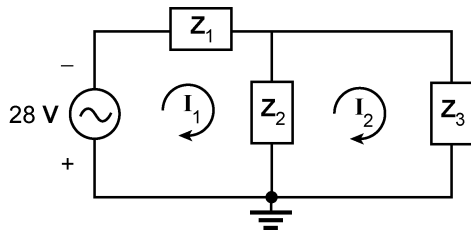
$$\begin{aligned} I_1(Z_4 + Z_6 + Z_7) - I_2Z_4 - I_4Z_6 &= E_1 \\ I_2(Z_1 + Z_2 + Z_4) - I_1Z_4 - I_3Z_2 &= 0 \\ I_3(Z_2 + Z_3 + Z_5) - I_2Z_2 - I_4Z_5 &= -E_2 \\ I_4(Z_5 + Z_6 + Z_8) - I_1Z_6 - I_3Z_5 &= 0 \end{aligned}$$

$$\begin{array}{ccccccc} (Z_4 + Z_6 + Z_7) I_1 & & -Z_4 I_2 & & +0 & & -Z_6 I_4 = E_1 \\ & -Z_4 I_1 & + (Z_1 + Z_2 + Z_4) I_2 & & -Z_2 I_3 & & +0 = 0 \\ & 0 & & -Z_2 I_2 & + (Z_2 + Z_3 + Z_5) I_3 & & -Z_5 I_4 = -E_2 \\ -Z_6 I_1 & & +0 & & -Z_5 I_3 & + (Z_5 + Z_6 + Z_7) I_4 & = 0 \end{array}$$

Applying determinants:

$$I_{R1} = I_{80\Omega} = \mathbf{0.68 \, \text{A} \angle -162.9^\circ}$$

12.



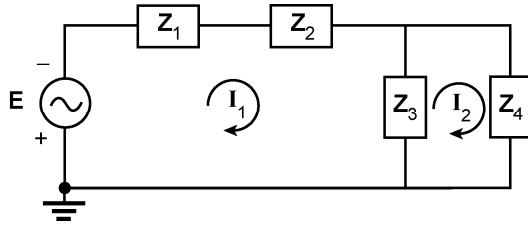
$$\begin{aligned} Z_1 &= 5 \, \text{k}\Omega \angle 0^\circ \\ Z_2 &= 10 \, \text{k}\Omega \angle 0^\circ \\ Z_3 &= 1 \, \text{k}\Omega + j4 \, \text{k}\Omega = 4.123 \, \text{k}\Omega \angle 75.96^\circ \end{aligned}$$

$$\begin{aligned} I_1(Z_1 + Z_2) - Z_2 I_2 &= -28 \, \text{V} \\ I_2(Z_2 + Z_3) - Z_2 I_1 &= 0 \end{aligned}$$

$$\begin{aligned} (Z_1 + Z_2) I_1 - Z_2 I_2 &= -28 \, \text{V} \\ -Z_2 I_1 + (Z_2 + Z_3) I_2 &= 0 \end{aligned}$$

$$I_L = I_2 = \frac{-Z_2 28 \, \text{V}}{Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_3} = \mathbf{-3.17 \times 10^{-3} \, \text{V} \angle 137.29^\circ}$$

13.



Source Conversion:

$$\begin{aligned}
 \mathbf{E} &= (I \angle \theta)(R_p \angle 0^\circ) \\
 &= (50 \text{ I})(40 \text{ k}\Omega \angle 0^\circ) \\
 &= 2 \times 10^6 \text{ I} \angle 0^\circ \\
 \mathbf{Z}_1 &= R_s = R_p = 40 \text{ k}\Omega \angle 0^\circ \\
 \mathbf{Z}_2 &= -j0.2 \text{ k}\Omega \\
 \mathbf{Z}_3 &= 8 \text{ k}\Omega \angle 0^\circ \\
 \mathbf{Z}_4 &= 4 \text{ k}\Omega \angle 90^\circ
 \end{aligned}$$

$$\mathbf{I}_1(\mathbf{Z}_1 + \mathbf{Z}_2 + \mathbf{Z}_3) - \mathbf{Z}_3 \mathbf{I}_2 = -\mathbf{E}$$

$$\mathbf{I}_2(\mathbf{Z}_3 + \mathbf{Z}_4) - \mathbf{Z}_3 \mathbf{I}_1 = 0$$

$$(\mathbf{Z}_1 + \mathbf{Z}_2 + \mathbf{Z}_3)\mathbf{I}_1 - \mathbf{Z}_3 \mathbf{I}_2 = -\mathbf{E}$$

$$-\mathbf{Z}_3 \mathbf{I}_1 + (\mathbf{Z}_3 + \mathbf{Z}_4)\mathbf{I}_2 = 0$$

$$\mathbf{I}_L = \mathbf{I}_2 = \frac{-\mathbf{Z}_3 \mathbf{E}}{(\mathbf{Z}_1 + \mathbf{Z}_2 + \mathbf{Z}_3)(\mathbf{Z}_3 + \mathbf{Z}_4) - \mathbf{Z}_3^2} = 42.91 \text{ I} \angle 149.31^\circ$$

14. $6\mathbf{V}_x - \mathbf{I}_1 1 \text{ k}\Omega - 10 \text{ V} \angle 0^\circ = 0$

$$10 \text{ V} \angle 0^\circ - \mathbf{I}_2 4 \text{ k}\Omega - \mathbf{I}_2 2 \text{ k}\Omega = 0$$

$$\mathbf{V}_x = \mathbf{I}_2 2 \text{ k}\Omega$$

$$-\mathbf{I}_1 1 \text{ k}\Omega + \mathbf{I}_2 12 \text{ k}\Omega = 10 \text{ V} \angle 0^\circ$$

$$-\mathbf{I}_2 6 \text{ k}\Omega = -10 \text{ V} \angle 0^\circ$$

$$\mathbf{I}_2 = \mathbf{I}_{2\text{k}\Omega} = \frac{10 \text{ V} \angle 0^\circ}{6 \text{ k}\Omega} = 1.67 \text{ mA} \angle 0^\circ = \mathbf{I}_{2\text{k}\Omega}$$

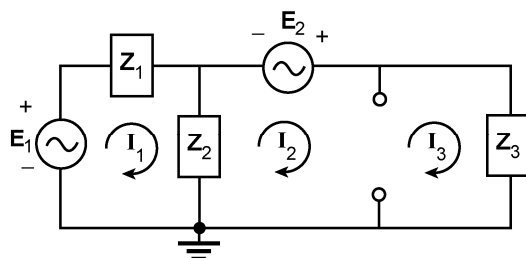
$$-\mathbf{I}_1 1 \text{ k}\Omega + (1.667 \text{ mA} \angle 0^\circ)(12 \text{ k}\Omega) = 10 \text{ V} \angle 0^\circ$$

$$-\mathbf{I}_1 1 \text{ k}\Omega + 20 \text{ V} \angle 0^\circ = 10 \text{ V} \angle 0^\circ$$

$$-\mathbf{I}_1 1 \text{ k}\Omega = -10 \text{ V} \angle 0^\circ$$

$$\mathbf{I}_1 = \mathbf{I}_{1\text{k}\Omega} = \frac{10 \text{ V} \angle 0^\circ}{1 \text{ k}\Omega} = 10 \text{ mA} \angle 0^\circ$$

15.



$$\begin{aligned} E_1 &= 5 \text{ V } \angle 0^\circ \\ E_2 &= 20 \text{ V } \angle 0^\circ \\ Z_1 &= 2.2 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 5 \text{ k}\Omega \angle 90^\circ \\ Z_3 &= 10 \text{ k}\Omega \angle 0^\circ \\ I &= 4 \text{ mA } \angle 0^\circ \end{aligned}$$

$$\begin{aligned} E_1 - I_1 Z_1 - Z_2(I_1 - I_2) &= 0 \\ -Z_2(I_2 - I_1) + E_2 - I_3 Z_3 &= 0 \end{aligned}$$

$$I_3 - I_2 = I$$

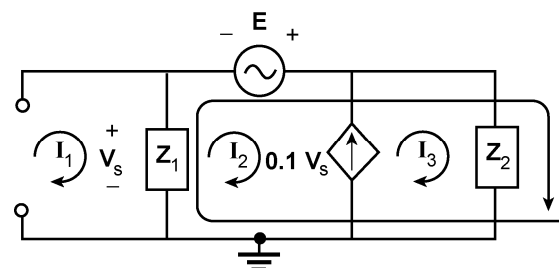
Substituting, we obtain:

$$\begin{aligned} I_1(Z_1 + Z_2) - I_2 Z_2 &= E_1 \\ I_1 Z_2 - I_2(Z_2 + Z_3) &= I Z_3 - E_2 \end{aligned}$$

Determinants:

$$\begin{aligned} I_1 &= 1.39 \text{ mA } \angle -126.48^\circ, I_2 = 1.341 \text{ mA } \angle -10.56^\circ, I_3 = 2.693 \text{ mA } \angle -174.8^\circ \\ I_{10\text{k}\Omega} &= I_3 = \mathbf{2.69 \text{ mA } \angle -174.8^\circ} \end{aligned}$$

16.



$$\begin{aligned} Z_1 &= 1 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 4 \text{ k}\Omega + j6 \text{ k}\Omega \\ E &= 10 \text{ V } \angle 0^\circ \end{aligned}$$

$$\begin{aligned} -Z_1(I_2 - I_1) + E - I_3 Z_2 &= 0 \\ I_1 &= 6 \text{ mA } \angle 0^\circ, 0.1 V_s = I_3 - I_2, V_s = (I_1 - I_2)Z_1 \end{aligned}$$

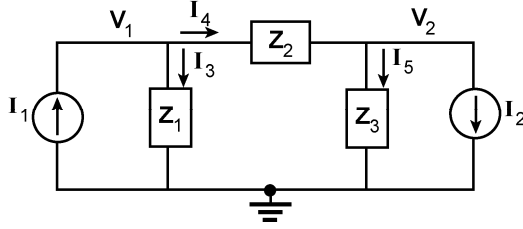
Substituting:

$$\begin{aligned} (1 \text{ k}\Omega)I_2 + (4 \text{ k}\Omega + j6 \text{ k}\Omega)I_3 &= 16 \text{ V } \angle 0^\circ \\ (99 \Omega)I_2 + I_3 &= 0.6 \text{ V } \angle 0^\circ \end{aligned}$$

Determinants:

$$I_3 = I_{6 \text{ k}\Omega} = \mathbf{1.38 \text{ mA } \angle -56.31^\circ}$$

17.



$$\begin{aligned} Z_1 &= 4 \, \Omega \angle 0^\circ \\ Z_2 &= 5 \, \Omega \angle 90^\circ \\ Z_3 &= 2 \, \Omega \angle -90^\circ \\ I_1 &= 3 \, \text{A} \angle 0^\circ \\ I_2 &= 5 \, \text{A} \angle 30^\circ \end{aligned}$$

$$I_1 = I_3 + I_4$$

$$I_1 = \frac{V_1}{Z_1} + \frac{V_1 - V_2}{Z_2} \Rightarrow V_1 \left[\frac{1}{Z_1} + \frac{1}{Z_2} \right] - V_2 \left[\frac{1}{Z_2} \right] = I_1$$

$$\text{or } V_1[Y_1 + Y_2] - V_2[Y_2] = I_1$$

$$I_4 = I_5 + I_2$$

$$\frac{V_1 - V_2}{Z_2} = \frac{V_2}{Z_3} + I_2 \Rightarrow V_2 \left[\frac{1}{Z_2} + \frac{1}{Z_3} \right] - V_1 \left[\frac{1}{Z_2} \right] = -I_2$$

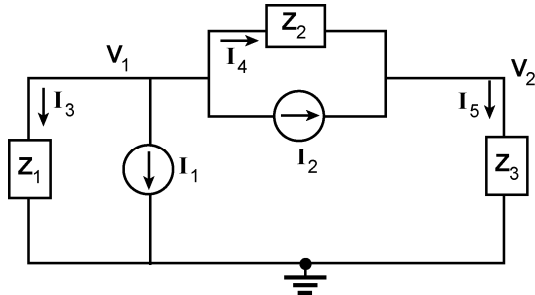
$$\text{or } V_2[Y_2 + Y_3] - V_1[Y_2] = -I_2$$

$$\begin{aligned} [Y_1 + Y_2]V_1 - Y_2 V_2 &= I_1 \\ -Y_2 V_1 + [Y_2 + Y_3]V_2 &= -I_2 \end{aligned}$$

$$V_1 = \frac{[Y_2 + Y_3]I_1 - Y_2 I_2}{Y_1 Y_2 + Y_1 Y_3 + Y_2 Y_3} = 14.68 \, \text{V} \angle 68.89^\circ$$

$$V_2 = \frac{-[Y_1 + Y_2]I_2 + Y_2 I_1}{Y_1 Y_2 + Y_1 Y_3 + Y_2 Y_3} = 12.97 \, \text{V} \angle 155.88^\circ$$

18.



$$\begin{aligned} Z_1 &= 3 \, \Omega + j4 \, \Omega = 5 \, \Omega \angle 53.13^\circ \\ Z_2 &= 2 \, \Omega \angle 0^\circ \\ Z_3 &= 6 \, \Omega \angle 0^\circ \parallel 8 \, \Omega \angle -90^\circ \\ &= 4.8 \, \Omega \angle -36.87^\circ \\ I_1 &= 0.6 \, \text{A} \angle 20^\circ \\ I_2 &= 4 \, \text{A} \angle 80^\circ \end{aligned}$$

$$0 = I_1 + I_3 + I_4 + I_2$$

$$0 = I_1 + \frac{V_1}{Z_1} + \frac{V_1 - V_2}{Z_2} + I_2$$

$$V_1 \left[\frac{1}{Z_1} + \frac{1}{Z_2} \right] - V_2 \left[\frac{1}{Z_2} \right] = -I_1 - I_2$$

$$\text{or } V_1[Y_1 + Y_2] - V_2[Y_2] = -I_1 - I_2$$

$$I_2 + I_4 = I_5$$

$$I_2 + \frac{V_1 - V_2}{Z_2} = \frac{V_2}{Z_3}$$

$$\mathbf{V}_2 \left[\frac{1}{\mathbf{Z}_2} + \frac{1}{\mathbf{Z}_3} \right] - \mathbf{V}_1 \left[\frac{1}{\mathbf{Z}_2} \right] = +\mathbf{I}_2$$

$$\text{or } \mathbf{V}_2[\mathbf{Y}_2 + \mathbf{Y}_3] - \mathbf{V}_1[\mathbf{Y}_2] = \mathbf{I}_2$$

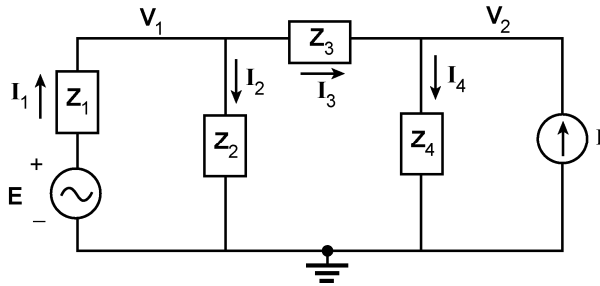
$$\text{and } \begin{aligned} [\mathbf{Y}_1 + \mathbf{Y}_2]\mathbf{V}_1 - \mathbf{Y}_2\mathbf{V}_2 &= -\mathbf{I}_1 - \mathbf{I}_2 \\ -\mathbf{Y}_2\mathbf{V}_1 + [\mathbf{Y}_2 + \mathbf{Y}_3]\mathbf{V}_2 &= \mathbf{I}_2 \end{aligned}$$

Applying determinants:

$$\mathbf{V}_1 = \frac{-[\mathbf{Y}_2 + \mathbf{Y}_3][\mathbf{I}_1 + \mathbf{I}_2] + \mathbf{Y}_2\mathbf{I}_2}{\mathbf{Y}_1\mathbf{Y}_2 + \mathbf{Y}_1\mathbf{Y}_3 + \mathbf{Y}_2\mathbf{Y}_3} = 5.12 \text{ V } \angle -79.36^\circ$$

$$\mathbf{V}_2 = \frac{\mathbf{Y}_1\mathbf{I}_2 - \mathbf{I}_1\mathbf{Y}_2}{\mathbf{Y}_1\mathbf{Y}_2 + \mathbf{Y}_1\mathbf{Y}_3 + \mathbf{Y}_2\mathbf{Y}_3} = 2.71 \text{ V } \angle 39.96^\circ$$

19.



$$\begin{aligned} \mathbf{Z}_1 &= 5 \, \Omega \, \angle 0^\circ \\ \mathbf{Z}_2 &= 6 \, \Omega \, \angle 90^\circ \\ \mathbf{Z}_3 &= 4 \, \Omega \, \angle -90^\circ \\ \mathbf{Z}_4 &= 2 \, \Omega \, \angle 0^\circ \\ \mathbf{E} &= 30 \text{ V } \angle 50^\circ \\ \mathbf{I} &= 0.04 \text{ A } \angle 90^\circ \end{aligned}$$

$$\mathbf{I}_1 = \mathbf{I}_2 + \mathbf{I}_3$$

$$\frac{\mathbf{E}_1 - \mathbf{V}_1}{\mathbf{Z}_1} = \frac{\mathbf{V}_1}{\mathbf{Z}_2} + \frac{(\mathbf{V}_1 - \mathbf{V}_2)}{\mathbf{Z}_3} \Rightarrow \mathbf{V}_1 \left[\frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2} + \frac{1}{\mathbf{Z}_3} \right] - \frac{\mathbf{V}_2}{\mathbf{Z}_3} = \frac{\mathbf{E}_1}{\mathbf{Z}_1}$$

$$\text{or } \mathbf{V}_1[\mathbf{Y}_1 + \mathbf{Y}_2 + \mathbf{Y}_3] - \mathbf{Y}_3\mathbf{V}_2 = \mathbf{E}_1\mathbf{Y}_1$$

$$\mathbf{I}_3 + \mathbf{I} = \mathbf{I}_4$$

$$\frac{\mathbf{V}_1 - \mathbf{V}_2}{\mathbf{Z}_3} + \mathbf{I} = \frac{\mathbf{V}_2}{\mathbf{Z}_4} \Rightarrow \mathbf{V}_2 \left[\frac{1}{\mathbf{Z}_3} + \frac{1}{\mathbf{Z}_4} \right] - \frac{\mathbf{V}_1}{\mathbf{Z}_3} = \mathbf{I}$$

$$\text{or } \mathbf{V}_2[\mathbf{Y}_3 + \mathbf{Y}_4] - \mathbf{V}_1\mathbf{Y}_3 = \mathbf{I}$$

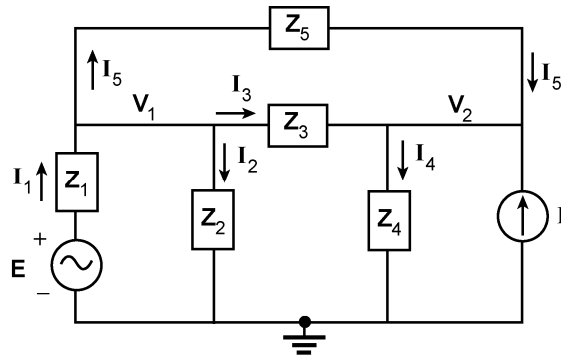
resulting in

$$\begin{aligned} \mathbf{V}_1[\mathbf{Y}_1 + \mathbf{Y}_2 + \mathbf{Y}_3] - \mathbf{V}_2\mathbf{Y}_3 &= \mathbf{E}_1\mathbf{Y}_1 \\ -\mathbf{V}_1[\mathbf{Y}_3] + \mathbf{V}_2[\mathbf{Y}_3 + \mathbf{Y}_4] &= +\mathbf{I} \end{aligned}$$

Using determinants:

$$\mathbf{V}_1 = 19.86 \text{ V } \angle 43.8^\circ \text{ and } \mathbf{V}_2 = 8.94 \text{ V } \angle 106.9^\circ$$

20.



$$\begin{aligned} Z_1 &= 10 \, \Omega \angle 0^\circ \\ Z_2 &= 10 \, \Omega \angle 0^\circ \\ Z_3 &= 4 \, \Omega \angle 90^\circ \\ Z_4 &= 2 \, \Omega \angle 0^\circ \\ Z_5 &= 8 \, \Omega \angle -90^\circ \\ E &= 50 \, \text{V} \angle 120^\circ \\ I &= 0.8 \, \text{A} \angle 70^\circ \end{aligned}$$

$$I_1 = I_2 + I_5$$

$$\frac{E - V_1}{Z_1} = \frac{V_1}{Z_2} + \frac{(V_1 - V_2)}{Z_5} + \frac{V_1 - V_2}{Z_3} \Rightarrow V_1 \left[\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} + \frac{1}{Z_5} \right] - V_2 \left[\frac{1}{Z_3} + \frac{1}{Z_5} \right] = \frac{E}{Z_1}$$

$$\text{or } V_1[Y_1 + Y_2 + Y_3 + Y_5] - V_2[Y_3 + Y_5] = E_1 Y_1$$

$$I_3 + I_5 = I_4 + I$$

$$\frac{V_1 - V_2}{Z_3} + \frac{V_1 - V_2}{Z_5} = \frac{V_2}{Z_4} + I \Rightarrow V_2 \left[\frac{1}{Z_3} + \frac{1}{Z_4} + \frac{1}{Z_5} \right] - V_1 \left[\frac{1}{Z_3} + \frac{1}{Z_5} \right] = -I$$

$$\text{or } V_2[Y_3 + Y_4 + Y_5] - V_1[Y_3 + Y_5] = -I$$

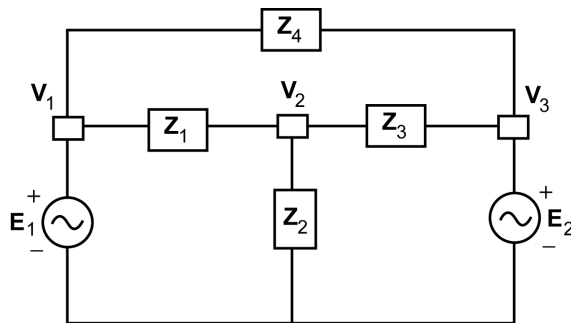
resulting in

$$\begin{aligned} V_1[Y_1 + Y_2 + Y_3 + Y_5] - V_2[Y_3 + Y_5] &= E_1 Y_1 \\ -V_1[Y_3 + Y_5] + V_2[Y_3 + Y_4 + Y_5] &= -I \end{aligned}$$

Applying determinants:

$$V_1 = 19.78 \, \text{V} \angle 132.48^\circ \text{ and } V_2 = 13.37 \, \text{V} \angle 98.78^\circ$$

21.



$$\begin{aligned} Z_1 &= 15 \, \Omega \angle 0^\circ \\ Z_2 &= 10 \, \Omega \angle -90^\circ \\ Z_3 &= 15 \, \Omega \angle 0^\circ \\ Z_4 &= 3 \, \Omega + j4 \, \Omega \end{aligned}$$

$$V_2 \left[\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} \right] - \frac{1}{Z_1} V_1 - \frac{1}{Z_3} V_3 = 0$$

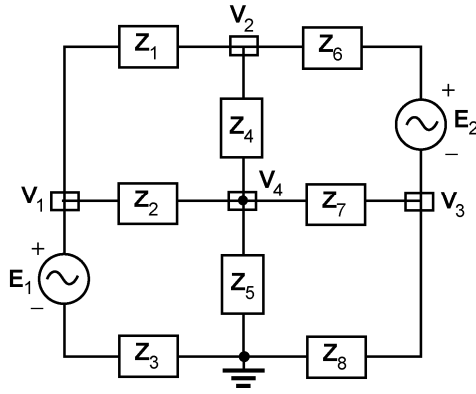
$$V_2 \left[\frac{1}{15 \, \Omega} + \frac{1}{-j10 \, \Omega} + \frac{1}{15 \, \Omega} \right] - \frac{1}{15 \, \Omega} [220 \, \text{V} \angle 0^\circ] - \frac{1}{15 \, \Omega} [100 \, \text{V} \angle 90^\circ] = 0$$

$$V_2 \, 133.34 \times 10^{-3} + j100 \times 10^{-3} = 14.67 + j6.67$$

$$V_2 = \frac{16.05 \, \text{V} \angle 24.55^\circ}{166.67 \times 10^{-3} \angle 36.37^\circ} = 96.30 \, \text{V} \angle -12.32^\circ$$

$$V_1 = E_1 = 220 \, \text{V} \angle 0^\circ, V_3 = E_2 = 100 \, \text{V} \angle 90^\circ$$

22.



$$\begin{aligned} E_1 &= 25 \text{ V } \angle 0^\circ \\ E_2 &= 75 \text{ V } \angle 20^\circ \end{aligned}$$

$$\begin{aligned} Z_1 &= 10 \, \Omega + j20 \, \Omega \\ Z_2 &= 6 \, \Omega \angle 0^\circ \\ Z_3 &= 5 \, \Omega \angle 0^\circ \\ Z_4 &= 20 \, \Omega \angle -90^\circ \\ Z_5 &= 10 \, \Omega \angle 0^\circ \\ Z_6 &= 80 \, \Omega \angle 0^\circ \\ Z_7 &= 15 \, \Omega \angle 90^\circ \\ Z_8 &= 5 \, \Omega - j20 \, \Omega \end{aligned}$$

$$\begin{aligned} V_1: \quad & \frac{V_1 - V_2}{Z_1} + \frac{V_1 - V_4}{Z_2} + \frac{V_1 - E_1}{Z_3} = 0 \\ V_2: \quad & \frac{V_2 - V_1}{Z_1} + \frac{V_2 - V_4}{Z_4} + \frac{V_2 - E_2 - V_3}{Z_6} = 0 \\ V_3: \quad & \frac{V_3 + E_2 - V_2}{Z_6} + \frac{V_3 - V_4}{Z_7} + \frac{V_3}{Z_8} = 0 \\ V_4: \quad & \frac{V_4 - V_1}{Z_2} + \frac{V_4 - V_2}{Z_4} + \frac{V_4 - V_3}{Z_7} + \frac{V_4}{Z_5} = 0 \end{aligned}$$

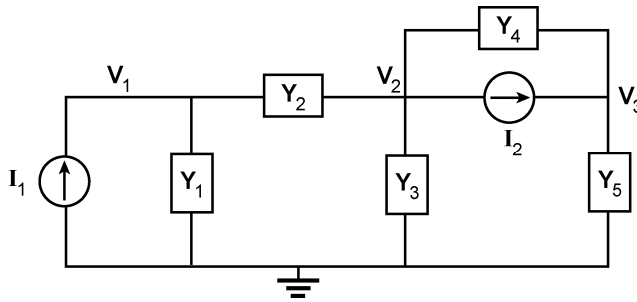
Rearranging:

$$\begin{aligned} V_1 \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} \right) - \frac{V_2}{Z_1} - \frac{V_4}{Z_2} &= \frac{E_1}{Z_3} \\ V_2 \left(\frac{1}{Z_1} + \frac{1}{Z_4} + \frac{1}{Z_6} \right) - \frac{V_1}{Z_1} - \frac{V_4}{Z_4} - \frac{V_3}{Z_6} &= \frac{E_2}{Z_6} \\ V_3 \left(\frac{1}{Z_6} + \frac{1}{Z_7} + \frac{1}{Z_8} \right) - \frac{V_2}{Z_6} - \frac{V_4}{Z_7} &= -\frac{E_2}{Z_6} \\ V_4 \left(\frac{1}{Z_2} + \frac{1}{Z_4} + \frac{1}{Z_7} + \frac{1}{Z_5} \right) - \frac{V_1}{Z_2} - \frac{V_2}{Z_4} - \frac{V_3}{Z_7} &= 0 \end{aligned}$$

Setting up and then using determinants:

$$\begin{aligned} V_1 &= 14.62 \text{ V } \angle -5.86^\circ, \quad V_2 = 35.03 \text{ V } \angle -37.69^\circ \\ V_3 &= 32.4 \text{ V } \angle -73.34^\circ, \quad V_4 = 5.67 \text{ V } \angle 23.53^\circ \end{aligned}$$

23.



$$\begin{aligned} \mathbf{Y}_1 &= \frac{1}{4 \Omega \angle 0^\circ} \\ &= 0.25 \text{ S } \angle 0^\circ \end{aligned}$$

$$\begin{aligned} \mathbf{Y}_2 &= \frac{1}{1 \Omega \angle 90^\circ} \\ &= 1 \text{ S } \angle -90^\circ \end{aligned}$$

$$\begin{aligned} \mathbf{Y}_3 &= \frac{1}{5 \Omega \angle 0^\circ} \\ &= 0.2 \text{ S } \angle 0^\circ \end{aligned}$$

$$\begin{aligned} \mathbf{Y}_4 &= \frac{1}{4 \Omega \angle -90^\circ} \\ &= 0.25 \text{ S } \angle 90^\circ \end{aligned}$$

$$\begin{aligned} \mathbf{Y}_5 &= \frac{1}{8 \Omega \angle 90^\circ} \\ &= 0.125 \text{ S } \angle -90^\circ \end{aligned}$$

$$\mathbf{I}_1 = 2 \text{ A } \angle 30^\circ$$

$$\mathbf{I}_2 = 3 \text{ A } \angle 150^\circ$$

$$\begin{aligned} \mathbf{V}_1[\mathbf{Y}_1 + \mathbf{Y}_2] - \mathbf{Y}_2 \mathbf{V}_2 &= \mathbf{I}_1 \\ \mathbf{V}_2[\mathbf{Y}_2 + \mathbf{Y}_3 + \mathbf{Y}_4] - \mathbf{Y}_2 \mathbf{V}_1 - \mathbf{Y}_4 \mathbf{V}_3 &= -\mathbf{I}_2 \\ \mathbf{V}_3[\mathbf{Y}_4 + \mathbf{Y}_5] - \mathbf{Y}_4 \mathbf{V}_2 &= \mathbf{I}_2 \end{aligned}$$

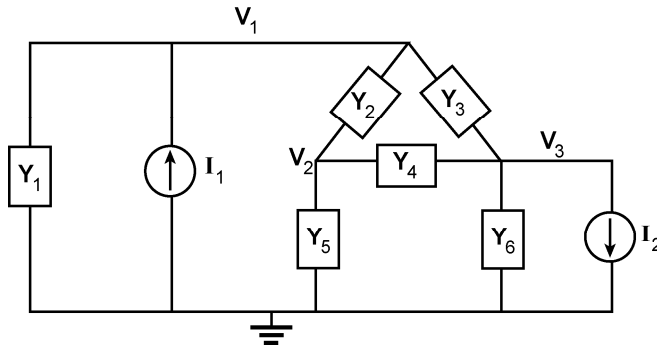
$$\begin{aligned} [\mathbf{Y}_1 + \mathbf{Y}_2]\mathbf{V}_1 &\quad - \mathbf{Y}_2 \mathbf{V}_2 &\quad + 0 &= \mathbf{I}_1 \\ -\mathbf{Y}_2 \mathbf{V}_1 + [\mathbf{Y}_2 + \mathbf{Y}_3 + \mathbf{Y}_4]\mathbf{V}_2 &\quad - \mathbf{Y}_4 \mathbf{V}_3 &= -\mathbf{I}_2 \\ 0 &\quad - \mathbf{Y}_4 \mathbf{V}_2 + [\mathbf{Y}_4 + \mathbf{Y}_5]\mathbf{V}_3 &= \mathbf{I}_2 \end{aligned}$$

$$\begin{aligned} \mathbf{V}_1 &= \frac{\mathbf{I}_1[(\mathbf{Y}_2 + \mathbf{Y}_3 + \mathbf{Y}_4)(\mathbf{Y}_4 + \mathbf{Y}_5) - \mathbf{Y}_4^2] - \mathbf{I}_2[\mathbf{Y}_2 \mathbf{Y}_5]}{[\mathbf{Y}_1 + \mathbf{Y}_2][(\mathbf{Y}_2 + \mathbf{Y}_3 + \mathbf{Y}_4)(\mathbf{Y}_4 + \mathbf{Y}_5) - \mathbf{Y}_4^2] - \mathbf{Y}_2^2(\mathbf{Y}_4 + \mathbf{Y}_5) = \mathbf{Y}_\Delta} \\ &= 5.74 \text{ V } \angle 122.76^\circ \end{aligned}$$

$$\mathbf{V}_2 = \frac{\mathbf{I}_1 \mathbf{Y}_2 (\mathbf{Y}_4 + \mathbf{Y}_5) - \mathbf{I}_2 \mathbf{Y}_5 (\mathbf{Y}_1 + \mathbf{Y}_2)}{\mathbf{Y}_\Delta} = 4.04 \text{ V } \angle 145.03^\circ$$

$$\mathbf{V}_3 = \frac{\mathbf{I}_2[(\mathbf{Y}_1 + \mathbf{Y}_2)(\mathbf{Y}_3 + \mathbf{Y}_4) - \mathbf{Y}_2^2] - \mathbf{Y}_2 \mathbf{Y}_4 \mathbf{I}_1}{\mathbf{Y}_\Delta} = 25.94 \text{ V } \angle 78.07^\circ$$

24.



$$\begin{aligned} V_1[Y_1 + Y_2 + Y_3] - Y_2V_2 - Y_3V_3 &= I_1 \\ V_2[Y_2 + Y_4 + Y_5] - Y_2V_1 - Y_4V_3 &= 0 \\ V_3[Y_3 + Y_4 + Y_6] - Y_3V_1 - Y_4V_2 &= -I_2 \end{aligned}$$

$$\begin{aligned} [Y_1 + Y_2 + Y_3]V_1 &\quad - Y_2V_2 &\quad - Y_3V_3 &= I_1 \\ -Y_2V_1 + [Y_2 + Y_4 + Y_5]V_2 &\quad - Y_4V_3 &= 0 \\ -Y_3V_1 &\quad - Y_4V_2 + [Y_3 + Y_4 + Y_6]V_3 &= -I_2 \end{aligned}$$

$$Y_1 = \frac{1}{4 \Omega \angle 0^\circ} = 0.25 \text{ S } \angle 0^\circ$$

$$Y_2 = \frac{1}{6 \Omega \angle 0^\circ} = 0.167 \text{ S } \angle 0^\circ$$

$$Y_3 = \frac{1}{8 \Omega \angle 0^\circ} = 0.125 \text{ S } \angle 0^\circ$$

$$Y_4 = \frac{1}{2 \Omega \angle -90^\circ} = 0.5 \text{ S } \angle 90^\circ$$

$$Y_5 = \frac{1}{5 \Omega \angle 90^\circ} = 0.2 \text{ S } \angle -90^\circ$$

$$Y_6 = \frac{1}{4 \Omega \angle 90^\circ} = 0.25 \text{ S } \angle -90^\circ$$

$$I_1 = 4 \text{ A } \angle 0^\circ$$

$$I_2 = 6 \text{ A } \angle 90^\circ$$

$$V_1 = \frac{I_1[(Y_2 + Y_4 + Y_5)(Y_3 + Y_4 + Y_6) - Y_4^2] - I_2[Y_2Y_4 + Y_3(Y_3 + Y_4 + Y_5)]}{Y_\Delta = (Y_1 + Y_2 + Y_3)[(Y_2 + Y_4 + Y_5)(Y_3 + Y_4 + Y_6) - Y_4^2] - Y_2[Y_2(Y_3 + Y_4 + Y_6) + Y_3Y_4] - Y_3[Y_2Y_4 + Y_3(Y_2 + Y_4 + Y_5)]}$$

$$= 15.13 \text{ V } \angle 1.29^\circ$$

$$V_2 = \frac{I_1[(Y_2)(Y_3 + Y_4 + Y_6) + Y_3Y_4] + I_2[Y_4(Y_1 + Y_2 + Y_3) - Y_2Y_3]}{Y_\Delta} = 17.24 \text{ V } \angle 3.73^\circ$$

$$V_3 = \frac{I_1[(Y_3)(Y_2 + Y_4 + Y_5) + Y_2Y_4] + I_2[Y_2^2 - (Y_1 + Y_2 + Y_3)(Y_2 + Y_4 + Y_5)]}{Y_\Delta}$$

$$= 10.59 \text{ V } \angle -0.11^\circ$$

25. Left node: V_1
 $\sum I_i = \sum I_o$

$$4I_x = I_x + 5 \text{ mA } \angle 0^\circ + \frac{V_1 - V_2}{2 \text{ k}\Omega}$$

Right node: V_2
 $\sum I_i = \sum I_o$

$$8 \text{ mA } \angle 0^\circ = \frac{V_2}{1 \text{ k}\Omega} + \frac{V_2 - V_1}{2 \text{ k}\Omega} + 4I_x$$

$$\text{Insert } I_x = \frac{V_1}{4 \text{ k}\Omega \angle -90^\circ}$$

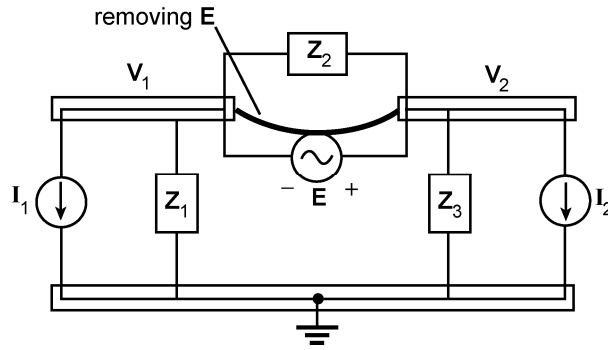
Rearrange, reduce and 2 equations with 2 unknowns result:

$$V_1[1.803 \angle 123.69^\circ] + V_2 = 10$$

$$V_1[2.236 \angle 116.57^\circ] + 3 V_2 = 16$$

Determinants: $V_1 = 4.37 \text{ V } \angle -128.66^\circ$
 $V_2 = V_{1\text{k}\Omega} = 2.25 \text{ V } \angle 17.63^\circ$

26.



$$\begin{aligned} Z_1 &= 1 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 2 \text{ k}\Omega \angle 90^\circ \\ Z_3 &= 3 \text{ k}\Omega \angle -90^\circ \\ I_1 &= 12 \text{ mA} \angle 0^\circ \\ I_2 &= 4 \text{ mA} \angle 0^\circ \\ E &= 10 \text{ V} \angle 0^\circ \end{aligned}$$

$$\begin{aligned} \sum I_i &= \sum I_o \\ 0 &= I_1 + \frac{V_1}{Z_1} + \frac{V_2}{Z_3} + I_2 \\ \text{and } \frac{V_1}{Z_1} + \frac{V_2}{Z_3} &= -I_1 - I_2 \\ \text{with } V_2 - V_1 &= E \end{aligned}$$

Substituting and rearranging:

$$V_1 \left[\frac{1}{Z_1} + \frac{1}{Z_3} \right] = -I_1 - I_2 - \frac{E}{Z_3}$$

and solving for V_1 :

$$\begin{aligned} V_1 &= 15.4 \text{ V} \angle 178.2^\circ \\ \text{with } V_2 = V_C &= 5.41 \text{ V} \angle 174.87^\circ \end{aligned}$$

27. Left node: V_1

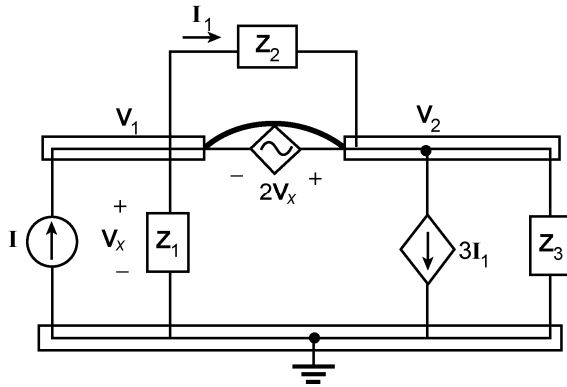
$$\begin{aligned} \sum I_i &= \sum I_o \\ 2 \text{ mA} \angle 0^\circ &= 12 \text{ mA} \angle 0^\circ + \frac{V_1}{2 \text{ k}\Omega} + \frac{V_1 - V_2}{1 \text{ k}\Omega} \\ \text{and } 1.5 V_1 - V_2 &= -10 \end{aligned}$$

Right node: V_2

$$\begin{aligned} \sum I_i &= \sum I_o \\ 0 &= 2 \text{ mA} \angle 0^\circ + \frac{V_2 - V_1}{1 \text{ k}\Omega} - \frac{V_2 - 6 \text{ V}_x}{3.3 \text{ k}\Omega} \\ \text{and } 2.7 V_1 - 3.7 V_2 &= -6.6 \end{aligned}$$

$$\begin{aligned} \text{Using determinants: } V_1 &= V_{2\text{k}\Omega} = -10.67 \text{ V} \angle 0^\circ = 10.67 \text{ V} \angle 180^\circ \\ V_2 &= -6 \text{ V} \angle 0^\circ = 6 \text{ V} \angle 180^\circ \end{aligned}$$

28.



$$\begin{aligned} Z_1 &= 2 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 1 \text{ k}\Omega \angle 0^\circ \\ Z_3 &= 1 \text{ k}\Omega \angle 0^\circ \\ I &= 5 \text{ mA} \angle 0^\circ \end{aligned}$$

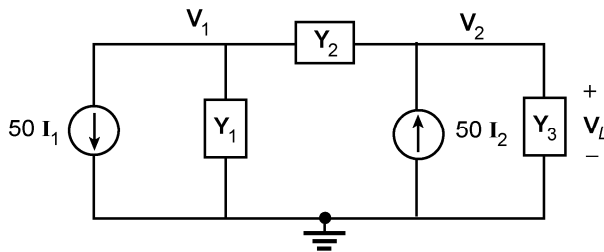
$$\begin{aligned} V_1: I &= \frac{V_1}{Z_1} + 3I_1 + \frac{V_2}{Z_3} \\ \text{with } I_1 &= \frac{V_1 - V_2}{Z_2} \\ \text{and } V_2 - V_1 &= 2V_x = 2V_1 \text{ or } V_2 = 3V_1 \end{aligned}$$

Substituting will result in:

$$\begin{aligned} V_1 \left[\frac{1}{Z_1} + \frac{3}{Z_2} \right] + 3V_1 \left[\frac{1}{Z_3} - \frac{3}{Z_2} \right] &= I \\ \text{or } V_1 \left[\frac{1}{Z_1} - \frac{6}{Z_2} + \frac{3}{Z_3} \right] &= I \end{aligned}$$

$$\begin{aligned} \text{and } V_1 &= V_x = -2 \text{ V} \angle 0^\circ \\ \text{with } V_2 &= -6 \text{ V} \angle 0^\circ \end{aligned}$$

29.



$$\begin{aligned} I_1 &= \frac{E_i \angle \theta}{R_1 \angle 0^\circ} = 1 \times 10^{-3} E_i \\ Y_1 &= \frac{1}{50 \text{ k}\Omega} = 0.02 \text{ mS} \angle 0^\circ \\ Y_2 &= \frac{1}{1 \text{ k}\Omega} = 1 \text{ mS} \angle 0^\circ \\ Y_3 &= 0.02 \text{ mS} \angle 0^\circ \\ I_2 &= (V_1 - V_2)Y_2 \end{aligned}$$

$$\begin{aligned} V_1(Y_1 + Y_2) - Y_2 V_2 &= -50I_1 \\ V_2(Y_2 + Y_3) - Y_2 V_1 &= 50I_2 = 50(V_1 - V_2)Y_2 = 50Y_2 V_1 - 50Y_2 V_2 \end{aligned}$$

$$\begin{aligned} (Y_1 + Y_2)V_1 - Y_2 V_2 &= -50I_1 \\ -51Y_2 V_1 + (51Y_2 + Y_3)V_2 &= 0 \end{aligned}$$

$$V_L = V_2 = \frac{-(50)(51)Y_2 I_1}{(Y_1 + Y_2)(51Y_2 + Y_3) - 51Y_2^2} = -2451.92 E_i$$

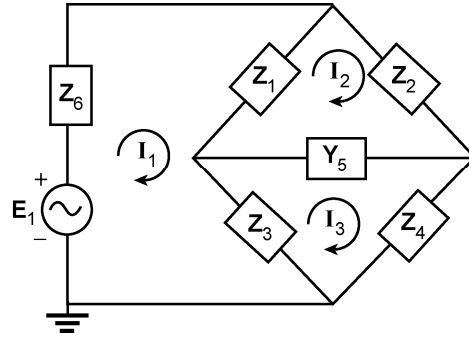
30. a. yes

$$\frac{\underline{Z}_1}{\underline{Z}_3} = \frac{\underline{Z}_2}{\underline{Z}_4}$$

$$\frac{5 \times 10^3 \angle 0^\circ}{2.5 \times 10^3 \angle 90^\circ} = \frac{8 \times 10^3 \angle 0^\circ}{4 \times 10^3 \angle 90^\circ}$$

$$2 \angle -90^\circ = 2 \angle -90^\circ \text{ (balanced)} \checkmark$$

- b. $\underline{Z}_1 = 5 \text{ k}\Omega \angle 0^\circ$, $\underline{Z}_2 = 8 \text{ k}\Omega \angle 0^\circ$
 $\underline{Z}_3 = 2.5 \text{ k}\Omega \angle 90^\circ$, $\underline{Z}_4 = 4 \text{ k}\Omega \angle 90^\circ$
 $\underline{Z}_5 = 5 \text{ k}\Omega \angle -90^\circ$, $\underline{Z}_6 = 1 \text{ k}\Omega \angle 0^\circ$



$$\begin{aligned} \underline{I}_1[\underline{Z}_1 + \underline{Z}_3 + \underline{Z}_6] - \underline{Z}_1 \underline{I}_2 - \underline{Z}_3 \underline{I}_3 &= \underline{E} \\ \underline{I}_2[\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_5] - \underline{Z}_1 \underline{I}_1 - \underline{Z}_5 \underline{I}_3 &= 0 \\ \underline{I}_3[\underline{Z}_3 + \underline{Z}_4 + \underline{Z}_5] - \underline{Z}_3 \underline{I}_1 - \underline{Z}_5 \underline{I}_2 &= 0 \end{aligned}$$

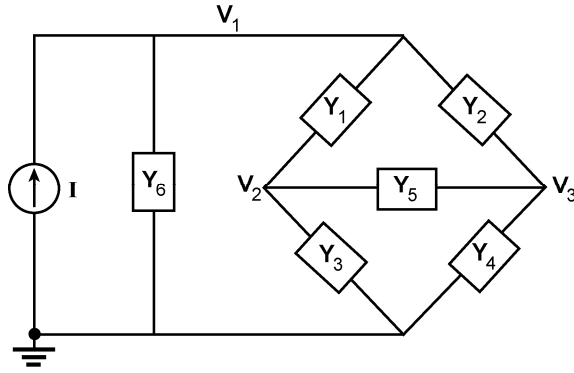
$$\begin{aligned} [\underline{Z}_1 + \underline{Z}_3 + \underline{Z}_6] \underline{I}_1 &\quad - \underline{Z}_1 \underline{I}_2 &\quad - \underline{Z}_3 \underline{I}_3 &= \underline{E} \\ -\underline{Z}_1 \underline{I}_1 &+ [\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_5] \underline{I}_2 &\quad - \underline{Z}_5 \underline{I}_3 &= 0 \\ -\underline{Z}_3 \underline{I}_1 &\quad - \underline{Z}_5 \underline{I}_2 &+ [\underline{Z}_3 + \underline{Z}_4 + \underline{Z}_5] \underline{I}_3 &= 0 \end{aligned}$$

$$\underline{I}_2 = \frac{\underline{E}[\underline{Z}_1(\underline{Z}_3 + \underline{Z}_4 + \underline{Z}_5) + \underline{Z}_3 \underline{Z}_5]}{\underline{Z}_\Delta = (\underline{Z}_1 + \underline{Z}_3 + \underline{Z}_6)[(\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_5)(\underline{Z}_3 + \underline{Z}_4 + \underline{Z}_5) - \underline{Z}_5^2] - \underline{Z}_1[\underline{Z}_1(\underline{Z}_3 + \underline{Z}_4 + \underline{Z}_5) - \underline{Z}_3 \underline{Z}_5] - \underline{Z}_3[\underline{Z}_1 \underline{Z}_5 + \underline{Z}_3(\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_5)]}$$

$$\underline{I}_3 = \frac{\underline{E}[\underline{Z}_1 \underline{Z}_5 + \underline{Z}_3(\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_5)]}{\underline{Z}_\Delta}$$

$$\underline{I}_{Z_5} = \underline{I}_2 - \underline{I}_3 = \frac{\underline{E}[\underline{Z}_1 \underline{Z}_4 - \underline{Z}_3 \underline{Z}_2]}{\underline{Z}_\Delta} = \frac{E[20 \times 10^6 \angle 90^\circ - 20 \times 10^6 \angle 90^\circ]}{\underline{Z}_\Delta} = 0 \text{ A}$$

c.



$$\begin{aligned} V_1[Y_1 + Y_2 + Y_6] - Y_1V_2 - Y_2V_3 &= I \\ V_2[Y_1 + Y_3 + Y_5] - Y_1V_1 - Y_5V_3 &= 0 \\ V_3[Y_2 + Y_4 + Y_5] - Y_2V_1 - Y_5V_2 &= 0 \end{aligned}$$

$$\begin{aligned} [Y_1 + Y_2 + Y_6]V_1 & - Y_1V_2 & - Y_2V_3 &= I \\ -Y_1V_1 + [Y_1 + Y_3 + Y_5]V_2 & - Y_5V_3 & &= 0 \\ -Y_2V_1 & - Y_5V_2 + [Y_2 + Y_4 + Y_5]V_3 & &= 0 \end{aligned}$$

$$\begin{aligned} I &= \frac{E_s}{R_s} = \frac{10 \text{ V } \angle 0^\circ}{1 \text{ k}\Omega \angle 0^\circ} \\ &= 10 \text{ mA } \angle 0^\circ \end{aligned}$$

$$\begin{aligned} Y_1 &= \frac{1}{5 \text{ k}\Omega \angle 0^\circ} \\ &= 0.2 \text{ mS } \angle 0^\circ \end{aligned}$$

$$\begin{aligned} Y_2 &= \frac{1}{8 \text{ k}\Omega \angle 0^\circ} \\ &= 0.125 \text{ mS } \angle 0^\circ \end{aligned}$$

$$\begin{aligned} Y_3 &= \frac{1}{2.5 \text{ k}\Omega \angle 90^\circ} \\ &= 0.4 \text{ mS } \angle -90^\circ \end{aligned}$$

$$\begin{aligned} Y_4 &= \frac{1}{4 \text{ k}\Omega \angle 90^\circ} \\ &= 0.25 \text{ mS } \angle -90^\circ \end{aligned}$$

$$\begin{aligned} Y_5 &= \frac{1}{5 \text{ k}\Omega \angle -90^\circ} \\ &= 0.2 \text{ mS } \angle 90^\circ \end{aligned}$$

$$\begin{aligned} Y_6 &= \frac{1}{1 \text{ k}\Omega \angle 0^\circ} \\ V_2 &= 1 \text{ mS } \angle 0^\circ \end{aligned}$$

$$\begin{aligned} V_2 &= \frac{I[Y_1(Y_2 + Y_4 + Y_5) + Y_2Y_5]}{Y_\Delta = (Y_1 + Y_2 + Y_6)[(Y_1 + Y_3 + Y_5)(Y_2 + Y_4 + Y_5) - Y_5^2] - Y_1[Y_1(Y_2 + Y_4 + Y_5) + Y_2Y_5] - Y_2[Y_1Y_5 + Y_2(Y_1 + Y_3 + Y_5)]} \\ V_3 &= \frac{I[Y_1Y_5 + Y_2(Y_1 + Y_3 + Y_5)]}{Y_\Delta} \end{aligned}$$

$$\begin{aligned} V_{Z_5} &= V_2 - V_3 = \frac{I[Y_1Y_4 - Y_4Y_3]}{Y_\Delta} = \frac{I[0.05 \times 10^{-3} \angle -90^\circ - 0.05 \times 10^{-3} \angle -90^\circ]}{Y_\Delta} \\ &= 0 \text{ V} \end{aligned}$$

31. a. $\frac{Z_1}{Z_3} = \frac{Z_2}{Z_4}$

$$\frac{4 \times 10^3 \angle 0^\circ}{4 \times 10^3 \angle 90^\circ} \stackrel{?}{=} \frac{4 \times 10^3 \angle 0^\circ}{4 \times 10^3 \angle -90^\circ}$$

$$1 \angle -90^\circ \neq 1 \angle 90^\circ \text{ (not balanced)}$$

b. The solution to 26(b) resulted in

$$I_3 = I_{X_C} = \frac{E(Z_1Z_5 + Z_3(Z_1 + Z_2 + Z_5))}{Z_\Delta}$$

$$\begin{aligned} \text{where } Z_\Delta &= (Z_1 + Z_3 + Z_6)[(Z_1 + Z_2 + Z_5)(Z_3 + Z_4 + Z_5) - Z_5^2] \\ &\quad - Z_1[Z_1(Z_3 + Z_4 + Z_5) - Z_3Z_5] - Z_3[Z_1Z_5 + Z_3(Z_1 + Z_2 + Z_5)] \end{aligned}$$

$$\text{and } Z_1 = 5 \text{ k}\Omega \angle 0^\circ, Z_2 = 8 \text{ k}\Omega \angle 0^\circ, Z_3 = 2.5 \text{ k}\Omega \angle 90^\circ$$

$$Z_4 = 4 \text{ k}\Omega \angle 90^\circ, Z_5 = 5 \text{ k}\Omega \angle -90^\circ, Z_6 = 1 \text{ k}\Omega \angle 0^\circ$$

$$\text{and } I_{X_C} = 1.76 \text{ mA } \angle -71.54^\circ$$

c. The solution to 26(c) resulted in

$$\mathbf{V}_3 = \mathbf{V}_{x_C} = \frac{\mathbf{I}[\mathbf{Y}_1\mathbf{Y}_5 + \mathbf{Y}_2(\mathbf{Y}_1 + \mathbf{Y}_3 + \mathbf{Y}_5)]}{\mathbf{Y}_\Delta}$$

$$\text{where } \mathbf{Y}_\Delta = (\mathbf{Y}_1 + \mathbf{Y}_2 + \mathbf{Y}_6)[(\mathbf{Y}_1 + \mathbf{Y}_3 + \mathbf{Y}_5)(\mathbf{Y}_2 + \mathbf{Y}_4 + \mathbf{Y}_5) - \mathbf{Y}_5^2] \\ - \mathbf{Y}_1[\mathbf{Y}_1(\mathbf{Y}_2 + \mathbf{Y}_4 + \mathbf{Y}_5) + \mathbf{Y}_2\mathbf{Y}_5] \\ - \mathbf{Y}_2[\mathbf{Y}_1\mathbf{Y}_5 + \mathbf{Y}_2(\mathbf{Y}_1 + \mathbf{Y}_3 + \mathbf{Y}_5)]$$

$$\text{with } \mathbf{Y}_1 = 0.2 \text{ mS } \angle 0^\circ, \mathbf{Y}_2 = 0.125 \text{ mS } \angle 0^\circ, \mathbf{Y}_3 = 0.4 \text{ mS } \angle -90^\circ \\ \mathbf{Y}_4 = 0.25 \text{ mS } \angle -90^\circ, \mathbf{Y}_5 = 0.2 \text{ mS } \angle 90^\circ$$

$$\text{Source conversion: } \mathbf{Y}_6 = 1 \text{ mS } \angle 0^\circ, \mathbf{I} = 10 \text{ mA } \angle 0^\circ \\ \text{and } \mathbf{V}_3 = \mathbf{V}_{x_C} = \mathbf{7.03 V } \angle -18.46^\circ$$

$$32. \quad \mathbf{Z}_1\mathbf{Z}_4 = \mathbf{Z}_3\mathbf{Z}_2$$

$$(R_1 - jX_C)(R_x + jX_{L_x}) = R_3R_2 \quad X_C = \frac{1}{\omega C} = \frac{1}{(10^3 \text{ rad/s})(1 \mu\text{F})} = 1 \text{ k}\Omega$$

$$(1 \text{ k}\Omega - j1 \text{ k}\Omega)(R_x + jX_{L_x}) = (0.1 \text{ k}\Omega)(0.1 \text{ k}\Omega) = 10 \text{ k}\Omega$$

$$\text{and } R_x + jX_{L_x} = \frac{10 \times 10^3 \Omega}{1 \times 10^3 - j1 \times 10^3} = \frac{10 \times 10^3}{1.414 \times 10^3 \angle -45^\circ} = 5 \Omega + j5 \Omega$$

$$\therefore R_x = 5 \Omega, L_x = \frac{X_{L_x}}{\omega} = \frac{5 \Omega}{10^3 \text{ rad/s}} = 5 \text{ mH}$$

$$33. \quad X_{C_1} = \frac{1}{\omega C} = \frac{1}{(1000 \text{ rad/s})(3 \mu\text{F})} = \frac{1}{3} \text{ k}\Omega$$

$$\mathbf{Z}_1 = R_1 \parallel X_{C_1} \angle -90^\circ = (2 \text{ k}\Omega \angle 0^\circ) \parallel 2 \frac{1}{3} \text{ k}\Omega \angle -90^\circ = 328.8 \Omega \angle -80.54^\circ$$

$$\mathbf{Z}_2 = R_2 \angle 0^\circ = 0.5 \text{ k}\Omega \angle 0^\circ, \mathbf{Z}_3 = R_3 \angle 0^\circ = 4 \text{ k}\Omega \angle 0^\circ$$

$$\mathbf{Z}_4 = R_x + jX_{L_x} = 1 \text{ k}\Omega + j6 \text{ k}\Omega$$

$$\frac{\mathbf{Z}_1}{\mathbf{Z}_3} = \frac{\mathbf{Z}_2}{\mathbf{Z}_4}$$

$$\frac{328.8 \Omega \angle -80.54^\circ}{4 \text{ k}\Omega \angle 0^\circ} \stackrel{?}{=} \frac{0.5 \text{ k}\Omega \angle 0^\circ}{6.083 \Omega \angle 80.54^\circ}$$

$$82.2 \times 10^{-3} \angle -80.54^\circ \stackrel{?}{=} 82.2 \times 10^{-3} \angle -80.54^\circ \text{ (balanced)}$$

34. Apply Eq. 17.6.

35. For balance:

$$R_1(R_x + jX_{L_x}) = R_2(R_3 + jX_{L_3})$$

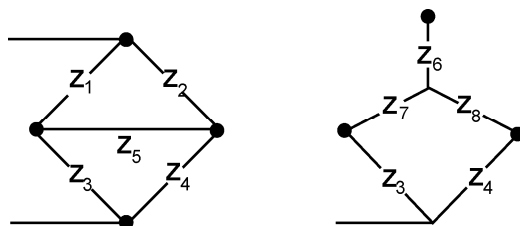
$$R_1R_x + jR_1X_{L_x} = R_2R_3 + jR_2X_{L_3}$$

$$\therefore R_1R_x = R_2R_3 \text{ and } R_x = \frac{R_2R_3}{R_1}$$

$$R_1X_{L_x} = R_2X_{L_3} \text{ and } R_1\omega L_x = R_2\omega L_3$$

$$\text{so that } L_x = \frac{R_2L_3}{R_1}$$

36. a.



$$Z_1 = 8 \Omega \angle -90^\circ = -j8 \Omega$$

$$Z_2 = 4 \Omega \angle 90^\circ = +j4 \Omega$$

$$Z_3 = 8 \Omega \angle 90^\circ = +j8 \Omega$$

$$Z_4 = 6 \Omega \angle -90^\circ = -j6 \Omega$$

$$Z_5 = 5 \Omega \angle 0^\circ$$

$$Z_6 = \frac{Z_1Z_2}{Z_1 + Z_2 + Z_5} = 5 \Omega \angle 38.66^\circ$$

$$Z_7 = \frac{Z_1Z_5}{Z_1 + Z_2 + Z_5} = 6.25 \Omega \angle -51.34^\circ$$

$$Z_8 = \frac{Z_2Z_5}{Z_1 + Z_2 + Z_5} = 3.125 \Omega \angle 128.66^\circ$$

$$Z' = Z_7 + Z_3 = 3.9 \Omega + j3.12 \Omega = 4.99 \Omega \angle 38.66^\circ$$

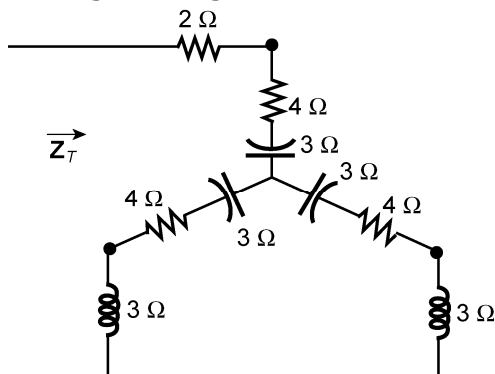
$$Z'' = Z_8 + Z_4 = -1.95 \Omega - j3.56 \Omega = 4.06 \Omega \angle -118.71^\circ$$

$$Z' \parallel Z'' = 10.13 \Omega \angle -67.33^\circ = 3.90 \Omega - j9.35 \Omega$$

$$Z_T = Z_6 + Z' \parallel Z'' = 7.80 \Omega - j6.23 \Omega = 9.98 \Omega \angle -38.61^\circ$$

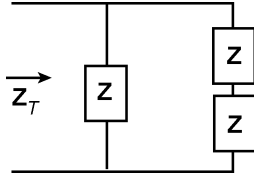
$$I = \frac{E}{Z_T} = \frac{120 \text{ V} \angle 0^\circ}{9.98 \Omega \angle -38.61^\circ} = 12.02 \text{ A} \angle 38.61^\circ$$

37. $Z_Y = \frac{Z_\Delta}{3} = \frac{12 \Omega - j9 \Omega}{3} = 4 \Omega - j3 \Omega$



$$\begin{aligned}
 \mathbf{Z}_T &= 2\ \Omega + 4\ \Omega + j3\ \Omega + [4\ \Omega - j3\ \Omega + j3\ \Omega] \parallel [4\ \Omega - j3\ \Omega + j3\ \Omega] / \\
 &= 6\ \Omega - j3\ \Omega + 2\ \Omega \\
 &= 8\ \Omega - j3\ \Omega = 8.544\ \Omega \angle -20.56^\circ \\
 \mathbf{I} &= \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{60\ \text{V} \angle 0^\circ}{8.544\ \Omega \angle -20.56^\circ} = \mathbf{7.02\ A} \angle \mathbf{20.56^\circ}
 \end{aligned}$$

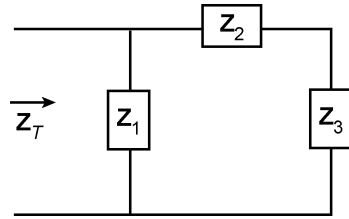
38.



$$\begin{aligned}
 \mathbf{Z}_\Delta &= 3\mathbf{Z}_Y = 3(3\ \Omega \angle 90^\circ) = 9\ \Omega \angle 90^\circ \\
 \mathbf{Z} &= 9\ \Omega \angle 90^\circ \parallel (12\ \Omega - j16\ \Omega) \\
 &= 9\ \Omega \angle 90^\circ \parallel 20\ \Omega \angle 53.13^\circ \\
 &= 12.96\ \Omega \angle 67.13^\circ
 \end{aligned}$$

$$\begin{aligned}
 \mathbf{Z}_T &= \mathbf{Z} \parallel 2\mathbf{Z} = \frac{2\mathbf{Z}^2}{\mathbf{Z} + 2\mathbf{Z}} = \frac{2}{3}\mathbf{Z} = \frac{2}{3} [12.96\ \Omega \angle 67.13^\circ] = 8.64\ \Omega \angle 67.13^\circ \\
 \mathbf{I} &= \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{100\ \text{V} \angle 0^\circ}{8.64\ \Omega \angle 67.13^\circ} = \mathbf{11.57\ A} \angle \mathbf{-67.13^\circ}
 \end{aligned}$$

39. $\mathbf{Z}_\Delta = 3\mathbf{Z}_Y = 3(5\ \Omega) = 15\ \Omega$

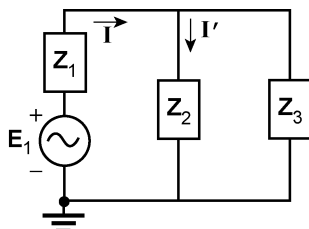


$$\begin{aligned}
 \mathbf{Z}_1 &= 15\ \Omega \angle 0^\circ \parallel 5\ \Omega \angle -90^\circ \\
 &= 4.74\ \Omega \angle -71.57^\circ \\
 \mathbf{Z}_2 &= 15\ \Omega \angle 0^\circ \parallel 6\ \Omega \angle 90^\circ \\
 &= 5.57\ \Omega \angle 68.2^\circ = 2.07\ \Omega + j5.17\ \Omega \\
 \mathbf{Z}_3 &= \mathbf{Z}_1 = 4.74\ \Omega \angle -71.57^\circ \\
 &= 1.5\ \Omega - j4.5\ \Omega
 \end{aligned}$$

$$\begin{aligned}
 \mathbf{Z}_T &= \mathbf{Z}_1 \parallel (\mathbf{Z}_2 + \mathbf{Z}_3) = (4.74\ \Omega \angle -71.57^\circ) \parallel (2.07\ \Omega + j5.17\ \Omega + 1.5\ \Omega - j4.5\ \Omega) \\
 &= (4.74\ \Omega \angle -71.57^\circ) \parallel (3.63\ \Omega \angle 10.63^\circ) \\
 &= 2.71\ \Omega \angle -23.87^\circ \\
 \mathbf{I} &= \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{100\ \text{V} \angle 0^\circ}{2.71\ \Omega \angle -23.87^\circ} = \mathbf{36.9\ A} \angle \mathbf{23.87^\circ}
 \end{aligned}$$

Chapter 18

1.

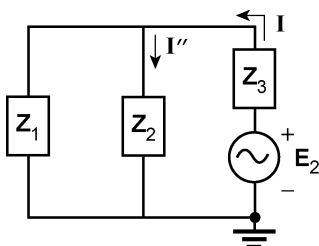


$$\mathbf{Z}_1 = 3 \Omega \angle 0^\circ, \mathbf{Z}_2 = 8 \Omega \angle 90^\circ, \mathbf{Z}_3 = 6 \Omega \angle -90^\circ$$

$$\mathbf{Z}_2 \parallel \mathbf{Z}_3 = 8 \Omega \angle 90^\circ \parallel 6 \Omega \angle -90^\circ = 24 \Omega \angle -90^\circ$$

$$\mathbf{I} = \frac{\mathbf{E}_1}{\mathbf{Z}_1 + \mathbf{Z}_2 \parallel \mathbf{Z}_3} = \frac{30 \text{ V} \angle 30^\circ}{3 \Omega - j24 \Omega} = 1.24 \text{ A} \angle 112.875^\circ$$

$$\mathbf{I}' = \frac{\mathbf{Z}_3 \mathbf{I}}{\mathbf{Z}_2 + \mathbf{Z}_3} = \frac{(6 \Omega \angle -90^\circ)(1.24 \text{ A} \angle 112.875^\circ)}{2 \Omega \angle 90^\circ} = 3.72 \text{ A} \angle -67.125^\circ$$



$$\mathbf{Z}_1 \parallel \mathbf{Z}_2 = 3 \Omega \angle 0^\circ \parallel 8 \Omega \angle 90^\circ = 2.809 \Omega \angle 20.556^\circ$$

$$\mathbf{I} = \frac{\mathbf{E}_2}{\mathbf{Z}_3 + \mathbf{Z}_1 \parallel \mathbf{Z}_2} = \frac{60 \text{ V} \angle 10^\circ}{-j6 \Omega + 2.630 \Omega + j0.986 \Omega} = 10.597 \text{ A} \angle 72.322^\circ$$

$$\mathbf{I}'' = \frac{\mathbf{Z}_1 \mathbf{I}}{\mathbf{Z}_1 + \mathbf{Z}_2} = \frac{(3 \Omega \angle 0^\circ)(10.597 \text{ A} \angle 72.322^\circ)}{3 \Omega + j8 \Omega} = 3.721 \text{ A} \angle 2.878^\circ$$

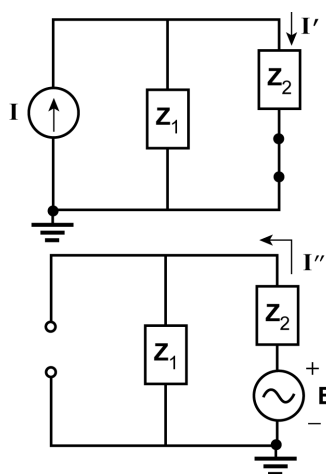
$$\mathbf{I}_{L_1} = \mathbf{I}' + \mathbf{I}'' = 3.72 \text{ A} \angle -67.125^\circ + 3.721 \text{ A} \angle 2.878^\circ$$

$$= 1.446 \text{ A} - j3.427 \text{ A} + 3.716 \text{ A} + j0.187 \text{ A}$$

$$= 5.162 \text{ A} - j3.24 \text{ A}$$

$$= \mathbf{6.09 \text{ A} \angle -32.12^\circ}$$

2.



$$\mathbf{Z}_1 = 8 \Omega \angle 90^\circ, \mathbf{Z}_2 = 5 \Omega \angle -90^\circ$$

$$\mathbf{I} = 0.3 \text{ A} \angle 60^\circ, \mathbf{E} = 10 \text{ V} \angle 0^\circ$$

$$\mathbf{I}' = \frac{\mathbf{Z}_2 \mathbf{I}}{\mathbf{Z}_1 + \mathbf{Z}_2} = \frac{(8 \Omega \angle 90^\circ)(0.3 \text{ A} \angle 60^\circ)}{+j8 \Omega - j5 \Omega} = \frac{2.4 \text{ A} \angle 150^\circ}{3 \angle 90^\circ}$$

$$= 0.8 \text{ A} \angle 60^\circ$$

$$\mathbf{I}'' = \frac{\mathbf{E}}{\mathbf{Z}_1 + \mathbf{Z}_2} = \frac{10 \text{ V} \angle 0^\circ}{+j8 \Omega - j5 \Omega} = \frac{10 \text{ A} \angle 0^\circ}{3 \angle 90^\circ}$$

$$= 3.33 \text{ A} \angle -90^\circ$$

$$\mathbf{I}_C = \mathbf{I}' - \mathbf{I}''$$

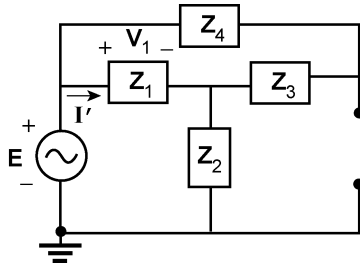
$$= 0.8 \text{ A} \angle 60^\circ - 3.33 \text{ A} \angle -90^\circ$$

$$= (0.4 \text{ A} + j0.69 \text{ A}) + j3.33 \text{ A}$$

$$= 0.4 \text{ A} + j4.02 \text{ A}$$

$$= \mathbf{4.04 \text{ A} \angle 84.32^\circ}$$

3. E:



$$\mathbf{Z}_1 = 3 \, \Omega \angle 90^\circ, \mathbf{Z}_2 = 7 \, \Omega \angle -90^\circ$$

$$\mathbf{E} = 10 \, \text{V} \angle 90^\circ$$

$$\mathbf{Z}_3 = 6 \, \Omega \angle -90^\circ, \mathbf{Z}_4 = 4 \, \Omega \angle 0^\circ$$

$$\mathbf{Z}' = \mathbf{Z}_1 \parallel (\mathbf{Z}_3 + \mathbf{Z}_4)$$

$$= 3 \, \Omega \angle 90^\circ \parallel (4 \, \Omega - j6 \, \Omega)$$

$$= 3 \, \Omega \angle 90^\circ \parallel 7.21 \, \Omega \angle -56.31^\circ$$

$$= 4.33 \, \Omega \angle 70.56^\circ$$

$$\mathbf{V}_1 = \frac{\mathbf{Z}' \mathbf{E}}{\mathbf{Z}' + \mathbf{Z}_2}$$

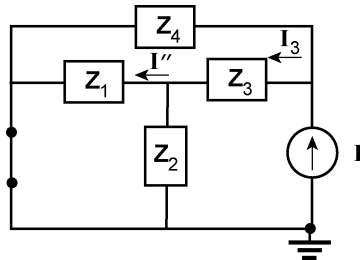
$$= \frac{(4.33 \, \Omega \angle 70.56^\circ)(10 \, \text{V} \angle 90^\circ)}{(1.44 \, \Omega + j4.08 \, \Omega) - j7 \, \Omega}$$

$$= \frac{43.3 \, \text{V} \angle 160.56^\circ}{3.26 \angle -63.75^\circ} = 13.28 \, \text{V} \angle 224.31^\circ$$

$$\mathbf{I}' = \frac{\mathbf{V}_1}{\mathbf{Z}_1} = \frac{13.28 \, \text{V} \angle 224.31^\circ}{3 \, \Omega \angle 90^\circ}$$

$$= 4.43 \, \text{A} \angle 134.31^\circ$$

I:



$$\mathbf{Z}'' = \mathbf{Z}_3 + \mathbf{Z}_1 \parallel \mathbf{Z}_2$$

$$= -j6 \, \Omega + 3 \, \Omega \angle 90^\circ \parallel 7 \, \Omega \angle -90^\circ$$

$$= -j6 \, \Omega + 5.25 \, \Omega \angle 90^\circ$$

$$= -j6 \, \Omega + j5.25 \, \Omega$$

$$= -j0.75 \, \Omega = 0.75 \, \Omega \angle -90^\circ$$

$$\text{CDR: } \mathbf{I}_3 = \frac{\mathbf{Z}_4 \mathbf{I}}{\mathbf{Z}_4 + \mathbf{Z}''} = \frac{(4 \, \Omega \angle 0^\circ)(0.6 \, \text{A} \angle 120^\circ)}{4 \, \Omega - j0.75 \, \Omega} = \frac{2.4 \, \text{A} \angle 120^\circ}{4.07 \angle -10.62^\circ}$$

$$= 0.59 \, \text{A} \angle 130.62^\circ$$

$$\mathbf{I}'' = \frac{\mathbf{Z}_2 \mathbf{I}_3}{\mathbf{Z}_2 + \mathbf{Z}_1} = \frac{(7 \, \Omega \angle -90^\circ)(0.59 \, \text{A} \angle 130.62^\circ)}{-j7 \, \Omega + j3 \, \Omega} = \frac{4.13 \, \text{A} \angle 40.62^\circ}{4 \angle -90^\circ}$$

$$= 1.03 \, \text{A} \angle 130.62^\circ$$

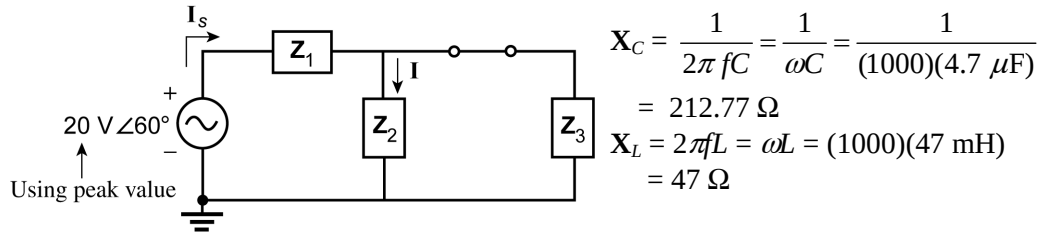
$$\mathbf{I}_L = \mathbf{I}' - \mathbf{I}'' \text{ (direction of } \mathbf{I}') \text{}$$

$$= 4.43 \, \text{A} \angle 134.31^\circ - 1.03 \, \text{A} \angle 130.62^\circ$$

$$= (-3.09 \, \text{A} + j3.17 \, \text{A}) - (-0.67 \, \text{A} + j0.78 \, \text{A}) = -2.42 \, \text{A} + j2.39 \, \text{A}$$

$$= \mathbf{3.40 \, A} \angle 135.36^\circ$$

4. AC:



$$Z_1 = 212.77 \Omega \angle -90^\circ, Z_2 = 47 \Omega \angle 0^\circ, Z_3 = 22 \Omega + j47 \Omega = 51.89 \Omega \angle 64.92^\circ$$

$$Z_2 \parallel Z_3 = 29.23 \Omega \angle 30.66^\circ$$

$$Z_T = Z_1 + Z_2 \parallel Z_3 = -j212.77 \Omega + 25.14 \Omega + j14.91 \Omega$$

$$= 25.14 \Omega - j197.86 \Omega = 199.45 \Omega \angle -82.76^\circ$$

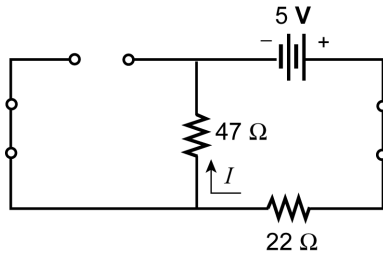
$$I_s = \frac{E}{Z_T} = \frac{20 \text{ V} \angle 60^\circ}{199.45 \Omega \angle -82.76^\circ} = 0.1 \text{ A} \angle 142.76^\circ$$

$$I = \frac{Z_3 I_s}{Z_3 + Z_2} = \frac{(51.89 \Omega \angle 64.92^\circ)(0.1 \text{ A} \angle 142.76^\circ)}{22 \Omega + j47 \Omega + 47 \Omega} = \frac{5.19 \text{ A} \angle 207.68^\circ}{83.49 \angle 34.26^\circ}$$

$$I = 62.16 \text{ mA} \angle 173.42^\circ$$

$$\text{and } i = 62.16 \times 10^{-3} \sin(1000t + 173.42^\circ)$$

DC:

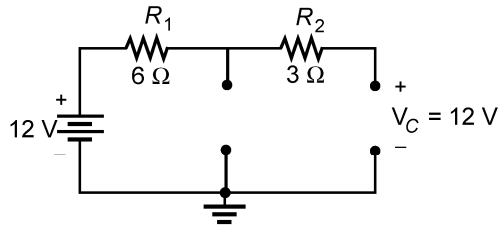


$$I = \frac{5 \text{ V}}{22 \Omega + 47 \Omega} = \frac{5 \text{ V}}{69 \Omega}$$

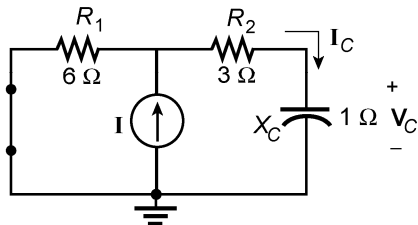
$$= 72.46 \text{ mA}$$

$$i = -72.46 \text{ mA} + 62.16 \times 10^{-3} \sin(1000t + 173.42^\circ)$$

5. DC:



AC:



$$I_C = \frac{(6 \Omega \angle 0^\circ)(I)}{6 \Omega + 3 \Omega - j1 \Omega}$$

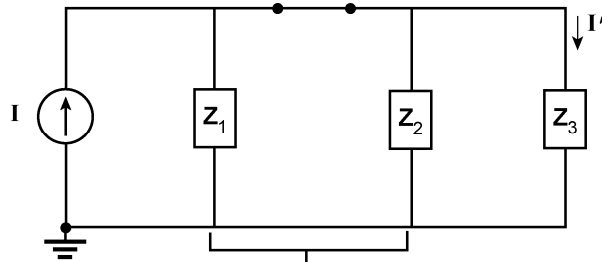
$$= \frac{(6 \Omega \angle 0^\circ)(4 \text{ A} \angle 0^\circ)}{9 \Omega - j1 \Omega}$$

$$= \frac{24 \text{ A} \angle 0^\circ}{9.055 \angle -6.34^\circ}$$

$$= 2.65 \text{ A} \angle 6.34^\circ$$

$$\begin{aligned} V_C &= I_C X_C = (2.65 \text{ A } \angle 6.34^\circ)(1 \Omega \angle -90^\circ) = 2.65 \text{ V } \angle -83.66^\circ \\ &= 12 \text{ V} + 2.65 \text{ V } \angle -83.66^\circ \\ v_C &= 12 \text{ V} + 3.75 \sin(\omega t - 83.66^\circ) \end{aligned}$$

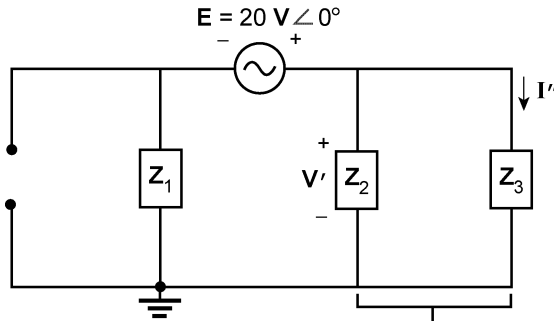
6.



$$\begin{aligned} E &= 20 \text{ V } \angle 0^\circ \\ Z_1 &= 10 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 5 \text{ k}\Omega - j5 \text{ k}\Omega \\ &= 7.071 \text{ k}\Omega \angle -45^\circ \\ Z_3 &= 5 \text{ k}\Omega \angle 90^\circ \\ I &= 5 \text{ mA } \angle 0^\circ \end{aligned}$$

$$Z' = Z_1 \parallel Z_2 = 10 \text{ k}\Omega \angle 0^\circ \parallel 7.071 \text{ k}\Omega \angle -45^\circ = 4.472 \text{ k}\Omega \angle -26.57^\circ$$

$$\begin{aligned} \text{(CDR)} \quad I' &= \frac{Z' I}{Z' + Z_3} = \frac{(4.472 \text{ k}\Omega \angle -26.57^\circ)(5 \text{ mA } \angle 0^\circ)}{4 \text{ k}\Omega - j2 \text{ k}\Omega + j5 \text{ k}\Omega} = \frac{22.36 \text{ mA } \angle -26.57^\circ}{5 \angle 36.87^\circ} \\ &= 4.472 \text{ mA } \angle -63.44^\circ \end{aligned}$$

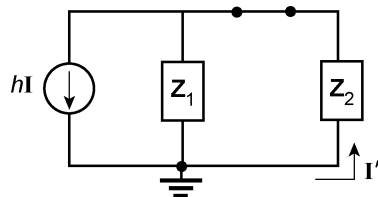


$$\begin{aligned} Z'' &= Z_2 \parallel Z_3 \\ &= 7.071 \text{ k}\Omega \angle -45^\circ \parallel 5 \text{ k}\Omega \angle 90^\circ \\ &= 7.071 \text{ k}\Omega \angle 45^\circ \end{aligned}$$

$$\begin{aligned} \text{(VDR)} \quad V' &= \frac{Z'' E}{Z'' + Z_1} = \frac{(7.071 \text{ k}\Omega \angle 45^\circ)(20 \text{ V } \angle 0^\circ)}{(5 \text{ k}\Omega + j5 \text{ k}\Omega) + (10 \text{ k}\Omega)} = \frac{141.42 \text{ V } \angle 45^\circ}{15.81 \angle 18.435^\circ} \\ &= 8.945 \text{ V } \angle 26.565^\circ \\ I'' &= \frac{V'}{Z_3} = \frac{8.945 \text{ V } \angle 26.565^\circ}{5 \text{ k}\Omega \angle 90^\circ} = 1.789 \text{ mA } \angle -63.435^\circ = 0.8 \text{ mA} - j1.6 \text{ mA} \end{aligned}$$

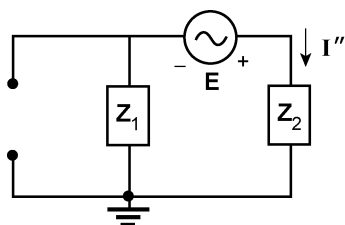
$$\begin{aligned} I &= I' + I'' = (2 \text{ mA} - j4 \text{ mA}) + (0.8 \text{ mA} - j1.6 \text{ mA}) = 2.8 \text{ mA} - j5.6 \text{ mA} \\ &= \mathbf{6.26 \text{ mA } \angle -63.43^\circ} \end{aligned}$$

7.



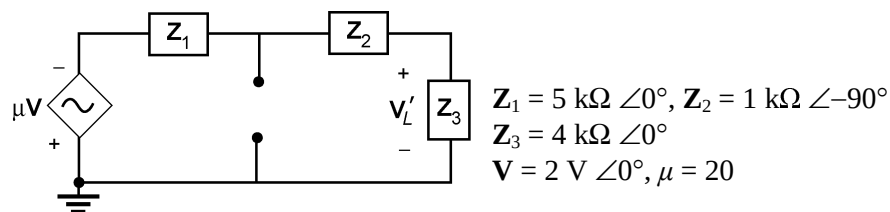
$$\begin{aligned} Z_1 &= 20 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 10 \text{ k}\Omega \angle 90^\circ \\ I &= 2 \text{ mA } \angle 0^\circ \\ E &= 10 \text{ V } \angle 0^\circ \end{aligned}$$

$$I' = \frac{Z_1(hI)}{Z_1 + Z_2} = \frac{(20 \text{ k}\Omega \angle 0^\circ)(100)(2 \text{ mA } \angle 0^\circ)}{20 \text{ k}\Omega + j10 \text{ k}\Omega} = 0.179 \text{ A } \angle -26.57^\circ$$



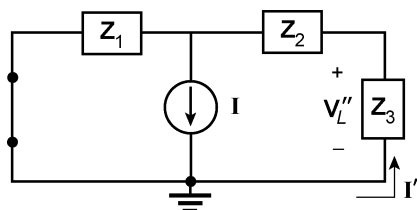
$$\begin{aligned} \mathbf{I}'' &= \frac{\mathbf{E}}{\mathbf{Z}_1 + \mathbf{Z}_2} = \frac{10 \text{ V } \angle 0^\circ}{22.36 \text{ k}\Omega \angle 26.57^\circ} \\ &= 0.447 \text{ mA } \angle -26.57^\circ \\ \mathbf{I}_L &= \mathbf{I}' - \mathbf{I}'' \text{ (direction of } \mathbf{I}') \\ &= 179 \text{ mA } \angle -26.57^\circ - 0.447 \text{ mA } \angle -26.57^\circ \\ &= \mathbf{178.55 \text{ mA } \angle -26.57^\circ} \end{aligned}$$

8. μV :



$$\mathbf{V}'_L = \frac{-\mathbf{Z}_3(\mu\mathbf{V})}{\mathbf{Z}_1 + \mathbf{Z}_2 + \mathbf{Z}_3} = \frac{-(4 \text{ k}\Omega \angle 0^\circ)(20)(2 \text{ V } \angle 0^\circ)}{5 \text{ k}\Omega - j1 \text{ k}\Omega + 4 \text{ k}\Omega} = -17.67 \text{ V } \angle 6.34^\circ$$

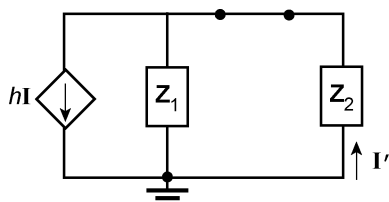
$\mathbf{I}$:



$$\begin{aligned} \text{CDR: } \mathbf{I}' &= \frac{\mathbf{Z}_1 \mathbf{I}}{\mathbf{Z}_1 + \mathbf{Z}_2 + \mathbf{Z}_3} \\ &= \frac{(5 \text{ k}\Omega \angle 0^\circ)(2 \text{ mA } \angle 0^\circ)}{9.056 \text{ k}\Omega \angle -6.34^\circ} \\ &= 1.104 \text{ mA } \angle 6.34^\circ \end{aligned}$$

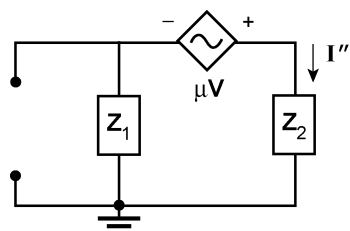
$$\begin{aligned} \mathbf{V}''_L &= -\mathbf{I}' \mathbf{Z}_3 = -(1.104 \text{ mA } \angle 6.34^\circ)(4 \text{ k}\Omega \angle 0^\circ) = -4.416 \text{ V } \angle 6.34^\circ \\ \mathbf{V}_L &= \mathbf{V}'_L + \mathbf{V}''_L = -17.67 \text{ V } \angle 6.34^\circ - 4.416 \text{ V } \angle 6.34^\circ = \mathbf{-22.09 \text{ V } \angle 6.34^\circ} \end{aligned}$$

9.



$$\begin{aligned} \mathbf{Z}_1 &= 20 \text{ k}\Omega \angle 0^\circ \\ \mathbf{Z}_2 &= 5 \text{ k}\Omega + j5 \text{ k}\Omega \end{aligned}$$

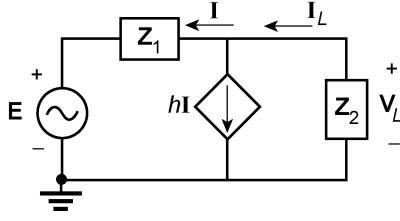
$$\mathbf{I}' = \frac{\mathbf{Z}_1(h\mathbf{I})}{\mathbf{Z}_1 + \mathbf{Z}_2} = \frac{(20 \text{ k}\Omega \angle 0^\circ)(100)(1 \text{ mA } \angle 0^\circ)}{20 \text{ k}\Omega + 5 \text{ k}\Omega + j5 \text{ k}\Omega} = 78.45 \text{ mA } \angle -11.31^\circ$$



$$\begin{aligned} \mathbf{I}'' &= \frac{\mu\mathbf{V}}{\mathbf{Z}_1 + \mathbf{Z}_2} = \frac{(20)(10 \text{ V } \angle 0^\circ)}{25.495 \text{ k}\Omega \angle 11.31^\circ} \\ &= 7.845 \text{ mA } \angle -11.31^\circ \end{aligned}$$

$$\begin{aligned} \mathbf{I}_L &= \mathbf{I}' - \mathbf{I}'' \text{ (direction of } \mathbf{I}') \\ &= 78.45 \text{ mA } \angle -11.31^\circ - 7.845 \text{ mA } \angle -11.31^\circ \\ &= \mathbf{70.61 \text{ mA } \angle -11.31^\circ} \end{aligned}$$

10.



$$\mathbf{Z}_1 = 2 \text{ k}\Omega \angle 0^\circ, \mathbf{Z}_2 = 2 \text{ k}\Omega \angle 0^\circ$$

$$\mathbf{V}_L = -\mathbf{I}_L \mathbf{Z}_2$$

$$\mathbf{I}_L = h\mathbf{I} + \mathbf{I} = (h+1)\mathbf{I}$$

$$\mathbf{V}_L = -(h+1)\mathbf{I}\mathbf{Z}_2$$

$$\text{and by KVL: } \mathbf{V}_L = \mathbf{I}\mathbf{Z}_1 + \mathbf{E}$$

$$\text{so that } \mathbf{I} = \frac{\mathbf{V}_L - \mathbf{E}}{\mathbf{Z}_1}$$

$$\mathbf{V}_L = -(h+1)\mathbf{I}\mathbf{Z}_2 = -(h+1) \left[\frac{\mathbf{V}_L - \mathbf{E}}{\mathbf{Z}_1} \right] \mathbf{Z}_2$$

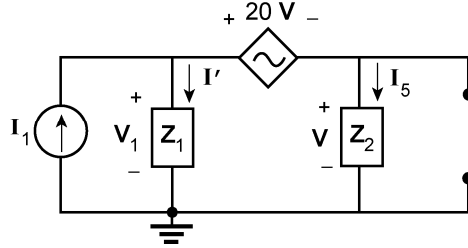
Subt. for $\mathbf{Z}_1, \mathbf{Z}_2$

$$\mathbf{V}_L = -(h+1)(\mathbf{V}_L - \mathbf{E})$$

$$\mathbf{V}_L(2+h) = \mathbf{E}(h+1)$$

$$\mathbf{V}_L = \frac{(h+1)}{(h+2)} \mathbf{E} = \frac{51}{52} (20 \text{ V} \angle 53^\circ) = \mathbf{19.62 \text{ V} } \angle 53^\circ$$

11. $\mathbf{I}_1$:



$$\mathbf{I}_1 = 1 \text{ mA} \angle 0^\circ$$

$$\mathbf{Z}_1 = 2 \text{ k}\Omega \angle 0^\circ$$

$$\mathbf{Z}_2 = 5 \text{ k}\Omega \angle 0^\circ$$

$$\text{KVL: } \mathbf{V}_1 - 20 \text{ V} - \mathbf{V} = 0$$

$$\mathbf{I}' = \frac{\mathbf{V}_1}{\mathbf{Z}_1} \therefore \mathbf{I}' = \frac{21 \text{ V}}{\mathbf{Z}_1} \text{ or } \mathbf{V} = \frac{\mathbf{Z}_1 \mathbf{I}'}{21}$$

$$\mathbf{V}_1 = 21 \text{ V}$$

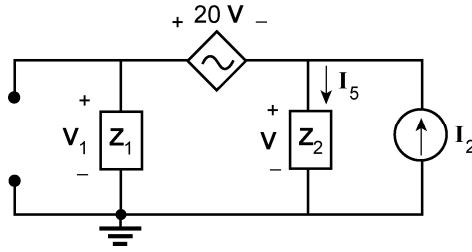
$$\mathbf{V} = \mathbf{I}_5 \mathbf{Z}_2 = [\mathbf{I}_1 - \mathbf{I}'] \mathbf{Z}_2$$

$$\frac{\mathbf{Z}_1}{21} \mathbf{I}' = \mathbf{I}_1 \mathbf{Z}_2 - \mathbf{I}' \mathbf{Z}_2$$

$$\mathbf{I}' \left[\frac{\mathbf{Z}_1}{21} + \mathbf{Z}_2 \right] = \mathbf{I}_1 \mathbf{Z}_2$$

$$\text{and } \mathbf{I}' = \frac{\mathbf{Z}_2}{\frac{\mathbf{Z}_1}{21} + \mathbf{Z}_2} [\mathbf{I}_1] = \frac{(5 \text{ k}\Omega \angle 0^\circ)(1 \text{ mA} \angle 0^\circ)}{\left(\frac{2 \text{ k}\Omega \angle 0^\circ}{21} \right) + 5 \text{ k}\Omega \angle 0^\circ} = 0.981 \text{ mA} \angle 0^\circ$$

I_2 :



$$V_1 = 20 \text{ V} + V = 21 \text{ V}$$

$$I'' = \frac{V_1}{Z_1} = \frac{21 \text{ V}}{Z_1} \Rightarrow V = \frac{Z_1}{21} I''$$

$$I_5 = \frac{V}{Z_2} = \frac{Z_1}{21 Z_2} I''$$

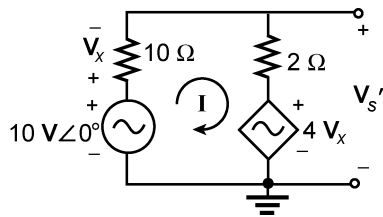
$$I'' = I_2 - I_5 = I_2 - \frac{Z_1}{21 Z_2} I''$$

$$I'' \left[1 + \frac{Z_1}{21 Z_2} \right] = I_2$$

$$I'' = \frac{I_2}{1 + \frac{Z_1}{21 Z_2}} = \frac{2 \text{ mA } \angle 0^\circ}{1 + \frac{2 \text{ k}\Omega}{21(5 \text{ k}\Omega)}} = 1.963 \text{ mA } \angle 0^\circ$$

$$I = I' + I'' = 0.981 \text{ mA } \angle 0^\circ + 1.963 \text{ mA } \angle 0^\circ = 2.94 \text{ mA } \angle 0^\circ$$

12. E_1 :



$$10 \text{ V } \angle 0^\circ - I 10 \Omega - I 2 \Omega - 4 V_x = 0$$

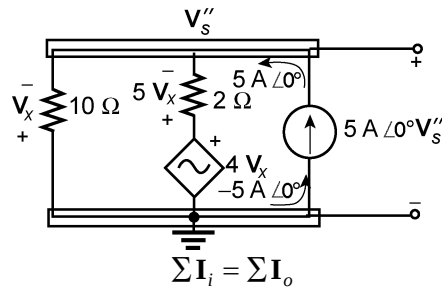
with $V_x = I 10 \Omega$

Solving for I :

$$I = \frac{10 \text{ V } \angle 0^\circ}{52 \Omega} = 192.31 \text{ mA } \angle 0^\circ$$

$$V_s' = 10 \text{ V } \angle 0^\circ - I(10 \Omega) = 10 \text{ V} - (192.31 \text{ mA } \angle 0^\circ)(10 \Omega \angle 0^\circ) = 8.08 \text{ V } \angle 0^\circ$$

I:



$$5 \text{ A } \angle 0^\circ + \frac{V_x}{10 \Omega} + \frac{5 V_x}{2 \Omega} =$$

$$5 \text{ A} + 0.1 V_x + 2.5 V_x = 0$$

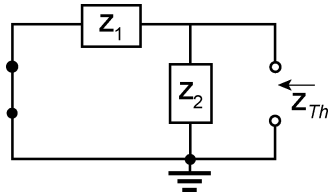
$$2.6 V_x = -5 \text{ A}$$

$$V_x = -\frac{5}{2.6} \text{ V} = -1.923 \text{ V}$$

$$V_s'' = -V_x = -(-1.923 \text{ V}) = 1.923 \text{ V } \angle 0^\circ$$

$$V_s = V_s' + V_s'' = 8.08 \text{ V } \angle 0^\circ + 1.923 \text{ V } \angle 0^\circ = \mathbf{10 \text{ V } \angle 0^\circ}$$

13. Z_{Th} :

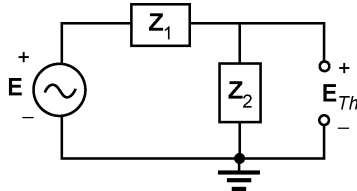


$$Z_1 = 3 \Omega \angle 0^\circ, Z_2 = 4 \Omega \angle 90^\circ$$

$$E = 100 \text{ V } \angle 0^\circ$$

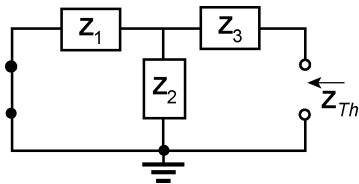
$$Z_{Th} = Z_1 \parallel Z_2 = (3 \Omega \angle 0^\circ \parallel 4 \Omega \angle 90^\circ) \\ = \mathbf{2.4 \Omega \angle 36.87^\circ = 1.92 \Omega + j1.44 \Omega}$$

E_{Th} :



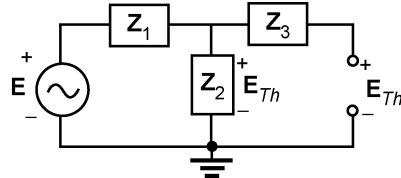
$$E_{Th} = \frac{Z_2 E}{Z_2 + Z_1} = \frac{(4 \Omega \angle 90^\circ)(100 \text{ V } \angle 0^\circ)}{5 \Omega \angle 53.13^\circ} \\ = \mathbf{80 \text{ V } \angle 36.87^\circ}$$

14. Z_{Th} :



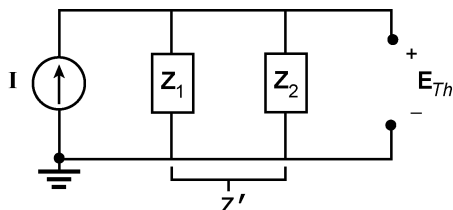
$$Z_{Th} = Z_3 + Z_1 \parallel Z_2 \\ = +j6 \text{ k}\Omega + (2 \text{ k}\Omega \angle 0^\circ \parallel 3 \text{ k}\Omega \angle -90^\circ) \\ = +j6 \text{ k}\Omega + 1.664 \text{ k}\Omega \angle -33.69^\circ \\ = +j6 \text{ k}\Omega + 1.385 \text{ k}\Omega - j0.923 \text{ k}\Omega \\ = 1.385 \text{ k}\Omega + j5.077 \text{ k}\Omega \\ = \mathbf{5.26 \text{ k}\Omega \angle 74.74^\circ}$$

E_{Th} :



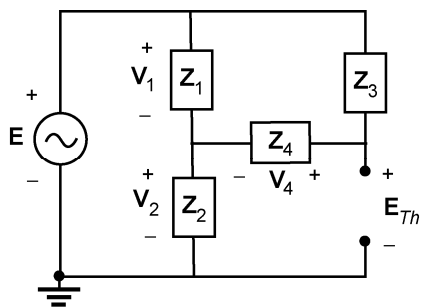
$$E_{Th} = \frac{Z_2 E}{Z_2 + Z_1} = \frac{(3 \text{ k}\Omega \angle -90^\circ)(20 \text{ V } \angle 0^\circ)}{2 \text{ k}\Omega - j3 \text{ k}\Omega} \\ = \frac{60 \text{ V } \angle -90^\circ}{3.606 \angle -56.31^\circ} = \mathbf{16.64 \text{ V } \angle -33.69^\circ}$$

15. From #31. $Z_{Th} = Z_1 \parallel Z_2$
 $Z_{Th} = Z_N = 21.31 \Omega \angle 32.2^\circ$



$$\begin{aligned} E_{Th} &= IZ' = IZ_{Th} \\ &= (0.1 \text{ A } \angle 0^\circ)(21.31 \Omega \angle 32.12^\circ) \\ &= 2.13 \text{ V } \angle 32.2^\circ \end{aligned}$$

16. From #31. $Z_{Th} = Z_N = 6.81 \Omega \angle -54.23^\circ = 3.98 \Omega - j5.53 \Omega$



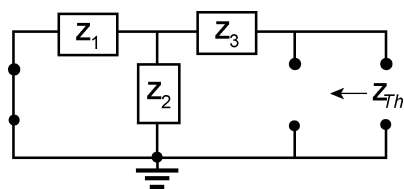
$$\begin{aligned} Z_1 &= 2 \Omega \angle 0^\circ, Z_3 = 8 \Omega \angle -90^\circ \\ Z_2 &= 4 \Omega \angle 90^\circ, Z_4 = 10 \Omega \angle 0^\circ \\ E &= 50 \text{ V } \angle 0^\circ \\ E_{Th} &= V_2 + V_4 \\ V_2 &= \frac{Z_2 E}{Z_2 + Z_1 \parallel (Z_3 + Z_4)} \\ &= \frac{(4 \Omega \angle 90^\circ)(50 \text{ V } \angle 0^\circ)}{+j4 \Omega + 2 \Omega \angle 0^\circ \parallel (10 \Omega - j8 \Omega)} \\ &= 47.248 \text{ V } \angle 24.7^\circ \end{aligned}$$

$$V_1 = E - V_2 = 50 \text{ V } \angle 0^\circ - 47.248 \text{ V } \angle 24.7^\circ = 20.972 \text{ V } \angle -70.285^\circ$$

$$V_4 = \frac{Z_4 V_1}{Z_4 + Z_3} = \frac{(10 \Omega \angle 0^\circ)(20.972 \text{ V } \angle -70.285^\circ)}{10 \Omega - j8 \Omega} = 16.377 \text{ V } \angle -31.625^\circ$$

$$\begin{aligned} E_{Th} &= V_2 + V_4 = 47.248 \text{ V } \angle 24.7^\circ + 16.377 \text{ V } \angle -31.625^\circ \\ &= (42.925 \text{ V} + j19.743 \text{ V}) + (13.945 \text{ V} - j8.587 \text{ V}) \\ &= 56.870 \text{ V} + j11.156 \text{ V} = 57.95 \text{ V } \angle 11.10^\circ \end{aligned}$$

17. Z_{Th} :

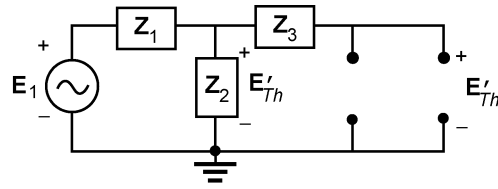


$$\begin{aligned} Z_1 &= 10 \Omega \angle 0^\circ, Z_2 = 8 \Omega \angle 90^\circ \\ Z_3 &= 8 \Omega \angle -90^\circ \end{aligned}$$

$$\begin{aligned} Z_{Th} &= Z_3 + Z_1 \parallel Z_2 \\ &= -j8 \Omega + 10 \Omega \angle 0^\circ \parallel 8 \Omega \angle 90^\circ \\ &= -j8 \Omega + 6.247 \Omega \angle 51.34^\circ \\ &= -j8 \Omega + 3.902 \Omega + j4.878 \Omega \\ &= 3.902 \Omega - j3.122 \Omega \\ &= 5.00 \Omega \angle -38.66^\circ \end{aligned}$$

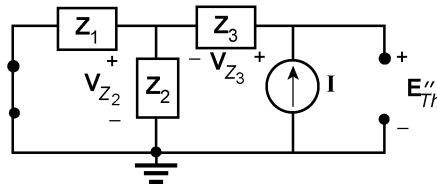
E_{Th} : Superposition:

(E_1)



$$\begin{aligned} E'_{Th} &= \frac{(8 \Omega \angle 90^\circ)(120 \text{ V} \angle 0^\circ)}{10 \Omega + j8 \Omega} \\ &= \frac{960 \text{ V} \angle 90^\circ}{12.806 \angle 38.66^\circ} \\ &= 74.965 \text{ V} \angle 51.34^\circ \end{aligned}$$

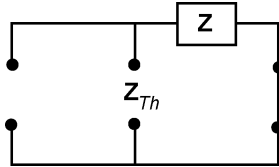
(I)



$$\begin{aligned} E''_{Th} &= V_{Z_2} + V_{Z_3} \\ &= IZ_3 + I(Z_1 \parallel Z_2) \\ &= I(Z_3 + Z_1 \parallel Z_2) \\ &= (0.5 \text{ A} \angle 60^\circ)(-j8 \Omega + 10 \Omega \angle 0^\circ \parallel 8 \Omega \angle 90^\circ) \\ &= (0.5 \text{ A} \angle 60^\circ)(-j8 \Omega + 3.902 \Omega + j4.878 \Omega) \\ &= (0.5 \text{ A} \angle 60^\circ)(3.902 \Omega - j3.122 \Omega) \\ &= (0.5 \text{ A} \angle 60^\circ)(4.997 \Omega \angle -38.663^\circ) \\ &= 2.499 \text{ V} \angle 21.337^\circ \end{aligned}$$

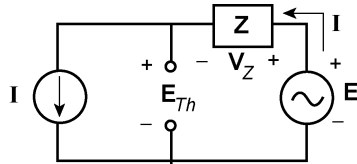
$$\begin{aligned} E_{Th} &= E'_{Th} + E''_{Th} \\ &= 74.965 \text{ V} \angle 51.34^\circ + 2.449 \text{ V} \angle 21.337^\circ \\ &= (46.83 \text{ V} + j58.538 \text{ V}) + (2.328 \text{ V} + j0.909 \text{ V}) \\ &= 49.158 \text{ V} + j59.447 \text{ V} = \mathbf{77.14 \text{ V} \angle 50.41^\circ} \end{aligned}$$

18. Z_{Th} :



$$Z_{Th} = Z = 10 \Omega - j10 \Omega = \mathbf{14.14 \Omega \angle -45^\circ}$$

E_{Th} :

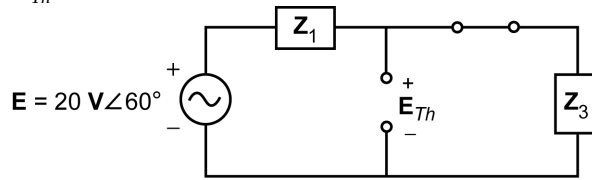


$$\begin{aligned} E_{Th} &= E - V_Z \\ &= 20 \text{ V} \angle 40^\circ - IZ \\ &= 20 \text{ V} \angle 40^\circ - (0.6 \text{ A} \angle 90^\circ)(14.14 \Omega \angle -45^\circ) \\ &= 20 \text{ V} \angle 40^\circ - 8.484 \text{ V} \angle 45^\circ \\ &= (15.321 \text{ V} + j12.856 \text{ V}) - (6 \text{ V} + j6 \text{ V}) \\ &= 9.321 \text{ V} + j6.856 \text{ V} \\ &= \mathbf{11.57 \text{ V} \angle 36.34^\circ} \end{aligned}$$

19. a.

AC:

E_{Th} :



$$Z_1 = \frac{1}{\omega C} \angle -90^\circ = 212.77 \, \Omega \angle -90^\circ$$

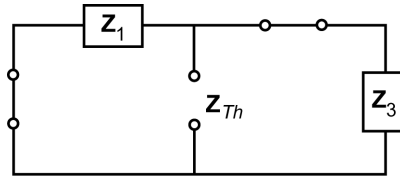
$$Z_3 = 22 \, \Omega + \omega L \angle 90^\circ$$

$$= 22 \, \Omega + j47 \, \Omega$$

$$= 51.89 \, \Omega \angle 64^\circ$$

$$E_{Th} = \frac{Z_3 E}{Z_3 + Z_1} = \frac{(51.89 \, \Omega \angle 64.92^\circ)(20 \, \text{V} \angle 60^\circ)}{22 \, \Omega + j47 \, \Omega - j212.77 \, \Omega} = 6.21 \, \text{V} \angle 207.36^\circ$$

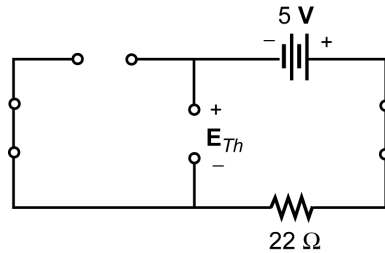
Z_{Th} :



$$Z_{Th} = Z_1 \parallel Z_2 = \frac{(212.77 \, \Omega \angle -90^\circ)(51.89 \, \Omega \angle 64.92^\circ)}{-j212.77 \, \Omega + 22 \, \Omega + j47 \, \Omega}$$

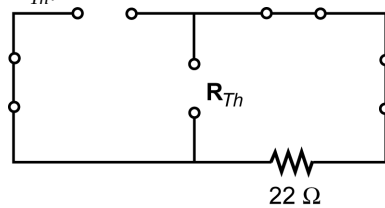
$$= 66.04 \, \Omega \angle 57.36^\circ = 35.62 \, \Omega + j55.61 \, \Omega$$

DC: E_{Th} :



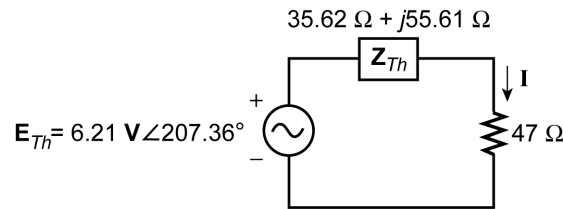
$$E_{Th} = -5 \, \text{V}$$

R_{Th} :



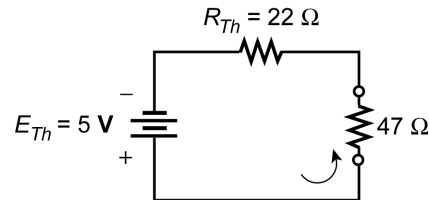
$$R_{Th} = 22 \, \Omega$$

b. AC:



$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{E}_{Th}}{\mathbf{Z}_{Th} + \mathbf{R}_L} \\ &= \frac{6.21 \text{ V } \angle 207.36^\circ}{35.62 \Omega + j55.61 \Omega + 47 \Omega} \\ &= \frac{6.21 \text{ V } \angle 207.36^\circ}{82.62 \Omega + j55.61 \Omega} \\ &= \frac{6.21 \text{ V } \angle 207.36^\circ}{99.59 \Omega \angle 33.94^\circ} \\ &= \mathbf{62.36 \text{ mA } \angle 173.42^\circ} \end{aligned}$$

DC:

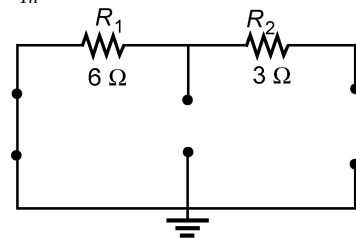


$$\begin{aligned} I &= \frac{5 \text{ V}}{22 \Omega + 47 \Omega} = \frac{5 \text{ V}}{69 \Omega} \\ &= \mathbf{72.46 \text{ mA}} \end{aligned}$$

$$i = -72.46 \text{ mA} + 62.36 \times 10^{-3} \sin(1000t + 173.42^\circ)$$

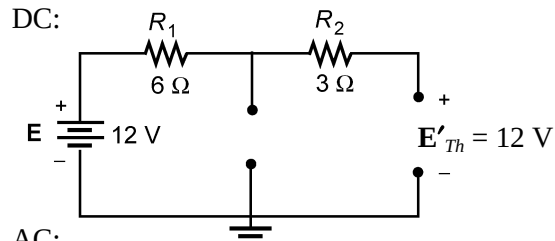
matching the results of Problem 4.

20. a. $\mathbf{Z}_{Th}$:

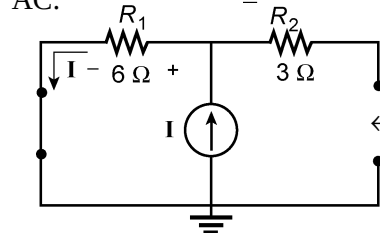


$$\leftarrow \mathbf{Z}_{Th} = \mathbf{Z}_{R_1} + \mathbf{Z}_{R_2} = 6 \Omega + 3 \Omega = \mathbf{9 \Omega}$$

DC:



AC:

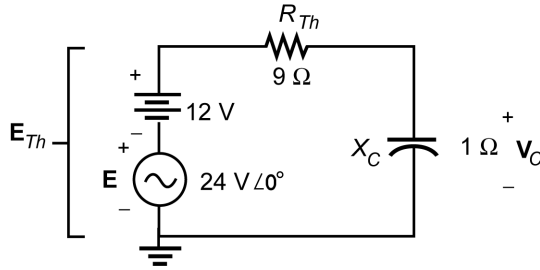


$$\leftarrow \mathbf{E}''_{Th} = \mathbf{I} \mathbf{Z}_{R_1} = (4 \text{ A } \angle 0^\circ)(6 \Omega \angle 0^\circ) = \mathbf{24 \text{ V } \angle 0^\circ}$$

$$\mathbf{E}_{Th} = 12 \text{ V} + 24 \text{ V } \angle 0^\circ$$

(DC) (AC)

b.

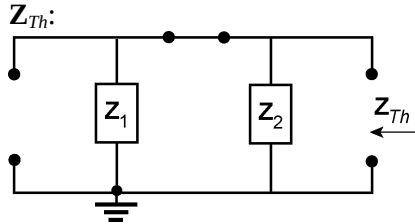


$$\text{DC: } V_C = 12 \text{ V}$$

$$\begin{aligned} \text{AC: } V_C &= \frac{Z_C E}{Z_C + Z_{R_{Th}}} \\ &= \frac{(1 \Omega \angle -90^\circ)(24 \text{ V} \angle 0^\circ)}{-j1 \Omega + 9 \Omega} \\ &= \frac{24 \text{ V} \angle -90^\circ}{9.055 \angle -6.34^\circ} \\ V_C &= 2.65 \text{ V} \angle -83.66^\circ \end{aligned}$$

$$\begin{aligned} v_C &= 12 \text{ V} + 2.65 \text{ V} \angle -83.66^\circ \\ &= 12 \text{ V} + 3.75 \sin(\omega t - 83.66^\circ) \end{aligned}$$

21. a.

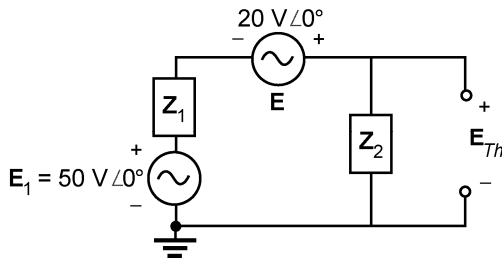


$$\begin{aligned} Z_1 &= 10 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 5 \text{ k}\Omega - j5 \text{ k}\Omega \\ &= 7.071 \text{ k}\Omega \angle -45^\circ \end{aligned}$$

$$Z_{Th} = Z_1 \parallel Z_2 = (10 \text{ k}\Omega \angle 0^\circ) \parallel (7.071 \text{ k}\Omega \angle -45^\circ) = 4.47 \text{ k}\Omega \angle -26.57^\circ$$

Source conversion:

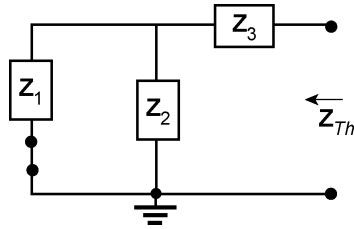
$$E_1 = (I \angle \theta)(R_1 \angle 0^\circ) = (5 \text{ mA} \angle 0^\circ)(10 \text{ k}\Omega \angle 0^\circ) = 50 \text{ V} \angle 0^\circ$$



$$\begin{aligned} E_{Th} &= \frac{Z_2(E + E_1)}{Z_2 + Z_1} \\ &= \frac{(7.071 \text{ k}\Omega \angle -45^\circ)(20 \text{ V} \angle 0^\circ + 50 \text{ V} \angle 0^\circ)}{(5 \text{ k}\Omega - j5 \text{ k}\Omega) + (10 \text{ k}\Omega)} \\ &= \frac{(7.071 \text{ k}\Omega \angle -45^\circ)(70 \text{ V} \angle 0^\circ)}{(15 \text{ k}\Omega - j5 \text{ k}\Omega)} \\ &= \frac{494.97 \text{ V} \angle -45^\circ}{15.811 \angle -18.435^\circ} \\ &= 31.31 \text{ V} \angle -26.57^\circ \end{aligned}$$

$$\begin{aligned} \text{b. } I &= \frac{E_{Th}}{Z_{Th} + Z_L} = \frac{31.31 \text{ V} \angle -26.565^\circ}{4.472 \text{ k}\Omega \angle -26.565^\circ + 5 \text{ k}\Omega \angle 90^\circ} \\ &= \frac{31.31 \text{ V} \angle -26.565^\circ}{4 \text{ k}\Omega - j2 \text{ k}\Omega + j5 \text{ k}\Omega} = \frac{31.31 \text{ V} \angle -26.565^\circ}{4 \text{ k}\Omega + j3 \text{ k}\Omega} \\ &= \frac{31.31 \text{ V} \angle -26.565^\circ}{5 \text{ k}\Omega \angle 36.87^\circ} = 6.26 \text{ mA} \angle 63.44^\circ \end{aligned}$$

22.

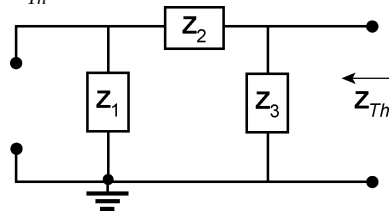


$$\begin{aligned} Z_1 &= 10 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 10 \text{ k}\Omega \angle 0^\circ \\ Z_3 &= 1 \text{ k}\Omega \angle -90^\circ \end{aligned}$$

$$Z_{Th} = Z_3 + Z_1 \parallel Z_2 = 5 \text{ k}\Omega - j1 \text{ k}\Omega \cong 5.1 \text{ k}\Omega \angle -11.31^\circ$$

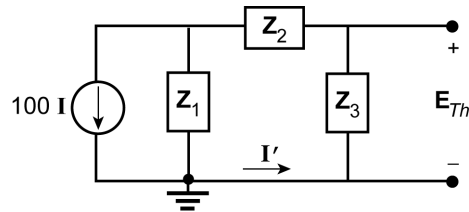
$$E_{Th}: (\text{VDR}) \quad E_{Th} = \frac{Z_2(20 \text{ V})}{Z_2 + Z_1} = \frac{(10 \text{ k}\Omega \angle 0^\circ)(20 \text{ V})}{10 \text{ k}\Omega + 10 \text{ k}\Omega} = 10 \text{ V}$$

23. Z_{Th} :



$$\begin{aligned} Z_1 &= 40 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 0.2 \text{ k}\Omega \angle -90^\circ \\ Z_3 &= 5 \text{ k}\Omega \angle 0^\circ \end{aligned}$$

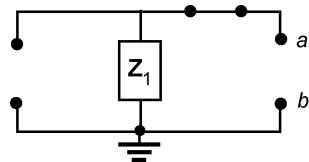
$$Z_{Th} = Z_3 \parallel (Z_1 + Z_2) = 5 \text{ k}\Omega \angle 0^\circ \parallel (40 \text{ k}\Omega - j0.2 \text{ k}\Omega) = 4.44 \text{ k}\Omega \angle -0.03^\circ$$



$$\begin{aligned} I' &= \frac{Z_1(100 \text{ I})}{Z_1 + Z_2 + Z_3} \\ &= \frac{(40 \text{ k}\Omega \angle 0^\circ)(100 \text{ I})}{45 \text{ k}\Omega \angle -0.255^\circ} \\ &= 88.89 \text{ I} \angle 0.255^\circ \end{aligned}$$

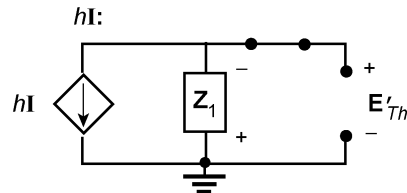
$$E_{Th} = -I'Z_3 = -(88.89 \text{ I} \angle 0.255^\circ)(5 \text{ k}\Omega \angle 0^\circ) = -444.45 \times 10^3 \text{ I} \angle 0.26^\circ$$

24. Z_{Th} :



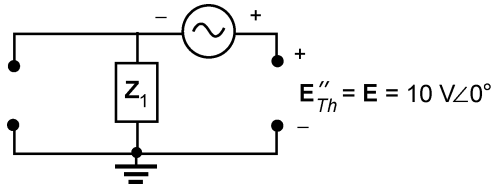
$$\leftarrow Z_{Th} = Z_1 = 20 \text{ k}\Omega \angle 0^\circ$$

E_{Th} :



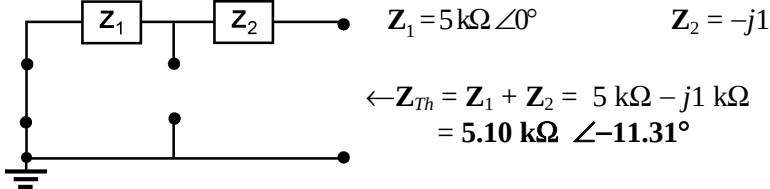
$$\begin{aligned} E'_{Th} &= -(hI)(Z_1) \\ &= -(100)(2 \text{ mA} \angle 0^\circ)(20 \text{ k}\Omega \angle 0^\circ) \\ &= -4 \text{ kV} \angle 0^\circ \end{aligned}$$

E:



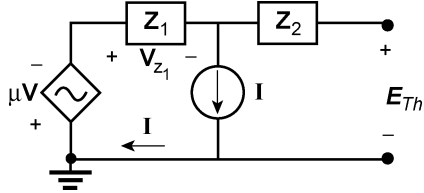
$$\begin{aligned} E_{Th} &= E'_{Th} + E''_{Th} \\ &= -4 \text{ kV } \angle 0^\circ + 10 \text{ V } \angle 0^\circ \\ &= -3990 \text{ V } \angle 0^\circ \end{aligned}$$

25. Z_{Th} :



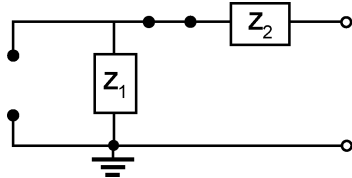
$$\begin{aligned} \leftarrow Z_{Th} &= Z_1 + Z_2 = 5 \text{ k}\Omega - j1 \text{ k}\Omega \\ &= 5.10 \text{ k}\Omega \angle -11.31^\circ \end{aligned}$$

E_{Th} :



$$\begin{aligned} E_{Th} &= -[\mu V + V_{Z_1}] \\ &= -\mu V - IZ_1 \\ &= -(20)(2 \text{ V } \angle 0^\circ) - (2 \text{ mA } \angle 0^\circ)(5 \text{ k}\Omega \angle 0^\circ) \\ &= -50 \text{ V } \angle 0^\circ \end{aligned}$$

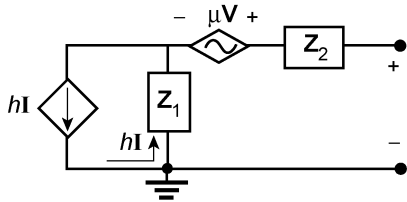
26. Z_{Th} :



$$\begin{aligned} Z_1 &= 20 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 5 \text{ k}\Omega \angle 0^\circ \end{aligned}$$

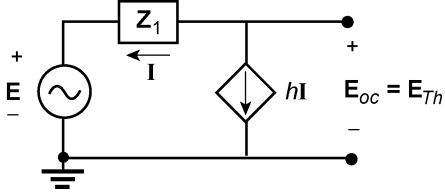
$$\leftarrow Z_{Th} = Z_1 + Z_2 = 25 \text{ k}\Omega \angle 0^\circ$$

E_{Th} :



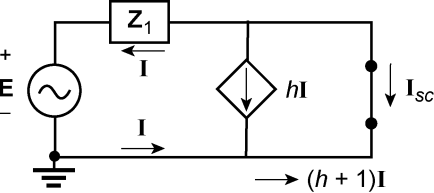
$$\begin{aligned} E_{Th} &= \mu V - (hI)(Z_1) \\ &= (20)(10 \text{ V } \angle 0^\circ) - (100)(1 \text{ mA } \angle 0^\circ)(20 \text{ k}\Omega \angle 0^\circ) \\ &= -1800 \text{ V } \angle 0^\circ \end{aligned}$$

27. E_{Th} : (E_{oc})



$$\begin{aligned} hI &= -I & Z_1 &= 2 \text{ k}\Omega \angle 0^\circ \\ \therefore I &= 0 \\ \text{and } hI &= 0 \\ \text{with } E_{oc} &= E_{Th} = E = 20 \text{ V} \angle 53^\circ \end{aligned}$$

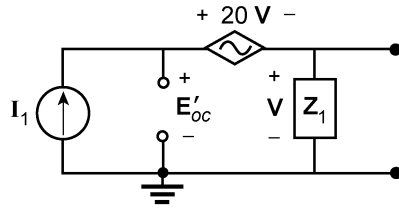
I_{sc} :



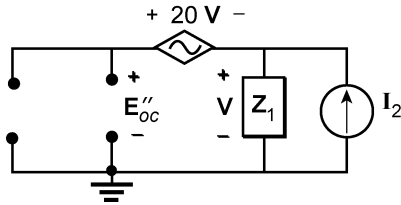
$$\begin{aligned} I_{sc} &= -(h+1)I \\ &= -(h+1)(10 \text{ mA} \angle 53^\circ) \\ &= -510 \text{ mA} \angle 53^\circ \end{aligned}$$

$$Z_{Th} = \frac{E_{oc}}{I_{sc}} = \frac{20 \text{ V} \angle 53^\circ}{-510 \text{ mA} \angle 53^\circ} = -39.22 \Omega \angle 0^\circ \text{ (negative resistance)}$$

28. E_{Th} :

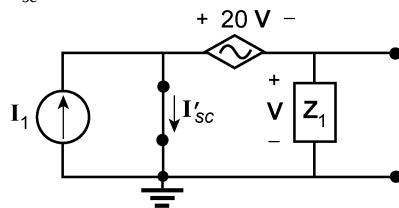


$$\begin{aligned} E'_{oc} &= 21 \text{ V} & Z_1 &= 5 \text{ k}\Omega \angle 0^\circ \\ V &= I_1 Z_1 = (1 \text{ mA} \angle 0^\circ)(5 \text{ k}\Omega \angle 0^\circ) \\ &= 5 \text{ V} \angle 0^\circ \\ E'_{oc} &= E'_{Th} = 21(5 \text{ V} \angle 0^\circ) \\ &= 105 \text{ V} \angle 0^\circ \end{aligned}$$

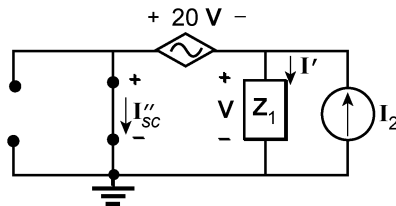


$$\begin{aligned} V &= I_2 Z_1 \\ &= (2 \text{ mA} \angle 0^\circ)(5 \text{ k}\Omega \angle 0^\circ) \\ &= 10 \text{ V} \angle 0^\circ \\ E''_{oc} &= E''_{Th} = V + 20 \text{ V} = 21 \text{ V} = 210 \text{ V} \angle 0^\circ \end{aligned}$$

I_{sc} :



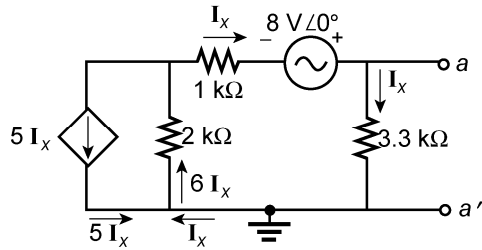
$$I'_{sc} = I_1$$



$$\begin{aligned} 20 \text{ V} &= V \therefore V = 0 \text{ V} \\ &\text{and } I' = 0 \text{ A} \\ \therefore I''_{sc} &= I_2 \end{aligned}$$

$$\begin{aligned} I_{sc} &= I'_{sc} + I''_{sc} = 3 \text{ mA} \angle 0^\circ \\ E_{oc} &= E'_{oc} + E''_{oc} = 315 \text{ V} \angle 0^\circ = E_{Th} \\ Z_{Th} &= \frac{E_{oc}}{I_{sc}} = \frac{315 \text{ V} \angle 0^\circ}{3 \text{ mA} \angle 0^\circ} = 105 \text{ k}\Omega \angle 0^\circ \end{aligned}$$

29. E_{oc} :
(E_{Th})

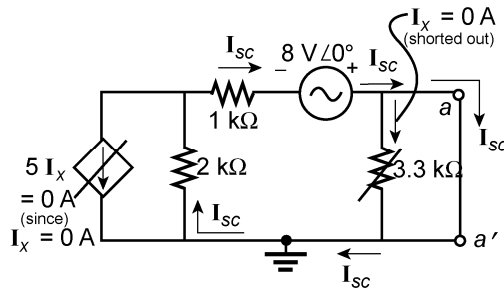


$$\text{KVL: } -6 I_x(2 \text{ k}\Omega) - I_x(1 \text{ k}\Omega) + 8 \text{ V } \angle 0^\circ - I_x(3.3 \text{ k}\Omega) = 0$$

$$I_x = \frac{8 \text{ V } \angle 0^\circ}{16.3 \text{ k}\Omega} = 0.491 \text{ mA } \angle 0^\circ$$

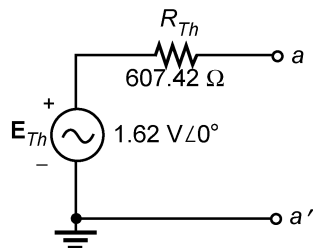
$$E_{oc} = E_{Th} = I_x(3.3 \text{ k}\Omega) = 1.62 \text{ V } \angle 0^\circ$$

I_{sc} :



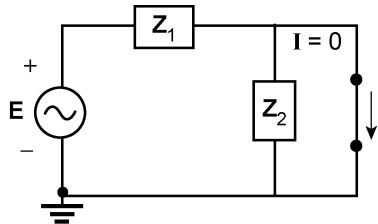
$$I_{sc} = \frac{8 \text{ V}}{3 \text{ k}\Omega} = 2.667 \text{ mA } \angle 0^\circ$$

$$Z_{Th} = \frac{E_{oc}}{I_{sc}} = \frac{1.62 \text{ V } \angle 0^\circ}{2.667 \text{ mA } \angle 0^\circ} = 607.42 \Omega \angle 0^\circ$$



30. From Problem 13: $Z_N = Z_{Th} = 1.92 \Omega + j1.44 \Omega = 2.4 \Omega \angle 36.87^\circ$

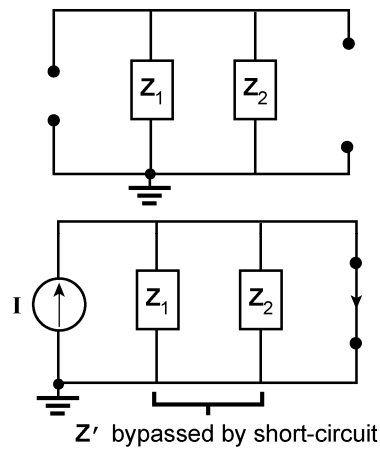
I_N :



$$Z_1 = 3 \Omega \angle 0^\circ, Z_2 = 4 \Omega \angle 90^\circ$$

$$I_{sc} = I_N = \frac{E}{Z_1} = \frac{100 \text{ V } \angle 0^\circ}{3 \Omega \angle 0^\circ} = 33.33 \text{ A } \angle 0^\circ$$

31.



$$Z_1 = 20 \Omega + j20 \Omega = 28.284 \Omega \angle 45^\circ$$

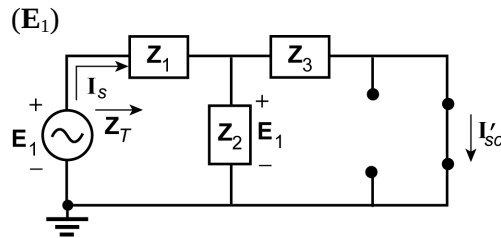
$$Z_2 = 68 \Omega \angle 0^\circ$$

$$\begin{aligned} \leftarrow Z_N &= Z_1 \parallel Z_2 \\ &= (28.284 \Omega \angle 45^\circ) \parallel (68 \Omega \angle 0^\circ) \\ &= 21.31 \Omega \angle 32.2^\circ \end{aligned}$$

$$\leftarrow I_{sc} = I = I_N = 0.1 \text{ A} \angle 0^\circ$$

32. From Problem 17: $Z_N = Z_{Th} = 5.00 \Omega \angle -38.66^\circ$

I_N : Superposition:



$$\begin{aligned} Z_T &= Z_1 + Z_2 \parallel Z_3 \\ &= 10 \Omega + 8 \Omega \angle 90^\circ \parallel 8 \Omega \angle -90^\circ \\ &= 10 \Omega + \frac{64 \Omega \angle 0^\circ}{0} \\ &= \text{very large impedance} \end{aligned}$$

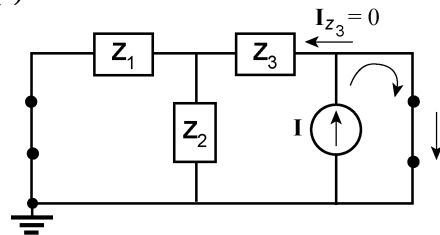
$$I_s = \frac{E}{Z_T} = 0 \text{ A}$$

$$\text{and } V_{Z_1} = 0 \text{ V}$$

$$\text{with } V_{Z_2} = V_{Z_3} = E_1 = 120 \text{ V} \angle 0^\circ$$

$$\begin{aligned} \text{so that } I'_{sc} &= \frac{E_1}{Z_3} = \frac{120 \text{ V} \angle 0^\circ}{8 \Omega \angle -90^\circ} \\ &= 15 \text{ A} \angle 90^\circ \end{aligned}$$

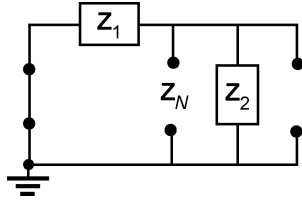
(I)



$$I''_{sc} = I = 0.5 \text{ A} \angle 60^\circ$$

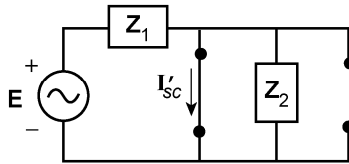
$$\begin{aligned} I_N &= I'_{sc} + I''_{sc} = +j15 \text{ A} + 0.5 \text{ A} \angle 60^\circ = +j15 \text{ A} + 0.25 \text{ A} + j0.433 \text{ A} \\ &= 0.25 \text{ A} + j15.433 \text{ A} = 15.44 \text{ A} \angle 89.07^\circ \end{aligned}$$

33. a. Z_N :



$$\begin{aligned} E &= 20 \text{ V } \angle 0^\circ, I_2 = 0.4 \text{ A } \angle 20^\circ \\ Z_1 &= 6 \Omega + j8 \Omega = 10 \Omega \angle 53.13^\circ \\ Z_2 &= \Omega - j12 \Omega = 15 \Omega \angle -53.13^\circ \\ Z_N &= Z_1 \parallel Z_2 = (10 \Omega \angle 53.13^\circ) \parallel (15 \Omega \angle -53.13^\circ) \\ &= 9.66 \Omega \angle 14.93^\circ \end{aligned}$$

I_N :
(E)

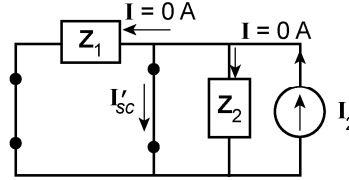


$$I'_{sc} = E/Z_1 = 20 \text{ V } \angle 0^\circ / 10 \Omega \angle 53.13^\circ$$

$$= 2 \text{ A } \angle -53.13^\circ$$

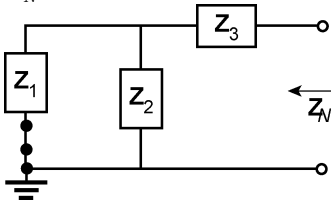
$$\begin{aligned} I_N &= I'_{sc} + I''_{sc} = 2 \text{ A } \angle -53.13^\circ + 0.4 \text{ A } \angle 20^\circ \\ &= 2.15 \text{ A } \angle -42.87^\circ \end{aligned}$$

(I_2)



$$I''_{sc} = I_2 = 0.4 \text{ A } \angle 20^\circ$$

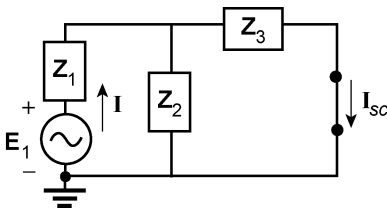
34. Z_N :



$$\begin{aligned} E_1 &= 120 \text{ V } \angle 30^\circ, Z_1 = 3 \Omega \angle 0^\circ \\ Z_2 &= 8 \Omega - j8 \Omega, Z_3 = 4 \Omega \angle 90^\circ \end{aligned}$$

$$\begin{aligned} Z_N &= Z_3 + Z_1 \parallel Z_2 \\ &= 4 \Omega \angle 90^\circ + (3 \Omega \angle 0^\circ) \parallel (8 \Omega - j8 \Omega) \\ &= 4.37 \Omega \angle 55.67^\circ = 2.47 \Omega + j3.61 \Omega \end{aligned}$$

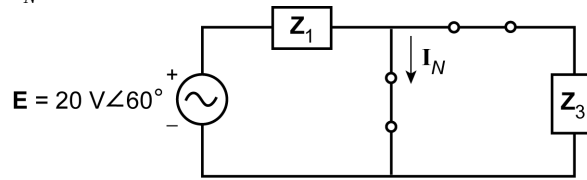
I_N :



$$\begin{aligned} I &= \frac{E_1}{Z_T} = \frac{120 \text{ V } \angle 30^\circ}{Z_1 + Z_2 \parallel Z_3} \\ &= \frac{120 \text{ V } \angle 30^\circ}{3 \Omega + (8 \Omega - j8 \Omega) \parallel 4 \Omega \angle 90^\circ} \\ &= \frac{120 \text{ V } \angle 30^\circ}{6.65 \Omega \angle 46.22^\circ} \\ &= 18.05 \text{ A } \angle -16.22^\circ \end{aligned}$$

$$I_{sc} = I_N = \frac{Z_2(I)}{Z_2 + Z_3} = \frac{(8 \Omega - j8 \Omega)(18.05 \text{ A } \angle -16.22^\circ)}{8 \Omega - j8 \Omega + j4 \Omega} = 22.83 \text{ A } \angle -34.65^\circ$$

35. a. AC:
 I_N :



$$Z_1 = 212.77 \, \Omega \angle -90^\circ$$

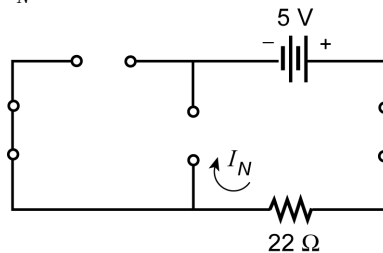
$$Z_3 = 22 \, \Omega + j47 \, \Omega$$

$$= 51.89 \, \Omega \angle 64^\circ$$

$$I_N = \frac{E}{Z_1} = \frac{20 \, \text{V} \angle 60^\circ}{212.77 \, \Omega \angle -90^\circ} = 94 \, \text{mA} \angle 150^\circ$$

$$Z_N = Z_{Th} \text{ (problem 19)} = 66.04 \, \Omega \angle 57.36^\circ = 35.62 \, \Omega + j55.61 \, \Omega$$

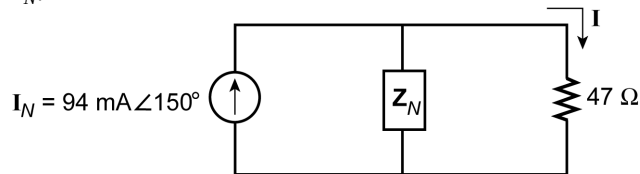
- DC:
 I_N :



$$I_N = \frac{5 \, \text{V}}{22 \, \Omega} = 227.27 \, \text{mA}$$

$$R_N = R_{Th} = \text{(problem 19)} = 22 \, \Omega$$

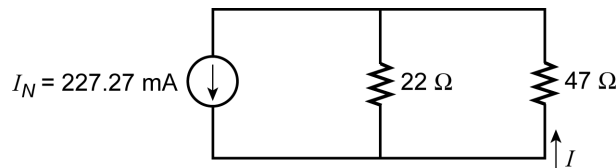
- b. AC:
 I_N :



$$I = \frac{Z_N (I_N)}{Z_N + 47 \, \Omega} = \frac{(66.04 \, \Omega \angle 57.36^\circ)(94 \, \text{mA} \angle 150^\circ)}{35.62 \, \Omega + j55.61 \, \Omega + 47 \, \Omega}$$

$$= \frac{6.21 \, \text{A} \angle 207.36^\circ}{99.08 \, \Omega \angle 34.14^\circ} = 62.68 \, \text{mA} \angle 173.22^\circ$$

- DC:

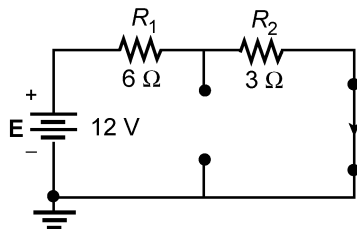


$$I = \frac{22 \, \Omega (227.27 \, \text{mA})}{22 \, \Omega + 47 \, \Omega} = 72.46 \, \text{mA}$$

and $i = -72.46 \, \text{mA} + 62.68 \times 10^{-3} \sin(1000t + 173.22^\circ)$
 Same as Problem 4 and 19.

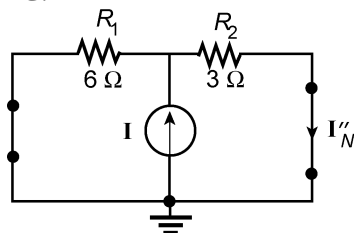
36. a. From #20 $\mathbf{Z}_N = \mathbf{Z}_{Th} = 9\ \Omega \angle 0^\circ$

DC:



$$I'_N = \frac{E}{R_T} = \frac{12\text{ V}}{9\ \Omega} = 1.33\text{ A}$$

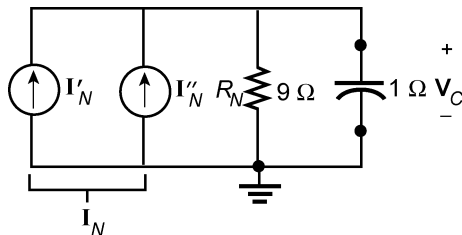
AC:



$$\begin{aligned} I''_N &= \frac{R_1 \mathbf{I}}{R_1 + R_2} = \frac{(6\ \Omega \angle 0^\circ)(4\text{ A} \angle 0^\circ)}{9\ \Omega \angle 0^\circ} \\ &= \frac{24\text{ V} \angle 0^\circ}{9\ \Omega \angle 0^\circ} = 2.67\text{ A} \angle 0^\circ \end{aligned}$$

$$\mathbf{I}_N = 1.33\text{ A} + 2.67\text{ A} \angle 0^\circ$$

b.



$$\begin{aligned} \text{DC: } V_C &= IR \\ &= (1.33\text{ A})(9\ \Omega) \\ &= 12\text{ V} \end{aligned}$$

$$\begin{aligned} \text{AC: } \mathbf{Z}' &= 9\ \Omega \angle 0^\circ \parallel 1\ \Omega \angle -90^\circ \\ &= 0.994\ \Omega \angle -83.66^\circ \end{aligned}$$

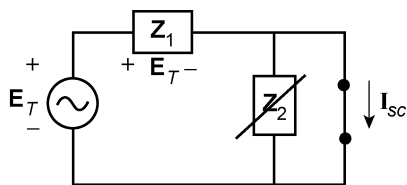
$$\begin{aligned} \mathbf{V}_C &= \mathbf{I} \mathbf{Z}' = (2.667\text{ A} \angle 0^\circ)(0.994\ \Omega \angle -83.66^\circ) \\ &= 2.65\text{ V} \angle -83.66^\circ \end{aligned}$$

$$\mathbf{V}_C = 12\text{ V} + 2.65\text{ V} \angle -83.66^\circ$$

37. a. Note Problem 21(a): $\mathbf{Z}_N = \mathbf{Z}_{Th} = 4.47\text{ k}\Omega \angle -26.57^\circ$

Using the same source conversion: $\mathbf{E}_1 = 50\text{ V} \angle 0^\circ$

Defining $\mathbf{E}_T = \mathbf{E}_1 + \mathbf{E} = 50\text{ V} \angle 0^\circ + 20\text{ V} \angle 0^\circ = 70\text{ V} \angle 0^\circ$



$$\begin{aligned} \mathbf{Z}_1 &= 10\text{ k}\Omega \angle 0^\circ \\ \mathbf{Z}_2 &= 5\text{ k}\Omega - j5\text{ k}\Omega = 7.071\text{ k}\Omega \angle -45^\circ \end{aligned}$$

$$\mathbf{I}_{sc} = \frac{\mathbf{E}_T}{\mathbf{Z}_1} = \frac{70\text{ V} \angle 0^\circ}{10\text{ k}\Omega \angle 0^\circ} = 7\text{ mA} \angle 0^\circ$$

$$\mathbf{I}_N = \mathbf{I}_{sc} = 7\text{ mA} \angle 0^\circ$$

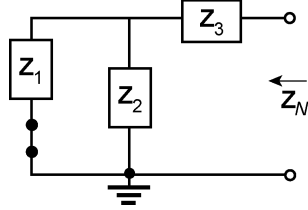
b.
$$\mathbf{I} = \frac{\mathbf{Z}_N(\mathbf{I}_N)}{\mathbf{Z}_N + \mathbf{Z}_L} = \frac{(4.472 \text{ k}\Omega \angle -26.565^\circ)(7 \text{ mA} \angle 0^\circ)}{4.472 \text{ k}\Omega \angle -26.565^\circ + 5 \text{ k}\Omega \angle 90^\circ}$$

$$= \frac{31.30 \text{ mA} \angle -26.565^\circ}{4 - j2 + j5} = \frac{31.30 \text{ mA} \angle -26.565^\circ}{4 + j3}$$

$$= \frac{31.30 \text{ mA} \angle -26.565^\circ}{5 \angle 36.87^\circ} = \mathbf{6.26 \text{ mA} \angle 63.44^\circ}$$
 as obtained in Problem 21.

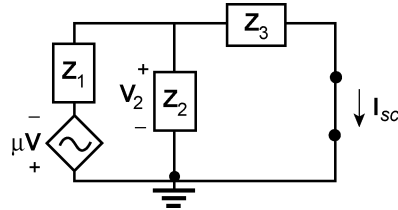
38.

$\mathbf{Z}_N$:



$$\begin{aligned} \mathbf{Z}_1 &= 10 \text{ k}\Omega \angle 0^\circ, \mathbf{Z}_2 = 10 \text{ k}\Omega \angle 0^\circ \\ \mathbf{Z}_3 &= -j1 \text{ k}\Omega \\ \mathbf{Z}_N &= \mathbf{Z}_3 + \mathbf{Z}_1 \parallel \mathbf{Z}_2 = 5 \text{ k}\Omega - j1 \text{ k}\Omega \\ &= \mathbf{5.1 \text{ k}\Omega \angle -11.31^\circ} \end{aligned}$$

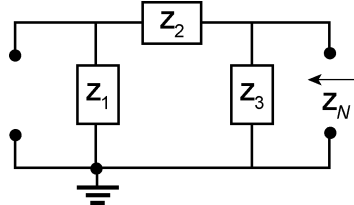
$\mathbf{I}_N$:



$$\begin{aligned} \mathbf{V}_2 &= \frac{-(\mathbf{Z}_2 \parallel \mathbf{Z}_3)20 \text{ V}}{(\mathbf{Z}_2 \parallel \mathbf{Z}_3) + \mathbf{Z}_1} \\ &= \frac{-(0.995 \text{ k}\Omega \angle -84.29^\circ)(20 \text{ V})}{0.1 \text{ k}\Omega - j0.99 \text{ k}\Omega + 10 \text{ k}\Omega} \\ \mathbf{V}_2 &= -1.961 \text{ V} \angle -78.69^\circ \end{aligned}$$

$$\mathbf{I}_N = \mathbf{I}_{sc} = \frac{\mathbf{V}_2}{\mathbf{Z}_3} = \frac{-1.961 \text{ V} \angle -78.69^\circ}{1 \text{ k}\Omega \angle -90^\circ} = \mathbf{-1.96 \times 10^{-3} \text{ V} \angle 11.31^\circ}$$

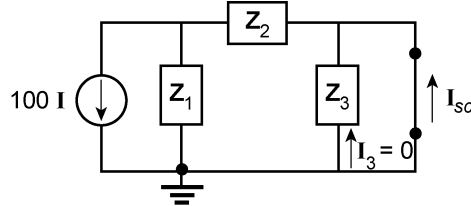
39. $\mathbf{Z}_N$:



$$\begin{aligned} \mathbf{Z}_1 &= 40 \text{ k}\Omega \angle 0^\circ, \mathbf{Z}_2 = 0.2 \text{ k}\Omega \angle -90^\circ \\ \mathbf{Z}_3 &= 5 \text{ k}\Omega \angle 0^\circ \end{aligned}$$

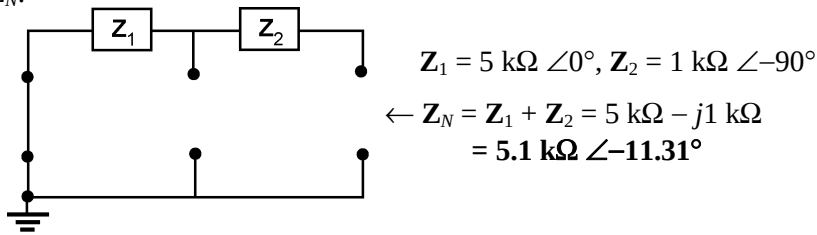
$$\begin{aligned} \mathbf{Z}_N &= \mathbf{Z}_3 \parallel (\mathbf{Z}_1 + \mathbf{Z}_2) \\ &= 5 \text{ k}\Omega \angle 0^\circ \parallel (40 \text{ k}\Omega - j0.2 \text{ k}\Omega) \\ &= \mathbf{4.44 \text{ k}\Omega \angle -0.03^\circ} \end{aligned}$$

$\mathbf{I}_N$:

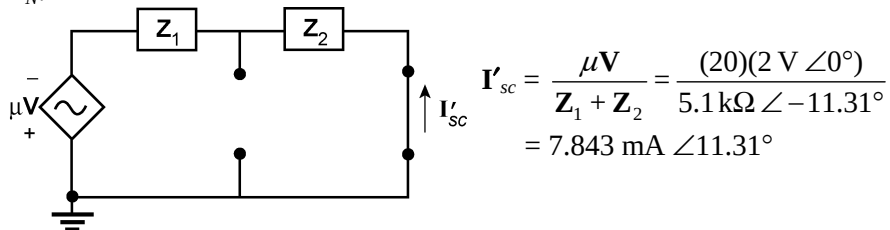


$$\begin{aligned} \mathbf{I}_N = \mathbf{I}_{sc} &= \frac{\mathbf{Z}_1(100 \text{ I})}{\mathbf{Z}_1 + \mathbf{Z}_2} \\ &= \frac{(40 \text{ k}\Omega \angle 0^\circ)(100 \text{ I})}{40 \text{ k}\Omega \angle -0.286^\circ} \\ &= \mathbf{100 \text{ I} \angle 0.29^\circ} \end{aligned}$$

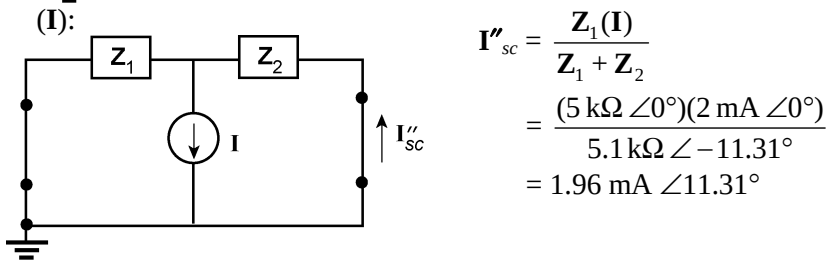
40. Z_N :



I_N :



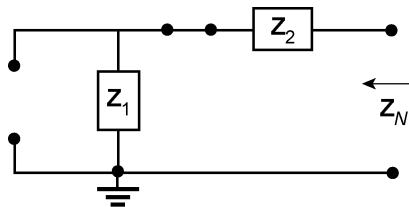
(I):



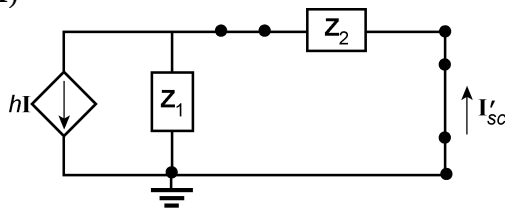
$$I_N = I'_{sc} + I''_{sc} = 7.843 \text{ mA} \angle 11.31^\circ + 1.96 \text{ mA} \angle 11.31^\circ$$

$$= 9.81 \text{ mA} \angle 11.31^\circ$$

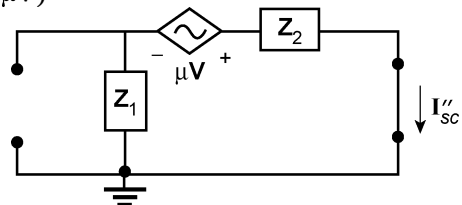
41. Z_N :



I_N : (hI)

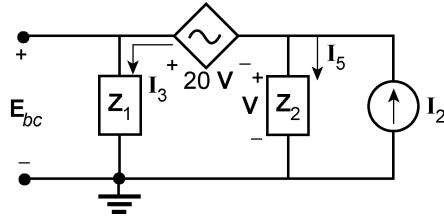


(μV)



$$I_N \text{ (direction of } I'_{sc}) = I'_{sc} - I''_{sc} = 80 \text{ mA} \angle 0^\circ - 8 \text{ mA} \angle 0^\circ = 72 \text{ mA} \angle 0^\circ$$

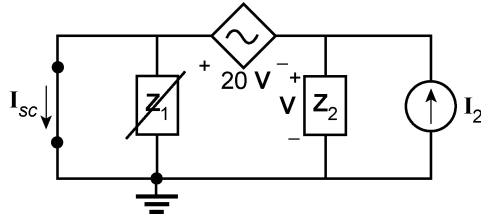
42.



$$\begin{aligned} Z_1 &= 2 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 5 \text{ k}\Omega \angle 0^\circ \end{aligned}$$

$$\begin{aligned} I_2 &= I_3 + I_5 \\ V &= I_5 Z_2 = (I_2 - I_3) Z_2 \\ E_{oc} = E_{Th} &= 21 \text{ V} = 21(I_2 - I_3) Z_2 \\ &= 21 \left(I_2 - \frac{E_{oc}}{Z_1} \right) Z_2 \end{aligned}$$

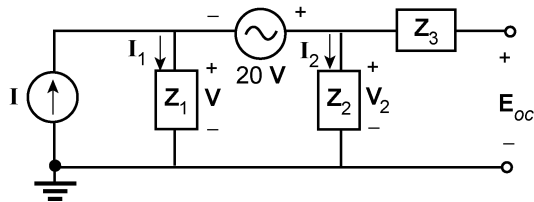
$$\begin{aligned} E_{oc} \left[1 + 21 \frac{Z_2}{Z_1} \right] &= 21 Z_2 I_2 \\ E_{oc} &= \frac{21 Z_2 I_2}{1 + 21 \frac{Z_2}{Z_1}} = \frac{21(5 \text{ k}\Omega \angle 0^\circ)(2 \text{ mA} \angle 0^\circ)}{1 + 21 \left(\frac{5 \text{ k}\Omega \angle 0^\circ}{2 \text{ k}\Omega \angle 0^\circ} \right)} \\ E_{Th} = E_{oc} &= 3.925 \text{ V} \angle 0^\circ \end{aligned}$$



$$\begin{aligned} 20 \text{ V} &\neq -V \therefore V = 0 \\ \text{and } I_N = I_{sc} = I_2 &= 2 \text{ mA} \angle 0^\circ \end{aligned}$$

$$Z_N = \frac{E_{oc}}{I_{sc}} = \frac{3.925 \text{ V} \angle 0^\circ}{2 \text{ mA} \angle 0^\circ} = 1.96 \text{ k}\Omega$$

43.



$$\begin{aligned} Z_1 &= 1 \text{ k}\Omega \angle 0^\circ \\ Z_2 &= 3 \text{ k}\Omega \angle 0^\circ \\ Z_3 &= 4 \text{ k}\Omega \angle 0^\circ \end{aligned}$$

$$V_2 = 21 \text{ V} = E_{oc} \Rightarrow V = \frac{E_{oc}}{21}$$

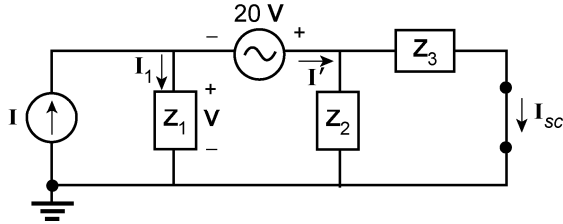
$$I = I_1 + I_2, I_1 = \frac{V}{Z_1} = \frac{E_{oc}}{21 Z_1}$$

$$I_2 = \frac{E_{oc}}{Z_2}, I = I_1 + I_2 = \frac{E_{oc}}{21Z_1} + \frac{E_{oc}}{Z_2} = E_{oc} \left[\frac{1}{21Z_1} + \frac{1}{Z_2} \right]$$

$$I = E_{oc} \left[\frac{Z_2 + 21Z_1}{21Z_1Z_2} \right]$$

$$\text{and } E_{oc} = \frac{21Z_1Z_2I}{Z_2 + 21Z_1} = \frac{(21)(1\text{ k}\Omega \angle 0^\circ)(3\text{ k}\Omega \angle 0^\circ)(2\text{ mA} \angle 0^\circ)}{3\text{ k}\Omega + 21(1\text{ k}\Omega \angle 0^\circ)}$$

$$E_{Th} = E_{oc} = 5.25\text{ V} \angle 0^\circ$$



$$I_{sc} = \frac{V_3}{Z_3} = \frac{21\text{ V}}{Z_3} \Rightarrow V = \frac{Z_3}{21} I_{sc}$$

$$V = I_1 Z_1$$

$$I = I_1 + I'$$

$$I_{sc} = \frac{Z_2 I'}{Z_2 + Z_3} \Rightarrow I' = \left(\frac{Z_2 + Z_3}{Z_2} \right) I_{sc}$$

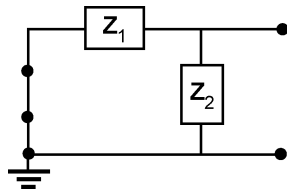
$$I = I_1 + I' = \frac{V}{Z_1} + \left(\frac{Z_2 + Z_3}{Z_2} \right) I_{sc} = \left[\frac{Z_3}{21Z_1} + \frac{Z_2 + Z_3}{Z_2} \right] I_{sc}$$

$$I_{sc} = \frac{I}{\frac{Z_3}{21Z_1} + \frac{Z_2 + Z_3}{Z_2}} = \frac{2\text{ mA} \angle 0^\circ}{\frac{4\text{ k}\Omega}{21\text{ k}\Omega} + \frac{7\text{ k}\Omega}{3\text{ k}\Omega}} = 0.79\text{ mA} \angle 0^\circ$$

$$\therefore I_N = 0.79\text{ mA} \angle 0^\circ$$

$$Z_N = \frac{E_{oc}}{I_{sc}} = \frac{5.25\text{ V} \angle 0^\circ}{0.79\text{ mA} \angle 0^\circ} = 6.65\text{ k}\Omega \angle 0^\circ$$

44.



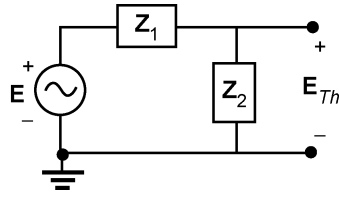
$$Z_1 = 3\Omega + j4\Omega, Z_2 = -j6\Omega$$

$$\leftarrow Z_{Th} = Z_1 \parallel Z_2$$

$$= 5\Omega \angle 53.13^\circ \parallel 6\Omega \angle -90^\circ$$

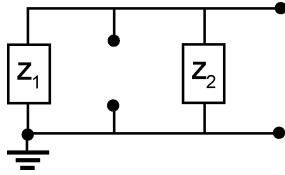
$$= 8.32\Omega \angle -3.18^\circ$$

$$Z_L = 8.32\Omega \angle 3.18^\circ = 8.31\Omega - j0.46\Omega$$

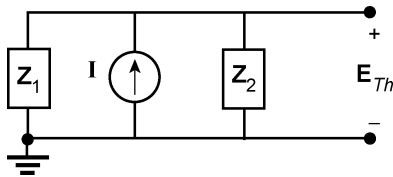


$$\begin{aligned}
 E_{Th} &= \frac{Z_2 E}{Z_2 + Z_1} \\
 &= \frac{(6 \Omega \angle -90^\circ)(120 \text{ V} \angle 0^\circ)}{3.61 \Omega \angle -33.69^\circ} \\
 &= \mathbf{199.45 \text{ V} \angle -56.31^\circ} \\
 P_{\max} &= \frac{E_{Th}^2}{4R_{Th}} = \frac{(3.124 \text{ V})^2}{4(8.31 \Omega)} = \mathbf{1198.2 \text{ W}}
 \end{aligned}$$

45.

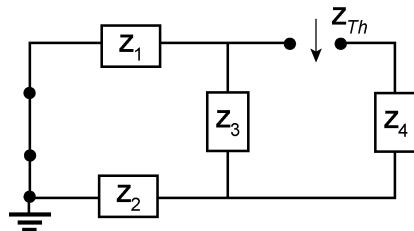


$$\begin{aligned}
 Z_1 &= 3 \Omega + j4 \Omega = 5 \Omega \angle 53.13^\circ \\
 Z_2 &= 2 \Omega \angle 0^\circ \\
 \leftarrow Z_N = Z_{Th} &= Z_1 \parallel Z_2 \\
 &= 5 \Omega \angle 53.13^\circ \parallel 2 \Omega \angle 0^\circ \\
 &= \frac{10 \Omega \angle 53.13^\circ}{2 + 3 + j4} \\
 &= \frac{10 \Omega \angle 53.13^\circ}{5 + j4} \\
 &= \frac{10 \Omega \angle 53.13^\circ}{6.403 \angle 38.66^\circ} \\
 &= \mathbf{1.56 \Omega \angle 14.47^\circ} \\
 Z_{Th} &= 1.56 \Omega \angle 14.47^\circ \\
 &= 1.51 \Omega + j0.39 \Omega \\
 Z_L &= \mathbf{1.51 \Omega - j0.39 \Omega}
 \end{aligned}$$



$$\begin{aligned}
 E_{Th} &= I(Z_1 \parallel Z_2) \\
 &= (2 \text{ A} \angle 30^\circ)(1.562 \Omega \angle 14.47^\circ) \\
 &= \mathbf{3.12 \text{ V} \angle 44.47^\circ} \\
 P_{\max} &= \frac{E_{Th}^2}{4R_{Th}} = \frac{(3.12 \text{ V})^2}{4(1.51 \Omega)} = \mathbf{1.61 \text{ W}}
 \end{aligned}$$

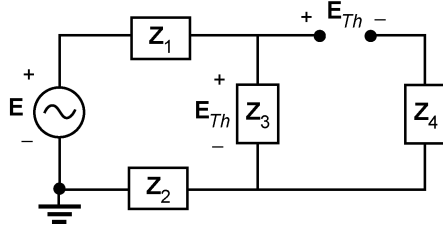
46. Z_{Th} :



$$\begin{aligned}
 Z_1 &= 4 \Omega \angle 90^\circ, Z_2 = 10 \Omega \angle 0^\circ \\
 Z_3 &= 5 \Omega \angle -90^\circ, Z_4 = 6 \Omega \angle -90^\circ \\
 E &= 60 \text{ V} \angle 60^\circ
 \end{aligned}$$

$$\begin{aligned}
 Z_{Th} &= Z_4 + Z_3 \parallel (Z_1 + Z_2) = -j6 \Omega + (5 \Omega \angle -90^\circ) \parallel (10 \Omega + j4 \Omega) \\
 &= 2.475 \Omega - j4.754 \Omega \\
 &= 11.04 \Omega \angle -77.03^\circ \\
 Z_L &= \mathbf{11.04 \Omega \angle 77.03^\circ}
 \end{aligned}$$

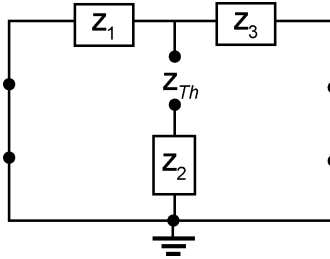
E_{Th} :



$$\begin{aligned} E_{Th} &= \frac{Z_3(E)}{Z_3 + Z_1 + Z_2} \\ &= \frac{(5 \Omega \angle -90^\circ)(60 \text{ V} \angle 60^\circ)}{-j5 \Omega + j4 \Omega + 10 \Omega} \\ &= 29.85 \text{ V} \angle -24.29^\circ \end{aligned}$$

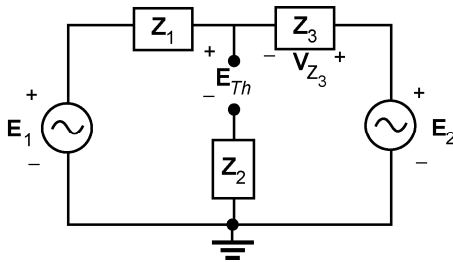
$$P_{\max} = E_{Th}^2 / 4R_{Th} = (29.85 \text{ V})^2 / 4(2.475 \Omega) = 90 \text{ W}$$

47.



$$\begin{aligned} Z_1 &= 3 \Omega + j4 \Omega = 5 \Omega \angle 53.13^\circ \\ Z_2 &= -j8 \Omega \\ Z_3 &= 12 \Omega + j9 \Omega \end{aligned}$$

$$\begin{aligned} Z_{Th} &= Z_2 + Z_1 \parallel Z_3 = -j8 \Omega + (5 \Omega \angle 53.13^\circ) \parallel (15 \Omega \angle 36.87^\circ) \\ &= 5.71 \Omega \angle -64.30^\circ = 2.475 \Omega - j5.143 \Omega \\ Z_L &= 5.71 \Omega \angle 64.30^\circ = 2.48 \Omega + j5.15 \Omega \end{aligned}$$



$$\begin{aligned} E_{Th} + V_{Z_3} - E_2 &= 0 \\ E_{Th} &= E_2 - V_{Z_3} \\ V_{Z_3} &= \frac{Z_3(E_2 - E_1)}{Z_3 + Z_1} \\ &= 168.97 \text{ V} \angle 112.53^\circ \end{aligned}$$

$$E_{Th} = E_2 - V_{Z_3} = 200 \text{ V} \angle 90^\circ - 168.97 \text{ V} \angle 112.53^\circ = 78.24 \text{ V} \angle 34.16^\circ$$

$$P_{\max} = E_{Th}^2 / 4R_{Th} = (78.24 \text{ V})^2 / 4(2.475 \Omega) = 618.33 \text{ W}$$

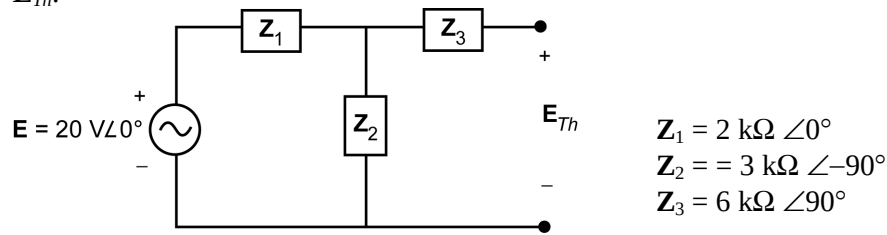
48. $I = \frac{E \angle 0^\circ}{R_1 \angle 0^\circ} = \frac{1 \text{ V} \angle 0^\circ}{1 \text{ k}\Omega \angle 0^\circ} = 1 \text{ mA} \angle 0^\circ$

$$Z_{Th} = 40 \text{ k}\Omega \angle 0^\circ$$

$$E_{Th} = (50 \text{ I})(40 \text{ k}\Omega \angle 0^\circ) = (50)(1 \text{ mA} \angle 0^\circ)(40 \text{ k}\Omega \angle 0^\circ) = 2000 \text{ V} \angle 0^\circ$$

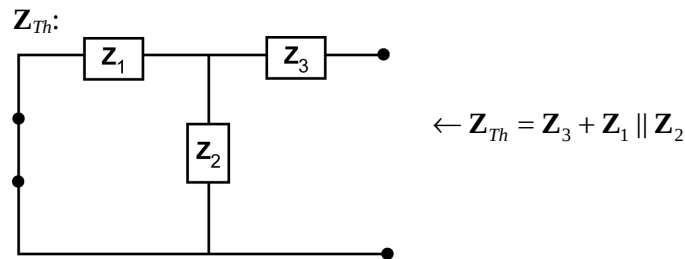
$$P_{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{(2 \text{ kV})^2}{4(40 \text{ k}\Omega)} = 25 \text{ W}$$

49. E_{Th} :



$$E_{Th} = \frac{Z_2 E}{Z_2 + Z_1} = \frac{(3 \text{ k}\Omega \angle 90^\circ)(20 \text{ V} \angle 0^\circ)}{-j3 \text{ k}\Omega + 2 \text{ k}\Omega}$$

$$= \frac{60 \text{ V} \angle -90^\circ}{3.61 \angle -56.31^\circ} = \mathbf{16.62 \text{ V} \angle -33.69^\circ}$$



$$Z_{Th} = +j6 \text{ k}\Omega + \frac{(2 \text{ k}\Omega \angle 0^\circ)(3 \text{ k}\Omega \angle -90^\circ)}{2 \text{ k}\Omega - j3 \text{ k}\Omega}$$

$$= +j6 \text{ k}\Omega + 1.66 \text{ k}\Omega \angle -33.69^\circ$$

$$= +j6 \text{ k}\Omega + 1.38 \text{ k}\Omega - j920.8 \Omega$$

$$= 1.38 \text{ k}\Omega + j5.08 \text{ k}\Omega$$

$$= 5.26 \text{ k}\Omega \angle +74.80^\circ$$

$$\therefore Z_L = 5.36 \text{ k}\Omega \angle -74.80^\circ = \mathbf{1.38 \text{ k}\Omega - j5.08 \text{ k}\Omega}$$

b. $P_{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{(16.62 \text{ V})^2}{4(1.38 \text{ k}\Omega)} = \mathbf{50.04 \text{ mW}}$

50. From #20, $Z_{Th} = 9 \Omega$, $E_{Th} = 12 \text{ V} + 24 \text{ V} \angle 0^\circ$

a. $\therefore Z_L = 9 \Omega$

b. $P_{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{(12 \text{ V})^2}{4(9 \Omega)} + \frac{(24 \text{ V})^2}{4(9 \Omega)} = 4 \text{ W} + 16 \text{ W} = \mathbf{20 \text{ W}}$

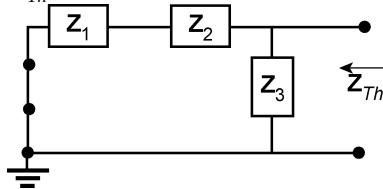
or $E_{Th} = \sqrt{V_0^2 + V_{1\text{eff}}^2} = 26.833 \text{ V}$

and $P_{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{(26.833 \text{ V})^2}{4(9 \Omega)} = \mathbf{20 \text{ W}}$

51. a. Problem 21(a):
 $\mathbf{Z}_{Th} = 4.47 \text{ k}\Omega \angle -26.57^\circ = 4 \text{ k}\Omega - j2 \text{ k}\Omega$
 $\mathbf{Z}_L = 4 \text{ k}\Omega + j2 \text{ k}\Omega$
 $\mathbf{E}_{Th} = 31.31 \text{ V} \angle -26.57^\circ$
- b. $P_{\max} = E_{Th}^2 / 4R_{Th} = (31.31 \text{ V})^2 / 4(4 \text{ k}\Omega) = 61.27 \text{ mW}$

52. a. $\mathbf{Z}_{Th} = 2 \text{ k}\Omega \angle 0^\circ \parallel 2 \text{ k}\Omega \angle -90^\circ = 1 \text{ k}\Omega - j1 \text{ k}\Omega$
 $R_L = \sqrt{R_{Th}^2 + (X_{Th} + X_{Load})^2}$
 $= \sqrt{(1 \text{ k}\Omega)^2 + (-1 \text{ k}\Omega + 2 \text{ k}\Omega)^2}$
 $= \sqrt{(1 \text{ k}\Omega)^2 + (1 \text{ k}\Omega)^2}$
 $= 1.41 \text{ k}\Omega$

- b. $R_{av} = (R_{Th} + R_{Load}) / 2 = (1 \text{ k}\Omega + 1.41 \text{ k}\Omega) / 2 = 1.21 \text{ k}\Omega$
 $P_{\max} = \frac{E_{Th}^2}{4R_{av}} = \frac{(50 \text{ V})^2}{4(1.21 \text{ k}\Omega)} = 516.53 \text{ mW}$

53. a. $\mathbf{Z}_{Th}$:
- 

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(10 \text{ kHz})(4 \text{ nF})}$$

$$\cong 3978.87 \Omega$$

$$X_L = 2\pi fL = 2\pi(10 \text{ kHz})(30 \text{ mH})$$

$$\cong 1884.96 \Omega$$

$$\mathbf{Z}_1 = 1 \text{ k}\Omega \angle 0^\circ, \mathbf{Z}_2 = 1884.96 \Omega \angle 90^\circ$$

$$\mathbf{Z}_3 = 3978.87 \Omega \angle -90^\circ$$

$$\mathbf{Z}_{Th} = (\mathbf{Z}_1 + \mathbf{Z}_2) \parallel \mathbf{Z}_3 = (1 \text{ k}\Omega + j1884.96 \Omega) \parallel 3978.87 \Omega \angle -90^\circ$$

$$= 2133.79 \Omega \angle 62.05^\circ \parallel 3978.87 \Omega \angle -90^\circ$$

$$= 3658.65 \Omega \angle 36.52^\circ$$

$$\therefore \mathbf{Z}_L = 3658.65 \Omega \angle -36.52^\circ = 2940.27 \Omega - j2177.27 \Omega$$

$$C = \frac{1}{2\pi fX_C} = \frac{1}{2\pi(10 \text{ kHz})(2177.27 \Omega)} = 7.31 \text{ nF}$$

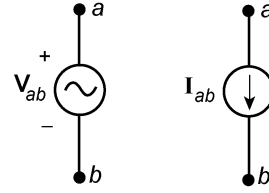
- b. $R_L = R_{Th} = 2940.27 \Omega$

c. $\mathbf{E}_{Th} = \frac{\mathbf{Z}_3(\mathbf{E})}{\mathbf{Z}_3 + \mathbf{Z}_1 + \mathbf{Z}_2} = \frac{(3978.87 \Omega \angle -90^\circ)(2 \text{ V} \angle 0^\circ)}{1 \text{ k}\Omega + j1884.96 \Omega - j3978.87 \Omega} = 3.43 \text{ V} \angle -25.53^\circ$

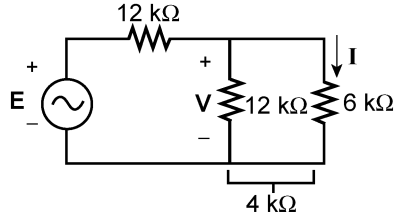
$$P_{\max} = E_{Th}^2 / 4R_{Th} = (3.43 \text{ V})^2 / 4(2940.27 \Omega) = 1 \text{ mW}$$

54.
$$\mathbf{I}_{ab} = \frac{(4 \text{ k}\Omega \angle 0^\circ)(4 \text{ mA} \angle 0^\circ)}{4 \text{ k}\Omega + 8 \text{ k}\Omega} = 1.33 \text{ mA} \angle 0^\circ$$

$$\mathbf{V}_{ab} = (\mathbf{I}_{ab})(8 \text{ k}\Omega \angle 0^\circ) = 10.67 \text{ V} \angle 0^\circ$$



55. a.

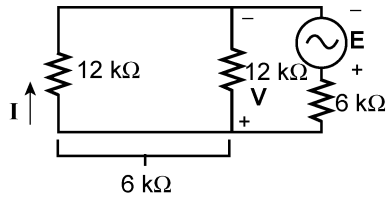


$$\mathbf{V} = \frac{4 \text{ k}\Omega(\mathbf{E})}{4 \text{ k}\Omega + 12 \text{ k}\Omega} = \frac{1}{4}(20 \text{ V} \angle 0^\circ)$$

$$= 5 \text{ V} \angle 0^\circ$$

$$\mathbf{I} = \frac{5 \text{ V} \angle 0^\circ}{6 \text{ k}\Omega} = 0.83 \text{ mA} \angle 0^\circ$$

b.

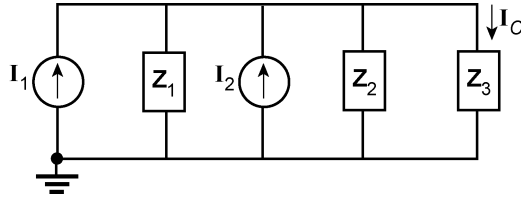


$$\mathbf{V} = \frac{6 \text{ k}\Omega(\mathbf{E})}{6 \text{ k}\Omega + 6 \text{ k}\Omega} = \frac{1}{2}(20 \text{ V} \angle 0^\circ)$$

$$= 10 \text{ V} \angle 0^\circ$$

$$\mathbf{I} = \frac{10 \text{ V} \angle 0^\circ}{12 \text{ k}\Omega} = 0.83 \text{ mA} \angle 0^\circ$$

56.



$$\mathbf{I}_1 = \frac{100 \text{ V} \angle 0^\circ}{2 \text{ k}\Omega \angle 0^\circ} = 50 \text{ mA} \angle 0^\circ$$

$$\mathbf{I}_2 = \frac{50 \text{ V} \angle 0^\circ}{4 \text{ k}\Omega \angle 90^\circ}$$

$$= 12.5 \text{ mA} \angle -90^\circ$$

$$\mathbf{Z}_1 = 2 \text{ k}\Omega \angle 0^\circ$$

$$\mathbf{Z}_2 = 4 \text{ k}\Omega \angle 90^\circ$$

$$\mathbf{Z}_3 = 4 \text{ k}\Omega \angle -90^\circ$$

$$\mathbf{I}_T = \mathbf{I}_1 - \mathbf{I}_2 = (50 \text{ mA} \angle 0^\circ - 12.5 \text{ mA} \angle -90^\circ) = 50 \text{ mA} + j12.5 \text{ mA}$$

$$= 51.54 \text{ mA} \angle 14.04^\circ$$

$$\mathbf{Z}' = \mathbf{Z}_1 \parallel \mathbf{Z}_2 = (2 \text{ k}\Omega \angle 0^\circ) \parallel (4 \text{ k}\Omega \angle 90^\circ) = 1.79 \text{ k}\Omega \angle 26.57^\circ$$

$$\mathbf{I}_C = \frac{\mathbf{Z}'\mathbf{I}_T}{\mathbf{Z}' + \mathbf{Z}_3} = \frac{(1.79 \text{ k}\Omega \angle 26.57^\circ)(51.54 \text{ mA} \angle 14.04^\circ)}{1.6 \text{ k}\Omega + j0.8 \text{ k}\Omega - j4 \text{ k}\Omega}$$

$$= 25.77 \text{ mA} \angle 104.04^\circ$$

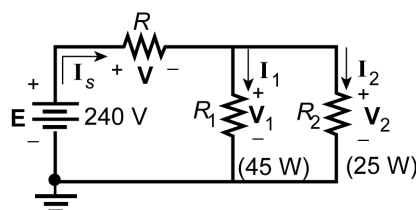
Chapter 19

1. a. $P_T = 60 \text{ W} + 45 \text{ W} + 25 \text{ W} = \mathbf{130 \text{ W}}$

b. $Q_T = \mathbf{0 \text{ VARS}}, S_T = P_T = \mathbf{130 \text{ VA}}$

c. $S_T = EI_s, I_s = \frac{S_T}{E} = \frac{130 \text{ VA}}{240 \text{ V}} = \mathbf{0.542 \text{ A}}$

d.



$$P = I_s^2 R, R = \frac{P}{I_s^2} = \frac{60 \text{ W}}{(0.542 \text{ A})^2} = 204.2 \Omega$$

$$V = I_s R = (0.542 \text{ A})(204.2 \Omega) = 110.68 \text{ V}$$

$$V_1 = V_2 = E - V = 240 \text{ V} - 110.68 \text{ V} = 129.32 \text{ V}$$

$$P_1 = \frac{V_1^2}{R_1}, R_1 = \frac{V_1^2}{P_1} = \frac{(129.32 \text{ V})^2}{45 \text{ W}} = \mathbf{371.6 \Omega}$$

$$P_2 = \frac{V_2^2}{R_2}, R_2 = \frac{V_2^2}{P_2} = \frac{(129.32 \text{ V})^2}{25 \text{ W}} = \mathbf{668.9 \Omega}$$

e. $I_1 = \frac{V_1}{R_1} = \frac{129.32 \text{ V}}{371.6 \Omega} = \mathbf{0.348 \text{ A}}, I_2 = \frac{V_2}{R_2} = \frac{129.32 \text{ V}}{668.9 \Omega} = \mathbf{0.193 \text{ A}}$

2. a. $Z_T = 3 \Omega - j5 \Omega + j9 \Omega = 3 \Omega + j4 \Omega = 5 \Omega \angle 53.13^\circ$

$$\mathbf{I} = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{50 \text{ V} \angle 0^\circ}{5 \Omega \angle 53.13^\circ} = 10 \text{ A} \angle -53.13^\circ$$

R: $P = I^2 R = (10 \text{ A})^2 3 \Omega = \mathbf{300 \text{ W}}$

L: $P = \mathbf{0 \text{ W}}$

C: $P = \mathbf{0 \text{ W}}$

b. R: $Q = \mathbf{0 \text{ VAR}}$

C: $Q_C = I^2 X_C = (10 \text{ A})^2 5 \Omega = \mathbf{500 \text{ VAR}}$

L: $Q_L = I^2 X_L = (10 \text{ A})^2 9 \Omega = \mathbf{900 \text{ VAR}}$

c. R: $S = \mathbf{300 \text{ VA}}$

C: $S = \mathbf{500 \text{ VA}}$

L: $S = \mathbf{900 \text{ VA}}$

d. $P_T = 300 \text{ W}$

$$Q_T = Q_L - Q_C = \mathbf{400 \text{ VAR}(L)}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = EI = (50 \text{ V})(10 \text{ A}) = \mathbf{500 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{300 \text{ W}}{500 \text{ VA}} = \mathbf{0.6 \text{ lagging}}$$

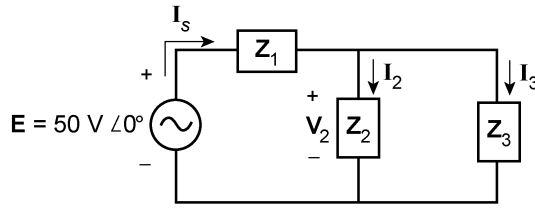
e. —

- f. $W_R = \frac{VI}{f_1} : W_R = 2 \left[\frac{VI}{f_2} \right] = 2 \left[\frac{VI}{2f_1} \right] = \frac{VI}{f_1}$
 $V = IR = (10 \text{ A})(3 \Omega) = 30 \text{ V}$
 $W_R = \frac{(30 \text{ V})(10 \text{ A})}{60 \text{ Hz}} = 5 \text{ J}$
- g. $V_C = IX_C = (10 \text{ A})(5 \Omega) = 50 \text{ V}$
 $W_C = \frac{VI}{\omega_1} = \frac{(50 \text{ V})(10 \text{ A})}{(2\pi)(60 \text{ Hz})} = 1.33 \text{ J}$
 $V_L = IX_L = (10 \text{ A})(9 \Omega) = 90 \text{ V}$
 $W_L = \frac{VI}{\omega_1} = \frac{(90 \text{ V})(10 \text{ A})}{376.8} = 2.39 \text{ J}$
3. a. $P_T = 0 + 100 \text{ W} + 300 \text{ W} = \mathbf{400 \text{ W}}$
 $Q_T = 200 \text{ VAR}(L) - 600 \text{ VAR}(C) + 0 = \mathbf{-400 \text{ VAR}(C)}$
 $S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{565.69 \text{ VA}}$
 $F_p = \frac{P_T}{S_T} = \frac{400 \text{ W}}{565.69 \text{ VA}} = \mathbf{0.707 \text{ (leading)}}$
- b. —
- c. $P_T = EI_s \cos \theta_T$
 $400 \text{ W} = (100 \text{ V})I_s(0.7071)$
 $I_s = \frac{400 \text{ W}}{70.71 \text{ V}} = 5.66 \text{ A}$
 $\mathbf{I_s = 5.66 \text{ A} \angle 135^\circ}$
4. a. $P_T = 600 \text{ W} + 500 \text{ W} + 100 \text{ W} = \mathbf{1200 \text{ W}}$
 $Q_T = 1200 \text{ VAR}(L) + 600 \text{ VAR}(L) - 1800(C) = \mathbf{0 \text{ VAR}}$
 $S_T = P_T = \mathbf{1200 \text{ VA}}$
- b. $F_p = \frac{P_T}{S_T} = \frac{1200 \text{ W}}{1200 \text{ VA}} = 1$
- c. —
- d. $I_s = \frac{S_T}{E} = \frac{1200 \text{ VA}}{200 \text{ V}} = 6 \text{ A}, 1 \Rightarrow 0^\circ$
 $\mathbf{I_s = 6 \text{ A} \angle 0^\circ}$
5. a. $P_T = 200 \text{ W} + 100 \text{ W} + 0 + 50 \text{ W} = \mathbf{350 \text{ W}}$
 $Q_T = 50 \text{ VAR}(L) + 100 \text{ VAR}(L) - 200 \text{ VAR}(C) - 400 \text{ VAR}(C) = \mathbf{-450 \text{ VAR}(C)}$
 $S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{570.09 \text{ VA}}$

- b. $F_p = \frac{P_T}{S_T} = \frac{350 \text{ W}}{570.09 \text{ VA}} = \mathbf{0.614 \text{ (leading)}}$
- c. —
- d. $P_T = EI_s \cos \theta_T$
 $350 \text{ W} = (50 \text{ V})I_s(0.614)$
 $I_s = \frac{350 \text{ W}}{30.7 \text{ V}} = 11.4 \text{ A}$
 $\mathbf{I_s = 11.4 A \angle 52.12^\circ}$
6. a. $\mathbf{I_R = \frac{60 \text{ V} \angle 30^\circ}{20 \Omega \angle 0^\circ} = 3 \text{ A} \angle 30^\circ}$
 $P = I^2 R = (3 \text{ A})^2 20 \Omega = \mathbf{180 \text{ W}}$
 $Q_R = \mathbf{0 \text{ VAR}}$
 $S = P = \mathbf{180 \text{ VA}}$
- b. $\mathbf{I_L = \frac{60 \text{ V} \angle 30^\circ}{10 \Omega \angle 90^\circ} = 6 \text{ A} \angle -60^\circ}$
 $P_L = \mathbf{0 \text{ W}}$
 $Q_L = I^2 X_L = (6 \text{ A})^2 10 \Omega = \mathbf{360 \text{ VAR}(L)}$
 $S = Q = \mathbf{360 \text{ VA}}$
- c. $P_T = 180 \text{ W} + 400 \text{ W} = \mathbf{580 \text{ W}}$
 $Q_T = 600 \text{ VAR}(L) + 360 \text{ VAR}(L) = \mathbf{960 \text{ VAR}(L)}$
 $S_T = \sqrt{(580 \text{ W})^2 + (960 \text{ VAR})^2} = \mathbf{1121.61 \text{ VA}}$
 $F_p = \frac{P_T}{S_T} = \frac{580 \text{ W}}{1121.61 \text{ VA}} = \mathbf{0.517 \text{ (lagging)}}$ $\theta = 58.87^\circ$
- d. $S_T = EI_s$
 $I_s = \frac{S_T}{E} = \frac{1121.61 \text{ VA}}{60 \text{ V}} = 18.69 \text{ A}$
 $\theta_{I_s} = 30^\circ - 58.87^\circ = -28.87^\circ$
 $\mathbf{I_s = 18.69 A \angle -28.87^\circ}$
7. a. $R: P = \frac{E^2}{R} = \frac{(20 \text{ V})^2}{2 \Omega} = \mathbf{200 \text{ W}}$
 $P_{L,C} = \mathbf{0 \text{ W}}$
- b. $R: Q = \mathbf{0 \text{ VAR}}$
 $C: Q_C = \frac{E^2}{X_C} = \frac{(20 \text{ V})^2}{5 \Omega} = \mathbf{80 \text{ VAR}(C)}$
 $L: Q_L = \frac{E^2}{X_L} = \frac{(20 \text{ V})^2}{4 \Omega} = \mathbf{100 \text{ VAR}(L)}$

- c. $R: S = \mathbf{200\ VA}$
 $C: S = \mathbf{80\ VA}$
 $L: S = \mathbf{100\ VA}$
- d. $P_T = 200\ \text{W} + 0 + 0 = \mathbf{200\ W}$
 $Q_T = 0 + 80\ \text{VAR}(C) + 100\ \text{VAR}(L) = \mathbf{20\ VAR}(L)$
 $S_T = \sqrt{(200\ \text{W})^2 + (20\ \text{VAR})^2} = \mathbf{200\ VA}$
 $F_p = \frac{P_T}{S_T} = \frac{200\ \text{W}}{200.998\ \text{VA}} = \mathbf{0.995\ (lagging)} \Rightarrow 5.73^\circ$
- e. —
- f. $I_s = \frac{S_T}{E} = \frac{200.998\ \text{VA}}{20\ \text{V}} = 10.05\ \text{A}$
 $\mathbf{I_s = 10.05\ A \angle -5.73^\circ}$
8. a. $R - L: \mathbf{I} = \frac{50\ \text{V} \angle 60^\circ}{5\ \Omega \angle 53.13^\circ} = 10\ \text{A} \angle 6.87^\circ$
 $P_R = I^2 R = (10\ \text{A})^2 3\ \Omega = \mathbf{300\ W}$
 $P_L = \mathbf{0\ W}$
 $P_C = \mathbf{0\ W}$
- b. $Q_R = \mathbf{0\ VAR}$
 $Q_L = I^2 X_L = (10\ \text{A})^2 4\ \Omega = \mathbf{400\ VAR}$
 $\mathbf{I_C} = \frac{50\ \text{V} \angle 60^\circ}{10\ \Omega \angle -90^\circ} = 5\ \text{A} \angle 150^\circ$
 $Q_C = I^2 X_C = (5\ \text{A})^2 10\ \Omega = \mathbf{250\ VAR}$
- c. $S_R = P = \mathbf{300\ VA}$
 $S_L = Q_L = \mathbf{400\ VA}$
 $S_C = Q_C = \mathbf{250\ VA}$
- d. $P_T = P_R = \mathbf{300\ W}$
 $Q_T = 400\ \text{VAR}(L) - 250\ \text{VAR}(C) = \mathbf{150\ VAR}(L)$
 $S_T = \sqrt{(300\ \text{W})^2 + (150\ \text{VAR})^2} = \mathbf{335.41\ VA}$
 $F_p = \frac{P_T}{S_T} = \frac{300\ \text{W}}{335.41\ \text{VA}} = \mathbf{0.894\ (lagging)}$
- e. —
- f. $I_s = \frac{S_T}{E} = \frac{335.41\ \text{VA}}{50\ \text{V}} = 6.71\ \text{A}$
 $0.894 \Rightarrow 26.62^\circ\ \text{lagging}$
 $\theta = 60^\circ - 26.62^\circ = 33.38^\circ$
 $\mathbf{I_s = 6.71\ A \angle 33.38^\circ}$

9. a–c.



$$X_L = \omega L = (400 \text{ rad/s})(0.1 \text{ H}) = 40 \, \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{(400 \text{ rad/s})(100 \, \mu\text{F})} = 25 \, \Omega$$

$$\mathbf{Z}_1 = 40 \, \Omega \angle 90^\circ, \mathbf{Z}_2 = 25 \, \Omega \angle -90^\circ$$

$$\mathbf{Z}_3 = 30 \, \Omega \angle 0^\circ$$

$$\begin{aligned} \mathbf{Z}_T &= \mathbf{Z}_1 + \mathbf{Z}_2 \parallel \mathbf{Z}_3 = +j40 \, \Omega + (25 \, \Omega \angle -90^\circ) \parallel (30 \, \Omega \angle 0^\circ) \\ &= +j40 \, \Omega + 19.21 \, \Omega \angle -50.19^\circ \\ &= +j40 \, \Omega + 12.3 \, \Omega - j14.76 \, \Omega \\ &= 12.3 \, \Omega + j25.24 \, \Omega \\ &= 28.08 \, \Omega \angle 64.02^\circ \end{aligned}$$

$$\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{50 \text{ V} \angle 0^\circ}{28.08 \, \Omega \angle 64.02^\circ} = 1.78 \text{ A} \angle -64.02^\circ$$

$$\begin{aligned} \mathbf{V}_2 &= \mathbf{I}_s(\mathbf{Z}_2 \parallel \mathbf{Z}_3) = (1.78 \text{ A} \angle -64.02^\circ)(19.21 \, \Omega \angle -50.19^\circ) \\ &= 34.19 \text{ V} \angle -114.21^\circ \end{aligned}$$

$$\mathbf{I}_2 = \frac{\mathbf{V}_2}{\mathbf{Z}_2} = \frac{34.19 \text{ V} \angle -114.21^\circ}{25 \, \Omega \angle -90^\circ} = 1.37 \text{ A} \angle -24.21^\circ$$

$$\mathbf{I}_3 = \frac{\mathbf{V}_2}{\mathbf{Z}_3} = \frac{34.19 \text{ V} \angle -114.21^\circ}{30 \, \Omega \angle 0^\circ} = 1.14 \text{ A} \angle -114.21^\circ$$

$$\mathbf{Z}_1: \quad P = 0 \text{ W}, Q_L = I_s^2 X_L = (1.78 \text{ A})^2 40 \, \Omega = \mathbf{126.74 \text{ VAR}(L)}, S = \mathbf{126.74 \text{ VA}}$$

$$\mathbf{Z}_2: \quad P = 0 \text{ W}, Q_C = I_2^2 X_C = (1.37 \text{ A})^2 25 \, \Omega = \mathbf{46.92 \text{ VAR}(C)}, S = \mathbf{46.92 \text{ VA}}$$

$$\mathbf{Z}_3: \quad P = I_3^2 R = (1.14 \text{ A})^2 30 \, \Omega = \mathbf{38.99 \text{ W}}, Q_R = 0 \text{ VAR}, S = \mathbf{38.99 \text{ VA}}$$

$$\begin{aligned} \text{d.} \quad P_T &= 0 + 0 + 38.99 \text{ W} = \mathbf{38.99 \text{ W}} \\ Q_T &= +126.74 \text{ VAR}(L) - 46.92 \text{ VAR}(C) + 0 = \mathbf{79.82 \text{ VAR}(L)} \\ S_T &= \sqrt{P_T^2 + Q_T^2} = \mathbf{88.83 \text{ VA}} \\ F_p &= \frac{P_T}{S_T} = \frac{38.99 \text{ W}}{88.83 \text{ VA}} = \mathbf{0.439 \text{ (lagging)}} \end{aligned}$$

e. —

$$\text{f.} \quad W_R = \frac{V_R I_R}{2f_1} = \frac{V_2 I_3}{2f_1} = \frac{(34.19 \text{ V})(1.14 \text{ A})}{2(63.69 \text{ Hz})} = \mathbf{0.31 \text{ J}}$$

$$f_1 = \frac{\omega_1}{2\pi} = \frac{400 \text{ rad/s}}{6.28} = 63.69 \text{ Hz}$$

$$\text{g.} \quad W_L = \frac{V_L I_L}{\omega_1} = \frac{(I_s X_L) I_s}{\omega_1} = \frac{I_s^2 X_L}{\omega_1} = \frac{(1.78 \text{ A})^2 40 \, \Omega}{400 \text{ rad/s}} = \mathbf{0.32 \text{ J}}$$

$$W_C = \frac{V_C I_C}{\omega_1} = \frac{V_2 I_2}{\omega_1} = \frac{(34.19 \text{ V})(1.37 \text{ A})}{400 \text{ rad/s}} = \mathbf{0.12 \text{ J}}$$

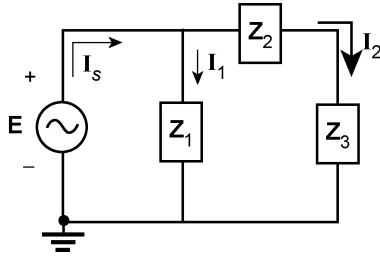
10. a. $I_s = \frac{S_T}{E} = \frac{10,000 \text{ VA}}{200 \text{ V}} = 50 \text{ A}$
 $0.5 \Rightarrow 60^\circ \text{ leading}$
 $\therefore \mathbf{I}_s \text{ leads } \mathbf{E} \text{ by } 60^\circ$
 $\mathbf{Z}_T = \frac{\mathbf{E}}{\mathbf{I}_s} = \frac{200 \text{ V } \angle 0^\circ}{50 \text{ A } \angle 60^\circ} = 4 \Omega \angle -60^\circ = 2 \Omega - j3.464 \Omega = R - jX_C$
- b. $F_p = \frac{P_T}{S_T} \Rightarrow P_T = F_p S_T = (0.5)(10,000 \text{ VA}) = \mathbf{5000 \text{ W}}$
11. a. $I = \frac{S_T}{E} = \frac{5000 \text{ VA}}{120 \text{ V}} = 41.67 \text{ A}$
 $F_p = 0.8 \Rightarrow 36.87^\circ \text{ (lagging)}$
 $\mathbf{E} = 120 \text{ V } \angle 0^\circ, \mathbf{I} = 41.67 \text{ A } \angle -36.87^\circ$
 $\mathbf{Z} = \frac{\mathbf{E}}{\mathbf{I}} = \frac{120 \text{ V } \angle 0^\circ}{41.67 \text{ A } \angle -36.87^\circ} = 2.88 \Omega \angle 36.87^\circ = \mathbf{2.30 \Omega + j1.73 \Omega} = R + jX_L$
- b. $P = S \cos \theta = (5000 \text{ VA})(0.8) = \mathbf{4000 \text{ W}}$
12. a. $P_T = 0 + 300 \text{ W} = \mathbf{300 \text{ W}}$
 $Q_T = 600 \text{ VAR}(C) + 200(L) = \mathbf{400 \text{ VAR}(C)}$
 $S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{500 \text{ VA}}$
 $F_p = \frac{P_T}{S_T} = \frac{300 \text{ W}}{500 \text{ VA}} = \mathbf{0.6 \text{ (leading)}}$
- b. $I_s = \frac{S_T}{E} = \frac{500 \text{ VA}}{30 \text{ V}} = 16.67 \text{ A}$
 $F_p = 0.6 \Rightarrow 53.13^\circ$
 $\mathbf{I}_s = \mathbf{16.67 \text{ A } \angle 53.13^\circ}$
- c. —
- d. Load: $600 \text{ VAR}(C), 0 \text{ W}$
 $R = 0, L = 0, Q_C = I^2 X_C \Rightarrow X_C = \frac{Q_C}{I^2} = \frac{600 \text{ VAR}}{(16.67 \text{ A})^2} = \mathbf{2.159 \Omega}$
 Load: $200 \text{ VAR}(L), 300 \text{ W}$
 $C = 0, R = P/I^2 = 300 \text{ W}/(16.67 \text{ A})^2 = \mathbf{1.079 \Omega}$
 $X_L = \frac{Q_L}{I^2} = \frac{200 \text{ VAR}}{(16.67 \text{ A})^2} = \mathbf{0.7197 \Omega}$
 $\mathbf{Z}_T = j2.159 \Omega + 1.0796 \Omega + j0.7197 \Omega$
 $= \mathbf{1.08 \Omega - j1.44 \Omega}$

13. a. $P_T = 0 + 300 \text{ W} + 600 \text{ W} = \mathbf{900 \text{ W}}$
 $Q_T = 500 \text{ VAR}(C) + 0 + 500 \text{ VAR}(L) = \mathbf{0 \text{ VAR}}$
 $S_T = P_T = \mathbf{900 \text{ VA}}$
 $F_p = \frac{P_T}{S_T} = \mathbf{1}$

b. $I_s = \frac{S_T}{E} = \frac{900 \text{ VA}}{100 \text{ V}} = 9 \text{ A}, \mathbf{I_s = 9 \text{ A } \angle 0^\circ}$

c. —

d.



Z_1 : $Q_C = \frac{V^2}{X_C} \Rightarrow X_C = \frac{V^2}{Q_C} = \frac{10^4}{500} = \mathbf{20 \Omega}$

$I_1 = \frac{E}{Z_1} = \frac{100 \text{ V } \angle 0^\circ}{20 \Omega \angle -90^\circ} = 5 \text{ A } \angle 90^\circ$

$I_2 = I_s - I_1 = 9 \text{ A} - j5 \text{ A} = 10.296 \text{ A } \angle -29.05^\circ$

Z_2 : $R = \frac{P}{I^2} = \frac{300 \text{ W}}{(10.296 \text{ A})^2} = \frac{300}{106} = \mathbf{2.83 \Omega}$

$X_{L,C} = \mathbf{0 \Omega}$

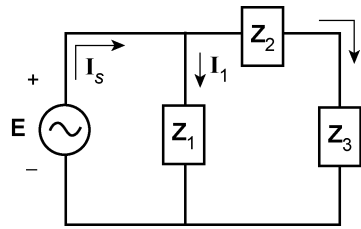
Z_3 : $R = \frac{P}{I_2^2} = \frac{600 \text{ W}}{(10.296 \text{ A})^2} = \mathbf{5.66 \Omega}$

$X_L = \frac{Q}{I_2^2} = \frac{500}{(10.296 \text{ A})^2} = \mathbf{4.72 \Omega}, X_C = \mathbf{0 \Omega}$

14. a. $P_T = 200 \text{ W} + 30 \text{ W} + 0 = \mathbf{230 \text{ W}}$
 $Q_T = 0 + 40 \text{ VAR}(L) + 100 \text{ VAR}(L) = \mathbf{140 \text{ VAR}(L)}$
 $S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{269.26 \text{ VA}}$
 $F_p = \frac{P_T}{S_T} = \frac{230 \text{ W}}{269.26 \text{ VA}} = \mathbf{0.854 \text{ (lagging)}} \Rightarrow 31.35^\circ$

b. $I_s = \frac{S_T}{E} = \frac{269.26 \text{ VA}}{100 \text{ V}} = 2.6926 \text{ A}$
 $\mathbf{I_s = 2.69 \text{ A } \angle -31.35^\circ}$

c.

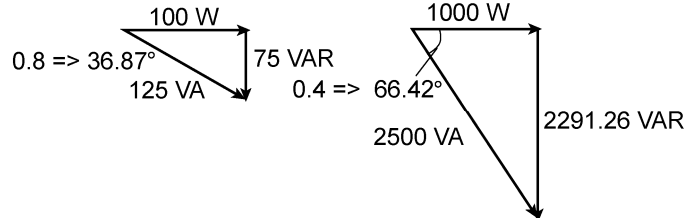


$$\begin{aligned} Z_1: \quad R &= \frac{V^2}{P} = \frac{10^4}{200} = 50 \, \Omega \\ X_L, X_C &= 0 \, \Omega \\ I_1 &= \frac{100 \, \text{V} \angle 0^\circ}{50 \, \Omega \angle 0^\circ} = 2 \, \text{A} \angle 0^\circ \\ I_2 &= I_s - I_1 \\ &= 2.6926 \, \text{A} \angle -31.35^\circ - 2 \, \text{A} \angle 0^\circ \\ &= 2.299 \, \text{A} - j1.40 \, \text{A} - 2.0 \, \text{A} \\ &= 0.299 \, \text{A} - j1.40 \, \text{A} \\ &= 1.432 \, \text{A} \angle -77.94^\circ \end{aligned}$$

$$\begin{aligned} Z_2: \quad R &= \frac{P}{I_2^2} = \frac{30 \, \text{W}}{(1.432 \, \text{A})^2} = \mathbf{14.63 \, \Omega}, \quad X_L = \frac{Q}{I_2^2} = \frac{40 \, \text{VAR}}{(1.432 \, \text{A})^2} = \mathbf{19.50 \, \Omega} \\ X_C &= \mathbf{0 \, \Omega} \end{aligned}$$

$$Z_3: \quad X_L = \frac{Q}{I_2^2} = \frac{100 \, \text{VAR}}{(1.432 \, \text{A})^2} = \mathbf{48.76 \, \Omega}, \quad R = \mathbf{0 \, \Omega}, \quad X_C = \mathbf{0 \, \Omega}$$

15. a. $P_T = 100 \, \text{W} + 1000 \, \text{W} = \mathbf{1100 \, W}$



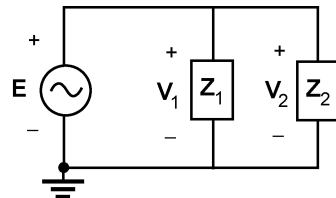
$$Q_T = 75 \, \text{VAR}(C) + 2291.26 \, \text{VAR}(C) = \mathbf{2366.26 \, \text{VAR}(C)}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{2609.44 \, \text{VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{1100 \, \text{W}}{2609.44 \, \text{VA}} = \mathbf{0.422 \, (\text{leading})} \Rightarrow 65.04^\circ$$

b. $S_T = EI \Rightarrow E = \frac{S_T}{I} = \frac{2609.44 \, \text{VA}}{5 \, \text{A}} = 521.89 \, \text{V}$
 $\mathbf{E = 521.89 \, V \angle -65.07^\circ}$

c.



$$I_{Z_1} = \frac{S}{V_1} = \frac{S}{E} = \frac{125 \, \text{VA}}{521.89 \, \text{V}} = 0.2395 \, \text{A}$$

$$I_{Z_2} = \frac{S}{V_2} = \frac{S}{E} = \frac{2500 \, \text{VA}}{521.89 \, \text{V}} = 4.79 \, \text{A}$$

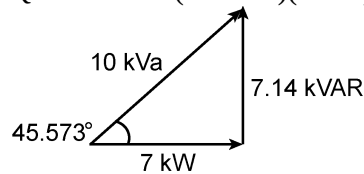
$$\mathbf{Z_1:} \quad R = \frac{P}{I_{Z_1}^2} = \frac{100 \text{ W}}{(0.2395)^2} = \mathbf{1743.38 \, \Omega}$$

$$Q = I_{Z_1}^2 X_C \Rightarrow X_C = \frac{Q}{I_{Z_1}^2} = \frac{75 \text{ VAR}}{(0.2395 \text{ A})^2} = \mathbf{1307.53 \, \Omega}$$

$$\mathbf{Z_2:} \quad R = \frac{P}{I_{Z_1}^2 X_C} = \frac{1000 \text{ W}}{(4.790 \text{ A})^2} = \mathbf{43.59 \, \Omega}$$

$$X_C = \frac{Q}{I_{Z_1}^2 X_C} = \frac{2291.26 \text{ VAR}}{(4.790 \text{ A})^2} = \mathbf{99.88 \, \Omega}$$

16. a. $0.7 \Rightarrow 45.573^\circ$
 $P = S \cos \theta = (10 \text{ kVA})(0.7) = 7 \text{ kW}$
 $Q = S \sin \theta = (10 \text{ kVA})(0.714) = 7.14 \text{ kVAR}(L)$



- b. $Q_C = 7.14 \text{ kVAR} = \frac{V^2}{X_C}$
 $X_C = \frac{V^2}{Q_C} = \frac{(208 \text{ V})^2}{7.14 \text{ kVAR}} = 6.059 \, \Omega$
 $X_C = \frac{1}{2\pi fC} \Rightarrow C = \frac{1}{2\pi fX_C} = \frac{1}{(2\pi)(60 \text{ Hz})(6.059 \, \Omega)} = \mathbf{438 \, \mu F}$

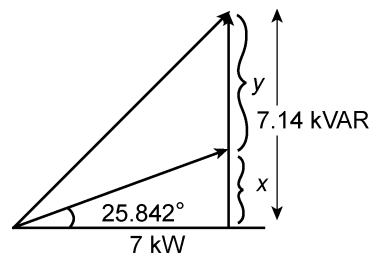
- c. Uncompensated:

$$I_s = \frac{S_T}{E} = \frac{10,000 \text{ VA}}{208 \text{ V}} = \mathbf{48.08 \text{ A}}$$

Compensated:

$$I_s = \frac{S_T}{E} = \frac{P_T}{E} = \frac{7,000 \text{ W}}{208 \text{ V}} = \mathbf{33.65 \text{ A}}$$

- d.



$$\cos \theta = 0.9$$

$$\theta = \cos^{-1} 0.9 = 25.842^\circ$$

$$\tan \theta = \frac{x}{7 \text{ kW}}$$

$$x = (7 \text{ kW})(\tan 25.842^\circ)$$

$$= (7 \text{ kW})(0.484)$$

$$= 3.39 \text{ kVAR}$$

$$y = (7.14 - 3.39) \text{ kVAR}$$

$$= 3.75 \text{ kVAR}$$

$$Q_C = 3.75 \text{ kVAR} = \frac{V^2}{X_C}$$

$$X_C = \frac{V^2}{Q_C} = \frac{(208 \text{ V})^2}{3.75 \text{ kVAR}} = 11.537 \Omega$$

$$C = \frac{1}{2\pi f X_C} = \frac{1}{(2\pi)(60 \text{ Hz})(11.537 \Omega)} = \mathbf{230 \mu F}$$

Uncompensated:

$$I_s = \mathbf{48.08 \text{ A}}$$

Compensated:

$$S_T = \sqrt{(7 \text{ kW})^2 + (3.39 \text{ kVAR})^2} = 7.778 \text{ kVA}$$

$$I_s = \frac{S_T}{E} = \frac{7.778 \text{ kVA}}{208 \text{ V}} = \mathbf{37.39 \text{ A}}$$

$$\Delta I_s = 48.08 \text{ A} - 37.39 \text{ A} = \mathbf{10.69 \text{ A}}$$

17. a. $P_T = 5 \text{ kW}, Q_T = 6 \text{ kVAR(L)}$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{7.81 \text{ kVA}}$$

b. $F_p = \frac{P_T}{S_T} = \frac{5 \text{ kW}}{7.81 \text{ kVA}} = \mathbf{0.640 \text{ (lagging)}}$

c. $I_s = \frac{S_T}{E} = \frac{7,810 \text{ VA}}{120 \text{ V}} = \mathbf{65.08 \text{ A}}$

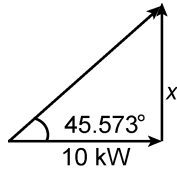
d. $X_C = \frac{1}{2\pi f C}, Q_C = I^2 X_C = \frac{E^2}{X_C} = \frac{(120 \text{ V})^2}{X_C}$
and $X_C = \frac{(120 \text{ V})^2}{Q_C} = \frac{14,400}{6000} = 2.4 \Omega$

$$C = \frac{1}{2\pi f X_C} = \frac{1}{(2\pi)(60 \text{ Hz})(2.4 \Omega)} = \mathbf{1105 \mu F}$$

e. $S_T = EI_s = P_T$

$$\therefore I_s = \frac{P_T}{E} = \frac{5000 \text{ W}}{120 \text{ V}} = \mathbf{41.67 \text{ A}}$$

18. a. Load 1: $P = 20,000 \text{ W}$, $Q = 0 \text{ VAR}$
 Load 2: $\theta = \cos^{-1}0.7 = 45.573^\circ$



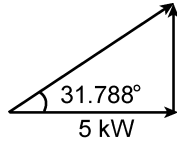
$$\tan \theta = \frac{x}{10 \text{ kW}}$$

$$x = (10 \text{ kW}) \tan 45.573^\circ$$

$$= (10 \text{ kW})(1.02)$$

$$= \mathbf{10,202 \text{ VAR}(L)}$$

Load 3: $\theta = \cos^{-1}0.85 = 31.788^\circ$



$$\tan \theta = \frac{x}{5 \text{ kW}}$$

$$x = (5 \text{ kW}) \tan 31.788^\circ$$

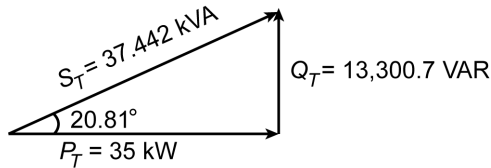
$$= (5 \text{ kW})(0.62)$$

$$= 3098.7 \text{ VAR}(L)$$

$$P_T = 20,000 \text{ W} + 10,000 \text{ W} + 5,000 \text{ W} = \mathbf{35 \text{ kW}}$$

$$Q_T = 0 + 10,202 \text{ VAR} + 3098.7 \text{ VAR} = \mathbf{13,300.7 \text{ VAR}(L)}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{37,442 \text{ VA} = 37.442 \text{ kVA}}$$



b. $Q_C = Q_L = 13,300.7 \text{ VAR}$

$$X_C = \frac{E^2}{Q_C} = \frac{(10^3 \text{ V})^2}{13,300.7 \text{ VAR}} = 75.184 \Omega$$

$$C = \frac{1}{2\pi f X_C} = \frac{1}{(2\pi)(60 \text{ Hz})(75.184 \Omega)} = \mathbf{35.28 \mu\text{F}}$$

c. Uncompensated:

$$I_s = \frac{S_T}{E} = \frac{37.442 \text{ kVA}}{1 \text{ kV}} = \mathbf{37.44 \text{ A}}$$

Compensated:

$$S_T = P_T = 35 \text{ kW}$$

$$I_s = \frac{S_T}{E} = \frac{35 \text{ kW}}{1 \text{ kV}} = \mathbf{35 \text{ A}}$$

$$\Delta I_s = 37.44 \text{ A} - 35 \text{ A} = \mathbf{2.44 \text{ A}}$$

19. a. $Z_T = R_1 + R_2 + R_3 + jX_L - jX_C$
 $= 2 \Omega + 3 \Omega + 1 \Omega + j3 \Omega - j12 \Omega = 6 \Omega - j9 \Omega = 10.82 \Omega \angle -56.31^\circ$
 $I = \frac{E}{Z_T} = \frac{50 \text{ V} \angle 0^\circ}{10.82 \Omega \angle -56.31^\circ} = 4.62 \text{ A} \angle 56.31^\circ$

- $P = VI \cos \theta = (50 \text{ V})(4.62 \text{ A}) \cos 56.31^\circ = \mathbf{128.14 \text{ W}}$
- b. a-b: $P = I^2 R = (4.62 \text{ A})^2 2 \Omega = \mathbf{42.69 \text{ W}}$
 b-c: $P = I^2 R = (4.62 \text{ A})^2 3 \Omega = \mathbf{64.03 \text{ W}}$
 a-c: $42.69 \text{ W} + 64.03 \text{ W} = \mathbf{106.72 \text{ W}}$
 a-d: $\mathbf{106.72 \text{ W}}$
 c-d: $\mathbf{0 \text{ W}}$
 d-e: $\mathbf{0 \text{ W}}$
 f-e: $P = I^2 R = (4.62 \text{ A})^2 1 \Omega = \mathbf{21.34 \text{ W}}$
20. a. $S_T = 660 \text{ VA} = EI_s$
 $I_s = \frac{660 \text{ VA}}{120 \text{ V}} = 5.5 \text{ A}$
 $\theta = \cos^{-1} 0.6 = 53.13^\circ$
 $\therefore \mathbf{E = 120 \text{ V } \angle 0^\circ, \mathbf{I_s = 5.5 \text{ A } \angle -53.13^\circ}}$
 $P = EI \cos \theta = (120 \text{ V})(5.5 \text{ A})(0.6) = \mathbf{396 \text{ W}}$
 $\text{Wattmeter} = \mathbf{396 \text{ W}}, \text{Ammeter} = \mathbf{5.5 \text{ A}}, \text{Voltmeter} = \mathbf{120 \text{ V}}$
- b. $\mathbf{Z_T = \frac{E}{I} = \frac{120 \text{ V } \angle 0^\circ}{5.5 \text{ A } \angle -53.13^\circ} = \mathbf{21.82 \Omega \angle 53.13^\circ} = \mathbf{13.09 \Omega + j17.46 \Omega} = R + jX_L}$
21. a. $R = \frac{P}{I^2} = \frac{80 \text{ W}}{(4 \text{ A})^2} = \mathbf{5 \Omega}, \mathbf{Z_T = \frac{E}{I} = \frac{200 \text{ V}}{4 \text{ A}} = 50 \Omega}$
 $X_L = \sqrt{Z_T^2 - R^2} = \sqrt{(50 \Omega)^2 - (5 \Omega)^2} = 49.75 \Omega$
 $L = \frac{X_L}{2\pi f} = \frac{49.75 \Omega}{(2\pi)(60 \text{ Hz})} = \mathbf{132.03 \text{ mH}}$
- b. $R = \frac{P}{I^2} = \frac{90 \text{ W}}{(3 \text{ A})^2} = \mathbf{10 \Omega}$
- c. $R = \frac{P}{I^2} = \frac{60 \text{ W}}{(2 \text{ A})^2} = \mathbf{15 \Omega}, \mathbf{Z_T = \frac{E}{I} = \frac{200 \text{ V}}{2 \text{ A}} = 100 \Omega}$
 $X_L = \sqrt{Z_T^2 - R^2} = \sqrt{(100 \Omega)^2 - (15 \Omega)^2} = 98.87 \Omega$
 $L = \frac{X_L}{2\pi f} = \frac{98.87 \Omega}{376.8} = \mathbf{262.39 \text{ mH}}$
22. a. $X_L = 2\pi fL = (6.28)(50 \text{ Hz})(0.08 \text{ H}) = 25.12 \Omega$
 $Z_T = \sqrt{R^2 + X_L^2} = \sqrt{(4 \Omega)^2 + (25.12 \Omega)^2} = 25.44 \Omega$
 $I = \frac{E}{Z_T} = \frac{60 \text{ V}}{25.44 \Omega} = 2.358 \text{ A}$
 $P = I^2 R = (2.358 \text{ A})^2 4 \Omega = \mathbf{22.24 \text{ W}}$

$$\begin{aligned}
 \text{b.} \quad I &= \sqrt{\frac{P}{R}} = \sqrt{\frac{30 \text{ W}}{7 \, \Omega}} = \mathbf{2.07 \text{ A}} \\
 Z_T &= \frac{E}{I} = \frac{60 \text{ V}}{2.07 \text{ A}} = 28.99 \, \Omega \\
 X_L &= \sqrt{(28.99 \, \Omega)^2 - (7 \, \Omega)^2} = 28.13 \, \Omega \\
 L &= \frac{X_L}{2\pi f} = \frac{28.13 \, \Omega}{(2\pi)(50 \text{ Hz})} = \mathbf{89.54 \text{ mH}}
 \end{aligned}$$

$$\begin{aligned}
 \text{c.} \quad P &= I^2 R = (1.7 \text{ A})^2 10 \, \Omega = \mathbf{28.9 \text{ W}} \\
 Z_T &= \frac{E}{I} = \frac{60 \text{ V}}{1.7 \text{ A}} = 35.29 \, \Omega \\
 X_L &= \sqrt{(35.29 \, \Omega)^2 - (10 \, \Omega)^2} = 33.84 \, \Omega \\
 L &= \frac{X_L}{2\pi f} = \frac{33.84 \, \Omega}{314} = \mathbf{107.77 \text{ mH}}
 \end{aligned}$$

Chapter 20

$$1. \quad a. \quad \omega_s = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \text{ H})(16 \mu\text{F})}} = \mathbf{250 \text{ rad/s}}$$

$$f_s = \frac{\omega_s}{2\pi} = \frac{250 \text{ rad/s}}{2\pi} = \mathbf{39.79 \text{ Hz}}$$

$$b. \quad \omega_s = \frac{1}{\sqrt{(0.51 \text{ H})(0.16 \mu\text{F})}} = \mathbf{3496.50 \text{ rad/s}}$$

$$f_s = \frac{\omega_s}{2\pi} = \frac{3496.50 \text{ rad/s}}{2\pi} = \mathbf{556.49 \text{ Hz}}$$

$$c. \quad \omega_s = \frac{1}{\sqrt{(0.27 \text{ mH})(7.5 \mu\text{F})}} = \mathbf{22,173 \text{ rad/s}}$$

$$f_s = \frac{\omega_s}{2\pi} = \frac{22,173 \text{ rad/s}}{2\pi} = \mathbf{3528.93 \text{ Hz}}$$

$$2. \quad a. \quad X_C = \mathbf{30 \Omega} \quad b. \quad Z_{T_s} = \mathbf{2 \Omega} \quad c. \quad I = \frac{E}{Z_{T_s}} = \frac{50 \text{ mV}}{2 \Omega} = \mathbf{25 \text{ mA}}$$

$$d. \quad V_R = IR = (25 \text{ mA})(10 \Omega) = \mathbf{250 \text{ mV}} = E$$

$$V_L = IX_L = (25 \text{ mA})(30 \Omega) = \mathbf{750 \text{ mV}}$$

$$V_C = IX_C = (25 \text{ mA})(30 \Omega) = \mathbf{750 \text{ mV}}$$

$$V_L = V_C$$

$$e. \quad Q_s = \frac{X_L}{R} = \frac{30 \Omega}{2 \Omega} = \mathbf{15 \text{ (med } Q)}} \quad f. \quad P = I^2 R = (25 \text{ mA})^2 2 \Omega = \mathbf{1.25 \text{ mW}}$$

$$3. \quad a. \quad X_L = \mathbf{2 \text{ k}\Omega}$$

$$b. \quad I = \frac{E}{Z_{T_s}} = \frac{12 \text{ V}}{100 \Omega} = \mathbf{120 \text{ mA}}$$

$$c. \quad V_R = IR = (120 \text{ mA})(100 \Omega) = \mathbf{12 \text{ V}} = E$$

$$V_L = IX_L = (120 \text{ mA})(2 \text{ k}\Omega) = \mathbf{240 \text{ V}}$$

$$V_C = IX_C = (120 \text{ mA})(2 \text{ k}\Omega) = \mathbf{240 \text{ V}}$$

$$V_L = V_C = 20 V_R$$

$$d. \quad Q_s = \frac{X_L}{R} = \frac{20000 \Omega}{100 \Omega} = \mathbf{20 \text{ (high } Q)}}}$$

$$e. \quad X_L = 2\pi fL, L = \frac{X_L}{2\pi f} = \frac{2 \text{ k}\Omega}{2\pi(5 \text{ kHz})} = \mathbf{63.7 \text{ mH}}$$

$$X_C = \frac{1}{2\pi fC}, C = \frac{1}{2\pi fX_C} = \frac{1}{2\pi(5 \text{ kHz})(2 \text{ k}\Omega)} = \mathbf{15,920 \text{ pF}}$$

$$\text{f. } BW = \frac{f_s}{Q_s} = \frac{5 \text{ kHz}}{20} = \mathbf{250 \text{ Hz}}$$

$$\text{g. } f_2 = f_s + \frac{BW}{2} = 5 \text{ kHz} + \frac{0.25 \text{ kHz}}{2} = \mathbf{5.13 \text{ kHz}}$$

$$f_1 = f_s - \frac{BW}{2} = 5 \text{ kHz} - \frac{0.25 \text{ kHz}}{2} = \mathbf{4.88 \text{ kHz}}$$

$$4. \quad \text{a. } f_s = \frac{1}{2\pi\sqrt{LC}} \Rightarrow L = \frac{1}{(2\pi f_s)^2 C} = \frac{1}{(2\pi \cdot 1.8 \text{ kHz})^2 \cdot 2 \mu\text{F}} = \mathbf{3.91 \text{ mH}}$$

$$\text{b. } X_L = 2\pi fL = 2\pi(1.8 \text{ kHz})(3.91 \text{ mH}) = \mathbf{44.2 \Omega}$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(1.8 \text{ kHz})(2 \mu\text{F})} = \mathbf{44.2 \Omega}$$

$$X_L = X_C$$

$$\text{c. } E_{\text{rms}} = (0.707)(20 \text{ mV}) = 14.14 \text{ mV}$$

$$I_{\text{rms}} = \frac{E_{\text{rms}}}{R} = \frac{14.14 \text{ mV}}{4.7 \Omega} = \mathbf{3.01 \text{ mA}}$$

$$\text{d. } P = I^2 R = (3.01 \text{ mA})^2 \cdot 4.7 \Omega = \mathbf{42.58 \mu\text{W}}$$

$$\text{e. } S_T = P_T = \mathbf{42.58 \mu\text{VA}}$$

$$\text{f. } F_p = \mathbf{1}$$

$$\text{g. } Q_s = \frac{X_L}{R} = \frac{44.2 \Omega}{4.7 \Omega} = \mathbf{9.4}$$

$$BW = \frac{f_s}{Q_s} = \frac{1.8 \text{ kHz}}{9.4} = \mathbf{191.49 \text{ Hz}}$$

$$\begin{aligned} \text{h. } f_2 &= \frac{1}{2\pi} \left[\frac{R}{2L} + \frac{1}{2} \sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}} \right] \\ &= \frac{1}{2\pi} \left[\frac{4.7 \Omega}{2(3.91 \text{ mH})} + \frac{1}{2} \sqrt{\left(\frac{4.7 \Omega}{3.91 \text{ mH}}\right)^2 + \frac{4}{(3.91 \text{ mH})(2 \mu\text{F})}} \right] \\ &= \frac{1}{2\pi} [601.02 + 11.324 \times 10^3] \\ &= \mathbf{1897.93 \text{ Hz}} \end{aligned}$$

$$\begin{aligned} f_1 &= \frac{1}{2\pi} \left[-\frac{R}{2L} + \frac{1}{2} \sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}} \right] \\ &= \frac{1}{2\pi} [-601.02 + 11.324 \times 10^3] \\ &= \mathbf{1.71 \text{ kHz}} \end{aligned}$$

$$P_{\text{HPF}} = \frac{1}{2} P_{\text{max}} = \frac{1}{2} (42.58 \mu\text{W}) = \mathbf{21.29 \mu\text{W}}$$

5. a. $BW = f_s/Q_s = 6000 \text{ Hz}/15 = \mathbf{400 \text{ Hz}}$
- b. $f_2 = f_s + \frac{BW}{2} = 6000 \text{ Hz} + 200 \text{ Hz} = \mathbf{6200 \text{ Hz}}$
 $f_1 = f_s - \frac{BW}{2} = 6000 \text{ Hz} - 200 \text{ Hz} = \mathbf{5800 \text{ Hz}}$
- c. $Q_s = \frac{X_L}{R} \Rightarrow X_L = Q_s R = (15)(3 \Omega) = \mathbf{45 \Omega} = X_C$
- d. $P_{\text{HPF}} = \frac{1}{2} P_{\text{max}} = \frac{1}{2} (I^2 R) = \frac{1}{2} (0.5 \text{ A})^2 3 \Omega = \mathbf{375 \text{ mW}}$
6. a. $L = \frac{X_L}{2\pi f} = \frac{200 \Omega}{2\pi(10^4 \text{ Hz})} = 3.185 \text{ mH}$
 $BW = \frac{R}{2\pi L} = \frac{5 \Omega}{2\pi(3.185 \text{ mH})} \cong \mathbf{250 \text{ Hz}}$
or $Q_s = \frac{X_L}{R} = \frac{X_C}{R} = \frac{200 \Omega}{5 \Omega} = 40, BW = \frac{f_s}{Q_s} = \frac{10,000 \text{ Hz}}{40} = \mathbf{250 \text{ Hz}}$
- b. $f_2 = f_s + BW/2 = 10,000 \text{ Hz} + 250 \text{ Hz}/2 = \mathbf{10,125 \text{ Hz}}$
 $f_1 = f_s - BW/2 = 10,000 \text{ Hz} - 125 \text{ Hz} = \mathbf{9,875 \text{ Hz}}$
- c. $Q_s = \frac{X_L}{R} = \frac{200 \Omega}{5 \Omega} = \mathbf{40}$
- d. $\mathbf{I} = \frac{E \angle 0^\circ}{R \angle 0^\circ} = \frac{30 \text{ V} \angle 0^\circ}{5 \Omega \angle 0^\circ} = 6 \text{ A} \angle 0^\circ, \mathbf{V}_L = (I \angle 0^\circ)(X_L \angle 90^\circ)$
 $= (6 \text{ A} \angle 0^\circ)(200 \Omega \angle 90^\circ)$
 $= \mathbf{1200 \text{ V} \angle 90^\circ}$
 $\mathbf{V}_C = (I \angle 0^\circ)(X_C \angle -90^\circ) = \mathbf{1200 \text{ V} \angle -90^\circ}$
- e. $P = I^2 R = (6 \text{ A})^2 5 \Omega = \mathbf{180 \text{ W}}$
7. a. $BW = \frac{f_s}{Q_s} \Rightarrow Q_s = f_s/BW = 2000 \text{ Hz}/200 \text{ Hz} = \mathbf{10}$
- b. $Q_s = \frac{X_L}{R} \Rightarrow X_L = Q_s R = (10)(2 \Omega) = \mathbf{20 \Omega}$
- c. $L = \frac{X_L}{2\pi f} = \frac{20 \Omega}{(6.28)(2 \text{ kHz})} = \mathbf{1.59 \text{ mH}}$
 $C = \frac{1}{2\pi f X_C} = \frac{1}{(6.28)(2 \text{ kHz})(20 \Omega)} = \mathbf{3.98 \mu\text{F}}$

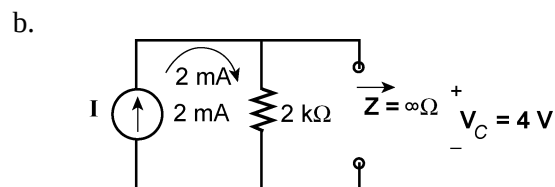
- d. $f_2 = f_s + BW/2 = 2000 \text{ Hz} + 100 \text{ Hz} = \mathbf{2100 \text{ Hz}}$
 $f_1 = f_s - BW/2 = 2000 \text{ Hz} - 100 \text{ Hz} = \mathbf{1900 \text{ Hz}}$
8. a. $BW = 6000 \text{ Hz} - 5400 \text{ Hz} = \mathbf{600 \text{ Hz}}$
- b. $BW = f_s/Q_s \Rightarrow f_s = Q_s BW = (9.5)(600 \text{ Hz}) = \mathbf{5700 \text{ Hz}}$
- c. $Q_s = \frac{X_L}{R} \Rightarrow X_L = X_C = Q_s R = (9.5)(2 \Omega) = \mathbf{19 \Omega}$
- d. $L = \frac{X_L}{2\pi f} = \frac{19 \Omega}{2\pi(5700 \text{ Hz})} = \mathbf{0.53 \text{ mH}}$
 $C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(5.7 \text{ kHz})(19 \Omega)} = \mathbf{1.47 \mu F}$
9. $I_M = \frac{E}{R} \Rightarrow R = \frac{E}{I_M} = \frac{5 \text{ V}}{500 \text{ mA}} = \mathbf{10 \Omega}$
 $BW = f_s/Q_s \Rightarrow Q_s = f_s/BW = 8400 \text{ Hz}/120 \text{ Hz} = 70$
 $Q_s = \frac{X_L}{R} \Rightarrow X_L = Q_s R = (70)(10 \Omega) = \mathbf{700 \Omega}$
 $X_C = X_L = \mathbf{700 \Omega}$
 $L = \frac{X_L}{2\pi f} = \frac{700 \Omega}{(2\pi)(8.4 \text{ kHz})} = \mathbf{13.26 \text{ mH}}$
 $C = \frac{1}{2\pi f X_C} = \frac{1}{(2\pi)(8.4 \text{ kHz})(0.7 \text{ k}\Omega)} = \mathbf{27.07 \text{ nF}}$
 $f_2 = f_s + BW/2 = 8400 \text{ Hz} + 120 \text{ Hz}/2 = \mathbf{8.46 \text{ kHz}}$
 $f_1 = f_s - BW/2 = 8400 \text{ Hz} - 60 \text{ Hz} = \mathbf{8.34 \text{ kHz}}$
10. $Q_s = \frac{X_L}{R} \Rightarrow X_L = Q_s R = 20(2 \Omega) = \mathbf{40 \Omega} = X_C$
 $BW = \frac{f_s}{Q_s} \Rightarrow f_s = Q_s BW = (20)(400 \text{ Hz}) = \mathbf{8 \text{ kHz}}$
 $L = \frac{X_L}{2\pi f} = \frac{40 \Omega}{2\pi(8 \text{ kHz})} = \mathbf{795.77 \mu H}$
 $C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(8 \text{ kHz})(40 \Omega)} = \mathbf{497.36 \text{ nF}}$
 $f_2 = f_s + BW/2 = 8000 \text{ Hz} + 400 \text{ Hz}/2 = \mathbf{8200 \text{ Hz}}$
 $f_1 = f_s - BW/2 = 8000 \text{ Hz} - 200 \text{ Hz} = \mathbf{7800 \text{ Hz}}$
11. a. $f_s = \frac{\omega_s}{2\pi} = \frac{2\pi \times 10^6 \text{ rad/s}}{2\pi} = \mathbf{1 \text{ MHz}}$
- b. $\frac{f_2 - f_1}{f_s} = 0.16 \Rightarrow BW = f_2 - f_1 = 0.16 f_s = 0.16(1 \text{ MHz}) = \mathbf{160 \text{ kHz}}$

$$\begin{aligned}
 \text{c. } P &= \frac{V_R^2}{R} \Rightarrow R = \frac{V_R^2}{P} = \frac{(120 \text{ V})^2}{20 \text{ W}} = \mathbf{720 \, \Omega} \\
 BW &= \frac{R}{2\pi L} \Rightarrow L = \frac{R}{2\pi BW} = \frac{720 \, \Omega}{(6.28)(160 \text{ kHz})} = \mathbf{0.716 \text{ mH}} \\
 f_s &= \frac{1}{2\pi\sqrt{LC}} \Rightarrow C = \frac{1}{4\pi^2 f_s^2 L} = \frac{1}{4\pi^2 (10^6 \text{ Hz})^2 (0.716 \text{ mH})} = \mathbf{35.38 \text{ pF}} \\
 \text{d. } Q_\ell &= \frac{X_L}{R_\ell} = 80 \Rightarrow R_p = \frac{X_L}{80} = \frac{2\pi f_s L}{80} = \frac{2\pi (10^6 \text{ Hz})(0.716 \text{ mH})}{80} = \mathbf{56.23 \, \Omega}
 \end{aligned}$$

$$\begin{aligned}
 12. \text{ a. } Q_\ell &= \frac{X_L}{R_\ell} \\
 R_\ell &= \frac{X_L}{Q_\ell} = \frac{2\pi fL}{Q_\ell} = \frac{2\pi (1 \text{ MHz})(100 \, \mu\text{H})}{12.5} = 50.27 \, \Omega \\
 \frac{f_2 - f_1}{f_s} &= \frac{1}{Q_s} = 0.2 \\
 Q_s &= \frac{1}{0.2} = 5 = \frac{X_L}{R} = \frac{2\pi fL}{R} = \frac{2\pi (1 \text{ MHz})(100 \, \mu\text{H})}{R} = \frac{628.32 \, \Omega}{R} \\
 R &= \frac{628.32 \, \Omega}{5} = 125.66 \\
 R &= R_d + R_\ell \\
 125.66 \, \Omega &= R_d + 50.27 \, \Omega \\
 \text{and } R_d &= 125.66 \, \Omega - 50.27 \, \Omega = \mathbf{75.39 \, \Omega}
 \end{aligned}$$

$$\begin{aligned}
 \text{c. } X_C &= \frac{1}{2\pi fC} = X_L \\
 C &= \frac{1}{2\pi fX_C} = \frac{1}{2\pi (1 \text{ MHz})(628.32 \, \Omega)} = \mathbf{253.3 \text{ pF}}
 \end{aligned}$$

$$13. \text{ a. } f_p = \frac{1}{2\pi\sqrt{LC}} = \frac{2}{2\pi\sqrt{(0.1 \text{ mH})(10 \text{ nF})}} = \mathbf{159.16 \text{ kHz}}$$



$$\begin{aligned}
 \text{c. } I_L &= \frac{V_L}{X_L} = \frac{4 \text{ V}}{2\pi f_p L} = \frac{4 \text{ V}}{100 \, \Omega} = \mathbf{40 \text{ mA}} \\
 I_C &= \frac{V_L}{X_C} = \frac{4 \text{ V}}{1/2\pi f_p C} = \frac{4 \text{ V}}{100 \, \Omega} = \mathbf{40 \text{ mA}}
 \end{aligned}$$

$$d. \quad Q_p = \frac{R_s}{X_{L_p}} = \frac{2 \text{ k}\Omega}{2\pi f_p L} = \frac{2 \text{ k}\Omega}{100 \Omega} = \mathbf{20}$$

$$14. \quad a. \quad f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(4.7 \text{ mH})(30 \text{ nF})}} = \mathbf{13.4 \text{ kHz}}$$

$$b. \quad Q_\ell = \frac{X_L}{R_\ell} = \frac{2\pi f L}{R_\ell} = \frac{2\pi(13.4 \text{ kHz})(4.7 \text{ mH})}{8 \Omega} = \mathbf{49.46} \geq 10 \text{ (yes)}$$

$$c. \quad \text{Since } Q_\ell \geq 10, f_p \cong f_s = \mathbf{13.4 \text{ kHz}}$$

$$d. \quad X_L = 2\pi f_p L = 2\pi(13.4 \text{ kHz})(4.7 \text{ mH}) = \mathbf{395.72 \Omega}$$

$$X_C = \frac{1}{2\pi f_p C} = \frac{1}{2\pi(13.4 \text{ kHz})(30 \text{ nF})} = \mathbf{395.91 \Omega}$$

$$X_L = X_C$$

$$e. \quad Z_{T_p} = Q_\ell^2 R_\ell = (49.46)^2 8 \Omega = \mathbf{19.57 \text{ k}\Omega}$$

$$f. \quad V_C = I Z_{T_p} = (10 \text{ mA})(19.57 \text{ k}\Omega) = \mathbf{195.7 \text{ V}}$$

$$g. \quad Q_\ell \geq 10, \quad Q_p = Q_\ell = \mathbf{49.46}$$

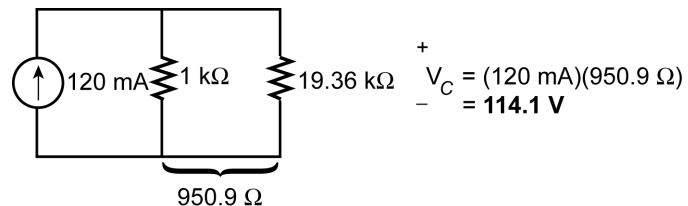
$$BW = \frac{f_p}{Q_p} = \frac{13.4 \text{ kHz}}{49.46} = \mathbf{270.9 \text{ Hz}}$$

$$h. \quad I_L = I_C = Q_\ell I_T = (49.46)(10 \text{ mA}) = \mathbf{494.6 \text{ mA}}$$

$$15. \quad a. \quad f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(200 \mu\text{H})(120 \mu\text{F})}} = \mathbf{1.027 \text{ MHz}}$$

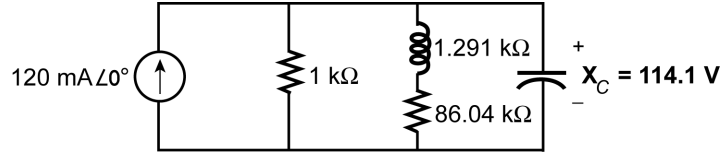
$$b. \quad Q_\ell = \frac{X_L}{R_\ell} \Rightarrow R_\ell = \frac{X_L}{Q_\ell} = \frac{2\pi(1.027 \text{ MHz})(200 \mu\text{H})}{15} = 86.04 \Omega$$

$$Z_p = Q_\ell^2 R_\ell = (15)^2 86.04 \Omega = 19.36 \text{ k}\Omega$$



$$c. \quad P = I^2 R = (120 \text{ mA})^2 (950.9 \Omega) = \mathbf{13.69 \text{ W}}$$

d.



$$X_L = 2\pi fL = 2\pi(1.027 \text{ MHz})(200 \mu\text{H}) = 1.291 \text{ k}\Omega$$

$$|V_{R_\ell}| = \frac{86.04 \Omega \angle 0^\circ (114.1 \text{ V})}{86.04 \Omega + j1.291 \text{ k}\Omega} = 7.587 \text{ V}$$

$$P = V_{R_\ell}^2 / R_\ell = (7.587 \text{ V})^2 / 86.04 \Omega = \mathbf{669 \text{ mW}}$$

$$13.69 \text{ W} : 669 \text{ mW} \cong 20:1$$

16. a. $Q_\ell = \frac{X_L}{R_L} = \frac{100 \Omega}{20 \Omega} = 5 \leq 10$

$$\therefore \frac{X_L}{R_\ell^2 + X_L^2} = \frac{1}{X_C} \Rightarrow X_C = \frac{R_\ell^2 + X_L^2}{X_L} = \frac{(20 \Omega)^2 + (100 \Omega)^2}{100 \Omega} = \mathbf{104 \Omega}$$

b. $Z_T = R_s \parallel R_p = R_s \parallel \frac{R_\ell^2 + X_L^2}{R_\ell} = 1000 \Omega \parallel \frac{10,400 \Omega}{20} = \mathbf{342.11 \Omega}$

c. $\mathbf{E} = \mathbf{I}Z_{T_p} = (5 \text{ mA} \angle 0^\circ)(342.11 \Omega \angle 0^\circ) = 1.711 \text{ V} \angle 0^\circ$

$$\mathbf{I}_C = \frac{\mathbf{E}}{X_C \angle -90^\circ} = \frac{1.711 \text{ V} \angle 0^\circ}{104 \Omega \angle -90^\circ} = \mathbf{16.45 \text{ mA} \angle 90^\circ}$$

$$\mathbf{Z}_L = 20 \Omega + j100 \Omega = 101.98 \Omega \angle 78.69^\circ$$

$$\mathbf{I}_L = \frac{\mathbf{E}}{\mathbf{Z}_L} = \frac{1.711 \text{ V} \angle 0^\circ}{101.98 \Omega \angle 78.69^\circ} = \mathbf{16.78 \text{ mA} \angle -78.69^\circ}$$

d. $L = \frac{X_L}{2\pi f} = \frac{100 \Omega}{2\pi(20 \text{ kHz})} = \mathbf{795.77 \mu\text{H}}$

$$C = \frac{1}{2\pi fX_C} = \frac{1}{2\pi(20 \text{ kHz})(104 \Omega)} = \mathbf{76.52 \text{ nF}}$$

e. $Q_p = \frac{R}{X_C} = \frac{342.11 \Omega}{104 \Omega} = \mathbf{3.29}$

$$BW = f_p / Q_p = 20,000 \text{ Hz} / 3.29 = \mathbf{6079.03 \text{ Hz}}$$

17. $Q_\ell = 35 = \frac{X_L}{R_\ell} \Rightarrow R_\ell = \frac{X_L}{35} = \frac{2\pi \left(\frac{2 \times 10^6}{2\pi} \text{ Hz} \right) (1 \text{ mH})}{35} = \frac{2000 \Omega}{35} = 57.14 \Omega$

$$Q_\ell \geq 10 : Q_p = \frac{f_p}{BW} = \frac{2 \times 10^6 / 2\pi \text{ Hz}}{100,000 \text{ Hz}} = 20$$

$$Q_p = 20 = \frac{R \parallel Q_\ell^2 R_\ell}{X_L} = \frac{R \parallel (35)^2 57.14 \Omega}{2000}$$

$$\text{And } 40,000 = R \parallel 70,000$$

$$\text{So } R = 93.33 \text{ k}\Omega \Rightarrow \text{use } R = \mathbf{91 \text{ k}\Omega} \text{ (standard value)}$$

$$Q_p \geq 10, X_C = X_L = 2000 \Omega = \frac{1}{2\pi fC} = \frac{1}{2\pi \left(\frac{2 \times 10^6}{2\pi} \text{ Hz} \right) C}$$

$$C = 250 \text{ pF} \Rightarrow \text{use } C = \mathbf{240 \text{ pF}} \text{ (standard value)}$$

$$18. \quad a. \quad f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(80 \mu\text{H})(0.03 \mu\text{F})}} = \mathbf{102.73 \text{ kHz}}$$

$$f_p = f_s \sqrt{1 - \frac{R_\ell^2 C}{L}} = 102.73 \text{ kHz} \sqrt{1 - \frac{(1.5 \Omega)^2 0.03 \mu\text{F}}{80 \mu\text{H}}} = 102.73 \text{ kHz}(.99958)$$

$$= \mathbf{102.69 \text{ kHz}}$$

$$f_m = f_s \sqrt{1 - \frac{1}{4} \left[\frac{R_\ell^2 C}{L} \right]} = 102.73 \text{ kHz}(0.99989) = \mathbf{102.72 \text{ kHz}}$$

$$\text{Since } f_s \cong f_p \cong f_m \Rightarrow \text{high } Q_p$$

$$b. \quad X_L = 2\pi f_p L = 2\pi(102.69 \text{ kHz})(80 \mu\text{H}) = \mathbf{51.62 \Omega}$$

$$X_C = \frac{1}{2\pi f_p C} = \frac{1}{2\pi(102.69 \text{ kHz})(0.03 \mu\text{F})} = \mathbf{51.66 \Omega}$$

$$X_L \cong X_C$$

$$c. \quad Z_{T_p} = R_s \parallel Q_\ell^2 R_\ell$$

$$Q_\ell = \frac{X_L}{R_\ell} = \frac{51.62 \Omega}{1.5 \Omega} = 34.41$$

$$Z_{T_p} = 10 \text{ k}\Omega \parallel (34.41)^2 1.5 \Omega = 10 \text{ k}\Omega \parallel 1.776 \text{ k}\Omega = \mathbf{1.51 \text{ k}\Omega}$$

$$d. \quad Q_p = \frac{R_s \parallel Q_\ell^2 R_\ell}{X_L} = \frac{Z_{T_p}}{X_L} = \frac{1.51 \text{ k}\Omega}{51.62 \Omega} = \mathbf{29.25}$$

$$BW = \frac{f_p}{Q_p} = \frac{102.69 \text{ kHz}}{29.25} = \mathbf{3.51 \text{ kHz}}$$

$$e. \quad \text{Converting the voltage source to a current source:}$$

$$I_s = \frac{E}{R_s} = \frac{100 \text{ V}}{10 \text{ k}\Omega} = 10 \text{ mA}$$

$$\text{And } R_s = R_p = 10 \text{ k}\Omega$$

$$\text{Then } I_T = \frac{R_s I_s}{R_s + Q_\ell^2 R_\ell} = \frac{10 \text{ k}\Omega(10 \text{ mA})}{10 \text{ k}\Omega + 1.78 \text{ k}\Omega} = 8.49 \text{ mA}$$

$$I_C = I_L \cong Q_\ell I_T = (34.41)(8.49 \text{ mA}) = \mathbf{292.14 \text{ mA}}$$

f. $V_C = IZ_{T_p} = (10 \text{ mA})(1.51 \text{ k}\Omega) = \mathbf{15.1 \text{ V}}$

19. a. $f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(0.5 \text{ mH})(1 \mu\text{F})}} = \mathbf{7.12 \text{ kHz}}$

$$f_p = f_s \sqrt{1 - \frac{R_\ell^2 C}{L}} = 7.12 \text{ kHz} \sqrt{1 - \frac{(8 \Omega)^2 (1 \mu\text{F})}{0.5 \text{ mH}}} = 7.12 \text{ kHz}(0.9338) = \mathbf{6.65 \text{ kHz}}$$

$$f_m = f_s \sqrt{1 - \frac{1}{4} \left[\frac{R_\ell^2 C}{L} \right]} = 7.12 \text{ kHz} \sqrt{1 - \frac{1}{4} \left[\frac{(8 \Omega)^2 (1 \mu\text{F})}{0.5 \text{ mH}} \right]} = 7.12 \text{ kHz} (0.9839)$$

$= \mathbf{7.01 \text{ kHz}}$

Low Q_p

b. $X_L = 2\pi f_p L = 2\pi(6.647 \text{ kHz})(0.5 \text{ mH}) = 20.88 \Omega$

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(6.647 \text{ kHz})(1 \mu\text{F})} = 23.94 \Omega$$

$X_C > X_L$ (low Q)

c. $Z_{T_p} = R_s \parallel R_p = R_s \parallel \frac{R_\ell^2 + X_L^2}{R_\ell} = 500 \Omega \parallel \frac{(8 \Omega)^2 + (20.88 \Omega)^2}{8 \Omega} = 500 \Omega \parallel 62.5 \Omega$

$= \mathbf{55.56 \Omega}$

d. $Q_p = \frac{Z_{T_p}}{X_{L_p}} = \frac{55.56 \Omega}{23.94 \Omega} = \mathbf{2.32}$

$$BW = \frac{f_p}{Q_p} = \frac{6.647 \text{ kHz}}{2.32} = \mathbf{2.87 \text{ kHz}}$$

e. One method: $V_C = IZ_{T_p} = (40 \text{ mA})(55.56 \Omega) = 2.22 \text{ V}$

$$I_C = \frac{V_C}{X_C} = \frac{2.22 \text{ V}}{23.94 \Omega} = \mathbf{92.73 \text{ mA}}$$

$$I_L = \frac{|V_C|}{|R_\ell + jX_L|} = \frac{2.22 \text{ V}}{|8 + j20.88|} = \frac{2.22 \text{ V}}{22.36 \Omega} = \mathbf{99.28 \text{ mA}}$$

f. $V_C = \mathbf{2.22 \text{ V}}$

20. a. $Z_{T_p} = \frac{R_\ell^2 + X_L^2}{R_\ell} = 50 \text{ k}\Omega$

$$(50 \Omega)^2 + X_L^2 = (50 \text{ k}\Omega)(50 \Omega)$$

$$X_L = \sqrt{250 \times 10^4 - 2.5 \times 10^3} = \mathbf{1580.3 \Omega}$$

- b. $Q = \frac{X_L}{R_\ell} = \frac{1580.3}{50} = 31.61 \geq 10$
 $\therefore X_C = X_L = \mathbf{1580.3 \Omega}$
- c. $X_L = 2\pi f_p L \Rightarrow f_p = \frac{X_L}{2\pi L} = \frac{1580.3 \Omega}{2\pi(16 \text{ mH})} = \mathbf{15.72 \text{ kHz}}$
- d. $X_C = \frac{1}{2\pi f_p C} \Rightarrow C = \frac{1}{2\pi f_s X_C} = \frac{1}{2\pi(15.72 \text{ kHz})(1580.3 \Omega)} = \mathbf{6.4 \text{ nF}}$
21. a. $Q_\ell = 20 > 10 \therefore f_p = f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(200 \text{ mH})(10 \text{ nF})}} = \mathbf{3558.81 \text{ Hz}}$
- b. $Q_\ell = \frac{X_L}{R_\ell} = \frac{2\pi fL}{R_\ell} \Rightarrow R_\ell = \frac{2\pi fL}{Q_\ell} = \frac{2\pi(3558.81 \text{ Hz})(0.2 \text{ H})}{20} = 223.61 \Omega$
 $Z_{T_p} = R_s \parallel R_p = R_s \parallel Q_\ell^2 R_\ell = 40 \text{ k}\Omega \parallel (20)^2 223.61 \Omega$
 $Z_{T_p} = 27.64 \text{ k}\Omega$
 Converting the voltage source to a current source:
 $I_s = \frac{E}{R_s} = \frac{200 \text{ V}}{40 \text{ k}\Omega} = 5 \text{ mA}$
 $R_p = R_s = 40 \text{ k}\Omega$
 $V_C = IZ_{T_p} = (5 \text{ mA})(27.64 \text{ k}\Omega) = \mathbf{138.2 \text{ V}}$
- c. $P = I^2 R = (5 \text{ mA})^2 27.64 \text{ k}\Omega = \mathbf{691 \text{ mW}}$
- d. $Q_p = \frac{R}{X_L} = \frac{R_s \parallel R_p}{X_L} = \frac{27.64 \text{ k}\Omega}{2\pi(3558.81 \text{ Hz})(0.2 \text{ H})} = \mathbf{6.18}$
 $BW = \frac{f_p}{Q_p} = \frac{3558.81 \text{ Hz}}{6.18} = \mathbf{575.86 \text{ Hz}}$
22. a. Ratio of X_C to R_ℓ suggests high Q system.
 $\therefore X_L = \mathbf{400 \Omega} = X_C$
- b. $Q_\ell = \frac{X_L}{R_\ell} = \frac{400 \Omega}{8 \Omega} = \mathbf{50}$
- c. $Q_p = \frac{R}{X_L} = \frac{R_s \parallel R_p}{X_L} = \frac{R_s \parallel Q_\ell^2 R_\ell}{X_L} = \frac{20 \text{ k}\Omega \parallel (50)^2 8 \Omega}{400 \Omega} = \frac{10 \text{ k}\Omega}{400 \Omega} = \mathbf{25}$
 $BW = \frac{f_p}{Q_p} \Rightarrow f_p = Q_p BW = (25)(1000 \text{ Hz}) = \mathbf{25 \text{ kHz}}$

- d. $V_{C_{\max}} = IZ_{T_p} = (0.1 \text{ mA})(10 \text{ k}\Omega) = \mathbf{1 \text{ V}}$
- e. $f_2 = f_p + BW/2 = 25 \text{ kHz} + \frac{1 \text{ kHz}}{2} = \mathbf{25.5 \text{ kHz}}$
 $f_1 = f_p - BW/2 = 25 \text{ kHz} - \frac{1 \text{ kHz}}{2} = \mathbf{24.5 \text{ kHz}}$
23. a. $X_C = \frac{R_\ell^2 + X_L^2}{X_L} \Rightarrow X_L^2 - X_L X_C + R_\ell^2 = 0$
 $X_L^2 - 100 X_L + 144 = 0$
 $X_L = \frac{-(-100) \pm \sqrt{(100)^2 - 4(1)(144)}}{2}$
 $= 50 \Omega \pm \frac{\sqrt{10^4 - 576}}{2} = 50 \Omega \pm 48.54 \Omega$
 $X_L = \mathbf{98.54 \Omega}$ or ~~1.46Ω~~
- b. $Q_\ell = \frac{X_L}{R_\ell} = \frac{98.54 \Omega}{12 \Omega} = \mathbf{8.21}$
- c. $Q_p = \frac{R_s \parallel R_p}{X_{L_p}} = \frac{40 \text{ k}\Omega \parallel \frac{R_\ell^2 + X_L^2}{R_\ell}}{X_C} = \frac{40 \text{ k}\Omega \parallel \frac{(12 \Omega)^2 + (98.54 \Omega)^2}{12 \Omega}}{100 \Omega}$
 $= \frac{40 \text{ k}\Omega \parallel 821.18 \Omega}{100 \Omega} = \frac{804.66 \Omega}{100 \Omega} = \mathbf{8.05}$
 $BW = f_p/Q_p \Rightarrow f_p = Q_p BW = (8.05)(1 \text{ kHz}) = \mathbf{8.05 \text{ kHz}}$
- d. $V_{C_{\max}} = IZ_{T_p} = (6 \text{ mA})(804.66 \Omega) = \mathbf{4.83 \text{ V}}$
- e. $f_2 = f_p + BW/2 = 8.05 \text{ kHz} + \frac{1 \text{ kHz}}{2} = \mathbf{8.55 \text{ kHz}}$
 $f_1 = f_p - BW/2 = 8.05 \text{ kHz} - \frac{1 \text{ kHz}}{2} = \mathbf{7.55 \text{ kHz}}$
24. a. $f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(0.5 \text{ mH})(30 \text{ nF})}} = \mathbf{41.09 \text{ kHz}}$
 $f_p = f_s \sqrt{1 - \frac{R_\ell^2 C}{L}} = 41.09 \text{ kHz} \sqrt{1 - \frac{(6 \Omega)^2 30 \text{ nF}}{0.5 \text{ mH}}} = 41.09 \text{ kHz}(0.9978) = \mathbf{41 \text{ kHz}}$
 $f_m = f_s \sqrt{1 - \frac{1}{4} \left[\frac{R_\ell^2 C}{L} \right]} = 41.09 \text{ kHz} \sqrt{1 - \frac{1}{4} \left[\frac{(6 \Omega)^2 (30 \text{ nF})}{0.5 \text{ mH}} \right]} = 41.09 \text{ kHz}(0.9995)$
 $= \mathbf{41.07 \text{ kHz}}$
High Q_p

$$\text{b. } \mathbf{I} = \frac{80 \text{ V } \angle 0^\circ}{20 \text{ k}\Omega \angle 0^\circ} = 4 \text{ mA } \angle 0^\circ, R_s = 20 \text{ k}\Omega$$

$$Q_\ell = \frac{X_L}{R_\ell} = \frac{2\pi fL}{R_\ell} = \frac{2\pi(41 \text{ kHz})(0.5 \text{ mH})}{6 \Omega} = \mathbf{21.47} \text{ (high } Q \text{ coil)}$$

$$Q_p = \frac{R_s \parallel R_p}{X_{L_p}} = \frac{R_s \parallel \frac{R_\ell^2 + X_L^2}{R_\ell}}{\frac{R_\ell^2 + X_L^2}{X_L}} = \frac{20 \text{ k}\Omega \parallel \frac{(6 \Omega)^2 + (128.81 \Omega)^2}{6 \Omega}}{\frac{(6 \Omega)^2 + (128.81 \Omega)^2}{128.81 \Omega}}$$

$$= \frac{20 \text{ k}\Omega \parallel 2.771 \text{ k}\Omega}{129.09 \Omega} = \frac{2.434 \text{ k}\Omega}{129.09 \Omega} = \mathbf{18.86} \text{ (high } Q_p)$$

$$\text{c. } Z_{T_p} = R_s \parallel R_p = 20 \text{ k}\Omega \parallel 2.771 \text{ k}\Omega = \mathbf{2.43 \text{ k}\Omega}$$

$$\text{d. } V_C = IZ_{T_p} = (4 \text{ mA})(2.43 \text{ k}\Omega) = \mathbf{9.74 \text{ V}}$$

$$\text{e. } BW = \frac{f_p}{Q_p} = \frac{41 \text{ kHz}}{18.86} = \mathbf{2.17 \text{ kHz}}$$

$$\text{f. } X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(41 \text{ kHz})(30 \text{ nF})} = \mathbf{129.39 \Omega}$$

$$I_C = \frac{V_C}{X_C} = \frac{9.736 \text{ V}}{129.39 \Omega} = \mathbf{75.25 \text{ mA}}$$

$$I_L = \frac{V_C}{|R + jX_L|} = \frac{9.736 \text{ V}}{6 \Omega + j128.81 \Omega} = \frac{9.736 \text{ V}}{128.95 \Omega} = \mathbf{75.50 \text{ mA}}$$

$$25. \quad Q_\ell = \frac{X_L}{R_\ell} = \frac{2\pi f_p L}{R_\ell} \Rightarrow R_\ell = \frac{2\pi f_p L}{Q_\ell} = \frac{2\pi(20 \text{ kHz})(2 \text{ mH})}{80} = \mathbf{3.14 \Omega}$$

$$BW = f_p/Q_p \Rightarrow Q_p = f_p/BW = 20 \text{ kHz}/1.8 \text{ kHz} = \mathbf{11.11}$$

$$\text{High } Q \therefore f_p \cong f_s = \frac{1}{2\pi\sqrt{LC}} \Rightarrow C = \frac{1}{4\pi^2 f_p^2 L} = \frac{1}{4\pi^2 (20 \text{ kHz})^2 2 \text{ mH}} = \mathbf{31.66 \text{ nF}}$$

$$Q_p = \frac{R}{X_C} \Rightarrow R = Q_p X_C = \frac{Q_p}{2\pi f_p C} = \frac{11.11}{2\pi(20 \text{ kHz})(31.66 \text{ nF})} = \mathbf{2.79 \text{ k}\Omega}$$

$$R_p = Q_\ell^2 R_\ell = (80)^2 3.14 \Omega = 20.1 \text{ k}\Omega$$

$$R = R_s \parallel R_p = \frac{R_s R_p}{R_s + R_p} \Rightarrow R_s = \frac{R_p R}{R_p - R} = \frac{(20.1 \text{ k}\Omega)(2.793 \text{ k}\Omega)}{20.1 \text{ k}\Omega - 2.793 \text{ k}\Omega} = \mathbf{3.24 \text{ k}\Omega}$$

$$\begin{aligned}
26. \quad V_{C_{\max}} &= IZ_{T_p} \Rightarrow Z_{T_p} = \frac{V_{C_{\max}}}{I} = \frac{1.8 \text{ V}}{0.2 \text{ mA}} = 9 \text{ k}\Omega \\
Q_p &= \frac{R}{X_L} = \frac{R_s \parallel R_p}{X_L} = \frac{R_p}{X_L} \Rightarrow X_L = \frac{R_p}{Q_p} = \frac{9 \text{ k}\Omega}{30} = 300 \text{ }\Omega = X_C \\
BW &= \frac{f_p}{Q_p} \Rightarrow f_p = Q_p BW = (30)(500 \text{ Hz}) = \mathbf{15 \text{ kHz}} \\
L &= \frac{X_L}{2\pi f} = \frac{300 \text{ }\Omega}{2\pi(15 \text{ kHz})} = \mathbf{3.18 \text{ mH}} \\
C &= \frac{1}{2\pi f X_C} = \frac{1}{2\pi(15 \text{ kHz})(300 \text{ }\Omega)} = \mathbf{35.37 \text{ nF}} \\
Q_p &= Q_\ell (R_s = \infty \text{ }\Omega) = \frac{X_L}{R_\ell} \Rightarrow R_\ell = \frac{X_L}{Q_p} = \frac{300 \text{ }\Omega}{30} = \mathbf{10 \text{ }\Omega}
\end{aligned}$$

$$\begin{aligned}
27. \quad a. \quad f_s &= \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(200 \text{ }\mu\text{H})(2 \text{ nF})}} = 251.65 \text{ kHz} \\
Q_\ell &= \frac{X_L}{R_\ell} = \frac{2\pi(251.65 \text{ kHz})(200 \text{ }\mu\text{H})}{20 \text{ }\Omega} = 15.81 \geq 10 \\
\therefore f_p &= f_s = \mathbf{251.65 \text{ kHz}}
\end{aligned}$$

$$b. \quad Z_{T_p} = R_s \parallel Q_\ell^2 R_\ell = 40 \text{ k}\Omega \parallel (15.81)^2 20 \text{ }\Omega = \mathbf{4.44 \text{ k}\Omega}$$

$$c. \quad Q_p = \frac{R_s \parallel Q_\ell^2 R_\ell}{X_L} = \frac{4.444 \text{ k}\Omega}{316.23 \text{ }\Omega} = \mathbf{14.05}$$

$$d. \quad BW = \frac{f_p}{Q_p} = \frac{251.65 \text{ kHz}}{14.05} = \mathbf{17.91 \text{ kHz}}$$

$$e. \quad \mathbf{20 \text{ }\mu\text{H}, 20 \text{ nF}}$$

f_s the same since product LC the same

$$f_s = 251.65 \text{ kHz}$$

$$Q_\ell = \frac{X_L}{R_\ell} = \frac{2\pi(251.65 \text{ kHz})(20 \text{ }\mu\text{H})}{20 \text{ }\Omega} = 1.581$$

Low Q_ℓ :

$$f_p = f_s \sqrt{1 - \frac{R_\ell^2 C}{L}} = (251.65 \text{ kHz}) \sqrt{1 - \frac{(20 \text{ }\Omega)^2 (20 \text{ nF})}{20 \text{ }\mu\text{H}}}$$

$$= (251.65 \text{ kHz})(0.775) = \mathbf{194.93 \text{ kHz}}$$

$$X_L = 2\pi f_p L = 2\pi(194.93 \text{ kHz})(20 \text{ }\mu\text{H}) = 24.496 \text{ }\Omega$$

$$R_p = \frac{R_\ell^2 + X_L^2}{R_\ell} = \frac{(20 \text{ }\Omega)^2 + (24.496 \text{ }\Omega)^2}{20 \text{ }\Omega} = 50 \text{ }\Omega$$

$$Z_{T_p} = R_s \parallel R_p = 40 \text{ k}\Omega \parallel 50 \text{ }\Omega = \mathbf{49.94 \text{ }\Omega}$$

$$Q_p = \frac{R}{X_L} = \frac{49.94 \Omega}{24.496 \Omega} = \mathbf{2.04}$$

$$BW = \frac{f_p}{Q_p} = \frac{194.93 \text{ kHz}}{2.04} = \mathbf{95.55 \text{ kHz}}$$

f. **0.4 mH, 1 nF**

$f_s = 251.65 \text{ kHz}$ since LC product the same

$$Q_\ell = \frac{X_L}{R_\ell} = \frac{2\pi(251.65 \text{ kHz})(0.4 \text{ mH})}{20 \Omega} = 31.62 \geq 10$$

$$\therefore f_p = f_s = \mathbf{251.65 \text{ kHz}}$$

$$Z_{T_p} = R_s \parallel Q_\ell^2 R_\ell = 40 \text{ k}\Omega \parallel (31.62)^2 20 \Omega = 40 \text{ k}\Omega \parallel (\cong 20 \text{ k}\Omega) \cong \mathbf{13.33 \text{ k}\Omega}$$

$$Q_p = \frac{R_s \parallel Q_\ell^2 R_\ell}{X_L} = \frac{13.33 \text{ k}\Omega}{632.47 \Omega} = \mathbf{21.08}$$

$$BW = \frac{f_p}{Q_p} = \frac{251.65 \text{ kHz}}{21.08} = \mathbf{11.94 \text{ kHz}}$$

g. Network $\frac{L}{C} = \frac{200 \mu\text{H}}{2 \text{ nF}} = \mathbf{100 \times 10^3}$

part (e) $\frac{L}{C} = \frac{20 \mu\text{H}}{20 \text{ nF}} = \mathbf{1 \times 10^3}$

part (f) $\frac{L}{C} = \frac{0.4 \text{ mH}}{1 \text{ nF}} = \mathbf{400 \times 10^3}$

h. Yes, as $\frac{L}{C}$ ratio increased BW decreased.

Also, $V_p = IZ_{T_p}$ and for a fixed I , Z_{T_p} and therefore V_p will increase with increase in the L/C ratio.

Chapter 21

1. a. left: $d_1 = \frac{3}{16}'' = 0.1875'', d_2 = 1''$

$$\begin{aligned}\text{Value} &= 10^3 \times 10^{0.1875''/1''} \\ &= 10^3 \times 1.54 \\ &= \mathbf{1.54 \text{ kHz}}\end{aligned}$$

right: $d_1 = \frac{3}{4}'' = 0.75'', d_2 = 1''$

$$\begin{aligned}\text{Value} &= 10^3 \times 10^{0.75''/1''} \\ &= 10^3 \times 5.623 \\ &= \mathbf{5.62 \text{ kHz}}\end{aligned}$$

b. bottom: $d_1 = \frac{5}{16}'' = 0.3125'', d_2 = \frac{15}{16}'' = 0.9375''$

$$\begin{aligned}\text{Value} &= 10^{-1} \times 10^{0.3125''/0.9375''} = 10^{-1} \times 10^{0.333} \\ &= 10^{-1} \times 2.153 \\ &= \mathbf{0.22 \text{ V}}\end{aligned}$$

top: $d_1 = \frac{11}{16}'' = 0.6875'', d_2 = 0.9375''$

$$\begin{aligned}\text{Value} &= 10^{-1} \times 10^{0.6875''/0.9375''} = 10^{-1} \times 10^{0.720} \\ &= 10^{-1} \times 5.248 \\ &= \mathbf{0.52 \text{ V}}\end{aligned}$$

2. a. **5** b. **-4** c. **8** d. **-6**

e. **1.30** f. **3.94** g. **4.75** h. **-0.498**

3. a. **1000** b. **10^{12}** c. **1.59** d. **1.1**

e. **10^{10}** f. **1513.56** g. **10.02** h. **1,258,925.41**

4. a. **11.51** b. **-9.21** c. **2.996** d. **9.07**

5. $\log_{10} 48 = \mathbf{1.68}$
 $\log_{10} 8 + \log_{10} 6 = 0.903 + 0.778 = \mathbf{1.68}$

6. $\log_{10} 0.2 = \mathbf{-0.699}$
 $\log_{10} 18 - \log_{10} 90 = 1.255 - 1.954 = \mathbf{-0.699}$

7. $\log_{10} 0.5 = \mathbf{-0.30}$
 $-\log_{10} 2 = -(0.301) = \mathbf{-0.30}$

8. $\log_{10} 27 = \mathbf{1.43}$
 $3 \log_{10} 3 = 3(0.4771) = \mathbf{1.43}$

$$9. \quad \text{a.} \quad \text{bels} = \log_{10} \frac{P_2}{P_1} = \log_{10} \frac{280 \text{ mW}}{4 \text{ mW}} = \log_{10} 70 = \mathbf{1.85}$$

$$\text{b.} \quad \text{dB} = 10 \log_{10} \frac{P_2}{P_1} = 10(\log_{10} 70) = 10(1.845) = \mathbf{18.45}$$

$$10. \quad \text{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

$$6 \text{ dB} = 10 \log_{10} \frac{100 \text{ W}}{P_1}$$

$$0.6 = \log_{10} x$$

$$x = 3.981 = \frac{100 \text{ W}}{P_1}$$

$$P_1 = \frac{100 \text{ W}}{3.981} = \mathbf{25.12 \text{ W}}$$

$$11. \quad \text{dB} = 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{40 \text{ W}}{2 \text{ W}} = 10 \log_{10} 20 = \mathbf{13.01}$$

$$12. \quad \text{dB}_m = 10 \log_{10} \frac{P}{1 \text{ mW}}$$

$$\text{dB}_m = 10 \log_{10} \frac{120 \text{ mW}}{1 \text{ mW}} = 10 \log_{10} 120 = \mathbf{20.79}$$

$$13. \quad \text{dB}_v = 20 \log_{10} \frac{V_2}{V_1} = 20 \log_{10} \frac{8.4 \text{ V}}{0.1 \text{ V}} = 20 \log_{10} 84 = \mathbf{38.49}$$

$$14. \quad \text{dB}_v = 20 \log_{10} \frac{V_2}{V_1}$$

$$22 = 20 \log_{10} \frac{V_o}{20 \text{ mV}}$$

$$1.1 = \log_{10} x$$

$$x = 12.589 = \frac{V_o}{20 \text{ mV}}$$

$$V_o = \mathbf{251.79 \text{ mV}}$$

$$15. \quad \text{dB}_s = 20 \log_{10} \frac{P}{0.0002 \mu\text{bar}}$$

$$\text{dB}_s = 20 \log_{10} \frac{0.001 \mu\text{bar}}{0.0002 \mu\text{bar}} = \mathbf{13.98}$$

$$\text{dB}_s = 20 \log_{10} \frac{0.016 \mu\text{bar}}{0.0002 \mu\text{bar}} = \mathbf{38.06}$$

$$\text{Increase} = \mathbf{24.08 \text{ dB}_s}$$

16. $60 \text{ dB}_s \Rightarrow 90 \text{ dB}_s$
quiet loud

$$60 \text{ dB}_s = 20 \log_{10} \frac{P_1}{0.002 \mu\text{bar}} = 20 \log_{10} x$$

$$3 = \log_{10} x$$

$$x = \mathbf{1000}$$

$$90 \text{ dB}_s = 20 \log_{10} \frac{P_2}{0.002 \mu\text{bar}} = 20 \log_{10} y$$

$$4.5 = \log_{10} y$$

$$y = \mathbf{31.623 \times 10^3}$$

$$\frac{x}{y} = \frac{\frac{P_1}{0.002 \mu\text{bar}}}{\frac{P_2}{0.002 \mu\text{bar}}} = \frac{P_1}{P_2} = \frac{10^3}{31.623 \times 10^3}$$

$$\text{and } P_2 = \mathbf{31.62 P_1}$$

18. a. $8 \text{ dB} = 20 \log_{10} \frac{V_2}{0.775 \text{ V}}$

$$0.4 = \log_{10} \frac{V_2}{0.775 \text{ V}}$$

$$\frac{V_2}{0.775 \text{ V}} = 2.512$$

$$V_2 = (2.512)(0.775 \text{ V}) = 1.947 \text{ V}$$

$$P = \frac{V^2}{R} = \frac{(1.947 \text{ V})^2}{600 \Omega} = \mathbf{6.32 \text{ mW}}$$

- b. $-5 \text{ dB} = 20 \log_{10} \frac{V_2}{0.775 \text{ V}}$

$$-0.25 = \log_{10} \frac{V_2}{0.775 \text{ V}}$$

$$\frac{V_2}{0.775 \text{ V}} = 0.562$$

$$V_2 = (0.562)(0.775 \text{ V}) = 0.436 \text{ V}$$

$$P = \frac{V^2}{R} = \frac{(0.436 \text{ V})^2}{600 \Omega} = \mathbf{0.32 \text{ mW}}$$

19. a. $A_v = \frac{V_o}{V_i} = \frac{X_C}{\sqrt{R^2 + X_C^2}} \angle -90^\circ + \tan^{-1} X_C/R = \frac{1}{\sqrt{\left(\frac{R}{X_C}\right)^2 + 1}} \angle -\tan^{-1} R/X_C$

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(2.2 \text{ k}\Omega)(0.02 \mu\text{F})} = \mathbf{3617.16 \text{ Hz}}$$

$$f = f_c: \quad A_v = \frac{V_o}{V_i} = \mathbf{0.707}$$

$$f = 0.1f_c: \quad \text{At } f_c, X_C = R = 2.2 \text{ k}\Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(0.1f_c)C} = \frac{1}{0.1} \left[\frac{1}{2\pi f_c C} \right] = 10[2.2 \text{ k}\Omega] = 22 \text{ k}\Omega$$

$$A_v = \frac{1}{\sqrt{\left(\frac{R}{X_C}\right)^2 + 1}} = \frac{1}{\sqrt{\left(\frac{2.2 \text{ k}\Omega}{22 \text{ k}\Omega}\right)^2 + 1}} = \frac{1}{\sqrt{(0.1)^2 + 1}} = \mathbf{0.995}$$

$$f = 0.5f_c = \frac{1}{2}f_c: \quad X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi\left(\frac{f_c}{2}\right)C} = 2 \left[\frac{1}{2\pi f_c C} \right] = 2[2.2 \text{ k}\Omega] = 4.4 \text{ k}\Omega$$

$$A_v = \frac{1}{\sqrt{\left(\frac{2.2 \text{ k}\Omega}{4.4 \text{ k}\Omega}\right)^2 + 1}} = \frac{1}{\sqrt{(0.5)^2 + 1}} = \mathbf{0.894}$$

$$f = 2f_c: \quad X_C = \frac{1}{2\pi(2f_c)C} = \frac{1}{2} \left[\frac{1}{2\pi f_c C} \right] = \frac{1}{2}[2.2 \text{ k}\Omega] = 1.1 \text{ k}\Omega$$

$$A_v = \frac{1}{\sqrt{\left(\frac{2.2 \text{ k}\Omega}{1.1 \text{ k}\Omega}\right)^2 + 1}} = \frac{1}{\sqrt{(2)^2 + 1}} = \mathbf{0.447}$$

$$f = 10f_c: \quad X_C = \frac{1}{2\pi(10f_c)C} = \frac{1}{10} \left[\frac{1}{2\pi f_c C} \right] = \frac{1}{10}[2.2 \text{ k}\Omega] = 0.22 \text{ k}\Omega$$

$$A_v = \frac{1}{\sqrt{\left(\frac{2.2 \text{ k}\Omega}{0.22 \text{ k}\Omega}\right)^2 + 1}} = \frac{1}{\sqrt{(10)^2 + 1}} = \mathbf{0.0995}$$

b. $\theta = -\tan^{-1} R/X_C$

$$f = f_c: \quad \theta = -\tan^{-1} 1 = \mathbf{-45^\circ}$$

$$f = 0.1f_c: \quad \theta = -\tan^{-1} 2.2 \text{ k}\Omega/22 \text{ k}\Omega = -\tan^{-1} \frac{1}{10} = \mathbf{-5.71^\circ}$$

$$f = 0.5f_c: \quad \theta = -\tan^{-1} 2.2 \text{ k}\Omega/4.4 \text{ k}\Omega = -\tan^{-1} \frac{1}{2} = \mathbf{-26.57^\circ}$$

$$f = 2f_c: \quad \theta = -\tan^{-1} 2.2 \text{ k}\Omega/1.1 \text{ k}\Omega = -\tan^{-1} 2 = \mathbf{-63.43^\circ}$$

$$f = 10f_c: \quad \theta = -\tan^{-1} 2.2 \text{ k}\Omega/0.22 \text{ k}\Omega = -\tan^{-1} 10 = \mathbf{-84.29^\circ}$$

20. a. $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(1 \text{ k}\Omega)(0.01 \mu\text{F})} = 15.915 \text{ kHz}$

$$f = 2f_c = 31.83 \text{ kHz}$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(31.83 \text{ kHz})(0.01 \mu\text{F})} = 500 \Omega$$

$$A_v = \frac{V_o}{V_i} = \frac{X_C}{\sqrt{R^2 + X_C^2}} = \frac{500 \Omega}{\sqrt{(1 \text{ k}\Omega)^2 + (0.5 \text{ k}\Omega)^2}} = 0.4472$$

$$V_o = 0.4472V_i = 0.4472(10 \text{ mV}) = \mathbf{4.47 \text{ mV}}$$

$$\begin{aligned}
 \text{b. } f &= \frac{1}{10} f_c = \frac{1}{10} (15,915 \text{ kHz}) = 1.5915 \text{ kHz} \\
 X_C &= \frac{1}{2\pi fC} = \frac{1}{2\pi (1.5915 \text{ kHz})(0.01 \mu\text{F})} = 10 \text{ k}\Omega \\
 A_v &= \frac{V_o}{V_i} = \frac{X_C}{\sqrt{R^2 + X_C^2}} = \frac{10 \text{ k}\Omega}{\sqrt{(1 \text{ k}\Omega)^2 + (10 \text{ k}\Omega)^2}} = 0.995 \\
 V_o &= 0.995 V_i = 0.995(10 \text{ mV}) = \mathbf{9.95 \text{ mV}}
 \end{aligned}$$

$$\begin{aligned}
 \text{c. } &\text{Yes, at } f = f_c, V_o = 7.07 \text{ mV} \\
 &\text{at } f = \frac{1}{10} f_c, V_o = 9.95 \text{ mV (much higher)} \\
 &\text{at } f = 2f_c, V_o = 4.47 \text{ mV (much lower)}
 \end{aligned}$$

$$\begin{aligned}
 21. \quad f_c &= 500 \text{ Hz} = \frac{1}{2\pi RC} = \frac{1}{2\pi (1.2 \text{ k}\Omega)C} \\
 C &= \frac{1}{2\pi Rf_c} = \frac{1}{2\pi (1.2 \text{ k}\Omega)(500 \text{ Hz})} = \mathbf{0.265 \mu\text{F}} \\
 A_v &= \frac{V_o}{V_i} = \frac{1}{\sqrt{\left(\frac{R}{X_C}\right)^2 + 1}}
 \end{aligned}$$

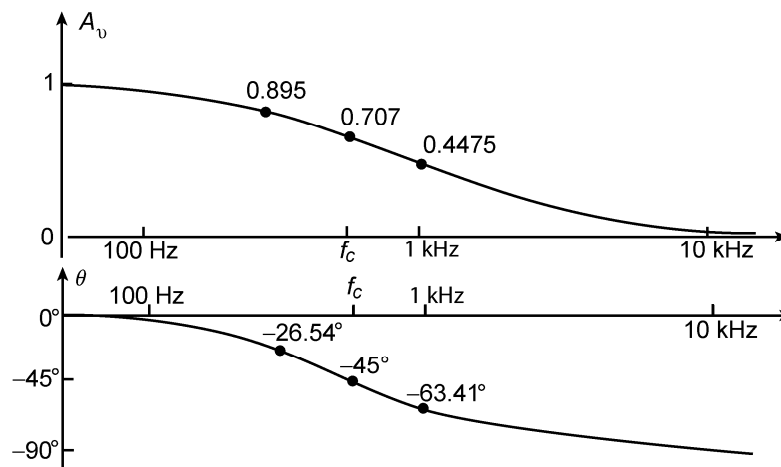
$$\text{At } f = 250 \text{ Hz, } X_C = 2402.33 \Omega \text{ and } A_v = 0.895$$

$$\text{At } f = 1000 \text{ Hz, } X_C = 600.58 \Omega \text{ and } A_v = 0.4475$$

$$\theta = -\tan^{-1} R/X_C$$

$$\text{At } f = 250 \text{ Hz} = \frac{1}{2} f_c, \theta = -26.54^\circ$$

$$\text{At } f = 1 \text{ kHz} = 2f_c, \theta = -63.41^\circ$$



22. a. $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(4.7 \text{ k}\Omega)(500 \text{ pF})} = \mathbf{67.73 \text{ kHz}}$

b. $f = 0.1 f_c = 0.1(67.726 \text{ kHz}) \cong 6.773 \text{ kHz}$
 $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(6.773 \text{ kHz})(500 \text{ pF})} = 46.997 \text{ k}\Omega$
 $A_v = \frac{V_o}{V_i} = \frac{X_C}{\sqrt{R^2 + X_C^2}} = \frac{46.997 \text{ k}\Omega}{\sqrt{(4.7 \text{ k}\Omega)^2 + (46.997 \text{ k}\Omega)^2}} = \mathbf{0.995} \cong 1$

c. $f = 10f_c = 677.26 \text{ kHz}$
 $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(677.26 \text{ kHz})(500 \text{ pF})} \cong 470 \text{ }\Omega$
 $A_v = \frac{V_o}{V_i} = \frac{X_C}{\sqrt{R^2 + X_C^2}} = \frac{470 \text{ }\Omega}{\sqrt{(4.7 \text{ k}\Omega)^2 + (470 \text{ }\Omega)^2}} = \mathbf{0.0995} \cong 0.1$

d. $A_v = \frac{V_o}{V_i} = 0.01 = \frac{X_C}{\sqrt{R^2 + X_C^2}}$
 $\sqrt{R^2 + X_C^2} = \frac{X_C}{0.01} = 100 X_C$
 $R^2 + X_C^2 = 10^4 X_C^2$
 $R^2 = 10^4 X_C^2 - X_C^2 = 9,999 X_C^2$
 $X_C = \frac{R}{\sqrt{9,999}} = \frac{4.7 \text{ k}\Omega}{99.995} \cong 47 \text{ }\Omega$
 $X_C = \frac{1}{2\pi fC} \Rightarrow f = \frac{1}{2\pi X_C C} = \frac{1}{2\pi(47 \text{ }\Omega)(500 \text{ pF})} = \mathbf{6.77 \text{ MHz}}$

23. a. $A_v = \frac{V_o}{V_i} = \frac{R}{\sqrt{R^2 + X_C^2}} \angle \tan^{-1} X_C/R = \frac{1}{\sqrt{1 + \left(\frac{X_C}{R}\right)^2}} \angle \tan^{-1} X_C/R$
 $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(2.2 \text{ k}\Omega)(0.02 \text{ }\mu\text{F})} = \mathbf{3.62 \text{ kHz}}$
 $f = f_c: \quad A_v = \frac{V_o}{V_i} = \mathbf{0.707}$
 $f = 2f_c: \quad \text{At } f_c, X_C = R = 2.2 \text{ k}\Omega$
 $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(2f_c)C} = \frac{1}{2} \left[\frac{1}{2\pi f_c C} \right] = \frac{1}{2} [2.2 \text{ k}\Omega] = 1.1 \text{ k}\Omega$
 $A_v = \frac{1}{\sqrt{1 + \left(\frac{1.1 \text{ k}\Omega}{2.2 \text{ k}\Omega}\right)^2}} = \mathbf{0.894}$

$$f = \frac{1}{2}f_c: \quad X_C = \frac{1}{2\pi\left(\frac{f_c}{2}\right)C} = 2\left[\frac{1}{2\pi f_c C}\right] = 2[2.2 \text{ k}\Omega] = 4.4 \text{ k}\Omega$$

$$A_v = \frac{1}{\sqrt{1 + \left(\frac{4.4 \text{ k}\Omega}{2.2 \text{ k}\Omega}\right)^2}} = \mathbf{0.447}$$

$$f = 10f_c: \quad X_C = \frac{1}{2\pi(10f_c)C} = \frac{1}{10}\left[\frac{1}{2\pi f_c C}\right] = \frac{2.2 \text{ k}\Omega}{10} = 0.22 \text{ k}\Omega$$

$$A_v = \frac{1}{\sqrt{1 + \left(\frac{0.22 \text{ k}\Omega}{2.2 \text{ k}\Omega}\right)^2}} = \mathbf{0.995}$$

$$f = \frac{1}{10}f_c: \quad X_C = \frac{1}{2\pi\left(\frac{f_c}{10}\right)C} = 10\left[\frac{1}{2\pi f_c C}\right] = 10[2.2 \text{ k}\Omega] = 22 \text{ k}\Omega$$

$$A_v = \frac{1}{\sqrt{1 + \left(\frac{22 \text{ k}\Omega}{2.2 \text{ k}\Omega}\right)^2}} = \mathbf{0.0995}$$

b. $f = f_c, \quad \theta = \mathbf{45^\circ}$

$$f = 2f_c, \quad \theta = \tan^{-1}(X_C/R) = \tan^{-1} 1.1 \text{ k}\Omega/2.2 \text{ k}\Omega = \tan^{-1} \frac{1}{2} = \mathbf{26.57^\circ}$$

$$f = \frac{1}{2}f_c, \quad \theta = \tan^{-1} \frac{4.4 \text{ k}\Omega}{2.2 \text{ k}\Omega} = \tan^{-1} 2 = \mathbf{63.43^\circ}$$

$$f = 10f_c, \quad \theta = \tan^{-1} \frac{0.22 \text{ k}\Omega}{2.2 \text{ k}\Omega} = \mathbf{5.71^\circ}$$

$$f = \frac{1}{10}f_c, \quad \theta = \tan^{-1} \frac{22 \text{ k}\Omega}{2.2 \text{ k}\Omega} = \mathbf{84.29^\circ}$$

24. a. $f = f_c: A_v = \frac{V_o}{V_i} = \mathbf{0.707}$

b. $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(10 \text{ k}\Omega)(1000 \text{ pF})} = 15.915 \text{ kHz}$

$$f = 4f_c = 4(15.915 \text{ kHz}) = 63.66 \text{ kHz}$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(63.66 \text{ kHz})(1000 \text{ pF})} = 2.5 \text{ k}\Omega$$

$$A_v = \frac{V_o}{V_i} = \frac{R}{\sqrt{R^2 + X_C^2}} = \frac{10 \text{ k}\Omega}{\sqrt{(10 \text{ k}\Omega)^2 + (2.5 \text{ k}\Omega)^2}} = \mathbf{0.970} \text{ (significant rise)}$$

c. $f = 100f_c = 100(15.915 \text{ kHz}) = 1591.5 \text{ kHz} \cong 1.592 \text{ MHz}$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(1.592 \text{ MHz})(1000 \text{ pF})} = 99.972 \Omega$$

$$A_v = \frac{R}{\sqrt{R^2 + X_C^2}} = \frac{10 \text{ k}\Omega}{\sqrt{(10 \text{ k}\Omega)^2 + (99.972 \Omega)^2}} = 0.99995 \cong 1$$

d. At $f = f_c$, $V_o = 0.707V_i = 0.707(10 \text{ mV}) = 7.07 \text{ mV}$

$$P_o = \frac{V_o^2}{R} = \frac{(7.07 \text{ mV})^2}{10 \text{ k}\Omega} \cong 5 \text{ nW}$$

25. $A_v = \frac{V_o}{V_i} = \frac{1}{\sqrt{1 + \left(\frac{X_C}{R}\right)^2}} \angle \tan^{-1} X_C/R$

$$f_c = \frac{1}{2\pi RC} \Rightarrow R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi(2 \text{ kHz})(0.1 \mu\text{F})} = 795.77 \Omega$$

$$R = 795.77 \Omega \Rightarrow \underbrace{750 \Omega + 47 \Omega}_{\text{nominal values}} = 797 \Omega$$

$$\therefore f_c = \frac{1}{2\pi(797 \Omega)(0.1 \mu\text{F})} = \mathbf{1996.93 \text{ Hz}} \text{ using nominal values}$$

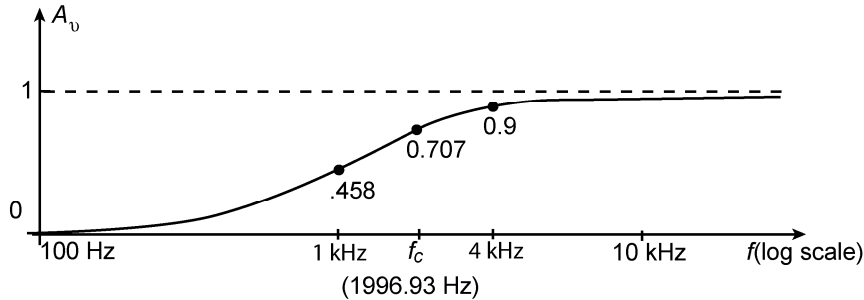
At $f = 1 \text{ kHz}$, $A_v = 0.458$

$f = 4 \text{ kHz}$, $A_v \cong 0.9$

$$\theta = \tan^{-1} \frac{X_C}{R}$$

$f = 1 \text{ kHz}$, $\theta = 63.4^\circ$

$f = 4 \text{ kHz}$, $\theta = 26.53^\circ$



26. a. $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(100 \text{ k}\Omega)(20 \text{ pF})} = \mathbf{79.58 \text{ kHz}}$

b. $f = 0.01f_c = 0.01(79.577 \text{ kHz}) = 0.7958 \text{ kHz} \cong 796 \text{ Hz}$
 $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(796 \text{ Hz})(20 \text{ pF})} = 9.997 \text{ M}\Omega$
 $A_v = \frac{V_o}{V_i} = \frac{R}{\sqrt{R^2 + X_C^2}} = \frac{100 \text{ k}\Omega}{\sqrt{(100 \text{ k}\Omega)^2 + (9.997 \text{ M}\Omega)^2}} = \mathbf{0.01} \cong 0$

c. $f = 100f_c = 100(79.577 \text{ kHz}) \cong 7.96 \text{ MHz}$
 $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(7.96 \text{ MHz})(20 \text{ pF})} = 999.72 \text{ }\Omega$
 $A_v = \frac{V_o}{V_i} = \frac{R}{\sqrt{R^2 + X_C^2}} = \frac{100 \text{ k}\Omega}{\sqrt{(100 \text{ k}\Omega)^2 + (999.72 \text{ }\Omega)^2}} = \mathbf{0.99995} \cong 1$

d. $A_v = \frac{V_o}{V_i} = 0.5 = \frac{R}{\sqrt{R^2 + X_C^2}}$
 $\sqrt{R^2 + X_C^2} = 2R$
 $R^2 + X_C^2 = 4R^2$
 $X_C^2 = 4R^2 - R^2 = 3R^2$
 $X_C = \sqrt{3R^2} = \sqrt{3}R = \sqrt{3}(100 \text{ k}\Omega) = 173.2 \text{ k}\Omega$
 $X_C = \frac{1}{2\pi fC} \Rightarrow f = \frac{1}{2\pi X_C C} = \frac{1}{2\pi(173.2 \text{ k}\Omega)(20 \text{ pF})}$
 $f = \mathbf{45.95 \text{ kHz}}$

27. a. low-pass section: $f_{c_1} = \frac{1}{2\pi RC} = \frac{1}{2\pi(0.1 \text{ k}\Omega)(2 \text{ }\mu\text{F})} = \mathbf{795.77 \text{ Hz}}$
high-pass section: $f_{c_2} = \frac{1}{2\pi RC} = \frac{1}{2\pi(10 \text{ k}\Omega)(8200 \text{ pF})} = \mathbf{1.94 \text{ Hz}}$

For the analysis to follow, it is assumed $(R_2 + jX_{C_2}) \parallel R_1 \cong R_1$ for all frequencies of interest.

At $f_{c_1} = 795.77 \text{ Hz}$:

$$V_{R_1} = 0.707 V_i$$

$$X_{C_2} = \frac{1}{2\pi fC_2} = 24.39 \text{ k}\Omega$$

$$|V_o| = \frac{24.39 \text{ k}\Omega(V_{R_1})}{\sqrt{(10 \text{ k}\Omega)^2 + (24.39 \text{ k}\Omega)^2}} = 0.925 V_{R_1}$$

$$V_o = (0.925)(0.707 V_i) = \mathbf{0.654 V_i}$$

At $f_{c_2} = 1.94 \text{ kHz}$:

$$V_o = 0.707 V_{R_1}$$

$$X_{C_1} = \frac{1}{2\pi f C_1} = 41 \Omega$$

$$|V_{R_1}| = \frac{R_1 V_i}{\sqrt{R_1^2 + X_{C_1}^2}} = \frac{100 \Omega (V_i)}{\sqrt{(100 \Omega)^2 + (41 \Omega)^2}} = 0.925 V_i$$

$$|V_o| = (0.707)(0.925 V_i) = \mathbf{0.64 V_i}$$

$$\text{At } f = 795.77 \text{ Hz} + \frac{(1.94 \text{ kHz} - 795.77 \text{ Hz})}{2} = 1.37 \text{ kHz}$$

$$X_{C_1} = 58.1 \Omega, X_{C_2} = 14.17 \text{ k}\Omega$$

$$|V_{R_1}| = \frac{100 \Omega (V_i)}{\sqrt{(100 \Omega)^2 + (58.1 \Omega)^2}} = 0.864 V_i$$

$$|V_o| = \frac{14.17 \text{ k}\Omega (V_{R_1})}{\sqrt{(10 \text{ k}\Omega)^2 + (14.17 \text{ k}\Omega)^2}} = 0.817 V_{R_1}$$

$$V_o = 0.817(0.864 V_i) = 0.706 V_i$$

$$\text{and } A_v = \frac{V_o}{V_i} = \mathbf{0.706} (\cong \text{maximum value})$$

After plotting the points it was determined that the gain should also be determined at $f = 500 \text{ Hz}$ and 4 kHz :

$$f = 500 \text{ Hz:} \quad X_{C_1} = 159.15 \Omega, X_{C_2} = 38.82 \text{ k}\Omega,$$

$$V_{R_1} = 0.532 V_i, V_o = 0.968 V_{R_1}$$

$$V_o = \mathbf{0.515 V_i}$$

$$f = 4 \text{ kHz:} \quad X_{C_1} = 19.89 \Omega, X_{C_2} = 4.85 \text{ k}\Omega,$$

$$V_{R_1} = 0.981 V_i, V_o = 0.437 V_{R_1}$$

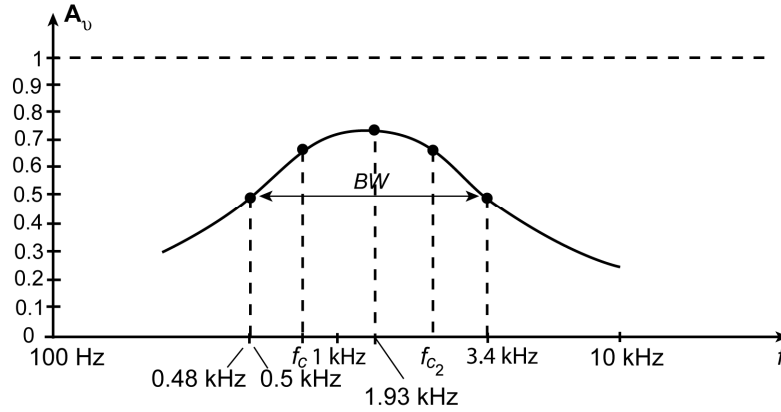
$$V_o = \mathbf{0.429 V_i}$$

b. Using $0.707(.706) \cong 0.5$ to define the bandwidth

$$BW \cong 3.4 \text{ kHz} - 0.48 \text{ kHz} = 2.92 \text{ kHz}$$

$$\text{and } BW \cong \mathbf{2.9 \text{ kHz}}$$

$$\text{with } f_{\text{center}} = 480 \text{ Hz} + \left(\frac{2.9 \text{ kHz}}{2} \right) = \mathbf{1930 \text{ Hz}}$$



28. $f_1 = \frac{1}{2\pi R_1 C_1} = 4 \text{ kHz}$

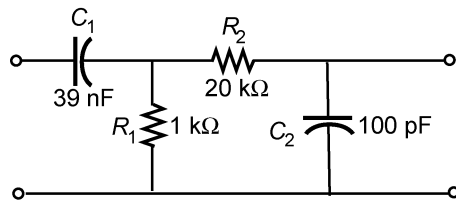
Choose $R_1 = 1 \text{ k}\Omega$

$$C_1 = \frac{1}{2\pi f_1 R_1} = \frac{1}{2\pi(4 \text{ kHz})(1 \text{ k}\Omega)} = 39.8 \text{ nF} \therefore \text{Use } \mathbf{39 \text{ nF}}$$

$$f_2 = \frac{1}{2\pi R_2 C_2} = 80 \text{ kHz}$$

Choose $R_2 = 20 \text{ k}\Omega$

$$C_2 = \frac{1}{2\pi f_2 R_2} = \frac{1}{2\pi(80 \text{ kHz})(20 \text{ k}\Omega)} = 99.47 \text{ pF} \therefore \text{Use } 100 \text{ pF}$$



$$\text{Center frequency} = 4 \text{ kHz} + \frac{80 \text{ kHz} - 4 \text{ kHz}}{2} = 42 \text{ kHz}$$

$$\text{At } f = 42 \text{ kHz}, X_{C_1} = 97.16 \Omega, X_{C_2} = 37.89 \text{ k}\Omega$$

Assuming $Z_2 \gg Z_1$

$$|V_{R_1}| = \frac{R_1(V_i)}{\sqrt{R_1^2 + X_{C_1}^2}} = 0.995V_i$$

$$|V_o| = \frac{X_{C_2}(V_{R_1})}{\sqrt{R_2^2 + X_{C_2}^2}} = 0.884V_i$$

$$V_o = 0.884V_{R_1} = 0.884(0.995V_i) = \mathbf{0.88 V_i}$$

$$\text{as } f = f_1: V_{R_1} = 0.707V_i, X_{C_2} = 221.05 \text{ k}\Omega$$

$$\text{and } V_o = 0.996V_{R_1}$$

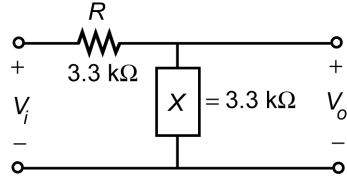
$$\text{so that } V_o = 0.996V_{R_1} = 0.996(0.707V_i) = 0.704V_i$$

Although $A_v = 0.88$ is less than the desired level of 1, f_1 and f_2 do define a band of frequencies for which $A_v \geq 0.7$ and the power to the load is significant.

29. a. $f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(4.7 \text{ mH})(560 \text{ pF})}} = \mathbf{98.1 \text{ kHz}}$
- b. $Q_s = \frac{X_L}{R + R_\ell} = \frac{2\pi(98.1 \text{ kHz})(4.7 \text{ mH})}{160 \Omega + 12 \Omega} = \mathbf{16.84}$
 $BW = \frac{f_s}{Q_s} = \frac{98.1 \text{ kHz}}{16.84} = \mathbf{5.83 \text{ kHz}}$
- c. At $f = f_s$: $V_{o_{\max}} = \frac{R}{R + R_\ell} V_i = \frac{160 \Omega (1 \text{ V})}{172 \Omega} = 0.93 \text{ V}$ and $A_v = \frac{V_o}{V_i} = \mathbf{0.93}$
 Since $Q_s \geq 10$, $f_1 = f_s - \frac{BW}{2} = 98.1 \text{ kHz} - \frac{5.83 \text{ kHz}}{2} = 95.19 \text{ kHz}$
 $f_2 = f_s + \frac{BW}{2} = 101.02 \text{ kHz}$
 At $f = 95.19 \text{ kHz}$: $X_L = 2\pi fL = 2\pi(95.19 \text{ kHz})(4.7 \text{ mH}) = 2.81 \text{ k}\Omega$
 $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(95.19 \text{ kHz})(560 \text{ pF})} = 2.99 \text{ k}\Omega$
 $\mathbf{V_o} = \frac{160 \Omega (1 \text{ V} \angle 0^\circ)}{172 + j2.81 \text{ k}\Omega - j2.99 \text{ k}\Omega} = \frac{160 \text{ V} \angle 0^\circ}{172 - j180}$
 $= \frac{160 \text{ V} \angle 0^\circ}{248.97 \angle -46.30^\circ} = \mathbf{0.643 \text{ V} \angle 46.30^\circ}$
 At $f = 101.02 \text{ kHz}$: $X_L = 2\pi fL = 2\pi(101.02 \text{ kHz})(4.7 \text{ mH}) = 2.98 \text{ k}\Omega$
 $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(101.02 \text{ kHz})(560 \text{ pF})} = 2.81 \text{ k}\Omega$
 $\mathbf{V_o} = \frac{160 (1 \text{ V} \angle 0^\circ)}{172 + j2.98 \text{ k}\Omega - j2.81 \text{ k}\Omega} = \frac{160 \text{ V} \angle 0^\circ}{172 + j170}$
 $= \frac{160 \text{ V} \angle 0^\circ}{241.83 \angle 44.66^\circ} = \mathbf{0.66 \text{ V} \angle -44.66^\circ}$
- d. $f = f_s$: $V_{o_{\max}} = \mathbf{0.93 \text{ V}}$
 $f = f_1 = 95.19 \text{ kHz}$, $V_o = 0.707(0.93 \text{ V}) = \mathbf{0.66 \text{ V}}$
 $f = f_2 = 101.02 \text{ kHz}$, $V_o = 0.707(0.93 \text{ V}) = \mathbf{0.66 \text{ V}}$
30. a. $f_p = \frac{1}{2\pi\sqrt{LC}} \sqrt{1 - \frac{R_\ell^2 C}{L}} \cong \mathbf{159.15 \text{ kHz}}$
 $Q_\ell = \frac{X_L}{R_\ell} = \frac{2\pi f_p L}{R_\ell} = \frac{2\pi(159.15 \text{ kHz})(1 \text{ mH})}{16 \Omega} = 62.5 \gg 10$
 $Z_{T_p} = Q_\ell^2 R_\ell = (62.5)^2 16 \Omega = 62.5 \text{ k}\Omega \gg 4 \text{ k}\Omega$
 and $V_o \cong V_i$ at resonance.

However, $R = 3.3 \text{ k}\Omega$ affects the shape of the resonance curve and $BW = f_p / Q_\ell$ cannot be applied.

For $A_v = \frac{V_o}{V_i} = 0.707$, $|X| = R$ for the following configuration



For frequencies near f_p , $X_L \gg R_\ell$ and $Z_L = R_\ell + jX_L \cong X_L$ and $X = X_L \parallel X_C$.

For frequencies near f_p but less than f_p

$$X = \frac{X_C X_L}{X_C - X_L}$$

and for $A_v = 0.707$

$$\frac{X_C X_L}{X_C - X_L} = R$$

Substituting $X_C = \frac{1}{2\pi f_1 C}$ and $X_L = 2\pi f_1 L$

the following equation can be derived:

$$f_1^2 + \frac{1}{2\pi RC} f_1 - \frac{1}{4\pi^2 LC} = 0$$

For this situation:

$$\frac{1}{2\pi RC} = \frac{1}{2\pi(3.3 \text{ k}\Omega)(0.001 \mu\text{F})} = 48.23 \times 10^3$$

$$\frac{1}{4\pi^2 LC} = \frac{1}{4\pi^2(1 \text{ mH})(0.001 \mu\text{F})} = 2.53 \times 10^{10}$$

and solving the quadratic equation, $f_1 = 135.83 \text{ kHz}$

and $\frac{BW}{2} = f_p - f_1 = 159.15 \text{ kHz} - 135.83 \text{ kHz} = 22.32 \text{ kHz}$

so that $f_2 = f_p + \frac{BW}{2} = 159.15 \text{ kHz} + 18.75 \text{ kHz} = 177.9 \text{ kHz}$

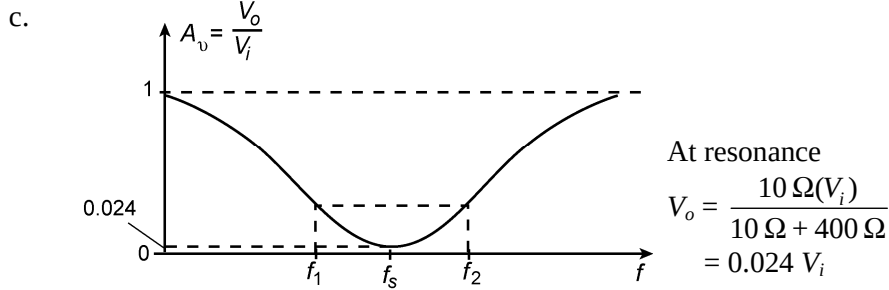
b. $Q_p = \frac{f_p}{BW} = \frac{159.15 \text{ kHz}}{44.64 \text{ kHz}} = \mathbf{3.57}$
 $BW = 2(18.75 \text{ kHz}) = \mathbf{37.5 \text{ kHz}}$

$$31. \quad a. \quad Q_s = \frac{X_L}{R + R_\ell} = \frac{5000 \, \Omega}{390 \, \Omega + 10 \, \Omega} = \frac{5000 \, \Omega}{400 \, \Omega} = 12.5$$

$$b. \quad BW = \frac{f_s}{Q_s} = \frac{5000 \, \text{Hz}}{12.5} = \mathbf{400 \, \text{Hz}}$$

$$f_1 = 5000 \, \text{Hz} - \frac{400 \, \text{Hz}}{2} = \mathbf{4.8 \, \text{kHz}}$$

$$f_2 = 5000 \, \text{Hz} + \frac{400 \, \text{Hz}}{2} = \mathbf{5.20 \, \text{kHz}}$$



d. At resonance, $10 \, \Omega \parallel 2 \, \text{k}\Omega = 9.95 \, \Omega$

$$V_o = \frac{9.95 \, \Omega (V_i)}{9.95 \, \Omega + 400 \, \Omega} \cong 0.024 V_i \text{ as above!}$$

$$32. \quad a. \quad Q_\ell = \frac{X_L}{R_\ell} = \frac{400 \, \Omega}{10 \, \Omega} = 40$$

$$Z_{Tp} = Q_\ell^2 R_\ell = (40)^2 20 \, \Omega = 32 \, \text{k}\Omega \gg 1 \, \text{k}\Omega$$

$$\text{At resonance, } V_o = \frac{32 \, \text{k}\Omega V_i}{32 \, \text{k}\Omega + 1 \, \text{k}\Omega} = 0.97 V_i$$

$$\text{and } A_v = \frac{V_o}{V_i} = 0.97$$

For the low cutoff frequency note solution to Problem 30:

$$f_1^2 + \frac{1}{2\pi RC} f_1 - \frac{1}{4\pi^2 LC} = 0$$

$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi (20 \, \text{kHz})(400 \, \Omega)} = 19.9 \, \text{nF}$$

$$L = \frac{X_L}{2\pi f} = \frac{400 \, \Omega}{2\pi (20 \, \text{kHz})} = 3.18 \, \text{mH}$$

Substituting into the above equation and solving

$$f_1 = 16.4 \, \text{kHz}$$

$$\text{with } \frac{BW}{2} = 20 \, \text{kHz} - 16.4 \, \text{kHz} = 3.6 \, \text{kHz}$$

$$\text{and } BW = 2(3.6 \, \text{kHz}) = \mathbf{7.2 \, \text{kHz}}$$

$$Q_p = \frac{f_p}{BW} = \frac{20 \, \text{kHz}}{7.2 \, \text{kHz}} = \mathbf{2.78}$$

b. –

c. At resonance

$$Z_{T_p} = 32 \text{ k}\Omega \parallel 100 \text{ k}\Omega = 24.24 \text{ k}\Omega$$

$$\text{with } V_o = \frac{24.24 \text{ k}\Omega V_i}{24.24 \text{ k}\Omega + 1 \text{ k}\Omega} = 0.96 V_i$$

$$\text{and } A_v = \frac{V_o}{V_i} = 0.96 \text{ vs } 0.97 \text{ above}$$

At frequencies to the right and left of f_p , the impedance Z_{T_p} will decrease and be affected less and less by the parallel $100 \text{ k}\Omega$ load. The characteristics, therefore, are only slightly affected by the $100 \text{ k}\Omega$ load.

d. At resonance

$$Z_{T_p} = 32 \text{ k}\Omega \parallel 20 \text{ k}\Omega = 12.31 \text{ k}\Omega$$

$$\text{with } V_o = \frac{12.31 \text{ k}\Omega V_i}{12.31 \text{ k}\Omega + 1 \text{ k}\Omega} = 0.925 V_i \text{ vs } 0.97 V_i \text{ above}$$

At frequencies to the right and left of f_p , the impedance of each frequency will actually be less due to the parallel $20 \text{ k}\Omega$ load. The effect will be to narrow the resonance curve and decrease the bandwidth with an increase in Q_p .

$$33. \quad a. \quad f_p = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(400 \mu\text{H})(120 \text{ pF})}} = \mathbf{726.44 \text{ kHz}} \text{ (band-stop)}$$

$$X_{L_s} \angle 90^\circ + (X_{L_p} \angle 90^\circ \parallel X_C \angle -90^\circ) = 0$$

$$jX_{L_s} + \frac{(X_{L_p} \angle 90^\circ)(X_C \angle -90^\circ)}{jX_{L_p} - jX_C} = 0$$

$$jX_{L_s} + \frac{X_{L_p} X_C}{j(X_{L_p} - X_C)} = 0$$

$$jX_{L_s} - j \frac{X_{L_p} X_C}{(X_{L_p} - X_C)} = 0$$

$$X_{L_s} - \frac{X_{L_p} X_C}{X_{L_p} - X_C} = 0$$

$$X_{L_s} X_C - X_{L_s} X_{L_p} + X_{L_p} X_C = 0$$

$$\frac{\omega L_s}{\omega C} - \omega L_s + \frac{\omega L_p}{\omega C} = 0$$

$$L_s L_p \omega^2 - \frac{1}{C} [L_s + L_p] = 0$$

$$\omega = \sqrt{\frac{L_s + L_p}{C L_s L_p}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{L_s + L_p}{C L_s L_p}} = \frac{1}{2\pi} \sqrt{\frac{460 \times 10^{-6}}{28.8 \times 10^{-19}}} = \mathbf{2.01 \text{ MHz}} \text{ (pass-band)}$$

34. a. $f_s = \frac{1}{2\pi\sqrt{LC}} \Rightarrow L_s = \frac{1}{4\pi^2 f_s^2 C} = \frac{1}{4\pi^2 (100 \text{ kHz})^2 (200 \text{ pF})} = 12.68 \text{ mH}$

$$X_L = 2\pi f L = 2\pi (30 \text{ kHz}) (12.68 \text{ mH}) = 2388.91 \Omega$$

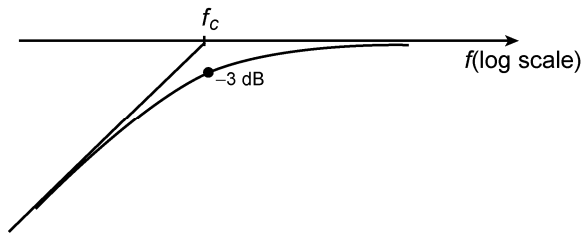
$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi (30 \text{ kHz}) (200 \text{ pF})} = 26.54 \text{ k}\Omega$$

$$X_C - X_L = 26.54 \text{ k}\Omega - 2388.91 \Omega = 24.15 \text{ k}\Omega(C)$$

$$X_{L_p} = X_{C(\text{net})} = 24.15 \text{ k}\Omega$$

$$L_p = \frac{X_L}{2\pi f} = \frac{24.15 \text{ k}\Omega}{2\pi (30 \text{ kHz})} = \mathbf{128.19 \text{ mH}}$$

35. a, b. $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi (0.47 \text{ k}\Omega) (0.047 \mu\text{F})} = \mathbf{7.2 \text{ kHz}}$



c. $f = \frac{1}{2} f_c: A_{v\text{dB}} = 20 \log_{10} \frac{1}{\sqrt{1 + (f_c/f)^2}} = 20 \log_{10} \frac{1}{\sqrt{1 + (2)^2}} = \mathbf{-7 \text{ dB}}$

$$f = 2f_c: A_{v\text{dB}} = 20 \log_{10} \frac{1}{\sqrt{1 + (0.5)^2}} = \mathbf{-0.969 \text{ dB}}$$

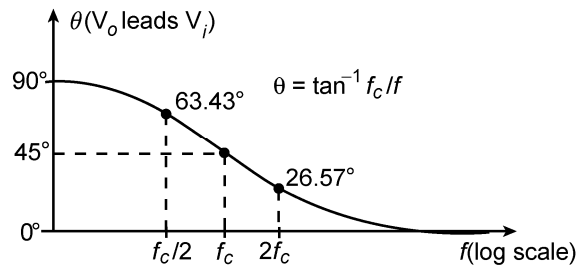
$$f = \frac{1}{10} f_c: A_{v\text{dB}} = 20 \log_{10} \frac{1}{\sqrt{1 + (10)^2}} = \mathbf{-20.04 \text{ dB}}$$

$$f = 10f_c: A_{v\text{dB}} = 20 \log_{10} \frac{1}{\sqrt{1 + (0.1)^2}} = \mathbf{-0.043 \text{ dB}}$$

d. $f = \frac{1}{2} f_c: A_v = \frac{1}{\sqrt{1 + (f_c/f)^2}} = \frac{1}{\sqrt{1 + (2)^2}} = \mathbf{0.447}$

$$f = 2f_c: A_v = \frac{1}{\sqrt{1 + (0.5)^2}} = \mathbf{0.894}$$

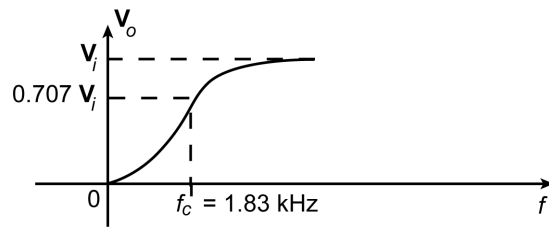
e.



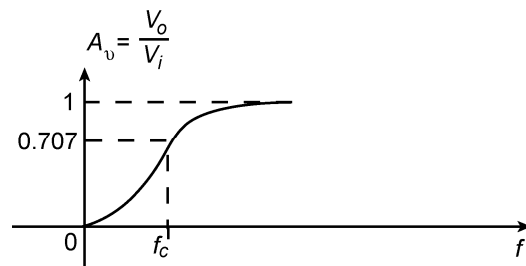
36. a. $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(6.8 \text{ k}\Omega \parallel 12 \text{ k}\Omega)0.02 \mu\text{F}} = \frac{1}{2\pi(4.34 \text{ k}\Omega)(0.02 \mu\text{F})} = \mathbf{1.83 \text{ kHz}}$

$$\frac{V_o}{V_i} = \frac{1}{\sqrt{1 + (f_c/f)^2}}$$

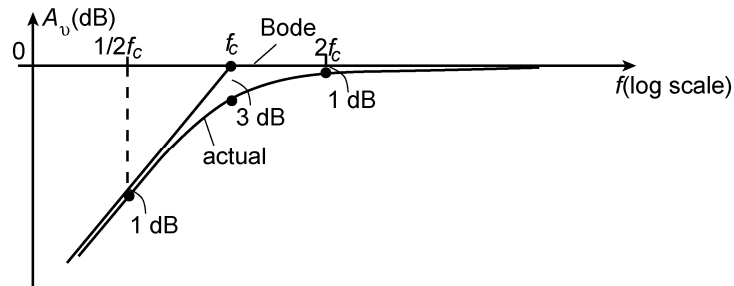
and $V_o = \left(\frac{1}{\sqrt{1 + (f_c/f)^2}} \right) V_i$



b.



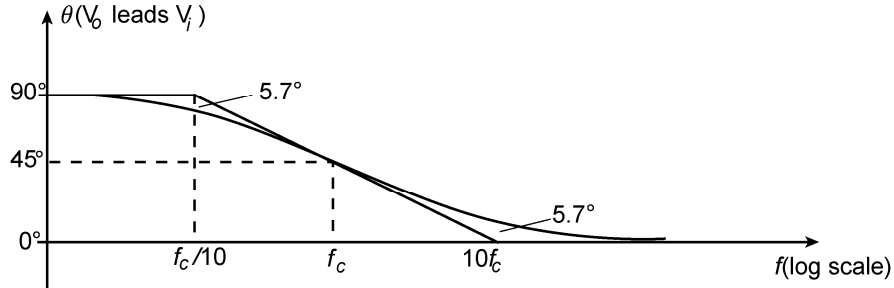
c. & d.



- e. Remember the log scale! $1.5f_c$ is not midway between f_c and $2f_c$

$$\begin{aligned} A_{v_{dB}} &= 20 \log_{10} A_v \\ -1.5 &= 20 \log_{10} A_v \\ -0.075 &= \log_{10} A_v \\ A_v &= \frac{V_o}{V_i} = \mathbf{0.84} \end{aligned}$$

f. $\theta = \tan^{-1} f_c/f$



37. a, b. $A_v = \frac{V_o}{V_i} = A_v \angle \theta = \frac{1}{\sqrt{1 + (f/f_c)^2}} \angle -\tan^{-1} f/f_c$

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(12 \text{ k}\Omega)(1 \text{ nF})} = \mathbf{13.26 \text{ kHz}}$$

c. $f = f_c/2 = 6.63 \text{ kHz}$

$$A_{v_{dB}} = 20 \log_{10} \frac{1}{\sqrt{1 + (0.5)^2}} = \mathbf{-0.97 \text{ dB}}$$

$f = 2f_c = 26.52 \text{ kHz}$

$$A_{v_{dB}} = 20 \log_{10} \frac{1}{\sqrt{1 + (2)^2}} = \mathbf{-6.99 \text{ dB}}$$

$f = f_c/10 = 1.326 \text{ kHz}$

$$A_{v_{dB}} = 20 \log_{10} \frac{1}{\sqrt{1 + (0.1)^2}} = \mathbf{-0.04 \text{ dB}}$$

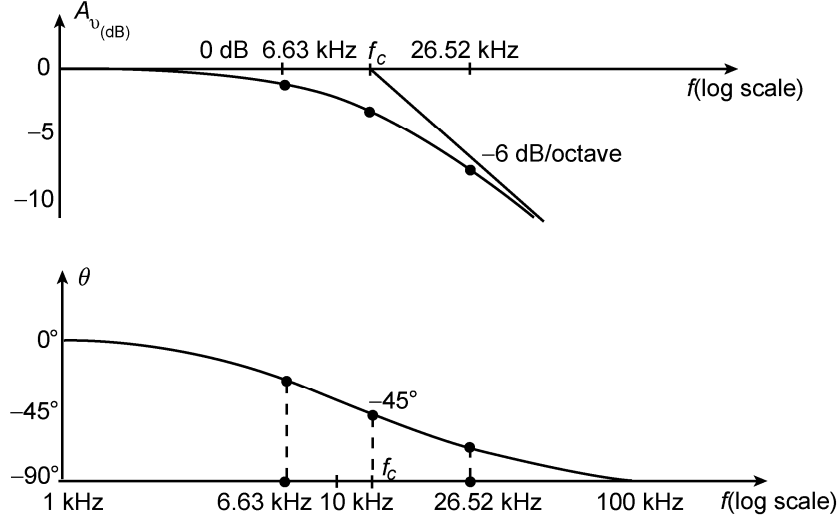
$f = 10f_c = 132.6 \text{ kHz}$

$$A_{v_{dB}} = 20 \log_{10} \frac{1}{\sqrt{1 + (10)^2}} = \mathbf{-20.04 \text{ dB}}$$

d. $f = f_c/2$: $A_v = \frac{1}{\sqrt{1 + (0.5)^2}} = \mathbf{0.894}$

$f = 2f_c$: $A_v = \frac{1}{\sqrt{1 + (2)^2}} = \mathbf{0.447}$

e. $\theta = \tan^{-1} f/f_c$
 $f = f_c/2: \quad \theta = -\tan^{-1} 0.5 = -26.57^\circ$
 $f = f_c: \quad \theta = -\tan^{-1} 1 = -45^\circ$
 $f = 2f_c: \quad \theta = -\tan^{-1} 2 = -63.43^\circ$



38. a. $R_2 \parallel X_C = \frac{(R_2)(-jX_C)}{R_2 - jX_C} = -j \frac{R_2 X_C}{R_2 - jX_C}$

$$\mathbf{V}_o = \frac{\left(\frac{-jR_2 X_C}{R_2 - jX_C} \right) \mathbf{V}_i}{R_1 - j \frac{R_2 X_C}{R_2 - jX_C}} = -j \frac{R_2 X_C \mathbf{V}_i}{R_1(R_2 - jX_C) - jR_2 X_C}$$

$$= \frac{-jR_2 X_C \mathbf{V}_i}{R_1 R_2 - jR_1 X_C - jR_2 X_C} = \frac{-jR_2 X_C \mathbf{V}_i}{R_1 R_2 - j(R_1 + R_2) X_C}$$

$$= \frac{R_2 X_C \mathbf{V}_i}{jR_1 R_2 + (R_1 + R_2) X_C} = \frac{R_2 \mathbf{V}_i}{j \frac{R_1 R_2}{X_C} + (R_1 + R_2)}$$

$$= \frac{R_2 \mathbf{V}_i}{R_1 + R_2 + j \frac{R_1 R_2}{X_C}} = \frac{\left(\frac{R_2}{R_1 + R_2} \right) \mathbf{V}_i}{1 + j \left(\frac{R_1 R_2}{R_1 + R_2} \right) \frac{1}{X_C}}$$

and $\mathbf{A}_v = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{\frac{R_2}{R_1 + R_2}}{1 + j \omega \left(\frac{R_1 R_2}{R_1 + R_2} \right) C}$

or $\mathbf{A}_v = \frac{R_2}{R_1 + R_2} \left[\frac{1}{1 + j 2\pi f (R_1 \parallel R_2) C} \right]$

$$\text{defining } f_c = \frac{1}{2\pi(R_1 \parallel R_2)C}$$

$$\mathbf{A}_v = \frac{R_2}{R_1 + R_2} \left[\frac{1}{1 + j f/f_c} \right]$$

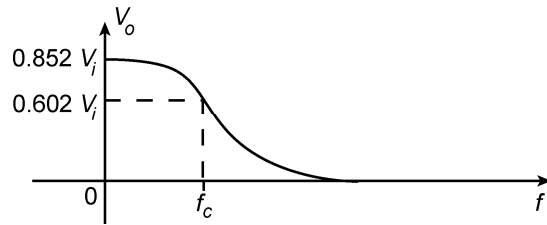
$$\text{and } \mathbf{A}_v = \frac{R_2}{R_1 + R_2} \left[\frac{1}{\sqrt{1 + (f/f_c)^2}} \angle -\tan^{-1} f/f_c \right]$$

$$\text{with } |V_o| = \frac{R_2}{R_1 + R_2} \left[\frac{1}{\sqrt{1 + (f/f_c)^2}} \right] |V_i|$$

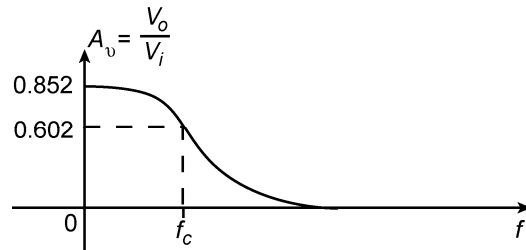
$$\text{for } f \ll f_c, V_o = \frac{R_2}{R_1 + R_2} V_i = \frac{27 \text{ k}\Omega}{4.7 \text{ k}\Omega + 27 \text{ k}\Omega} V_i = 0.852 V_i$$

$$\text{at } f = f_c: V_o = 0.852[0.707] V_i = 0.602 V_i$$

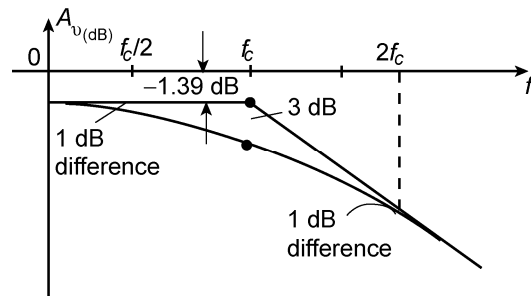
$$f_c = \frac{1}{2\pi(R_1 \parallel R_2)C} = \frac{1}{2\pi(4.7 \text{ k}\Omega \parallel 27 \text{ k}\Omega)0.039 \mu\text{F}} = \mathbf{1.02 \text{ kHz}}$$



b.



c. & d.



$$\begin{aligned} -20 \log_{10} \frac{R_1 + R_2}{R_2} &= -20 \log_{10} \frac{4.7 \text{ k}\Omega + 27 \text{ k}\Omega}{27 \text{ k}\Omega} \\ &= -20 \log_{10} 1.174 = -1.39 \text{ dB} \end{aligned}$$

e. $A_{v_{dB}} \cong -1.39 \text{ dB} - 0.5 \text{ dB} = -1.89 \text{ dB}$

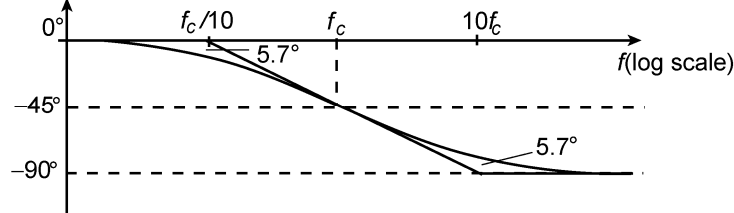
$$A_{v_{dB}} = 20 \log_{10} A_v$$

$$-1.89 = 20 \log_{10} A_v$$

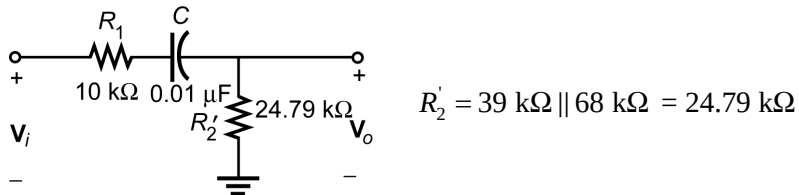
$$0.0945 = \log_{10} A_v$$

$$A_v = \frac{V_o}{V_i} = \mathbf{0.80}$$

f. $\theta = -\tan^{-1} f/f_c$



39.

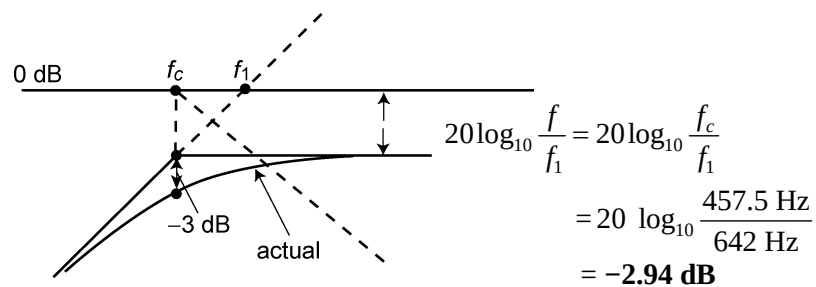


a. From Section 21.11,

$$A_v = \frac{V_o}{V_i} = \frac{j f/f_1}{1 + j f/f_c}$$

$$f_1 = \frac{1}{2\pi R_2' C} = \frac{1}{2\pi (24.79 \text{ k}\Omega)(0.01 \mu\text{F})} = \mathbf{642.01 \text{ Hz}}$$

$$f_c = \frac{1}{2\pi (R_1 + R_2') C} = \frac{1}{2\pi (10 \text{ k}\Omega + 24.79 \text{ k}\Omega)(0.01 \mu\text{F})} = \mathbf{457.47 \text{ Hz}}$$



b. $\theta = 90^\circ - \tan^{-1} \frac{f}{f_1} = + \tan^{-1} \frac{f_1}{f}$

$f = f_1: \quad \theta = 45^\circ$

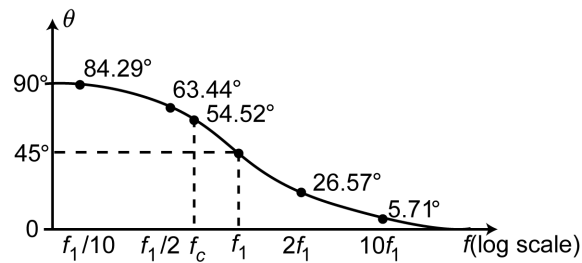
$f = f_c: \quad \theta = 54.52^\circ$

$f = \frac{1}{2}f_1 = 321 \text{ Hz}, \theta = 63.44^\circ$

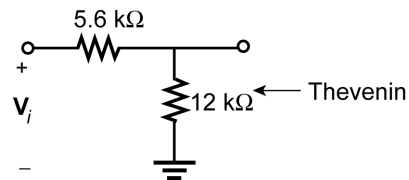
$f = \frac{1}{10}f_1 = 64.2 \text{ Hz}, \theta = 84.29^\circ$

$f = 2f_1 = 1,284 \text{ Hz}, \theta = 26.57^\circ$

$f = 10f_1 = 6420 \text{ Hz}, \theta = 5.71^\circ$

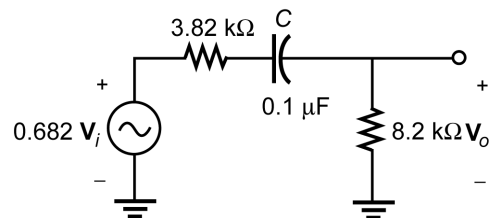


40. a.



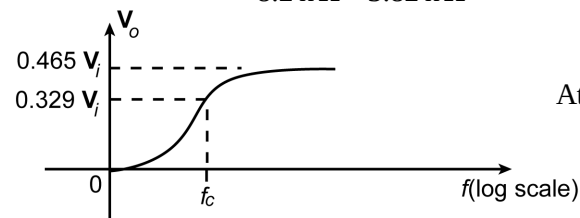
$$V_{Th} = \frac{12 \text{ k}\Omega V_i}{12 \text{ k}\Omega + 5.6 \text{ k}\Omega} = 0.682 V_i$$

$$R_{Th} = 5.6 \text{ k}\Omega \parallel 12 \text{ k}\Omega = 3.82 \text{ k}\Omega$$



$f = \infty \text{ Hz: } (C \Rightarrow \text{short circuit})$

$$V_o = \frac{8.2 \text{ k}\Omega (0.682 V_i)}{8.2 \text{ k}\Omega + 3.82 \text{ k}\Omega} = 0.465 V_i$$



At $f_c: V_o = 0.707(0.465 V_i) = 0.329 V_i$

$$\text{voltage-divider rule: } \mathbf{V}_o = \frac{R_2(0.682 \mathbf{V}_i)}{R_1 + R_2 - jX_C} = \frac{0.682 R_2 \mathbf{V}_i}{R_1 + R_2 - jX_C}$$

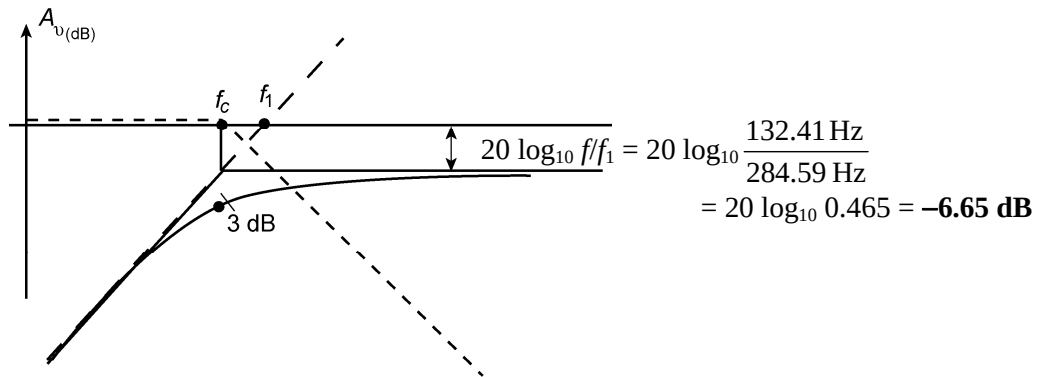
$$\text{and } \mathbf{A}_v = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{0.682 R_2}{R_1 + R_2 - jX_C} = \frac{j2\pi f(0.682 R_2)C}{1 + j2\pi f(R_1 + R_2)C}$$

$$\text{so that } \mathbf{A}_v = \frac{j f/f_1}{1 + j f/f_c} \text{ with } f_1 = \frac{1}{2\pi 0.682 R_2 C} = \frac{1}{2\pi 0.682(8.2 \text{ k}\Omega)(0.1 \mu\text{F})}$$

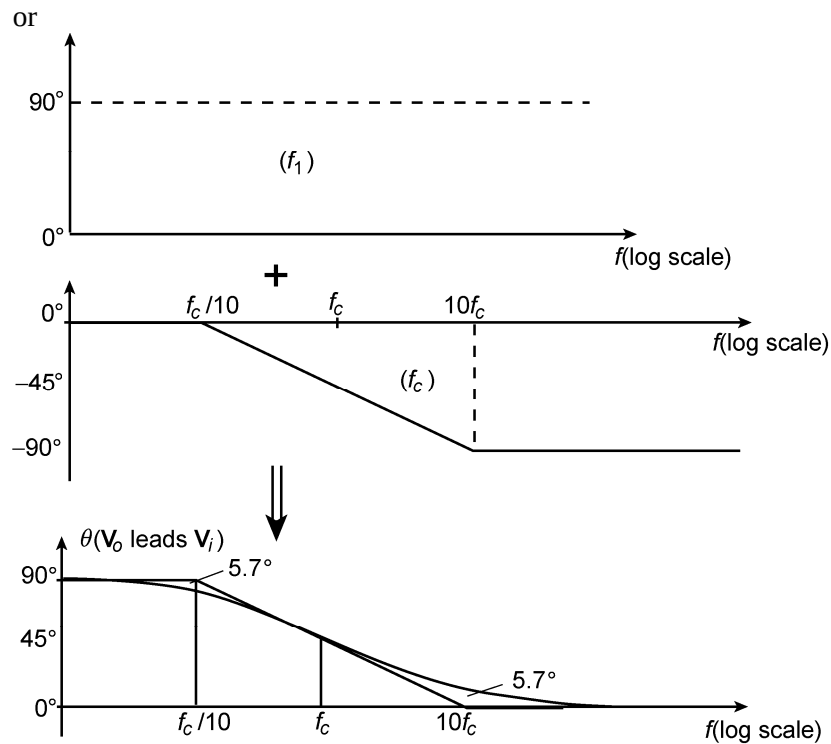
$$= 284.59 \text{ Hz}$$

$$\text{and } f_c = \frac{1}{2\pi(R_1 + R_2)C} = \frac{1}{2\pi(3.82 \text{ k}\Omega + 8.2 \text{ k}\Omega)(0.1 \mu\text{F})}$$

$$= 132.41 \text{ Hz}$$



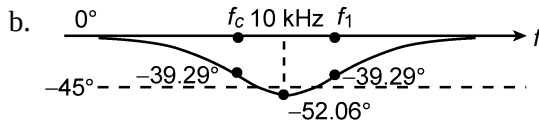
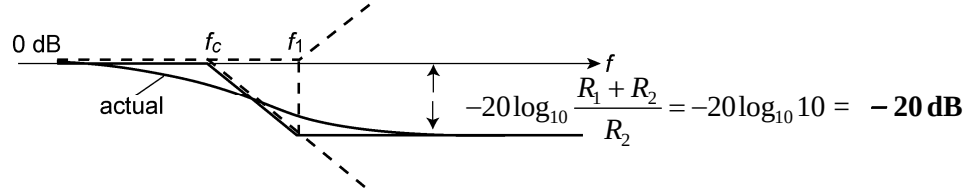
b. $\theta = 90^\circ - \tan^{-1} f/f_c = +\tan^{-1} f_c/f = \tan^{-1} 132.6 \text{ Hz}/f$



41. a.
$$\mathbf{A}_v = \frac{1 + j \frac{f}{f_1}}{1 + j \frac{f}{f_c}}$$

$$f_1 = \frac{1}{2\pi R_2 C} = \frac{1}{2\pi(10 \text{ k}\Omega)(820 \text{ pF})} = 19.41 \text{ kHz}$$

$$f_c = \frac{1}{2\pi(R_1 + R_2)C} = \frac{1}{2\pi(10 \text{ k}\Omega + 91 \text{ k}\Omega)(820 \text{ pF})} = 1.92 \text{ kHz}$$



$$\theta = \tan^{-1} f/f_1 - \tan^{-1} f/f_c$$

$$f = 10 \text{ kHz}$$

$$\theta = \tan^{-1} \frac{10 \text{ kHz}}{19.41 \text{ kHz}} - \tan^{-1} \frac{10 \text{ kHz}}{1.92 \text{ kHz}} = 27.25^\circ - 79.13^\circ = -51.88^\circ$$

$$f = f_c: (f_1 = 10 f_c)$$

$$\theta = \tan^{-1} \frac{f_c}{10 f_c} - \tan^{-1} \frac{f_c}{f_c} = \tan^{-1} 0.1 \tan^{-1} 1 = 5.71^\circ - 45^\circ = -39.29^\circ$$

42. a. R_1 no effect!
Note Section 21.12.

$$\mathbf{A}_v = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{1 + j(f/f_1)}{1 + j(f/f_c)}$$

$$f_1 = \frac{1}{2\pi(5.6 \text{ k}\Omega)(0.01 \mu\text{F})} = 2.84 \text{ kHz}$$

$$f_c = \frac{1}{2\pi(12 \text{ k}\Omega + 5.6 \text{ k}\Omega)(0.01 \mu\text{F})} = 904.3 \text{ Hz}$$

Note Fig. 21.65.

Asymptote at 0 dB from $0 \rightarrow f_c$

-6 dB/octave from f_c to f_1

$$-9.95 \text{ dB from } f_1 \text{ on } \left(-20 \log \frac{12 \text{ k}\Omega + 5.6 \text{ k}\Omega}{5.6 \text{ k}\Omega} = -9.95 \text{ dB} \right)$$

(b) Note Fig. 21.67.

From 0° to -26.50° at f_c and f_1

$$\theta = \tan^{-1} f/f_1 - \tan^{-1} f/f_c$$

At $f = 1500$ Hz (between f_c and f_1)

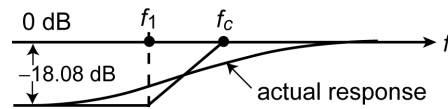
$$\begin{aligned}\theta &= \tan^{-1} 1500 \text{ Hz}/2.84 \text{ kHz} - \tan^{-1} 1500 \text{ Hz}/904.3 \text{ Hz} \\ &= 27.83^\circ - 58.92^\circ = -31.09^\circ\end{aligned}$$

43. a.
$$\mathbf{A}_v = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{1 - jf_1/f}{1 - jf/f_c}$$

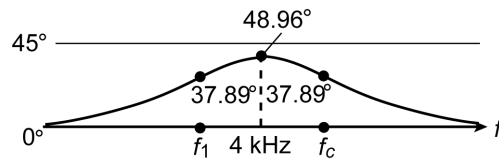
$$f_1 = \frac{1}{2\pi R_1 C} = \frac{1}{2\pi(3.3 \text{ k}\Omega)(0.051 \mu\text{F})} = 945.66 \text{ Hz}$$

$$f_c = \frac{1}{2\pi(R_1 \parallel R_2)C} = \frac{1}{2\pi(\underbrace{3.3 \text{ k}\Omega \parallel 0.47 \text{ k}\Omega}_{0.411 \text{ k}\Omega})(0.051 \mu\text{F})} = \mathbf{7.59 \text{ kHz}}$$

$$-20 \log_{10} \frac{R_1 + R_2}{R_2} = -20 \log_{10} \frac{3.3 \text{ k}\Omega + 0.47 \text{ k}\Omega}{0.47 \text{ k}\Omega} = -20 \log_{10} 8.02 = -18.08 \text{ dB}$$



b.



$$\theta = -\tan^{-1} \frac{f_1}{f} + \tan^{-1} \frac{f_c}{f}$$

$$f = f_1: \theta = -\tan^{-1} \frac{f_1}{f_1} + \tan^{-1} \frac{f_c}{f_1}$$

$$\begin{aligned}&= -\tan^{-1} 1 + \tan^{-1} \frac{7.59 \text{ kHz}}{945.66 \text{ Hz}} \\ &= -45^\circ + 82.89^\circ = \mathbf{+37.89^\circ}\end{aligned}$$

$$\begin{aligned}f = 4 \text{ kHz}: \theta &= -\tan^{-1} \frac{945.66 \text{ Hz}}{4 \text{ kHz}} + \tan^{-1} \frac{7.59 \text{ kHz}}{4 \text{ kHz}} \\ &= -13.28^\circ + 62.24^\circ = \mathbf{48.96^\circ}\end{aligned}$$

$$\begin{aligned}f = f_c: \theta &= -\tan^{-1} \frac{945.66 \text{ Hz}}{7.59 \text{ kHz}} + \tan^{-1} \frac{f_c}{f_c} \\ &= -\tan^{-1} 0.125 + \tan^{-1} 1 \\ &= -7.11^\circ + 45^\circ = \mathbf{+37.89^\circ}\end{aligned}$$

44. a. Note Section 21.13.

$$\mathbf{A}_v = \frac{1 - j(f_1/f)}{1 - j(f_c/f)}$$

$$f_1 = \frac{1}{2\pi R_1 C} = \frac{1}{2\pi(3.3 \text{ k}\Omega)(0.05 \text{ }\mu\text{F})} = \mathbf{964.58 \text{ Hz}}$$

$$f_c = \frac{1}{2\pi(R_1 \parallel R_2)C} = \frac{1}{2\pi(\underbrace{3.3 \text{ k}\Omega \parallel 0.5 \text{ k}\Omega}_{0.434 \text{ k}\Omega})0.05 \text{ }\mu\text{F}} = \mathbf{7334.33 \text{ Hz}}$$

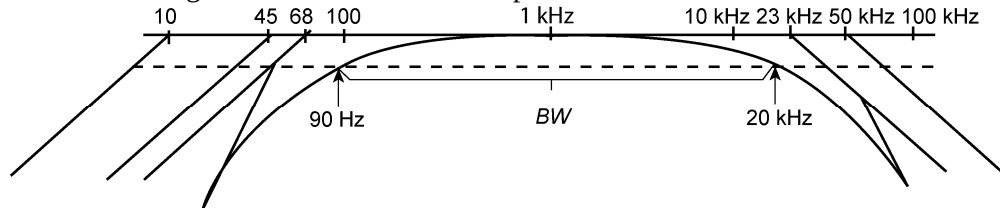
Note Fig. 21.72.

$$-20 \log_{10} \frac{R_1 + R_2}{R_2} = -20 \log_{10} \frac{3.3 \text{ k}\Omega + 0.5 \text{ k}\Omega}{0.5 \text{ k}\Omega} = -17.62 \text{ dB}$$

Asymptote at -17.62 dB from $0 \rightarrow f_1$
 $+6 \text{ dB/octave}$ from f_1 to f_c
 0 dB from f_c on

- b. $\theta = -\tan^{-1} f_1/f + \tan^{-1} f_c/f$
 Test at 3 kHz
 $\theta = -\tan^{-1} 964.58 \text{ Hz}/3.0 \text{ kHz} + \tan^{-1} 7334.33 \text{ Hz}/3.0 \text{ kHz}$
 $= -17.82^\circ + 67.75^\circ = 49.93^\circ \cong 50^\circ$

Therefore rising above 45° at and near the peak



50 kHz vs $23 \text{ kHz} \rightarrow$ drop about 1 dB at 23 kHz due to 50 kHz break.

Ignore effect of break frequency at 10 Hz .

Assume -2 dB drop at 68 Hz due to break frequency at 45 Hz .

Rough sketch suggests low cut-off frequency of 90 Hz .

Checking: Ignoring upper terms

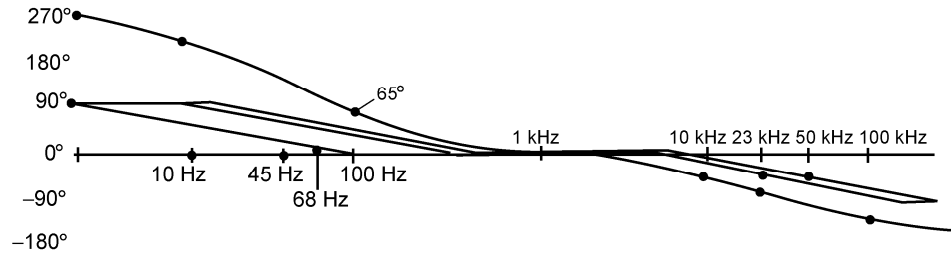
$$\begin{aligned} A'_{\text{dB}} &= -20 \log_{10} \sqrt{1 + \left(\frac{10 \text{ Hz}}{f}\right)^2} - 20 \log_{10} \sqrt{1 + \left(\frac{45 \text{ Hz}}{f}\right)^2} - 20 \log_{10} \sqrt{1 + \left(\frac{68 \text{ Hz}}{f}\right)^2} \\ &= -0.0532 \text{ dB} - 0.969 \text{ dB} - 1.96 \text{ dB} \\ &= -2.98 \text{ dB} \quad (\text{excellent}) \end{aligned}$$

High frequency cutoff: Try 20 kHz

$$\begin{aligned}
 A'_{\text{dB}} &= -20 \log_{10} \sqrt{1 + \left(\frac{f}{23 \text{ kHz}} \right)^2} - 20 \log_{10} \sqrt{1 + \left(\frac{f}{50 \text{ kHz}} \right)^2} \\
 &= -2.445 \text{ dB} - 0.6445 \text{ dB} \\
 &= -3.09 \text{ dB (excellent)}
 \end{aligned}$$

$$\therefore BW = 20 \text{ kHz} - 90 \text{ Hz} = \mathbf{19,910 \text{ Hz}} \cong 20 \text{ kHz}$$

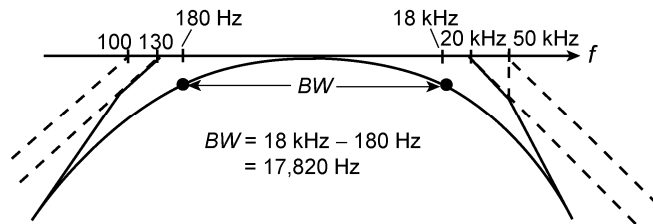
$$f_1 = \mathbf{90 \text{ Hz}}, f_2 = \mathbf{20 \text{ kHz}}$$



Testing: $f = 100 \text{ Hz}$

$$\begin{aligned}
 \theta &= \tan^{-1} \frac{10 \text{ Hz}}{f} + \tan^{-1} \frac{45 \text{ Hz}}{f} + \tan^{-1} \frac{68 \text{ Hz}}{f} - \tan^{-1} \frac{f}{23 \text{ kHz}} - \tan^{-1} \frac{f}{50 \text{ kHz}} \\
 &= \tan^{-1} 0.1 + \tan^{-1} 0.45 + \tan^{-1} 0.68 - \tan^{-1} 0.00435 - \tan^{-1} .002 \\
 &= 5.71^\circ + 24.23^\circ + 34.22^\circ - 0.249^\circ - 0.115^\circ \\
 &= \mathbf{63.8^\circ} \text{ vs about } 65^\circ \text{ on the plot}
 \end{aligned}$$

45. a.
$$A_{v_{\text{max}}} = \frac{1}{\left(1 - j \frac{100 \text{ Hz}}{f}\right) \left(1 - j \frac{130 \text{ Hz}}{f}\right) \left(1 + j \frac{f}{20 \text{ kHz}}\right) \left(1 + j \frac{f}{50 \text{ kHz}}\right)}$$



Proximity of 100 Hz to 130 Hz will raise lower cutoff frequency above 130 Hz:

Testing: $f = 180 \text{ Hz}$: (with lower terms only)

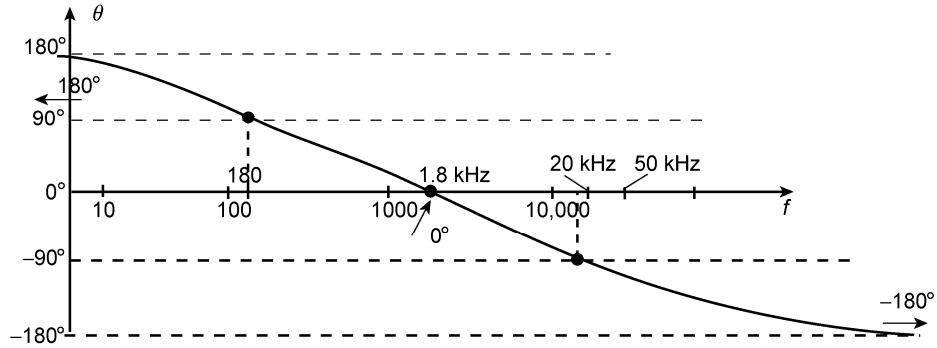
$$\begin{aligned}
 A_{v_{\text{dB}}} &= -20 \log_{10} \sqrt{1 + \left(\frac{100}{f} \right)^2} - 20 \log_{10} \sqrt{1 + \left(\frac{130}{f} \right)^2} \\
 &= -20 \log_{10} \sqrt{1 + \left(\frac{100}{180} \right)^2} - 20 \log_{10} \sqrt{1 + \left(\frac{130}{180} \right)^2} \\
 &= 1.17 \text{ dB} - 1.82 \text{ dB} = \mathbf{-2.99 \text{ dB}} \cong \mathbf{-3 \text{ dB}}
 \end{aligned}$$

Proximity of 50 kHz to 20 kHz will lower high cutoff frequency below 20 kHz:

Testing: $f = 18 \text{ kHz}$: (with upper terms only)

$$\begin{aligned} A_{v\text{dB}} &= -20 \log_{10} \sqrt{1 + \left(\frac{f}{20 \text{ kHz}} \right)^2} - 20 \log_{10} \sqrt{1 + \left(\frac{f}{50 \text{ kHz}} \right)^2} \\ &= -20 \log_{10} \sqrt{1 + \left(\frac{18 \text{ kHz}}{20 \text{ kHz}} \right)^2} - 20 \log_{10} \sqrt{1 + \left(\frac{13 \text{ kHz}}{20 \text{ kHz}} \right)^2} \\ &= -2.576 \text{ dB} - 0.529 \text{ dB} = \mathbf{-3.105 \text{ dB}} \end{aligned}$$

b.

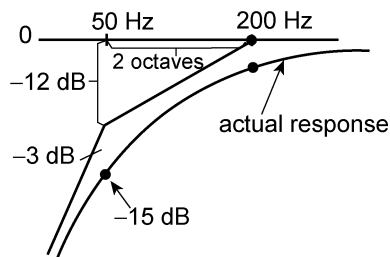


Testing: $f = 1.8 \text{ kHz}$:

$$\begin{aligned} \theta &= \tan^{-1} \frac{100}{1.8 \text{ kHz}} + \tan^{-1} \frac{130}{1.8 \text{ kHz}} - \tan^{-1} \frac{1.8 \text{ kHz}}{20 \text{ kHz}} - \tan^{-1} \frac{1.8 \text{ kHz}}{50 \text{ kHz}} \\ &= 3.18^\circ + 4.14^\circ - 5.14^\circ - 2.06^\circ \\ &= \mathbf{0.12^\circ \cong 0^\circ} \end{aligned}$$

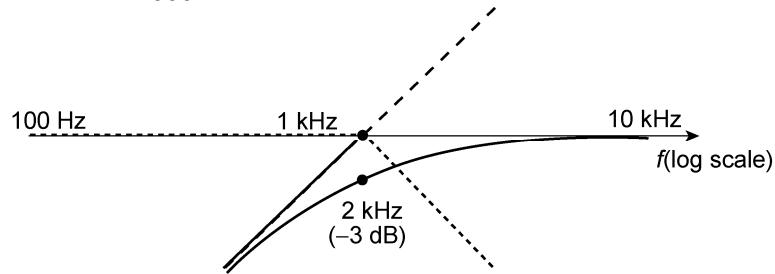
47. $f_{\text{low}} = f_{\text{high}} - BW = 36 \text{ kHz} - 35.8 \text{ kHz} = 0.2 \text{ kHz} = 200 \text{ Hz}$

$$A_v = \frac{-120}{\left(1 - j \frac{50}{f} \right) \left(1 - j \frac{200}{f} \right) \left(1 + j \frac{f}{36 \text{ kHz}} \right)}$$



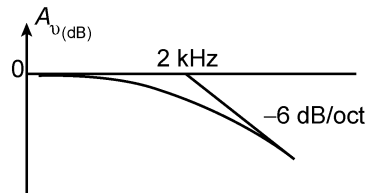
$$48. \quad \mathbf{A}_v = \frac{0.05}{0.05 - j\frac{100}{f}} = \frac{1}{1 - j\frac{100}{0.05f}} = \frac{1}{1 - j\frac{2000}{f}} = \frac{+jf}{+jf + 2000}$$

$$= \frac{+j\frac{f}{2000}}{1 + j\frac{f}{2000}} \text{ and } f_1 = 2000 \text{ Hz}$$

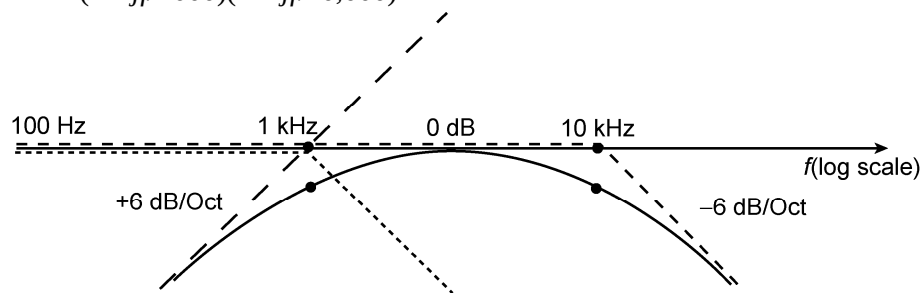


$$49. \quad \mathbf{A}_v = \frac{200}{200 + j0.1f} = \frac{1}{1 + j\frac{0.1f}{200}} = \frac{1}{1 + j\frac{f}{2000}}$$

$$A_{v_{dB}} = 20 \log_{20} \frac{1}{\sqrt{1 + \left(\frac{f}{2000}\right)^2}}, \quad \frac{f}{2000} = 1 \text{ and } f = 2 \text{ kHz}$$

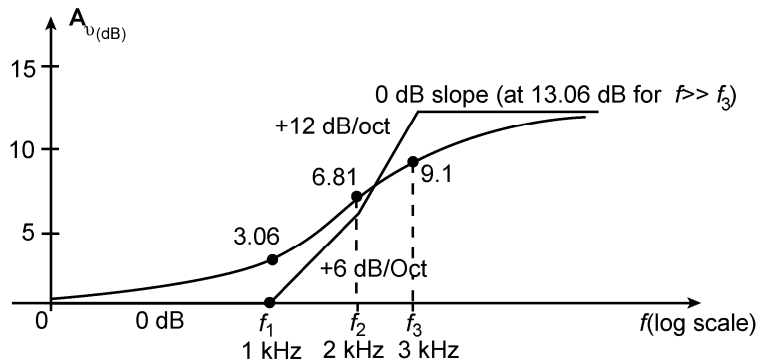


$$50. \quad \mathbf{A}_v = \frac{jf/1000}{(1 + jf/1000)(1 + jf/10,000)}$$

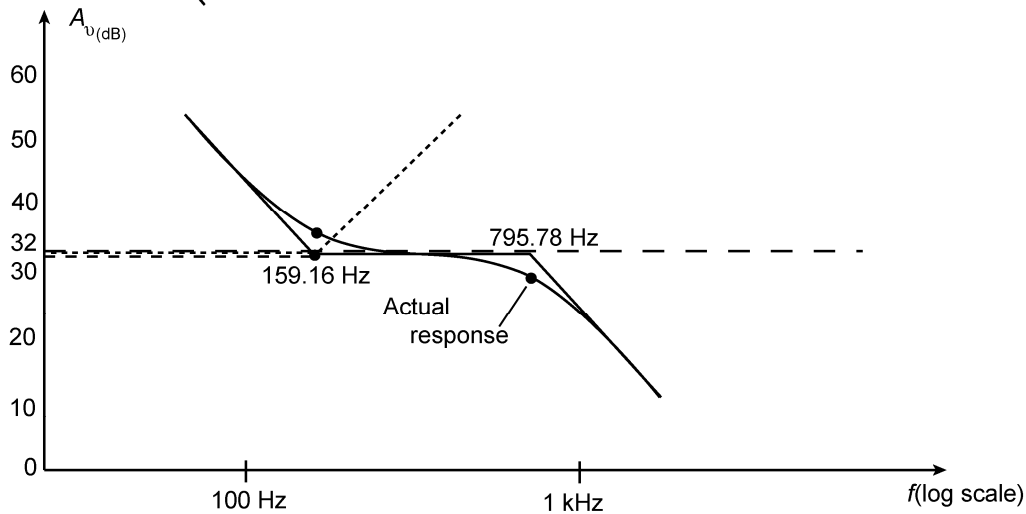
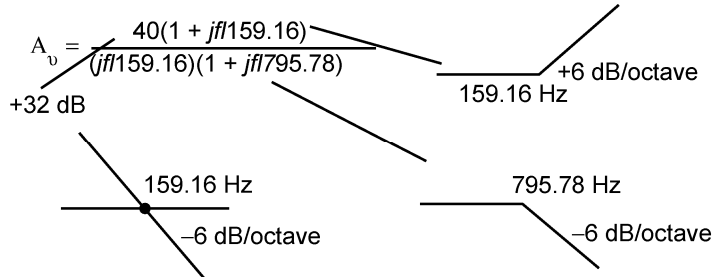


$$51. \quad A_v = \frac{\left(1 + j\frac{f}{1000}\right)\left(1 + j\frac{f}{2000}\right)}{\left(1 + j\frac{f}{3000}\right)^2}$$

$$A_{v_{dB}} = 20 \log_{10} \sqrt{1 + \left(\frac{f_1}{1000}\right)^2} + 20 \log_{10} \sqrt{1 + \left(\frac{f_2}{2000}\right)^2} + 40 \log_{10} \frac{1}{\sqrt{1 + \left(\frac{f_3}{3000}\right)^2}}$$



$$52. \quad \frac{j\omega}{1000} = j\frac{2\pi f}{1000} = j\frac{f}{\frac{1000}{2\pi}} = j\frac{f}{159.16 \text{ Hz}}, \quad \frac{j\omega}{5000} = j\frac{f}{\frac{5000}{2\pi}} = j\frac{f}{795.78 \text{ Hz}}$$



53. a. Woofer – 400 Hz:

$$X_L = 2\pi fL = 2\pi(400 \text{ Hz})(4.7 \text{ mH}) = 11.81 \Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(400 \text{ Hz})(39 \mu\text{F})} = 10.20 \Omega$$

$$R \parallel X_C = 8 \Omega \angle 0^\circ \parallel 10.20 \Omega \angle -90^\circ = 6.3 \Omega \angle -38.11^\circ$$

$$\mathbf{V}_o = \frac{(R \parallel X_C)(\mathbf{V}_i)}{(R \parallel X_C) + jX_L} = \frac{(6.3 \Omega \angle -38.11^\circ)(\mathbf{V}_i)}{(6.3 \Omega \angle -38.11^\circ) + j11.81 \Omega}$$

$$\mathbf{V}_o = 0.673 \angle -96.11^\circ \mathbf{V}_i$$

$$\text{and } A_v = \frac{V_o}{V_i} = \mathbf{0.673} \text{ vs desired } 0.707 \text{ (off by less than 5\%)}$$

Tweeter – 5 kHz:

$$X_L = 2\pi fL = 2\pi(5 \text{ kHz})(0.39 \text{ mH}) = 12.25 \Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(5 \text{ kHz})(2.7 \mu\text{F})} = 11.79 \Omega$$

$$R \parallel X_L = 8 \Omega \angle 0^\circ \parallel 12.25 \Omega \angle 90^\circ = 6.7 \Omega \angle 33.15^\circ$$

$$\mathbf{V}_o = \frac{(6.7 \Omega \angle 33.15^\circ)(\mathbf{V}_i)}{(6.7 \Omega \angle 33.15^\circ) - j11.79 \Omega}$$

$$\mathbf{V}_o = 0.678 \angle 88.54^\circ \mathbf{V}_i$$

$$\text{and } A_v = \frac{V_o}{V_i} = \mathbf{0.678} \text{ vs } 0.707 \text{ (off by less than 5\%)}$$

b. Woofer – 3 kHz:

$$X_L = 2\pi fL = 2\pi(3 \text{ kHz})(4.7 \text{ mH}) = 88.59 \Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(3 \text{ kHz})(39 \mu\text{F})} = 1.36 \Omega$$

$$R \parallel X_C = 8 \Omega \angle 0^\circ \parallel 1.36 \Omega \angle -90^\circ = 1.341 \Omega \angle -80.35^\circ$$

$$\mathbf{V}_o = \frac{(R \parallel X_C)(\mathbf{V}_i)}{(R \parallel X_C) + jX_L} = \frac{(1.341 \Omega \angle -80.35^\circ)(\mathbf{V}_i)}{(1.341 \Omega \angle -80.35^\circ) + j88.59 \Omega}$$

$$\mathbf{V}_o = 0.015 \angle -170.2^\circ \mathbf{V}_i$$

$$\text{and } A_v = \frac{V_o}{V_i} = \mathbf{0.015} \text{ vs desired } 0 \text{ (excellent)}$$

Tweeter – 3 kHz:

$$X_L = 2\pi fL = 2\pi(3 \text{ kHz})(0.39 \text{ mH}) = 7.35 \Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(3 \text{ kHz})(2.7 \mu\text{F})} = 19.65 \Omega$$

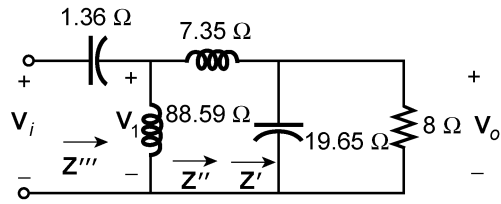
$$R \parallel X_L = 8 \Omega \angle 0^\circ \parallel 7.35 \Omega \angle 90^\circ = 5.42 \Omega \angle 47.42^\circ$$

$$\mathbf{V}_o = \frac{(R \parallel X_L)(\mathbf{V}_i)}{(R \parallel X_L) + jX_C} = \frac{(5.42 \Omega \angle 47.42^\circ)(\mathbf{V}_i)}{(5.42 \Omega \angle 47.42^\circ) - j19.65 \Omega}$$

$$\mathbf{V}_o = 0.337 \angle 124.24^\circ \mathbf{V}_i$$

$$\text{and } A_v = \frac{V_o}{V_i} = \mathbf{0.337} \text{ (acceptable since relatively close to cut frequency for tweeter)}$$

c. Mid-range speaker – 3 kHz:



$$\overbrace{Z' = 7.41 \, \Omega \angle -22.15^\circ}$$

$$\overbrace{Z'' = 8.24 \, \Omega \angle 33.58^\circ}$$

$$\overbrace{Z''' = 7.816 \, \Omega \angle 37.79^\circ}$$

$$V_1 = \frac{Z''' V_i}{Z''' - jX_C} = \frac{(7.816 \, \Omega \angle 37.79^\circ) V_i}{7.816 \, \Omega \angle 37.79^\circ - j1.36 \, \Omega} = 1.11 \angle 8.83^\circ V_i$$

$$V_o = \frac{Z' V_1}{Z' + jX_L} = \frac{(7.41 \, \Omega \angle -22.15^\circ) V_i}{7.41 \, \Omega \angle -22.15^\circ + j7.35 \, \Omega} = 0.998 \angle -46.9^\circ V_i$$

and $A_v = \frac{V_o}{V_i} = \mathbf{0.998}$ (excellent)

Chapter 22

1. a. $M = k\sqrt{L_p L_s} \Rightarrow L_s = \frac{M^2}{L_p k^2} = \frac{(40 \text{ mH})^2}{(50 \text{ mH})(0.8)^2} = \mathbf{50 \text{ mH}}$
- b. $e_p = N_p \frac{d\phi_p}{dt} = (20)(0.08 \text{ Wb/s}) = \mathbf{1.6 \text{ V}}$
 $e_s = kN_s \frac{d\phi_p}{dt} = (0.8)(80 \text{ t})(0.08 \text{ Wb/s}) = \mathbf{5.12 \text{ V}}$
- c. $e_p = L_p \frac{di_p}{dt} = (40 \text{ mH})(0.3 \times 10^3 \text{ A/s}) = \mathbf{12 \text{ V}}$
 $e_s = M \frac{di_p}{dt} = (80 \text{ mH})(0.03 \times 10^3 \text{ A/s}) = \mathbf{24 \text{ V}}$
2. a. $k = 1$
 - (a) $L_s = \frac{M^2}{L_p k^2} = \frac{(40 \text{ mH})^2}{(50 \text{ mH})(1)^2} = \mathbf{32 \text{ mH}}$
 - (b) $e_p = \mathbf{1.6 \text{ V}}, e_s = kN_s \frac{d\phi_p}{dt} = (1)(80 \text{ t})(0.08 \text{ Wb/s}) = \mathbf{6.4 \text{ V}}$
 - (c) $e_p = \mathbf{15 \text{ V}}, e_s = \mathbf{12 \text{ V}}$
- b. $k = 0.2$
 - (a) $L_s = \frac{M^2}{L_p k^2} = \frac{(40 \text{ mH})^2}{(50 \text{ mH})(0.2)^2} = \mathbf{0.8 \text{ H}}$
 - (b) $e_p = \mathbf{1.6 \text{ V}}, e_s = kN_s \frac{d\phi_p}{dt} = (0.2)(80 \text{ t})(0.08 \text{ Wb/s}) = \mathbf{1.28 \text{ V}}$
 - (c) $e_p = \mathbf{15 \text{ V}}, e_s = \mathbf{12 \text{ V}}$
3. a. $L_s = \frac{M^2}{L_p k^2} = \frac{(40 \text{ mH})^2}{(50 \text{ mH})(0.3)^2} = \mathbf{355.56 \text{ mH}}$
- b. $e_p = N_p \frac{d\phi_p}{dt} = (300 \text{ t})(0.08 \text{ Wb/s}) = \mathbf{24 \text{ V}}$
 $e_s = kN_s \frac{d\phi_p}{dt} = (0.9)(25 \text{ t})(0.08 \text{ Wb/s}) = \mathbf{1.8 \text{ V}}$
- c. e_p and e_s the same as problem 1: $e_p = \mathbf{15 \text{ V}}, e_s = \mathbf{12 \text{ V}}$

4. a. $E_s = \frac{N_s}{N_p} E_p = \frac{120 \text{ t}}{20 \text{ t}} (40 \text{ V}) = \mathbf{240 \text{ V}}$
b. $\Phi_{\max} = \frac{E_p}{4.4 f N_p} = \frac{40 \text{ V}}{4.44(60 \text{ Hz})(20 \text{ t})} = \mathbf{7.51 \text{ mWb}}$
5. a. $E_s = \frac{N_s}{N_p} E_p = \frac{30 \text{ t}}{240 \text{ t}} (40 \text{ V}) = \mathbf{5 \text{ V}}$
b. $\Phi_{m(\max)} = \frac{E_p}{4.44 f N_p} = \frac{40 \text{ V}}{(4.44)(60 \text{ Hz})(240 \text{ t})} = \mathbf{625.63 \mu\text{Wb}}$
6. $E_p = \frac{N_p}{N_s} E_s = \frac{60 \text{ t}}{720 \text{ t}} (240 \text{ V}) = \mathbf{20 \text{ V}}$
7. $f = \frac{E_p}{(4.44) N_p \Phi_{m(\max)}} = \frac{40 \text{ V}}{(4.44)(20 \text{ t})(3.75 \text{ mWb})} = \mathbf{120 \text{ Hz}}$
8. a. $I_L = a I_p = \left(\frac{1}{5} \right) (2 \text{ A}) = \mathbf{0.4 \text{ A}}$
 $V_L = I_L Z_L = \left(\frac{2}{5} \text{ A} \right) (2 \Omega) = \mathbf{0.8 \text{ V}}$
b. $Z_{\text{in}} = a^2 Z_L = \left(\frac{1}{5} \right)^2 2 \Omega = \mathbf{0.08 \Omega}$
9. $Z_p = \frac{V_g}{I_p} = \frac{120 \text{ V}}{4 \text{ A}} = \mathbf{30 \Omega}$
10. $V_g = a V_L = \left(\frac{1}{4} \right) (600 \text{ V}) = \mathbf{150 \text{ V}}$
 $I_p = \frac{V_g}{Z_i} = \frac{150 \text{ V}}{4 \Omega} = \mathbf{37.5 \text{ A}}$
11. $I_L = I_s = \frac{V_L}{Z_L} = \frac{240 \text{ V}}{20 \Omega} = 12 \text{ A}$
 $\frac{I_s}{I_p} = a = \frac{N_p}{N_s} \Rightarrow \frac{12 \text{ A}}{0.05 \text{ A}} = \frac{N_p}{50}$
 $N_p = \frac{50(12)}{0.05} = \mathbf{12,000 \text{ turns}}$

12. a. $a = \frac{N_p}{N_s} = \frac{400 \text{ t}}{1200 \text{ t}} = \frac{1}{3}$

$$Z_i = a^2 Z_L = \left(\frac{1}{3}\right)^2 [12 \Omega + j12 \Omega] = 1.333 \Omega + j1.333 \Omega = 1.885 \Omega \angle 45^\circ$$

$$I_p = V_g / Z_i = 100 \text{ V} / 1.885 \Omega = \mathbf{53.05 \text{ A}}$$

b. $I_L = a I_p = \frac{1}{3} (53.05 \text{ A}) = \mathbf{17.68 \text{ A}}$, $V_L = I_L Z_L = (17.68 \text{ A})(16.97 \Omega) = \mathbf{300 \text{ V}}$

13. a. $Z_p = a^2 Z_L \Rightarrow a = \sqrt{\frac{Z_p}{Z_L}}$

$$Z_p = \frac{V_p}{I_p} = \frac{10 \text{ V}}{20 \text{ V} / 72 \Omega} = 36 \Omega$$

$$a = \sqrt{\frac{36 \Omega}{4 \Omega}} = \mathbf{3}$$

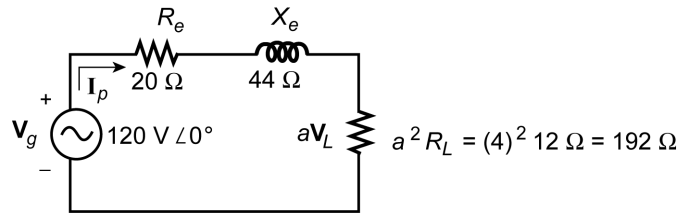
b. $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{1}{3} \Rightarrow V_s = \frac{1}{3} V_p = \frac{1}{3} (10 \text{ V}) = \mathbf{3\frac{1}{3} \text{ V}}$

$$P = \frac{V_s^2}{Z_s} = \frac{(3.33 \text{ V})^2}{4 \Omega} = \mathbf{2.78 \text{ W}}$$

14. a. $R_e = R_p + a^2 R_s = 4 \Omega + (4)^2 1 \Omega = \mathbf{20 \Omega}$

b. $X_e = X_p + a^2 X_s = 12 \Omega + (4)^2 2 \Omega = \mathbf{44 \Omega}$

c.



d. $I_p = \frac{V_g}{Z_p} = \frac{120 \text{ V} \angle 0^\circ}{20 \Omega + 192 \Omega + j44 \Omega} = \frac{120 \text{ V} \angle 0^\circ}{212 \Omega + j44 \Omega} = \mathbf{0.554 \text{ A} \angle -11.73^\circ}$

e. $aV_L = \frac{a^2 R_L V_g}{(R_e + a^2 R_L) + jX_e} = I_p a^2 R_L$

or $V_L = a I_p R_L \angle 0^\circ = (4)(0.554 \text{ A} \angle -11.73^\circ)(20 \Omega \angle 0^\circ) = \mathbf{26.59 \text{ V} \angle -11.73^\circ}$

f. —

g. $V_L = \frac{N_s}{N_p} V_g = \frac{1}{4} (120 \text{ V}) = \mathbf{30 \text{ V}}$

15. a. $a = \frac{N_p}{N_s} = \frac{4 \text{ t}}{1 \text{ t}} = 4$
 $R_e = R_p + a^2 R_s = 4 \Omega + (4)^2 1 \Omega = 20 \Omega$
 $X_e = X_p + a^2 X_s = 12 \Omega + (4)^2 2 \Omega = 44 \Omega$
 $\mathbf{Z}_p = \mathbf{Z}_{R_e} + \mathbf{Z}_{X_e} + a^2 \mathbf{Z}_{X_L} = 20 \Omega + j44 \Omega + j(4)^2 20 \Omega$
 $= 20 \Omega + j44 \Omega + j320 \Omega = 20 \Omega + j364 \Omega = \mathbf{364.55 \Omega \angle 86.86^\circ}$
- b. $\mathbf{I}_p = \frac{\mathbf{V}_g}{\mathbf{Z}_p} = \frac{120 \text{ V} \angle 0^\circ}{364.55 \Omega \angle 86.86^\circ} = 329.17 \text{ mA} \angle -86.86^\circ$
- c. $\mathbf{V}_{R_e} = (I \angle \theta)(R_e \angle 0^\circ) = (329.17 \text{ mA} \angle -86.86^\circ)(20 \Omega \angle 0^\circ)$
 $= \mathbf{6.58 \text{ V} \angle -86.86^\circ}$
 $\mathbf{V}_{X_e} = (I \angle \theta)(X_e \angle 90^\circ) = (329.17 \text{ mA} \angle -86.86^\circ)(44 \Omega \angle 90^\circ)$
 $= \mathbf{14.48 \text{ V} \angle 3.14^\circ}$
 $\mathbf{V}_{X_L} = \mathbf{I}(a^2 \mathbf{Z}_{X_L}) = (329.17 \text{ mA} \angle -86.86^\circ)(320 \Omega \angle 90^\circ)$
 $= \mathbf{105.33 \text{ V} \angle 3.14^\circ}$
16. a. $a = N_p/N_s = 4 \text{ t}/1 \text{ t} = 4$, $R_e = R_p + a^2 R_s = 4 \Omega + (4)^2 1 \Omega = 20 \Omega$
 $X_e = X_p + a^2 X_s = 12 \Omega + (4)^2 2 \Omega = 44 \Omega$
 $\mathbf{Z}_p = R_e + jX_e - ja^2 X_C = 20 \Omega + j44 \Omega - j(4)^2 20 \Omega$
 $= 20 \Omega + j44 \Omega - j320 \Omega = 20 \Omega - j276 \Omega = \mathbf{276.72 \Omega \angle -85.86^\circ}$
- b. $\mathbf{I}_p = \frac{\mathbf{V}_g}{\mathbf{Z}_p} = \frac{120 \text{ V} \angle 0^\circ}{276.72 \Omega \angle -85.86^\circ} = \mathbf{0.43 \text{ A} \angle 85.86^\circ}$
- c. $\mathbf{V}_{R_e} = (I_p \angle \theta)(R_e \angle 0^\circ) = (0.43 \text{ A} \angle 85.86^\circ)(20 \Omega \angle 0^\circ) = \mathbf{8.6 \text{ V} \angle 85.86^\circ}$
 $\mathbf{V}_{X_e} = (I_p \angle \theta)(X_e \angle 90^\circ) = (0.43 \text{ A} \angle 85.86^\circ)(44 \Omega \angle 90^\circ) = \mathbf{18.92 \text{ V} \angle 175.86^\circ}$
 $\mathbf{V}_{X_C} = (I_p \angle \theta)(a^2 X_C \angle -90^\circ) = (0.43 \text{ A} \angle 85.86^\circ)(320 \Omega \angle -90^\circ) = \mathbf{137.60 \text{ V} \angle -4.14^\circ}$
17. —
18. Coil 1: $L_1 - M_{12}$
Coil 2: $L_2 - M_{12}$
 $L_T = L_1 + L_2 - 2M_{12} = 4 \text{ H} + 7 \text{ H} - 2(1 \text{ H}) = \mathbf{9 \text{ H}}$
19. $L_{T(+) } = L_1 + L_2 + 2M_{12}$
 $M_{12} = k\sqrt{L_1 L_2} = (0.8)\sqrt{(200 \text{ mH})(600 \text{ mH})} = \mathbf{277 \text{ mH}}$
 $L_{T(+) } = 200 \text{ mH} + 600 \text{ mH} + 2(277 \text{ mH}) = \mathbf{1.35 \text{ H}}$

$$\begin{aligned}
20. \quad M_{23} &= k\sqrt{L_2 L_3} = 1\sqrt{(1\text{ H})(4\text{ H})} = 2\text{ H} \\
\text{Coil 1:} \quad L_1 + M_{12} - M_{13} &= 2\text{ H} + 0.2\text{ H} - 0.1\text{ H} = 2.1\text{ H} \\
\text{Coil 2:} \quad L_2 + M_{12} - M_{23} &= 1\text{ H} + 0.2\text{ H} - 2\text{ H} = -0.8\text{ H} \\
\text{Coil 3:} \quad L_3 - M_{23} - M_{13} &= 4\text{ H} - 2\text{ H} - 0.1\text{ H} = 1.9\text{ H} \\
L_T &= 2.1\text{ H} - 0.8\text{ H} + 1.9\text{ H} = \mathbf{3.2\text{ H}}
\end{aligned}$$

$$\begin{aligned}
21. \quad \mathbf{E}_1 - \mathbf{I}_1[\mathbf{Z}_{R_1} + \mathbf{Z}_{L_1}] - \mathbf{I}_2[\mathbf{Z}_m] &= 0 \\
\mathbf{I}_2[\mathbf{Z}_{L_2} + \mathbf{Z}_{R_L}] + \mathbf{I}_1[\mathbf{Z}_m] &= 0 \\
\hline
\mathbf{I}_1(\mathbf{Z}_{R_1} + \mathbf{Z}_{L_1}) + \mathbf{I}_2(\mathbf{Z}_m) &= \mathbf{E}_1 \\
\mathbf{I}_1(\mathbf{Z}_m) + \mathbf{I}_2(\mathbf{Z}_{L_2} + \mathbf{Z}_{R_L}) &= 0 \quad X_m = -\omega M \angle 90^\circ \\
\hline
\end{aligned}$$

$$\begin{aligned}
22. \quad \mathbf{Z}_i &= \mathbf{Z}_p + \frac{(\omega M)^2}{\mathbf{Z}_s + \mathbf{Z}_L} = R_p + jX_{L_p} + \frac{(\omega M)^2}{R_s + jX_{L_s} + R_L} \\
R_p &= 2\ \Omega, \quad X_{L_p} = \omega L_p = (10^3\text{ rad/s})(8\text{ H}) = 8\text{ k}\Omega \\
R_s &= 1\ \Omega, \quad X_{L_s} = \omega L_s = (10^3\text{ rad/s})(2\text{ H}) = 2\text{ k}\Omega \\
M &= k\sqrt{L_p L_s} = 0.05\sqrt{(8\text{ H})(2\text{ H})} = 0.2\text{ H} \\
\mathbf{Z}_i &= 2\ \Omega + j8\text{ k}\Omega + \frac{(10^3\text{ rad/s} \cdot 0.2\text{ H})^2}{1\ \Omega + j2\text{ k}\Omega + 20\ \Omega} \\
&= 2\ \Omega + j8\text{ k}\Omega + \frac{4 \times 10^4\ \Omega}{21 + j2 \times 10^3} \\
&= 2\ \Omega + j8\text{ k}\Omega + 0.21\ \Omega - j19.99\ \Omega = 2.21\ \Omega + j7980\ \Omega \\
\mathbf{Z}_i &= \mathbf{7980\ \Omega \angle 89.98^\circ}
\end{aligned}$$

$$\begin{aligned}
23. \quad \text{a.} \quad a &= \frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{2400\text{ V}}{120\text{ V}} = \mathbf{20} \\
\text{b.} \quad 10,000\text{ VA} &= V_s I_s \Rightarrow I_s = \frac{10,000\text{ VA}}{V_s} = \frac{10,000\text{ VA}}{120\text{ V}} = \mathbf{83.33\text{ A}} \\
\text{c.} \quad I_p &= \frac{10,000\text{ VA}}{V_p} = \frac{10,000\text{ VA}}{2400\text{ V}} = \mathbf{4.17\text{ A}} \\
\text{d.} \quad a &= \frac{V_p}{V_s} = \frac{120\text{ V}}{2400\text{ V}} = 0.05 = \frac{1}{20} \\
I_s &= \frac{10,000\text{ VA}}{2400\text{ V}} = \mathbf{4.17\text{ A}}, \quad I_p = \mathbf{83.33\text{ A}}
\end{aligned}$$

$$\begin{aligned}
24. \quad I_s = I_1 = 2 \text{ A}, E_p = V_L = 40 \text{ V} \\
E_s = V_s - V_L = 200 \text{ V} - 40 \text{ V} = \mathbf{160 \text{ V}} \\
V_g I_1 = V_L I_L \Rightarrow I_L = V_g / V_L \cdot I_1 = \frac{200 \text{ V}}{40 \text{ V}} (2 \text{ A}) = \mathbf{10 \text{ A}} \\
I_p + I_1 = I_L \Rightarrow I_p = I_L - I_1 = 10 \text{ A} - 2 \text{ A} = \mathbf{8 \text{ A}}
\end{aligned}$$

$$\begin{aligned}
25. \quad a. \quad \mathbf{E}_s &= \frac{N_s}{N_p} \mathbf{E}_p \\
&= \frac{25 \text{ t}}{100 \text{ t}} (100 \text{ V} \angle 0^\circ) = \mathbf{25 \text{ V} \angle 0^\circ = V_L} \\
\mathbf{I}_s &= \frac{\mathbf{E}_s}{\mathbf{Z}_L} = \frac{25 \text{ V} \angle 0^\circ}{5 \Omega \angle 0^\circ} = \mathbf{5 \text{ A} \angle 0^\circ = \mathbf{I}_L}
\end{aligned}$$

$$b. \quad \mathbf{Z}_i = a^2 \mathbf{Z}_L = \left(\frac{N_p}{N_s} \right)^2 \mathbf{Z}_L = \left(\frac{100 \text{ t}}{25 \text{ t}} \right)^2 5 \Omega \angle 0^\circ = (4)^2 5 \Omega \angle 0^\circ = \mathbf{80 \Omega \angle 0^\circ}$$

$$c. \quad \mathbf{Z}_{1/2} = \frac{1}{4} \mathbf{Z}_i = \frac{1}{4} (80 \Omega \angle 0^\circ) = \mathbf{20 \Omega \angle 0^\circ}$$

$$\begin{aligned}
26. \quad a. \quad \mathbf{E}_2 &= \frac{N_2}{N_1} \mathbf{E}_1 = \frac{15 \text{ t}}{90 \text{ t}} (60 \text{ V} \angle 0^\circ) = \mathbf{10 \text{ V} \angle 0^\circ} \\
\mathbf{E}_3 &= \frac{N_3}{N_1} \mathbf{E}_1 = \frac{45 \text{ t}}{90 \text{ t}} (60 \text{ V} \angle 0^\circ) = \mathbf{30 \text{ V} \angle 0^\circ} \\
\mathbf{I}_2 &= \frac{\mathbf{E}_2}{\mathbf{Z}_2} = \frac{10 \text{ V} \angle 0^\circ}{8 \Omega \angle 0^\circ} = \mathbf{1.25 \text{ A} \angle 0^\circ} \\
\mathbf{I}_3 &= \frac{\mathbf{E}_3}{\mathbf{Z}_3} = \frac{30 \text{ V} \angle 0^\circ}{5 \Omega \angle 0^\circ} = \mathbf{6 \text{ A} \angle 0^\circ}
\end{aligned}$$

$$\begin{aligned}
b. \quad \frac{1}{R_1} &= \frac{1}{(N_1/N_2)^2 R_2} + \frac{1}{(N_1/N_3)^2 R_3} \\
&= \frac{1}{(90 \text{ t}/15 \text{ t})^2 8 \Omega} + \frac{1}{(90 \text{ t}/45 \text{ t})^2 5 \Omega} \\
\frac{1}{R_1} &= \frac{1}{288 \Omega} + \frac{1}{20 \Omega} = 0.05347 \text{ S} \\
R_1 &= \mathbf{18.70 \Omega}
\end{aligned}$$

$$\begin{aligned}
27. \quad a. \quad \mathbf{E}_2 &= \frac{N_2}{N_1} \mathbf{E}_1 = \left(\frac{40 \text{ t}}{120 \text{ t}} \right) (120 \text{ V} \angle 60^\circ) = \mathbf{40 \text{ V} \angle 60^\circ} \\
\mathbf{I}_2 &= \frac{\mathbf{E}_2}{\mathbf{Z}_2} = \frac{40 \text{ V} \angle 60^\circ}{12 \Omega \angle 0^\circ} = \mathbf{3.33 \text{ A} \angle 60^\circ} \\
\mathbf{E}_3 &= \frac{N_3}{N_1} \mathbf{E}_1 = \left(\frac{30 \text{ t}}{120 \text{ t}} \right) (120 \text{ V} \angle 60^\circ) = \mathbf{30 \text{ V} \angle 60^\circ} \\
\mathbf{I}_3 &= \frac{\mathbf{E}_3}{\mathbf{Z}_3} = \frac{30 \text{ V} \angle 60^\circ}{10 \Omega \angle 0^\circ} = \mathbf{3 \text{ A} \angle 60^\circ}
\end{aligned}$$

$$\begin{aligned}
\text{b. } \frac{1}{R_1} &= \frac{1}{(N_1/N_2)^2 R_2} + \frac{1}{(N_1/N_3)^2 R_3} \\
&= \frac{1}{(120 \text{ t}/40 \text{ t})^2 12 \Omega} + \frac{1}{(120 \text{ t}/30 \text{ t})^2 10 \Omega} \\
\frac{1}{R_1} &= \frac{1}{108 \Omega} + \frac{1}{160 \Omega} = 0.0155 \text{ S} \\
R_1 &= \frac{1}{0.0155 \text{ S}} = \mathbf{64.52 \Omega}
\end{aligned}$$

$$28. \quad \mathbf{Z}_M = \mathbf{Z}_{M_{12}} = \omega M_{12} \angle 90^\circ$$

$$\begin{aligned}
&\mathbf{E} - \mathbf{I}_1 \mathbf{Z}_1 - \mathbf{I}_1 \mathbf{Z}_{L_1} - \mathbf{I}_1(-\mathbf{Z}_m) - \mathbf{I}_2(+\mathbf{Z}_m) - \mathbf{I}_1 \mathbf{Z}_{L_2} + \mathbf{I}_2 \mathbf{Z}_{L_2} - \mathbf{I}_1(-\mathbf{Z}_m) = 0 \\
&\mathbf{E} - \mathbf{I}_1(\mathbf{Z}_1 + \mathbf{Z}_{L_1} - \mathbf{Z}_m + \mathbf{Z}_{L_2} - \mathbf{Z}_m) - \mathbf{I}_2(\mathbf{Z}_m - \mathbf{Z}_{L_2}) = 0 \\
\text{or } &\mathbf{I}_1(\mathbf{Z}_1 + \mathbf{Z}_{L_1} + \mathbf{Z}_{L_2} - 2 \mathbf{Z}_m) + \mathbf{I}_2(\mathbf{Z}_m - \mathbf{Z}_{L_2}) = \mathbf{E} \\
&\hline
&-\mathbf{I}_2 \mathbf{Z}_2 - \mathbf{Z}_{L_2} (\mathbf{I}_2 - \mathbf{I}_1) - \mathbf{I}_1(+\mathbf{Z}_m) = 0 \\
\text{or } &\mathbf{I}_1(\mathbf{Z}_m - \mathbf{Z}_{L_2}) + \mathbf{I}_2(\mathbf{Z}_2 + \mathbf{Z}_{L_2}) = 0 \\
&\hline
\end{aligned}$$

$$29. \quad \mathbf{E}_1 - \mathbf{I}_1 \mathbf{Z}_1 - \mathbf{I}_1 \mathbf{Z}_{L_1} - \mathbf{I}_2(-\mathbf{Z}_{M_{12}}) - \mathbf{I}_3(+\mathbf{Z}_{M_{13}}) = 0$$

$$\text{or } \mathbf{E}_1 - \mathbf{I}_1[\mathbf{Z}_1 + \mathbf{Z}_{L_1}] + \mathbf{I}_2 \mathbf{Z}_{M_{12}} - \mathbf{I}_3 \mathbf{Z}_{M_{13}} = 0$$

$$\hline -\mathbf{I}_2(\mathbf{Z}_2 + \mathbf{Z}_3 + \mathbf{Z}_{L_2}) + \mathbf{I}_3 \mathbf{Z}_2 - \mathbf{I}_1(-\mathbf{Z}_{M_{12}}) = 0$$

$$\text{or } -\mathbf{I}_2(\mathbf{Z}_2 + \mathbf{Z}_3 + \mathbf{Z}_{L_2}) + \mathbf{I}_3 \mathbf{Z}_2 + \mathbf{I}_1 \mathbf{Z}_{M_{12}} = 0$$

$$\hline -\mathbf{I}_3(\mathbf{Z}_2 + \mathbf{Z}_4 + \mathbf{Z}_{L_3}) + \mathbf{I}_2 \mathbf{Z}_2 - \mathbf{I}_1(+\mathbf{Z}_{M_{13}}) = 0$$

$$\text{or } -\mathbf{I}_3(\mathbf{Z}_2 + \mathbf{Z}_4 + \mathbf{Z}_{L_3}) + \mathbf{I}_2 \mathbf{Z}_2 - \mathbf{I}_1 \mathbf{Z}_{M_{13}} = 0$$

$$\begin{aligned}
\therefore &[\mathbf{Z}_1 + \mathbf{Z}_{L_1}] \mathbf{I}_1 - \mathbf{Z}_{M_{12}} \mathbf{I}_2 + \mathbf{Z}_{M_{13}} \mathbf{I}_3 = \mathbf{E}_1 \\
&\mathbf{Z}_{M_{12}} \mathbf{I}_1 - [\mathbf{Z}_2 + \mathbf{Z}_3 + \mathbf{Z}_{L_2}] \mathbf{I}_2 + \mathbf{Z}_2 \mathbf{I}_3 = 0 \\
&\mathbf{Z}_{M_{13}} \mathbf{I}_1 \mathbf{I}_2 + [\mathbf{Z}_2 + \mathbf{Z}_4 + \mathbf{Z}_{L_3}] \mathbf{I}_3 = 0 \\
&\hline
\end{aligned}$$

Chapter 23

1.
 - a. $E_\phi = E_L / \sqrt{3} = 208 \text{ V} / 1.732 = \mathbf{120.1 \text{ V}}$
 - b. $V_\phi = E_\phi = \mathbf{120.1 \text{ V}}$
 - c. $I_\phi = \frac{V_\phi}{R_\phi} = \frac{120.1 \text{ V}}{10 \Omega} = \mathbf{12.01 \text{ A}}$
 - d. $I_L = I_\phi = \mathbf{12.01 \text{ A}}$
2.
 - a. $E_\phi = E_L / \sqrt{3} = 208 \text{ V} / 1.732 = \mathbf{120.1 \text{ V}}$
 - b. $V_\phi = E_\phi = \mathbf{120.1 \text{ V}}$
 - c. $\mathbf{Z_\phi = 12 \Omega - j16 \Omega = 20 \Omega \angle -53.13^\circ}$
 $I_\phi = \frac{V_\phi}{Z_\phi} = \frac{120.1 \text{ V}}{20 \Omega} \cong \mathbf{6 \text{ A}}$
 - d. $I_L = I_\phi = \mathbf{6 \text{ A}}$
3.
 - a. $E_\phi = \mathbf{120.1 \text{ V}}$
 - b. $V_\phi = \mathbf{120.1 \text{ V}}$
 - c. $\mathbf{Z_\phi = (10 \Omega \angle 0^\circ \parallel (10 \Omega \angle -90^\circ)) = 7.071 \Omega \angle -45^\circ}$
 $I_\phi = \frac{V_\phi}{Z_\phi} = \frac{120.1 \text{ V}}{7.071 \Omega} = \mathbf{16.98 \text{ A}}$
 - d. $I_L = \mathbf{16.98 \text{ A}}$
4.
 - a. $\theta_2 = \mathbf{-120^\circ}, \theta_3 = \mathbf{120^\circ}$
 - b. $\mathbf{V_{an} = 120 \text{ V} \angle 0^\circ, V_{bn} = 120 \text{ V} \angle -120^\circ, V_{cn} = 120 \text{ V} \angle 120^\circ}$
 - c. $\mathbf{I_{an} = \frac{V_{an}}{Z_{an}} = \frac{120 \text{ V} \angle 0^\circ}{20 \Omega \angle 0^\circ} = 6 \text{ A} \angle 0^\circ}$
 $\mathbf{I_{bn} = \frac{V_{bn}}{Z_{bn}} = \frac{120 \text{ V} \angle -120^\circ}{20 \Omega \angle 0^\circ} = 6 \text{ A} \angle -120^\circ}$
 $\mathbf{I_{cn} = \frac{V_{cn}}{Z_{cn}} = \frac{120 \text{ V} \angle 120^\circ}{20 \Omega \angle 0^\circ} = 6 \text{ A} \angle 120^\circ}$
 - d. $I_L = I_\phi = \mathbf{6 \text{ A}}$
 - e. $V_L = \sqrt{3} V_\phi = \sqrt{3} (120 \text{ V}) = \mathbf{207.8 \text{ V}}$
5.
 - a. $\theta_2 = \mathbf{-120^\circ}, \theta_3 = \mathbf{+120^\circ}$
 - b. $\mathbf{V_{an} = 120 \text{ V} \angle 0^\circ, V_{bn} = 120 \text{ V} \angle -120^\circ, V_{cn} = 120 \text{ V} \angle 120^\circ}$
 - c. $\mathbf{Z_\phi = 9 \Omega + j12 \Omega = 15 \Omega \angle 53.13^\circ}$
 $\mathbf{I_{an} = \frac{120 \text{ V} \angle 0^\circ}{15 \Omega \angle 53.13^\circ} = 8 \text{ A} \angle -53.13^\circ, I_{bn} = \frac{120 \text{ V} \angle -120^\circ}{15 \Omega \angle 53.13^\circ} = 8 \text{ A} \angle -173.13^\circ}$
 $\mathbf{I_{cn} = \frac{120 \text{ V} \angle 120^\circ}{15 \Omega \angle 53.13^\circ} = 8 \text{ A} \angle 66.87^\circ}$

$$e. \quad I_L = I_\phi = \mathbf{8 \text{ A}}$$

$$f. \quad E_L = \sqrt{3} E_\phi = (1.732)(120 \text{ V}) = \mathbf{207.85 \text{ V}}$$

6. a, b. The same as problem 4.

$$c. \quad \mathbf{Z}_\phi = 6 \Omega \angle 0^\circ \parallel 8 \Omega \angle -90^\circ = 4.8 \Omega \angle -36.87^\circ$$

$$\mathbf{I}_{an} = \frac{\mathbf{V}_{an}}{\mathbf{Z}_{an}} = \frac{120 \text{ V} \angle 0^\circ}{4.8 \Omega \angle -36.87^\circ} = \mathbf{25 \text{ A} \angle 36.87^\circ}$$

$$\mathbf{I}_{bn} = \frac{\mathbf{V}_{bn}}{\mathbf{Z}_{bn}} = \frac{120 \text{ V} \angle -120^\circ}{4.8 \Omega \angle -36.87^\circ} = \mathbf{25 \text{ A} \angle -83.13^\circ}$$

$$\mathbf{I}_{cn} = \frac{\mathbf{V}_{cn}}{\mathbf{Z}_{cn}} = \frac{120 \text{ V} \angle 120^\circ}{4.8 \Omega \angle -36.87^\circ} = \mathbf{25 \text{ A} \angle 156.87^\circ}$$

$$d. \quad I_L = I_\phi = \mathbf{25 \text{ A}}$$

$$e. \quad V_L = \sqrt{3} V_\phi = \sqrt{3} (120 \text{ V}) = \mathbf{207.84 \text{ V}}$$

$$7. \quad V_\phi = V_{an} = V_{bn} = V_{cn} = \frac{V_L}{\sqrt{3}} = \frac{220 \text{ V}}{1.732} = \mathbf{127.0 \text{ V}}$$

$$\mathbf{Z}_\phi = 10 \Omega - j10 \Omega = 14.42 \Omega \angle -45^\circ$$

$$I_\phi = I_{an} = I_{bn} = I_{cn} = \frac{V_\phi}{Z_\phi} = \frac{127 \text{ V}}{14.142 \Omega} = \mathbf{8.98 \text{ A}}$$

$$I_L = I_{Aa} = I_{Bb} = I_{Cc} = I_\phi = \mathbf{8.98 \text{ A}}$$

$$8. \quad \mathbf{Z}_\phi = 12 \Omega + j16 \Omega = 20 \Omega \angle 53.13^\circ$$

$$I_\phi = \frac{V_\phi}{Z_\phi} = \frac{50 \text{ V}}{20 \Omega} = \mathbf{2.5 \text{ A}}$$

$$\mathbf{Z}_{T_\phi} = 13 \Omega + j16 \Omega = 20.62 \Omega \angle 50.91^\circ$$

$$V_\phi = I_\phi Z_{T_\phi} = (2.5 \text{ A})(20.62 \Omega) = 51.55 \text{ V}$$

$$V_L = \sqrt{3} V_\phi = (\sqrt{3})(51.55 \text{ V}) = \mathbf{89.29 \text{ V}}$$

$$9. \quad a. \quad \mathbf{E}_{AN} = \frac{22 \text{ kV}}{\sqrt{3}} \angle -30^\circ = \mathbf{12.7 \text{ kV} \angle -30^\circ}$$

$$\mathbf{E}_{BN} = \frac{22 \text{ kV}}{\sqrt{3}} \angle -150^\circ = \mathbf{12.7 \text{ kV} \angle -150^\circ}$$

$$\mathbf{E}_{CN} = \frac{22 \text{ kV}}{\sqrt{3}} \angle 90^\circ = \mathbf{12.7 \text{ kV} \angle 90^\circ}$$

$$\begin{aligned}
 \text{b, c. } \mathbf{I}_{Aa} = \mathbf{I}_{an} &= \frac{\mathbf{E}_{AN}}{\mathbf{Z}_{AN}} = \frac{12.7 \text{ kV } \angle -30^\circ}{(30 \Omega + j40 \Omega) + (0.4 \text{ k}\Omega + j1 \text{ k}\Omega)} \\
 &= \frac{12.7 \text{ kV } \angle -30^\circ}{430 \Omega + j1040 \Omega} = \frac{12.7 \text{ kV } \angle -30^\circ}{1125.39 \Omega \angle 67.54^\circ} \\
 &= \mathbf{11.29 \text{ A } \angle -97.54^\circ} \\
 \mathbf{I}_{Bb} = \mathbf{I}_{bn} &= \frac{\mathbf{E}_{BN}}{\mathbf{Z}_{BN}} = \frac{12.7 \text{ kV } \angle -150^\circ}{1125.39 \Omega \angle 67.54^\circ} = \mathbf{11.29 \text{ A } \angle -217.54^\circ} \\
 \mathbf{I}_{Cc} = \mathbf{I}_{cn} &= \frac{\mathbf{E}_{CN}}{\mathbf{Z}_{CN}} = \frac{12.7 \text{ kV } \angle 90^\circ}{1125.39 \Omega \angle 67.54^\circ} = \mathbf{11.29 \text{ A } \angle 22.46^\circ}
 \end{aligned}$$

$$\begin{aligned}
 \text{d. } \mathbf{V}_{an} = \mathbf{I}_{an} \mathbf{Z}_{an} &= (11.29 \text{ A } \angle -97.54^\circ)(400 + j1000) \\
 &= (11.29 \text{ A } \angle -97.54^\circ)(1077.03 \Omega \angle 68.2^\circ) \\
 &= \mathbf{12.16 \text{ kV } \angle -29.34^\circ} \\
 \mathbf{V}_{bn} = \mathbf{I}_{bn} \mathbf{Z}_{bn} &= (11.29 \text{ A } \angle -217.54^\circ)(1077.03 \Omega \angle 68.2^\circ) \\
 &= \mathbf{12.16 \text{ kV } \angle -149.34^\circ} \\
 \mathbf{V}_{cn} = \mathbf{I}_{cn} \mathbf{Z}_{cn} &= (11.29 \text{ A } \angle 22.46^\circ)(1077.03 \Omega \angle 68.2^\circ) \\
 &= \mathbf{12.16 \text{ kV } \angle 90.66^\circ}
 \end{aligned}$$

10. a. $E_\phi = E_L / \sqrt{3} = 208 \text{ V} / 1.732 = \mathbf{120.1 \text{ V}}$ b. $V_\phi = E_L = \mathbf{208 \text{ V}}$
- c. $I_\phi = \frac{V_\phi}{Z_\phi} = \frac{208 \text{ V}}{20 \Omega} = \mathbf{10.4 \text{ A}}$ d. $I_L = \sqrt{3} I_\phi = (1.732)(10.4 \text{ A}) = \mathbf{18 \text{ A}}$
11. a. $E_\phi = E_L / \sqrt{3} = 208 \text{ V} / 1.732 = \mathbf{120.1 \text{ V}}$ b. $V_\phi = E_L = \mathbf{208 \text{ V}}$
- c. $\mathbf{Z}_\phi = 6.8 \Omega + j14 \Omega = 15.564 \Omega \angle 64.09^\circ}$
 $I_\phi = \frac{V_\phi}{Z_\phi} = \frac{208 \text{ V}}{15.564 \Omega} = \mathbf{13.36 \text{ A}}$
- d. $I_L = \sqrt{3} I_\phi = (1.732)(13.36 \text{ A}) = \mathbf{23.14 \text{ A}}$
12. $\mathbf{Z}_\phi = 18 \Omega \angle 0^\circ \parallel 18 \Omega \angle -90^\circ = 12.728 \Omega \angle -45^\circ}$
- a. $E_\phi = V_L / \sqrt{3} = 208 \text{ V} / \sqrt{3} = \mathbf{120.09 \text{ V}}$ b. $V_\phi = \mathbf{208 \text{ V}}$
- c. $I_\phi = \frac{V_\phi}{Z_\phi} = \frac{208 \text{ V}}{12.728 \Omega} = \mathbf{16.34 \text{ A}}$
- d. $I_L = \sqrt{3} I_\phi = (1.732)(16.34 \text{ A}) = \mathbf{28.30 \text{ A}}$
13. a. $\theta_2 = \mathbf{-120^\circ}, \theta_3 = \mathbf{+120^\circ}$
- b. $\mathbf{V}_{ab} = 208 \text{ V } \angle 0^\circ, \mathbf{V}_{bc} = 208 \text{ V } \angle -120^\circ, \mathbf{V}_{ca} = 208 \text{ V } \angle 120^\circ}$

- c. —
- d.
$$\mathbf{I}_{ab} = \frac{\mathbf{V}_{ab}}{\mathbf{Z}_{ab}} = \frac{208 \text{ V } \angle 0^\circ}{22 \Omega \angle 0^\circ} = \mathbf{9.46 \text{ A } \angle 0^\circ}$$
$$\mathbf{I}_{bc} = \frac{\mathbf{V}_{bc}}{\mathbf{Z}_{bc}} = \frac{208 \text{ V } \angle 120^\circ}{22 \Omega \angle 0^\circ} = \mathbf{9.46 \text{ A } \angle -120^\circ}$$
$$\mathbf{I}_{ca} = \frac{\mathbf{V}_{ca}}{\mathbf{Z}_{ca}} = \frac{208 \text{ V } \angle 120^\circ}{22 \Omega \angle 0^\circ} = \mathbf{9.46 \text{ A } \angle 120^\circ}$$
- e.
$$I_L = \sqrt{3} I_\phi = (1.732)(9.46 \text{ A}) = \mathbf{16.38 \text{ A}}$$
- f.
$$E_\phi = E_L / \sqrt{3} = 208 \text{ V} / 1.732 = \mathbf{120.1 \text{ V}}$$
14. a. $\theta_2 = -120^\circ, \theta_3 = +120^\circ$
- b. $\mathbf{V}_{ab} = 208 \text{ V } \angle 0^\circ, \mathbf{V}_{bc} = 208 \text{ V } \angle -120^\circ, \mathbf{V}_{ca} = 208 \text{ V } \angle 120^\circ$
- c. —
- d.
$$\mathbf{Z}_\phi = 100 \Omega - j100 \Omega = 141.42 \Omega \angle -45^\circ$$
$$\mathbf{I}_{ab} = \frac{\mathbf{V}_{ab}}{\mathbf{Z}_{ab}} = \frac{208 \text{ V } \angle 0^\circ}{141.42 \Omega \angle -45^\circ} = \mathbf{1.47 \text{ A } \angle 45^\circ}$$
$$\mathbf{I}_{bc} = \frac{\mathbf{V}_{bc}}{\mathbf{Z}_{bc}} = \frac{208 \text{ V } \angle -120^\circ}{141.42 \Omega \angle -45^\circ} = \mathbf{1.47 \text{ A } \angle -75^\circ}$$
$$\mathbf{I}_{ca} = \frac{\mathbf{V}_{ca}}{\mathbf{Z}_{ca}} = \frac{208 \text{ V } \angle 120^\circ}{141.42 \Omega \angle -45^\circ} = \mathbf{1.47 \text{ A } \angle 165^\circ}$$
- e.
$$I_L = \sqrt{3} I_\phi = (1.732)(1.471 \text{ A}) = \mathbf{2.55 \text{ A}}$$
- f.
$$E_\phi = E_L / \sqrt{3} = 208 \text{ V} / 1.732 = \mathbf{120.1 \text{ V}}$$
15. a, b. The same as problem 13.
- c. —
- d.
$$\mathbf{Z}_\phi = 3 \Omega \angle 0^\circ \parallel 4 \Omega \angle 90^\circ = 2.4 \Omega \angle 36.87^\circ$$
$$\mathbf{I}_{ab} = \frac{\mathbf{V}_{ab}}{\mathbf{Z}_{ab}} = \frac{208 \text{ V } \angle 0^\circ}{2.4 \Omega \angle 36.87^\circ} = \mathbf{86.67 \text{ A } \angle -36.87^\circ}$$
$$\mathbf{I}_{bc} = \frac{\mathbf{V}_{bc}}{\mathbf{Z}_{bc}} = \frac{208 \text{ V } \angle -120^\circ}{2.4 \Omega \angle 36.87^\circ} = \mathbf{86.67 \text{ A } \angle -156.87^\circ}$$
$$\mathbf{I}_{ca} = \frac{\mathbf{V}_{ca}}{\mathbf{Z}_{ca}} = \frac{208 \text{ V } \angle 120^\circ}{2.4 \Omega \angle 36.87^\circ} = \mathbf{86.67 \text{ A } \angle 83.13^\circ}$$

$$e. \quad I_L = \sqrt{3} I_\phi = (1.732)(86.67 \text{ A}) = \mathbf{150.11 \text{ A}}$$

$$f. \quad E_\phi = \mathbf{120.1 \text{ V}}$$

$$16. \quad V_{ab} = V_{bc} = V_{ca} = \mathbf{220 \text{ V}}$$

$$\mathbf{Z}_\phi = 10 \Omega + j10 \Omega = 14.142 \Omega \angle 45^\circ$$

$$I_{ab} = I_{bc} = I_{ca} = \frac{V_\phi}{Z_\phi} = \frac{220 \text{ V}}{14.142 \Omega} = \mathbf{15.56 \text{ A}}$$

$$17. \quad a. \quad \mathbf{I}_{ab} = \frac{\mathbf{V}_{ab}}{\mathbf{Z}_{ab}} = \frac{16 \text{ kV} \angle 0^\circ}{300 \Omega + j1000 \Omega} = \frac{16 \text{ kV} \angle 0^\circ}{1044.03 \Omega \angle 73.30^\circ}$$

$$\mathbf{I}_{ab} = \mathbf{15.33 \text{ A} \angle -73.30^\circ}$$

$$\mathbf{I}_{bc} = \frac{\mathbf{V}_{bc}}{\mathbf{Z}_{bc}} = \frac{16 \text{ kV} \angle -120^\circ}{1044.03 \Omega \angle 73.30^\circ} = \mathbf{15.33 \text{ A} \angle -193.30^\circ}$$

$$\mathbf{I}_{ca} = \frac{\mathbf{V}_{ca}}{\mathbf{Z}_{ca}} = \frac{16 \text{ kV} \angle 120^\circ}{1044.03 \Omega \angle 73.30^\circ} = \mathbf{15.33 \text{ A} \angle 46.7^\circ}$$

$$b. \quad \mathbf{I}_{Aa} - \mathbf{I}_{ab} + \mathbf{I}_{ca} = 0$$

$$\begin{aligned} \mathbf{I}_{Aa} &= \mathbf{I}_{ab} - \mathbf{I}_{ca} = 15.33 \text{ A} \angle -73.30^\circ - 15.33 \text{ A} \angle 46.7^\circ \\ &= (4.41 \text{ A} - j14.68 \text{ A}) - (10.51 \text{ A} + j11.16 \text{ A}) \\ &= 4.41 \text{ A} - 10.51 \text{ A} - j(14.68 \text{ A} + 11.16 \text{ A}) \\ &= -6.11 \text{ A} - j25.84 \text{ A} = \mathbf{26.55 \text{ A} \angle -103.30^\circ} \end{aligned}$$

$$\mathbf{I}_{Bb} + \mathbf{I}_{ab} = \mathbf{I}_{bc}$$

$$\begin{aligned} \mathbf{I}_{Bb} &= \mathbf{I}_{bc} - \mathbf{I}_{ab} = 15.33 \text{ A} \angle -193.30^\circ - 15.33 \text{ A} \angle -73.30^\circ \\ &= \mathbf{26.55 \text{ A} \angle 136.70^\circ} \end{aligned}$$

$$\mathbf{I}_{Cc} + \mathbf{I}_{bc} = \mathbf{I}_{ca}$$

$$\begin{aligned} \mathbf{I}_{Cc} &= \mathbf{I}_{ca} - \mathbf{I}_{bc} = 15.33 \text{ A} \angle 46.7^\circ - 15.33 \text{ A} \angle -193.30^\circ \\ &= \mathbf{26.55 \text{ A} \angle 16.70^\circ} \end{aligned}$$

$$\begin{aligned} c. \quad \mathbf{E}_{AB} &= \mathbf{I}_{Aa}(10 \Omega + j20 \Omega) + \mathbf{V}_{ab} - \mathbf{I}_{Bb}(22.361 \Omega \angle 63.43^\circ) \\ &= (26.55 \text{ A} \angle -103.30^\circ)(22.361 \Omega \angle 63.43^\circ) + 16 \text{ kV} \angle 0^\circ \\ &\quad - (26.55 \text{ A} \angle 136.70^\circ)(22.361 \Omega \angle 63.43^\circ) \\ &= (455.65 \text{ V} - j380.58 \text{ V}) + 16,000 \text{ V} - (-557.42 \text{ V} - j204.32 \text{ V}) \\ &= 17.01 \text{ kV} - j176.26 \text{ V} \\ &= \mathbf{17.01 \text{ kV} \angle -0.59^\circ} \end{aligned}$$

$$\begin{aligned} \mathbf{E}_{BC} &= \mathbf{I}_{Bb}(22.361 \Omega \angle 63.43^\circ) + \mathbf{V}_{bc} - \mathbf{I}_{Cc}(22.361 \Omega \angle 63.53^\circ) \\ &= (26.55 \text{ A} \angle 136.70^\circ)(22.361 \Omega \angle 63.53^\circ) + 16 \text{ kV} \angle -120^\circ \\ &\quad - (26.55 \text{ A} \angle 16.70^\circ)(22.361 \Omega \angle 63.53^\circ) \\ &= \mathbf{17.01 \text{ kV} \angle -120.59^\circ} \end{aligned}$$

$$\begin{aligned} \mathbf{E}_{CA} &= \mathbf{I}_{Cc}(22.361 \Omega \angle 63.43^\circ) + \mathbf{V}_{ca} - \mathbf{I}_{Aa}(22.361 \Omega \angle 63.43^\circ) \\ &= \mathbf{17.01 \text{ kV} \angle 119.41^\circ} \end{aligned}$$

18. a. $E_\phi = E_L = \mathbf{208\text{ V}}$ b. $V_\phi = \frac{E_L}{\sqrt{3}} = \frac{208\text{ V}}{1.732} = \mathbf{120.1\text{ V}}$
- c. $I_\phi = \frac{V_\phi}{Z_\phi} = \frac{120.1\text{ V}}{30\ \Omega} = \mathbf{4.00\text{ A}}$ d. $I_L = I_\phi \cong \mathbf{4\text{ A}}$
19. a. $E_\phi = E_L = \mathbf{208\text{ V}}$ b. $V_\phi = E_L \sqrt{3} = \mathbf{120.09\text{ V}}$
- c. $I_\phi = \frac{V_\phi}{Z_\phi} = \frac{120.09\text{ V}}{16.971\ \Omega} = \mathbf{7.08\text{ A}}$ d. $I_L = I_\phi = \mathbf{7.08\text{ A}}$
20. a, b. The same as problem 18.
- c. $Z_\phi = 15\ \Omega \angle 0^\circ \parallel 20\ \Omega \angle -90^\circ = 12\ \Omega \angle -36.87^\circ$
 $I_\phi = \frac{V_\phi}{Z_\phi} = \frac{120.1\text{ V}}{12\ \Omega} \cong \mathbf{10\text{ A}}$
- d. $I_L = I_\phi \cong \mathbf{10\text{ A}}$
21. $V_{an} = V_{bn} = V_{cn} = \frac{120\text{ V}}{\sqrt{3}} = \frac{120\text{ V}}{1.732} = \mathbf{69.28\text{ V}}$
 $I_{an} = I_{bn} = I_{cn} = \frac{69.28\text{ V}}{24\ \Omega} = \mathbf{2.89\text{ A}}$
 $I_{Aa} = I_{Bb} = I_{Cc} = \mathbf{2.89\text{ A}}$
22. $V_{an} = V_{bn} = V_{cn} = \frac{120\text{ V}}{\sqrt{3}} = \mathbf{69.28\text{ V}}$
 $Z_\phi = 10\ \Omega + j20\ \Omega = 22.36\ \Omega \angle 63.43^\circ$
 $I_{an} = I_{bn} = I_{cn} = \frac{V_\phi}{Z_\phi} = \frac{69.28\text{ V}}{22.36\ \Omega} = \mathbf{3.10\text{ A}}$
 $I_{Aa} = I_{Bb} = I_{Cc} = I_\phi = \mathbf{3.10\text{ A}}$
23. $V_{an} = V_{bn} = V_{cn} = \mathbf{69.28\text{ V}}$
 $Z_\phi = 20\ \Omega \angle 0^\circ \parallel 15\ \Omega \angle -90^\circ = 12\ \Omega \angle -53.13^\circ$
 $I_{an} = I_{bn} = I_{cn} = \frac{69.28\text{ V}}{12\ \Omega} = \mathbf{5.77\text{ A}}$
 $I_{Aa} = I_{Bb} = I_{Cc} = \mathbf{5.77\text{ A}}$
24. a. $E_\phi = E_L = \mathbf{440\text{ V}}$ b. $V_\phi = E_L = E_\phi = \mathbf{440\text{ V}}$
- c. $I_\phi = \frac{V_\phi}{Z_\phi} = \frac{440\text{ V}}{220\ \Omega} = \mathbf{2\text{ A}}$ d. $I_L = \sqrt{3} I_\phi = (1.732)(2\text{ A}) = \mathbf{3.46\text{ A}}$

25. a. $E_\phi = E_L = \mathbf{440\text{ V}}$ b. $V_\phi = E_L = \mathbf{440\text{ V}}$
- c. $\mathbf{Z_\phi = 12\ \Omega - j9\ \Omega = 15\ \Omega \angle -36.87^\circ}$

$$I_\phi = \frac{V_\phi}{Z_\phi} = \frac{440\text{ V}}{15\ \Omega} = \mathbf{29.33\text{ A}}$$
- d. $I_L = \sqrt{3} I_\phi = (1.732)(29.33\text{ A}) = \mathbf{50.8\text{ A}}$
26. a, b. The same as problem 24.
- c. $\mathbf{Z_\phi = 22\ \Omega \angle 0^\circ \parallel 22\ \Omega \angle 90^\circ = 15.56\ \Omega \angle 45^\circ}$

$$I_\phi = \frac{V_\phi}{Z_\phi} = \frac{440\text{ V}}{15.56\ \Omega} = \mathbf{28.28\text{ A}}$$
- d. $I_L = \sqrt{3} I_\phi = (1.732)(28.28\text{ A}) = \mathbf{48.98\text{ A}}$
27. a. $\theta_2 = \mathbf{-120^\circ}, \theta_3 = \mathbf{+120^\circ}$
- b. $\mathbf{V_{ab} = 100\text{ V} \angle 0^\circ, V_{bc} = 100\text{ V} \angle -120^\circ, V_{ca} = 100\text{ V} \angle 120^\circ}$
- c. —
- d.
$$\mathbf{I_{ab} = \frac{V_{ab}}{Z_{ab}} = \frac{100\text{ V} \angle 0^\circ}{20\ \Omega \angle 0^\circ} = 5\text{ A} \angle 0^\circ}$$

$$\mathbf{I_{bc} = \frac{V_{bc}}{Z_{bc}} = \frac{100\text{ V} \angle -120^\circ}{20\ \Omega \angle 0^\circ} = 5\text{ A} \angle -120^\circ}$$

$$\mathbf{I_{ca} = \frac{V_{ca}}{Z_{ca}} = \frac{100\text{ V} \angle 120^\circ}{20\ \Omega \angle 0^\circ} = 5\text{ A} \angle 120^\circ}$$
- e. $I_{Aa} = I_{Bb} = I_{Cc} = \sqrt{3} (5\text{ A}) = \mathbf{8.66\text{ A}}$
28. a. $\theta_2 = \mathbf{-120^\circ}, \theta_3 = \mathbf{+120^\circ}$
- b. $\mathbf{V_{ab} = 100\text{ V} \angle 0^\circ, V_{bc} = 100\text{ V} \angle -120^\circ, V_{ca} = 100\text{ V} \angle 120^\circ}$
- c. —
- d. $\mathbf{Z_\phi = 12\ \Omega + j16\ \Omega = 20\ \Omega \angle 53.13^\circ}$

$$\mathbf{I_{ab} = \frac{V_{ab}}{Z_{ab}} = \frac{100\text{ V} \angle 0^\circ}{20\ \Omega \angle 53.13^\circ} = 5\text{ A} \angle -53.13^\circ}$$

$$\mathbf{I_{bc} = \frac{V_{bc}}{Z_{bc}} = \frac{100\text{ V} \angle -120^\circ}{20\ \Omega \angle 53.13^\circ} = 5\text{ A} \angle -173.13^\circ}$$

$$\mathbf{I}_{ca} = \frac{\mathbf{V}_{ca}}{\mathbf{Z}_{ca}} = \frac{100 \text{ V } \angle 120^\circ}{20 \Omega \angle 53.13^\circ} = 5 \text{ A } \angle 66.87^\circ$$

$$\text{e. } I_{Aa} = I_{Bb} = I_{Cc} = \sqrt{3} I_\phi = (1.732)(5 \text{ A}) = \mathbf{8.66 \text{ A}}$$

$$29. \text{ a. } \theta_2 = \mathbf{-120^\circ}, \theta_3 = \mathbf{120^\circ}$$

$$\text{b. } \mathbf{V}_{ab} = 100 \text{ V } \angle 0^\circ, \mathbf{V}_{bc} = 100 \text{ V } \angle -120^\circ, \mathbf{V}_{ca} = 100 \text{ V } \angle 120^\circ$$

$$\text{c. } -$$

$$\text{d. } \mathbf{Z}_\phi = 20 \Omega \angle 0^\circ \parallel 20 \Omega \angle -90^\circ = 14.14 \Omega \angle -45^\circ$$

$$\mathbf{I}_{ab} = \frac{100 \text{ V } \angle 0^\circ}{14.14 \Omega \angle -45^\circ} = \mathbf{7.07 \text{ A } \angle 45^\circ}$$

$$\mathbf{I}_{bc} = \frac{100 \text{ V } \angle -120^\circ}{14.14 \Omega \angle -45^\circ} = \mathbf{7.07 \text{ A } \angle -75^\circ}$$

$$\mathbf{I}_{ca} = \frac{100 \text{ V } \angle 120^\circ}{14.14 \Omega \angle -45^\circ} = \mathbf{7.07 \text{ A } \angle 165^\circ}$$

$$\text{e. } I_{Aa} = I_{Bb} = I_{Cc} = (\sqrt{3})(7.07 \text{ A}) = \mathbf{12.25 \text{ A}}$$

$$30. \quad P_T = 3I_\phi^2 R_\phi = 3(6 \text{ A})^2 12 \Omega = \mathbf{1296 \text{ W}}$$

$$Q_T = 3I_\phi^2 X_\phi = 3(6 \text{ A})^2 16 \Omega = \mathbf{1728 \text{ VAR}(C)}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{2160 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{1296 \text{ W}}{2160 \text{ VA}} = \mathbf{0.6 \text{ (leading)}}$$

$$31. \quad V_\phi = 120 \text{ V}, I_\phi = 120 \text{ V}/20 \Omega = 6 \text{ A}$$

$$P_T = 3I_\phi^2 R_\phi = 3(6 \text{ A})^2 20 \Omega = \mathbf{2160 \text{ W}}$$

$$Q_T = \mathbf{0 \text{ VAR}}$$

$$S_T = P_T = \mathbf{2160 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{2160 \text{ W}}{2160 \text{ VA}} = \mathbf{1}$$

$$32. \quad P_T = 3I_\phi^2 R_\phi = 3(8.98 \text{ A})^2 10 \Omega = \mathbf{2419.21 \text{ W}}$$

$$Q_T = 3I_\phi^2 X_\phi = 3(8.98 \text{ A})^2 10 \Omega = \mathbf{2419.21 \text{ VAR}(C)}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{3421.28 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{2419.21 \text{ W}}{3421.28 \text{ VA}} = \mathbf{0.7071 \text{ (leading)}}$$

$$33. \quad V_\phi = 208 \text{ V}$$

$$P_T = 3 \left(\frac{V_\phi^2}{R_\phi} \right) = 3 \cdot \frac{(208 \text{ V})^2}{18 \Omega} = \mathbf{7210.67 \text{ W}}$$

$$Q_T = 3 \left(\frac{V_\phi^2}{X_\phi} \right) = 3 \cdot \frac{(208 \text{ V})^2}{18 \Omega} = \mathbf{7210.67 \text{ VAR}(C)}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{10,197.42 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{7210.67 \text{ W}}{10,197.42 \text{ VA}} = \mathbf{0.707 \text{ (leading)}}$$

$$34. \quad P_T = 3I_\phi^2 R_\phi = 3(1.471 \text{ A})^2 100 \Omega = \mathbf{649.15 \text{ W}}$$

$$Q_T = 3I_\phi^2 X_\phi = 3(1.471 \text{ A})^2 100 \Omega = \mathbf{649.15 \text{ VAR}(C)}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{918.04 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{649.15 \text{ W}}{918.04 \text{ VA}} = \mathbf{0.7071 \text{ (leading)}}$$

$$35. \quad P_T = 3I_\phi^2 R_\phi = 3(15.56 \text{ A})^2 10 \Omega = \mathbf{7.26 \text{ kW}}$$

$$Q_T = 3I_\phi^2 X_\phi = 3(15.56 \text{ A})^2 10 \Omega = \mathbf{7.26 \text{ kVAR}}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{10.27 \text{ kVA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{7.263 \text{ kW}}{10.272 \text{ kVA}} = \mathbf{0.7071 \text{ (lagging)}}$$

$$36. \quad P_T = 3 \frac{V_\phi^2}{R_\phi} = \frac{3(120.1 \text{ V})^2}{15 \Omega} = \mathbf{2884.80 \text{ W}}$$

$$Q_T = 3 \frac{V_\phi^2}{X_\phi} = \frac{3(120.1 \text{ V})^2}{20 \Omega} = \mathbf{2163.60 \text{ VAR}(C)}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{3605.97 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{2884.80 \text{ W}}{3605.97 \text{ VA}} = \mathbf{0.8 \text{ (leading)}}$$

$$37. \quad \mathbf{Z_\phi = 10 \Omega + j20 \Omega = 22.36 \Omega \angle 63.43^\circ}$$

$$V_\phi = \frac{V_L}{\sqrt{3}} = \frac{120 \text{ V}}{1.732} = 69.28 \text{ V}$$

$$I_\phi = \frac{V_\phi}{Z_\phi} = \frac{69.28 \text{ V}}{22.36 \Omega} = 3.098 \text{ A}$$

$$P_T = 3I_\phi^2 R_\phi = 3(3.098 \text{ A})^2 10 \Omega = \mathbf{287.93 \text{ W}}$$

$$Q_T = 3I_\phi^2 X_\phi = 3(3.098 \text{ A})^2 20 \Omega = \mathbf{575.86 \text{ VAR}}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{643.83 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{287.93 \text{ W}}{643.83 \text{ VA}} = \mathbf{0.447 \text{ (lagging)}}$$

$$38. \quad P_T = 3 \frac{V_\phi^2}{R_\phi} = \frac{3(440 \text{ V})^2}{22 \Omega} = \mathbf{26.4 \text{ kW}}$$

$$Q_T = P_T = \mathbf{26.4 \text{ kVAR(L)}}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{37.34 \text{ kVA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{26.4 \text{ kW}}{37.34 \text{ kVA}} = \mathbf{0.707 \text{ (lagging)}}$$

$$39. \quad \mathbf{Z}_\phi = 12 \Omega + j16 \Omega = 20 \Omega \angle 53.13^\circ$$

$$I_\phi = \frac{V_\phi}{Z_\phi} = \frac{100 \text{ V}}{20 \Omega} = 5 \text{ A}$$

$$P_T = 3I_\phi^2 R_\phi = 3(5 \text{ A})^2 12 \Omega = \mathbf{900 \text{ W}}$$

$$Q_T = 3I_\phi^2 X_\phi = 3(5 \text{ A})^2 16 \Omega = \mathbf{1200 \text{ VAR(L)}}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{1500 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{900 \text{ W}}{1500 \text{ VA}} = \mathbf{0.6 \text{ (lagging)}}$$

$$40. \quad P_T = \sqrt{3} E_L I_L \cos \theta$$

$$4800 \text{ W} = (1.732)(200 \text{ V})I_L (0.8)$$

$$I_L = 17.32 \text{ A}$$

$$I_\phi = \frac{I_L}{\sqrt{3}} = \frac{17.32 \text{ A}}{1.732} = 10 \text{ A}$$

$$\theta = \cos^{-1} 0.8 = 36.87^\circ$$

$$\mathbf{Z}_\phi = \frac{\mathbf{V}_\phi}{\mathbf{I}_\phi} = \frac{200 \text{ V} \angle 0^\circ}{10 \text{ A} \angle -36.87^\circ} = 20 \Omega \angle 36.87^\circ = \mathbf{16 \Omega + j12 \Omega}$$

$$41. \quad P_T = \sqrt{3} E_L I_L \cos \theta$$

$$1200 \text{ W} = \sqrt{3} (208 \text{ V})I_L (0.6) \Rightarrow I_L = 5.55 \text{ A}$$

$$V_\phi = \frac{V_L}{\sqrt{3}} = \frac{208 \text{ V}}{1.732} = 120.1 \text{ V}$$

$$\theta = \cos^{-1} 0.6 = 53.13^\circ \text{ (leading)}$$

$$\mathbf{Z}_\phi = \frac{\mathbf{V}_\phi}{\mathbf{I}_\phi} = \frac{120.1 \text{ V} \angle 0^\circ}{5.55 \text{ A} \angle 53.13^\circ} = 21.64 \Omega \angle -53.13^\circ = \underbrace{12.98 \Omega}_R - j\underbrace{17.31 \Omega}_{X_C}$$

42. Δ : $\mathbf{Z}_\phi = 15 \Omega + j20 \Omega = 25 \Omega \angle 53.13^\circ$

$$I_\phi = \frac{V_\phi}{Z_\phi} = \frac{125 \text{ V}}{25 \Omega} = 5 \text{ A}$$

$$P_T = 3I_\phi^2 R_\phi = 3(5 \text{ A})^2 15 \Omega = \mathbf{1125 \text{ W}}$$

$$Q_T = 3I_\phi^2 X_\phi = 3(5 \text{ A})^2 20 \Omega = \mathbf{1500 \text{ VAR}(L)}$$

Y: $V_\phi = V_L / \sqrt{3} = 125 \text{ V} / 1.732 = \mathbf{72.17 \text{ V}}$

$$\mathbf{Z}_\phi = 3 \Omega - j4 \Omega = 5 \Omega \angle -53.13^\circ$$

$$I_\phi = \frac{V_\phi}{Z_\phi} = \frac{72.17 \text{ V}}{5 \Omega} = \mathbf{14.43 \text{ A}}$$

$$P_T = 3I_\phi^2 R_\phi = 3(14.43 \text{ A})^2 3 \Omega = \mathbf{1874.02 \text{ W}}$$

$$Q_T = 3I_\phi^2 X_\phi = 3(14.43 \text{ A})^2 4 \Omega = \mathbf{2498.7 \text{ VAR}}$$

$$P_T = 1125 \text{ W} + 1874.02 \text{ W} = \mathbf{2999.02 \text{ W}}$$

$$Q_T = 1500 \text{ VAR}(L) - 2498.7 \text{ VAR}(C) = \mathbf{998.7 \text{ VAR}(C)}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{3161 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{2999.02 \text{ W}}{3161 \text{ VA}} = \mathbf{0.949 \text{ (leading)}}$$

43. a. $E_\phi = \frac{16 \text{ kV}}{\sqrt{3}} = \mathbf{9,237.6 \text{ V}}$

b. $I_L = I_\phi = \mathbf{80 \text{ A}}$

c. $P_{\phi_L} = \frac{1200 \text{ kW}}{3} = 400 \text{ kW}$

$$P_{4\Omega} = (80 \text{ A})^2 4 \Omega = 25.6 \text{ kW}$$

$$P_T = 3P_\phi = 3(25.6 \text{ kW} + 400 \text{ kW}) = \mathbf{1276.8 \text{ kW}}$$

d. $F_p = \frac{P_T}{S_T}, S_T = \sqrt{3} V_L I_L = \sqrt{3} (16 \text{ kV})(80 \text{ A}) = 2,217.025 \text{ kVA}$

$$F_p = \frac{1,276.8 \text{ kW}}{2,217.025 \text{ kVA}} = \mathbf{0.576 \text{ lagging}}$$

e. $\theta_L = \cos^{-1} 0.576 = 54.83^\circ \text{ (lagging)}$

$$\mathbf{I_{Aa}} = \frac{\mathbf{E_{AN} \angle 0^\circ}}{\underbrace{\mathbf{Z_T \angle 54.83^\circ}}_{\uparrow}} \Rightarrow \underbrace{\mathbf{80 \text{ A}}}_{\text{given}} \angle -54.83^\circ$$

for entire load

- f. $\mathbf{V}_{an} = \mathbf{E}_{AN} - \mathbf{I}_{Aa}(4 \Omega + j20 \Omega)$
 $= 9237.6 \text{ V } \angle 0^\circ - (80 \text{ A } \angle -54.83^\circ)(20.396 \Omega \angle 78.69^\circ)$
 $= 9237.6 \text{ V } \angle 0^\circ - 1631.68 \text{ V } \angle 23.86^\circ$
 $= 9237.6 \text{ V} - (1492.22 \text{ V} + j660 \text{ V})$
 $= 7745.38 \text{ V} - j660 \text{ V}$
 $= \mathbf{7773.45 \text{ V } \angle -4.87^\circ}$
- g. $\mathbf{Z}_\phi = \frac{\mathbf{V}_{an}}{\mathbf{I}_{Aa}} = \frac{7773.45 \text{ V } \angle -4.87^\circ}{80 \text{ A } \angle -54.83^\circ} = 97.168 \Omega \angle 49.95^\circ$
 $= \underbrace{62.52 \Omega}_R + j \underbrace{74.38 \Omega}_{X_C}$
- h. $F_p(\text{entire system}) = \mathbf{0.576 \text{ (lagging)}}$
 $F_p(\text{load}) = \mathbf{0.643 \text{ (lagging)}}$
- i. $\eta = \frac{P_o}{P_i} = \frac{P_i - P_{\text{lost}}}{P_i} = \frac{1276.8 \text{ kW} - 3(25.6 \text{ kW})}{1276.8 \text{ kW}} = 0.9398 \Rightarrow \mathbf{93.98\%}$

44. a. —

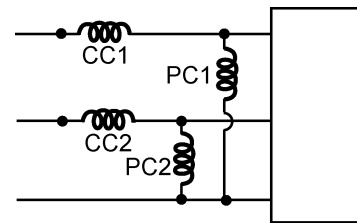
b. $V_\phi = \frac{220 \text{ V}}{\sqrt{3}} = 127.02 \text{ V}, \mathbf{Z}_\phi = 10 \Omega - j10 \Omega = 14.14 \Omega \angle -45^\circ$
 $I_\phi = \frac{V_\phi}{Z_\phi} = \frac{127.02 \text{ V}}{14.14 \Omega} = 8.98 \text{ A}$
 $P_T = 3I_\phi^2 R_\phi = 3(8.98 \text{ A})^2 10 \Omega = \mathbf{2419.2 \text{ W}}$
 Each wattmeter: $\frac{2419.2 \text{ W}}{3} = \mathbf{806.4 \text{ W}}$

45. b. $P_T = \mathbf{5899.64 \text{ W}}, P_{\text{meter}} = \mathbf{1966.55 \text{ W}}$

46. a. —

b. $P_T = P_\ell + P_h = 85 \text{ W} + 200 \text{ W} = \mathbf{285 \text{ W}}$

c. $0.2 \Rightarrow \frac{P_\ell}{P_h} = 0.5$
 $P_h = \frac{P_\ell}{0.5} = \frac{100 \text{ W}}{0.5} = \mathbf{200 \text{ W}}$
 $P_T = P_h - P_\ell = 200 \text{ W} - 100 \text{ W} = \mathbf{100 \text{ W}}$



48. a.
$$\mathbf{I}_{ab} = \frac{\mathbf{E}_{AB}}{R \angle 0^\circ} = \frac{208 \text{ V} \angle 0^\circ}{10 \Omega \angle 0^\circ} = \mathbf{20.8 \text{ A} \angle 0^\circ}$$

$$\mathbf{I}_{bc} = \frac{\mathbf{E}_{BC}}{R + jX_L} = \frac{208 \text{ V} \angle -120^\circ}{10 \Omega + j10 \Omega} = \frac{208 \text{ V} \angle -120^\circ}{14.142 \Omega \angle 45^\circ} = \mathbf{14.708 \text{ A} \angle -165^\circ}$$

$$\mathbf{I}_{ca} = \frac{\mathbf{E}_{CA}}{R - jX_C} = \frac{208 \text{ V} \angle 120^\circ}{10 \Omega - j10 \Omega} = \frac{208 \text{ V} \angle 120^\circ}{14.142 \Omega \angle -45^\circ} = \mathbf{14.708 \text{ A} \angle 165^\circ}$$

b.
$$\mathbf{I}_{Aa} + \mathbf{I}_{ca} - \mathbf{I}_{ab} = 0$$

$$\begin{aligned} \mathbf{I}_{Aa} &= \mathbf{I}_{ab} - \mathbf{I}_{ca} \\ &= 20.8 \text{ A} \angle 0^\circ - 14.708 \text{ A} \angle 165^\circ \\ &= 20.8 \text{ A} - (-14.207 \text{ A} + j3.807 \text{ A}) \\ &= 35.007 \text{ A} - j3.807 \text{ A} \\ &= \mathbf{35.213 \text{ A} \angle -6.207^\circ} \end{aligned}$$

$$\mathbf{I}_{Bb} + \mathbf{I}_{ab} - \mathbf{I}_{bc} = 0$$

$$\begin{aligned} \mathbf{I}_{Bb} &= \mathbf{I}_{bc} - \mathbf{I}_{ab} \\ &= 14.708 \text{ A} \angle -165^\circ - 20.8 \text{ A} \angle 0^\circ \\ &= (-14.207 \text{ A} - j3.807 \text{ A}) - 20.8 \text{ A} \\ &= -35.007 \text{ A} - j3.807 \text{ A} \\ &= \mathbf{35.213 \text{ A} \angle -173.79^\circ} \end{aligned}$$

$$\mathbf{I}_{Cc} + \mathbf{I}_{bc} - \mathbf{I}_{ca} = 0$$

$$\begin{aligned} \mathbf{I}_{Cc} &= \mathbf{I}_{ca} - \mathbf{I}_{bc} \\ &= 14.708 \text{ A} \angle 165^\circ - 14.708 \text{ A} \angle -165^\circ \\ &= (-14.207 \text{ A} + j3.807 \text{ A}) - (-14.207 \text{ A} - j3.807 \text{ A}) \\ &= \mathbf{7.614 \text{ A} \angle 90^\circ} \end{aligned}$$

c.
$$\begin{aligned} P_1 &= V_{ac} I_{Aa} \cos \theta_{V_{ac} I_{Aa}} \quad \mathbf{V}_{ca} = V_{ca} \angle \theta - 180^\circ = 208 \text{ V} \angle 120^\circ - 180^\circ \\ &= 208 \text{ V} \angle -60^\circ \\ \mathbf{I}_{Aa} &= 35.213 \text{ A} \angle -6.207^\circ \\ P_1 &= (208 \text{ V})(35.213 \text{ A}) \cos 53.793^\circ \\ &= \mathbf{4.326 \text{ kW}} \end{aligned}$$

$$\begin{aligned} P_2 &= V_{bc} I_{Bb} \cos \theta_{V_{bc} I_{Bb}} \quad \mathbf{V}_{bc} = 208 \text{ V} \angle -120^\circ \\ \mathbf{I}_{Bb} &= 35.213 \text{ A} \angle -173.79^\circ \\ P_2 &= (208 \text{ V})(35.213 \text{ A}) \cos 53.79^\circ \\ &= \mathbf{4.327 \text{ kW}} \end{aligned}$$

d.
$$P_T = P_1 + P_2 = 4.326 \text{ kW} + 4.327 \text{ kW} = \mathbf{8.653 \text{ kW}}$$

49. a. $V_\phi = E_\phi = \frac{E_L}{\sqrt{3}} = \mathbf{120.09\text{ V}}$
- b. $I_{an} = \frac{V_{an}}{Z_{an}} = \frac{120.09\text{ V}}{14.142\ \Omega} = \mathbf{8.49\text{ A}}$
 $I_{bn} = \frac{V_{bn}}{Z_{bn}} = \frac{120.09\text{ V}}{16.971\ \Omega} = \mathbf{7.08\text{ A}}$
 $I_{cn} = \frac{V_{cn}}{Z_{cn}} = \frac{120.09\text{ V}}{2.828\ \Omega} = \mathbf{42.47\text{ A}}$
- c. $P_T = I_{an}^2 10\ \Omega + I_{bn}^2 12\ \Omega + I_{cn}^2 2\ \Omega$
 $= (8.49\text{ A})^2 10\ \Omega + (7.08\text{ A})^2 12\ \Omega + (42.47\text{ A})^2 2\ \Omega$
 $= 720.80\text{ W} + 601.52\text{ W} + 3.61\text{ kW}$
 $= \mathbf{4.93\text{ kW}}$
 $Q_T = P_T = \mathbf{4.93\text{ kVAR}(L)}$
 $S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{6.97\text{ kVA}}$
 $F_p = \frac{P_T}{S_T} = \mathbf{0.707\text{ (lagging)}}$
- d. $\mathbf{E_{an} = 120.09\text{ V} \angle -30^\circ, \mathbf{E_{bn} = 120.09\text{ V} \angle -150^\circ, \mathbf{E_{cn} = 120.09\text{ V} \angle 90^\circ}}$
 $\mathbf{I_{an} = \frac{E_{an}}{Z_{an}} = \frac{120.09\text{ V} \angle -30^\circ}{10\ \Omega + j10\ \Omega} = \frac{120.09\text{ V} \angle -30^\circ}{14.142\ \Omega \angle 45^\circ} = \mathbf{8.49\text{ A} \angle -75^\circ}}$
 $\mathbf{I_{bn} = \frac{E_{bn}}{Z_{bn}} = \frac{120.09\text{ V} \angle -150^\circ}{12\ \Omega + j12\ \Omega} = \frac{120.09\text{ V} \angle -150^\circ}{16.971\ \Omega \angle 45^\circ} = \mathbf{7.08\text{ A} \angle -195^\circ}}$
 $\mathbf{I_{cn} = \frac{E_{cn}}{Z_{cn}} = \frac{120.09\text{ V} \angle 90^\circ}{2\ \Omega + j2\ \Omega} = \frac{120.09\text{ V} \angle 90^\circ}{2.828\ \Omega \angle 45^\circ} = \mathbf{42.47\text{ A} \angle 45^\circ}}$
- e. $\mathbf{I_N = I_{an} + I_{bn} + I_{cn}}$
 $= 8.49\text{ A} \angle -75^\circ + 7.08\text{ A} \angle -195^\circ + 42.47\text{ A} \angle 45^\circ$
 $= (2.02\text{ A} - j8.20\text{ A}) + (-6.84\text{ A} + j1.83\text{ A}) + (30.30\text{ A} + j30.30\text{ A})$
 $= 25.66\text{ A} - j23.93\text{ A}$
 $= \mathbf{35.09\text{ A} \angle -43.00^\circ}$
50. $\mathbf{Z_1 = 12\ \Omega - j16\ \Omega = 20\ \Omega \angle -53.13^\circ, \mathbf{Z_2 = 3\ \Omega + j4\ \Omega = 5\ \Omega \angle 53.13^\circ}}$
 $\mathbf{Z_3 = 20\ \Omega \angle 0^\circ}$

$$\begin{aligned} \mathbf{E_{AB} = 200\text{ V} \angle 0^\circ, \mathbf{E_{BC} = 200\text{ V} \angle -120^\circ, \mathbf{E_{CA} = 200\text{ V} \angle 120^\circ}} \\ \mathbf{Z_\Delta = Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_3} \\ = (20\ \Omega \angle -53.13^\circ)(5\ \Omega \angle 53.13^\circ) + (20\ \Omega \angle -53.13^\circ)(20\ \Omega \angle 0^\circ) \\ + (5\ \Omega \angle 53.13^\circ)(20\ \Omega \angle 0^\circ) \\ = 100\ \Omega \angle 0^\circ + 400\ \Omega \angle -53.13^\circ + 100\ \Omega \angle 53.13^\circ \\ = 100\ \Omega + (240\ \Omega - j320\ \Omega) + (60\ \Omega + j80\ \Omega) \\ = 400\ \Omega - j240\ \Omega \\ = 466.48\ \Omega \angle -30.96^\circ \end{aligned}$$

$$\mathbf{I}_{an} = \frac{\mathbf{E}_{AB}\mathbf{Z}_3 - \mathbf{E}_{CA}\mathbf{Z}_2}{\mathbf{Z}_{\Delta}} = \frac{(200 \text{ V } \angle 0^\circ)(20 \Omega \angle 0^\circ) - (200 \text{ V } \angle 120^\circ)(5 \Omega \angle 53.13^\circ)}{\mathbf{Z}_{\Delta}}$$

$$= \frac{4000 \text{ A } \angle 0^\circ - 1000 \text{ A } \angle 173.13^\circ}{466.48 \angle -30.96^\circ} = \mathbf{10.71 \text{ A } \angle 29.59^\circ}$$

$$\mathbf{I}_{bn} = \frac{\mathbf{E}_{BC}\mathbf{Z}_1 - \mathbf{E}_{AB}\mathbf{Z}_3}{\mathbf{Z}_{\Delta}} = \frac{(200 \text{ V } \angle -120^\circ)(20 \Omega \angle -53.13^\circ) - (200 \text{ V } \angle 0^\circ)(20 \Omega \angle 0^\circ)}{\mathbf{Z}_{\Delta}}$$

$$= \frac{4000 \text{ A } \angle -173.13^\circ - 4000 \text{ A } \angle 0^\circ}{466.48 \angle -30.96^\circ} = \mathbf{17.12 \text{ A } \angle -145.61^\circ}$$

$$\mathbf{I}_{cn} = \frac{\mathbf{E}_{CA}\mathbf{Z}_2 - \mathbf{E}_{BC}\mathbf{Z}_1}{\mathbf{Z}_{\Delta}} = \frac{(200 \text{ V } \angle 120^\circ)(5 \Omega \angle 53.13^\circ) - (200 \text{ V } \angle -120^\circ)(20 \Omega \angle -53.13^\circ)}{\mathbf{Z}_{\Delta}}$$

$$= \frac{1000 \text{ A } \angle 173.13^\circ - 4000 \text{ A } \angle -173.13^\circ}{466.48 \angle -30.96^\circ} = \mathbf{6.51 \text{ A } \angle 42.32^\circ}$$

$$P_T = I_{an}^2 12 \Omega + I_{bn}^2 4 \Omega + I_{cn}^2 20 \Omega$$

$$= 1376.45 \text{ W} + 1172.38 \text{ W} + 847.60 \text{ W} = \mathbf{3396.43 \text{ W}}$$

$$Q_T = I_{an}^2 16 \Omega + I_{bn}^2 3 \Omega = 1835.27 \text{ VAR}(C) + 879.28 \text{ VAR}(L) = \mathbf{955.99 \text{ VAR}(C)}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = \mathbf{3508.40 \text{ VA}}$$

$$F_p = \frac{P_T}{S_T} = \frac{3396.43 \text{ W}}{3508.40 \text{ VA}} = \mathbf{0.968 \text{ (leading)}}$$

Chapter 24

1.
 - a. **positive-going**
 - b. $V_b = 2 \text{ V}$
 - c. $t_p = 0.2 \text{ ms}$
 - d. Amplitude = $8 \text{ V} - 2 \text{ V} = 6 \text{ V}$
 - e.

$$\% \text{ tilt} = \frac{V_1 - V_2}{V} \times 100\%$$

$$V = \frac{8 \text{ V} + 7.5 \text{ V}}{2} = 7.75 \text{ V}$$

$$\% \text{ tilt} = \frac{8 \text{ V} - 7.5 \text{ V}}{7.75 \text{ V}} \times 100\% = \mathbf{6.5\%}$$
 - f.

$$\text{prf} = \frac{1}{T} = \frac{1}{(2.0 \text{ ms} - 0.4 \text{ ms})} = \frac{1}{1.6 \text{ ms}} = \mathbf{625 \text{ kHz}}$$
 - g.

$$\frac{t_p}{T} \times 100\% = \frac{0.2 \text{ ms}}{1.6 \text{ ms}} \times 100\% = \mathbf{12.5\%}$$
2.
 - a. **negative-going**
 - b. $+7 \text{ mV}$
 - c. $3 \mu\text{s}$
 - d. -8 mV (from base line level)
 - e.

$$V = \frac{-8 \text{ mV} - 7 \text{ mV}}{2} = \frac{-15 \text{ mV}}{2} = -7.5 \text{ mV}$$

$$\% \text{ Tilt} = \frac{V_1 - V_2}{V} \times 100\% = \frac{-8 \text{ mV} - (-7 \text{ mV})}{-7.5 \text{ mV}} \times 100\%$$

$$= \frac{-1 \text{ mV}}{-7.5 \text{ mV}} \times 100\% = \mathbf{13.3\%}$$
 - f.

$$T = 15 \mu\text{s} - 7 \mu\text{s} = 8 \mu\text{s}$$

$$\text{prf} = \frac{1}{T} = \frac{1}{8 \mu\text{s}} = \mathbf{125 \text{ kHz}}$$
 - g.

$$\text{Duty cycle} = \frac{t_p}{T} \times 100\% = \frac{3 \mu\text{s}}{8 \mu\text{s}} \times 100\% = \mathbf{37.5\%}$$
3.
 - a. **positive-going**
 - b. $V_b = 10 \text{ mV}$
 - c. $t_p = \left(\frac{8}{10}\right) 4 \text{ ms} = \mathbf{3.2 \text{ ms}}$
 - d. Amplitude = $(30 - 10) \text{ mV} = \mathbf{20 \text{ mV}}$
 - e.

$$\% \text{ tilt} = \frac{V_1 - V_2}{V} \times 100\%$$

$$V = \frac{30 \text{ mV} + 28 \text{ mV}}{2} = 29 \text{ mV}$$

$$\% \text{ tilt} = \frac{30 \text{ mV} - 28 \text{ mV}}{29 \text{ mV}} \times 100\% \cong \mathbf{6.9\%}$$

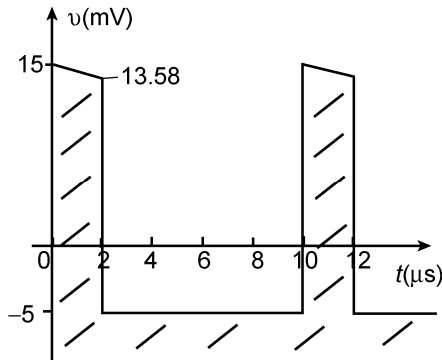
$$4. \quad t_r \cong (0.2 \text{ div.})(2 \text{ ms/div.}) = \mathbf{0.4 \text{ ms}}$$

$$t_f \cong (0.4 \text{ div.})(2 \text{ ms/div.}) = \mathbf{0.8 \text{ ms}}$$

$$5. \quad \text{tilt} = \frac{V_1 - V_2}{V} = 0.1 \text{ with } V = \frac{V_1 + V_2}{2}$$

Substituting V into top equation,

$$\frac{V_1 - V_2}{\frac{V_1 + V_2}{2}} = 0.1 \text{ leading to } V_2 = \frac{0.95 V_1}{1.05} \text{ or } V_2 = 0.905(15 \text{ mV}) = 13.58 \text{ mV}$$



$$6. \quad a. \quad t_r = 80\% \text{ of straight line segment}$$

$$= 0.8(2 \mu\text{s}) = \mathbf{1.6 \mu\text{s}}$$

$$b. \quad t_f = 80\% \text{ of } 4 \mu\text{s interval}$$

$$= 0.8(4 \mu\text{s}) = \mathbf{3.2 \mu\text{s}}$$

$$c. \quad \text{At 50\% level (10 mV)}$$

$$t_p = (8 - 1)\mu\text{s} = \mathbf{7 \mu\text{s}}$$

$$d. \quad \text{prf} = \frac{1}{T} = \frac{1}{20 \mu\text{s}} = \mathbf{50 \text{ kHz}}$$

$$7. \quad a. \quad T = (4.8 - 2.4)\text{div.} [50 \mu\text{s/div.}] = \mathbf{120 \mu\text{s}}$$

$$b. \quad f = \frac{1}{T} = \frac{1}{120 \mu\text{s}} = 8.33 \text{ kHz}$$

$$c. \quad \text{Maximum Amplitude: } (2.2 \text{ div.})(0.2 \text{ V/div.}) = 0.44 \text{ V} = \mathbf{440 \text{ mV}}$$

$$\text{Minimum Amplitude: } (0.4 \text{ div.})(0.2 \text{ V/div.}) = 0.08 \text{ V} = \mathbf{80 \text{ mV}}$$

$$8. \quad T = (3.6 - 2.0)\text{ms} = 1.6 \text{ ms}$$

$$\text{prf} = \frac{1}{T} = \frac{1}{1.6 \text{ ms}} = \mathbf{625 \text{ Hz}}$$

$$\text{Duty cycle} = \frac{t_p}{T} \times 100\% = \frac{0.2 \text{ ms}}{1.6 \text{ ms}} \times 100\% = \mathbf{12.5\%}$$

9. $T = (15 - 7)\mu s = 8 \mu s$

$$\text{prf} = \frac{1}{T} = \frac{1}{8 \mu s} = \mathbf{125 \text{ kHz}}$$

$$\text{Duty cycle} = \frac{t_p}{T} \times 100\% = \frac{(20-15)\mu s}{8\mu s} \times 100\% = \frac{5}{8} \times 100\% = \mathbf{62.5\%}$$

10. $T = (3.6 \text{ div.})(2 \text{ ms/div.}) = 7.2 \text{ ms}$

$$\text{prf} = \frac{1}{T} = \frac{1}{7.2 \text{ ms}} = \mathbf{138.89 \text{ Hz}}$$

$$\text{Duty cycle} = \frac{t_p}{T} \times 100\% = \frac{1.6 \text{ div.}}{3.6 \text{ div.}} \times 100\% = \mathbf{44.4\%}$$

11. a. $T = (9 - 1)\mu s = \mathbf{8\,\mu s}$

b. $t_p = (3 - 1)\mu s = \mathbf{2\,\mu s}$

c. $\text{prf} = \frac{1}{T} = \frac{1}{8 \mu\text{s}} = \mathbf{125 \text{ kHz}}$

d. $V_{av} = (\text{Duty cycle})(\text{Peak value}) + (1 - \text{Duty cycle})(V_b)$

$$\text{Duty cycle} = \frac{t_p}{T} \times 100\% = \frac{2 \mu s}{8 \mu s} \times 100\% = 25\%$$

$$\begin{aligned} V_{av} &= (0.25)(6 \text{ mV}) + (1 - 0.25)(-2 \text{ mV}) \\ &= 1.5 \text{ mV} - 1.5 \text{ mV} = \mathbf{0 \text{ V}} \end{aligned}$$

or

$$V_{av} = \frac{(2 \mu s)(6 \text{ mV}) - (2 \mu s)(6 \text{ mV})}{8 \mu s} = \mathbf{0 \text{ V}}$$

e. $V_{\text{eff}} = \sqrt{\frac{(36 \times 10^{-6})(2 \mu\text{s}) + (4 \times 10^{-6})(6 \mu\text{s})}{8 \mu\text{s}}} = 3.46 \text{ mV}$

12. Eq. 24.5 cannot be applied due to tilt in the waveform.
(Method of Section 13.6)

Between 2 and 3.6 ms

$$\begin{aligned} V_{av} &= \frac{(3.4 \text{ ms} - 2 \text{ ms})(2 \text{ V}) + (3.6 \text{ ms} - 3.4 \text{ ms})(7.5 \text{ V}) + \frac{1}{2}(3.6 \text{ ms} - 3.4 \text{ ms})(0.5 \text{ V})}{3.6 \text{ ms} - 2 \text{ ms}} \\ &= \frac{(1.4 \text{ ms})(2 \text{ V}) + (0.2 \text{ ms})(7.5 \text{ V}) + \frac{1}{2}(0.2 \text{ ms})(0.5 \text{ V})}{1.6 \text{ ms}} \\ &= \frac{2.8 \text{ V} + 1.5 \text{ V} + 0.05 \text{ V}}{1.6} = \mathbf{2.719 \text{ V}} \end{aligned}$$

13. Ignoring tilt and using 20 mV level to define t_p
 $t_p = (2.8 \text{ div.} - 1.2 \text{ div.})(2 \text{ ms/div.}) = 3.2 \text{ ms}$
 $T = (\text{at } 10 \text{ mV level}) = (4.6 \text{ div.} - 1 \text{ div.})(2 \text{ ms/div.}) = 7.2 \text{ ms}$
 $\text{Duty cycle} = \frac{t_p}{T} \times 100\% = \frac{3.2 \text{ ms}}{7.2 \text{ ms}} \times 100\% = 44.4\%$

$$\begin{aligned} V_{av} &= (\text{Duty cycle})(\text{peak value}) + (1 - \text{Duty cycle})(V_b) \\ &= (0.444)(30 \text{ mV}) + (1 - 0.444)(10 \text{ mV}) \\ &= 13.320 \text{ mV} + 5.560 \text{ mV} \\ &= \mathbf{18.88 \text{ mV}} \end{aligned}$$

14. $V_{av} = (\text{Duty cycle})(\text{Peak value}) + (1 - \text{Duty cycle})(V_b)$

$$\begin{aligned} \text{Duty cycle} &= \frac{t_p}{T} \text{ (decimal form)} \\ &= \frac{(8-1)\mu s}{20 \mu s} = 0.35 \end{aligned}$$

$$\begin{aligned} V_{av} &= (0.35)(20 \text{ mV}) + (1 - 0.35)(0) \\ &= 7 \text{ mV} + 0 \\ &= \mathbf{7 \text{ mV}} \end{aligned}$$

15. Using methods of Section 13.8:

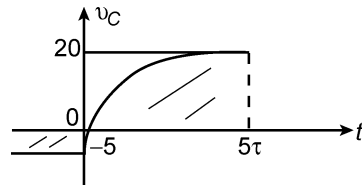
$$\begin{aligned} A_1 &= b_1 h_1 = [(0.2 \text{ div.})(50 \mu s/\text{div.})][(2 \text{ div.})(0.2 \text{ V/div.})] = 4 \mu sV \\ A_2 &= b_2 h_2 = [(0.2 \text{ div.})(50 \mu s/\text{div.})][(2.2 \text{ div.})(0.2 \text{ V/div.})] = 4.4 \mu sV \\ A_3 &= b_3 h_3 = [(0.2 \text{ div.})(50 \mu s/\text{div.})][(1.4 \text{ div.})(0.2 \text{ V/div.})] = 2.8 \mu sV \\ A_4 &= b_4 h_4 = [(0.2 \text{ div.})(50 \mu s/\text{div.})][(1 \text{ div.})(0.2 \text{ V/div.})] = 2.0 \mu sV \\ A_5 &= b_5 h_5 = [(0.2 \text{ div.})(50 \mu s/\text{div.})][(0.4 \text{ div.})(0.2 \text{ V/div.})] = 0.8 \mu sV \end{aligned}$$

$$V_{av} = \frac{(4 + 4.4 + 2.8 + 2.0 + 0.8)\mu sV}{120 \mu s} = \mathbf{117 \text{ mV}}$$

16. Using the defined polarity of Fig. 24.57 for v_C , $V_i = -5 \text{ V}$, $V_f = +20 \text{ V}$ and $\tau = RC = (10 \text{ k}\Omega)(0.02 \mu F) = 0.2 \text{ ms}$

$$\begin{aligned} \text{a. } v_C &= V_i + (V_f - V_i)(1 - e^{-t/\tau}) \\ &= -5 + (20 - (-5))(1 - e^{-t/0.2 \text{ ms}}) \\ &= -5 + 25(1 - e^{-t/0.2 \text{ ms}}) \\ &= -5 + 25 - 25e^{-t/0.2 \text{ ms}} \\ v_C &= \mathbf{20 \text{ V} - 25 \text{ V}e^{-t/0.2 \text{ ms}}} \end{aligned}$$

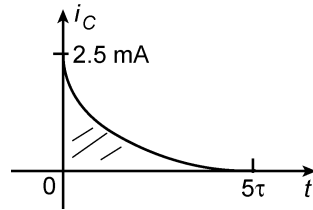
b.



c. $I_i = 0$

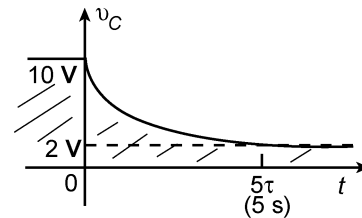
$$i_C = \frac{E - v_C}{R} = \frac{20 \text{ V} - [20 \text{ V} - 25 \text{ V} e^{-t/0.2 \text{ ms}}]}{10 \text{ k}\Omega} = 2.5 \text{ mA} e^{-t/0.2 \text{ ms}}$$

d.

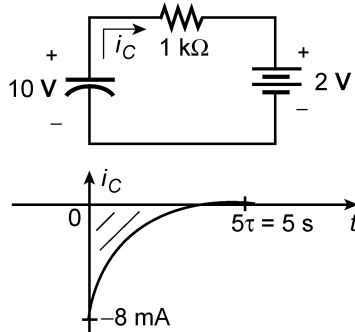


$$\begin{aligned} 17. \quad v_C &= V_i + (V_f - V_i)(1 - e^{-t/RC}) \\ &= 8 + (4 - 8)(1 - e^{-t/20 \text{ ms}}) \\ &= 8 - 4(1 - e^{-t/20 \text{ ms}}) \\ &= 8 - 4 + 4e^{-t/20 \text{ ms}} \\ &= 4 + 4e^{-t/20 \text{ ms}} \\ v_C &= 4 \text{ V}(1 + e^{-t/20 \text{ ms}}) \end{aligned} \quad \begin{aligned} \tau &= RC = (2 \text{ k}\Omega)(10 \mu\text{F}) \\ &= 20 \text{ ms} \end{aligned}$$

$$\begin{aligned} 18. \quad V_i &= 10 \text{ V}, V_f = 2 \text{ V}, \tau = RC = (1 \text{ k}\Omega)(1000 \mu\text{F}) = 1 \text{ s} \\ v_C &= V_i + (V_f - V_i)(1 - e^{-t/\tau}) \\ &= 10 \text{ V} + (2 \text{ V} - 10 \text{ V})(1 - e^{-t}) \\ &= 10 - 8(1 - e^{-t}) \\ &= 10 - 8 + 8e^{-t} \\ v_C &= 2 \text{ V} + 8 \text{ V} e^{-t} \end{aligned}$$



19. $V_i = 10 \text{ V}, I_i = 0 \text{ A}$

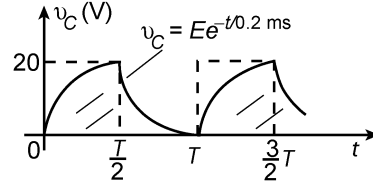


Using the defined direction of i_C

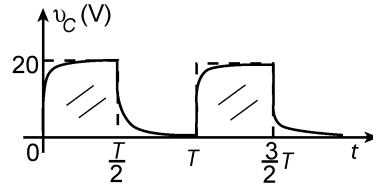
$$\begin{aligned} i_C &= \frac{-(10 \text{ V} - 2 \text{ V})}{1 \text{ k}\Omega} e^{-t/\tau} \\ \tau &= RC = (1 \text{ k}\Omega)(1000 \mu\text{F}) = 1 \text{ s} \\ i_C &= -\frac{8 \text{ V}}{1 \text{ k}\Omega} e^{-t} \\ \text{and } i_C &= -8 \text{ mA} e^{-t} \end{aligned}$$

20. $\tau = RC = (5 \text{ k}\Omega)(0.04 \text{ }\mu\text{F}) = 0.2 \text{ ms}$ (throughout)
 $v_C = E(1 - e^{-t/\tau}) = 20 \text{ V}(1 - e^{-t/0.2 \text{ ms}})$
 (Starting at $t = 0$ for each plot)

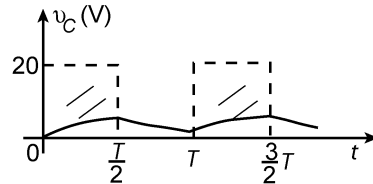
a. $T = \frac{1}{f} = \frac{1}{500 \text{ Hz}} = 2 \text{ ms}$
 $\frac{T}{2} = 1 \text{ ms}$
 $5\tau = 1 \text{ ms} = \frac{T}{2}$



b. $T = \frac{1}{f} = \frac{1}{100 \text{ Hz}} = 10 \text{ ms}$
 $\frac{T}{2} = 5 \text{ ms}$
 $5\tau = 1 \text{ ms} = \frac{1}{5}\left(\frac{T}{2}\right)$



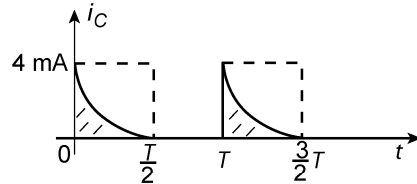
c. $T = \frac{1}{f} = \frac{1}{5 \text{ Hz}} = 0.2 \text{ ms}$
 $\frac{T}{2} = 0.1 \text{ ms}$
 $5\tau = 1 \text{ ms} = 10\left(\frac{T}{2}\right)$



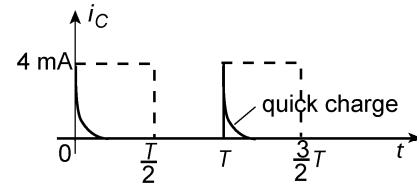
21. The mathematical expression for i_C is the same for each frequency!

$\tau = RC = (5 \text{ k}\Omega)(0.04 \text{ }\mu\text{F}) = 0.2 \text{ ms}$
 and $i_C = \frac{20 \text{ V}}{5 \text{ k}\Omega} e^{-t/0.2 \text{ ms}} = 4 \text{ mA} e^{-t/0.2 \text{ ms}}$

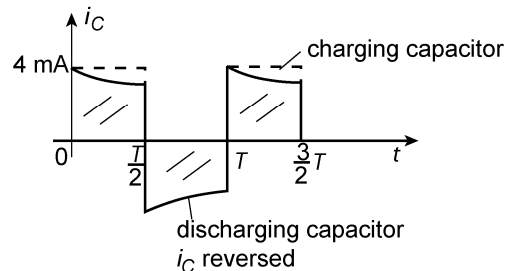
a. $T = \frac{1}{500 \text{ Hz}} = 2 \text{ ms}, \frac{T}{2} = 1 \text{ ms}$
 $5\tau = 5(0.2 \text{ ms}) = 1 \text{ ms} = \frac{T}{2}$



b. $T = \frac{1}{100 \text{ Hz}} = 10 \text{ ms}, \frac{T}{2} = 5 \text{ ms}$
 $5\tau = 1 \text{ ms} = \frac{1}{5}\left(\frac{T}{2}\right)$



c. $T = \frac{1}{5000 \text{ Hz}} = 0.2 \text{ ms}, \frac{T}{2} = 0.1 \text{ ms}$
 $5\tau = 1 \text{ ms} = 10\left(\frac{T}{2}\right)$



22. $\tau = 0.2 \text{ ms}$ as above

$$T = \frac{1}{500 \text{ Hz}} = 2 \text{ ms}$$

$$5\tau = 1 \text{ ms} = \frac{T}{2}$$

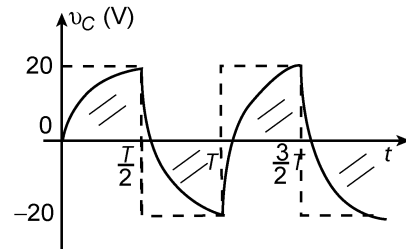
$$0 \rightarrow \frac{T}{2}: v_C = 20 \text{ V}(1 - e^{-t/0.2 \text{ ms}})$$

$$\frac{T}{2} \rightarrow T: V_i = 20 \text{ V}, V_f = -20 \text{ V}$$

$$\begin{aligned} v_C &= V_i + (V_f - V_i)(1 - e^{-t/\tau}) \\ &= 20 + (-20 - 20)(1 - e^{-t/0.2 \text{ ms}}) \\ &= 20 - 40(1 - e^{-t/0.2 \text{ ms}}) \\ &= 20 - 40 + 40e^{-t/0.2 \text{ ms}} \\ v_C &= \mathbf{-20 \text{ V} + 40 \text{ V}e^{-t/0.2 \text{ ms}}} \end{aligned}$$

$$T \rightarrow \frac{3}{2}T: V_i = -20 \text{ V}, V_f = +20 \text{ V}$$

$$\begin{aligned} v_C &= V_i + (V_f - V_i)(1 - e^{-t/\tau}) \\ &= -20 + (20 - (-20))(1 - e^{-t/\tau}) \\ &= -20 + 40(1 - e^{-t/\tau}) \\ &= -20 + 40 - 40e^{-t/\tau} \\ v_C &= \mathbf{20 \text{ V} - 40 \text{ V}e^{-t/0.2 \text{ ms}}} \end{aligned}$$



23. $v_C = V_i + (V_f - V_i)(1 - e^{-t/RC})$
 $V_i = 20 \text{ V}, V_f = 20 \text{ V}$
 $v_C = 20 + (20 - 20)(1 - e^{-t/RC})$
 $= \mathbf{20 \text{ V}}$ (for $0 \rightarrow \frac{T}{2}$)

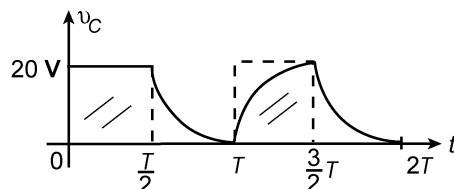
For $\frac{T}{2} \rightarrow T$, $v_i = 0 \text{ V}$ and $v_C = \mathbf{20 \text{ V}e^{-t/\tau}}$

$$\tau = RC = 0.2 \text{ ms}$$

with $\frac{T}{2} = 1 \text{ ms}$ and $5\tau = \frac{T}{2}$

For $T \rightarrow \frac{3}{2}T$, $v_i = 20 \text{ V}$
 $v_C = \mathbf{20 \text{ V}(1 - e^{-t/\tau})}$

For $\frac{3}{2}T \rightarrow 2T$, $v_i = 0 \text{ V}$
 $v_C = \mathbf{20 \text{ V}e^{-t/\tau}}$



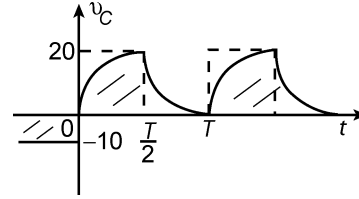
24. $\tau = RC = 0.2 \text{ ms}$

$$5\tau = 1 \text{ ms} = \frac{T}{2}$$

$$V_i = -10 \text{ V}, V_f = +20 \text{ V}$$

$$0 \rightarrow \frac{T}{2}:$$

$$\begin{aligned} v_C &= V_i + (V_f - V_i)(1 - e^{-t/\tau}) \\ &= -10 + (20 - (-10))(1 - e^{-t/\tau}) \\ &= -10 + 30(1 - e^{-t/\tau}) \\ &= -10 + 30 - 30e^{-t/\tau} \\ v_C &= +20 \text{ V} - 30 \text{ V}e^{-t/0.2 \text{ ms}} \end{aligned}$$



$$\frac{T}{2} \rightarrow T: \quad V_i = 20 \text{ V}, V_f = 0 \text{ V}$$

$$v_C = 20 \text{ V}e^{-t/0.2 \text{ ms}}$$

25. $\mathbf{Z}_p$: $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(10 \text{ kHz})(3 \text{ pF})} = 5.31 \text{ M}\Omega$

$$\mathbf{Z}_p = \frac{(9 \text{ M}\Omega \angle 0^\circ)(5.31 \text{ M}\Omega \angle -90^\circ)}{9 \text{ M}\Omega - j5.31 \text{ M}\Omega} = 4.573 \text{ M}\Omega \angle -59.5^\circ$$

$\mathbf{Z}_s$: $C_T = 18 \text{ pF} + 9 \text{ pF} = 27 \text{ pF}$

$$X_C = \frac{1}{2\pi fC_T} = \frac{1}{2\pi(10 \text{ kHz})(27 \text{ pF})} = 0.589 \text{ M}\Omega$$

$$\mathbf{Z}_s = \frac{(1 \text{ M}\Omega \angle 0^\circ)(0.589 \text{ M}\Omega \angle -90^\circ)}{1 \text{ M}\Omega - j0.589 \text{ M}\Omega} = 0.507 \text{ M}\Omega \angle -59.5^\circ$$

$$\begin{aligned} \mathbf{V}_{\text{scope}} &= \frac{\mathbf{Z}_s \mathbf{V}_i}{\mathbf{Z}_s + \mathbf{Z}_p} = \frac{(0.507 \text{ M}\Omega \angle -59.5^\circ)(100 \text{ V} \angle 0^\circ)}{(0.257 \text{ M}\Omega - j0.437 \text{ M}\Omega) + (2.324 \text{ M}\Omega - j3.939 \text{ M}\Omega)} \\ &= \frac{50.7 \times 10^6 \text{ V} \angle -59.5^\circ}{5.07 \times 10^6 \angle -59.5^\circ} = 10 \text{ V} \angle 0^\circ = \frac{1}{10}(100 \text{ V} \angle 0^\circ) \end{aligned}$$

$$\theta_{\mathbf{Z}_s} = \theta_{\mathbf{Z}_p} = -59.5^\circ$$

26. $\mathbf{Z}_p$: $X_C = \frac{1}{\omega C} = \frac{1}{(10^5 \text{ rad/s})(3 \text{ pF})} = 3.333 \text{ M}\Omega$

$$\mathbf{Z}_p = \frac{(9 \text{ M}\Omega \angle 0^\circ)(3.333 \text{ M}\Omega)}{9 \text{ M}\Omega - j3.333 \text{ M}\Omega} = 3.126 \text{ M}\Omega \angle -69.68^\circ$$

$\mathbf{Z}_s$: $X_C = \frac{1}{\omega C} = \frac{1}{(10^5 \text{ rad/s})(27 \text{ pF})} = 0.370 \text{ M}\Omega$

$$\mathbf{Z}_s = \frac{(1 \text{ M}\Omega \angle 0^\circ)(0.370 \text{ M}\Omega \angle -90^\circ)}{1 \text{ M}\Omega - j0.370 \text{ M}\Omega} = 0.347 \text{ M}\Omega \angle -69.68^\circ$$

$$\checkmark \theta_{\mathbf{Z}_p} = \theta_{\mathbf{Z}_s}$$

$$\begin{aligned}
\mathbf{V}_{\text{scope}} &= \frac{\mathbf{Z}_s \mathbf{V}_i}{\mathbf{Z}_s + \mathbf{Z}_p} = \frac{(0.347 \text{ M}\Omega \angle -69.68^\circ)(100 \text{ V} \angle 0^\circ)}{(0.121 \text{ M}\Omega - j0.325 \text{ M}\Omega) + (1.086 \text{ M}\Omega - j2.931 \text{ M}\Omega)} \\
&= \frac{34.70 \times 10^6 \text{ V} \angle -69.68^\circ}{3.470 \times 10^6 \angle -69.68^\circ} \\
&\cong \mathbf{10 \text{ V} \angle 0^\circ} = \frac{1}{10} (100 \text{ V} \angle 0^\circ)
\end{aligned}$$

Chapter 25

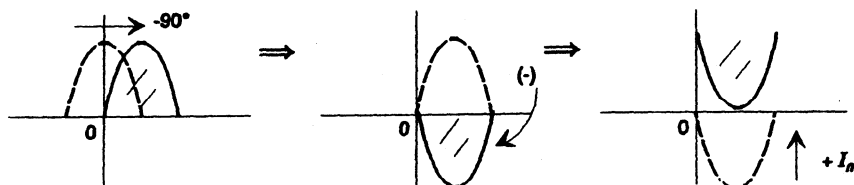
1. I: a. no b. no c. yes d. no e. yes
 II: a. yes b. yes c. yes d. yes e. no
 III: a. yes b. yes c. no d. yes e. yes
 IV: a. no b. no c. yes d. yes e. yes

2. b.
$$i = \frac{2I_m}{\pi} \left(1 + \frac{2}{3} \cos(2\omega t - 90^\circ) - \frac{2}{15} \cos(4\omega t - 90^\circ) + \frac{2}{35} \cos(6\omega t - 90^\circ) + \dots \right)$$

c.
$$\frac{2I_m}{\pi} - \frac{I_m}{2} = \frac{2I_m}{\pi} \left[1 - \frac{\pi}{4} \right]$$

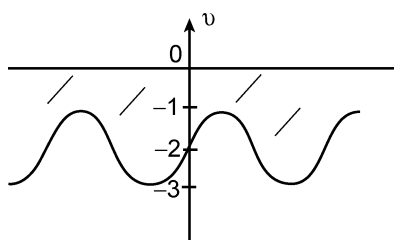
$$i = \frac{2I_m}{\pi} \left[1 - \frac{\pi}{4} + \frac{2}{3} \cos(2\omega t - 90^\circ) - \frac{2}{15} \cos(4\omega t - 90^\circ) + \frac{2}{35} \cos(6\omega t - 90^\circ) + \dots \right]$$

d.

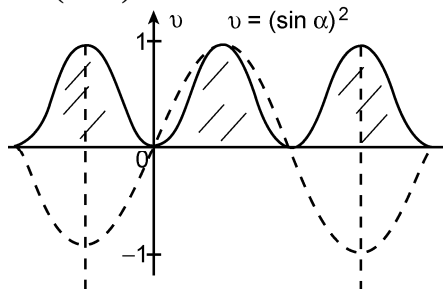


$$i = \frac{-2I_m}{\pi} \left[1 - \frac{\pi}{4} + \frac{2}{3} \cos(2\omega t - 90^\circ) - \frac{2}{15} \cos(4\omega t - 90^\circ) + \frac{2}{35} \cos(6\omega t - 90^\circ) + \dots \right]$$

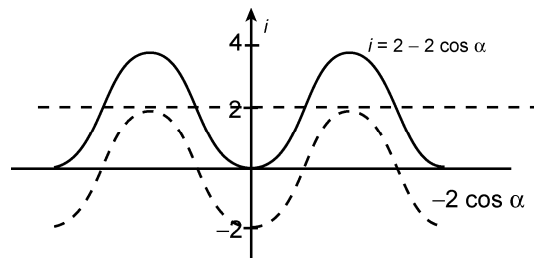
3. a. $v = -4 + 2 \sin \alpha$



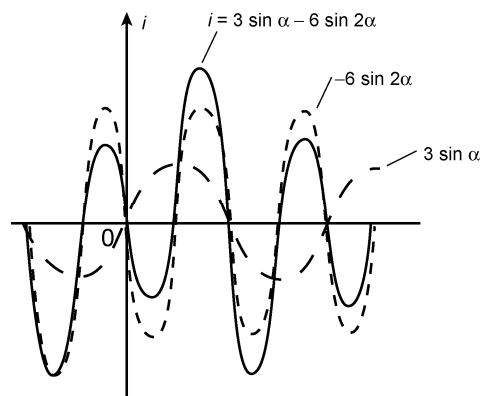
b. $v = (\sin \alpha)^2$



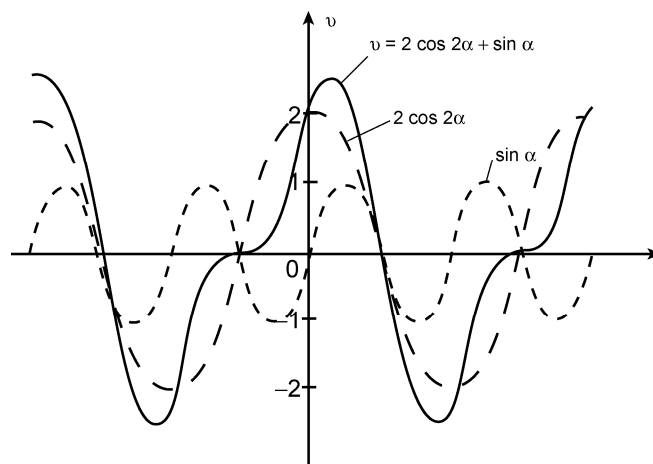
c. $i = 2 - 2 \cos \alpha$



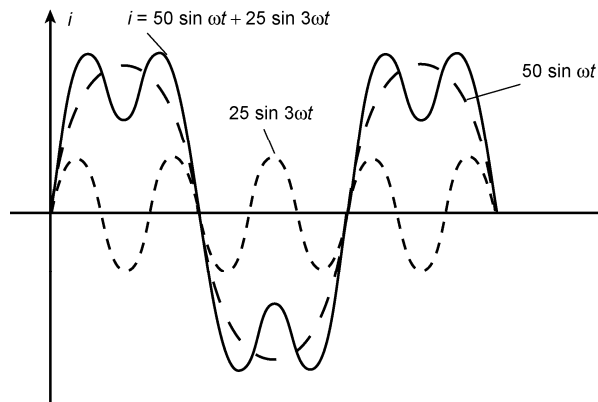
4. a.



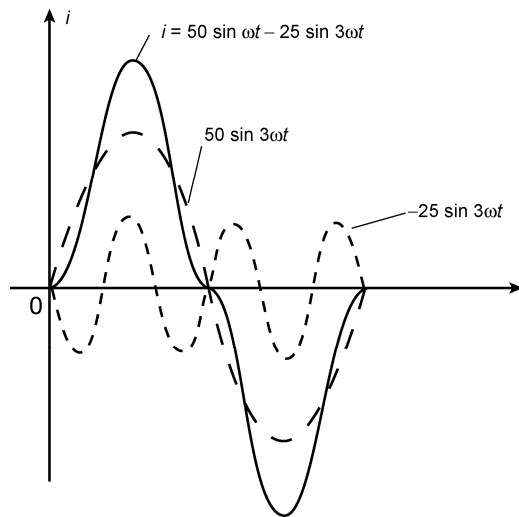
b.



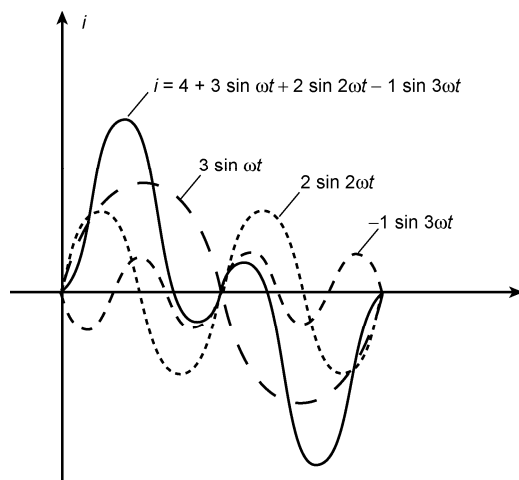
5. a.



b.



c.



6. a. $V_{av} = 100 \text{ V}$
 $V_{eff} = \sqrt{(100 \text{ V})^2 + \frac{(50 \text{ V})^2 + (25 \text{ V})^2}{2}} = 107.53 \text{ V}$
- b. $I_{av} = 3 \text{ A}$
 $I_{eff} = \sqrt{(3 \text{ A})^2 + \frac{(2 \text{ A})^2 + (0.8 \text{ A})^2}{2}} = 3.36 \text{ A}$
7. a. $V_{eff} = \sqrt{\frac{(20 \text{ V})^2 + (15 \text{ V})^2 + (10 \text{ V})^2}{2}} = 19.04 \text{ V}$
- b. $I_{eff} = \sqrt{\frac{(6 \text{ A})^2 + (2 \text{ A})^2 + (1 \text{ A})^2}{2}} = 4.53 \text{ A}$
8. $P_T = V_0 I_0 + V_1 I_1 \cos \theta_1 + \dots + V_n I_n \cos \theta_n$
 $= (100 \text{ V})(3 \text{ A}) + \frac{(50 \text{ V})(2 \text{ A})}{2} \cos 53^\circ + \frac{(25 \text{ V})(0.8 \text{ A})}{2} \cos 70^\circ$
 $= 300 + (50)(0.6018) + (10)(0.3420)$
 $= 333.52 \text{ W}$
9. $P = \frac{(20 \text{ V})(6 \text{ A})}{2} \cos 20^\circ + \frac{(15 \text{ V})(2 \text{ A})}{2} \cos 30^\circ + \frac{(10 \text{ V})(1 \text{ A})}{2} \cos 60^\circ$
 $= 60(0.9397) + 15(0.866) + 5(0.5)$
 $= 71.87 \text{ W}$
10. a. DC: $E = 18 \text{ V}, I_o = \frac{E}{R} = \frac{18 \text{ V}}{12 \Omega} = 1.5 \text{ A}$
 $\omega = 400 \text{ rad/s}; \quad X_L = \omega L = (400 \text{ rad/s})(0.02 \text{ H}) = 8 \Omega$
 $\mathbf{Z} = 12 \Omega + j8 \Omega = 14.42 \Omega \angle 33.69^\circ$
 $\mathbf{I} = \frac{\mathbf{E}}{\mathbf{Z}} = \frac{30 \text{ V} / \sqrt{2} \angle 0^\circ}{14.42 \Omega \angle 33.69^\circ} = \frac{2.08 \text{ A}}{\sqrt{2}} \angle -33.69^\circ$
 $i = 1.5 + \sqrt{2} \left(\frac{2.08}{\sqrt{2}} \right) \sin(400t - 33.69^\circ)$
 $i = 1.5 + 2.08 \sin(400t - 33.69^\circ)$
- b. $I_{eff} = \sqrt{(1.5 \text{ A})^2 + \frac{(2.08 \text{ A})^2}{2}} = 2.10 \text{ A}$

- c. DC: $v_R = E = 18 \text{ V}$, $\mathbf{V}_R = \left(\frac{2.08 \text{ A}}{\sqrt{2}} \angle -33.69^\circ \right) (12 \Omega \angle 0^\circ)$

$$= \frac{24.96 \text{ V}}{\sqrt{2}} \angle -33.69^\circ$$

$$v_R = 18 + \sqrt{2} \left(\frac{24.96}{\sqrt{2}} \right) \sin(400t - 33.69^\circ)$$

$$v_R = \mathbf{18 + 24.96 \sin(400t - 33.69^\circ)}$$
- d.
$$V_{R\text{eff}} = \sqrt{(18 \text{ V})^2 + \frac{(24.96 \text{ V})^2}{2}} = \mathbf{25.21 \text{ V}}$$
- e. DC: $V_L = 0 \text{ V}$
 $\omega = 400 \text{ rad/s}$: $\mathbf{V}_L = \left(\frac{2.08 \text{ A}}{\sqrt{2}} \angle -33.69^\circ \right) (8 \Omega \angle 90^\circ)$

$$= \frac{16.64 \text{ V}}{\sqrt{2}} \angle 56.31^\circ$$

$$v_L = \mathbf{0 + 16.64 \sin(400t + 56.31^\circ)}$$
- f.
$$V_{L\text{eff}} = \sqrt{0^2 + \frac{(16.64 \text{ V})^2}{2}} = \mathbf{11.77 \text{ V}}$$
- g.
$$P = I_{\text{eff}}^2 R = (2.101 \text{ A})^2 12 \Omega = \mathbf{52.97 \text{ W}}$$
11. a. DC: $I_{\text{DC}} = \frac{24 \text{ V}}{12 \Omega} = 2 \text{ A}$
 $\omega = 400 \text{ rad/s}$:

$$\mathbf{Z} = 12 \Omega + j(400 \text{ rad/s})(0.02 \text{ H}) = 12 \Omega + j8 \Omega = 14.422 \Omega \angle 33.69^\circ$$

$$\mathbf{I} = \frac{30 \text{ V} \angle 0^\circ}{14.422 \Omega \angle 33.69^\circ} = 2.08 \text{ A} \angle -33.69^\circ \text{ (peak values)}$$

 $\omega = 800 \text{ rad/s}$:

$$\mathbf{Z} = 12 \Omega + j(800 \text{ rad/s})(0.02 \text{ H}) = 12 \Omega + j16 \Omega = 20 \Omega \angle 53.13^\circ$$

$$\mathbf{I} = \frac{10 \text{ V} \angle 0^\circ}{20 \Omega \angle 53.13^\circ} = 0.5 \text{ A} \angle -53.13^\circ \text{ (peak values)}$$

$$i = \mathbf{2 + 2.08 \sin(400t - 33.69^\circ) + 0.5 \sin(800t - 53.13^\circ)}$$
- b.
$$I_{\text{eff}} = \sqrt{(2 \text{ A})^2 + \frac{(2.08 \text{ A})^2 + (0.5 \text{ A})^2}{2}} = \mathbf{2.51 \text{ A}}$$
- c.
$$v_R = iR = i(12 \Omega)$$

$$= \mathbf{24 + 24.96 \sin(400t - 33.69^\circ) + 6 \sin(800t - 53.13^\circ)}$$
- d.
$$V_{\text{eff}} = \sqrt{(24 \text{ V})^2 + \frac{(24.96 \text{ V})^2 + (6 \text{ V})^2}{2}} = \mathbf{30.09 \text{ V}}$$

- e. DC: $V_L = 0 \text{ V}$
 $\omega = 400 \text{ rad/s: } \mathbf{V}_L = (2.08 \text{ A } \angle -33.69^\circ)(8 \Omega \angle 90^\circ)$
 $\quad \quad \quad = 16.64 \text{ V } \angle 56.31^\circ$
 $\omega = 800 \text{ rad/s: } \mathbf{V}_L = (0.5 \text{ A } \angle -53.13^\circ)(16 \Omega \angle 90^\circ)$
 $\quad \quad \quad = 8 \text{ V } \angle 36.87^\circ$
 $\mathbf{v}_L = \mathbf{0} + 16.64 \sin(400t + 56.31^\circ) + 8 \sin(800t + 36.87^\circ)$
- f. $V_{\text{eff}} = \sqrt{(0)^2 + \frac{(16.64 \text{ V})^2 + (8 \text{ V})^2}{2}} = \mathbf{13.06 \text{ V}}$
- g. $P_T = I_{\text{eff}}^2 R = (2.508 \text{ A})^2 12 \Omega = \mathbf{75.48 \text{ W}}$
12. a. DC: $I = -\frac{60 \text{ V}}{12 \Omega} = -5 \text{ A}$
 $\omega = 300 \text{ rad/s: } X_L = \omega L = (300 \text{ rad/s})(0.02 \text{ H}) = 6 \Omega$
 $\mathbf{Z} = 12 \Omega + j6 \Omega = 13.42 \Omega \angle 26.57^\circ$
 $\mathbf{E} = (0.707)(20 \text{ V}) \angle 0^\circ = 14.14 \text{ V } \angle 0^\circ$
 $\mathbf{I} = \frac{\mathbf{E}}{\mathbf{Z}} = \frac{14.14 \text{ V } \angle 0^\circ}{13.42 \Omega \angle 26.57^\circ} = 1.054 \text{ A } \angle -26.57^\circ$
 $\omega = 600 \text{ rad/s: } X_L = \omega L = (600 \text{ rad/s})(0.02 \text{ H}) = 12 \Omega$
 $\mathbf{Z} = 12 \Omega + j12 \Omega = 16.97 \Omega \angle 45^\circ$
 $\mathbf{E} = -(0.707)(10 \text{ V}) \angle 0^\circ = -7.07 \text{ V } \angle 0^\circ$
 $\mathbf{I} = \frac{\mathbf{E}}{\mathbf{Z}} = -\frac{7.07 \text{ V } \angle 0^\circ}{16.97 \Omega \angle 45^\circ} = -0.417 \text{ A } \angle -45^\circ$
 $i = -5 + (1.414)(1.054)\sin(300t - 26.57^\circ) - (1.414)(0.417)\sin(600t - 45^\circ)$
 $i = \mathbf{-5 + 1.49 \sin(300t - 26.57^\circ) - 0.59 \sin(600t - 45^\circ)}$
- b. $I_{\text{eff}} = \sqrt{(5 \text{ A})^2 + \frac{(1.49 \text{ A})^2 + (0.59 \text{ A})^2}{2}} = \mathbf{5.13 \text{ A}}$
- c. DC: $V = IR = (-5 \text{ A})(12 \Omega) = -60 \text{ V}$
 $\omega = 300 \text{ rad/s: } \mathbf{V}_R = (1.054 \text{ A } \angle -26.57^\circ)(12 \Omega \angle 0^\circ)$
 $\quad \quad \quad = 12.648 \text{ V } \angle -26.57^\circ$
 $\omega = 600 \text{ rad/s: } \mathbf{V}_R = (-0.417 \text{ A } \angle -45^\circ)(12 \Omega \angle 0^\circ)$
 $\quad \quad \quad = -5 \text{ V } \angle -45^\circ$
 $\mathbf{v}_R = -60 + (1.414)(12.648)\sin(300t - 26.57^\circ) - (1.414)(5)\sin(600t - 45^\circ)$
 $\mathbf{v}_R = -60 + 17.88 \sin(300t - 26.57^\circ) - 7.07 \sin(600t - 45^\circ)$
- d. $V_{\text{Reff}} = \sqrt{(60 \text{ V})^2 + \frac{(17.88 \text{ V})^2 + (7.07 \text{ V})^2}{2}} = \mathbf{61.52 \text{ V}}$

- e. DC: $V_L = 0 \text{ V}$
 $\omega = 300 \text{ rad/s: } \mathbf{V_L = (1.054 \text{ A } \angle -26.57^\circ)(6 \Omega \angle 90^\circ) = 6.324 \text{ V } \angle 63.43^\circ}$
 $\omega = 600 \text{ rad/s: } \mathbf{V_L = (-0.417 \text{ A } \angle -45^\circ)(12 \Omega \angle 90^\circ) = -5 \text{ V } \angle 45^\circ}$
 $v_L = 0 + (1.414)(6.324)\sin(300t + 63.43^\circ) - (1.414)(5)\sin(600t + 45^\circ)$
 $v_L = \mathbf{8.94 \sin(300t + 63.43^\circ) - 7.07 \sin(600t + 45^\circ)}$
- f. $V_{L\text{eff}} = \sqrt{\frac{(8.94 \text{ V})^2 + (3.54 \text{ V})^2}{2}} = \mathbf{6.8 \text{ V}}$
- g. $P = I_{\text{eff}}^2 R = (5.13 \text{ A})^2 12 \Omega = \mathbf{315.8 \text{ W}}$
13. a. DC: $I = \mathbf{0 \text{ A}}$
 $\omega = 400 \text{ rad/s: } X_C = \frac{1}{\omega C} = \frac{1}{(400 \text{ rad/s})(125 \mu\text{F})} = 20 \Omega$
 $\mathbf{Z = 15 \Omega - j20 \Omega = 25 \Omega \angle -53.13^\circ}$
 $\mathbf{E = (0.707)(30 \text{ V}) \angle 0^\circ = 21.21 \text{ V } \angle 0^\circ}$
 $\mathbf{I = \frac{E}{Z} = \frac{21.21 \text{ V } \angle 0^\circ}{25 \Omega \angle -53.13^\circ} = 0.848 \text{ A } \angle 53.13^\circ}$
 $i = 0 + (1.414)(0.848)\sin(400t + 53.13^\circ)$
 $i = \mathbf{1.2 \sin(400t + 53.13^\circ)}$
- b. $I_{\text{eff}} = \sqrt{\frac{(1.2 \text{ A})^2}{2}} = \mathbf{0.85 \text{ A}}$ as above
- c. DC: $V_R = 0 \text{ V}$
 $\omega = 400 \text{ rad/s: } \mathbf{V_R = (0.848 \text{ A } \angle 53.13^\circ)(15 \Omega \angle 0^\circ) = 12.72 \text{ V } \angle 53.13^\circ}$
 $v_R = 0 + (1.414)(12.72)\sin(400t + 53.13^\circ)$
 $v_R = \mathbf{18 \sin(400t + 53.13^\circ)}$
- d. $V_{R\text{eff}} = \sqrt{\frac{(18 \text{ V})^2}{2}} = \mathbf{12.73 \text{ V}}$
- e. DC: $V_C = 18 \text{ V}$
 $\omega = 400 \text{ rad/s: } \mathbf{V_C = (0.848 \text{ A } \angle 53.13^\circ)(20 \Omega \angle -90^\circ)}$
 $\mathbf{= 16.96 \text{ V } \angle -36.87^\circ}$
 $v_C = 18 + (1.414)(16.96)\sin(400t - 36.87^\circ)$
 $v_C = \mathbf{18 + 23.98 \sin(400t - 36.87^\circ)}$
- f. $V_{C\text{eff}} = \sqrt{(18 \text{ V})^2 + \frac{(23.98 \text{ V})^2}{2}} = \mathbf{24.73 \text{ V}}$
- g. $P = I_{\text{eff}}^2 R = (0.848 \text{ A})^2 15 \Omega = \mathbf{10.79 \text{ W}}$

14. a.
$$e = \frac{200}{\pi} + \frac{400}{3\pi} \cos 2\omega t - \frac{400}{15\pi} \cos 4\omega t$$

$$= 63.69 + 42.46 \sin(2\omega t + 90^\circ) - 8.49 \sin(4\omega t + 90^\circ)$$

 $\omega = 377 \text{ rad/s:}$

$$e = 63.69 + 42.46 \sin(754t + 90^\circ) - 8.49 \sin(1508t + 90^\circ)$$

DC: $X_L = 0 \therefore V_L = 0 \text{ V}$
 $\omega = 754 \text{ rad/s: } X_C = \frac{1}{\omega C} = \frac{1}{(754 \text{ rad/s})(1 \mu\text{F})} = 1330 \Omega$

$$X_L = \omega L = (754 \text{ rad/s})(0.1 \text{ H}) = 75.4 \Omega$$

$$\mathbf{Z}' = (1 \text{ k}\Omega \angle 0^\circ) \parallel 75.4 \Omega \angle 90^\circ = 75.19 \Omega \angle 85.69^\circ$$

$$\mathbf{E} = (0.707)(42.46 \text{ V}) \angle 90^\circ = 30.02 \text{ V} \angle 90^\circ$$

$$\mathbf{V}_o = \frac{\mathbf{Z}'(\mathbf{E})}{\mathbf{Z}' + \mathbf{Z}_C} = \frac{(75.19 \Omega \angle 85.69^\circ)(30.02 \text{ V} \angle 90^\circ)}{75.19 \Omega \angle 85.69^\circ + 1330 \Omega \angle -90^\circ} = 1.799 \text{ V} \angle -94.57^\circ$$

 $\omega = 1508 \text{ rad/s: } X_C = \frac{1}{\omega C} = \frac{1}{(1508 \text{ rad/s})(1 \mu\text{F})} = 6631.13 \Omega$

$$X_L = \omega L = (1508 \text{ rad/s})(0.1 \text{ H}) = 150.8 \Omega$$

$$\mathbf{Z}' = (1 \text{ k}\Omega \angle 0^\circ) \parallel 150.8 \Omega \angle 90^\circ = 149.12 \Omega \angle 81.42^\circ$$

$$\mathbf{E} = (0.707)(8.49 \text{ V}) \angle 90^\circ = 6 \text{ V} \angle 90^\circ$$

$$\mathbf{V}_o = \frac{\mathbf{Z}'(\mathbf{E})}{\mathbf{Z}' + \mathbf{Z}_C} = \frac{(149.12 \Omega \angle 81.42^\circ)(6 \text{ V} \angle 90^\circ)}{149.12 \Omega \angle 81.42^\circ + 6631.13 \Omega \angle -90^\circ}$$

$$= 1.73 \text{ V} \angle -101.1^\circ$$

$$v_o = 0 + 1.414(1.799)\sin(754t - 94.57^\circ) - 1.414(1.73)\sin(1508t - 101.1^\circ)$$

$$v_o = 2.54 \sin(754t - 94.57^\circ) - 2.45 \sin(1508t - 101.1^\circ)$$
- b.
$$V_{\text{eff}} = \sqrt{\frac{(2.54 \text{ V})^2 + (2.45 \text{ V})^2}{2}} = 2.50 \text{ V}$$
- c.
$$P = \frac{(V_{\text{eff}})^2}{R} = \frac{(2.50 \text{ V})^2}{1 \text{ k}\Omega} = 6.25 \text{ mW}$$
15.
$$i = 0.318I_m + 0.500 I_m \sin \omega t - 0.212I_m \cos 2\omega t - 0.0424I_m \cos 4\omega t + \dots (I_m = 10 \text{ mA})$$

$$i = 3.18 \times 10^{-3} + 5 \times 10^{-3} \sin \omega t - 2.12 \times 10^{-3} \sin(2\omega t + 90^\circ)$$

$$- 0.424 \times 10^{-3} \sin(4\omega t + 90^\circ) + \dots$$

$$i \cong 3.18 \times 10^{-3} + 5 \times 10^{-3} \sin \omega t - 2.12 \times 10^{-3} \sin(2\omega t + 90^\circ)$$

DC: $I_o = 0 \text{ A}, V_o = 0 \text{ V}$
 $\omega = 377 \text{ rad/s; } X_L = \omega L = (377 \text{ rad/s})(1.2 \text{ mH}) = 0.452 \Omega$

$$X_C = \frac{1}{\omega C} = \frac{1}{(377 \text{ rad/s})(200 \mu\text{F})} = 13.26 \Omega$$

$$\mathbf{Z}' = 200 \Omega - j13.26 \Omega = 200.44 \Omega \angle -3.79^\circ$$

$$\mathbf{I} = (0.707)(5 \times 10^{-3}) \text{ A} \angle 0^\circ = 3.54 \text{ mA} \angle 0^\circ$$

$$\mathbf{I}_o = \frac{\mathbf{Z}_L \mathbf{I}}{\mathbf{Z}_L + \mathbf{Z}'} = \frac{(0.452 \Omega \angle 90^\circ)(3.54 \text{ mA} \angle 0^\circ)}{j0.452 \Omega + 200 \Omega - j13.26 \Omega} = 7.98 \mu\text{A} \angle 93.66^\circ$$

$$\mathbf{V}_o = (7.98 \mu\text{A} \angle 93.66^\circ)(200 \Omega \angle 0^\circ) = 1.596 \text{ mV} \angle 93.66^\circ$$

$$\omega = 754 \text{ rad/s: } X_L = \omega L = (754 \text{ rad/s})(1.2 \text{ mH}) = 0.905 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{(754 \text{ rad/s})(200 \mu\text{F})} = 6.63 \Omega$$

$$\mathbf{Z}' = 200 \Omega - j6.63 \Omega = 200.11 \Omega \angle -1.9^\circ$$

$$\mathbf{I} = (0.707)(2.12 \text{ mA}) \angle 90^\circ = 1.5 \text{ mA} \angle 90^\circ$$

$$\mathbf{I}_o = \frac{\mathbf{Z}_L \mathbf{I}}{\mathbf{Z}_L + \mathbf{Z}'} = \frac{(0.905 \Omega \angle 90^\circ)(1.5 \text{ mA} \angle 90^\circ)}{j0.905 \Omega + 200 \Omega - j6.63 \Omega} = 6.8 \mu\text{A} \angle 181.64^\circ$$

$$\mathbf{V}_o = (6.8 \mu\text{A} \angle 181.64^\circ)(200 \Omega \angle 0^\circ) = 1.36 \text{ mA} \angle 181.64^\circ$$

$$v_o = 0 + (1.414)(1.596 \times 10^{-3})\sin(377t + 93.66^\circ)$$

$$- (1.414)(1.360 \times 10^{-3})\sin(754t + 181.64^\circ)$$

$$v_o = 2.26 \times 10^{-3} \sin(377t + 93.66^\circ) + 1.92 \times 10^{-3} \sin(754t + 1.64^\circ)$$

16. a. $60 + 70 \sin \omega t + 20 \sin(2\omega t + 90^\circ) + 10 \sin(3\omega t + 60^\circ)$
 $+20 + 30 \sin \omega t - 20 \sin(2\omega t + 90^\circ) + 5 \sin(3\omega t + 90^\circ)$
DC: $60 + 20 = 80$
 ω : $70 + 30 = 100 \Rightarrow 100 \sin \omega t$
 2ω : 0
 3ω : $10 \angle 60^\circ + 5 \angle 90^\circ = 5 + j8.66 + j5 = 5 + j13.66 = 14.55 \angle 69.9^\circ$
Sum = **$80 + 100 \sin \omega t + 14.55 \sin(3\omega t + 69.9^\circ)$**

b. $20 + 60 \sin \alpha + 10 \sin(2\alpha - 180^\circ) + 5 \sin(3\alpha + 180^\circ)$
 $-5 + 10 \sin \alpha + 0 - 4 \sin(3\alpha - 30^\circ)$
DC: $20 - 5 = 15$
 α : $60 + 10 = 70 \Rightarrow 70 \sin \alpha$
 2α : $10 \sin(2\alpha - 180^\circ)$
 3α : $5 \angle 180^\circ - 4 \angle -30^\circ = -5 - [3.46 - j2] = -8.46 + j2$
 $= 8.69 \angle 166.7^\circ$
Sum = **$15 + 70 \sin \alpha + 10 \sin(2\alpha - 180^\circ) + 8.69 \sin(3\alpha + 166.7^\circ)$**

17. $i_T = i_1 + i_2$
 $= 10 + 30 \sin 20t - 0.5 \sin(40t + 90^\circ)$
 $+20 + 4 \sin(20t + 90^\circ) + 0.5 \sin(40t + 30^\circ)$
DC: $10 \text{ A} + 20 \text{ A} = 30 \text{ A}$
 $\omega = 20 \text{ rad/s: } 30 \text{ A} \angle 0^\circ + 4 \text{ A} \angle 90^\circ = 30 \text{ A} + j4 \text{ A} = 30.27 \text{ A} \angle 7.59^\circ$
 $\omega = 40 \text{ rad/s: } -0.5 \text{ A} \angle 90^\circ + 0.5 \text{ A} \angle 30^\circ$
 $= -j0.5 \text{ A} + 0.433 \text{ A} + j0.25 \text{ A}$
 $= 0.433 \text{ A} - j0.25 \text{ A} = 0.5 \text{ A} \angle -30^\circ$
 $i_T = 30 + 30.27 \sin(20t + 7.59^\circ) + 0.5 \sin(40t - 30^\circ)$

$$\begin{aligned}
 18. \quad e &= v_1 + v_2 \\
 &= 20 - 200 \sin 600t + 100 \sin(1200t + 90^\circ) + 75 \sin 1800t \\
 &\quad -10 + 150 \sin(600t + 30^\circ) + 0 + 50 \sin(1800t + 60^\circ) \\
 \text{DC: } &20 \text{ V} - 10 \text{ V} = 10 \text{ V} \\
 \omega: \quad &600 \text{ rad/s: } -200 \text{ V } \angle 0^\circ + 150 \text{ V } \angle 30^\circ = 102.66 \text{ V } \angle 133.07^\circ \\
 &1200 \text{ rad/s: } 100 \sin(1200t + 90^\circ) \\
 &1800 \text{ rad/s: } 75 \text{ V } \angle 0^\circ + 50 \text{ V } \angle 60^\circ = 108.97 \text{ V } \angle 23.41^\circ \\
 e &= \mathbf{10 + 102.66 \sin(600t + 133.07^\circ) + 100 \sin(1200t + 90^\circ) + 108.97 \sin(1800t + 23.41^\circ)}
 \end{aligned}$$

Chapter 26

1.
$$Z_i = \frac{E_i}{I_i}; I_i = \frac{V_R}{R} = \frac{1.05 - 1.00 \text{ V}}{47 \Omega} = \frac{50 \text{ mV}}{47 \Omega} = 1.064 \text{ mA}$$
$$Z_i = \frac{E_i}{I_i} = \frac{1.05 \text{ V}}{1.064 \text{ mA}} = 986.84 \Omega$$
2.
$$\mathbf{Z_i} = \frac{\mathbf{E_i}}{\mathbf{I_i}} = \frac{120 \text{ V} \angle 0^\circ}{6.2 \text{ A} \angle -10.8^\circ} = 19.35 \Omega \angle 10.8^\circ = 19 \Omega + j3.623 \Omega$$
$$f = 60 \text{ Hz: } R = \mathbf{19 \Omega}, L = \frac{X_L}{2\pi f} = \frac{3.623 \Omega}{2\pi(60 \text{ Hz})} = \mathbf{9.61 \text{ mH}}$$
3.
 - a.
$$I_{i_1} = \frac{E_{i_1}}{Z_{i_1}} = \frac{20 \text{ mV}}{2 \text{ k}\Omega} = \mathbf{10 \mu A}$$
 - b.
$$Z_{i_2} = \frac{E_{i_2}}{I_{i_2}} = \frac{1.8 \text{ V}}{0.4 \text{ mA}} = \mathbf{4.5 \text{ k}\Omega}$$
 - c.
$$E_{i_3} = I_{i_3} Z_{i_3} = (1.5 \text{ mA})(4.6 \text{ k}\Omega) = \mathbf{6.9 \text{ V}}$$
4.
$$I_o = \frac{E_g - E_o}{R_s} = \frac{4 \text{ V} - 3.8 \text{ V}}{2 \text{ k}\Omega} = \frac{0.2 \text{ V}}{2 \text{ k}\Omega} = 0.1 \text{ mA}(p - p)$$
$$Z_o = \frac{E_o}{I_o} = \frac{3.8 \text{ V}(p - p)}{0.1 \text{ mA}(p - p)} = \mathbf{38 \text{ k}\Omega}$$
5.
$$E_{o_{\text{peak}}} = E_{g_{\text{peak}}} - V_{R_{\text{peak}}} = 2 \text{ V} \angle 0^\circ - 40 \times 10^{-3} \text{ V} \angle 0^\circ = 1.96 \text{ V} \angle 0^\circ$$
$$I_{\text{peak}} = \frac{V_{R_{\text{peak}}}}{R_s} = \frac{40 \text{ mV}}{0.91 \text{ k}\Omega} = 43.96 \mu A$$
$$Z_o = \frac{E_o}{I_R} = \frac{1.96 \text{ V} \angle 0^\circ}{43.96 \mu A} = \mathbf{44.59 \text{ k}\Omega}$$
6.
$$E_{o_{\text{peak}}} = \sqrt{2} 0.6 \text{ V}_{(\text{rms})} = 0.849 \text{ V}$$
$$E_{o_{(p-p)}} = 2(E_{o_{(\text{peak})}}) = 2(0.849 \text{ V}) = 1.697 \text{ V}$$
$$I_o = \frac{E_g - E_o}{R_s} = \frac{18 \text{ V} - 1.697 \text{ V}}{2 \text{ k}\Omega} = 51.5 \mu A(p - p)$$
$$Z_o = \frac{E_o}{I_o} = \frac{1.697 \text{ V}(p - p)}{51.5 \mu A(p - p)} = \mathbf{32.95 \text{ k}\Omega}$$

7.
$$Z_o = \frac{E_{o_{p-p}}}{I_{o_{p-p}}} = \frac{E_{g_{p-p}} - V_{R_{p-p}}}{I_{o_{p-p}}} = \frac{0.8 \text{ V} - 0.4 \text{ V}}{40 \mu\text{A}} = \mathbf{10 \text{ k}\Omega}$$

$$V_{R_{p-p}} = 2 \text{ div} [0.2 \text{ V/div}] = 0.4 \text{ V}$$

$$E_{g_{p-p}} = 4 \text{ div} [0.2 \text{ V/div}] = 0.8 \text{ V}$$

$$I_{o_{p-p}} = \frac{V_{R_{p-p}}}{10 \text{ k}\Omega} = \frac{0.4 \text{ V}}{10 \text{ k}\Omega} = \mathbf{40 \mu\text{A}}$$
8.
$$\mathbf{E_i = I_i Z_i = (10 \mu\text{A} \angle 0^\circ)(1.8 \text{ k}\Omega \angle 0^\circ) = 18 \text{ mV} \angle 0^\circ}$$

$$E_{i(\text{peak})} = \sqrt{2} (18 \text{ mV}) = 25.46 \text{ mV}$$

$$E_{i(p-p)} = 2(25.46 \text{ mV}) = 50.92 \text{ mV}$$

$$\mathbf{A_{v_{NL}} = \frac{E_o}{E_i} = \frac{4.05 \text{ V} \angle 180^\circ}{50.92 \text{ mV} \angle 0^\circ} = 79.54 \angle 180^\circ = -79.54}$$
9. a.
$$\mathbf{A_v = \frac{E_o}{E_i} = A_{v_{NL}} \frac{R_L}{R_L + R_o} = (-3200) \frac{(5.6 \text{ k}\Omega)}{5.6 \text{ k}\Omega + 40 \text{ k}\Omega} = -392.98}$$

b.
$$\mathbf{A_{v_T} = \frac{E_o}{E_g} = \frac{E_o}{E_i} \cdot \frac{E_i}{E_g}}$$

with
$$\mathbf{E_i = \frac{Z_i E_g}{Z_i + R_g}} \text{ and } \mathbf{\frac{E_i}{E_g} = \frac{Z_i}{Z_i + R_g}}$$

$$\mathbf{A_{v_T} = \frac{E_o}{E_i} \cdot \frac{Z_i}{Z_i + R_g} = (-392.98) \frac{(2.2 \text{ k}\Omega)}{2.2 \text{ k}\Omega + 0.5 \text{ k}\Omega} = -320.21}$$
10.
$$\mathbf{A_{v_{NL}} = \frac{-1400 \text{ mV}}{1.2 \text{ mV} \angle 0^\circ} = -1200}$$

$$\mathbf{A_v = \frac{-192 \text{ mV}}{1.2 \text{ mV}} = -160}$$

$$\mathbf{R_o = R_L \left(\frac{A_{v_{NL}}}{A_v} - 1 \right)}$$

$$\mathbf{= 4.7 \text{ k}\Omega \left(\frac{-1200}{-160} - 1 \right)}$$

$$\mathbf{= 30.55 \text{ k}\Omega}$$
11. a.
$$\mathbf{A_v = \frac{E_o}{E_i} = A_{v_{NL}} \frac{R_L}{R_L + R_o}}$$

$$\mathbf{-160 = A_{v_{NL}} \frac{2 \text{ k}\Omega}{2 \text{ k}\Omega + 28 \text{ k}\Omega} = A_{v_{NL}} (0.0667)}$$

$$\mathbf{A_{v_{NL}} = -2398.8}$$

$$b. \quad E_o = -I_o R_L = -(4 \text{ mA})(2 \text{ k}\Omega) = -8 \text{ V}$$

$$A_v = \frac{E_o}{E_i} = -160$$

$$E_i = \frac{E_o}{-160} = \frac{-8 \text{ V}}{-160} = 50 \text{ mV}$$

$$c. \quad I_i = \frac{E_g - E_i}{R_g} = \frac{70 \text{ mV} - 50 \text{ mV}}{0.4 \text{ k}\Omega} = 50 \mu\text{A}$$

$$Z_i = \frac{E_i}{I_i} = \frac{50 \text{ mV}}{50 \mu\text{A}} = 1 \text{ k}\Omega$$

$$12. \quad a. \quad A_i = -A_{v_{NL}} \frac{R_i}{R_L + R_o} \\ = \frac{-(-3200)(2.2 \text{ k}\Omega)}{5.6 \text{ k}\Omega + 40 \text{ k}\Omega} \\ = 154.39$$

$$b. \quad A_{i_T} = -A_{v_T} \left(\frac{R_g + Z_i}{R_L} \right) \\ = - \left(\frac{A_v Z_i}{Z_i + R_g} \right) \left(\frac{R_g + Z_i}{R_L} \right) \\ A_{i_T} = -A_v \frac{Z_i}{R_L} = - \left(A_{v_{NL}} \frac{R_L}{R_L + R_o} \right) \frac{Z_i}{R_L} \\ = -A_{v_{NL}} \frac{Z_i}{R_L + R_o} \\ = \frac{-(-3200)(2.2 \text{ k}\Omega)}{5.6 \text{ k}\Omega + 40 \text{ k}\Omega} \\ = 154.39$$

$$c. \quad \text{Same result since } I_i = I_g$$

$$13. \quad a. \quad A_G = A_v^2 \frac{R_i}{R_L} \\ = (-392.98)^2 \frac{2.2 \text{ k}\Omega}{5.6 \text{ k}\Omega} \\ = 6.067 \times 10^4$$

$$A_G = -A_v A_i \\ = -(-392.98)(154.39) \\ = 6.067 \times 10^4$$

$$A_v = A_{v_{NL}} \frac{R_L}{R_L + R_o} \\ = (-3200) \left(\frac{5.6 \text{ k}\Omega}{5.6 \text{ k}\Omega + 40 \text{ k}\Omega} \right) \\ = -392.98$$

$$A_i = -A_{v_{NL}} \frac{R_i}{R_L + R_o} \\ = -(-3200) \left(\frac{2.2 \text{ k}\Omega}{5.6 \text{ k}\Omega + 40 \text{ k}\Omega} \right) \\ = 154.39$$

b. $\mathbf{A}_{v_r} = \mathbf{A}_v \frac{\mathbf{Z}_i}{\mathbf{Z}_i + R_g} = (-392.98) \left(\frac{2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 0.5 \text{ k}\Omega} \right) = \mathbf{-320.21}$

$\mathbf{A}_{i_r} = -\mathbf{A}_{v_r} \frac{R_g + \mathbf{Z}_i}{R_L} = -(-320.21) \left(\frac{0.5 \text{ k}\Omega + 2.2 \text{ k}\Omega}{5.6 \text{ k}\Omega} \right) = \mathbf{154.39}$

$\mathbf{A}_{G_T} = -\mathbf{A}_{v_r}^2 \left(\frac{R_g + R_i}{R_L} \right) = (-320.21)^2 \left(\frac{0.5 \text{ k}\Omega + 2.2 \text{ k}\Omega}{5.6 \text{ k}\Omega} \right) = \mathbf{4.94 \times 10^4}$

$\mathbf{A}_{G_T} = -\mathbf{A}_{v_r} \mathbf{A}_{i_r} = -(-320.21)(154.39) = \mathbf{494 \times 10^4}$

14. a. $\mathbf{A}_i = \frac{\mathbf{I}_o}{\mathbf{I}_i} = -\mathbf{A}_v \frac{Z_i}{R_L}$

$= \frac{-(-160)(0.75 \text{ k}\Omega)}{2 \text{ k}\Omega}$

$= \mathbf{60}$

b. $\mathbf{A}_{G_T} = \frac{P_L}{P_g} = \mathbf{A}_{v_r}^2 \left(\frac{R_g + R_i}{R_L} \right)$

$\mathbf{A}_{v_r} = \mathbf{A}_v \frac{Z_i}{Z_i + R_g}$

$= \frac{(-160)(0.75 \text{ k}\Omega)}{0.75 \text{ k}\Omega + 0.4 \text{ k}\Omega} = -104.35$

$\mathbf{A}_{G_T} = (104.35)^2 \left(\frac{0.4 \text{ k}\Omega + 0.75 \text{ k}\Omega}{2 \text{ k}\Omega} \right)$

$= \mathbf{6.261 \times 10^3}$

15. a. $\mathbf{A}_{v_r} = \mathbf{A}_{v_1} \cdot \mathbf{A}_{v_2} = (-30)(-50) = \mathbf{1500}$

b. $\mathbf{A}_{i_r} = \mathbf{A}_{v_r} \frac{\mathbf{Z}_{i_L}}{R_L} = (1500) \left(\frac{1 \text{ k}\Omega}{8 \text{ k}\Omega} \right) = \mathbf{187.5}$

c. $\mathbf{A}_{i_1} = -\mathbf{A}_{v_1} \frac{\mathbf{Z}_{i_1}}{R_{L_1}} = -(-30) \left(\frac{1 \text{ k}\Omega}{2 \text{ k}\Omega} \right) = \mathbf{15}$

$\mathbf{A}_{i_2} = -\mathbf{A}_{v_2} \frac{\mathbf{Z}_{i_2}}{R_{L_2}} = -(-50) \left(\frac{2 \text{ k}\Omega}{8 \text{ k}\Omega} \right) = \mathbf{12.5}$

d. $\mathbf{A}_{i_r} = \mathbf{A}_{i_1} \cdot \mathbf{A}_{i_2} = (15)(12.5) = \mathbf{187.5}$ as above

16. a. $\mathbf{A}_{v_r} = \mathbf{A}_{v_1} \cdot \mathbf{A}_{v_2} \cdot \mathbf{A}_{v_3}$

$-6912 = (-12)(A_{v_2})(-32)$

$\mathbf{A}_{v_2} = \mathbf{-18}$

$$\text{b. } A_{i_1} = \frac{-A_{v_1} Z_{i_1}}{R_{L_1}} = \frac{-A_{v_1} Z_{i_1}}{Z_{i_2}}$$

$$4 = \frac{-(-12)(1 \text{ k}\Omega)}{Z_{i_2}}$$

$$Z_{i_2} = 3 \text{ k}\Omega$$

$$\text{c. } A_{i_3} = \frac{-A_{v_3} Z_{i_3}}{R_{L_3}} = \frac{-(-32)(2 \text{ k}\Omega)}{2.2 \text{ k}\Omega}$$

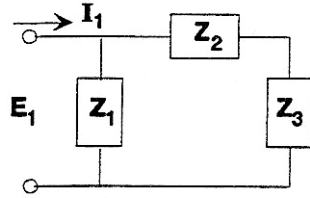
$$= 29.09$$

$$A_{i_T} = A_{i_1} \cdot A_{i_2} \cdot A_{i_3}$$

$$= (4)(26)(29.09)$$

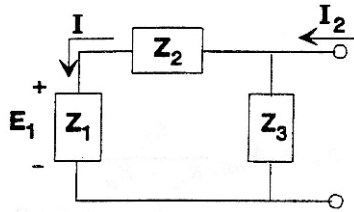
$$= 3.025 \times 10^3$$

17. a.



$$z_{11} = \left. \frac{E_1}{I_1} \right|_{I_2=0} = Z_1 \parallel (Z_2 + Z_3)$$

$$z_{11} = \frac{Z_1 Z_2 + Z_1 Z_3}{Z_1 + Z_2 + Z_3}$$



$$I = \frac{Z_3 I_2}{Z_1 + Z_2 + Z_3}$$

$$E_1 = I_1 Z_1 = \frac{(Z_3 I_2)(Z_1)}{Z_1 + Z_2 + Z_3}$$

$$z_{12} = \left. \frac{E_1}{I_2} \right|_{I_1=0} = \frac{Z_1 Z_3}{Z_1 + Z_2 + Z_3}$$

$$z_{21} = \left. \frac{E_2}{I_1} \right|_{I_2=0} \quad \text{Mirror image of } z_{12}$$

$$\therefore z_{21} = \frac{Z_1 Z_3}{Z_1 + Z_2 + Z_3}$$

$$z_{22} = \left. \frac{E_2}{I_2} \right|_{I_1=0} \quad \text{Mirror image of } z_{11}$$

$$\therefore z_{22} = Z_3 \parallel (Z_1 + Z_2)$$

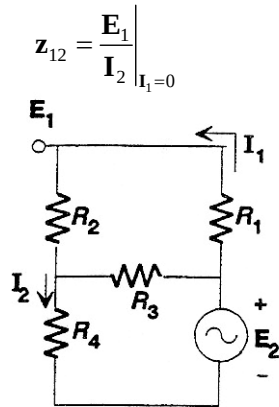
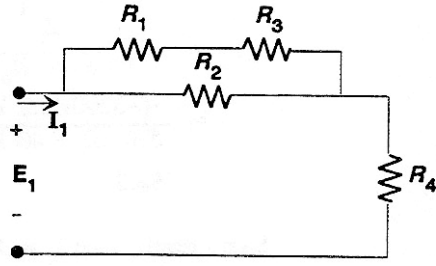
$$= \frac{Z_1 Z_3 + Z_2 Z_3}{Z_1 + Z_2 + Z_3}$$

b. —

18. a.
$$z_{11} = \left. \frac{E_1}{I_1} \right|_{I_2=0}$$

$$z_{11} = R_4 + R_2 \parallel (R_1 + R_3)$$

$$= R_4 + \frac{R_2(R_1 + R_3)}{R_1 + R_2 + R_3}$$

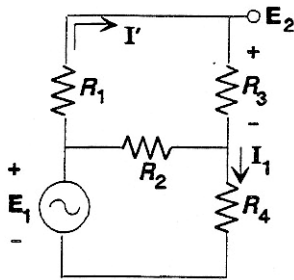


$$I' = \frac{R_2(I_2)}{(R_1 + R_2) + R_3}$$

$$E_1 = I'R_2 + I_2R_4$$

$$= \frac{R_2R_3I_2}{R_1 + R_2 + R_3} + R_4I_2$$

and
$$z_{12} = \frac{E_1}{I_2} = \frac{R_2R_3}{R_1 + R_2 + R_3} + R_4 = \frac{R_2R_3 + R_4(R_1 + R_2 + R_3)}{R_1 + R_2 + R_3}$$



$$E_2 = I'R_3 + I_1R_4$$

$$\text{CDR: } I' = \frac{R_2(I_1)}{(R_1 + R_3) + R_2}$$

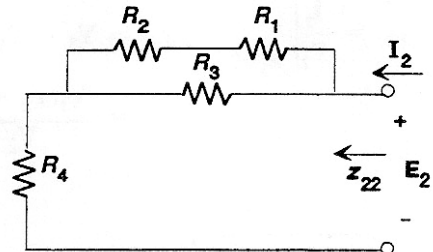
$$E_2 = \frac{R_2R_3I_1}{R_1 + R_2 + R_3} + I_1R_4$$

and
$$z_{21} = \frac{E_2}{I_1} = \frac{R_2R_3}{R_1 + R_2 + R_3} + R_4 = \frac{R_2R_3 + R_4(R_1 + R_2 + R_3)}{R_1 + R_2 + R_3}$$

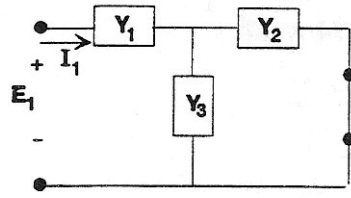
$$z_{22} = \left. \frac{E_2}{I_2} \right|_{I_1=0}$$

$$Z_{22} = R_4 + R_3 \parallel (R_1 + R_2)$$

$$= R_4 + \frac{R_3(R_1 + R_2)}{R_3 + (R_1 + R_2)}$$



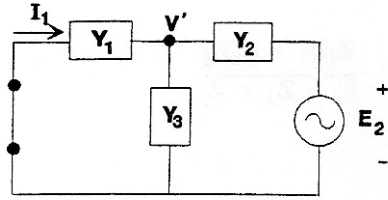
19. a.



$$y_{11} = \left. \frac{I_1}{E_1} \right|_{E_2=0} \quad Y_T = Y_1 \parallel (Y_2 + Y_3)$$

$$= \frac{Y_1(Y_2 + Y_3)}{Y_1 + Y_2 + Y_3}$$

$$= \frac{Y_1 Y_2 + Y_1 Y_3}{Y_1 + Y_2 + Y_3}$$



Nodal Analysis:

$$V'[Y_1 + Y_2 + Y_3] = E_2 Y_2$$

$$V' = I_1 / Y_1$$

and

$$\frac{-I_1}{Y_1} [Y_1 + Y_2 + Y_3] = E_2 Y_2$$

$$y_{12} = \left. \frac{I_1}{E_2} \right|_{E_1=0} = \frac{-Y_1 Y_2}{Y_1 + Y_2 + Y_3}$$

$$y_{21} = \left. \frac{I_2}{E_1} \right|_{E_2=0} \quad \text{Mirror image of } y_{12}$$

$$\therefore y_{21} = \frac{-Y_1 Y_2}{Y_1 + Y_2 + Y_3}$$

$$y_{22} = \left. \frac{I_2}{E_2} \right|_{E_1=0} \quad \text{Mirror image of } y_{11}$$

$$y_{22} = Y_T = Y_2 \parallel (Y_1 + Y_3)$$

$$= \frac{Y_1 Y_2 + Y_1 Y_3}{Y_1 + Y_2 + Y_3}$$

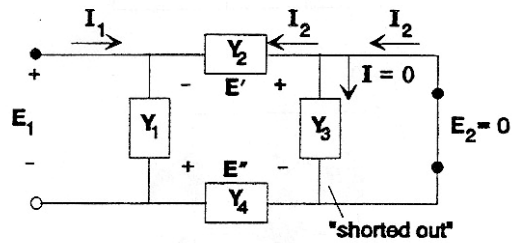
20. a.

$$y_{11} = \left. \frac{I_1}{E_1} \right|_{E_2=0}$$

$$y_{11} = Y_1 + Y_2 \parallel Y_4$$

$$= Y_1 + \frac{Y_2 Y_4}{Y_2 + Y_4}$$

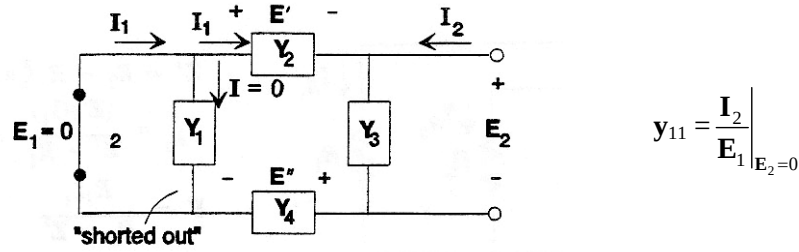
$$= \frac{Y_1(Y_2 + Y_4) + Y_2 Y_4}{Y_2 + Y_4}$$



$$y_{21} = \left. \frac{I_2}{E_1} \right|_{E_2=0} \quad (\text{using the above diagram})$$

$$E_1 = \frac{I_2}{Y_1} = -(E' + E'') = -\left(\frac{I_2}{Y_2} + \frac{I_2}{Y_4}\right) = -I_2 \left(\frac{1}{Y_2} + \frac{1}{Y_4}\right)$$

$$\text{and } E_1 = -I_2 \left(\frac{Y_4 + Y_2}{Y_4 Y_2}\right) \text{ with } y_{21} = \frac{I_2}{E_1} = -\frac{Y_2 Y_4}{Y_2 + Y_4}$$

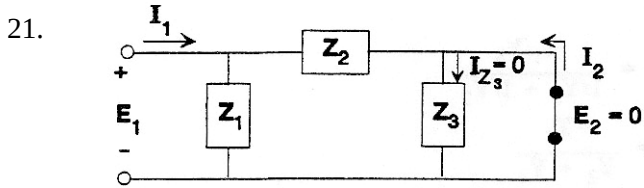


$$E_2 = \frac{I_1}{Y_3} = -(E' + E'') = -\left(\frac{I_1}{Y_2} + \frac{I_1}{Y_4}\right) = -I_1 \left(\frac{1}{Y_2} + \frac{1}{Y_4}\right)$$

$$\text{and } y_{12} = -\frac{Y_2 Y_4}{Y_2 + Y_4} = y_{21}$$

$$y_{22} = \left. \frac{I_2}{E_2} \right|_{E_1=0} \quad y_{22} = Y_3 + Y_2 \parallel Y_4 = Y_3 + \frac{Y_2 Y_4}{Y_2 + Y_4}$$

$$= \frac{Y_3(Y_2 + Y_4) + Y_2 Y_4}{Y_2 + Y_4}$$

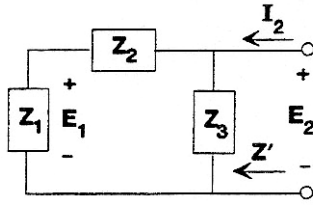


$$h_{11} = \left. \frac{E_1}{I_1} \right|_{E_2=0} = Z_T = Z_1 \parallel Z_2 = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

Using the above figure:

$$\text{CDR: } I_2 = \frac{-Z_1(I_1)}{Z_1 + Z_2}$$

$$h_{11} = \left. \frac{I_2}{I_1} \right|_{E_2=0} = \frac{-Z_1}{Z_1 + Z_2}$$



$$h_{12} = \left. \frac{E_1}{E_2} \right|_{I_1=0}$$

$$\text{VDR: } E_1 = \frac{Z_1 E_2}{Z_1 + Z_2}$$

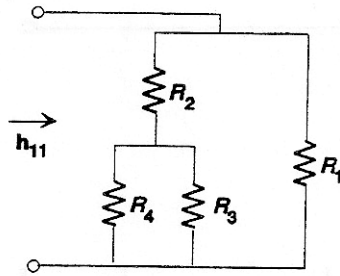
$$h_{12} = \left. \frac{E_1}{E_2} \right|_{I_1=0} = \frac{Z_1}{Z_1 + Z_2}$$

Using above figure:

$$h_{22} = \left. \frac{I_2}{E_2} \right|_{I_1=0} : Z' = Z_3 \parallel (Z_1 + Z_2) = \frac{Z_1 Z_3 + Z_2 Z_3}{Z_1 + Z_2 + Z_3}$$

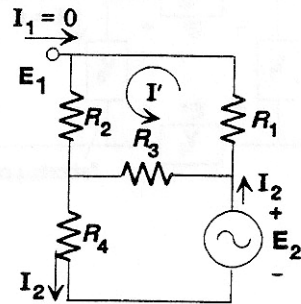
$$h_{22} = \frac{1}{Z'} = \frac{Z_1 + Z_2 + Z_3}{Z_1 Z_3 + Z_2 Z_3}$$

22. a.



$$h_{11} = \left. \frac{E_1}{I_1} \right|_{E_2=0}$$

$$= Z_i = R_1 \parallel (R_2 + R_3 \parallel R_4)$$



$$h_{12} = \left. \frac{E_1}{E_2} \right|_{I_1=0}$$

$$E_1 = I' R_2 + I_2 R_4$$

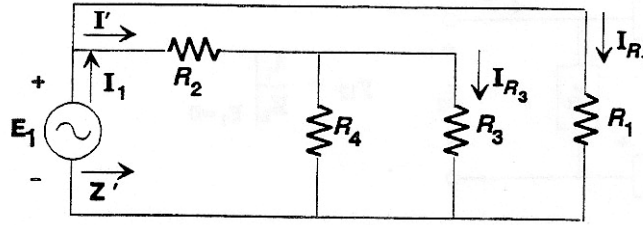
$$I' = \frac{R_3 (I_2)}{R_1 + R_2 + R_3}$$

$$\therefore E_1 = \frac{R_2 R_3 I_2}{R_1 + R_2 + R_3} + I_2 R_4$$

$$\text{and } I_2 = \frac{E_2}{Z'} = \frac{E_2}{R_4 + R_3 \parallel (R_1 + R_2)}$$

$$E_1 = \left[\frac{R_2 R_3}{R_1 + R_2 + R_3} + R_4 \right] \left[\frac{E_2}{R_4 + \frac{R_3 R_1 + R_2 R_3}{R_1 + R_2 + R_3}} \right]$$

$$\text{and } h_{12} = \frac{E_1}{E_2} = \frac{R_2 R_3 + R_4 (R_1 + R_2 + R_3)}{R_1 R_3 + R_2 R_3 + R_4 (R_1 + R_2 + R_3)}$$



$$Z' = R_2 + R_3 \parallel R_4$$

$$I_{R_1} = \frac{(Z')(I_1)}{Z' + R_1}$$

$$I' = \frac{R_1 I_1}{R_1 + Z'}$$

$$I_{R_3} = \frac{R_4 I'}{R_4 + R_3} = \frac{R_4}{R_4 + R_3} \left[\frac{R_1 (I_1)}{R_1 + Z'} \right]$$

$$= \frac{R_1 R_4 I_1}{(R_3 + R_4)(R_1 + Z')}$$

$$I_2 = -I_{R_1} - I_{R_3} = \frac{-Z' I_1}{Z' + R_1} - \frac{R_1 R_4 I_1}{(R_3 + R_4)(R_1 + Z')}$$

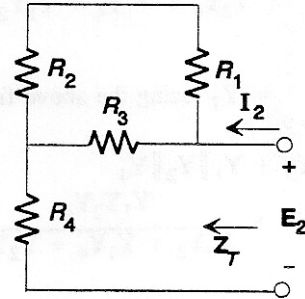
$$h_{22} = \frac{I_2}{I_1} \bigg|_{E_2=0} = \left[\frac{Z'}{Z' + R_1} + \frac{R_1 R_4}{(R_3 + R_4)(R_1 + Z')} \right]$$

$$= -\frac{1}{R_1 + Z'} \left[Z' + \frac{R_1 R_4}{R_3 + R_4} \right]$$

$$h_{22} = \frac{I_2}{E_2} \bigg|_{I_1=0} = \frac{1}{Z_T}$$

$$Z_T = R_4 + R_3 \parallel (R_1 + R_2)$$

$$h_{22} = \frac{1}{R_4 + R_3 \parallel (R_1 + R_2)}$$



A Y-Δ conversion would have simplified the problem to one similar to Fig 26.70.

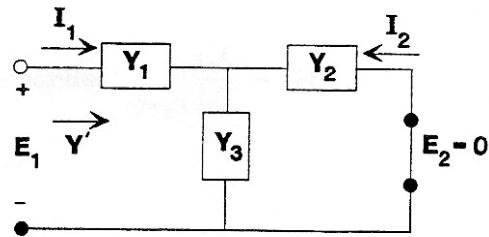
23. $h_{11} = \frac{E_1}{I_1} \bigg|_{E_2=0}$

$$Y' = Y_1 \parallel (Y_2 + Y_3)$$

$$Y' = \frac{Y_1 Y_2 + Y_1 Y_3}{Y_1 + Y_2 + Y_3}$$

$$h_{11} = \frac{1}{Y'} = \frac{Y_1 + Y_2 + Y_3}{Y_1 Y_2 + Y_1 Y_3}$$

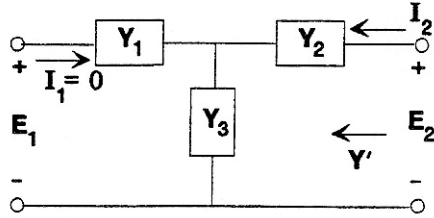
$$h_{21} = \frac{I_2}{I_1} \bigg|_{E_2=0}$$



From above figure:

$$\text{CDR: } I_2 = \frac{-Z_3 I_1}{Z_3 + Z_2} = \frac{-I_1/Y_3}{1/Y_3 + 1/Y_2}$$

$$\text{and } h_{21} = \left. \frac{I_2}{I_1} \right|_{E_2=0} = \frac{-1/Y_3}{1/Y_3 + 1/Y_2} = \frac{-Y_2}{Y_2 + Y_3}$$



$$h_{12} = \left. \frac{E_1}{E_2} \right|_{I_1=0}$$

$$\text{VDR: } E_1 = \frac{Z_3 E_2}{Z_3 + Z_2} = \frac{-1/Y_3 E_2}{1/Y_3 + 1/Y_2}$$

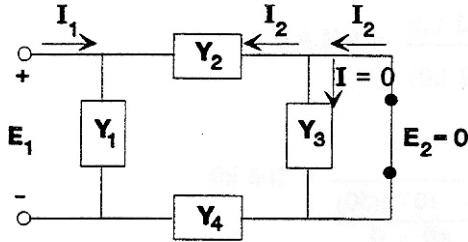
$$\text{And } E_1 = \frac{Y_2 E_2}{Y_2 + Y_3}$$

$$\text{with } h_{12} = \left. \frac{E_1}{E_2} \right|_{I_1=0} = \frac{Y_2}{Y_2 + Y_3}$$

$$h_{22} = \left. \frac{I_2}{E_2} \right|_{I_1=0} \quad Y' = \frac{Y_2 \cdot Y_3}{Y_2 + Y_3} \quad (\text{from above figure})$$

$$h_{22} = \left. \frac{I_2}{E_2} \right|_{I_1=0} = Y' = \frac{Y_2 Y_3}{Y_2 + Y_3}$$

24.

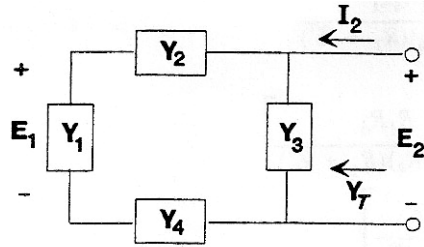


$$h_{11} = \left. \frac{E_1}{I_1} \right|_{E_2=0} = \frac{1}{Y_T} = \frac{1}{Y_1 + Y_2 \parallel Y_4}$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{E_2=0} : \text{CDR} \rightarrow I_2 = \frac{-Z_1(I_1)}{Z_1 + Z_2 + Z_4} = \frac{-1/Y_1(I_1)}{1/Y_1 + 1/Y_2 + 1/Y_4}$$

$$= \frac{-1/Y_1(I_1)}{\frac{Y_2 Y_4 + Y_1 Y_4 + Y_1 Y_2}{Y_1 Y_2 Y_4}}$$

$$\text{and } h_{21} = -\frac{Y_2 Y_4}{Y_2 Y_4 + Y_1 Y_4 + Y_1 Y_2}$$



$$h_{11} = \left. \frac{E_1}{E_2} \right|_{I_1=0}$$

$$\begin{aligned} \text{VDR: } E_1 &= \frac{Z_1(E_2)}{Z_1 + Z_2 + Z_4} \\ &= \frac{-1/Y_1(E_2)}{1/Y_1 + 1/Y_2 + 1/Y_4} \end{aligned}$$

$$\text{and } h_{12} = \frac{Y_2 Y_4}{Y_2 Y_4 + Y_1 Y_4 + Y_1 Y_2}$$

$$h_{22} = \left. \frac{I_2}{E_2} \right|_{I_1=0} = Y_T \text{ (using the above figure)}$$

$$\begin{aligned} Y_T &= Y_3 + Y_1 \parallel Y_2 \parallel Y_4 \\ &= Y_3 + \frac{Y_1 Y_2 Y_4}{Y_1 Y_2 + Y_1 Y_4 + Y_2 Y_4} \end{aligned}$$

25. a. Eq. 26.45:

$$A_i = \frac{h_f}{1 + h_o Z_L} = \frac{50}{1 + \left(\frac{1}{40 \text{ k}\Omega} \right) (2 \text{ k}\Omega)} = 47.62$$

b. Eq. 26.46:

$$\begin{aligned} A_v &= \frac{-h_f Z_L}{h_i (1 + h_o Z_L) - h_r h_f Z_L} \\ &= \frac{-50(2 \text{ k}\Omega)}{1 \text{ k}\Omega(1 + 0.05) - (4 \times 10^{-4})(50)(2 \text{ k}\Omega)} = -99 \end{aligned}$$

26. a.
$$Z_i = \frac{E_1}{I_1} = h_i - \frac{h_r h_f Z_L}{1 + h_o Z_L}$$

$$= 1 \text{ k}\Omega - \frac{(4 \times 10^{-4})(50)(2 \text{ k}\Omega)}{1 + \left(\frac{1}{40 \text{ k}\Omega} \right) (2 \text{ k}\Omega)} = 961.9 \text{ }\Omega$$

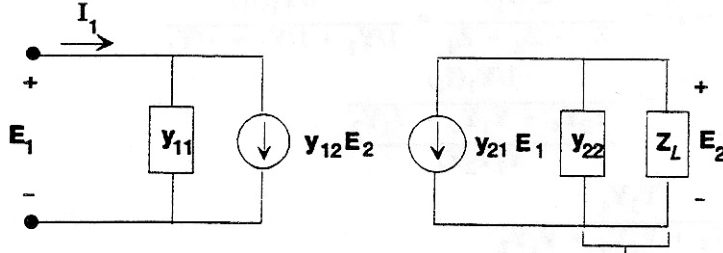
b.
$$Z_o = \frac{1}{h_o - \frac{h_r h_f}{h_i + R_s}} = \frac{1}{\frac{1}{40 \text{ k}\Omega} - \frac{(4 \times 10^{-4})(50)}{1 \text{ k}\Omega + 0}} = 200 \text{ k}\Omega$$

27. $\mathbf{Z}_{11} = 1 \text{ k}\Omega \angle 0^\circ$, $\mathbf{z}_{12} = 5 \text{ k}\Omega \angle 90^\circ$, $\mathbf{z}_{21} = 10 \text{ k}\Omega \angle 0^\circ$, $\mathbf{z}_{22} = 2 \text{ k}\Omega - j4 \text{ k}\Omega$, $\mathbf{Z}_L = 1 \text{ k}\Omega \angle 0^\circ$

$$\mathbf{Z}_i = \frac{\mathbf{E}}{\mathbf{I}} = \mathbf{z}_{11} - \frac{\mathbf{z}_{12}\mathbf{z}_{21}}{\mathbf{z}_{22} + \mathbf{Z}_L} = 1 \text{ k}\Omega - \frac{(5 \text{ k}\Omega \angle 90^\circ)(10 \text{ k}\Omega)}{2 \text{ k}\Omega - j4 \text{ k}\Omega + 1 \text{ k}\Omega} = 9,219.5 \Omega \angle -139.40^\circ$$

$$\mathbf{Z}_o = \frac{\mathbf{E}_2}{\mathbf{I}_2} = \mathbf{z}_{22} - \frac{\mathbf{z}_{12}\mathbf{z}_{21}}{\mathbf{R}_s + \mathbf{z}_{11}} = 2 \text{ k}\Omega - j4 \text{ k}\Omega - \frac{(5 \text{ k}\Omega \angle 90^\circ)(10 \text{ k}\Omega)}{1 \text{ k}\Omega + 1 \text{ k}\Omega} = 29.07 \text{ k}\Omega \angle -86.05^\circ$$

28.



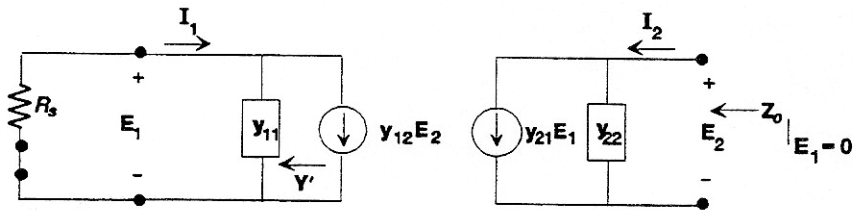
$$\begin{aligned} \frac{1}{y_{22}} \parallel \mathbf{Z}_L &= \frac{1/y_{22} \mathbf{Z}_L}{1/y_{22} + \mathbf{Z}_L} \\ &= \frac{\mathbf{Z}_L}{1 + y_{22} \mathbf{Z}_L} \end{aligned}$$

$$\mathbf{E}_2 = -y_{21}\mathbf{E}_1 \left[\frac{\mathbf{Z}_L}{1 + y_{22}\mathbf{Z}_L} \right]$$

$$\mathbf{I}_1 = \mathbf{E}_1 y_{11} + y_{12}\mathbf{E}_2 = \mathbf{E}_1 y_{11} + y_{12} \left[-y_{21}\mathbf{E}_1 \left(\frac{\mathbf{Z}_L}{1 + y_{22}\mathbf{Z}_L} \right) \right]$$

$$\frac{\mathbf{I}_1}{\mathbf{E}_1} = y_{11} - \frac{y_{12}y_{21}\mathbf{Z}_L}{1 + y_{22}\mathbf{Z}_L}$$

$$\text{and } \mathbf{Z}_i = \frac{\mathbf{E}_1}{\mathbf{I}_1} = \frac{1}{y_{11} - \frac{y_{12}y_{21}\mathbf{Z}_L}{1 + y_{22}\mathbf{Z}_L}}$$



$$\mathbf{Y}' = \mathbf{y}_{11} = \frac{1}{R_s}$$

$$\mathbf{E}_1 = \frac{-\mathbf{y}_{12}\mathbf{E}_2}{\mathbf{Y}'} = \frac{-\mathbf{y}_{12}\mathbf{E}_2}{\mathbf{y}_{11} + \frac{1}{R_s}} = \frac{-\mathbf{y}_{12}R_s\mathbf{E}_2}{\mathbf{y}_{11}R_s + 1}$$

$$\mathbf{I}_2 = \mathbf{y}_{21}\mathbf{E}_1 + \mathbf{y}_{22}\mathbf{E}_2 = \mathbf{y}_{22}\left[\frac{-\mathbf{y}_{12}R_s\mathbf{E}_2}{\mathbf{y}_{11}R_s + 1}\right] + \mathbf{y}_{22}\mathbf{E}_2$$

$$\frac{\mathbf{I}_2}{\mathbf{E}_2} = -\frac{\mathbf{y}_{12}\mathbf{y}_{21}R_s}{\mathbf{y}_{11}R_s + 1} + \mathbf{y}_{22}$$

$$\text{and } \mathbf{Z}_o = \left.\frac{\mathbf{E}_2}{\mathbf{I}_2}\right|_{\mathbf{E}_1=0} = \frac{1}{\mathbf{y}_{22} - \frac{\mathbf{y}_{12}\mathbf{y}_{21}R_s}{1 + \mathbf{y}_{11}R_s}}$$

$$29. \quad \mathbf{h}_{11} = \frac{\mathbf{z}_{11}\mathbf{z}_{22} - \mathbf{z}_{12}\mathbf{z}_{21}}{\mathbf{z}_{22}} = \frac{(4 \text{ k}\Omega)(4 \text{ k}\Omega) - (2 \text{ k}\Omega)(3 \text{ k}\Omega)}{4 \text{ k}\Omega} = \mathbf{2.5 \text{ k}\Omega}$$

$$\mathbf{h}_{12} = \frac{\mathbf{z}_{12}}{\mathbf{z}_{22}} = \frac{2 \text{ k}\Omega}{4 \text{ k}\Omega} = \mathbf{0.5}$$

$$\mathbf{h}_{12} = -\frac{\mathbf{z}_{21}}{\mathbf{z}_{22}} = \frac{3 \text{ k}\Omega}{4 \text{ k}\Omega} = \mathbf{-0.75}$$

$$\mathbf{h}_{22} = \frac{1}{\mathbf{z}_{22}} = \frac{1}{4 \text{ k}\Omega} = \mathbf{0.25 \text{ mS}}$$

$$30. \quad \text{a.} \quad \Delta_{\mathbf{h}} = \mathbf{h}_{11}\mathbf{h}_{22} - \mathbf{h}_{12}\mathbf{h}_{21} = (10^3)(20 \times 10^{-6}) - (2 \times 10^{-4})(100) \\ = 20 \times 10^{-3} - 20 \times 10^{-3} = 0$$

$$\mathbf{z}_{11} = \frac{\Delta_{\mathbf{h}}}{\mathbf{Z}_{22}} = \mathbf{0 \Omega}, \mathbf{z}_{12} = \frac{\mathbf{h}_{12}}{\mathbf{h}_{12}} = \frac{2 \times 10^{-4}}{20 \times 10^{-6} \text{ S}} = \mathbf{10 \Omega}$$

$$\mathbf{z}_{21} = \frac{-\mathbf{h}_{21}}{\mathbf{h}_{22}} = \frac{-100}{20 \times 10^{-6} \text{ S}} = \mathbf{-5 \text{ M}\Omega}, \mathbf{z}_{22} = \frac{1}{\mathbf{h}_{22}} = \mathbf{50 \Omega}$$

$$\text{b.} \quad \mathbf{y}_{11} = \frac{1}{\mathbf{h}_{11}} = \frac{1}{10^3 \Omega} = \mathbf{10^{-3} \text{ S}}, \mathbf{y}_{12} = \frac{-\mathbf{h}_{12}}{\mathbf{h}_{11}} = \frac{-2 \times 10^{-4}}{10^3 \Omega} = \mathbf{-2 \times 10^{-7} \text{ S}}$$

$$\mathbf{y}_{21} = \frac{\mathbf{h}_{21}}{\mathbf{h}_{11}} = \frac{100}{10^3 \Omega} = \mathbf{100 \times 10^{-3} \text{ S}}, \mathbf{y}_{22} = \frac{\Delta_{\mathbf{h}}}{\mathbf{h}_{11}} = \mathbf{0 \text{ S}}$$